Clair Field, 2016: geochemical evaluation of the 206/12a-5 (E-1) well

SOURCE ROCK SCREENING, MATURITY EVALUATION AND GEOCHEMICAL ANALYSIS AND EVALUATION OF FLOWED, SPUN OUT OIL AND EXTRACTED OILS, PLUS FLOWED, ISOTUBE AND ISOJAR GASES.

Report No. 1276

by
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Under Project no. APT UK 381

on behalf of
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Appendices are presented in full on the CD provided with the hard copy report.

CHAPTER 1
SUMMARY

This report contains the results of a geochemical evaluation of the BP 206/12a-5 (E-1) well drilled in the SW Clair segment of the Clair Field, West of Shetlands UK. Analyses have been undertaken on flowed oils, spun out oils, cuttings, cores, flowed gas, Isotube gases and Isojar gases. In addition drilling mud samples have been analysed for comparison with tested and extracted oil data.

All non-gas samples were analysed in the laboratories of APT AS (chemical analyses) in Norway and APT UK (microscopy work) in North Wales (with some additional oil analyses undertaken in the ALcontrol laboratory in North Wales). Gas samples have been analysed by ISOLAB BV in The Netherlands.

The following are the principal conclusions arising from the interpretation of the newly generated data:

MATURITY

- Both vitrinite reflectivity and spore colouration data for the interval from around 270m MDRT down to the Kimmeridge Clay Formation suggest that the section is immature to the threshold of the oil generation window.
- Limited maturity data was obtained in the Clair Group section.
- The maturity gradients extrapolated to surface suggest little or no recent uplift and erosion.

HYDROCARBON SOURCE POTENTIAL

- Organic rich samples are present in the Jurassic interval of the well.
- Oil prone source rocks are present in the Kimmeridge Clay Formation (KCF) and some samples in the Rona Member also appear to have some oil generative potential.

TESTED (FLOWED) OIL

- The flowed oils from the 206/12a-5 (E-1) well have measured API values averaging 25.8°.
- The flowed and spun out oils appear typical of oils previously analysed from the south west part of the Clair Field.
- The oils present in the 206/12a-5 (E-1) Cretaceous, Jurassic and Devonian intervals are likely to form a single continuous reservoir with homogenised oils in full communication.
- Two phases of oil are present, comprising an earlier heavily degraded phase on which later, lighter and un-degraded oil is superimposed, typical of Clair Field oils.
- The oil appears to have a slightly higher proportion of the later live oil compared to Clair Field oils in general and even with respect to the previously identified and different SW Clair Field oils.
- The oil has been generated from a source rock within the main oil window, with a generation temperature in the range of 135-140°C suggested.

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EXTRACTED (PROXY) OILS

- Good quality oil staining has been found in samples from the Cretaceous, Jurassic and Devonian.
- Oils extracted from cores are identical to the flowed oils.
- The deeper core samples from the Lower Clair Group Unit II/I are not oil stained and contain no migrant oils.
- The two phase oils are apparent in the extracted oils from the cores.

GASES

- Gases analysed from Cretaceous, Jurassic and Devonian reservoir intervals have been generated from a source rock in the latter part of the oil window at a vitrinite reflectivity equivalent of about 1.1-1.2%, and are of higher maturity than the oils in the well.
- On the basis of the data obtained it appears that the gases in the Cretaceous, Jurassic and Devonian reservoirs possibly form a continuous homogenised accumulation in full communication.
- The data from all of the tested gases (Cretaceous, Jurassic and Devonian) are very similar to data seen from the collective gases in the 206/13a-3 (HEX A) well.
- Comparison of the data from all the tested gases in the Clair area to date indicates that isotopically the C2+ gases are virtually identical but C1 (methane) is isotopically lighter than the 206/8-16 (SEG 0) and 206/9a-3 (SEG 5) wells but isotopically heavier than the 206/9a-4 (NEF).
- The well is located far to the south of previously drilled wells and penetrates through the basal Lower Clair Group into Basement. A particularly intriguing observation from the Isotube gas data is that while gas contents drop significantly at about the supposed field wide fwl at Ca. 2120m TVDSS gas contents then increase markedly in the basal Lower Clair Group below Ca. 2200m TVDSS into the Basement. This increase in gas abundance is mirrored with increases in wet gas contents, the methane becomes isotopically significantly heavier as does the methane deuterium isotope.

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CHAPTER 2
INTRODUCTION

2.1 Introduction

This study has been undertaken in response to a request from Tomieka Searcy (Petroleum Systems Analyst) for a geochemical analysis and evaluation programme for the 206/12a-5 well, Clair Field, West of Shetlands. A detailed proposal was generated and submitted on the 10th December 2014 (Proposal No. 2014-068). Notification that the proposal was accepted was received on 18th December 2014 under Maximo number 737742-01 and the APT UK internal project number 381 was assigned to the study. Changes to the BP Accenture accounting system in April 2015 resulted in the Maximo number changing to BP service order number 4300427354.

Subsequently a request for additional kerogen maturity work was made by Callum Leighton (Petroleum Systems Analyst) on an additional 10 cuttings samples. A proposal for the additional maturity work was submitted on the 11th November 2015 (Proposal No. 2015-52), and this was approved under BP service order number 4300573528. This report contains the results of these additional studies.

The study objectives as set out in the proposal were as follows:

- Obtain high quality geochemical data on:
  - tested fluids (oils and gases);
  - proxy oils extracted from core plugs;
  - mud gases (Isotubes and Isojars);
  - characterise potential source rock intervals.

- Evaluate the data to:
  - Establish the geochemical homogeneity (or otherwise), provenance and maturity of the hydrocarbons (oils and gases) encountered in well 206/12a-5;
  - Determine the amount and severity of biodegradation of any oils encountered;
  - Determine any similarities or differences between any oils encountered in well 206/12a-5 and the hydrocarbons identified in the nearby Cretaceous sands and Clair Groups;
  - Refine the hydrocarbon water contact and deepest residual oil through the investigation of changes in the amount and geochemical character of oils and gases recovered from the full extent of the analysed section;
  - Shed more light on the charge history of the different reservoir intervals and Clair Field petroleum system;
  - Determine source potential in any grey shale rich Mesozoic intervals.

In addition, as an aid in the interpretation of the data with respect to potential contamination from drilling filtrate; data were to be obtained from mud samples taken at various times during the drilling of the well.

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2.2 Well details and stratigraphy

The 206/12a-5 well is a vertical appraisal well drilled in the South West Clair Field, UK West of Shetlands (Figure 2.1). The well details as supplied by BP and used in this study are shown in Table 2.1 along with the sequence, chrono- and litho-stratigraphic tops used in the interpretation of the data. The stratigraphies recorded were supplied by BP and dated 18th June 2015.

2.3 Samples and analytical programme

The study aims have been carried out through the geochemical analyses of cuttings, cores, muds, oils, spun out oils and gases. The distribution of analysed samples throughout the well section is displayed diagrammatically as Figure 2.2.

Samples were received for analysis in several batches as follows:

- Batch 1: 17/12/2014 – 3 oils spun out from cores (1897.03m MDRT, 1922.08m MDRT and 1948.67m MDRT) arrived in the APT AS laboratories.
- Batch 2: 23/12/2014 – 1 box of Isojars (15 samples over interval 1120 to 2352m MDRT) and 3 boxes of Isotubes (93 samples 950 to 2349m MDRT) arrived in APT UK offices. They were then dispatched onward and arrived in the Isolab BV laboratories on 6th February 2015.
- Batch 3: 05/01/2015 – 33 COPL (interval; 1874.67 to 1997.67m MDRT) arrived in the APT AS laboratories.
- Batch 4: 05/01/2015 – 3 SWC (1984m MDRT, 2001m MDRT, 2022m MDRT and 2048m MDRT) arrived in the APT AS laboratories.
- Batch 5: 23/04/2015 – 12 Core chips (interval 1884.58 to 1967.35m MDRT) arrived in the APT UK laboratories.
- Batch 6: 13/08/2015 – 5 gas samples (2 DSTs and 3 MDTs, 1922.4m MDRT, 1956.0 m MDRT and 1984.5m MDRT) arrived in the Isolab BV laboratories.
- Batch 7: 13/08/2015 – 4 oils (1 DST sample and 3 MDTs, 1922.4m MDRT, 1956.0m MDRT and 1984.5m MDRT) arrived in the APT AS laboratories.
- Batch 8: 13/10/2015 – 2 muds (1983m MDRT and 2352m MDRT) arrived in the APT UK laboratories. They were then forwarded to the APT AS laboratories and received 29th October 2015.

Subsamples of 2 core plugs (1932.67m MDRT and 1928.61m MDRT) and 4 SWC (1984m MDRT, 2001m MDRT, 2022m MDRT and 2048m MDRT) from APT AS laboratories arrived in APT UK offices variously on 19th March 2015 and 18th May 2015 respectively. These were then sent for kerogen preparation.

A selection of cuttings samples (Batch 9) were received in the APT UK offices for extra microscopy work in six large boxes on 13th October 2015. Details are below.

- Box No. 1 (267 - 1111.5 m MDBRT)
- Box No. 2 (1111.5 - 1330 m MDBRT)
- Box No. 3 (1330 - 1590 m MDBRT)
- Box No. 4 (1590 - 1773 m MDBRT)
- Box No. 5 (1773 - 1851 m MDBRT)
- Box No. 9 (2109 - 2205 m MDBRT)

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Details of the tested oil and gas samples are given in Table 2.2. A full list of non-gas samples received excluding the cuttings are given in Table 2.3.

The full list of analysed cuttings, core, oil, spun out oils and muds along with the analytical programme undertaken are shown in Table 2.4. The non-gas sample analyses were mainly carried out by APT AS, with additional oils analyses carried out by ALcontrol and maturity analyses by APT UK.

The numbers of samples submitted for the various analytical procedures are summarised as follows:

Rocks and fluids

Analyses		Ctgs	Core	Oil	Spun out oil	Muds	Totals
Vitrinite reflectivity		25	15	0	0	0	**40**
Spore Colour Indices & kerogen composition		25	9	0	0	0	**34**
Total organic carbon (TOC) content*		0	14	0	0	0	**14**
Rock-Eval pyrolysis*		0	12	0	0	0	**12**
Non-quantitative solvent extraction (DCM)		0	0	0	0	2	**2**
Quantitative solvent extraction (DCM)		0	25	0	0	0	**25**
Whole oil gas chromatography (oils)		0	0	4	3	0	**7**
Whole extract gas chromatography		0	15	0	0	2	**17**
Topping loss (volatiles)		0	0	4	0	0	**4**
Fractionation (MPLC)		0	15	4	3	2	**24**
Asphaltene content (precipitated)		0	15	4	3	2	**24**
Alkane fraction gas chromatography		0	0	0	0	0	**0**
GC-MS	alkane fraction	0	7	4	3	2	**16**
	aromatic fraction	0	7	4	3	2	**16**
Carbon isotope ratios	whole oil/extract	0	2	4	3	0	**9**
	alkane fraction	0	2	4	3	0	**9**
	aromatic fraction	0	2	4	3	0	**9**
	polar fraction	0	2	4	3	0	**9**
	asphaltene fraction	0	2	4	2	0	**9**
Sulphur and metals content & total acid number		0	0	4	0	0	**4**
API gravity, kinematic viscosity & wax content		0	0	4	0	0	**4**

A list of all gas samples received is given in Table 2.5. The full list of analysed gases (flowed, Isotubes and Isojars) and the analytical programme undertaken is shown in Table 2.6. The gas analyses were wholly carried out by ISOLAB B.V.

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Gases

Analyses	Flowed gas	Isotubes	Isojars	Totals
Gas composition	5	57	15	**77**
Carbon isotope ratios, methane	5	52	15	**72**
Carbon isotope ratios, ethane to pentanes	5	33	9	**47**
Deuterium isotope ratios, methane	5	33	9	**47**
Carbon isotope ratios, carbon dioxide	5	-	-	**5**

Details of the analytical methods used in the APT AS and APT UK laboratories are shown in Appendix 12 and those in the ISOLAB BV laboratory in Appendix 13.

2.4 Reporting

All interpretation and comparison of the datasets were carried out by Emma Straughan, with extra technical guidance from Patrick Barnard. During the progress of the study data were reported directly to the client for interpretation and internal use as Excel workbooks, as and when analytical results became available.

This final report provides in-depth interpretations and discussions of the data by APT UK, as well as presenting a comprehensive compilation of the raw analytical data (as appendices); it is also intended as a formal presentation of the completed work for the purposes of archiving and future reference.

A draft copy of the final report was despatched to the client for comments on 10th March 2016. A second draft copy of the report was sent on the 6th May 2016 including additional maturity results. This final report was issued as a hard copy on the 10th June 2016.

2.5 Personnel

Personnel involved in the project have included:

Project administrator:	Emma Straughan
Data interpretation and report preparation:	Emma Straughan
Technical input:	Patrick Barnard
Maturity (microscopy) analyses supervised by:	Rick Harding and Edoardo Fiordalisi
Microscopy quality assurance:	Rick Harding
Chemical analyses (APT AS) supervised by:	Dr Kjell Urdal and Dr Per Erling Johansen of APT AS in Kjeller, Norway
Chemical analyses (ALcontrol) supervised by:	Gary Hogan of ALcontrol Oil & Fuel Analysis
Isotope analyses supervised by:	Bart van der Haven of ISOLAB B.V. in Neerijnen, The Netherlands

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Well name 206/12a-5
Alternative well name E-1
Basin Foula Basin
Field Clair
Field segment 12-1 East

Wellhead latitude (°) 60.60223
Wellhead longitude (°) -2.74322
Wellhead northing (mN) 6718807.6
Wellhead easting (mE) 514062.64

Rig datum, aMSL (m) 23.5
Water depth, bMSL (m) 163.6
RT-SEABED (m) 187.1
Total depth, MDBRT (m) 2352
Total depth, TVDSS (m) 2345.7

206/12a-5 formation tops				MDBRT, m	TVDSS, m
Tertiary	Westray Gp.	Pliocene	T190	187.10	163.60
	Stronsay Gp.	Eocene	T110	501.00	477.50
	Montrose-Moray Gp.	Balder	T50	986.50	963.00
Upper Cretaceous	Shetland Gp.	Late Maastrichtian	K90.4	1084.00	1060.50
			K90.3	1223.00	1199.40
		Early Maastrichtian	K90.2	1351.00	1330.40
		Campanian	K80.2	1544.00	1520.40
		Santonian	K76	1688.00	1664.40
		Coniacian	K74	1790.00	1766.40
		Turonian	K72	1855.00	1831.40
Lower Cretaceous	Cromer Knoll		K30	1919.00	1895.40
			K20	1924.00	1900.30
Jurassic	Humber Gp.	KCF		1929.00	1905.30
		Rona		1936.67	1913.01
Devonian	Lower Clair Gp.	Unit III		1967.00	1943.30
		Unit II/I		2015.67	1992.00
Basal conglomerate/BC sands				2063.50	2039.70
Precambrian	Basement	Lewisian Complex		2325.81	2296.51
			TD	2352.00	2322.20

Casing points	MDBRT, m	TVDSS, m	Drilling fluid
13 3/8"	1106.7	1106.68	DFS-150 LTOBM
9 5/8"	1828.0	1827.86	Carbo-Sea RDIF
7"	2348.6	2342.39	

Stratigraphy taken from completion log and tops table received 18th June 2015

TABLE 2.1 Well details and stratigraphic tops for well 206/12a-5

Well	Labels on samples in APT UK			Oil sample details from BP				Strat.	APTID
	Label depth 'MD (m)	Sampled depth 'MD (m)	Ex sample No/Cylinder	Sample Type	Test / comments	Sample No.	Sample taken		
206/12a-5	1727.6^	1936-1963*	13924-QA	Oil	DST/ main flow, bottom hole sample	2.404	09/02/2015 20:20	Rona	160301
	1922.4	1922.4	MPSR-3799	Oil	MDT / sample 20	1.24		K30	160302
	1956.0	1956.0	MPSR-3344	Oil	MDT / sample 12	1.15		Rona	160303
	1984.5	1984.5	MPSR-4608	Oil	MDT / sample 7	1.11		LCG III	160304
206/12a-5	1727.6^	1936-1963*	13924-QA	Isotube	DST/ main flow, bottom hole sample	2.404	09/02/2015 20:20	Rona	n/a
	1922.4	1922.4	MPSR-3799	Isotube	MDT / sample 20	1.24		K30	n/a
	1956.0	1956.0	MPSR-3344	Isotube	MDT / sample 12	1.15		Rona	n/a
	1984.5	1984.5	MPSR-4608	Isotube	MDT / sample 7	1.11		LCG III	n/a
	-	1936-1963*	A03185+	Isotube	DST / main flow, seperator gas sample, cylinder no. A01442	2.57	09/02/2015 21:18	Rona	n/a

^ - tool depth
* - perforated depth
+ - corelab transfer cylinder

TABLE 2.2 Tested oil and gas details for 206/12a-5

Well	MD (m)	TVDSS (m)	Sample type	APTID	Other ID	Mud type	Stratigraphy			Source	Analysed
206/12a-5	1874.67	1851.0	Core plug	146963		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	
	1878.67	1855.0	Core plug	146964		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	
	1884.58	1860.9	Core	154655		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	Y
	1884.67	1861.0	Core plug	146965		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	Y
	1888.67	1865.0	Core plug	146966		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	
	1892.66	1869.0	Core	154656		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	Y
	1892.67	1869.0	Core plug	146967		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	
	1896.67	1873.0	Core plug	146968		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	
	1897.03	1873.4	Spun out oils	146645	23A	OBM	Upr. Cretaceous	Turonian	K72	CL + TT	Y
	1900.67	1877.0	Core plug	146969		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	Y
	1902.70	1879.0	Core	154657		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	Y
	1910.00	1886.3	Core plug	146970		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	
	1911.17	1887.5	Core	154658		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	Y
	1912.65	1889.0	Core plug	146971		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	
	1916.67	1893.0	Core plug	146972		OBM	Upr. Cretaceous	Turonian	K72	CL + TT	Y
	1920.67	1897.0	Core plug	146973		OBM	Lwr. Cretaceous	Cromer Knoll	K30	CL + TT	
	1922.08	1898.4	Spun out oils	146643	44B	OBM	Lwr. Cretaceous	Cromer Knoll	K30	CL + TT	Y
	1922.40	1898.7	Oil (MDT)	160302	1.244	OBM	Lwr. Cretaceous	Cromer Knoll	K30	CL + TT	Y
	1924.67	1901.0	Core plug	146974		OBM	Lwr. Cretaceous	Cromer Knoll	K20	CL + TT	Y
	1928.61	1905.0	Core plug	146975		OBM	Lwr. Cretaceous	Cromer Knoll	K20	CL + TT	Y
	1932.42	1908.8	Core	154659		OBM	Jurassic	KCF		CL + TT	Y
	1932.67	1909.0	Core plug	146976		OBM	Jurassic	KCF		CL + TT	Y
	1936-1963	1912.3-1939.3	Oil (DST)	160301	2.404	OBM	Jurassic	Rona		CL + TT	Y

TABLE 2.3 Oil, spun out oils and cores received from 206/12a-5

Well	MD (m)	TVDSS (m)	Sample type	APTID	Other ID	Mud type	Stratigraphy			Source	Analysed
206/12a-5	1937.14	1913.5	Core	154660		OBM	Jurassic	Rona		CL + TT	Y
	1937.32	1913.7	Core	154661		OBM	Jurassic	Rona		CL + TT	Y
	1937.37	1913.7	Core plug	146977		OBM	Jurassic	Rona		CL + TT	
	1938.00	1914.3	Mud	163301		OBM	Jurassic	Rona		CL + TT	Y
	1940.67	1917.0	Core plug	146978		OBM	Jurassic	Rona		CL + TT	
	1944.05	1920.4	Core	154662		OBM	Jurassic	Rona		CL + TT	Y
	1944.67	1921.0	Core plug	146979		OBM	Jurassic	Rona		CL + TT	
	1948.67	1925.0	Spun out oils	146644	18G	OBM	Jurassic	Rona		CL + TT	Y
	1948.67	1925.0	Core plug	146980		OBM	Jurassic	Rona		CL + TT	Y
	1948.68	1925.0	Core	154663		OBM	Jurassic	Rona		CL + TT	Y
	1952.67	1929.0	Core plug	146981		OBM	Jurassic	Rona		CL + TT	
	1956.00	1932.3	Oil (MDT)	160303	1.154	OBM	Jurassic	Rona		CL + TT	Y
	1956.67	1933.0	Core plug	146982		OBM	Jurassic	Rona		CL + TT	Y
	1960.35	1936.7	Core	154664		OBM	Jurassic	Rona		CL + TT	Y
	1960.67	1937.0	Core plug	146983		OBM	Jurassic	Rona		CL + TT	
	1964.67	1941.0	Core plug	146984		OBM	Jurassic	Rona		CL + TT	
	1966.70	1943.0	Core	154665		OBM	Jurassic	Rona		CL + TT	Y
	1967.35	1943.7	Core	154666		OBM	Jurassic	Rona		CL + TT	Y
	1968.67	1945.0	Core plug	146985		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	
	1972.67	1949.0	Core plug	146986		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	
	1976.67	1953.0	Core plug	146987		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	
	1980.67	1957.0	Core plug	146988		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	
	1984.00	1960.3	SWC	153498		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	Y
	1984.50	1960.8	Oil (MDT)	160304	1.105	OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	Y

TABLE 2.3 Oil, spun out oils and cores received from 206/12a-5

Well	MD (m)	TVDSS (m)	Sample type	APTID	Other ID	Mud type	Stratigraphy			Source	Analysed
206/12a-5	1984.67	1961.0	Core plug	146989		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	
	1988.67	1965.0	Core plug	146990		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	Y
	1993.67	1970.0	Core plug	146991		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	
	1997.67	1974.0	Core plug	146992		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	Y
	2001.00	1977.3	SWC	153499		OBM	Devonian	Lower Clair Gp.	Unit III	CL + TT	Y
	2022.00	1998.3	SWC	153500		OBM	Devonian	Lower Clair Gp.	Unit II/I	CL + TT	Y
	2048.00	2024.3	SWC	153501		OBM	Devonian	Lower Clair Gp.	Unit II/I	CL + TT	Y
	2352.00	2336.8	Mud	163302		OBM	Precambrian	Basement		CL + TT	Y

MAT : Maturity analyses only

Mud type : OBM - oil based mud

Stratigraphy source : CL - Completion log
TT - tops table dated 18th June 2015

TABLE 2.3 Oil, spun out oils and cores received from 206/12a-5

Well	MD (m)	TVDSS (m)	Type	Other ID	Stratigraphy	Analysis reason	APTID	TOC	Rock-Eval	VR	SCI	Extraction	Fractionation	WE-GC	WO-GC	GC-MS	CI (blk/alk/arm)	CI (pol/asp)	Oil chem/phys
206/12a-5	270	246.5	Cuttings		Westray Gp.	Mat	n/a			Y	Y								
	360	336.5	Cuttings		Westray Gp.	Mat	n/a			Y	Y								
	480	456.5	Cuttings		Westray Gp.	Mat	n/a			Y	Y								
	540	516.5	Cuttings		Stronsay Gp.	Mat	n/a			Y	Y								
	570	546.5	Cuttings		Stronsay Gp.	Mat	n/a			Y	Y								
	600	576.5	Cuttings		Stronsay Gp.	Mat	n/a			Y	Y								
	720	696.5	Cuttings		Stronsay Gp.	Mat	n/a			Y	Y								
	760	736.5	Cuttings		Stronsay Gp.	Mat	n/a			Y	Y								
	1050	1026.5	Cuttings		Balder	Mat	n/a			Y	Y								
	1100	1076.5	Cuttings		Late Maastrictian K90.4	Mat	n/a			Y	Y								
	1150	1126.5	Cuttings		Late Maastrictian K90.4	Mat	n/a			Y	Y								
	1210	1186.4	Cuttings		Late Maastrictian K90.4	Mat	n/a			Y	Y								
	1250	1226.4	Cuttings		Late Maastrictian K90.3	Mat	n/a			Y	Y								
	1300	1276.4	Cuttings		Late Maastrictian K90.4	Mat	n/a			Y	Y								
	1360	1336.4	Cuttings		Early Maastrictian K90.2	Mat	n/a			Y	Y								
	1400	1376.4	Cuttings		Early Maastrictian K90.2	Mat	n/a			Y	Y								
	1450	1426.4	Cuttings		Early Maastrictian K90.2	Mat	n/a			Y	Y								
	1500	1476.4	Cuttings		Early Maastrictian K90.2	Mat	n/a			Y	Y								
	1550	1526.4	Cuttings		Campanian K80.2	Mat	n/a			Y	Y								
	1600	1576.4	Cuttings		Campanian K80.2	Mat	n/a			Y	Y								
	1650	1626.4	Cuttings		Campanian K80.2	Mat	n/a			Y	Y								
	1700	1676.4	Cuttings		Santonian K76	Mat	n/a			Y	Y								
	1749	1725.4	Cuttings		Santonian K76	Mat	n/a			Y	Y								
	1806	1782.4	Cuttings		Coniacian K74	Mat	n/a			Y	Y								
	1884.58	1860.9	Core		Turonian (K72)	SR/Mat	154655	Y		Y	Y	Y							
	1884.67	1861.0	Core plug		Turonian (K72)	OS	146965					Y	Y	Y					

TABLE 2.4 Analysed study samples and analyses undertaken - oils, spun out oils, cuttings, cores and muds

Well	MD (m)	TVDSS (m)	Type	Other ID	Stratigraphy	Analysis reason	APTID	TOC	Rock-Eval	VR	SCI	Extraction	Fractionation	WE-GC	WO-GC	GC-MS	CI (blk/alk/arm)	CI (pol/asp)	Oil chem/phys
206/12a-5	1892.66	1869.0	Core		Turonian (K72)	SR/Mat	154656	Y	Y	Y	Y	Y							
	1897.03	1873.4	Spun out oil	23A	Turonian (K72)	OS	146645						Y		Y	Y	Y	Y	
	1900.67	1877.0	Core plug		Turonian (K72)	OS	146969					Y	Y	Y					
	1902.70	1879.0	Core		Turonian (K72)	SR/Mat	154657	Y		Y	Y	Y							
	1911.17	1887.5	Core		Turonian (K72)	SR/Mat	154658	Y	Y	Y	Y	Y							
	1916.67	1893.0	Core plug		Turonian (K72)	OS	146972					Y	Y	Y		Y			
	1922.08	1898.4	Spun out oil	44B	Cromer Knoll (K30)	OS	146643						Y		Y	Y	Y	Y	
	1922.40	1898.7	Oil (MDT)	1.244	Cromer Knoll (K30)	Oil	160302						Y		Y	Y	Y	Y	Y
	1924.67	1901.0	Core plug		Cromer Knoll (K20)	OS	146974					Y	Y	Y					
	1928.61	1905.0	Core plug		Cromer Knoll (K20)	SR/Mat	146975	Y	Y	Y	Y								
	1932.42	1908.8	Core		KCF	SR	154659	Y	Y			Y							
	1932.67	1909.0	Core plug		KCF	SR/OS/Mat	146976	Y	Y	Y	Y	Y	Y	Y		Y	Y	Y	
	1936-1963	1912.3-1939.3	Oil (DST)	2.404	Rona	Oil	160301						Y		Y	Y	Y	Y	Y
	1937.14	1913.5	Core		Rona	SR	154660	Y	Y			Y							
	1937.32	1913.7	Core		Rona	SR	154661	Y	Y			Y	Y	Y		Y	Y	Y	
	1938.00	1914.3	Mud		Rona	Mud	163301						Y	Y		Y			
	1944.05	1920.4	Core		Rona	SR/Mat	154662	Y	Y	Y		Y							
	1948.67	1925.0	Spun out oil	18G	Rona	OS	146644						Y		Y	Y	Y	Y	
	1948.67	1925.0	Core plug		Rona	OS	146980					Y	Y	Y					
	1948.68	1925.0	Core		Rona	SR/Mat	154663	Y	Y	Y		Y							
	1956.00	1932.3	Oil (MDT)	1.154	Rona	Oil	160303						Y		Y	Y	Y	Y	Y
	1956.67	1933.0	Core plug		Rona	OS	146982					Y	Y	Y		Y			
	1960.35	1936.7	Core		Rona	SR/Mat	154664	Y	Y	Y		Y							
	1966.70	1943.0	Core		Rona	SR/Mat	154665	Y	Y	Y		Y							
	1967.35	1943.7	Core		Rona	SR/Mat	154666	Y	Y	Y		Y	Y	Y		Y			
	1984.00	1960.3	SWC		LCG unit III	OS	153498			Y	Y	Y	Y	Y		Y			

TABLE 2.4 Analysed study samples and analyses undertaken - oils, spun out oils, cuttings, cores and muds

Well	MD (m)	TVDSS (m)	Type	Other ID	Stratigraphy	Analysis reason	APTID	TOC	Rock-Eval	VR	SCI	Extraction	Fractionation	WE-GC	WO-GC	GC-MS	CI (blk/alk/arm)	CI (pol/asp)	Oil chem/phys
206/12a-5	1984.50	1960.8	Oil (MDT)	1.105	LCG unit III	Oil	160304						Y		Y	Y	Y	Y	Y
	1988.67	1965.0	Core plug		LCG unit III	OS	146990					Y	Y	Y					
	1997.67	1974.0	Core plug		LCG unit III	OS	146992					Y	Y	Y		Y			
	2001.00	1977.3	SWC		LCG unit III	Mat/OS	153499			Y		Y	Y	Y					
	2022.00	1998.3	SWC		LCG unit II/I	Mat/OS	153500			Y	Y	Y	Y	Y					
	2048.00	2024.3	SWC		LCG unit II/I	Mat/OS	153501			Y	Y	Y	Y	Y					
	2202.00	2176.2	Cuttings		Basal	Mat	n/a			Y	Y								
	2352.00	2336.8	Mud		Basement	Mud	163302						Y	Y		Y			

TABLE 2.4 Analysed study samples and analyses undertaken - oils, spun out oils, cuttings, cores and muds

SR : source rock
OS : oil stain/extracted oil
OIL : flowed/tested oil
MUD : drilling mud
Mat : Maturity

TOC : Source rock screening analyses - total organic carbon content (LECO)
Rock-Eval : Source rock screening analyses - Rock-Eval pyrolysis

VR : Maturity analysis - vitrinite reflectivity
SCI : Maturity analysis - spore colour indices

Extraction : Solvent extraction (D = quantitaive DCM extractiom)
Fractionation : Quantitative asphaltene precipitation and fractionation by MPLC

WE-GC : Whole extract gas chromatgraphy
WO-GC : Whole oil gas chromatgraphy; note also run on extracts where data for light hydrocarbons (<n-C12) are required
Alk-GC : Alkane fraction gas chromatgraphy

GC-MS : Combined alkane/aromatic fractions GC-MS

CI (blk/alk/arm) : Carbon isotope ratios of bulk, alkane and aromatic fractions
CI (pol/asp) : Carbon isotope ratios of polar and asphaltene fractions

Oil chem/phys : Oil analyses - metal, sulphur & wax contents, API gravity, viscosity and total acid number (TAN)

KCF : Kimmeridge Clay Formation
LCG : Lower Clair Group

TABLE 2.4 Analysed study samples and analyses undertaken - oils, spun out oils, cuttings, cores and muds

Well	MD (m)	TVDSS (m)	Sample type	Test	Other ID	Mud type	Stratigraphy			Analysed
206/12a-5	950	926.5	Isotube			KCL	Tertiary/Quaternary	Stronsay Gp.	-	Y
	970	946.5	Isotube			KCL	Tertiary/Quaternary	Stronsay Gp.	-	Y
	990	966.5	Isotube			KCL	Tertiary/Quaternary	Balder Fm.	-	Y
	1010	986.5	Isotube			KCL	Tertiary/Quaternary	Balder Fm.	-	Y
	1030	1006.5	Isotube			KCL	Tertiary/Quaternary	Balder Fm.	-	N
	1050	1026.5	Isotube			KCL	Tertiary/Quaternary	Balder Fm.	-	Y
	1070	1046.5	Isotube			KCL	Tertiary/Quaternary	Balder Fm.	-	N
	1090	1066.5	Isotube			KCL	Upper Cretaceous	Late Maastrichtian	K90.4	Y
	1111.5	1088.0	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	K90.4	Y
	1120	1096.5	Isojar		In	OBM	Upper Cretaceous	Late Maastrichtian	K90.4	N
	1120	1096.5	Isojar		Out	OBM	Upper Cretaceous	Late Maastrichtian	K90.4	Y
	1130	1106.5	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	K90.4	N
	1150	1126.5	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	K90.4	Y
	1170	1146.5	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	K90.4	N
	1190	1166.5	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	K90.4	Y
	1210	1186.4	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	K90.4	N
	1230	1206.4	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	K90.3	N
	1250	1226.4	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	-	Y
	1270	1246.4	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	-	N
	1290	1266.4	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	-	N
	1310	1286.4	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	-	Y
	1330	1306.4	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	-	N
	1350	1326.4	Isotube			OBM	Upper Cretaceous	Late Maastrichtian	-	N
	1370	1346.4	Isotube			OBM	Upper Cretaceous	Early Maastrichtian	K90.2	Y
	1390	1366.4	Isotube			OBM	Upper Cretaceous	Early Maastrichtian	K90.2	N
	1410	1386.4	Isotube			OBM	Upper Cretaceous	Early Maastrichtian	K90.2	Y

TABLE 2.5 Tested gases, Isotubes and Isojars received from 206/12a-5

Well	MD (m)	TVDSS (m)	Sample type	Test	Other ID	Mud type	Stratigraphy			Analysed
206/12a-5	1430	1406.4	Isotube			OBM	Upper Cretaceous	Early Maastrichtian	K90.2	N
	1450	1426.4	Isotube			OBM	Upper Cretaceous	Early Maastrichtian	K90.2	Y
	1470	1446.4	Isotube			OBM	Upper Cretaceous	Early Maastrichtian	K90.2	N
	1490	1466.4	Isotube			OBM	Upper Cretaceous	Early Maastrichtian	K90.2	Y
	1510	1486.4	Isotube			OBM	Upper Cretaceous	Early Maastrichtian	K90.2	N
	1530	1506.4	Isotube			OBM	Upper Cretaceous	Early Maastrichtian	K90.2	Y
	1550	1526.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	N
	1570	1546.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	Y
	1590	1566.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	N
	1610	1586.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	Y
	1630	1606.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	N
	1650	1626.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	Y
	1670	1646.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	N
	1690	1666.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	N
	1710	1686.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	N
	1730	1706.4	Isotube			OBM	Upper Cretaceous	Campanian	K90.2	N
	1750	1726.4	Isotube			OBM	Upper Cretaceous	Santonian	K90.2	Y
	1770	1746.4	Isotube			OBM	Upper Cretaceous	Santonian	K90.2	N
	1790	1766.4	Isotube			OBM	Upper Cretaceous	Santonian	K74	Y
	1810	1786.4	Isotube			OBM	Upper Cretaceous	Coniacian	K74	Y
	1830	1806.4	Isotube			OBM	Upper Cretaceous	Coniacian	K74	Y
	1832	1808.4	Isojar		In	OBM	Upper Cretaceous	Coniacian	K74	N
	1832	1808.4	Isojar		Out	OBM	Upper Cretaceous	Coniacian	K74	Y
	1833	1809.4	Isojar		In	OBM	Upper Cretaceous	Coniacian	K74	N
	1833	1809.4	Isojar		Out	OBM	Upper Cretaceous	Coniacian	K74	Y
	1850	1826.4	Isotube			OBM	Upper Cretaceous	Coniacian	K74	Y

TABLE 2.5 Tested gases, Isotubes and Isojars received from 206/12a-5

Well	MD (m)	TVDSS (m)	Sample type	Test	Other ID	Mud type	Stratigraphy			Analysed
206/12a-5	1870	1846.3	Isotube			OBM	Upper Cretaceous	Turonian	K72	Y
	1873	1849.3	Isojar		In	OBM	Upper Cretaceous	Turonian	K72	N
	1873	1849.3	Isojar		Out	OBM	Upper Cretaceous	Turonian	K72	Y
	1883	1859.3	Isojar		In	OBM	Upper Cretaceous	Turonian	K72	N
	1883	1859.3	Isojar		Out	OBM	Upper Cretaceous	Turonian	K72	Y
	1890	1866.3	Isotube			OBM	Upper Cretaceous	Turonian	K72	Y
	1910	1886.3	Isotube			OBM	Upper Cretaceous	Turonian	K72	Y
	1922.4	1898.7	Tested		MPSR-3799	OBM	Lower Cretaceous	Cromer Knoll	K30	Y
	1930	1906.3	Isotube			OBM	Upper Jurassic	Kimmeridge Clay Fm.		Y
	1933	1909.3	Isojar		In	OBM	Upper Jurassic	Kimmeridge Clay Fm.		N
	1933	1909.3	Isojar		Out	OBM	Upper Jurassic	Kimmeridge Clay Fm.		Y
	1936-1963	1912.3-1939.3	Tested	DST	BHS	OBM	Upper Jurassic	Rona		Y
	1936-1963	1912.3-1939.3	Tested	DST	SGS	OBM	Upper Jurassic	Rona		Y
	1950	1926.3	Isotube			OBM	Upper Jurassic	Rona		Y
	1956	1932.3	Tested	MDT	MPSR-3344	OBM	Upper Jurassic	Rona		Y
	1970	1946.3	Isotube			OBM	Devonian	Lower Clair Gp.	Unit III	Y
	1983	1959.3	Isojar		In	OBM	Devonian	Lower Clair Gp.	Unit III	N
	1983	1959.3	Isojar		Out	OBM	Devonian	Lower Clair Gp.	Unit III	Y
	1984.5	1960.8	Tested	MDT	MPSR-4608	OBM	Devonian	Lower Clair Gp.	Unit III	Y
	1990	1966.3	Isotube			OBM	Devonian	Lower Clair Gp.	Unit III	Y
	1998	1974.3	Isotube			OBM	Devonian	Lower Clair Gp.	Unit III	Y
	2016	1992.3	Isotube			OBM	Devonian	Lower Clair Gp.	Unit II/I	Y
	2025	2001.3	Isotube			OBM	Devonian	Lower Clair Gp.	Unit II/I	Y
	2033	2009.3	Isojar		In	OBM	Devonian	Lower Clair Gp.	Unit II/I	N
	2033	2009.3	Isojar		Out	OBM	Devonian	Lower Clair Gp.	Unit II/I	Y
	2034	2010.3	Isotube			OBM	Devonian	Lower Clair Gp.	Unit II/I	Y

TABLE 2.5 Tested gases, Isotubes and Isojars received from 206/12a-5

Well	MD (m)	TVDSS (m)	Sample type	Test	Other ID	Mud type	Stratigraphy			Analysed
206/12a-5	2043	2019.3	Isotube			OBM	Devonian	Lower Clair Gp.	Unit II/I	Y
	2052	2028.3	Isotube			OBM	Devonian	Lower Clair Gp.	Unit II/I	Y
	2061	2037.2	Isotube			OBM	Devonian	Lower Clair Gp.	Unit II/I	Y
	2070	2046.2	Isotube			OBM		Basal conglomerate		Y
	2077	2053.2	Isotube			OBM		Basal conglomerate		Y
	2079	2055.2	Isotube			OBM		Basal conglomerate		Y
	2083	2059.2	Isojar		In	OBM		Basal conglomerate		N
	2083	2059.2	Isojar		Out	OBM		Basal conglomerate		Y
	2088	2064.1	Isotube			OBM		Basal conglomerate		Y
	2097	2073.1	Isotube			OBM		Basal conglomerate		Y
	2106	2082.0	Isotube			OBM		Basal conglomerate		Y
	2115	2090.9	Isotube			OBM		Basal conglomerate		Y
	2124	2099.8	Isotube			OBM		Basal sand		Y
	2133	2108.7	Isotube			OBM		Basal sand		Y
	2133	2108.7	Isojar		In	OBM		Basal sand		N
	2133	2108.7	Isojar		Out	OBM		Basal sand		Y
	2142	2117.6	Isotube			OBM		Basal sand		Y
	2151	2126.5	Isotube			OBM		Basal sand		N
	2160	2135.3	Isotube			OBM		Basal sand		Y
	2169	2144.1	Isotube			OBM		Basal sand		N
	2178	2152.9	Isotube			OBM		Basal sand		N
	2183	2157.7	Isojar		In	OBM		Basal sand		N
	2183	2157.7	Isojar		Out	OBM		Basal sand		Y
	2187	2161.6	Isotube			OBM		Basal sand		Y
	2196	2170.4	Isotube			OBM		Basal sand		N
	2205	2179.1	Isotube			OBM		Basal sand		N

TABLE 2.5 Tested gases, Isotubes and Isojars received from 206/12a-5

Well	MD (m)	TVDSS (m)	Sample type	Test	Other ID	Mud type	Stratigraphy			Analysed
206/12a-5	2214	2187.8	Isotube			OBM		Basal conglomerate		N
	2223	2196.5	Isotube			OBM		Basal conglomerate		Y
	2232	2205.2	Isotube			OBM		Basal conglomerate		N
	2233	2206.1	Isojar		In	OBM		Basal conglomerate		N
	2233	2206.1	Isojar		Out	OBM		Basal conglomerate		Y
	2241	2213.9	Isotube			OBM		Basal conglomerate		N
	2250	2222.6	Isotube			OBM		Basal conglomerate		Y
	2259	2231.3	Isotube			OBM		Basal conglomerate		N
	2268	2240.0	Isotube			OBM		Basal conglomerate		N
	2277	2248.8	Isotube			OBM		Basal conglomerate		Y
	2283	2254.6	Isojar		In	OBM		Basal conglomerate		N
	2283	2254.6	Isojar		Out	OBM		Basal conglomerate		Y
	2286	2257.6	Isotube			OBM		Basal conglomerate		N
	2295	2266.3	Isotube			OBM		Basal conglomerate		N
	2304	2275.1	Isotube			OBM		Basal sand		Y
	2313	2283.9	Isotube			OBM		Basal sand		N
	2322	2292.8	Isotube			OBM		Basal sand		Y
	2331	2301.6	Isotube			OBM	Precambrian	Basement		Y
	2333	2303.6	Isojar		In	OBM	Precambrian	Basement		N
	2333	2303.6	Isojar		Out	OBM	Precambrian	Basement		Y
	2340	2310.5	Isotube			OBM	Precambrian	Basement		Y
	2349	2319.3	Isotube			OBM	Precambrian	Basement		N
	2352	2322.2	Isojar		In	OBM	Precambrian	Basement		N
	2352	2322.2	Isojar		Out	OBM	Precambrian	Basement		Y

Mud type : OBM - oil based mud
Sequence : LCG - Lower Clair Group

BHS : Bottom hole sample
SGS : Seperator gas sample

TABLE 2.5 Tested gases, Isotubes and Isojars received from 206/12a-5

Well	MD (m)	TVDSS (m)	Type	Other ID	Stratigraphy	HC-Composition	nonHC-Composition	CIR C1	CIR C2-C5	DIR C1	CIR CO2
206/12a-5	950	926.5	Isotube		Stronsay Gp.	Y					
	970	946.5	Isotube		Stronsay Gp.	Y					
	990	966.5	Isotube		Balder Fm.	Y					
	1010	986.5	Isotube		Balder Fm.	Y		Y			
	1050	1026.5	Isotube		Balder Fm.	Y		Y			
	1090	1066.5	Isotube		Late Maastrichtian (K90.4)	Y		Y			
	1111.5	1088.0	Isotube		Late Maastrichtian (K90.4)	Y		Y			
	1120	1096.5	Isojar	Out	Late Maastrichtian (K90.4)	Y		Y	Y	Y	
	1150	1126.5	Isotube		Late Maastrichtian (K90.4)	Y		Y			
	1190	1166.5	Isotube		Late Maastrichtian (K90.4)	Y		Y			
	1250	1226.4	Isotube		Late Maastrichtian (K90.3)	Y		Y			
	1310	1286.4	Isotube		Late Maastrichtian (K90.3)	Y		Y			
	1370	1346.4	Isotube		Early Maastrichtian (K90.2)	Y		Y	Y	Y	
	1410	1386.4	Isotube		Early Maastrichtian (K90.2)	Y		Y			
	1450	1426.4	Isotube		Early Maastrichtian (K90.2)	Y		Y	Y	Y	
	1490	1466.4	Isotube		Early Maastrichtian (K90.2)	Y		Y			
	1530	1506.4	Isotube		Early Maastrichtian (K90.2)	Y		Y	Y	Y	
	1570	1546.4	Isotube		Campanian (K80.2)	Y		Y			
	1610	1586.4	Isotube		Campanian (K80.2)	Y		Y			
	1650	1626.4	Isotube		Campanian (K80.2)	Y		Y	Y	Y	
	1750	1726.4	Isotube		Santonian (K76)	Y		Y	Y	Y	
	1790	1766.4	Isotube		Coniacian (K74)	Y		Y			
	1810	1786.4	Isotube		Coniacian (K74)	Y		Y	Y	Y	
	1830	1806.4	Isotube		Coniacian (K74)	Y		Y	Y	Y	
	1832	1808.4	Isojar	Out	Coniacian (K74)	Y		Y	Y	Y	
	1833	1809.4	Isojar	Out	Coniacian (K74)	Y		Y	Y	Y	
	1850	1826.4	Isotube		Coniacian (K74)	Y					
	1870	1846.3	Isotube		Turonian (K72)	Y					

TABLE 2.6 Analysed study samples and analyses undertaken - gases

Well	MD (m)	TVDSS (m)	Type	Other ID	Stratigraphy	HC-Composition	nonHC-Composition	CIR C1	CIR C2-C5	DIR C1	CIR CO2
206/12a-5	1873	1849.3	Isojar	Out	Turonian (K72)	Y		Y	Y	Y	
	1883	1859.3	Isojar	Out	Turonian (K72)	Y		Y	Y	Y	
	1890	1866.3	Isotube		Turonian (K72)	Y		Y	Y	Y	
	1910	1886.3	Isotube		Turonian (K72)	Y		Y			
	1922.4	1898.7	Tested	MDT - MPSR-3799	Turonian (K72)	Y	Y	Y	Y	Y	Y
	1930	1906.3	Isotube		Kimmeridge Clay Fm.	Y		Y	Y	Y	
	1933	1909.3	Isojar	Out	Kimmeridge Clay Fm.	Y		Y	Y	Y	
	1936-1963	1912.3-1939.3	Tested	DST (bottomhole sample)	Rona	Y	Y	Y	Y	Y	Y
	1936-1963	1912.3-1939.3	Tested	DST (seperator sample)	Rona	Y	Y	Y	Y	Y	Y
	1950	1926.3	Isotube		Rona	Y		Y			
	1956	1932.3	Tested	MDT - MPSR-3344	Rona	Y	Y	Y	Y	Y	Y
	1970	1946.3	Isotube		Lower Clair Gp., Unit III	Y		Y	Y	Y	
	1983	1959.3	Isojar	Out	Lower Clair Gp., Unit III	Y		Y	Y	Y	
	1984.5	1960.8	Tested	MDT - MPSR-4608	Lower Clair Gp., Unit III	Y	Y	Y	Y	Y	Y
	1990	1966.3	Isotube		Lower Clair Gp., Unit III	Y		Y			
	1998	1974.3	Isotube		Lower Clair Gp., Unit III	Y		Y	Y	Y	
	2016	1992.3	Isotube		Lower Clair Gp., Unit II/I	Y		Y	Y	Y	
	2025	2001.3	Isotube		Lower Clair Gp., Unit II/I	Y		Y	Y	Y	
	2033	2009.3	Isojar	Out	Lower Clair Gp., Unit II/I	Y		Y	Y	Y	
	2034	2010.3	Isotube		Lower Clair Gp., Unit II/I	Y		Y	Y	Y	
	2043	2019.3	Isotube		Lower Clair Gp., Unit II/I	Y		Y	Y	Y	
	2052	2028.3	Isotube		Lower Clair Gp., Unit II/I	Y		Y	Y	Y	
	2061	2037.2	Isotube		Lower Clair Gp., Unit II/I	Y		Y	Y	Y	
	2070	2046.2	Isotube		Basal conglomerate/sands	Y		Y			
	2077	2053.2	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2079	2055.2	Isotube		Basal conglomerate/sands	Y		Y			
	2083	2059.2	Isojar	Out	Basal conglomerate/sands	Y		Y	Y	Y	
	2088	2064.1	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	

TABLE 2.6 Analysed study samples and analyses undertaken - gases

Well	MD (m)	TVDSS (m)	Type	Other ID	Stratigraphy	HC-Composition	nonHC-Composition	CIR C1	CIR C2-C5	DIR C1	CIR CO2
206/12a-5	2097	2073.1	Isotube		Basal conglomerate/sands	Y		Y			
	2106	2082.0	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2115	2090.9	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2124	2099.8	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2133	2108.7	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2133	2108.7	Isojar	Out	Basal conglomerate/sands	Y		Y			
	2142	2117.6	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2160	2135.3	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2183	2157.7	Isojar	Out	Basal conglomerate/sands	Y		Y			
	2187	2161.6	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2223	2196.5	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2233	2206.1	Isojar	Out	Basal conglomerate/sands	Y		Y			
	2250	2222.6	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2277	2248.8	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2283	2254.6	Isojar	Out	Basal conglomerate/sands	Y		Y			
	2304	2275.1	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2322	2292.8	Isotube		Basal conglomerate/sands	Y		Y	Y	Y	
	2331	2301.6	Isotube		Basement	Y		Y	Y	Y	
	2333	2303.6	Isojar	Out	Basement	Y		Y			
	2340	2310.5	Isotube		Basement	Y		Y	Y	Y	
	2352	2322.2	Isojar	Out	Basement	Y		Y			

ANALYSES

HC-Composition : hydrocarbon gas composition
nonHC-Composition : non-hydrocarbon gas composition, including N_2, CO_2, H_2, H_2S
CIR C1 : methane carbon isotope ratio
CIR C2-C5 : ethane to pentane carbon isotope ratios
DIR C1 : methane deuterium isotope ratio
CIR CO2 : carbon dioxide carbon isotope ratio

TABLE 2.6 Analysed study samples and analyses undertaken - gases

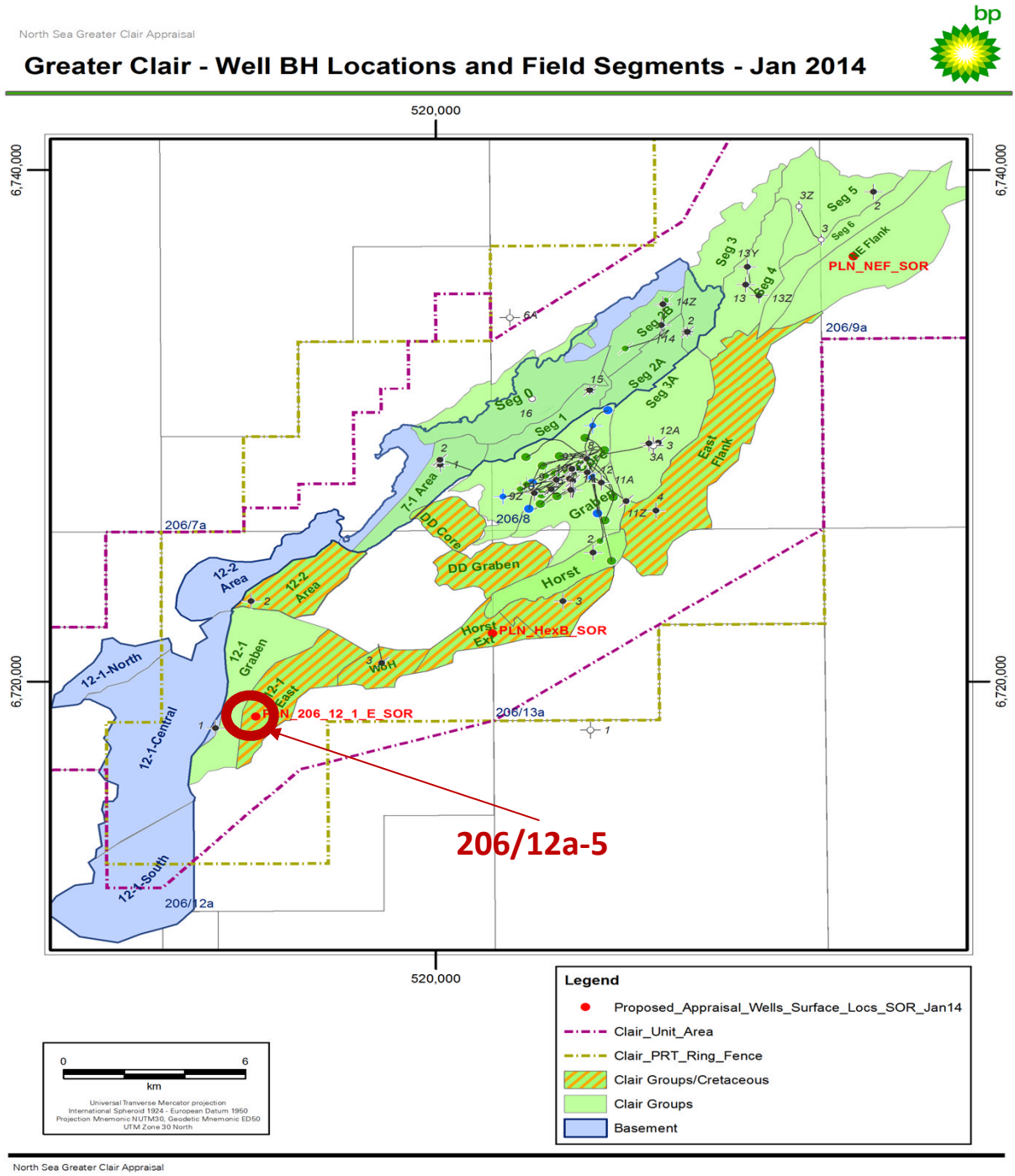

FIGURE 2.1 Field and well location map

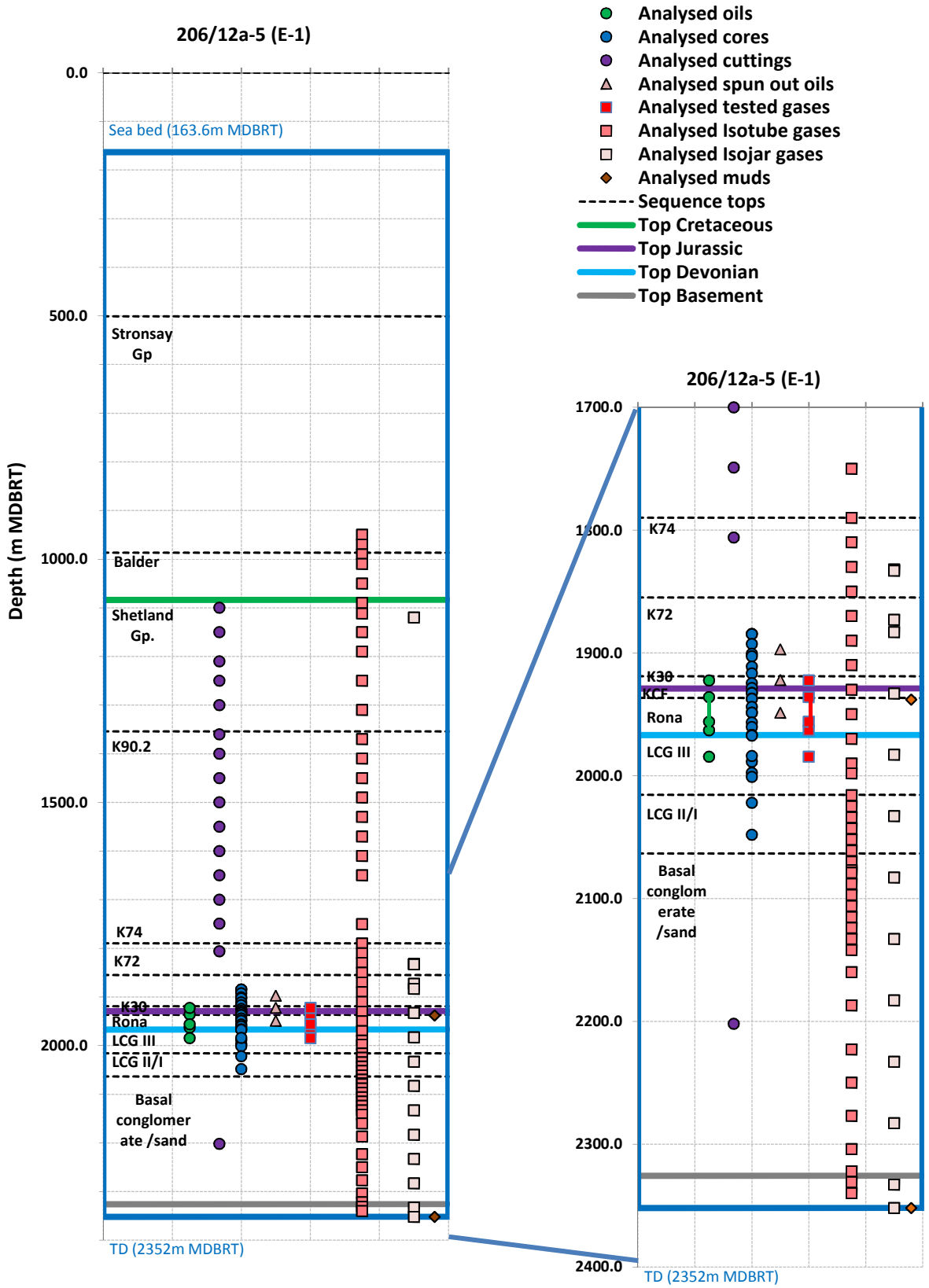

FIGURE 2.2 Stratigraphic distribution of analysed samples

CHAPTER 3
MATURITY EVALUATION

3.1 Samples and analyses

Vitrinite reflectivity and spore colouration analyses have been carried out on twenty five ditch cuttings, four SWCs and eleven core samples from the 206/12a-5 well. The work has been carried out on both whole rock and kerogen isolates; these analyses were supplemented by observation of the fluorescence of liptinitic macerals in blue short wave length light excitation.

Samples ranging from 270m MDRT (Pliocene Westray Group) to 2202m MDRT (Lower Clair Group (LCG) Unit II/I) have been analysed to determine a full kerogen maturity profile through the well section. In addition five core samples and four sidewall core samples of sandstones and siltstones containing occasional small, dark inclusions from between 1944.05m MDRT and 2001.0m MDRT (Rona Member through Upper Clair Group) were analysed to determine the nature and origin of the inclusions.

The results from the maturity analyses carried out are summarised in Table 3.1; depth plots of the spore colour indices and vitrinite reflectivity analyses are shown as Figures 3.1 and 3.2. The full maturity data sets are reported in Appendices 1 and 2 and details of the analytical techniques are provided in Appendix 12.

3.2 Maturity results

Kerogen composition

The Tertiary interval 270m to 1050m MDRT is commonly dominated by coarse grained clastic lithologies with variable proportions of poorly consolidated interbedded fine grained muds and clays. In view of this, the shallow nature of the section and the casing set at 259.4m (30”) and 1106.7m (13 3/8”), the samples from this interval were prepared for kerogen isolation using un-washed cuttings samples.

Samples from the Tertiary section gave generally good kerogen recovery which comprised variable amounts of fluorescent and non-fluorescent amorphous organic matter with often high abundances of humic vitrinitic organic matter. The kerogen overall through this interval was of sufficient abundance and quality for good reflectivity and spore colouration results.

Samples from the Shetland Group between 1100m and 1806m MDRT were dominated by humic organic matter being a mixture of variously coaly inertinitic, structured vitrinitic and amorphous vitrinitic (collinite) macerals mixed with varying but generally minor (trace up to 10% maximum) amounts of algal and cuticular material. Generally sufficient numbers of spores and pollen grains were seen in most samples but the fluorescence of the spores and pollens was rather variable from almost none to strong yellow and yellow-brown to orange-brown. It is to be noted that structured, un-oxidised organic matter in these samples is referred to as vitrinite for simplicity, although given the low maturity of the Mesozoic samples, it is perhaps more properly referred to as ‘huminite’.

The basal Shetland Group (Turonian – 4 samples) and Cromer Knoll Group (1 sample) section was represented by argillaceous core material. The Turonian section gave relatively good

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kerogen recovery but the Cromer Knoll Group sample was lean. In all cases the kerogen was dominantly amorphous non-fluorescent presumed of humic vitrinitic origin but the structured organic matter in all samples was dominantly coaly inertinitic organic matter with poor to fair kerogen preservation.

A core sample in the Jurassic Kimmeridge Clay Formation at 1932.67m MDRT gave good recovery of dominantly non-fluorescing amorphous kerogen of assumed humic and vitrinitic origin suitable for reflectivity studies but few palynomorphs were found.

Four samples were processed for kerogen recovery and maturity studies in the Clair Group section in this well but none gave recovery of kerogen or palynomorphs suitable for kerogen typing or spore colouration analysis.

Fluorescence

Yellow and yellow-orange miospores and yellow and orange-yellow dinocysts were observed through the Tertiary section indicative of low levels of maturity. Fluorescence and kerogen preservation in the Cretaceous section is rather variable; yellow to yellow-orange miospores grading to brown colours and some non-fluorescent spores are found in a number of samples but preservation is often rather poor and pyrite common. The overall trend is towards darker colours and decreasing fluorescence with depth but the preservational conditions appear to be the dominant control on fluorescence. Amorphous non-fluorescent ?humic kerogen is often the dominant kerogen component and oxidised coaly humic materials (inertinite) are generally strongly dominant over humic vitrinitic kerogens. Yellow to yellow-orange through to orange and orange-brown fluorescence colours have been observed in miospores and dinoflagellates over the interval from 1100m to 1932.67m MDRT (Kimmeridge Clay Formation sample) consistent with immaturity through to early or possibly later maturity.

Polished whole rock sandstone core samples between 1944.05m MDRT and 2001.0m MDRT were examined in blue light for their fluorescence properties but no exinite macerals were observed. However, it was noted that fluorescing fluids, possibly liquid hydrocarbons, were mobilised and flowed through the immersion oil when under UV excitation.

Spore colour indices

Spore colour index (SCI) analyses have been carried out on samples between 270m MDRT and 2202m MDRT covering the interval of Tertiary through Cretaceous and Jurassic age with the deepest samples within the Lower Clair Group. None of the deepest samples taken in the sandstones of the Rona Member and the Lower Clair Group provided meaningful data. The results are shown plotted against depth in Figure 3.1.

Miospores and pollens were present in sufficient abundance in the Tertiary section to record spore colour indices of 2.0 to 3.0 consistent with immaturity. SCI values through the basal Tertiary through Maastrichtian interval to about 1500m were spanning the immaturity to very earliest maturity zone (SCI 3.0 to 4.5). SCIs in the deeper Campanian to Kimmeridgian section (1550m to 1932.67m) were generally similarly on the cusp of maturity but several samples (particularly at 1550m and 1650m) seemed to indicate rather higher spore colours (up to SCI 6) possibly due to staining by minerals or darkened by the effects of a short pulse of higher heat flow. Samples below the Kimmeridge Clay Formation sample gave effectively no palynomorphs suitable for spore colour determination.

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Vitrinite reflectivity

Vitrinite reflectivity (VR) analysis has been carried out on the interval 270m to 2048m and has generally provided good results in the upper part down to the Kimmeridge Clay Formation interval at 1932.67m. Samples below this depth in the Rona Sandstone section contained small amounts of low reflecting possible bituminous material. No measurable kerogen was recorded in the Clair Group section.

Very low reflecting humic material was common in the shallowest Tertiary section with reflectivities of around 0.2% consistent with very shallow burial and little evidence of past deeper burial and subsequent uplift. Values increased steadily through to the Maastrichtian interval at around 1500m with reflectivity values rising to around 0.35%. Samples in the interval 1550m to 1806m mostly encountered plentiful humic organic matter, although vitrinite was not always common, and gave generally well defined histograms but rather higher values ranging from 0.5% to 0.6%. The Kimmeridge Clay Formation sample at 1932.67m gave a reflectivity of 0.42%. The results indicate that the Tertiary and Cretaceous section is immature.

No results have been obtained in the Pre Mesozoic section.

Organic petrography

Polished whole rock blocks have been prepared of five core plug sandstone samples from between 1944.05m and 1967.35 MDRT (Rona Member). Some were very heavily oil-stained while others had apparent coaly material included within the variably laminated sandstones. Reflectivities range from 0.35% to 0.62% Ro but weak UV light fluorescence in the 1967.35m MDRT sample suggests the presence of perhydrous vitrinite while higher reflectivities of around 0.6% in the samples at 1944.05m and 1948.68m are attributed to the presence of bituminite.

Organic petrographic studies have also been carried out on whole rock preparations of four sidewall core plugs in the Clair Group between 1984.0m and 2048m MDRT which were light grey sandstones containing darker grey flecks thought to be of possible organic origin. The results are shown in Table 3.1. The polished whole rock preparation of the sample at 2001m detected the presence of possible bitumen which on the basis of a limited number of measurements had a reflectivity of 0.79%. This and the other three samples were also submitted for kerogen isolation but failed to return sufficient kerogen for making a polished kerogen slide. The strew mount slides only contained possible drilling contaminants and no spores or acritarchs were observed.

These data indicate that the Tertiary through Jurassic sediments are presently immature for oil generation. The results obtained suggest that there has been virtually no uplift and erosion at unconformities or at the surface of the interval.

Regional comparison

The data obtained from the kerogen microscopy studies carried out in both the 206/12a-5 in the south westerly extent of the Clair Field and the 206/9a-4 in the far north east of the field, are comparable and consistent in providing near continuous kerogen maturity gradients (both spore colouration and vitrinite reflectivity) through the near surface Tertiary to Jurassic interval. The results suggest imposition of the maturity gradients resulting from burial alone with no evidence for significant additional mid to Late Tertiary burial and subsequent uplift and erosion.

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Well	Depth (m MD)	Sample type	Stratigraphy	Vitrinite Reflectivity			Spore Colour		UV Fluorescence		Composition (%)			Comments	Other analysis on sample	
				Type	*n*	Ave. %Ro	Type	SCI (*n*)	Form	Colour	Sap.	Vit.	Inert.		SR	Ext.
206/12a-5	270	DC	Westray Gp.	I	40	0.19	I	2.0 (7)	mio	yel		90	10	AOM dominates, but occasional fragments of peat are present, sometimes weathered.	No	No
	360	DC	Westray Gp.	I	40	0.18	I	2.0 (3)	mio	yel	80	10	10	AOM dominates with occasional peat like material present. Structured kerogen including inertinite more common.	No	No
							R	8.0 (2)	AOM	yel to orng						
	480	DC	Westray Gp.	I	40	0.22	I	2.5 (23)	mio	yel	30	60	10	AOM dominates with occasional peat like material present. Structured kerogen including inertinite more common.	No	No
							R	6.0 (3)	dino	yel-orng						
									AOM	orng-yel						
	540	DC	Stronsay Gp.	I	40	0.21	I	3.0 (14)	mio	yel		95	5	AOM dominates with occasional peat like material present. Structured kerogen including inertinite more common.	No	No
									dino	grnish yel						
									AOM	brn to nf						
	570	DC	Stronsay Gp.	I	40	0.23	I	3.0 (20)	mio	yel, yel-orng	15	75	10	AOM dominates with occasional peat like material present. Structured kerogen including inertinite more common.	No	No
									dino	orng-yel						
									AOM	yel to brn						
	600	DC	Stronsay Gp.	I	55	0.18	-	-	mio	yel	-	-	-	Peat sample. Vascular, telinite-like tissue with %Ro < 0.1 the primary component. No SCI slide available.	No	No
	720	DC	Stronsay Gp.	I	40	0.21	I	3.5 (9)	mio	orng	5	85	10	Sclerotinite at 0.4% sets upper reflectivity limit. Super abundant pyrite - vitinite may be largely perhydrous.	No	No
									dino	orng-yel						
									AOM	yel-orng						

TABLE 3.1 Summary of maturity data

Well	Depth (m MD)	Sample type	Stratigraphy	Vitrinite Reflectivity			Spore Colour		UV Fluorescence		Composition (%)			Comments	Other analysis on sample	
				Type	*n*	Ave. %Ro	Type	SCI (*n*)	Form	Colour	Sap.	Vit.	Inert.		SR	Ext.
206/12a-5	760	DC	Stronsay Gp.	I	40	0.27	I	3.0 (19)	mio	yel, occ brn		90	10	Good sample. Huminite increasingly abundant and generally well-preserved. Reworked Classopollis at 3.5 sets an upper limit for SCI.	No	No
							R	4.0-8.0 (5)	dino	yel						
	1050	DC	Balder	I	40	0.30	I	3.0 (26)	mio	yel-orng		85	15	Abundant well-preserved huminite. Very good sample.	No	No
							R	4.5 (2)	AOM	nf						
	1100	DC	Maastrichtian	I	47	0.24	I	2.5 (16)	mio	yel	10	15	75	Vitrinite rare but sufficient for reliable data; morphology is variable. Distinction from other macerals by reflectivity is clear.	No	No
							R	6.0 (6)								
	1150	DC	Maastrichtian	I	55	0.28	I	3.0 (28)	mio	yel to brn	10	35	55	Vitrinite rare and morphologically similar to semifusinite, but easily distinguished by reflectivity. Preservation is poor.	No	No
							R	5.0 (4)	dino	brn						
	1210	DC	Maastrichtian	I	49	0.30	I	3.5 (24)			5	70	25	Vitrinite rare, wispy to coaly. Abundant opaques with secondary woody debris, but dinocysts are more abundant than miospores. Aquillapollenites at 3.0-3.5.	No	No
				R	6	0.53	R	5.0 (8)								
	1250	DC	Maastrichtian	I	27	0.34	I	3.5 (18)			-	-	-	Relatively low recovery and vitrinite very rare. Generally wispy. Organic matter darker - possibly due to oil or mud staining.	No	No
							R	6.0 (7)								
	1300	DC	Maastrichtian	I	55	0.34	I	3.0 (21)			5	65	30	Increased AOM, but occasional good wispy and blocky vitrinite present. Aquillapollenites at 4.5 noted.	No	No
							R	6.0 (4)								

TABLE 3.1 Summary of maturity data

Well	Depth (m MD)	Sample type	Stratigraphy	Vitrinite Reflectivity			Spore Colour		UV Fluorescence		Composition (%)			Comments	Other analysis on sample	
				Type	*n*	Ave. %Ro	Type	SCI (*n*)	Form	Colour	Sap.	Vit.	Inert.		SR	Ext.
206/12a-5	1360	DC	Maastrichtian	I	38	0.32	I	3.5 (19)			10	60	30	Increased AOM, but occasional good wispy and blocky vitrinite present. 2.0-2.5 SCI miospores poorly preserved and thinned.	No	No
				L	4	0.18	R	7.0 (3)								
				R	13	0.50										
	1400	DC	Maastrichtian	I	46	0.32	I	3.5 (21)				80	20	Vitrinite slightly more abundant than above, generally wispy to irregular.	No	No
				R	9	0.50	R	7.5 (2)								
	1450	DC	Maastrichtian	I	55	0.35	I	4.0 (17)				90	10	Vitrinite slightly more abundant than above, generally wispy to irregular.	No	No
							R	6.5 (2)								
	1500	DC	Maastrichtian	I	40	0.36	I	3.5 (21)			5	90	5	Poor recovery but otherwise as above.	No	No
							R	6.0 (2)								
	1550	DC	Maastrichtian	I	5	0.51	I	4.0 (8)				90	10	Little structured organic matter which is generally fairly dark. Aquillapollenites present. Evidence of localised heating event.	No	No
							C	2.5 (1)								
	1600	DC	Campanian	I	20	0.57	I	4.5 (7)				70	30	Very lean sample. Particles quite small.	No	No
				R	13	0.79	R	6.5 (6)								
	1650	DC	Campanian	I	26	0.59	I	4.0 (6)				70	30	Particles are of moderate size, equi-dimensional and sub-rounded. Wide range of SCI from spores and dinocysts suggests a highly localised heating event.	No	No
				R	29	0.83	C	2.5 (3)								

TABLE 3.1 Summary of maturity data

Well	Depth (m MD)	Sample type	Stratigraphy	Vitrinite Reflectivity			Spore Colour		UV Fluorescence		Composition (%)			Comments	Other analysis on sample	
				Type	*n*	Ave. %Ro	Type	SCI (*n*)	Form	Colour	Sap.	Vit.	Inert.		SR	Ext.
206/12a-5	1700	DC	Campanian	I	12	0.67	R	6.5 (3)	mio	nf		80	20	Poor recovery. Mostly ragged brown finely disseminated organic matter. Palynomorphs rare.	No	No
				L	5	0.47			dino	nf						
				R	2	0.84										
	1749	DC	Santonian	I	28	0.50	I	4.5 (25)	mio	strong orng	5	80	15	All particles sub-rounded, wispy to blocky. Abundant opaques and common well-preserved dinocysts.	No	No
				L	8	0.37										
				R	19	0.69										
	1806	DC	Coniacian	R	3	0.69						60	40	Low recovery - mostly a mix of opaques and ragged brown finely disseminated organic matter.	No	No
				R	3	0.90										
	1884.58	COCH	Turonian				I	3.5 (14)	mio	yel to yel-orng	10	20	70	No vitrinite present in sample. Mostly small blocky opaques in SCI slide.	Yes	Yes
							R	5.5 (5)	AOM	wk orng-brn						
	1892.66	COCH	Turonian				I	3.0 (18)	mio	yel to yel-orng	5	75	20	Inertinite dominant with trace amounts of vitrinite present.	Yes	Yes
				R	20	0.42	R	6.0 (4)	Dino	yel						
	1902.70	COCH	Turonian	I	3	0.35	I	4.0 (6)	mio	yel-orng	10	50	40	Inertinite dominant with trace amounts of vitrinite present. Some staining in SCI slide; abundant dinocysts.	Yes	Yes
							R	7.0 (2)	Dino	yel to yel-orng						
									sph	yel						
									AOM	n.f						
	1911.17	COCH	Turonian				I	3.0 (1)	mio	yel	-	-	-	Vitrinite absent. 100% AOM. SCI slide dominated by pyrite rich non-fluorescing AOM.	Yes	Yes
									Dino	yel						
									sph	yel, occ. grn						
									AOM	n.f						

TABLE 3.1 Summary of maturity data

Well	Depth (m MD)	Sample type	Stratigraphy	Vitrinite Reflectivity			Spore Colour		UV Fluorescence		Composition (%)			Comments	Other analysis on sample	
				Type	*n*	Ave. %Ro	Type	SCI (*n*)	Form	Colour	Sap.	Vit.	Inert.		SR	Ext.
206/12a-5	1928.61	COCH	Cromer Knoll						mio	pl yel		100		Vitrinite absent in reflectivity slide. SCI slide barren.	Yes	Yes
									AOM	yel to wk yel						
	1932.67	COCH	KCF	I	55	0.42	I	4.5 (2)	mio	yel to orng-brn	5	90	5	Vitrinite abundant for KCF. Few miospores and algae present in SCI.	Yes	Yes
									AOM	wk yel						
	1944.05	COCH	Rona	B	5	0.58					-	-	-	Whole rock sample - heavily oil stained sandstone containing organic matter interpreted to be bituminite. No SCI slide made.	Yes	Yes
	1948.68	COCH	Rona	B	40	0.60					-	-	-	Whole rock sample - heavily oil stained sandstone containing organic matter interpreted to be bituminite. No SCI slide made.	Yes	Yes
	1960.35	COCH	Rona	I	40	0.43					-	-	-	Whole rock sample - heavily oil stained sandstone containing large coaly (?) particles, probably eutextinite. No SCI.	Yes	Yes
	1966.70	COCH	Rona	I	40	0.35					-	-	-	Whole rock sample - sandstone containing organic matter interpreted to be eutextinite or bituminite. No SCI slide made.	Yes	Yes
	1967.35	COCH	Rona	I	40	0.40					-	-	-	Whole rock block containing humic ?coal fragments. No SCI slide made.	Yes	Yes

TABLE 3.1 Summary of maturity data

Well	Depth (m MD)	Sample type	Stratigraphy	Vitrinite Reflectivity			Spore Colour		UV Fluorescence		Composition (%)			Comments	Other analysis on sample	
				Type	*n*	Ave. %Ro	Type	SCI (*n*)	Form	Colour	Sap.	Vit.	Inert.		SR	Ext.
206/12a-5	1984.0	SWC	LCG unit III								-	tr	-	Sandstone with black ?organic flecks. Acid processing gave insufficient organic residue for studies. Dark brown finely disseminated organics in kerogen slide.	No	Yes
	2001.0	SWC	LCG unit III								-	-	-	Whole rock sample - sandstone with flecks of dark ?organic material. Contains small, splintery and jagged particles believed to be bitumen. No SCI slide made.	No	Yes
				B	10	0.70										
				R	6	1.10										
	2022.0	SWC	LCG unit II/I										tr	Sandstone with black ?organic flecks. Kerogen processing yielded insufficient organic residue for studies. Trace amounts of small opaques in kerogen slide.	No	Yes
	2048.0	SWC	LCG unit II/I											Sandstone as above. Processing yielded insufficient organic residue for reflectivity studies. Trace amounts of light coloured organics in kerogen slide.	No	Yes
	2202.0	DC	LCG unit II/I								-	-	-	Both polished and strew mount slides are barren.	No	No

TABLE 3.1 Summary of maturity data

Population type: Vitrinite reflectivity

N.D.P. : No determination possible
I : Indigenous
C : Caved
L : Low reflecting
R : Reworked
B : Bitumen
E : Excinite
Z : Zooclast

Population type: Spore colouration

N.D.P. : No determination possible
I : Indigenous
C : Caved/bleached
R : Stained/reworked
e : estimate based on acritarch or overall kerogen colour/fluorescence

Colour

lt : Light
pal : Pale
brt: Bright
dl : Dull
wk : Weak
dk : Dark
brn : Brown
orng : Orange
yel : Yellow
gn: Green
n.f: non fluorescent

Morphology

amorph: Amorphous
paly: Palynomorphs
indet: Indeterminate
mio: Miospores
dino: Dinocysts
sph: Sphaeromorphs
bisa: Bisaccate
acrit: Acritarchs
AOM: Amorphous organic matter
FDOM: Finely disseminated organic matter

Kerogen types: six component system

FA: Fluoramorphinite - fluorescent amorphous material
HA: Hebamorphinite - non-fluorescent or weakly fluorescent amorphous material.
AL: Algal organic matter/phytoplankton - identifiable algae and phytoplankton, including dinocysts
HE: Herbaceous Organic Matter - identifiable species and fragments of spores, pollen and cuticles
WO: Woody Organic Matter - translucent, structured woody tissues
CO: Coaly Organic Matter - completely opaque, usually angular material
SP: Identifiable species and fragments of spores and pollen
CUT: Cuticular organic matter (cutinite)
Humic: Structured humic organic matter (huminite, vitrinite) - translucent, structured woody tissues.
Inert: Inert organic matter (inertinite)- completely opaque, usually angular material

Kerogen types: three component system

Sap: Includes alginite, sporinite, cuticle and liptodetrinite and organic matter interpreted to have originated as fluoramorphinite
Vit: Includes vitrinite and vitridetrinite and organic matter interpreted to have originated as hebamorphinite
Inert: Inert organic matter (inertinite)- completely opaque, usually angular material
Am: Amorphous organic matter (used as a modifier for sapropel and vitrinite)

TABLE 3.2 List of abbreviations used in maturity data tables

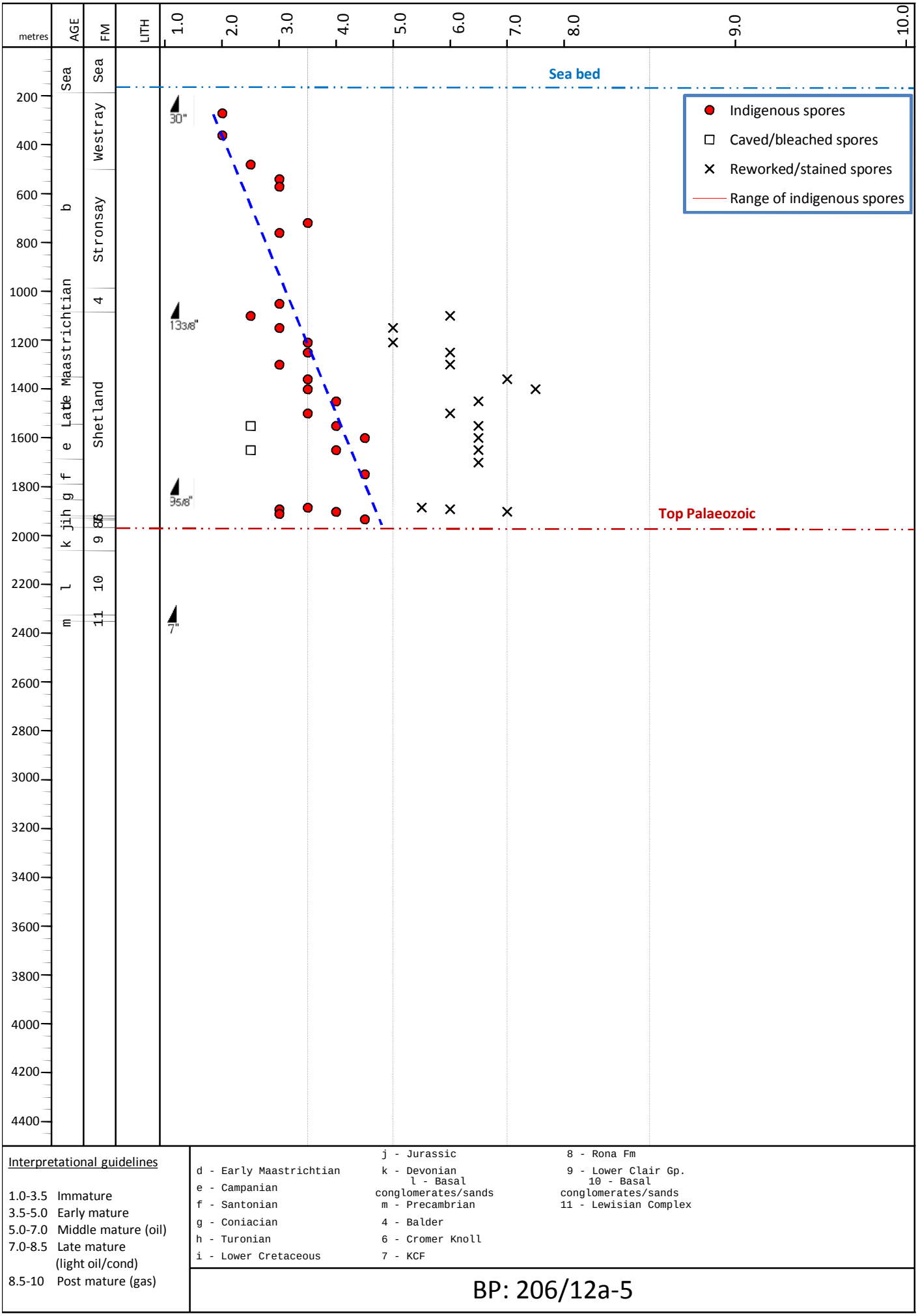

FIGURE 3.1 Spore colour index against depth

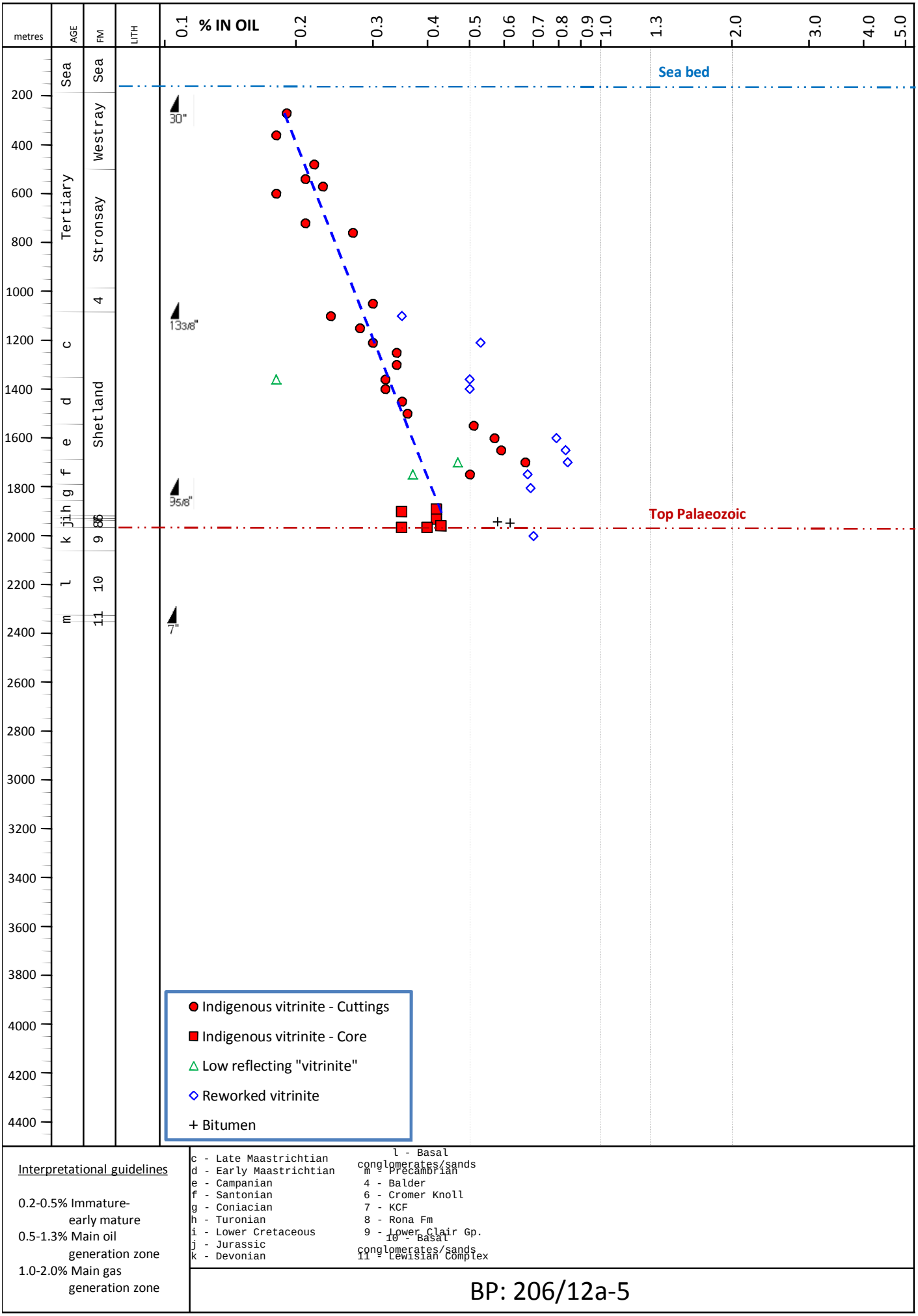

FIGURE 3.2 Vitrinite reflectivity against depth

CHAPTER 4
SOURCE ROCK SCREENING

4.1 Samples and analyses

At the request of the client, a selection of core samples have been analysed for source rock potential by total organic carbon (TOC) content and Rock-Eval pyrolysis analyses. The results from these analyses are reported below.

The results from the source rock screening analyses carried out are summarised in Table 4.1, with interpretative plots of the data shown as Figures 4.1 to 4.3. The full data sets are reported in Appendix 3 and details of the analytical techniques are provided in Appendix 12.

4.2 Source rock screening results

Most of the samples were initially analysed as received and the data were badly affected by drilling filtrate contamination. Consequently the samples were exhaustively solvent cleaned prior to running TOC and pyrolysis analysis. Fourteen core samples were selected for initial source rock screening and from this twelve were analysed further for Rock-Eval pyrolysis. The results of the analyses on the solvent cleaned samples are shown in Table 4.1 and used for interpretation; the full data set including pyrograms is included in Appendix 3.

Data have been obtained on samples from 1884.58m MDRT to 1967.35m MDRT, and represent the Late Cretaceous, Early Cretaceous, Jurassic and Lower Clair Group intervals. Data are shown against depth on Figure 4.1, illustrating the stratigraphic distribution of the analyses, and on interpretative Figures 4.2 and 4.3, illustrating the variations in source rock quality. A brief discussion of the results based on stratigraphic zonation is presented below.

Turonian (K72) (4 samples)
Four samples analysed from the K72 sequence are organically lean, with TOC's of 0.47% to 0.74%. Two samples at depths 1892.66m MDRT and 1911.17m MDRT with TOC values of 0.61 and 0.74% respectively were analysed using Rock-Eval pyrolysis. The pyrolysis yields are low suggesting no hydrocarbon source potential. Kerogen composition data (Table 3.1) indicate that vitrinite is the dominant component with only 5% being sapropelic consistent with the pyrolysis results.

Cromer Knoll (K20) (1 sample)
The single sample at 1928.61m MDRT has a very low TOC content of 0.05%. The sample was analysed for Rock-Eval pyrolysis as it was the only sample from the Cromer Knoll Group selected. The pyrolysis data confirm the sample has no hydrocarbon generative potential. Visual kerogen composition data (Table 3.1) indicate that the sample is 100% vitrinite consistent with the pyrolysis results.

Kimmeridge Clay Formation (2 samples)
Two core samples from the KCF (1932.42m and 1932.67m) have TOC contents of 2.11% and 3.35% and pyrolysis data reveal hydrogen indices of 285 and 324 suggesting moderate oil generative potential. Visual kerogen analysis reveals the presence of 90% non-fluorescent

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amorphous kerogen with 5% algal material with weak yellow to orange-brown fluorescence; a result poorly consistent with the pyrolysis results.

Rona (7 samples)

Seven samples of apparently rather arenaceous and variably shelly and ?oil stained sediments analysed from the Rona Member have varying TOC contents. Some are organically lean (1960.35m MDBRT – 0.88 TOC%), four samples have moderately high TOC% (1.15-3.51%TOC) and one sample at 1937.32m MDRT is organically rich with a TOC content of 7.21% while the sample at 1967.35 has a very high TOC of 49.40%.

All seven samples were analysed for Rock-Eval pyrolysis and range from poor to very good oil-prone source rocks (HI's ranging from 111 to 323). No kerogen typing or spore colour analyses have been performed on these samples. Whole rock reflectivity studies revealed rather variable results; the upper two samples studied apparently containing bituminite with higher reflectivity while the deeper samples contained apparently rather low reflecting vitrinitic material. The microscopy results for sample 1967.35m MDRT indicate that the sample is highly carbonaceous.

Summary

Organic rich samples with oil and gas generative potential have been detected in the Jurassic KCF and Rona Members in the 206/12a-5 well. The Kimmeridge Clay is a good oil prone source rock, while the Rona Member contains some good and very good potential oil source rocks. The Rona coaly sample at 1967.35m MDRT is potentially an excellent gas prone source rock.

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Well	MD (m)	TVDSS (m)	Type	Stratigraphy	APTID	TOC (%)	Rock-Eval pyrolysis summary					Source rock potential	Detailed analyses
							S2 (mg/g)	HI	OI	PI	Tmax (°C)		
206/12a-5	1884.58	1860.9	Core	Turonian (K72)	154655	0.49						n/a	n
	1892.66	1869.0	Core	Turonian (K72)	154656X	0.61	0.26	43	73	0.04	434	non source	n
	1902.70	1879.0	Core	Turonian (K72)	154657	0.47						n/a	n
	1911.17	1887.5	Core	Turonian (K72)	154658X	0.74	0.90	121	116	0.02	438	non source	n
	1928.61	1905.0	COPL	Cromer Knoll (K20)	146975X	0.05	0.02	39	314	0.00	438	non source	n
	1932.42	1908.8	Core	KCF	154659X	3.35	9.56	285	12	0.00	438	good	n
	1932.67	1909.0	COPL	KCF	146976X	2.11	6.84	324	28	0.01	437	good	y
	1937.14	1913.5	Core	Rona	154660X	1.50	1.67	111	16	0.01	427	poor	n
	1937.32	1913.7	Core	Rona	154661X	7.21	23.31	323	2	0.00	433	very good	y
	1944.05	1920.4	Core	Rona	154662X	2.83	6.21	219	4	0.01	438	good	n
	1948.68	1925.0	Core	Rona	154663X	3.51	7.80	222	6	0.03	434	good	n
	1960.35	1936.7	Core	Rona	154664X	0.88	1.29	147	17	0.04	418	poor	n
	1966.70	1943.0	Core	Rona	154665X	1.15	2.58	224	20	0.01	421	fair	n
	1967.35	1943.7	Core	Rona	154666X	49.40	101.44	205	3	0.00	430	excellent	y

TOC : total organic carbon content (%)
S2 : pyrolysis potential yield (S2 peak) (mg/g)
HI : hydrogen index (mg HC/g TOC)
OI : oxygen index (mg CO2/g TOC)
PI : production index
Tmax : pyrolysis Tmax - temperature of maximum S2 yield (°C)

Source quality	Potential Yield / S2 (mg/g)
non source/poor	<2
fair	2-6
good	6-20
very good	20-50
excellent	>50

TABLE 4.1 Summary of source rock screening data

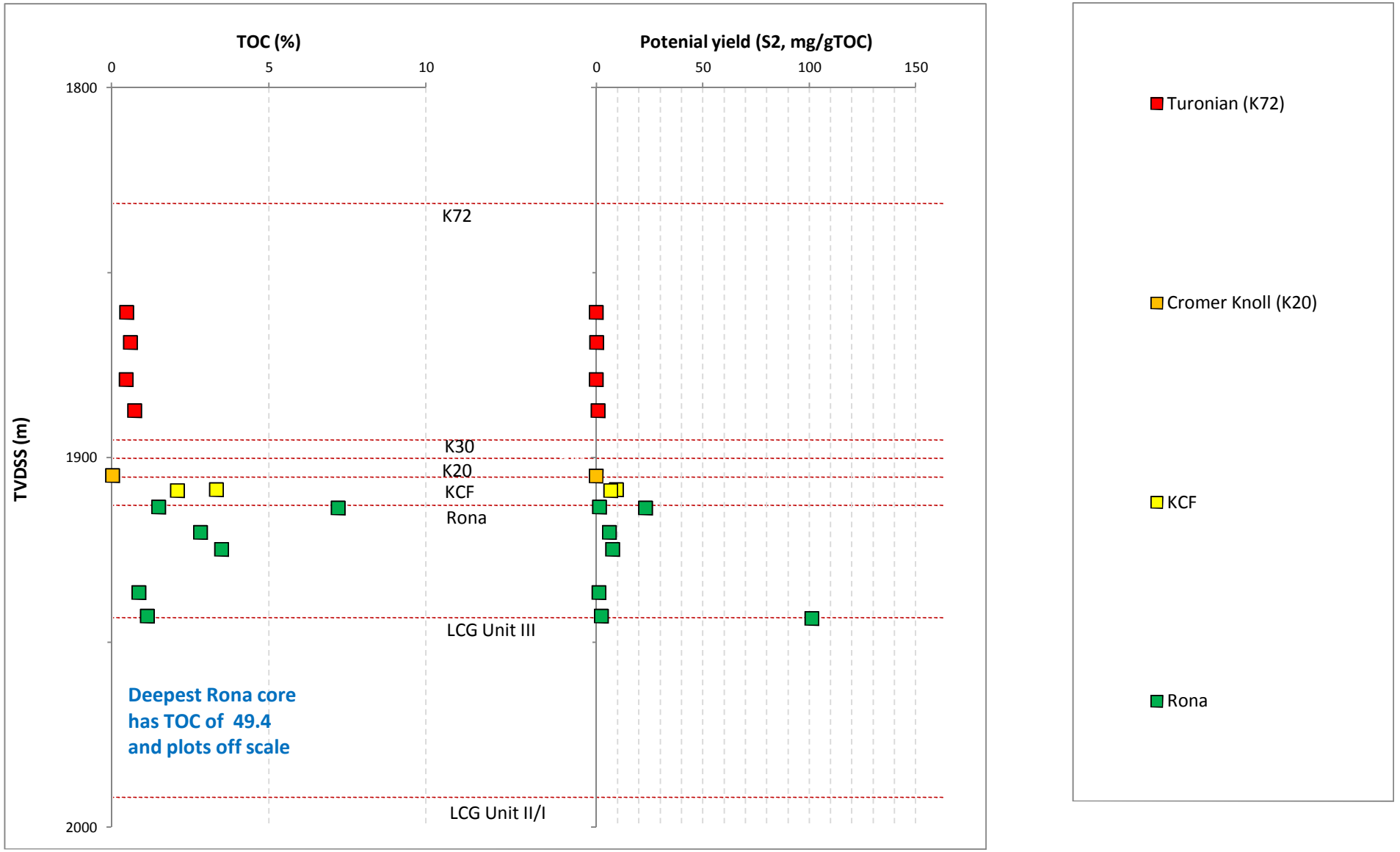

FIGURE 4.1 TOC contents and pyrolysis potential yields (S2) against depth

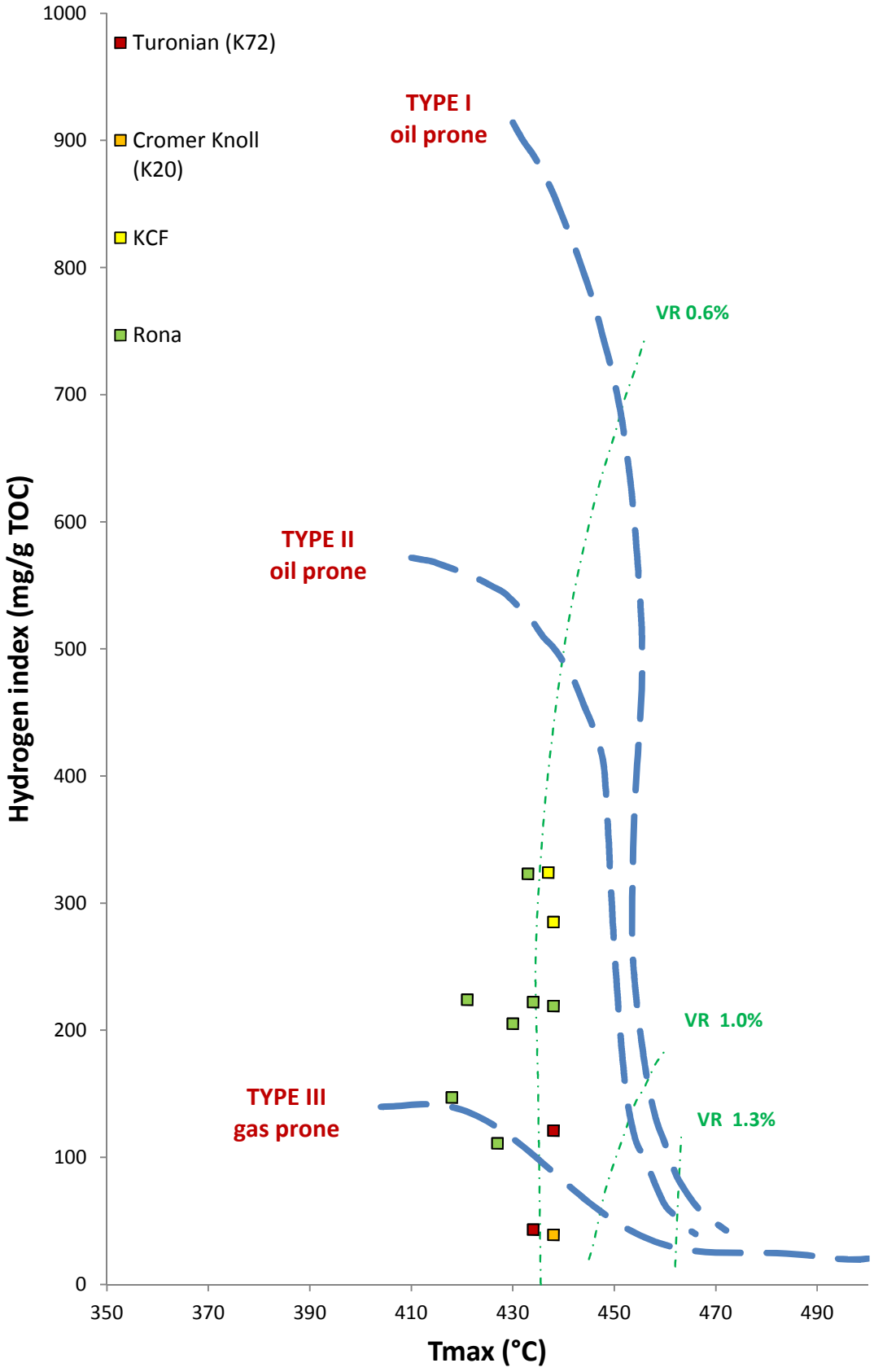

FIGURE 4.2 Pyrolysis data plot of hydrogen index (HI) against Tmax

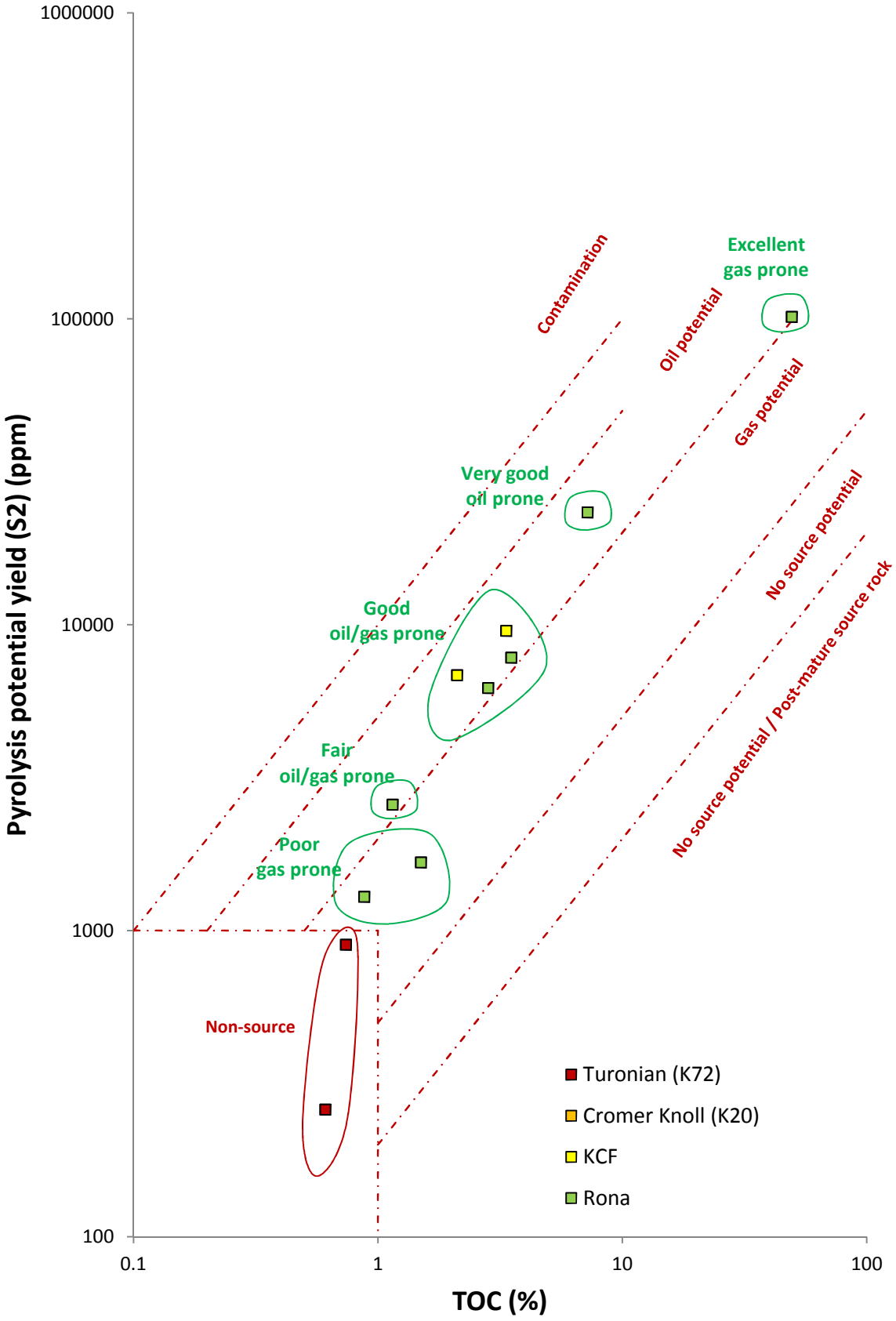

FIGURE 4.3 Pyrolysis data plot of potential yield (S2) against TOC

CHAPTER 5
FLOWED AND EXTRACTED OILS

5.1 Samples and analyses

Geochemical analyses have been undertaken on a number of different sample types. These include:

- Four flowed oil samples (1 DST and 3 MDTs) and three spun out oils from cores. The oils flowed from the Lower Cretaceous (Cromer Knoll Group), Jurassic (Rona) and the Devonian (Lower Clair Group).
- Cores selected by the client from intervals in the Cretaceous, Jurassic and Lower Clair Group. These samples included core plugs, cores and side wall cores some of which are coarse grained potential reservoir lithologies of varying porosity and others this study has demonstrated to have some hydrocarbon generative potential – possible source rocks. These samples have been extracted and analysed for the presence of migrant hydrocarbons.
- To aid in the interpretation of the data with respect to potential contamination from drilling filtrate a number of mud samples taken at various times during the drilling of the well have also been analysed in detail.

5.2 Discussion of geochemical analytical data

Following the broad outline of the analytical programme shown in Table 2.4, the data obtained from the flowed, spun out and extracted oils and muds are discussed below in terms of:

- chromatographic composition (whole oil and whole extract gas chromatography);
- bulk composition (MPLC fractionation);
- biomarkers and isotopes;
- elemental and bulk chemical data.

5.2.1 Whole oil and whole extract gas chromatography data

Bulk gas chromatography analyses were undertaken on the flowed oil samples (whole oil GC), three spun out oils, fifteen core extracts and two mud extracts (whole extract GCs). The full data sets are presented in Appendices 5 and 6 respectively, with summary data in Tables 5.1 and 5.2 and Figures 5.1 to 5.4.

Flowed oil quality

The whole oil GCs for the flowed oils are shown in Figure 5.1.

The traces from the four oils indicate good quality oil with a full range of compounds from the gasoline fraction up to n-C36+. The four oils have a very similar fingerprint, the 1956m MDRT Rona oil has some evidence of possible drilling fluid contamination in the n-C12 to n-C17 region.

The results suggest that the oils contain two indigenous phases present as has been demonstrated for previously analysed Clair Field oils. The two oil phases comprise a heavy biodegraded early oil phase characterised by the elevated unresolved complex mixture (UCM)

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'hump', upon which is superimposed a lower abundance of fresher second phase oil represented by the resolved normal and branched and cyclic alkanes. The oils have a hydrocarbon distribution typical of those seen in previous Clair Field oils analysed.

The whole oil GCs for the three spun out oils are shown in Figure 5.2. The spun out oils yielded near identical traces to the flowed oils, showing a full suite of normal alkanes up to n-C36+ and there is little evidence of filtrate contamination. The shallower spun out oil in the Turonian section has a lower concentration of the gasoline hydrocarbons as exemplified by the methylcyclohexane peak.

Extracted (proxy) oil quality and comparison to mud samples

The whole extract GCs for the extracted cores are shown in Figure 5.3, and those for the mud samples in Figure 5.4.

The two analysed mud samples yield near identical GC traces, with abundant normal and isoprenoid alkanes in the n-C12 to n-C18 range sitting above a tight elevated baseline hump. The muds contain minor amounts of heavier (or lighter) normal alkanes.

Out of the fifteen cores extracted and analysed for whole extract GC, only three showed no evidence of migrant hydrocarbons (1900.67m MDRT, 2022.00m MDRT and 2048.0m MDRT). Of the fifteen samples, nine were dominated by drilling filtrate contamination, closely matching the mud traces. Good GC traces with convincing evidence for the presence of migrant oil were found in the Turonian, Cromer Knoll, Rona and Lower Clair Group Unit III intervals.

The good clear migrant oil samples (1884.67m, 1916.67m, 1932.67, 1948.67 and 1956.67m) have elevated baselines and superimposed heavier normal alkanes representing the two phases of the typical Clair oil.

The two samples from the Lower Clair Group Unit II/I interval show no evidence of migrant oil. Baselines are flat and little or no heavier normal alkanes are apparent.

GC-MS evidence

Subsequent GC-MS analyses of the samples are dealt with in detail in the following sections. However, one of the ions collected in this analysis (m/e 183) exclusively shows the heavier n-C20 to n-C30 normal alkanes present in the analysed samples. Figure 5.5 shows all the whole oil/extract GCs with the matching m/e 183 ion trace (if available), while Figure 5.6 plots the absolute abundances of heavier n-alkanes against the ratio of n-C25:n-C20 using the GC-MS data.

Examination of these chromatogram pairs and the plot indicates the following:

- The muds contain little or no normal alkanes in the n-C23 to n-C30 range but do contain some normal alkanes in the n-C20 to n-C22 range.
- The flowed oils, spun out oils and oil stained core contain abundant normal alkanes in this range.
- Ratios of n-C20:n-C25 in the muds are generally high (142 and 147).
- Ratios of n-C20:n-C25 in the flowed oils, spun out oils and oil stained core generally are low (1.4-1.8), indicating drilling mud contamination is not dominant, even in the core. Two cores at 1937m and 1984m MDRT have ratios of 2.45 and 2.64, suggesting that although more drilling filtrate contamination is present it is still not dominant over migrant hydrocarbons.

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- The K72 and some of the KCF and Rona core extracts display the highest longer chain length alkanes (Figure 5.11) which might imply the presence of higher indigenous source derived hydrocarbons.

Comparison of flowed oils – gasoline range hydrocarbons

The analysed oils show a full range of hydrocarbons including the very light gasoline components although in rather diminished abundance to other Clair Field oils. These gasoline compounds can be used as very sensitive indicators to compare the oil with previously analysed oils from other wells in the Clair Field.

Figure 5.7a is a cross plot of absolute concentrations of selected cyclopentanes and cyclohexanes which illustrate variations in the Mango k1 ratio (Mango, 1987). The 206/12a-5 oils have k1 values of 1.04 to 1.05, the Turonian spun out sample (1897.03m, K72) is very different with a k1 of 1.02 (differences in gasoline range noted in whole oil GCs), while the other spun out oils (1922.08m - Cromer Knoll and 1948.67m - Rona) have k1 values of 1.05 identical to the flowed oils. Figure 5.7b plots all available Clair Field oil data for untopped oils for comparison with the 206/12a-5 samples. This shows that greater than 80% of all Clair oils have a k1 in the range 1.06-1.08 and that 1.08 is by far the most common measured value.

Figure 5.8 shows star plots of Halpern C7 correlation ratios (Halpern, 1995). The plot comparing the 206/12a-5 oils to the wider Clair Field indicates that the flowed oils fall mostly within the field delineated by previous Clair oil analyses, while the spun out oils fall fully within the field of standard Clair oils. Normalising the ratios to emphasise any differences shows no variation between the four flowed 206/12a-5 oils.

Both the Mango and Halpern ratios reveal that the oils in the 206/12a-5 well are behaving slightly away from the normal values for most of the Clair Field oils. It is noted that the oils from this well whether as flowed DST or MDT or spun out oils, are relatively deficient in the light hydrocarbons compared to typical Clair Field crude oils. It is considered that this is a genuine and important difference. As with the Mango plots, the Halpern plots also show that the Turonian spun out oil is different from the other spun out oils.

Comparison of flowed and proxy oils – mid range hydrocarbons

Figure 5.9a is a standard Shanmugam type cross plot of isoprenoid/normal alkane ratios. The flowed oils are essentially identical, and are easily differentiated from the mud samples that fall into a distinct separate field. The oil stained and possible source rock dominated cores also fall into one of the two fields, either non contaminated extracted oil plotting with the flowed oils or extracted oils that are dominated by drilling fluid which plot with the mud samples.

An expanded version of the plot (Figure 5.9b) displays the area delineated by all previously analysed flowed oils from the Clair Field. The flowed oil plots within the delineated Clair region indicating the oils in the 206/12a-5 well has isoprenoid ratios entirely typical for the Clair Field as a whole.

5.2.2 Bulk composition (MPLC fractionation)

The flowed oil samples has been fractionated by MPLC along with the three spun out oils, a selection of the more favourable core extracts and two muds for comparison. The full data are presented in Appendix 4 and are summarised on Table 5.3.

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A triangular plot of the fractionation data (Figure 5.10) shows small variations in the values obtained for the flowed oils. The majority of the results, however, fall into a small region coincident with, at the edge of or just outside of the area delineated by previous analyses of flowed oils from the Clair Field. Outliers that don't plot with the other groupings, include some samples that are clearly contaminated with base oils but two others from the KCF and Rona Member that might have a possible source rock extract component. The muds comprise >95% alkanes and plot well away from the Clair Field oils.

5.2.3 Biomarker and isotope data

Selected fractionated oil and rock extracts including the possible source rocks (4 flowed oils, 3 spun out oils, 7 core extracts and 2 muds) have been analysed for alkane and aromatic biomarkers using GC-MS; a sub-set of the flowed oils, spun out oils and 2 core extracts have also been analysed for carbon isotope ratios. These data are fully reported in Appendices 7 to 9, with key biomarker and isotope data shown on Tables 5.4 and 5.5. A series of cross, triangular and depth plots illustrating the results of the GC-MS biomarker and carbon isotope ratio analyses are shown as Figures 5.11 to 5.21.

Filtrate contamination

The data from the filtrate from the oil based drilling fluid (mud) samples have confirmed that only very low abundances of alkane and aromatic biomarkers are present in these samples (Figure 5.11). This confirms observations made using the whole oil/extract GCs that although some level of contamination is present in some samples, the biomarker data are largely unaffected by the contaminants.

The oils spun out from cores, the extracted cores which include those containing indigenous oil and those that might contain additional source derived hydrocarbons, have similar levels of biomarkers to the oils providing evidence for the presence of moderate to high levels of migrant oil.

Oil homogeneity and provenance

Figure 5.12 is a cross plot of biodegradation corrected diahopane and gammacerane to hopane content ratios. The flowed, spun out and extracted oils plot fairly tightly together and within although to one side of the area delineated by all previously analysed flowed oils from the Clair Field. The tight cluster of extracted oils with the flowed oils confirms the presence of oil staining in these samples. The mud samples yield quite different ratios and are easily distinguished from the migrant hydrocarbons.

A triangular plot of relative isosterane abundances (Figure 5.13) again shows that the flowed, spun out and extracted oils lie within a small field which is again coincident with previously analysed oils from the Clair Field. The muds are easily distinguished from the migrant oils.

The carbon isotope (CI) ratio data are shown on Figure 5.14. The flowed oils all plot in a tight cluster within the previously analysed Clair Field data. The spun out oils from the Cromer Knoll and Rona intervals and the extracted oil from the KCF plot with the flowed oils. The Turonian spun out oil plots away from the analysed group and on the edge of the delineated area of previously analysed oils which may or may not be significant. The difference between the spun out oils seen in the gasoline plots in Figures 5.7 and 5.8 is not observed in the carbon isotopes. The extracted oil from the Rona core plot away from all the other analysed samples; it

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is to be noted that this sample contained evidence of drilling fluid contamination which is known to have an adverse effect on carbon isotope data.

Overall, the biomarker and isotope data confirm the similarity of the flowed, spun out and extracted oils in 206/12a-5 to previously analysed Clair Field oils.

Maturity

Figures 5.15 to 5.17 are cross plots of maturity dependent biomarker parameters.

The cross plot of 29Ts/29-norhopane against Ts/Tm (Figure 5.15) shows a distinct split in which the data for the muds suggest they are hardly mature, while the flowed, spun out and extracted oils, plot in a cluster in the upper part of the middle mature zone of the oil window. While an area of the previously analysed Clair Field oils is not highlighted on the plot, the newly analysed samples are consistent with previously analysed Clair Field oils.

The steranes in Figure 5.16 confirm the split in the data. Although the results for the flowed, spun out and extracted oils are more variable than with the terpanes, the overall conclusions on maturity are similar, indicating the oils are main oil window mature and the muds extracts are early mature.

A cross plot of methyldibenzothiophene against triaromatic side chain cracking ratios is shown as Figure 5.17. This is a maturity related plot, cross plotting relevant maturity parameters which can be used to infer the temperature of the source at generation and maturity of the hydrocarbon fluids. Theoretical oil generation temperature isotherms determined from kinetics assigned to organofacies B source rocks in the BP Petroleum Toolkit, are displayed as a series of contours. The majority of the newly analysed samples plot closely together but interestingly on the boundary of the region outlined from the previous Clair Field oil analyses. The results suggest a generation temperature in the range of ~135-140°C. The extracted oils from one of the Rona and the Lower Clair Group sample which have been observed to contain contamination, plot away from the tight cluster of the other analysed samples and at inferred higher temperatures. The reason for this is not known.

Overall, the data for the flowed and spun out oils and the solvent extracted oils present in the 206/12a-5 well are consistent with the range of previously analysed Clair Field oils but appear to represent a sub set of those oils. The presence or absence of migrant oil and data quality is summarised in Table 5.6

Biodegradation - phase I & II oils

Figures 5.18 and 5.19 are two standard plots developed for studies of the Clair Field which have been shown to be useful to differentiate the oils found in different parts of the field. The cross plots show variations in the abundances of demethylated hopanes, attributable to the presence of the first phase of highly biodegraded oil, and of normal alkanes, which are attributable to the presence of the later, second phase of un-degraded oil. The same ratios and abundances are shown against true vertical depth (TVDSS) on Figures 5.20 and 5.21.

The 206/12a-5 flowed, spun out and the majority of the extracted cores (oils) fall within a cluster outside of the region delineated by data from samples analysed previously in the Clair Field but do plot tightly and to one side of an area where the SW Clair Field oils and data lie (Figure 5.18). The newly analysed samples indicate a level 2 on the scale devised by Wenger *et al.* (2002). These results imply a significantly higher proportion of the second phase (fresh oil)

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to early phase (heavily degraded) oil in the 206/12a-5 well, which is considerably less degraded than the previously analysed oils from the bulk of the Clair Field.

It has been shown that the absolute abundance of straight chain hydrocarbons to demethylated hopanes is a good measure of the relative amounts of the first (degraded) to second (fresh) oils in the typical Clair Field oil mix. The data from the 206/12a-5 oils are shown in Figure 5.19, where it can be seen that the oils, spun out oils, and core extracts fall into distinct groups of which the extracted oils group can be further subdivided by the relative amounts of drilling fluid filtrate contaminants they contain. All the extracts apart from the heavily drilling fluid contaminated samples and the fine grained possible source rock fall in the area delineated for SW Clair Field Cretaceous and Devonian oils.

The extracted oils from the possibly relatively lower permeability Cretaceous and Jurassic ?source rock cores appear to contain larger amounts of both the n-alkanes (from source rock extract) and demethylated hopanes (from being immersed for a long time in the first phase heavily degraded Clair Field oil).

5.2.4 Chemical and elemental composition data

The flowed oils have been analysed for bulk chemical composition (sulphur, metals and wax contents) plus physical properties (API, kinematic viscosity). These data, along with data for other metal and non-metal ions are reported fully in Appendix 10. A summary of the results is shown below.

	206/12a-5 (E-1)	Clair Field - average*
Ni/V	3.96	3.30
S%	0.31	0.36
Wax%	2.19 – 3.23	1.04**
API gravity	25.4 – 26.7	22**
Viscosity (cSt at 40°C)	29.8 – 34.5	41.72**

* - excluding contaminated data and obviously anomalous results in Robertson data
**-average of 206/9a-4 (NEF) and 206/12a-4 (HEX B) only

A cross plot of the sulphur and nickel/vanadium data along with the previously generated data from the 206/13a-3 (HEX A), 206/12a-4 (HEX B), 206/9a-4 (NEF), 206/8-16 (SEG 0) and 206/9a-3 & -3Z (SEG 5) studies is shown as Figures 5.22 which includes legacy Robertson and LGC data as well as the APT generated data. The new 206/12a-5 oils data are entirely in keeping with the previously analysed oils and form a tight cluster near the 206/8-16 (SEG 0) and 206/9a-3 & -3Z (SEG 5) oils.

It should be noted that the 206/12a-5 well oil samples submitted for bulk physical and chemical properties were supplied by Core Labs direct to the APT laboratories. It is understood that the samples were not thoroughly homogenised prior to sub sampling. There is therefore a possibility that the samples are not fully representative of the samples flowed from the wells most particularly in their wax contents. It has been reported elsewhere that there are discrepancies between the wax contents recorded by Corelabs and APT. It is believed both laboratories utilise ASTM methods for the determination so that any differences might be attributable to sampling conditions rather than laboratory error. This has not been verified at the time of preparation of this report (June 2016)

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Well	MDBRT, m	Sample type	Sequence	Mud type	APT ID	Whole oil GC light hydrocarbon ratios/indices					
						Aromaticity Index	Paraffinicity Index	Heptane Index	Isoheptane Index	Mango K1 ratio	Mango K2 ratio
206/12a-5	1897.03	Spun out	K72	OBM	146645	0.72	0.36	17.01	1.19	1.02	0.26
	1922.08	Spun out	K30	OBM	146643	0.92	0.53	20.56	1.42	1.05	0.25
	1922.40	Oil (MDT)	K30	OBM	160302	0.87	0.52	20.15	1.36	1.05	0.26
	1936-1963	Oil (DST)	Rona	OBM	160301	0.85	0.51	20.19	1.36	1.04	0.26
	1948.67	Spun out	Rona	OBM	146644	0.96	0.53	20.58	1.42	1.05	0.25
	1956.00	Oil (MDT)	Rona	OBM	160303	0.86	0.51	20.21	1.35	1.05	0.26
	1984.50	Oil (MDT)	LCG Unit III	OBM	160304	0.79	0.52	20.30	1.35	1.04	0.26

Well	MDBRT, m	Sample type	Sequence	Mud type	APT ID	Whole oil GC light hydrocarbon ratios/indices				
						Halpern C1	Halpern C2	Halpern C3	Halpern C4	Halpern C5
206/12a-5	1897.03	Spun out	K72	OBM	146645	0.13	0.53	0.21	0.13	0.12
	1922.08	Spun out	K30	OBM	146643	0.14	0.49	0.25	0.11	0.09
	1922.40	Oil (MDT)	K30	OBM	160302	0.12	0.48	0.25	0.15	0.06
	1936-1963	Oil (DST)	Rona	OBM	160301	0.12	0.48	0.25	0.15	0.06
	1948.67	Spun out	Rona	OBM	146644	0.14	0.50	0.25	0.11	0.09
	1956.00	Oil (MDT)	Rona	OBM	160303	0.12	0.49	0.25	0.15	0.06
	1984.50	Oil (MDT)	LCG Unit III	OBM	160304	0.11	0.49	0.25	0.15	0.06

TABLE 5.1 Whole oil gas chromatography data - key ratios and indices for flowed oil samples

Well	MDBRT, m	Sample type	Sequence	Mud type	APT ID	Whole oil GC ratios					
						CPI-1	Pr / n-C17	Ph / n-C18	(Pr/n-C17)/ (Ph/n-C18)	Pr / Ph	n-C17/ (n-C17 +n-C27)
206/12a-5	1897.03	Spun out	K72	OBM	146645	1.12	0.39	0.25	1.54	1.60	0.68
	1922.08	Spun out	K30	OBM	146643	1.12	0.42	0.30	1.41	1.66	0.71
	1922.40	Oil (MDT)	K30	OBM	160302	1.09	0.45	0.27	1.65	1.77	0.69
	1727.60	Oil (DST)	Rona	OBM	160301	1.08	0.45	0.28	1.61	1.72	0.69
	1948.67	Spun out	Rona	OBM	146644	1.13	0.41	0.27	1.52	1.64	0.70
	1956.00	Oil (MDT)	Rona	OBM	160303	1.11	0.45	0.29	1.58	1.95	0.74
	1984.50	Oil (MDT)	LGC Unit III	OBM	160304	1.07	0.45	0.27	1.64	1.81	0.70

TABLE 5.1 Whole oil gas chromatography data - key ratios and indices for flowed oil samples

Ratio :	Description / Definition
Aromaticity Index :	Tol/n-C7
Paraffinicity Index :	n-C7/MCyC6
Heptane Index :	n-C7*100/(CyC6+2-MC6+1,1-DMCyC5+3-MC6+c-1,3-DMCyC5+t-1,3-DMCyC5+t-1,2-DMCyC5+n-C7+MCyC6)
Isoheptane Index :	(2-MC6+3-MC6)/(c1,3-DMCyC5+t1,3-DMCyC5+t1,2-DMCyC5)
Mango K1 ratio :	Mango ratio K1 = (2-MC6 + 2,3-DMC5) / (2,4-DMC5 + 3-MC6)
Mango K2 ratio :	Mango ratio K2 = P3 / (P2+N2)
where	*P3 = 2,2-DMC5 + 2,4-DMC5 + 3,3-DMC5 + 2,3-DMC5*
	P2 = 2-MC6 + 3-MC6
	N2 = 1,1-DMCyC5 + c-1,3-DMCyC5 + t-1,3-DMCyC5
Halpern C1 :	2,2-DMC5 / P3
Halpern C2 :	2,3-DMC5 / P3
Halpern C3 :	2,4-DMC5 / P3
Halpern C4 :	3,3-DMC5 / P3
Halpern C5 :	3-EC5 / P3
CPI-1 :	Carbon Preference Index-1: 0.5 * [{ (C25+C27+C29+C31)/(C24+C26+C28+C30) } + { (C25+C27+C29+C31)/(C26+C28+C30+C32) }]
Pr /n-C17 :	Pristane / normal C17
Ph /n-C18 :	Phytane / normal C18
(Pr/n-C17) /(Ph/n-C18)) :	(Pristane / normal C17) / (Phytane / normal C18)
Pr /Ph :	Pristane / phytane
n-C17 /(n-C17+n-C27) :	normal C17 / (C17+C27) fraction

TABLE 5.1 Whole oil gas chromatography data - key ratios and indices for flowed oil samples

Well	MDBRT, m	Sample type	Sequence	Mud type	APTID	Whole extract GC ratios					
						CPI-1	Pr / n-C17	Ph / n-C18	(Pr/n-C17)/ (Ph/n-C18)	Pr / Ph	n-C17/ (n-C17 +n-C27)
206/12a-5	1884.67	COPL	K72	OBM	146965	1.13	0.36	0.26	1.37	1.42	0.79
	1900.67	COPL	K72	OBM	146969	1.13	0.48	0.51	0.95	2.69	1.00
	1916.67	COPL	K72	OBM	146972	1.09	0.37	0.24	1.52	1.58	0.69
	1924.67	COPL	K20	OBM	146974	1.26	0.57	0.64	0.89	1.04	0.83
	1932.67	COPL	KCF	OBM	146976	1.09	0.41	0.27	1.52	1.58	0.69
	1937.32	Core	Rona	OBM	154661	1.15	0.43	0.35	1.23	2.10	0.89
	1938.00	Mud	Rona	OBM	163301	0.95	0.50	0.55	0.91	2.58	1.00
	1948.67	COPL	Rona	OBM	146980	1.10	0.42	0.27	1.54	1.58	0.69
	1956.67	COPL	Rona	OBM	146982	1.10	0.42	0.27	1.55	1.59	0.70
	1967.35	Core	Rona	OBM	154666	1.13	0.44	0.31	1.40	1.97	0.79
	1984.00	SWC	LCG Unit III	OBM	153498	1.09	0.50	0.51	0.97	2.23	0.98
	1988.67	COPL	LCG Unit III	OBM	146990	1.10	0.39	0.25	1.55	1.63	0.74
	1997.67	COPL	LCG Unit III	OBM	146992	1.10	0.41	0.26	1.56	1.63	0.69
	2001.00	SWC	LCG Unit III	OBM	153499	1.10	0.49	0.53	0.94	2.58	0.99
	2022.00	SWC	LCG Unit II/I	OBM	153500	1.12	0.48	0.52	0.91	2.57	0.99
	2048.00	SWC	LCG Unit II/I	OBM	153501	1.03	0.51	0.54	0.95	2.45	1.00
	2352.0	Mud	Basement	OBM	163302	0.92	0.51	0.57	0.90	2.72	1.00

Ratio :	Description / Definition
CPI-1 :	Carbon Preference Index-1: 0.5 * [{ (C25+C27+C29+C31)/(C24+C26+C28+C30) } + { (C25+C27+C29+C31)/(C26+C28+C30+C32) }]
Pr /n-C17 :	Pristane / normal C17
Ph /n-C18 :	Phytane / normal C18
(Pr/n-C17) /(Ph/n-C18)) :	(Pristane / normal C17) / (Phytane / normal C18)
Pr /Ph :	Pristane / phytane
n-C17 /(n-C17+n-C27) :	normal C17 / (C17+C27) fraction

TABLE 5.2 Whole extract gas chromatography data - key ratios and indices

Well	MDBRT, m	Sample type	Sequence	Mud type	APT ID	Extract (ppm)	Volatiles (%)	MPLC & asphaltene precipitation data					
								Fraction (wt% of oil)				Ratios	
								Alks.	Aroms.	Pols.	Asph.	Alks/ aroms	HC/ non-HC
206/12a-5	1884.58	Core	K72	OBM	154655	1043							
	1884.67	Core plugs	K72	OBM	146965	250		34.8	22.3	18.6	24.3	1.56	1.33
	1892.66	Core	K72	OBM	154656	2499							
	1897.03	Spun out oils	K72	OBM	146645	n/a		50.6	20.3	22.4	6.7	2.49	2.44
	1900.67	Core plugs	K72	OBM	146969	3173		95.7	0.6	2.8	0.9	159.50	26.03
	1902.70	Core	K72	OBM	154657	2047							
	1911.17	Core	K72	OBM	154658	2384							
	1916.67	Core plugs	K72	OBM	146972	15766		53.3	20.8	19.2	6.7	2.56	2.86
	1922.08	Spun out oils	K30	OBM	146643	n/a		54.1	20.2	21.8	3.8	2.68	2.90
	1922.40	Oil (MDT)	K30	OBM	160302	n/a	11.6	52.3	28.5	14.9	4.3	1.84	4.21
	1924.67	Core plugs	K20	OBM	146974	20		7.0	6.2	9.1	77.8	1.13	0.15
	1932.42	Core	KCF	OBM	154659	7982							
	1932.67	Core plugs	KCF	OBM	146976	4902		44.0	20.9	17.5	17.6	2.11	1.85
	1936-1963	Oil (DST)	Rona	OBM	160301	n/a	7.4	50.5	26	18.3	5.3	1.94	3.24
	1937.14	Core	Rona	OBM	154660	8868							
	1937.32	Core	Rona	OBM	154661	11317		27.5	28.5	31.4	12.6	0.96	1.27
	1938.00	Mud	Rona	OBM	163301	-		95.4	0.1	3.5	1.0	954.00	21.22
	1944.05	Core	Rona	OBM	154662	65128							
	1948.67	Spun out oils	Rona	OBM	146644	n/a		53.6	20.1	21.6	4.8	2.67	2.79
	1948.67	Core plugs	Rona	OBM	146980	69263		47.0	23.3	21.4	8.3	2.02	2.37
	1948.68	Core	Rona	OBM	154663	65640							
	1956.00	Oil (MDT)	Rona	OBM	160303	n/a	7.2	54.9	25.6	14.5	5.1	2.14	4.11
	1956.67	Core plugs	Rona	OBM	146982	73633		56.2	26.7	7.4	9.7	2.10	4.85
	1960.35	Core	Rona	OBM	154664	46448							
	1966.70	Core	Rona	OBM	154665	30354							
	1967.35	Core	Rona	OBM	154666	102786		35.4	20.2	22.7	21.8	1.75	1.25
	1984.00	SWC	LCG Unit III	OBM	153498	11345		77.4	7.7	12.0	2.8	10.05	5.75
	1984.50	Oil (MDT)	LCG Unit III	OBM	160304	n/a	6.7	52.4	26.6	15.4	5.7	1.97	3.74

TABLE 5.3 Extraction and fractionation data

Well	MDBRT, m	Sample type	Sequence	Mud type	APT ID	Extract (ppm)	Volatiles (%)	MPLC & asphaltene precipitation data					
								Fraction (wt% of oil)				Ratios	
								Alks.	Aroms.	Pols.	Asph.	Alks/ aroms	HC/ non-HC
206/12a-5	1988.67	Core plugs	LCG Unit III	OBM	146990	17006		33.6	26.8	32.6	7.0	1.25	1.53
	1997.67	Core plugs	LCG Unit III	OBM	146992	21656		50.1	20.8	23.0	6.1	2.41	2.44
	2001.00	SWC	LCG Unit III	OBM	153499	16458		87.1	3.2	8.0	1.7	27.22	9.31
	2022.00	SWC	LCG Unit II/I	OBM	153500	38571		89.4	2.1	7.4	1.0	42.57	10.89
	2048.00	SWC	LCG Unit II/I	OBM	153501	12823		89.2	1.4	8.1	1.3	63.71	9.64
	2352.00	Mud	Basement	OBM	163302	-		95.7	0.5	2.9	1.0	191.40	24.67

MPLC : medium pressure liquid chromatography
Alks. : alkane (saturate) fraction (MPLC)
Aroms. : aromatic fraction (MPLC)
Pols. : polar fraction (MPLC)
Asph. : asphaltenes (precipitated)
HC : hydrocarbons (alkanes+aromatics)
non-HC : non-hydrocarbons (polars+asphaltenes)

TABLE 5.3 Extraction and fractionation data

Well	MDBRT, m	Sample type	Sequence	Mud type	APT ID	Maturity related biomarker ratios/fractions						
						Ts/Tm	C30 mor/ C30 hop	C32 S/R hop	tri/ (tri+hop)	Ts/(Ts + C30 hop)	29Ts/(29Ts +C29 hop)	29Ts/29ab
206/12a-5	1897.03	Spun out	K72	OBM	146645	1.01	0.14	1.22	0.12	0.17	0.28	0.39
	1916.67	COPL	K72	OBM	146972	1.09	0.14	1.31	0.14	0.18	0.29	0.41
	1922.08	Spun out	K30	OBM	146643	1.05	0.13	1.35	0.13	0.18	0.30	0.43
	1922.40	Oil (MDT)	K30	OBM	160302	1.09	0.16	1.29	0.14	0.19	0.30	0.42
	1932.67	COPL	KCF	OBM	146976	1.12	0.15	1.24	0.14	0.18	0.31	0.44
	1936-1963	Oil (DST)	Rona	OBM	160301	1.12	0.16	1.29	0.14	0.19	0.31	0.45
	1937.32	Core	Rona	OBM	154661	1.03	0.15	1.26	0.16	0.22	0.29	0.41
	1938.00	Mud	Rona	OBM	163301	0.68	0.14	1.17	0.05	0.06	0.17	0.20
	1948.67	Spun out	Rona	OBM	146644	0.94	0.14	1.28	0.13	0.17	0.30	0.43
	1956.00	Oil (MDT)	Rona	OBM	160303	1.16	0.16	1.29	0.14	0.19	0.29	0.40
	1956.67	COPL	Rona	OBM	146982	1.11	0.15	1.27	0.14	0.19	0.30	0.42
	1967.35	Core	Rona	OBM	154666	1.02	0.15	1.23	0.14	0.20	0.31	0.46
	1984.50	Oil (MDT)	LCG unit III	OBM	160304	1.03	0.15	1.22	0.14	0.19	0.30	0.43
	1984.00	SWC	LCG unit III	OBM	153498	0.95	0.14	1.25	0.13	0.17	0.28	0.40
	1997.67	COPL	LCG unit III	OBM	146992	1.05	0.16	1.26	0.14	0.19	0.31	0.45
	2352.00	Mud	Basement	OBM	163302	0.79	0.08	1.24	0.07	0.06	0.12	0.14

TABLE 5.4 Alkane and aromatic GC-MS biomarker data - key maturity ratios and fractions

Well	MDBRT, m	Sample type	Sequence	Mud type	APT ID	Maturity related biomarker ratios/fractions						
						αα S/S+R C29 ster	ββ/ αα+ββ C29 ster	αα S/S+R C28 ster	ββ/ αα+ββ C28 ster	MDR1	TASC1	MPI-1
206/12a-5	1897.03	Spun out	K72	OBM	146645	0.48	0.54	0.31	0.65	2.74	0.50	0.55
	1916.67	COPL	K72	OBM	146972	0.51	0.52	0.24	0.70	2.99	0.51	0.61
	1922.08	Spun out	K30	OBM	146643	0.50	0.52	0.36	0.68	2.88	0.51	0.60
	1922.40	Oil (MDT)	K30	OBM	160302	0.52	0.52	0.27	0.69	2.80	0.52	0.60
	1932.67	COPL	KCF	OBM	146976	0.45	0.53	0.27	0.68	2.72	0.47	0.58
	1936-1963	Oil (DST)	Rona	OBM	160301	0.51	0.52	0.25	0.69	2.84	0.52	0.61
	1937.32	Core	Rona	OBM	154661	0.48	0.46	0.26	0.62	2.96	0.72	0.53
	1938.00	Mud	Rona	OBM	163301	0.31	0.44	0.00	0.57	4.39	0.31	0.52
	1948.67	Spun out	Rona	OBM	146644	0.48	0.52	0.29	0.70	3.04	0.53	0.61
	1956.00	Oil (MDT)	Rona	OBM	160303	0.53	0.51	0.26	0.68	2.90	0.51	0.57
	1956.67	COPL	Rona	OBM	146982	0.52	0.52	0.31	0.67	2.95	0.50	0.60
	1967.35	Core	Rona	OBM	154666	0.50	0.50	0.30	0.65	3.02	0.67	0.54
	1984.50	Oil (MDT)	LCG unit III	OBM	160304	0.54	0.52	0.22	0.70	2.92	0.50	0.57
	1984.00	SWC	LCG unit III	OBM	153498	0.49	0.50	0.27	0.68	3.06	0.51	0.71
	1997.67	COPL	LCG unit III	OBM	146992	0.51	0.53	0.32	0.69	2.86	0.53	0.59
	2352.00	Mud	Basement	OBM	163302	0.40	0.40	0.00	0.62	3.59	0.29	0.44

TABLE 5.4 Alkane and aromatic GC-MS biomarker data - key maturity ratios and fractions

Ratio	Description	Fraction	Ion	Definition
Ts/Tm	C27 trisnorhopane ratio	Alkanes	191	27Ts / 27Tm
C30 mor/C30 hop	C30 moretane/hopane ratio	Alkanes	191	30ba / 30ab
C32 S/R hop	C32 hopane isomer ratio	Alkanes	191	32abS / 32abR
tri/(tri+hop)	Tricyclics / (Tricyclics + 17α hopanes) fraction	Alkanes	191	(19/3 + 20/3 + 21/3 + 23/3 + 24/3 + 25/3R + 25/3S + 26/3R + 26/3S + 28/3R + 28/3S + 29/3R + 29/3S) / (19/3 + 20/3 + 21/3 + 23/3 + 24/3 + 25/3R + 25/3S + 26/3R + 26/3S + 28/3R + 28/3S + 29/3R + 29/3S + 27Tm + 28ab + 29ab + 30ab + 31abS + 31abR + 32abS + 32abR + 33abS + 33abR + 34abS + 34abR + 35abS + 35abR)
Ts/(Ts+C30 hop)	Ts/C30 hopane fraction	Alkanes	191	27Ts / (27Ts + 30ab)
29Ts/(29Ts+C29 hop)	29Ts/C29 hopane fraction	Alkanes	191	29Ts / (29ab + 29Ts)
29Ts/29ab	29Ts/C29 hopane ratio	Alkanes	191	29Ts / 29ab
αα S/S+R C29 ster	C29 sterane isomerisation fraction	Alkanes	217	29aaS / (29aaS + 29aaR)
ββ/αα+ββ C29 ster	C29 isosterane fraction	Alkanes	217	(29bbR + 29bbS) / (29aaS + 29bbR + 29bbS + 29aaR)
αα S/S+R C28 ster	C28 sterane isomerisation fraction	Alkanes	217	28aaS / (28aaS + 28aaR)
ββ/αα+ββ C28 ster	C28 isosterane fraction	Alkanes	217	(28bbR + 28bbS) / (28aaS + 28bbR + 28bbS + 28aaR)
MDR1	Methyldibenzothiophene ratio 1	Aromatics	198	4-MDBT / 1-MDBT
TASC1	Triaromatic sidechain cracking 1	Aromatics	231	C20TA / (C20TA + RC28TA)
MPI-1	Methylphenanthrene Index 1	Aromatics	178, 192	1.5*(2-MP + 3-MP)/(P + 1-MP + 9-MP)

TABLE 5.4 Alkane and aromatic GC-MS biomarker data - key maturity ratios and fractions

Well	MDBRT, m	Sample type	Sequence	Mud type	APT ID	Biodegradation fractions/ratios				Source related biomarker ratios	
						C29 dmh/ (dmh+hop)	C30 dmh/ (dmh+hop)	C30 dmh/ hopane	10x C30 dmh/ Σ(C25-C30) n-alkanes	C30 G/ (hop+dmh)	C30 diahop/ (hop+dmh)
206/12a-5	1897.03	Spun out	K72	OBM	146645	0.47	0.25	0.48	0.15	0.06	0.11
	1916.67	COPL	K72	OBM	146972	0.48	0.24	0.44	0.11	0.07	0.11
	1922.08	Spun out	K30	OBM	146643	0.48	0.27	0.49	0.18	0.06	0.10
	1922.40	Oil (MDT)	K30	OBM	160302	0.49	0.26	0.51	0.15	0.07	0.11
	1932.67	COPL	KCF	OBM	146976	0.50	0.26	0.52	0.18	0.07	0.11
	1936-1963	Oil (DST)	Rona	OBM	160301	0.53	0.28	0.52	0.16	0.07	0.11
	1937.32	Core	Rona	OBM	154661	0.49	0.24	0.41	0.38	0.07	0.10
	1938.00	Mud	Rona	OBM	163301	0.24	0.07	0.15	0.41	0.10	0.05
	1948.67	Spun out	Rona	OBM	146644	0.51	0.26	0.49	0.19	0.07	0.10
	1956.00	Oil (MDT)	Rona	OBM	160303	0.49	0.27	0.52	0.16	0.08	0.11
	1956.67	COPL	Rona	OBM	146982	0.50	0.27	0.54	0.16	0.07	0.10
	1967.35	Core	Rona	OBM	154666	0.53	0.28	0.50	0.20	0.07	0.11
	1984.50	Oil (MDT)	LCG unit III	OBM	160304	0.52	0.29	0.55	0.16	0.07	0.11
	1984.00	SWC	LCG unit III	OBM	153498	0.50	0.26	0.47	0.17	0.07	0.10
	1997.67	COPL	LCG unit III	OBM	146992	0.53	0.28	0.54	0.16	0.07	0.11
	2352.00	Mud	Basement	OBM	163302	0.21	0.08	0.18	0.44	0.07	0.03

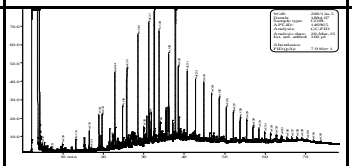
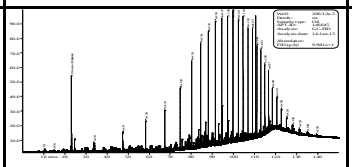
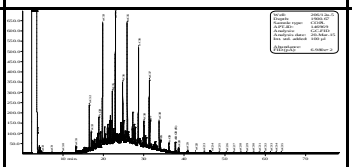
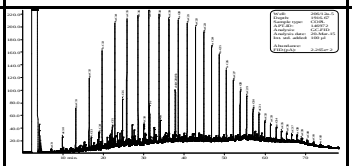
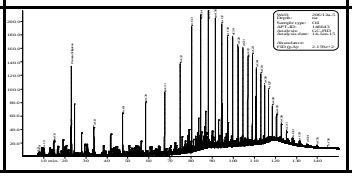
TABLE 5.5 Alkane and aromatic GC-MS biomarker and isotope data - key biodegradation and source ratios

Well	MDBRT, m	Sample type	Sequence	Mud type	APT ID	Source related biomarker ratios and isotope ratios					
						Bnh/ hop	29/30 hopanes	35/34 S hopanes	% C27:C28:C29 isosteranes	δ^{13}C alks	δ^{13}C aroms
206/12a-5	1897.03	Spun out	K72	OBM	146645	0.09	0.52	0.74	35:28:37	-30.40	-28.70
	1916.67	COPL	K72	OBM	146972	0.09	0.50	0.57	37:28:35		
	1922.08	Spun out	K30	OBM	146643	0.09	0.51	0.76	36:29:35	-30.60	-29.10
	1922.40	Oil (MDT)	K30	OBM	160302	0.10	0.52	0.59	35:28:37	-30.50	-29.20
	1932.67	COPL	KCF	OBM	146976	0.10	0.51	0.72	35:28:36	-30.70	-29.10
	1936-1963	Oil (DST)	Rona	OBM	160301	0.11	0.52	0.58	35:28:36	-30.40	-29.10
	1937.32	Core	Rona	OBM	154661	0.10	0.51	0.88	38:28:35	-30.30	-28.20
	1938.00	Mud	Rona	OBM	163301	0.04	0.39	1.13	19:34:47		
	1948.67	Spun out	Rona	OBM	146644	0.09	0.48	0.73	36:28:36	-30.50	-29.10
	1956.00	Oil (MDT)	Rona	OBM	160303	0.11	0.54	0.59	37:28:35	-30.50	-29.20
	1956.67	COPL	Rona	OBM	146982	0.11	0.54	0.56	35:28:36		
	1967.35	Core	Rona	OBM	154666	0.10	0.50	0.73	37:28:35		
	1984.50	Oil (MDT)	LCG unit III	OBM	160304	0.11	0.51	0.55	36:28:36	-30.40	-29.10
	1984.00	SWC	LCG unit III	OBM	153498	0.09	0.48	0.73	35:27:37		
	1997.67	COPL	LCG unit III	OBM	146992	0.11	0.53	0.65	35:29:35		
	2352.00	Mud	Basement	OBM	163302	0.04	0.50	0.93	28:30:42		

TABLE 5.5 Alkane and aromatic GC-MS biomarker and isotope data - key biodegradation and source ratios

Ratio	Description	Fraction	Ion	Definition
C29 norhop/(norhop+hop)	C29 biodegradation fraction	Alkanes	177, 191	25nor29ab / (25nor29ab + 29ab) [demethylated hopane from 177 ion]
C30 norhop/(norhop+hop)	C30 biodegradation fraction	Alkanes	177, 191	25nor30ab / (25nor30ab + 30ab) [demethylated hopane from 177 ion]
C30 norhop/hopane	C30 biodegradation ratio	Alkanes	191	25nor30ab / 30ab [demethylated hopane from 191 ion]
10x 25nor30ab / Σ(C25-C30) n-alkanes	Phase I oil/Phase II oil ratio	Alkanes	191, 183	10x 25nor30ab / Σ(C25-C30) n-alkanes [demethylated hopane from 191 ion, n-alkanes from 183 ion]
C30 G/(hop+dmh)	Degradation compensated gammacerane index	Alkanes	191	G / (25nor30ab + 30ab) [demethylated hopane from 191 ion]
C30 diahop/(hop+dmh)	Degradation compensated diahopane index	Alkanes	191	30d / (25nor30ab + 30ab) [demethylated hopane from 191 ion]
Bnh/hop	Bisnorhopane index	Alkanes	191	28ab / 30ab
29/30 hopanes	C29/C30 hopane ratio	Alkanes	191	29ab / 30ab
35/34 S hopanes	C35S/C34S homohopane ratio	Alkanes	191	35abS / 34abS
%C27:C28:C29 isosteranes	Percentage C27/C28/C29 iso-steranes	Alkanes	218	%(27bbR + 27bbS) / (27bbR + 27bbS + 28bbR + 28bbS + 29bbR + 29bbS) %(28bbR + 28bbS) / (27bbR + 27bbS + 28bbR + 28bbS + 29bbR + 29bbS) %(29bbR + 29bbS) / (27bbR + 27bbS + 28bbR + 28bbS + 29bbR + 29bbS)

TABLE 5.5 Alkane and aromatic GC-MS biomarker and isotope data - key biodegradation and source ratios

Well	MDBRT, m	Sample type	Sequence	Mud type	APTID	Fractionation data			Whole oil/extract GC	Data quality	Oil present
						Asphaltene (wt% of EOM/oil)	Alkanes/ aromatics	HC/ non-HC			
206/12a-5	1884.67	Core plugs	K72	OBM	146965	24.3	1.56	1.33		2	YES typical Clair oil
	1897.03	Spun out oils	K72	OBM	146645	6.7	2.49	2.44		1	N/A
	1900.67	Core plugs	K72	OBM	146969	0.9	159.50	26.03		3	NO
	1916.67	Core plugs	K72	OBM	146972	6.7	2.56	2.86		1	YES typical Clair oil
	1922.08	Spun out oils	K30	OBM	146643	3.8	2.68	2.90		1	N/A

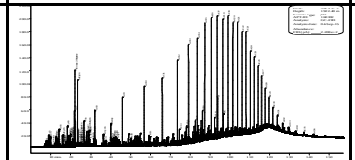
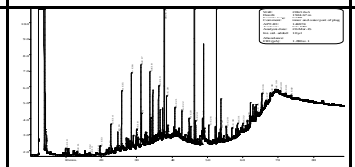
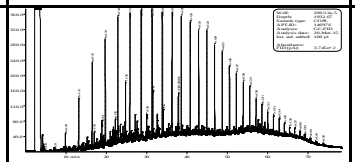
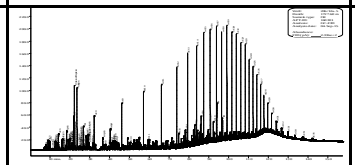
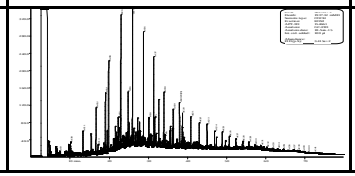
Abundance

0-200ppm - very low
200-500ppm - low
500-1000ppm - low-moderate
1000-2000ppm - moderate
2000-5000ppm - good
5000-10000ppm - high
>10000 - very high

Data quality

1 - Good data quality, representative sample obtained
2 - Moderate data quality, representative sample but with some visible contamination
3 - Poor data quality - non-representative sample, either due to low abundance and/or dominant contamination
i/s - insufficient material or data to assign data quality

TABLE 5.6 Fractionation and whole oil/extract GCs - sample quality

Well	MDBRT, m	Sample type	Sequence	Mud type	APTID	Fractionation data			Whole oil/extract GC	Data quality	Oil present
						Asphaltene (wt% of EOM/oil)	Alkanes/ aromatics	HC/ non-HC			
206/12a-5	1922.40	Oil (MDT)	K30	OBM	160302	4.3	1.84	4.21		1	N/A
	1924.67	Core plugs	K20	OBM	146974	77.8	1.13	0.15		2/3	YES heavily biodeg.
	1932.67	Core plugs	KCF	OBM	146976	17.6	2.11	1.85		1	YES typical Clair oil
	1936-1963	Oil (DST)	Rona	OBM	160301	5.3	1.94	3.24		1	N/A
	1937.32	Core		OBM	154661	12.6	0.96	1.27		2	YES typical Clair oil

Abundance

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1000-2000ppm - moderate
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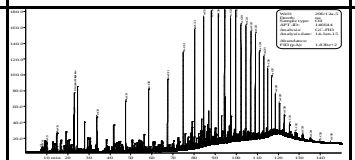
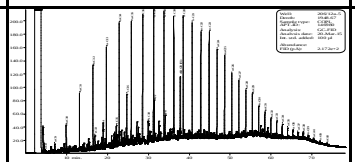
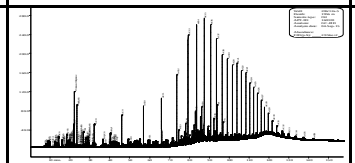
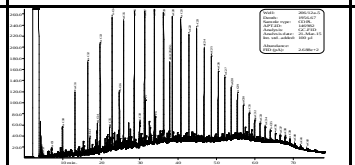
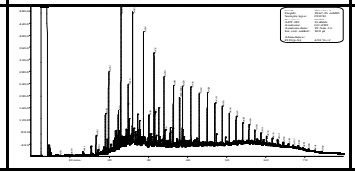
TABLE 5.6 Fractionation and whole oil/extract GCs - sample quality

Well	MDBRT, m	Sample type	Sequence	Mud type	APTID	Fractionation data			Whole oil/extract GC	Data quality	Oil present
						Asphaltene (wt% of EOM/oil)	Alkanes/ aromatics	HC/ non-HC			
206/12a-5	1948.67	Spun out oils	Rona	OBM	146644	4.8	2.67	2.79		1	N/A
	1948.67	Core plugs		OBM	146980	8.3	2.02	2.37		1	YES typical Clair oil
	1956.00	Oil (MDT)		OBM	160303	5.1	2.14	4.11		1	N/A
	1956.67	Core plugs		OBM	146982	9.7	2.10	4.85		1	YES typical Clair oil
	1967.35	Core		OBM	154666	21.8	1.75	1.25		2	YES typical Clair oil

Abundance
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2000-5000ppm - good
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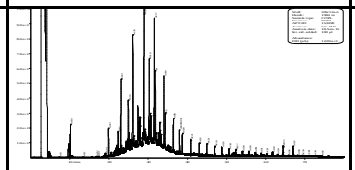
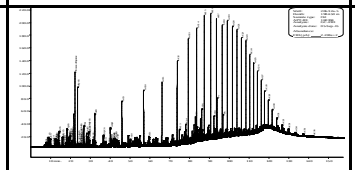
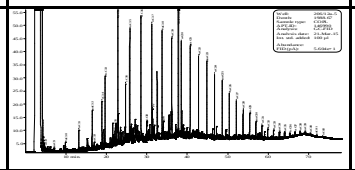
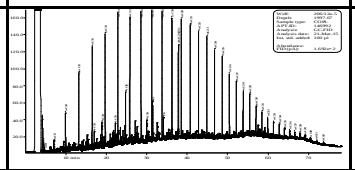
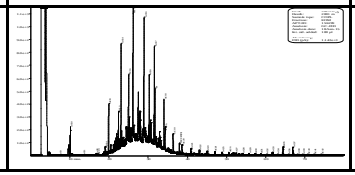
i/s - insufficient material or data to assign data quality

TABLE 5.6 Fractionation and whole oil/extract GCs - sample quality

Well	MDBRT, m	Sample type	Sequence	Mud type	APTID	Fractionation data			Whole oil/extract GC	Data quality	Oil present
						Asphaltene (wt% of EOM/oil)	Alkanes/ aromatics	HC/ non-HC			
206/12a-5	1984.00	SWC	LCG Unit III	OBM	153498	2.8	10.05	5.75		2	YES typical Clair oil
	1984.50	Oil (MDT)		OBM	160304	5.7	1.97	3.74		1	N/A
	1988.67	Core plugs		OBM	146990	7.0	1.25	1.53		1	YES typical Clair oil
	1997.67	Core plugs		OBM	146992	6.1	2.41	2.44		1	YES typical Clair oil
	2001.00	SWC		OBM	153499	1.7	27.22	9.31		3	NO

Abundance

0-200ppm - very low
200-500ppm - low
500-1000ppm - low-moderate
1000-2000ppm - moderate
2000-5000ppm - good
5000-10000ppm - high
≥10000 - very high

Data quality

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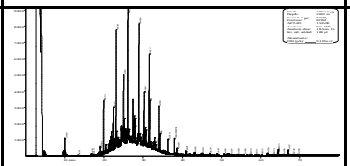
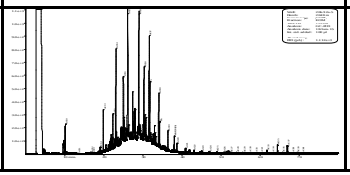
2 - Moderate data quality, representative sample but with some visible contamination
3 - Poor data quality - non-representative sample, either due to low abundance and/or dominant contamination
i/s - insufficient material or data to assign data quality

TABLE 5.6 Fractionation and whole oil/extract GCs - sample quality

Well	MDBRT, m	Sample type	Sequence	Mud type	APTID	Fractionation data			Whole oil/extract GC	Data quality	Oil present
						Asphaltene (wt% of EOM/oil)	Alkanes/ aromatics	HC/ non-HC			
206/12a-5	2022.00	SWC	LCG unit II/I	OBM	153500	1.0	42.57	10.89		3	NO
	2048.00	SWC		OBM	153501	1.3	63.71	9.64		3	NO

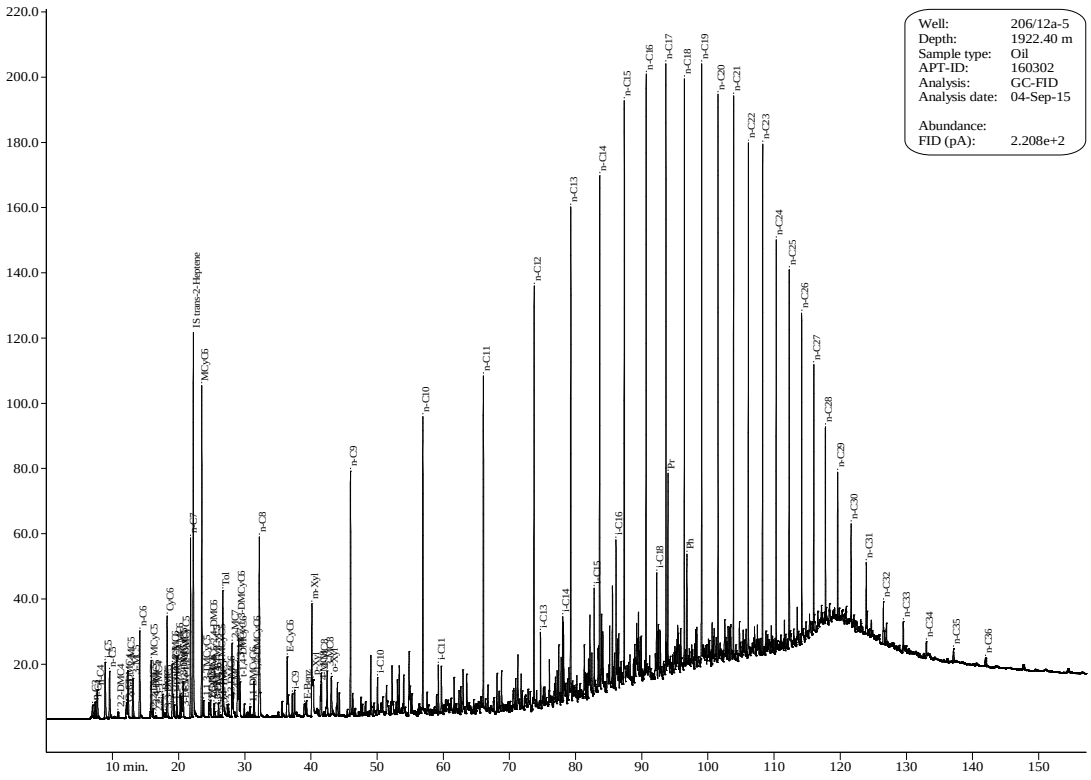
Abundance

0-200ppm - very low
200-500ppm - low
500-1000ppm - low-moderate
1000-2000ppm - moderate
2000-5000ppm - good
5000-10000ppm - high
>10000 - very high

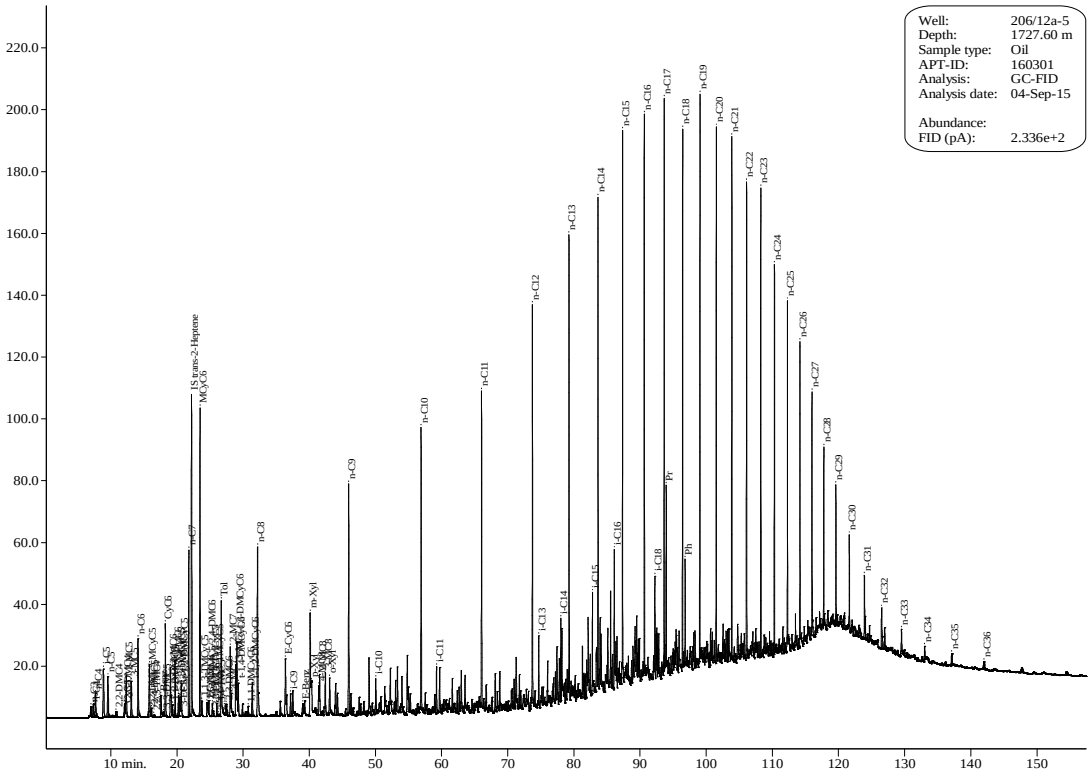
Data quality

1 - Good data quality, representative sample obtained
2 - Moderate data quality, representative sample but with some visible contamination
3 - Poor data quality - non-representative sample, either due to low abundance and/or dominant contamination
i/s - insufficient material or data to assign data quality

TABLE 5.6 Fractionation and whole oil/extract GCs - sample quality

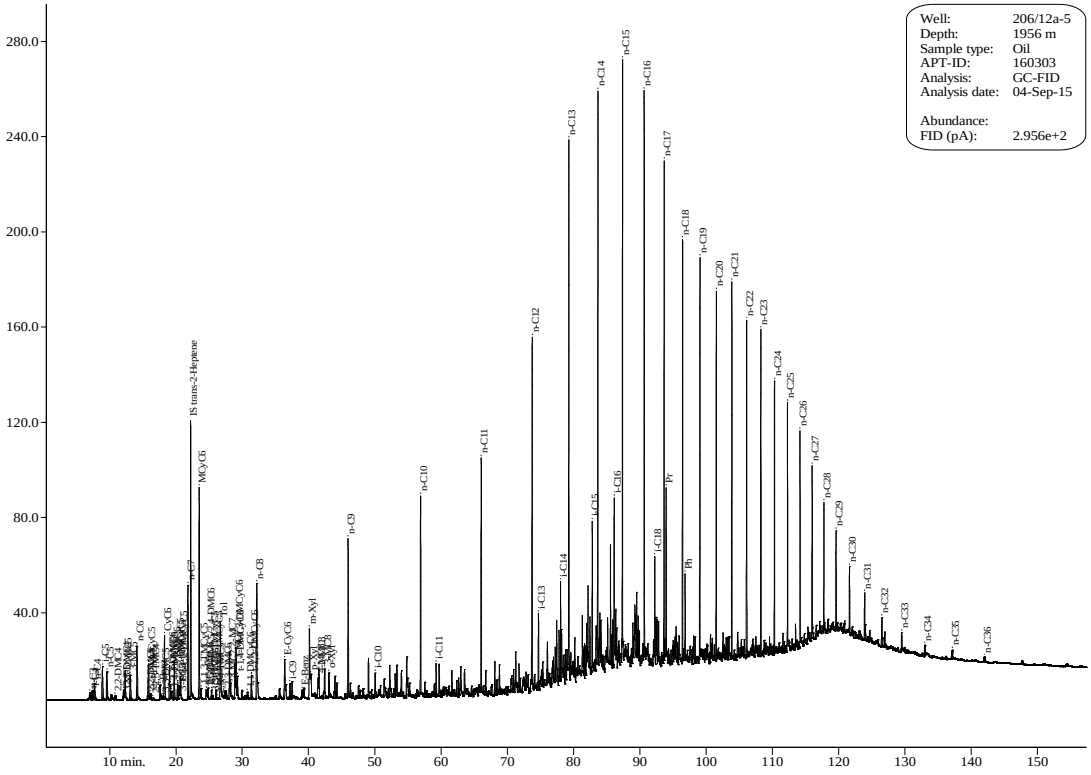


Clair Field: 206/12a-5, flowed oil, 1922.4m (K30)

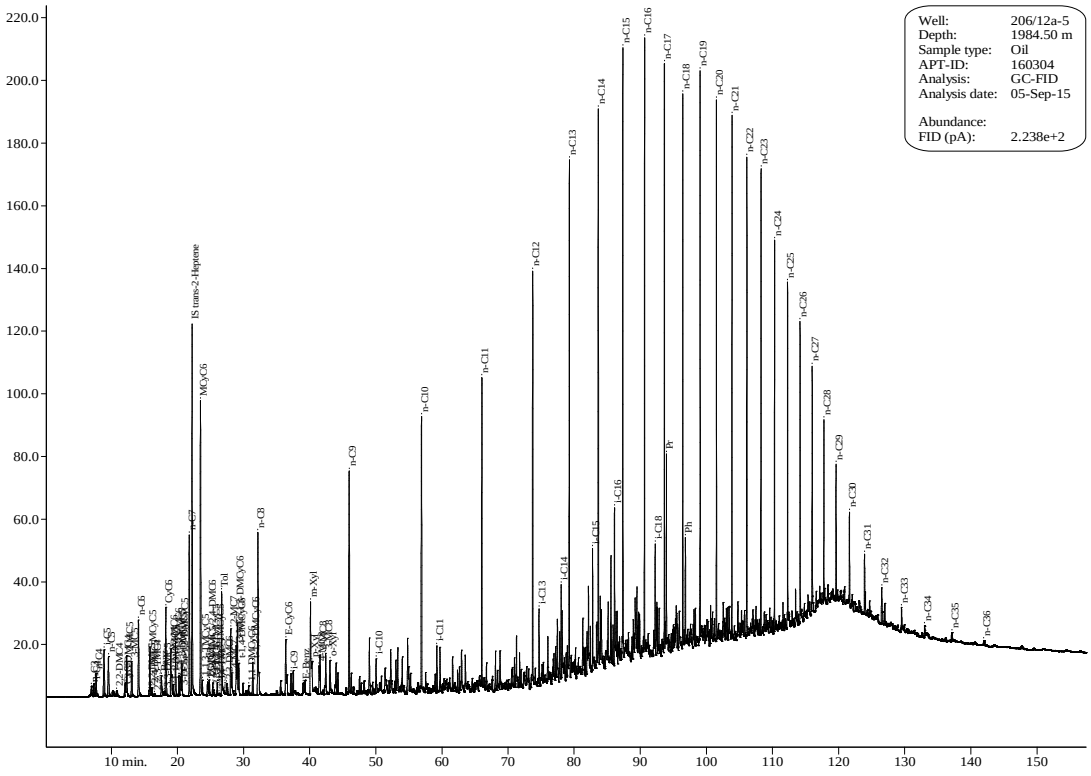


Clair Field: 206/12a-5, DST, 1936-1963m (Rona)

FIGURE 5.1 Whole oil gas chromatograms - tested oil

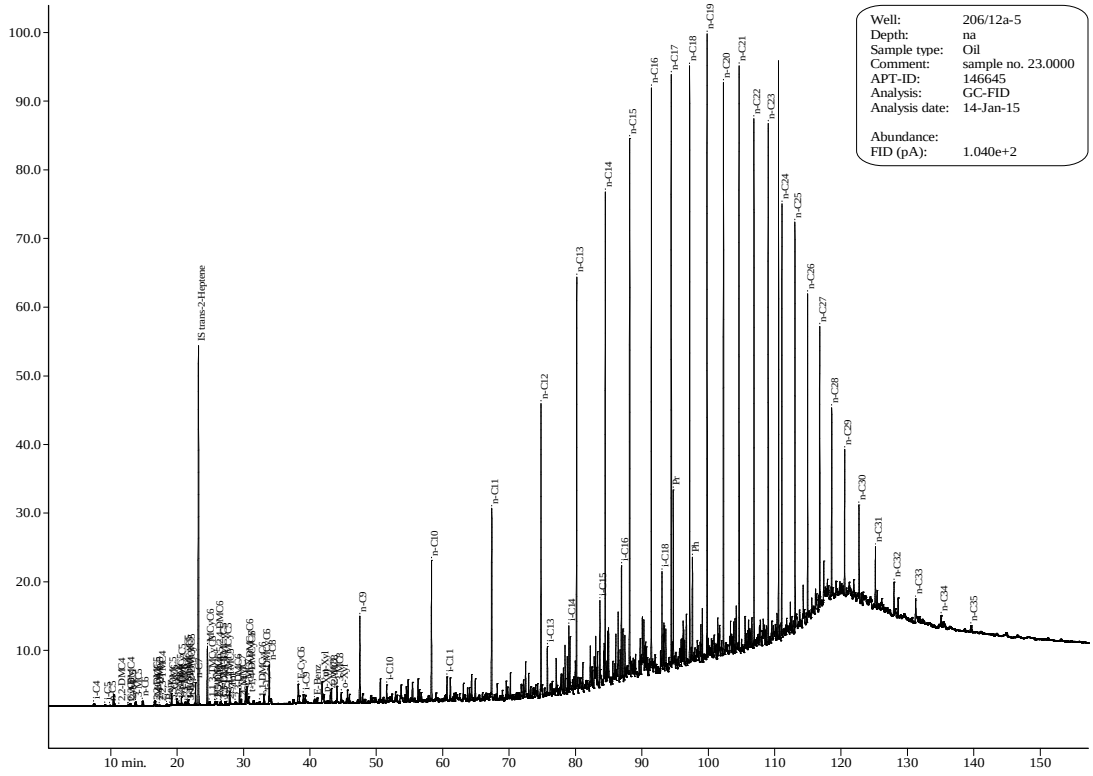


Clair Field: 206/12a-5, flowed oil, 1956.0m (Rona)

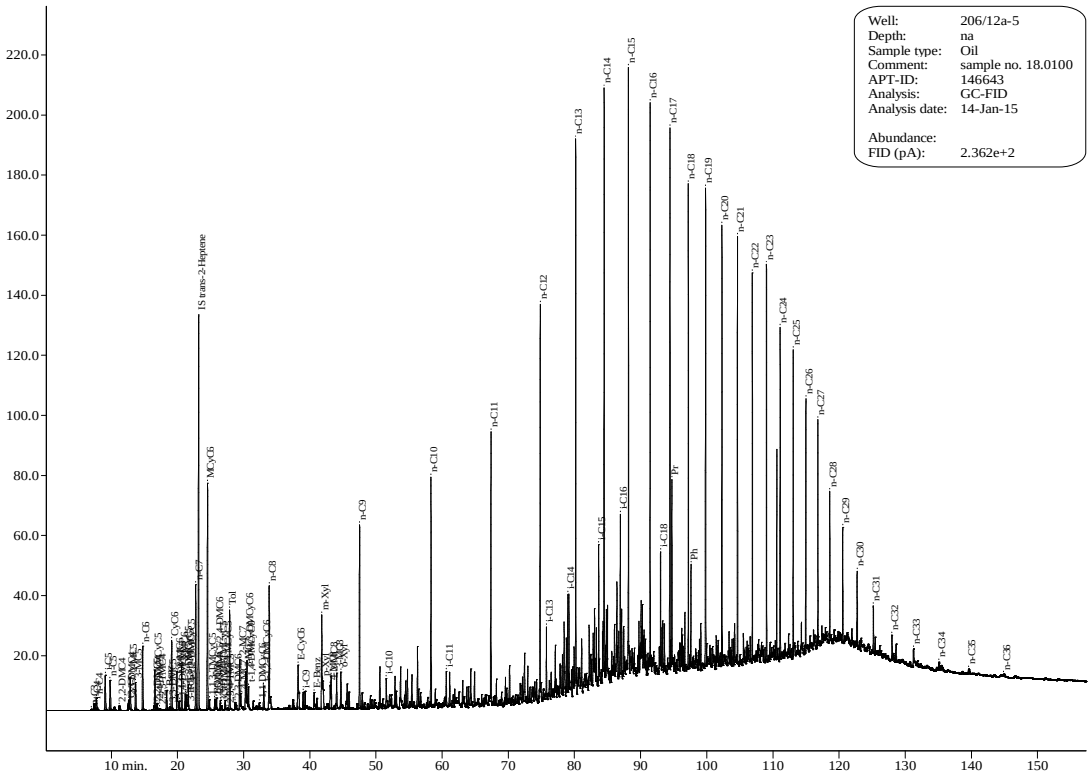


Clair Field: 206/12a-5, flowed oil, 1984.5m (LCG III)

FIGURE 5.1 Whole oil gas chromatograms - tested oil

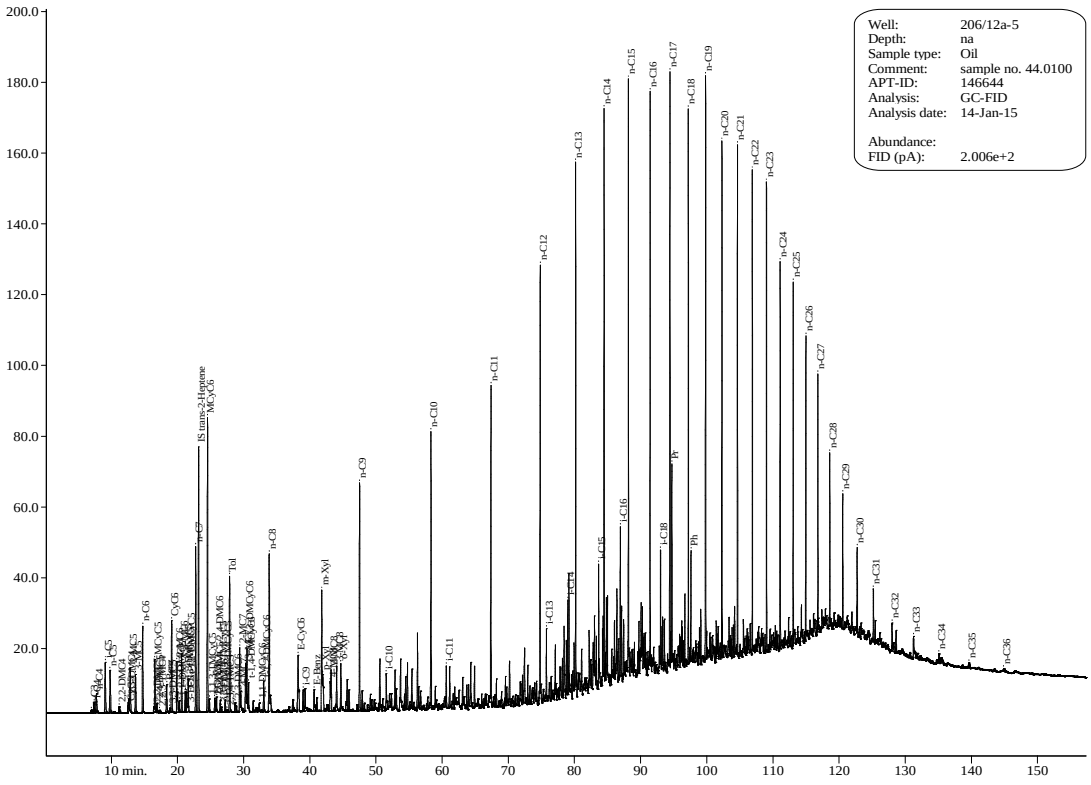


Clair Field: 206/12a-5, spun out oil, 1897.03m (K72)



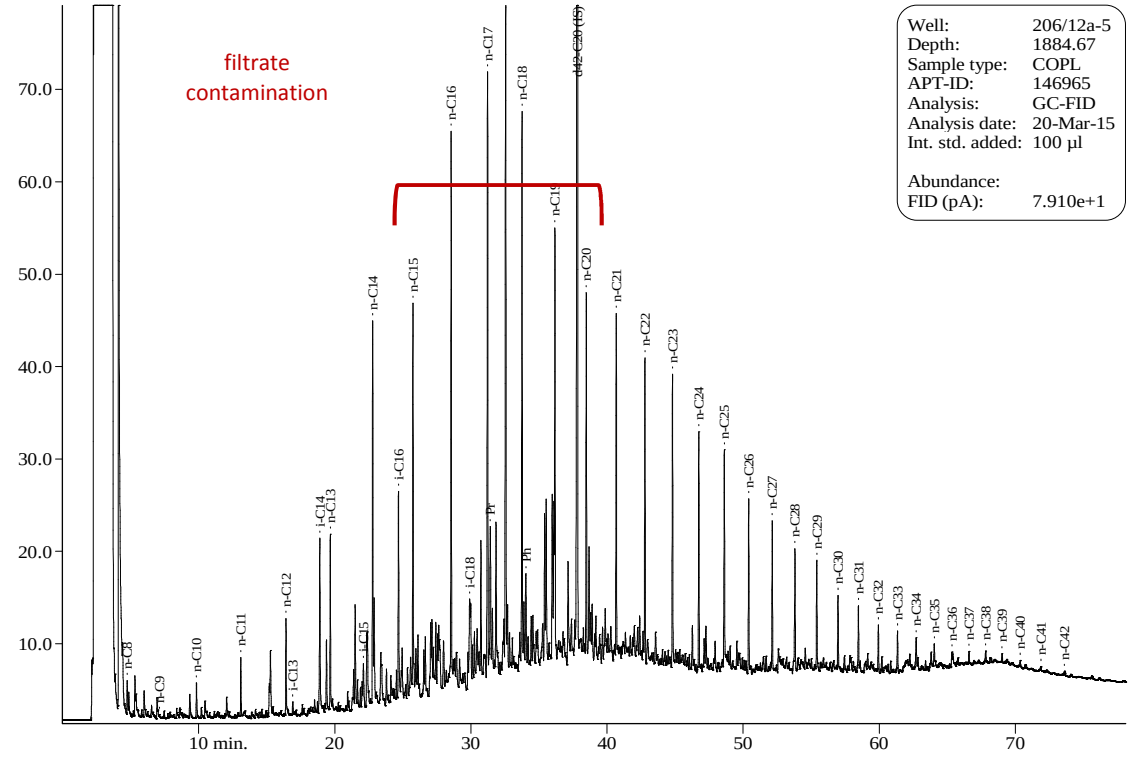
Clair Field: 206/12a-5, spun out oil, 1922.08m (K30)

FIGURE 5.2 Whole oil gas chromatograms - spun out oils

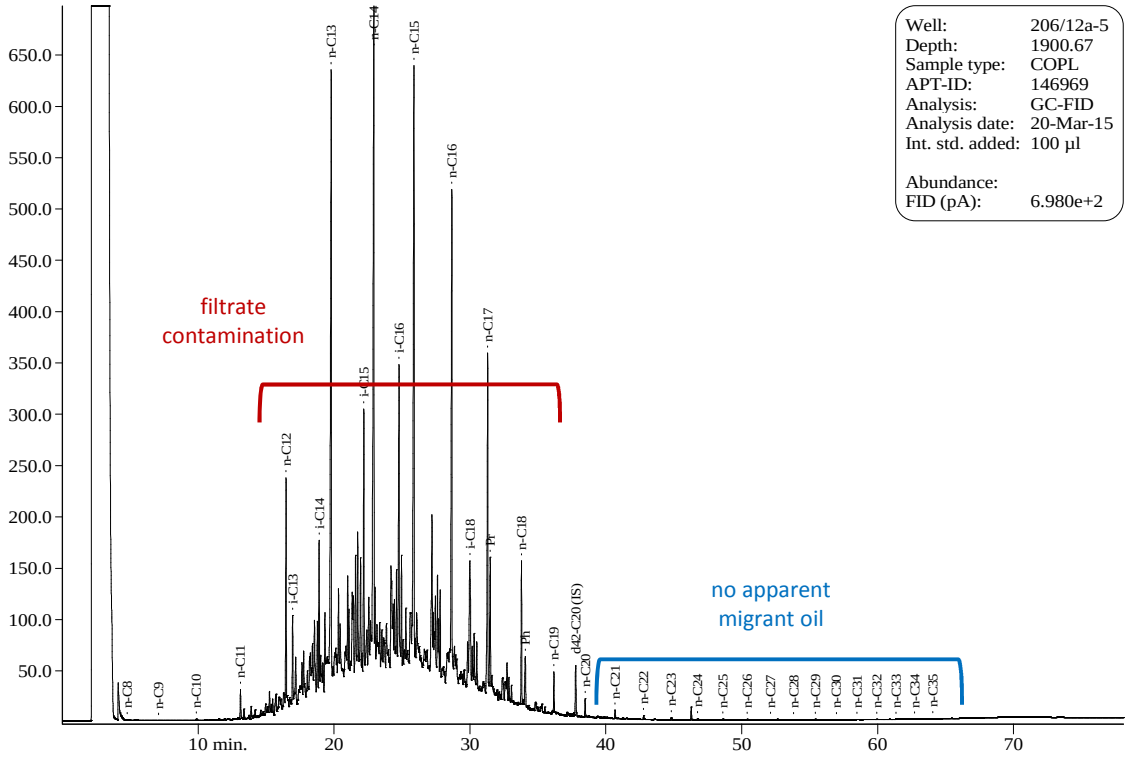


Clair Field: 206/12a-5, spun out oil, 1948.67m (Rona)

FIGURE 5.2 Whole oil gas chromatograms - spun out oils

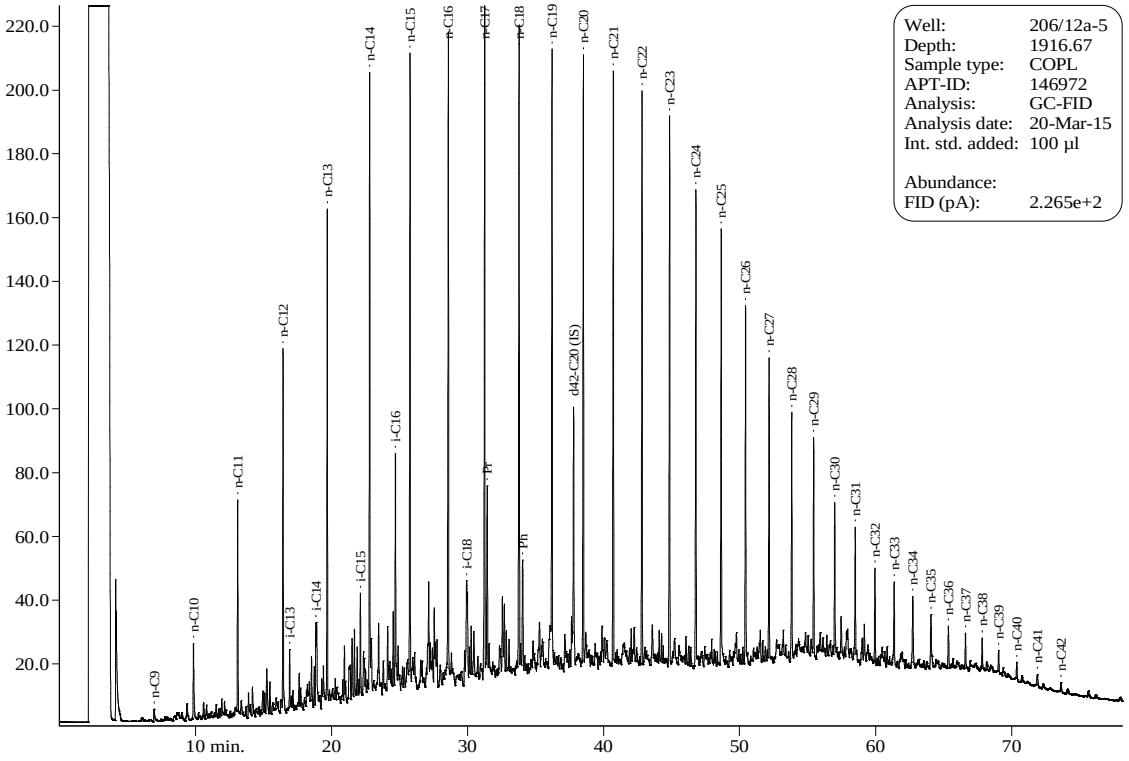


Clair Field: 206/12a-5, 1884.67m, COPL (K72)

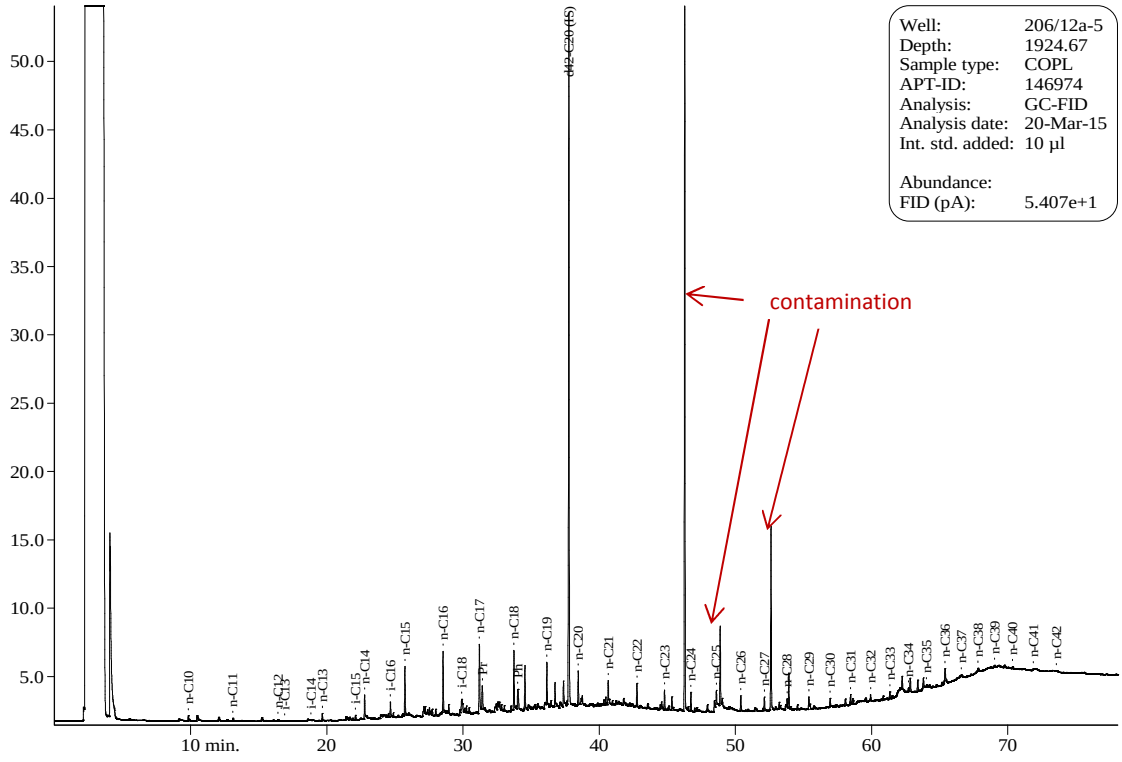


Clair Field: 206/12a-5, 1900.67m, COPL (K72)

FIGURE 5.3 Whole extract gas chromatograms - extracted cores

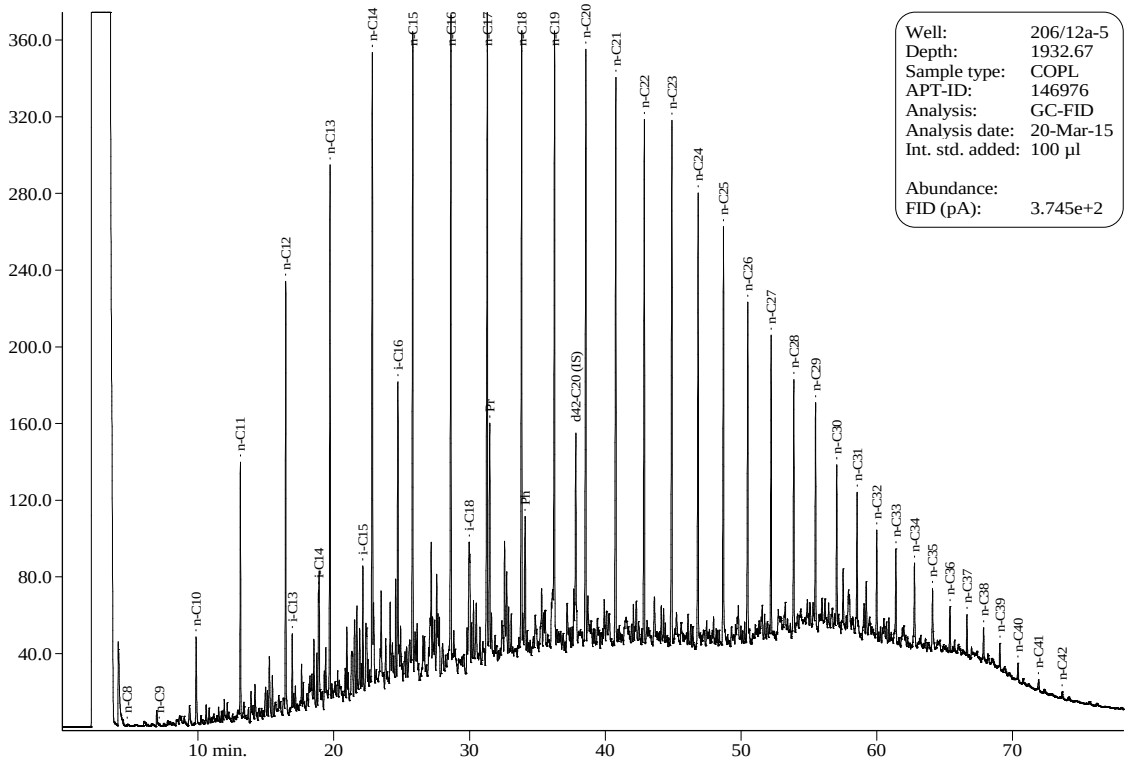


Clair Field: 206/12a-5, 1916.67m, COPL (K72)

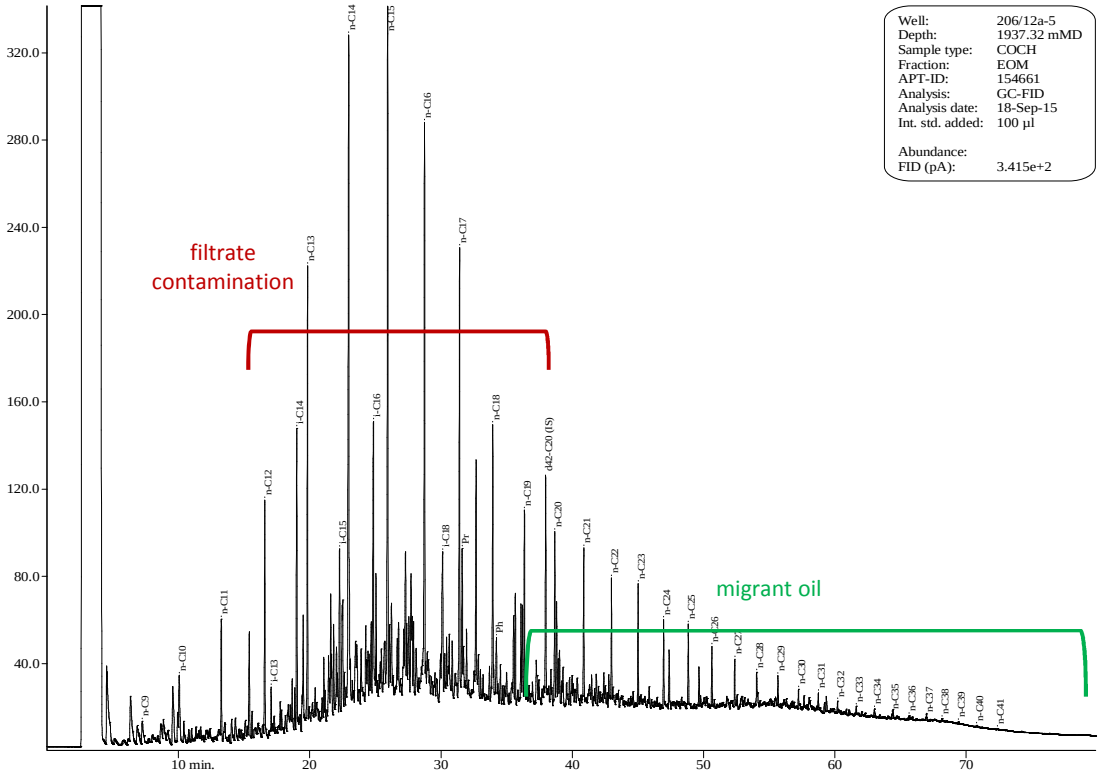


Clair Field: 206/12a-5, 1924.67m, COPL (K20)

FIGURE 5.3 Whole extract gas chromatograms - extracted cores

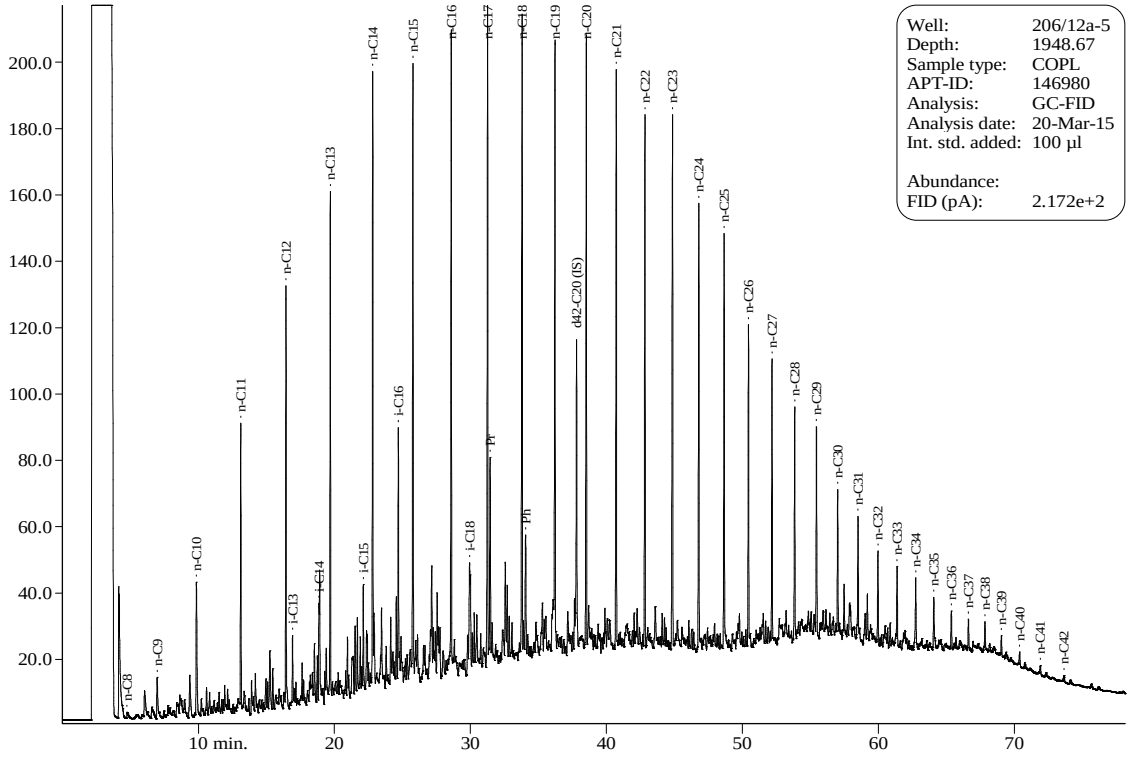


Clair Field: 206/12a-5, 1932.67m, COPL (K20)

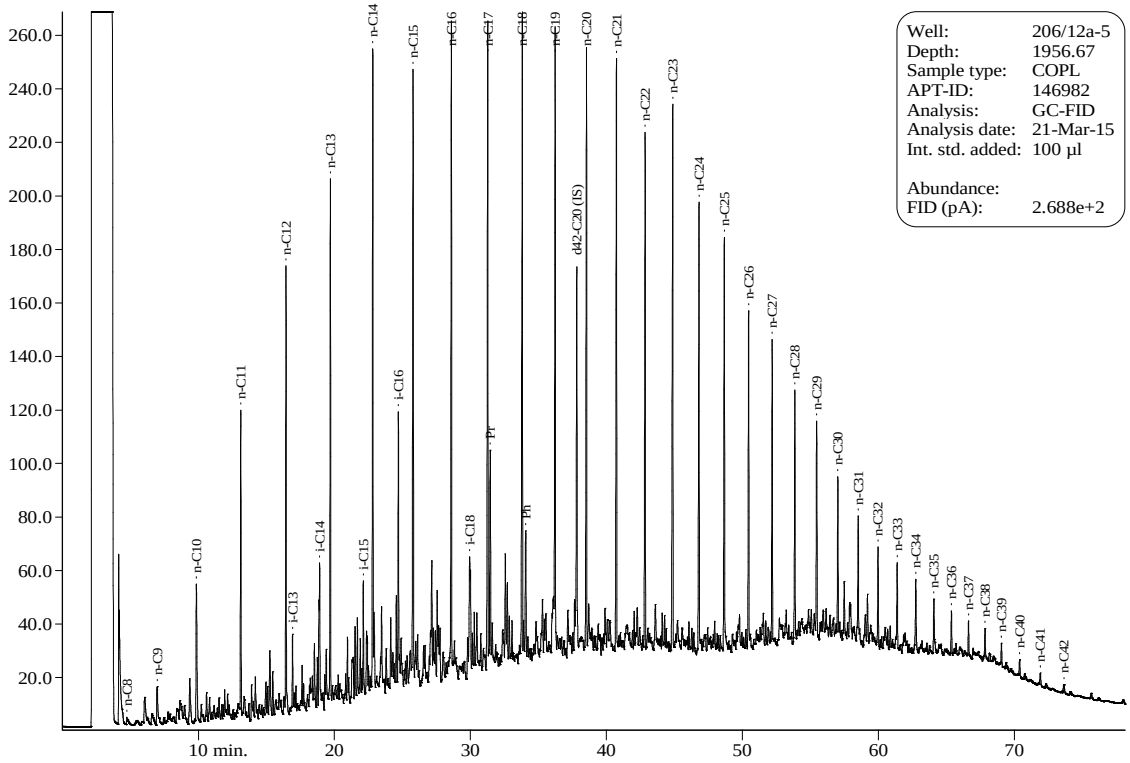


Clair Field: 206/12a-5, 1937.30m, Core (Rona)

FIGURE 5.3 Whole extract gas chromatograms - extracted cores

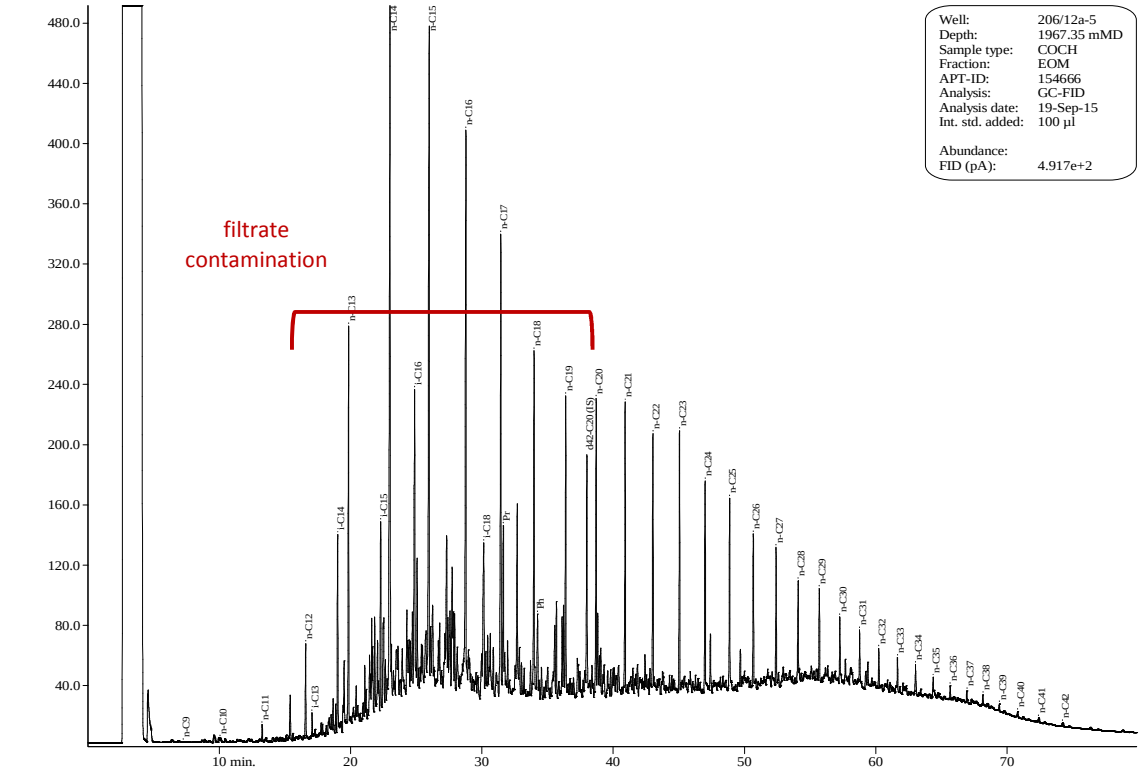


Clair Field: 206/12a-5, 1948.67m, COPL (Rona)

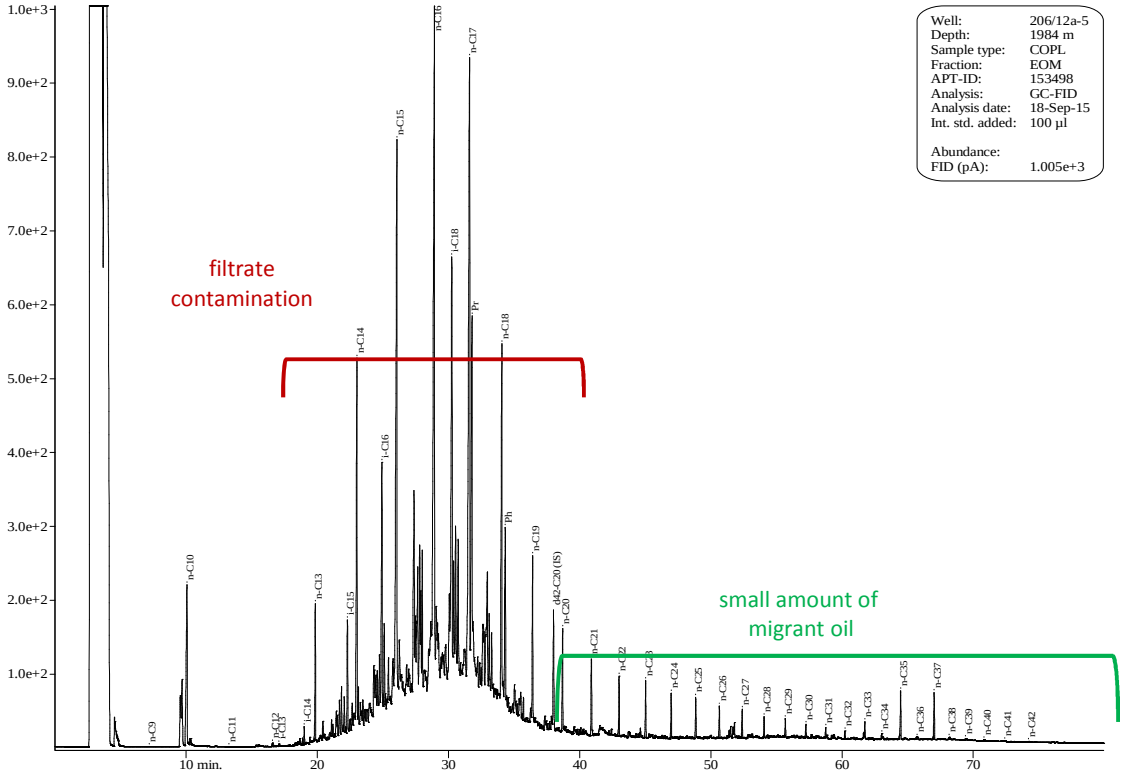


Clair Field: 206/12a-5, 1956.67m, COPL (Rona)

FIGURE 5.3 Whole extract gas chromatograms - extracted cores

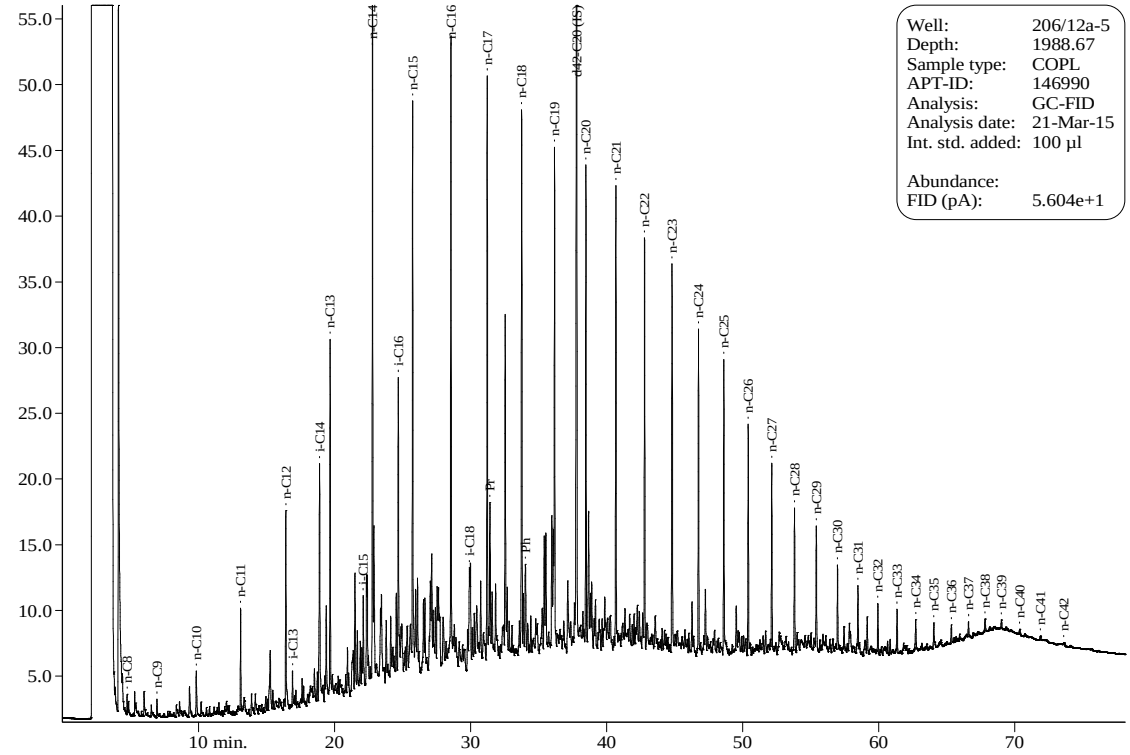


Clair Field: 206/12a-5, 1967.35m, Core (Rona)

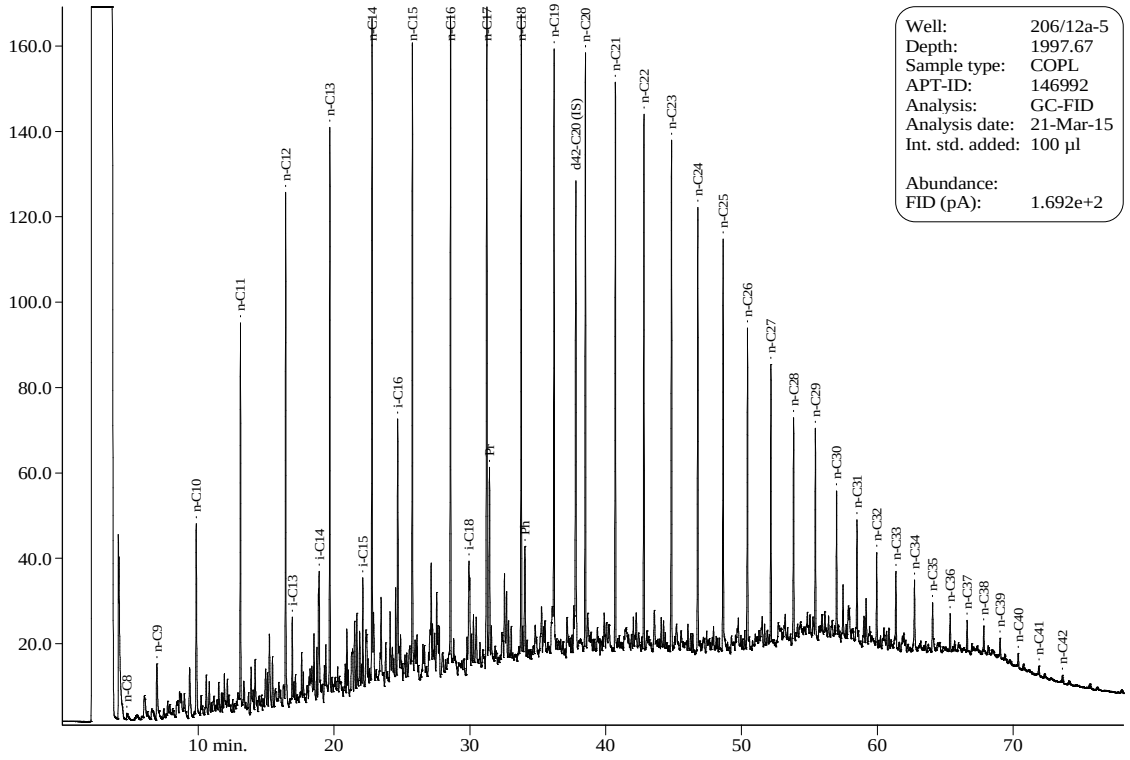


Clair Field: 206/12a-5, 1984.00m, SWC (LCG Unit III)

FIGURE 5.3 Whole extract gas chromatograms - extracted cores

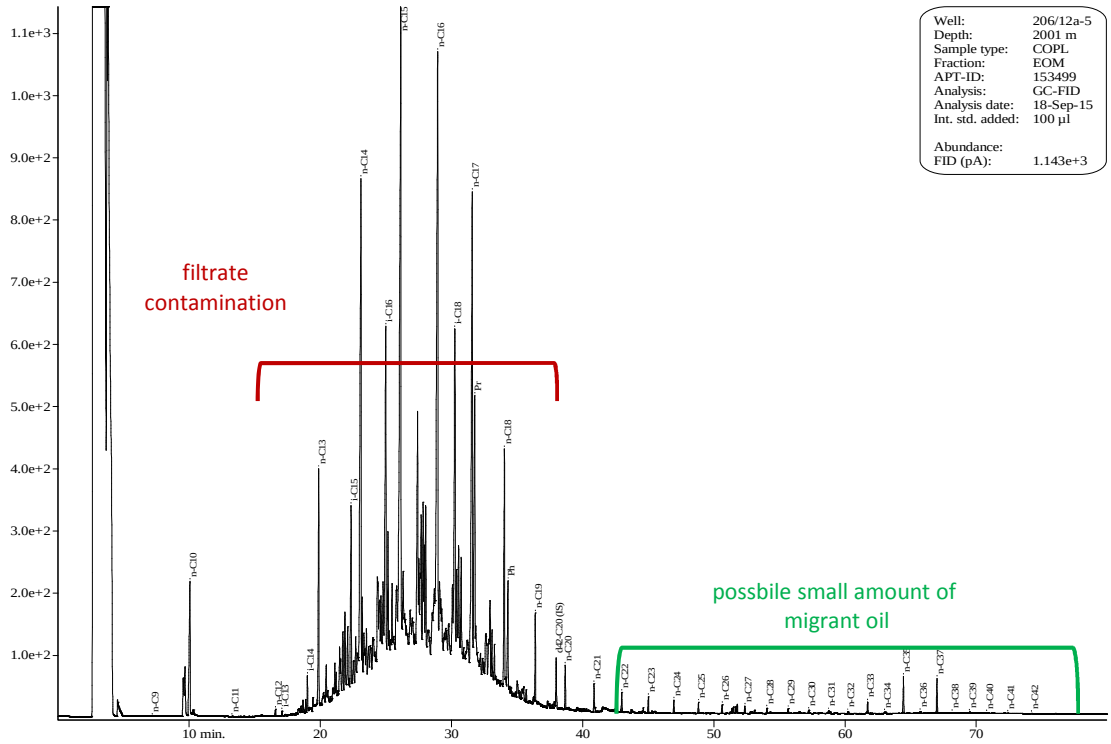


Clair Field: 206/12a-5, 1988.67m, COPL (LCG III)

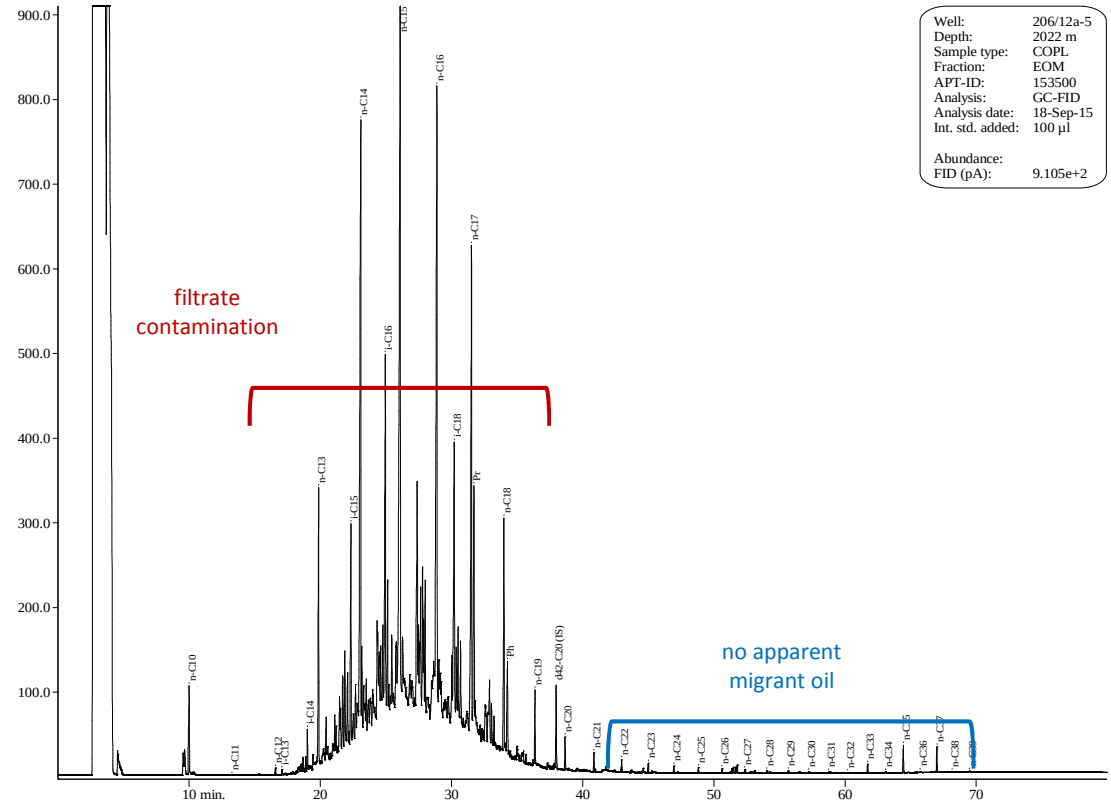


Clair Field: 206/12a-5, 1997.67m, COPL (LCG III)

FIGURE 5.3 Whole extract gas chromatograms - extracted cores

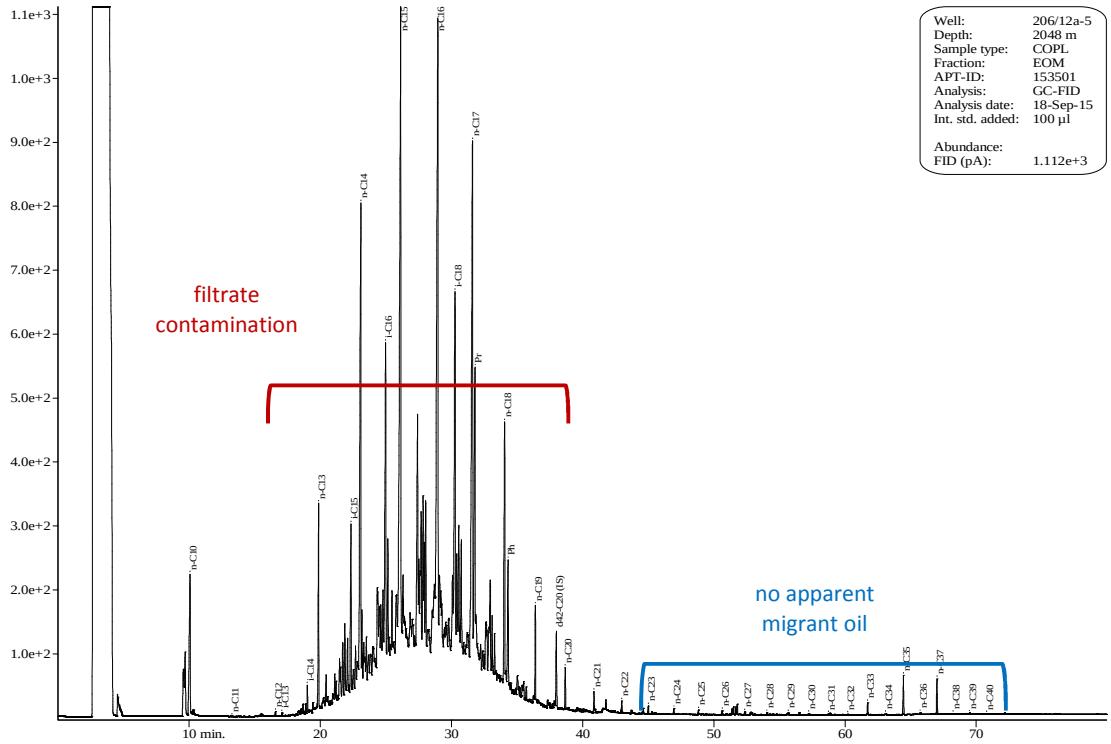


Clair Field: 206/12a-5, 2001.00m, SWC (LCG Unit III)



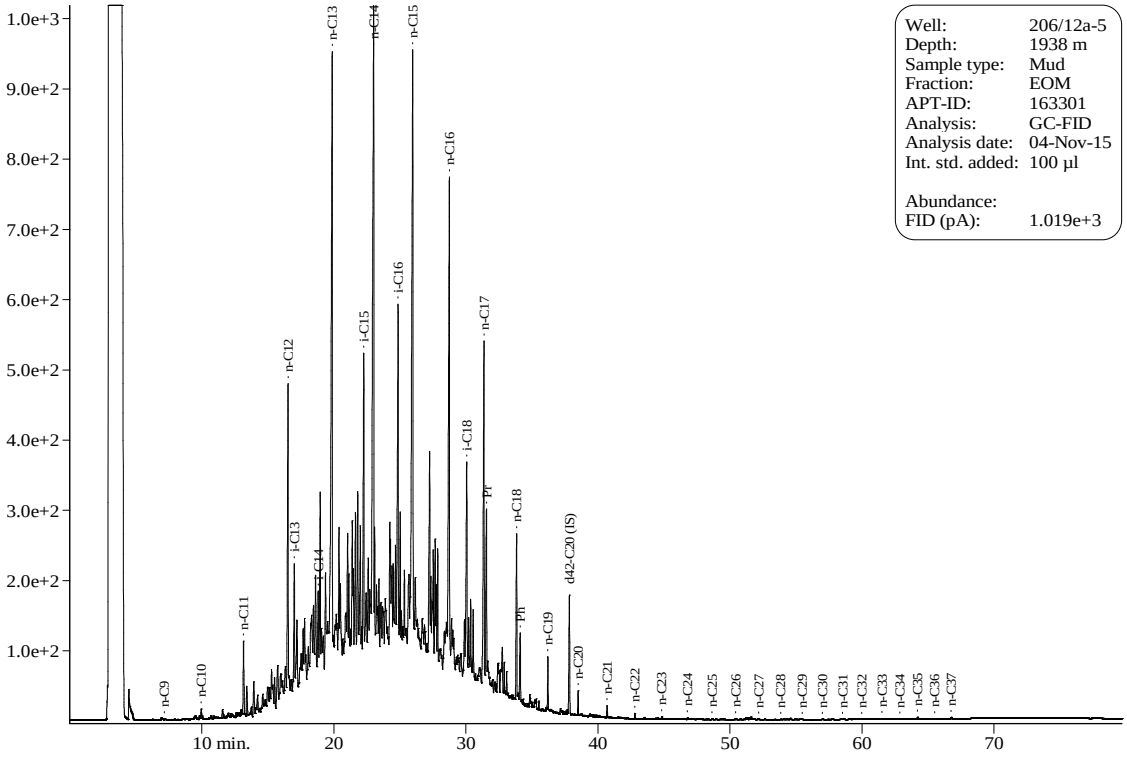
Clair Field: 206/12a-5, 2022.00m, SWC (LCG Unit II/I)

FIGURE 5.3 Whole extract gas chromatograms - extracted cores

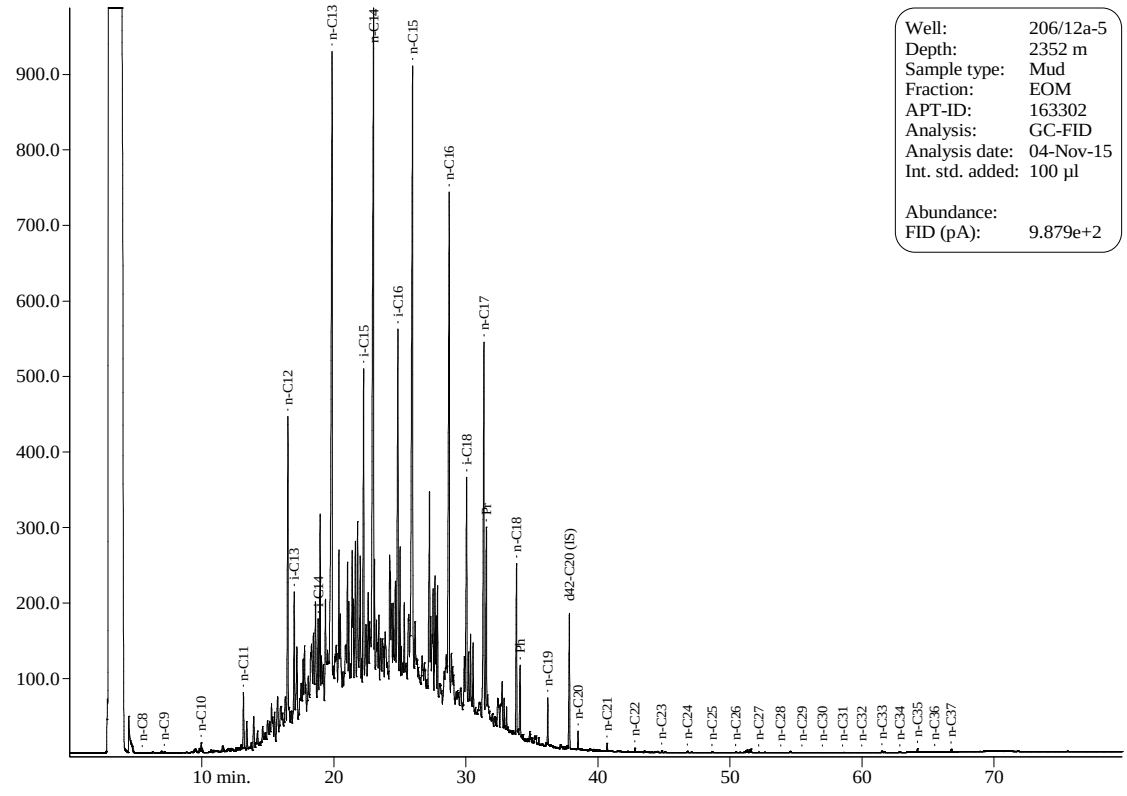


Clair Field: 206/12a-5, 2048.00m, SWC (LCG Unit II/I)

FIGURE 5.3 Whole extract gas chromatograms - extracted cores

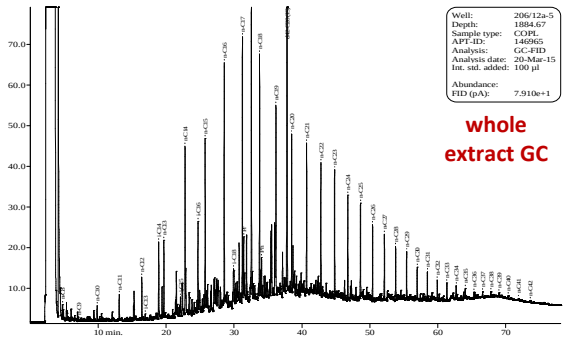


Clair Field: 206/12a-5, drilling mud, 1958.00m



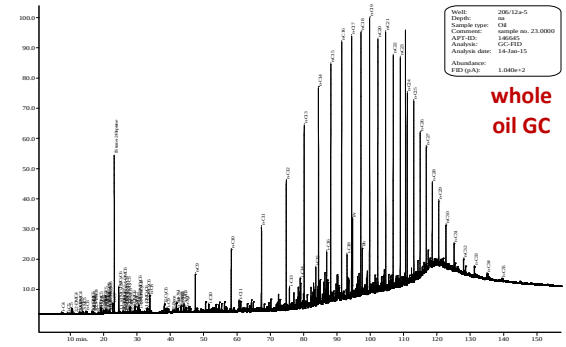
Clair Field: 206/12a-5, drilling mud, 2352.00m

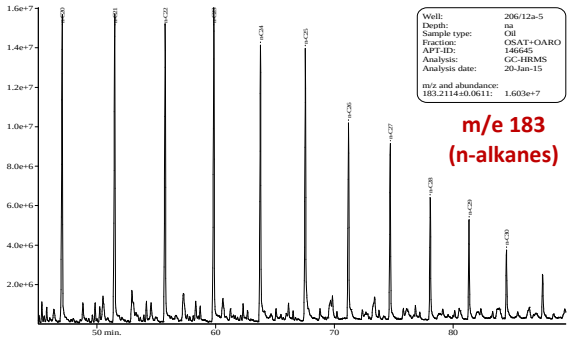
FIGURE 5.4 Whole extract gas chromatograms - muds



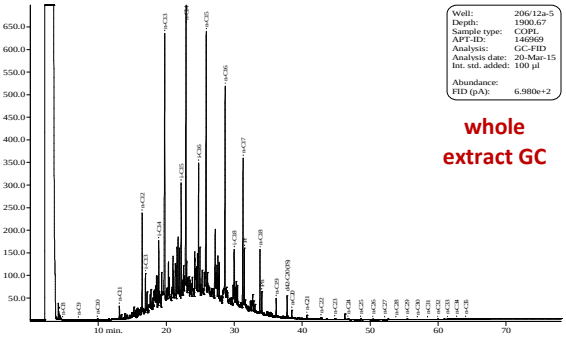
no GC-MS analysis

Clair Field: 206/12a-5, 1884.67m, COPL (K72)



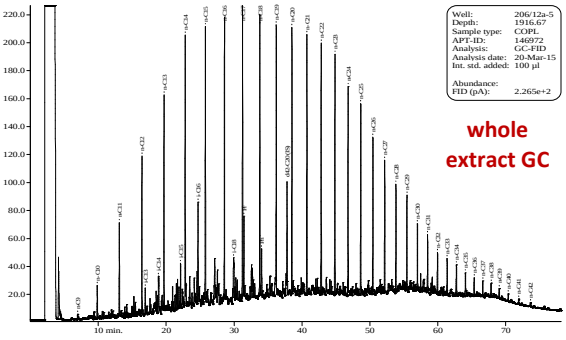


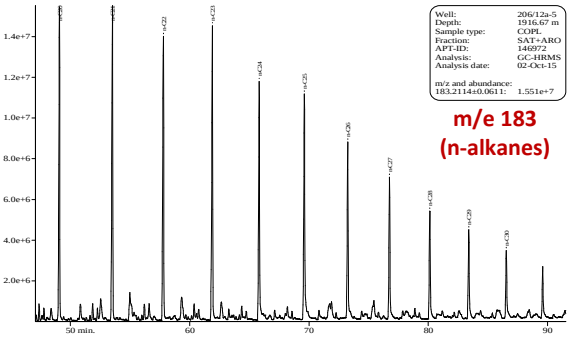
Clair Field: 206/12a-5, spun out oil, 1897.03m (K72)



no GC-MS analysis

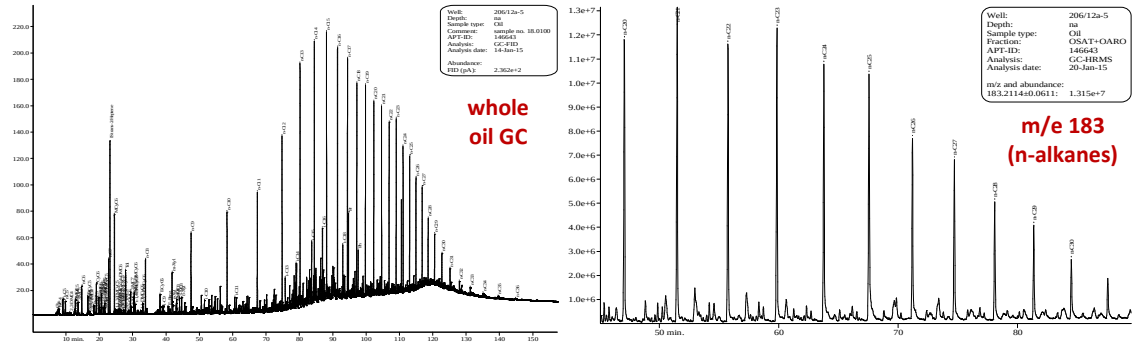
Clair Field: 206/12a-5, 1900.67m, COPL (K72)



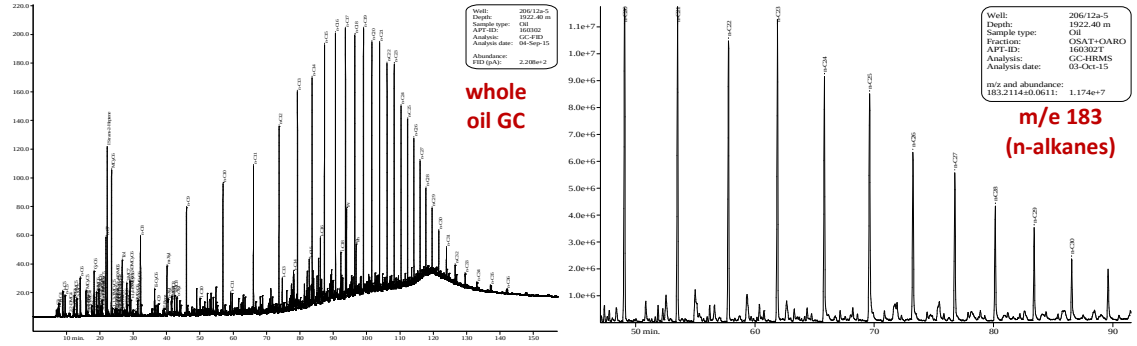


Clair Field: 206/12a-5, 1916.67m, COPL (K72)

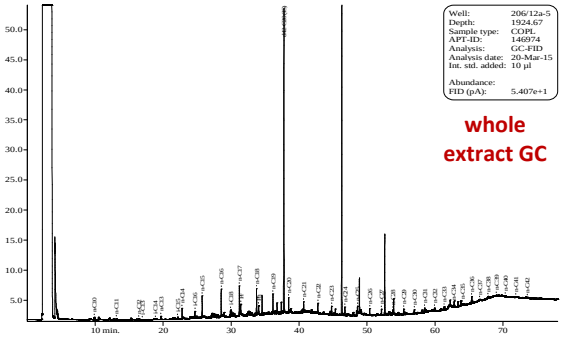
FIGURE 5.5 Comparison of whole oil/extract gas chromatograms and equivalent m/e 183 fragmentograms



Clair Field: 206/12a-5, spun out oil, 1922.08m (K30)

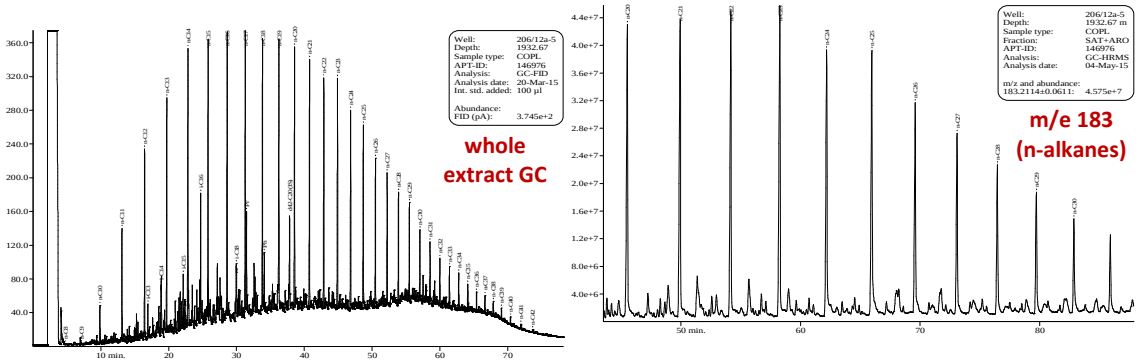


Clair Field: 206/12a-5, flowed oil, 1922.4m (K30)



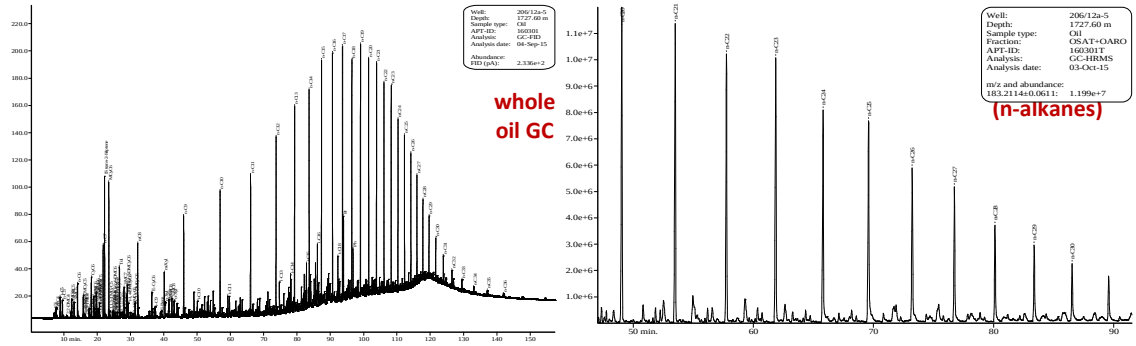
no GC-MS analysis

Clair Field: 206/12a-5, 1924.67m, COPL (K20)

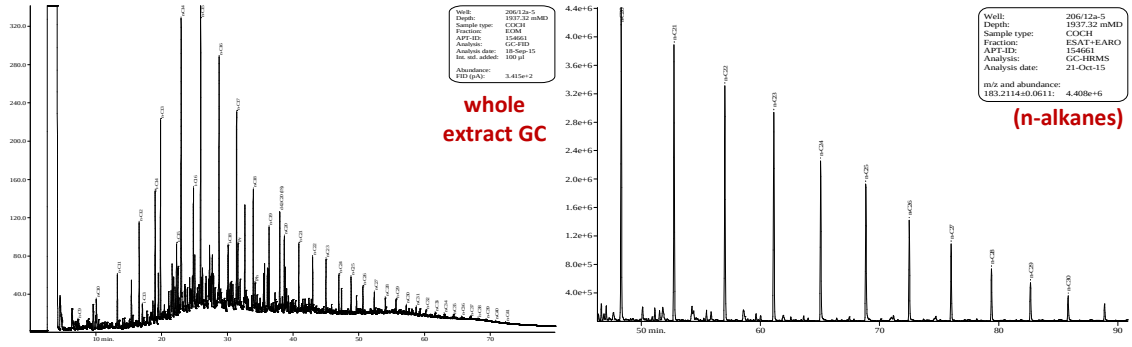


Clair Field: 206/12a-5, 1932.67m, COPL (K20)

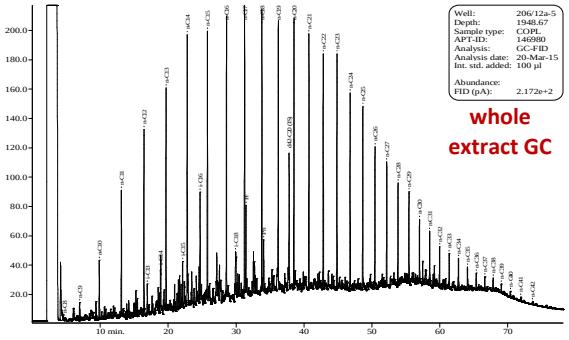
FIGURE 5.5 Comparison of whole oil/extract gas chromatograms and equivalent m/e 183 fragmentograms



Clair Field: 206/12a-5, DST, 1936-1963m (Rona)

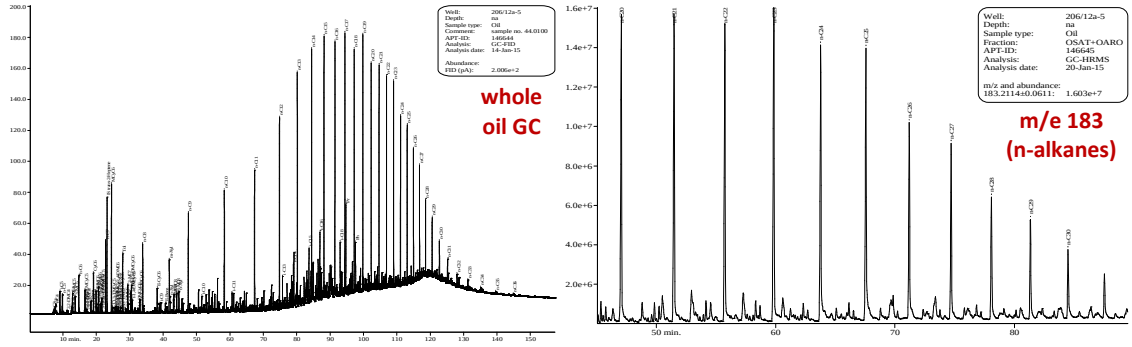


Clair Field: 206/12a-5, 1937.32m, Core (Rona)



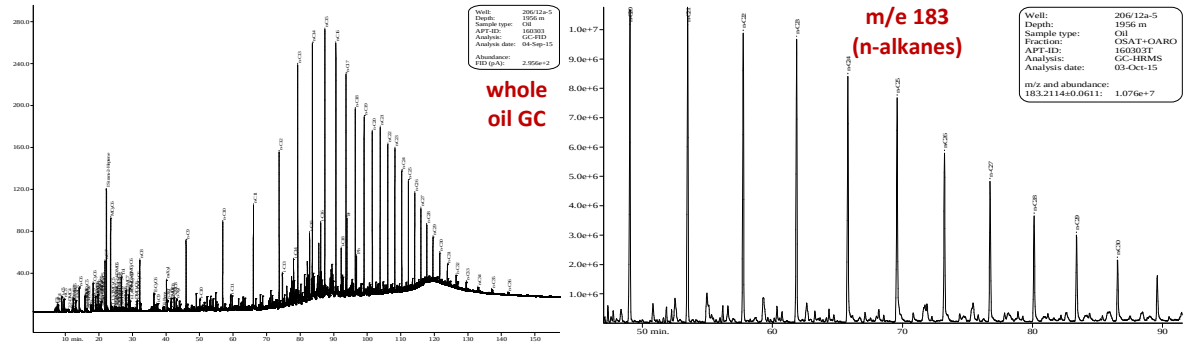
no GC-MS analysis

Clair Field: 206/12a-5, 1948.67m, COPL (Rona)

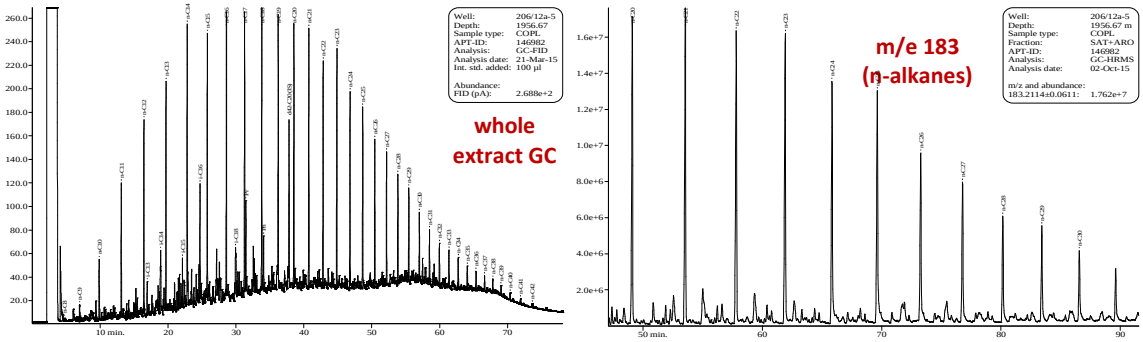


Clair Field: 206/12a-5, spun out oil, 1948.67m (Rona)

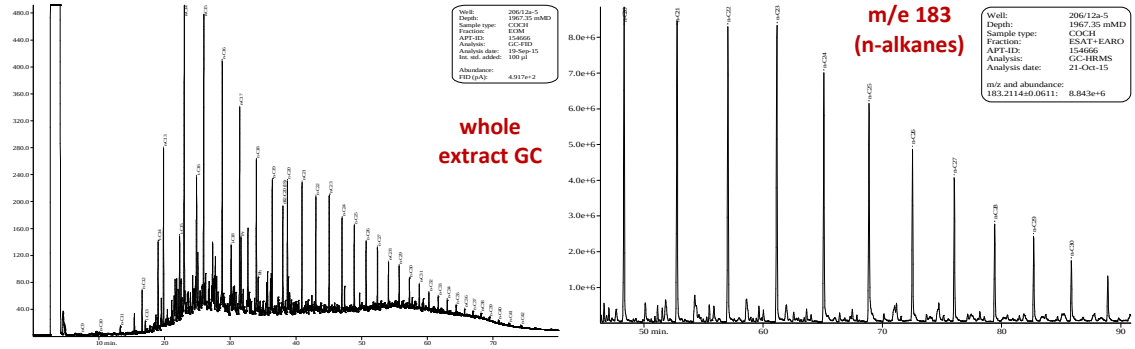
FIGURE 5.5 Comparison of whole oil/extract gas chromatograms and equivalent m/e 183 fragmentograms



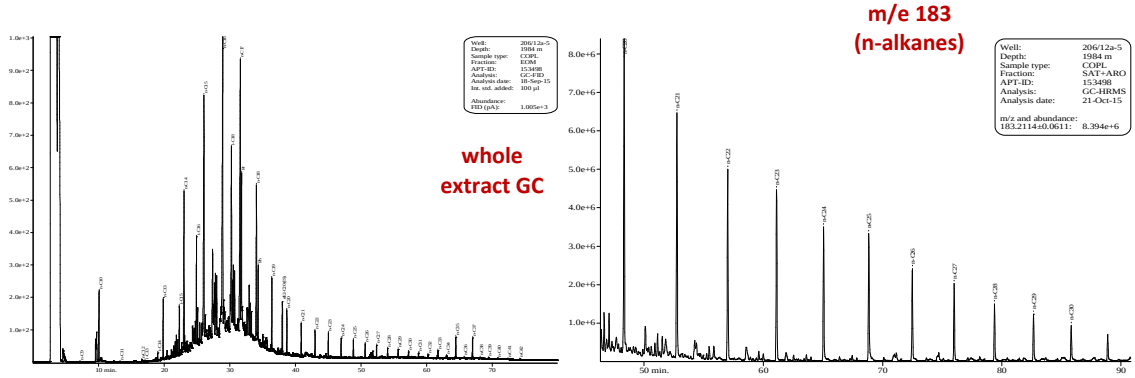
Clair Field: 206/12a-5, flowed oil, 1956.0m (Rona)



Clair Field: 206/12a-5, 1956.67m, COPL (Rona)

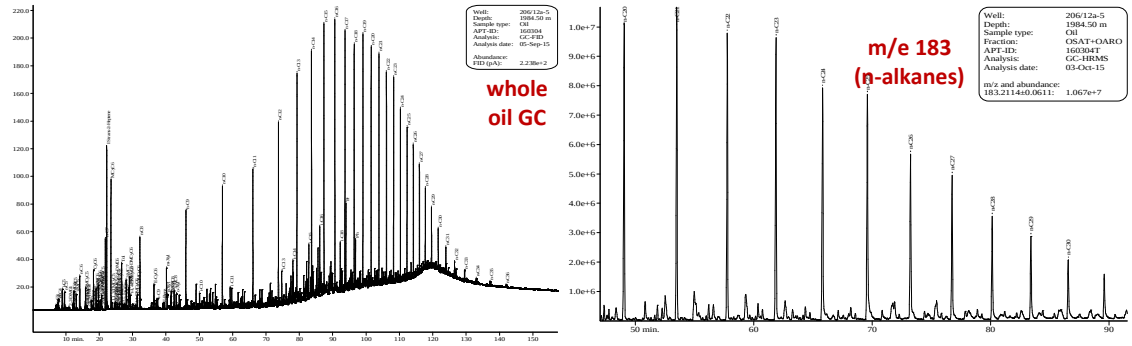


Clair Field: 206/12a-5, 1967.35m, Core (Rona)

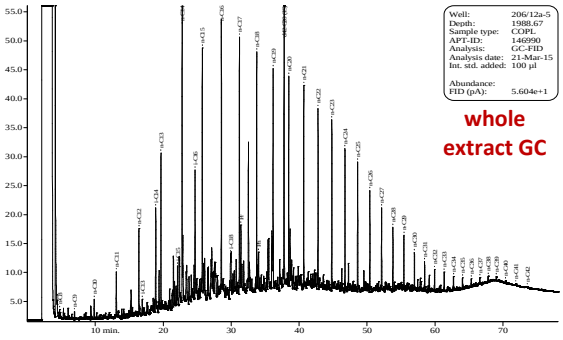


Clair Field: 206/12a-5, 1984.00m, SWC (LCG Unit III)

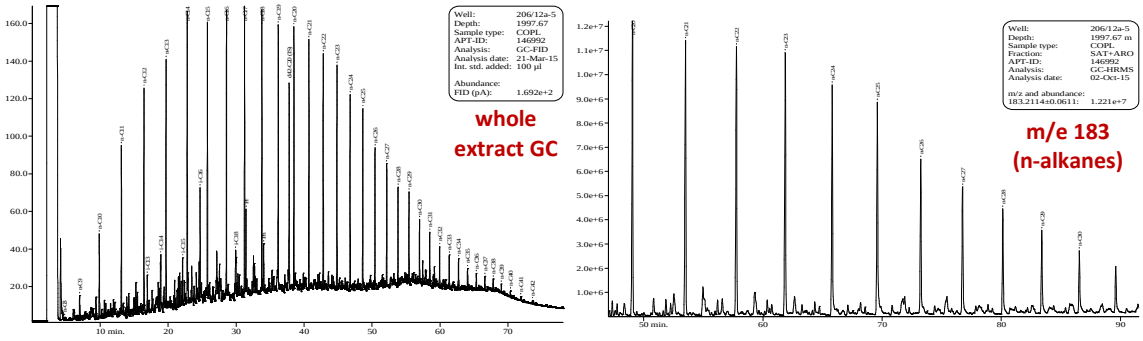
FIGURE 5.5 Comparison of whole oil/extract gas chromatograms and equivalent m/e 183 fragmentograms



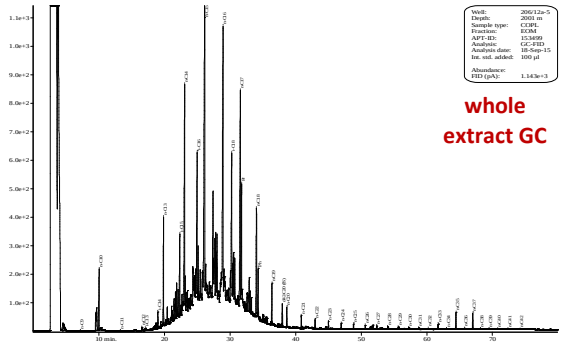
Clair Field: 206/12a-5, flowed oil, 1984.5m (LCG III)



Clair Field: 206/12a-5, 1988.67m, COPL (LCG III)

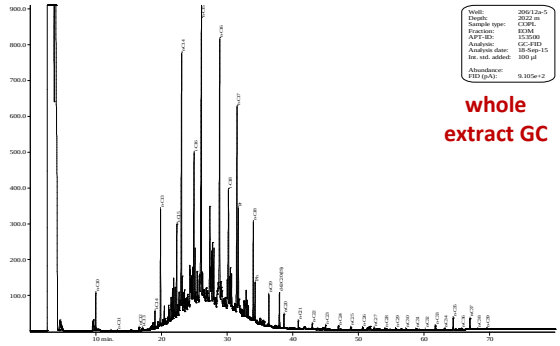


Clair Field: 206/12a-5, 1997.67m, COPL (LCG III)

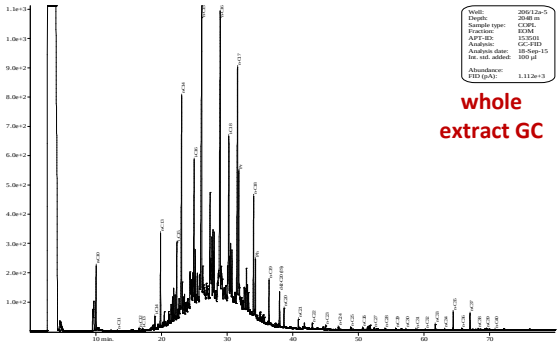


Clair Field: 206/12a-5, 2001.00m, SWC (LCG Unit III)

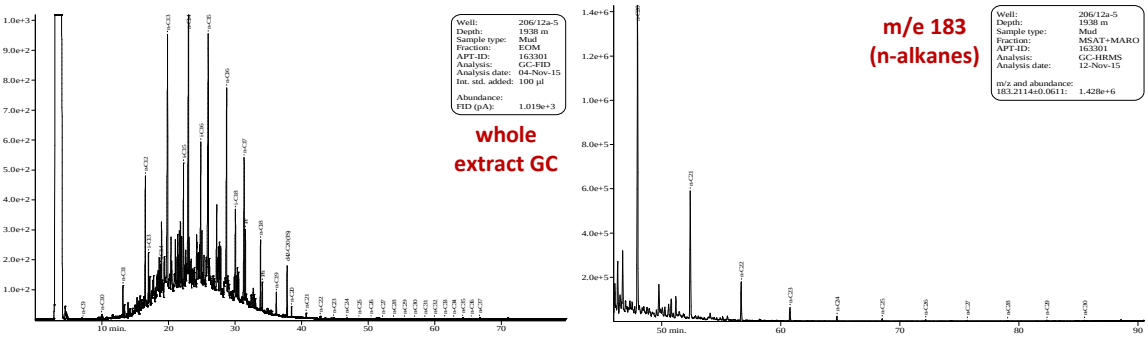
FIGURE 5.5 Comparison of whole oil/extract gas chromatograms and equivalent m/e 183 fragmentograms



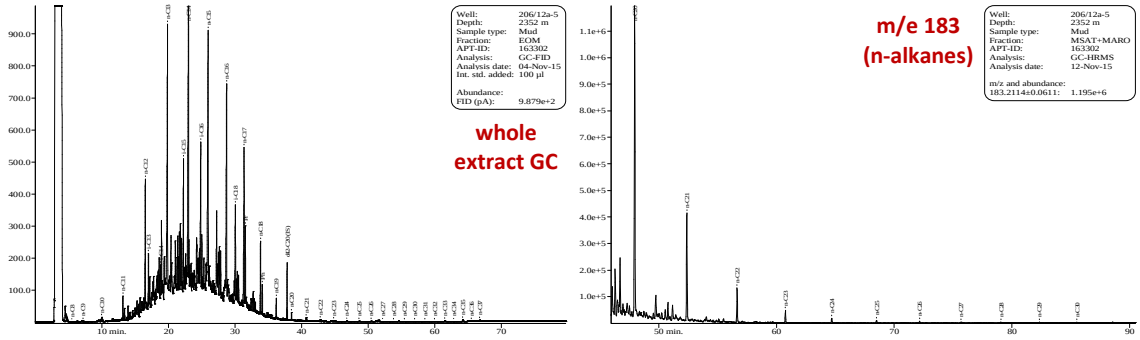
Clair Field: 206/12a-5, 2022.00m, SWC (LCG Unit II/I)



Clair Field: 206/12a-5, 2048.00m, SWC (LCG Unit II/I)



Clair Field: 206/12a-5, drilling mud, 1958.00m



Clair Field: 206/12a-5, drilling mud, 2352.00m

FIGURE 5.5 Comparison of whole oil/extract gas chromatograms and equivalent m/e 183 fragmentograms

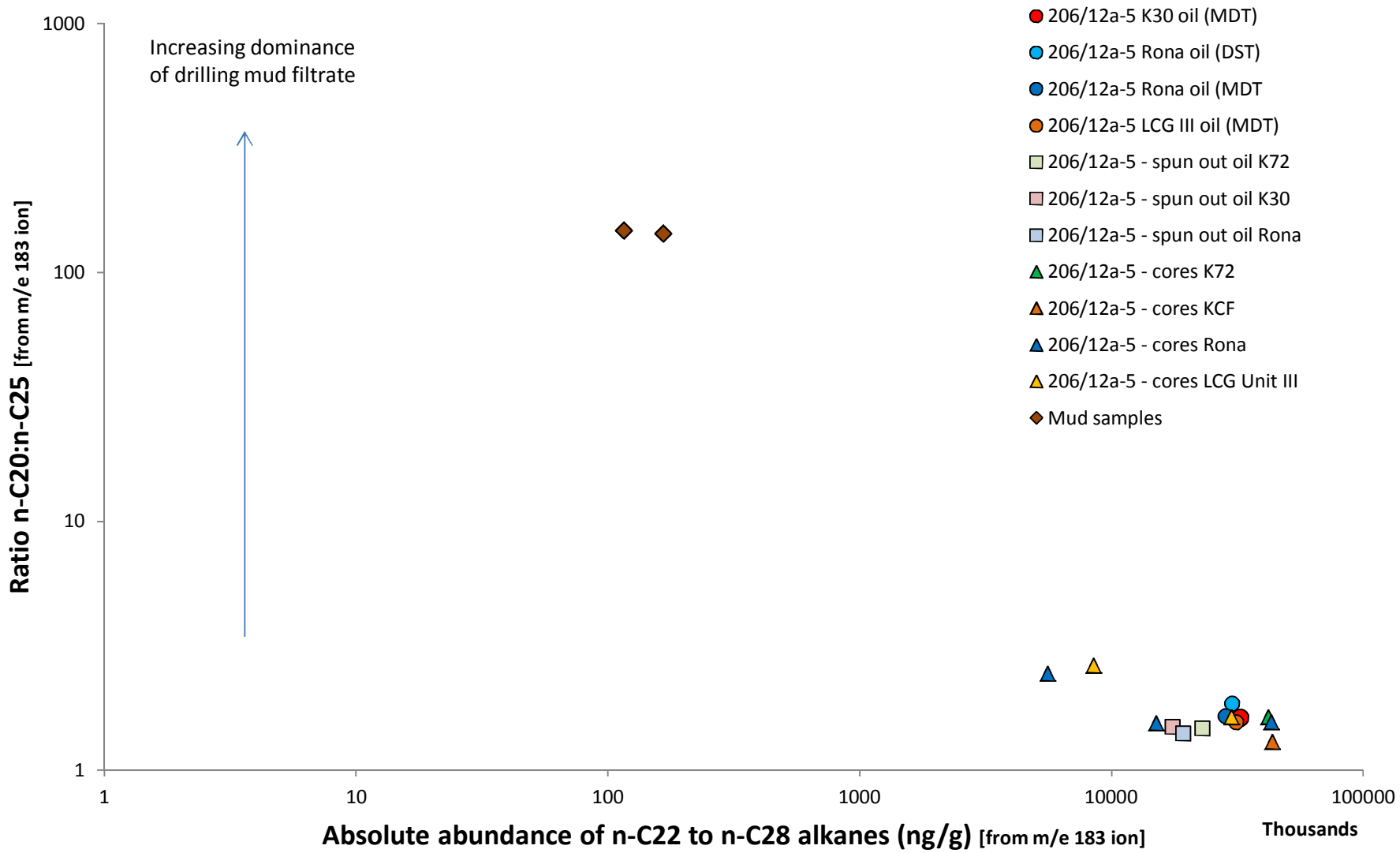


FIGURE 5.6 Cross plot of n-C20:n-C25 ratio against absolute abundance of n-C22 to n-C28 normal alkanes (m/e 183 data)

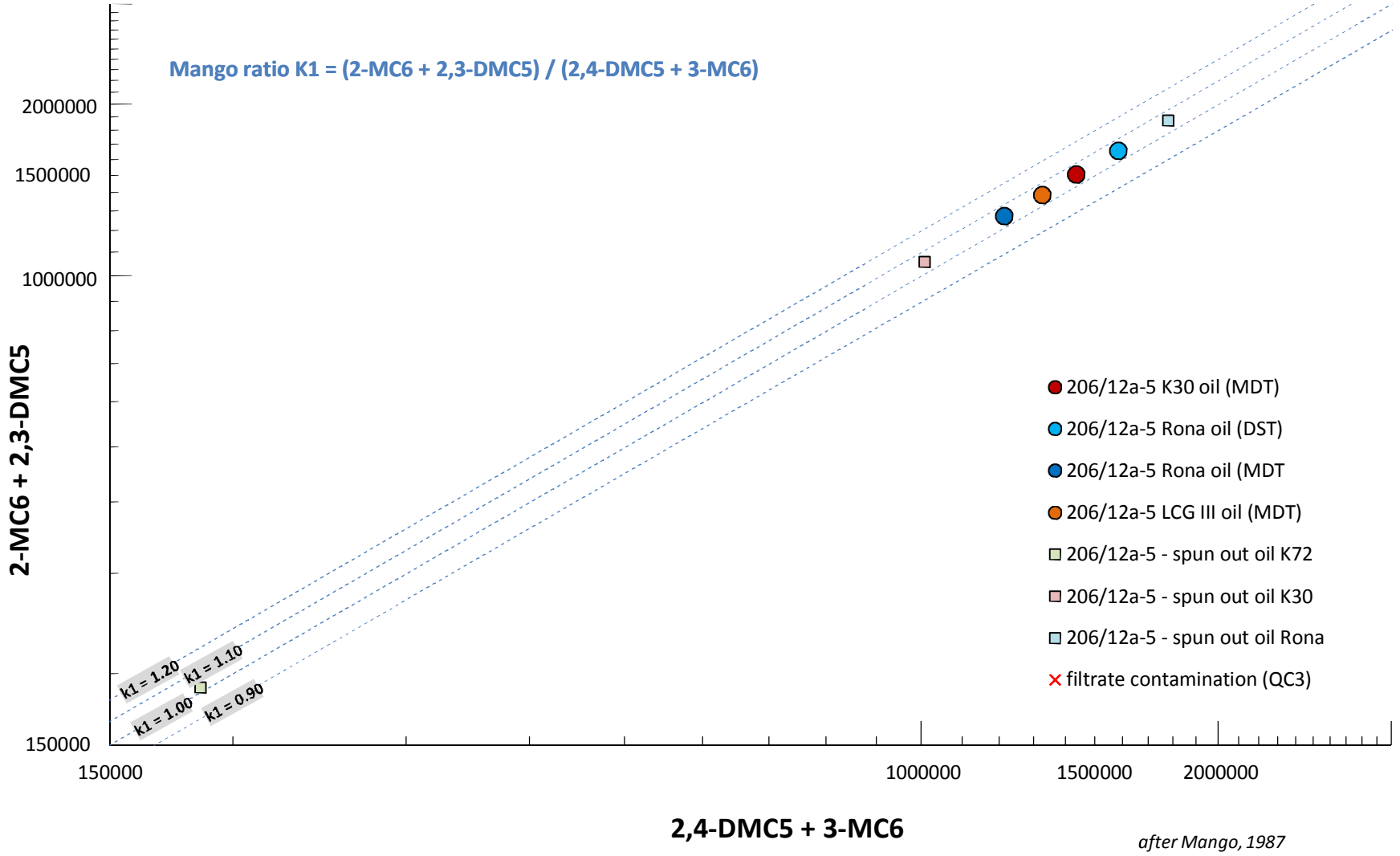


FIGURE 5.7a Cross plot of Mango parameters (k1): oils

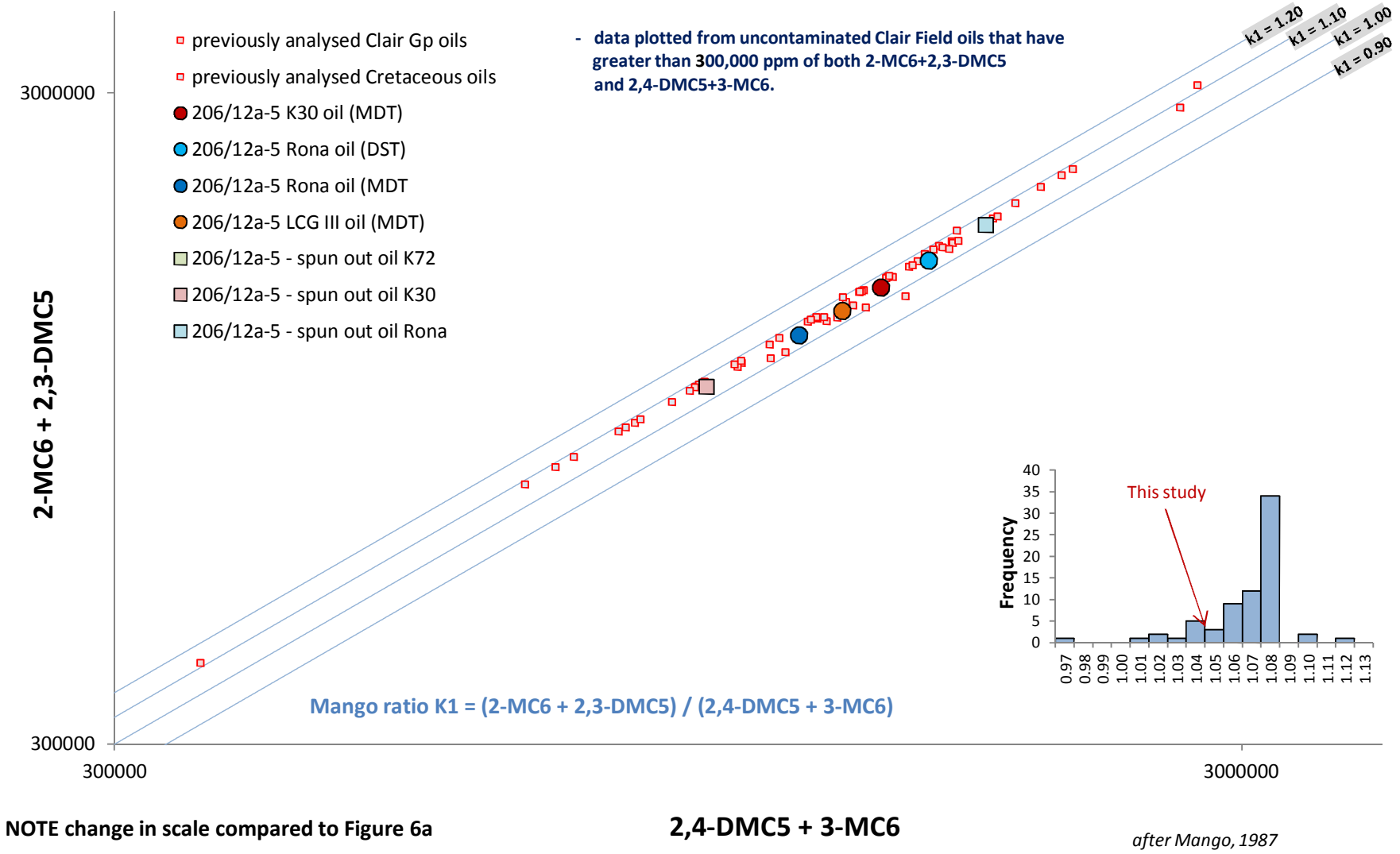


FIGURE 5.7b Cross plot of Mango parameters (k1) : comparison with Greater Clair Field oils

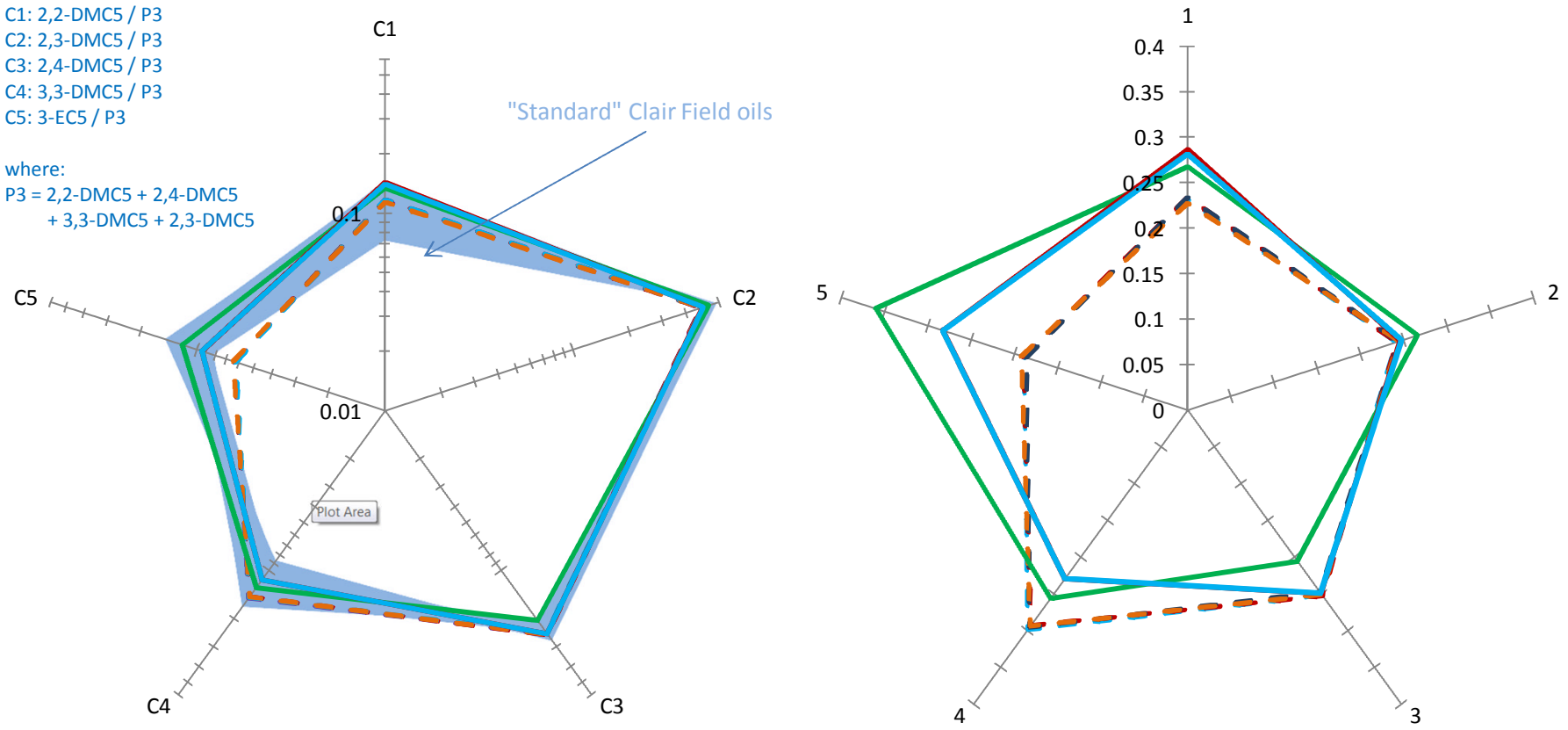
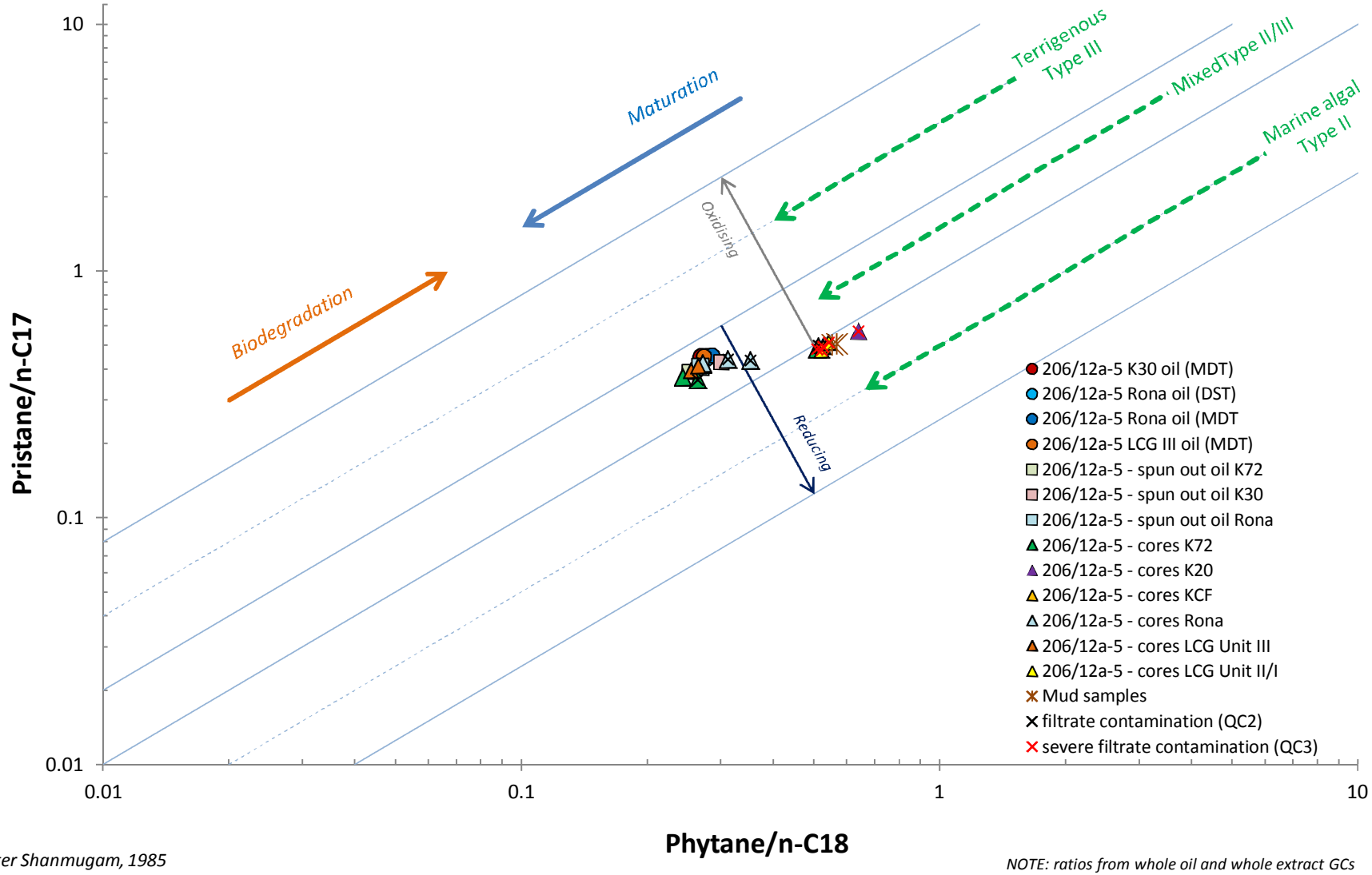


FIGURE 5.8 Halpern C7 oil correlation star diagram (C7-OCSD) (after Halpern, 1995): oils



after Shanmugam, 1985

NOTE: ratios from whole oil and whole extract GCs

FIGURE 5.9a Cross plot of isoprenoids/normal alkane ratios

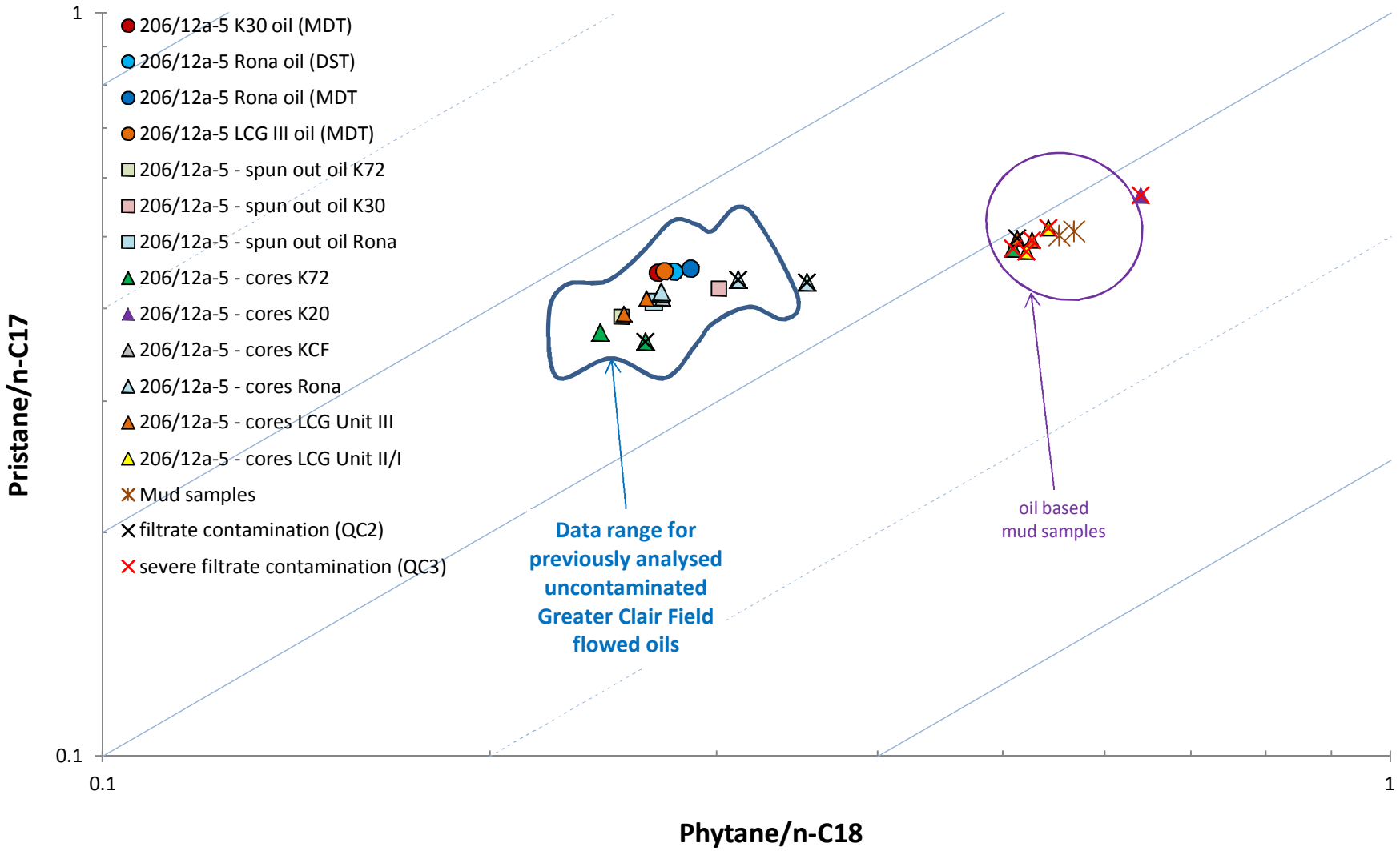


FIGURE 5.9b Cross plot of isoprenoids/normal alkane ratios - expanded scale

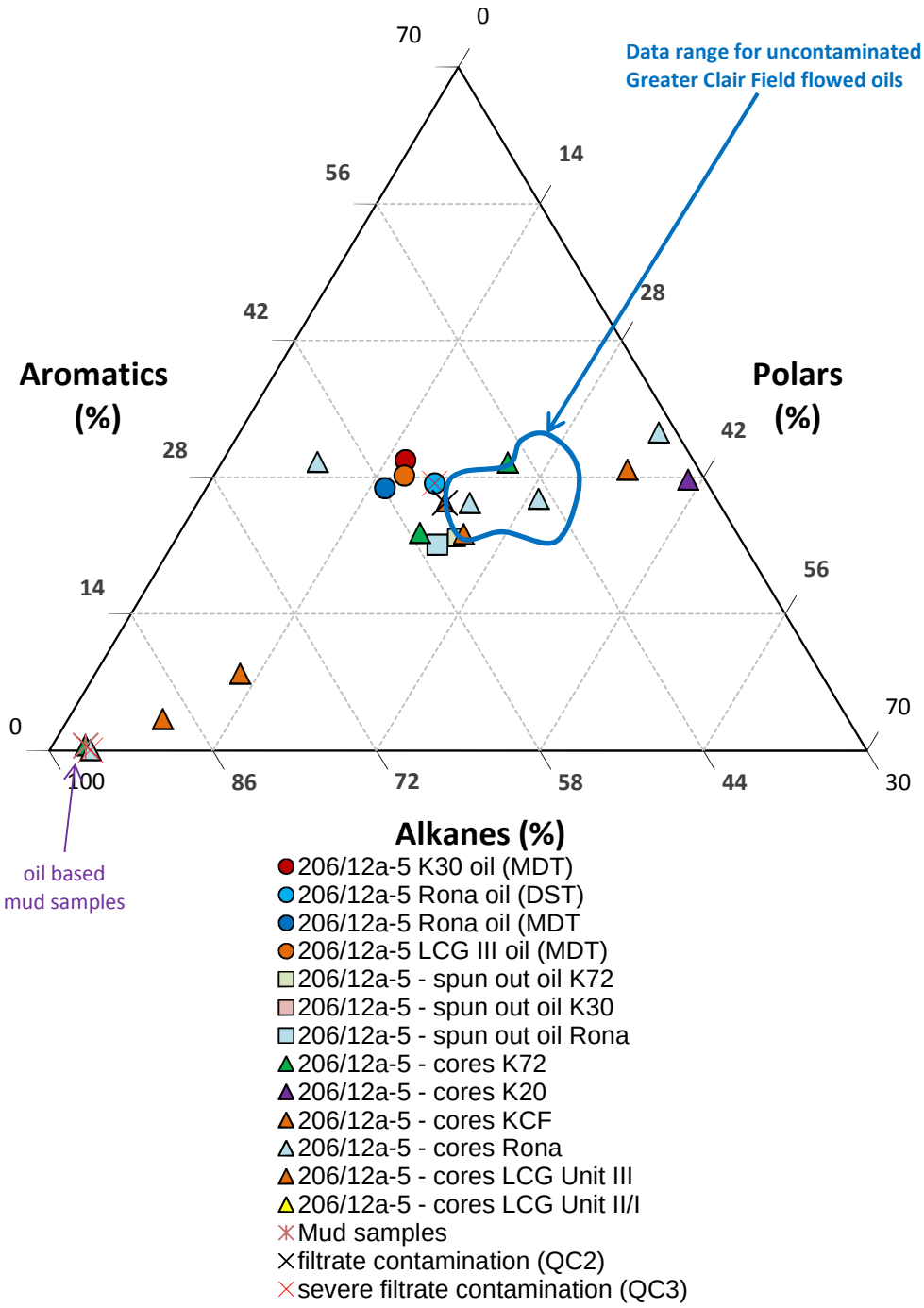


FIGURE 5.10 Ternary plot of fractionation data

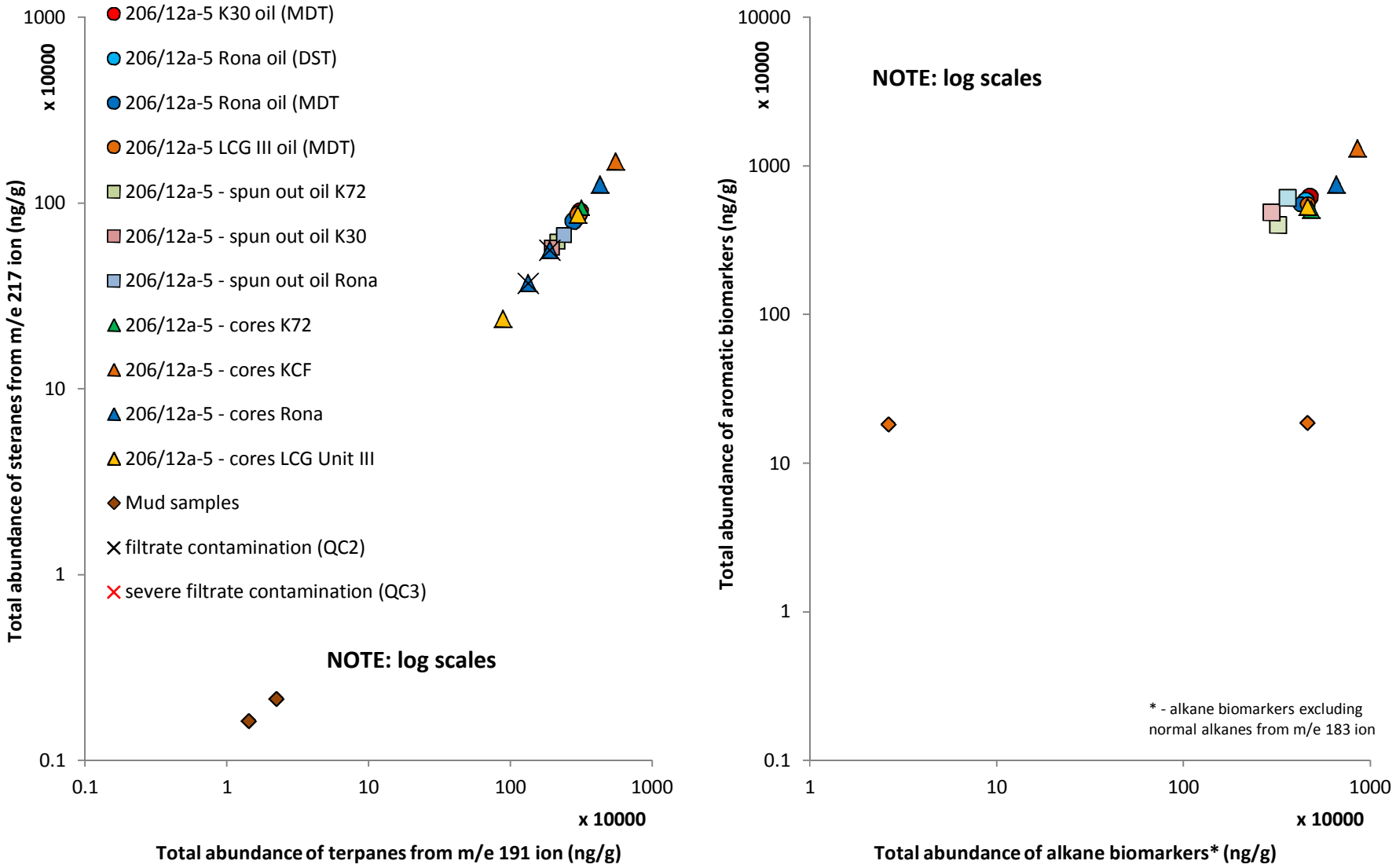


FIGURE 5.11 Cross plots of absolute abundances of biomarkers

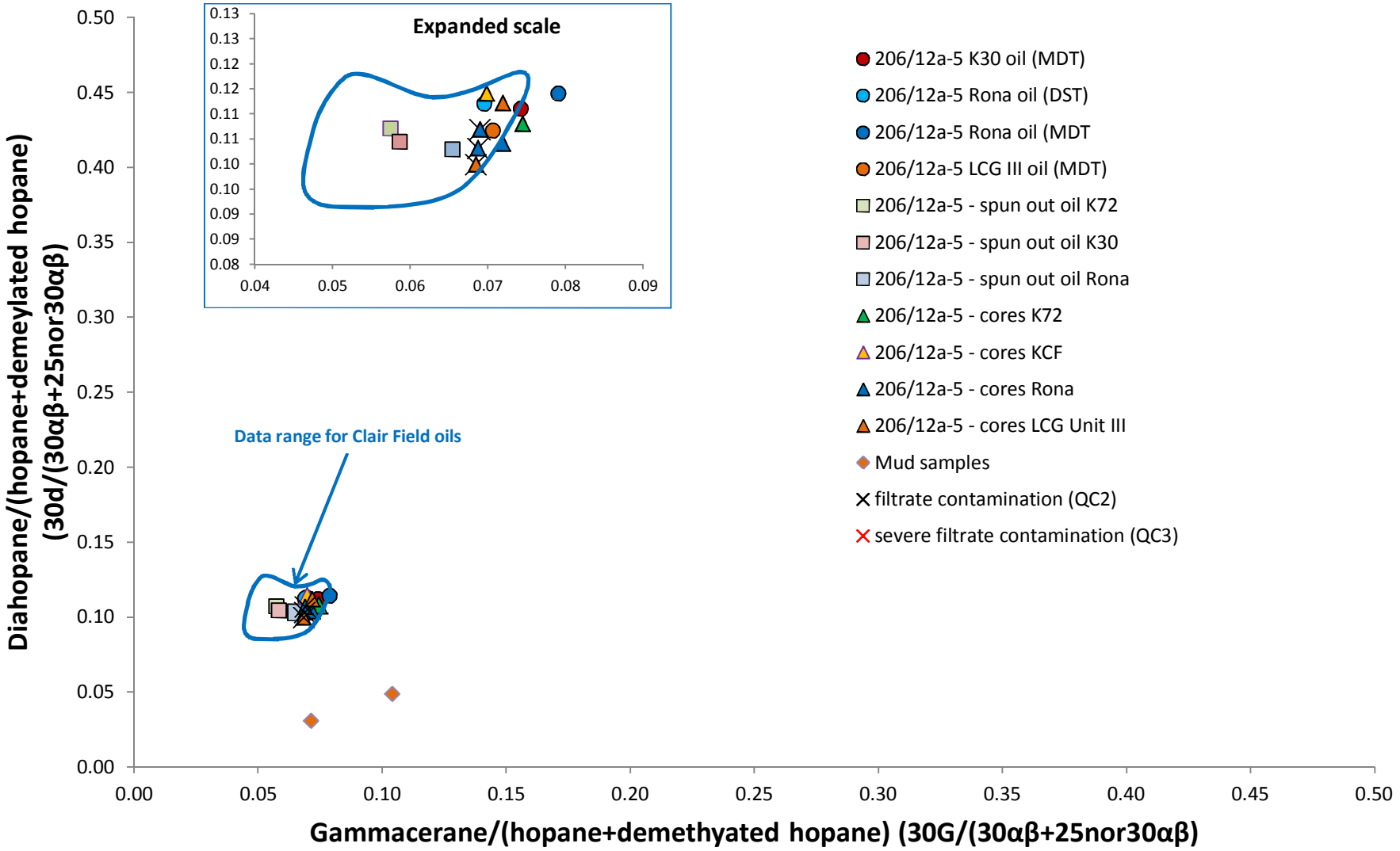


FIGURE 5.12 Cross plot of diahopane/(hopane+demethylated hopane) against gammacerane/(hopane+demethylated hopane) biomarker ratios

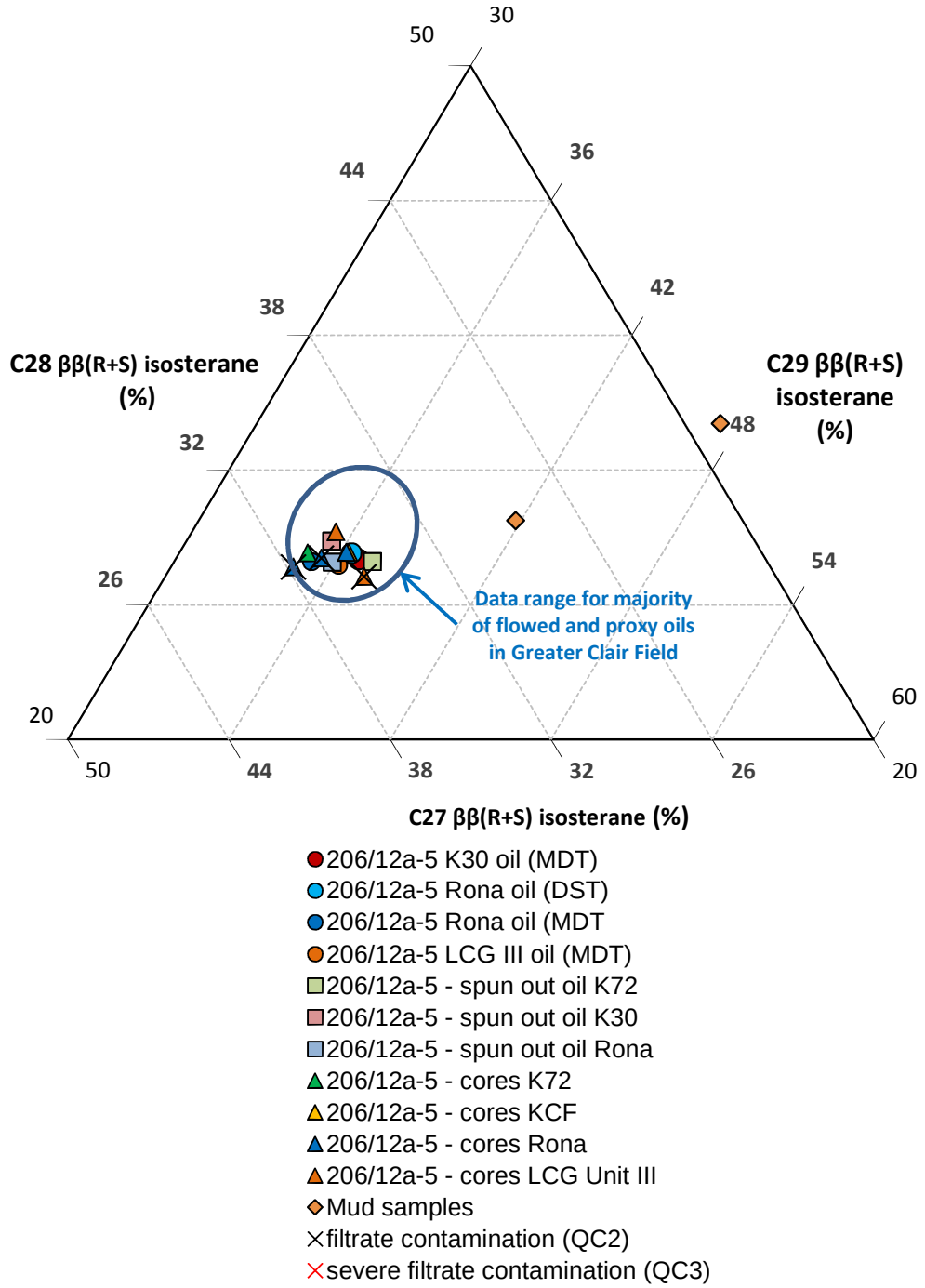


FIGURE 5.13 Ternary plot of C27-C29 αββ(R+S) isosteranes

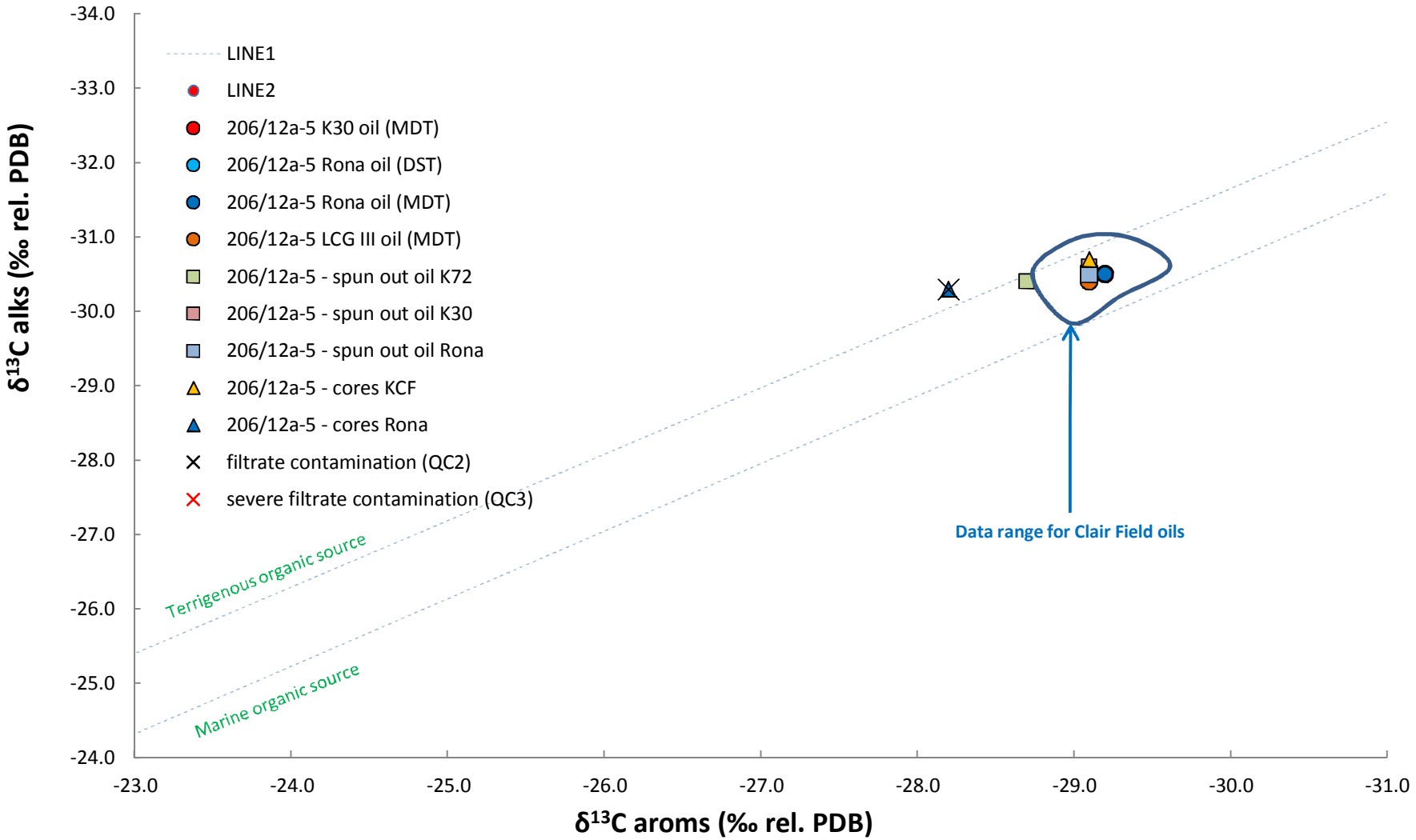


FIGURE 5.14 Cross plot of carbon isotope ratios

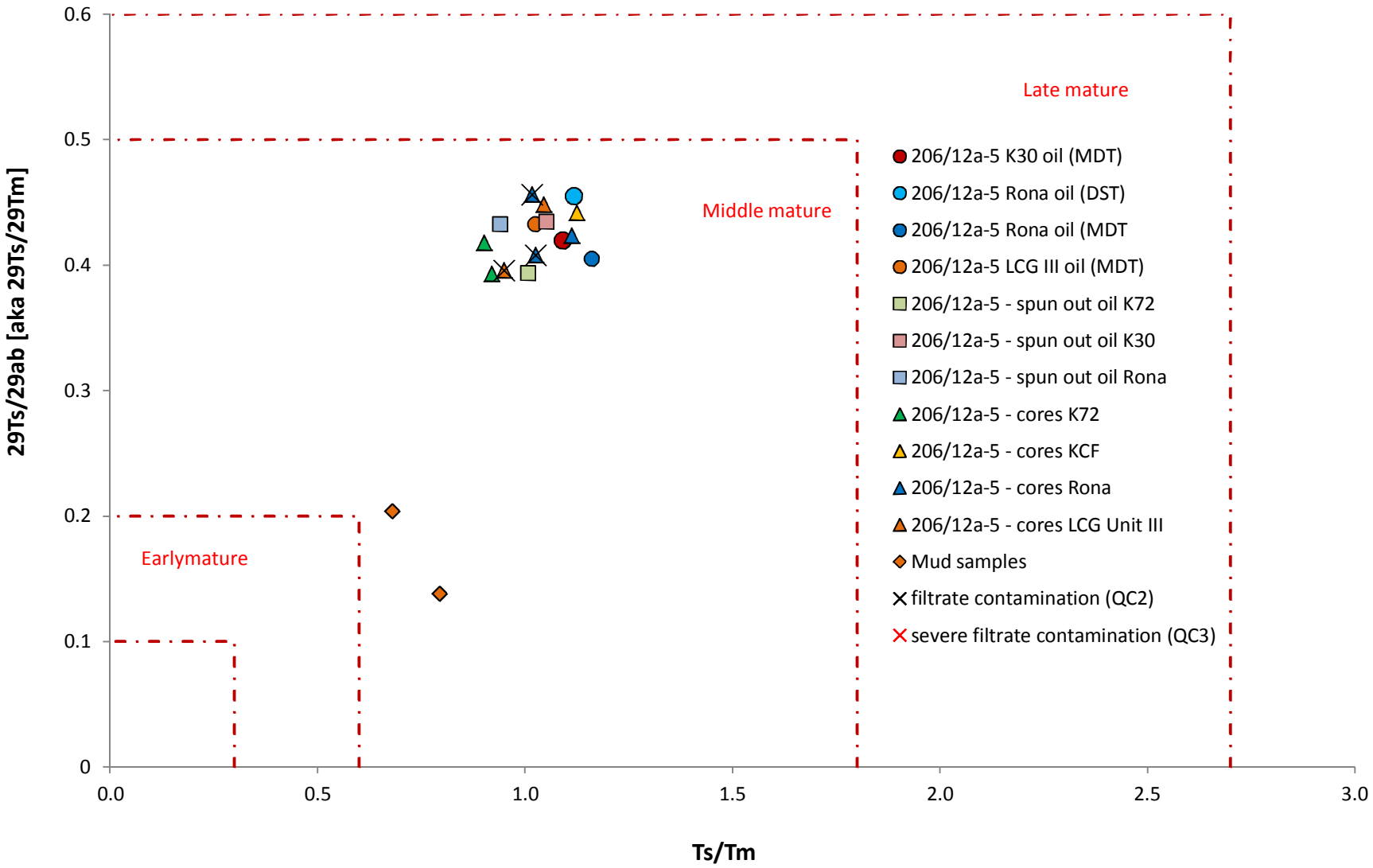


FIGURE 5.15 Cross plot of 29Ts/29ab ratio against Ts/Tm ratio

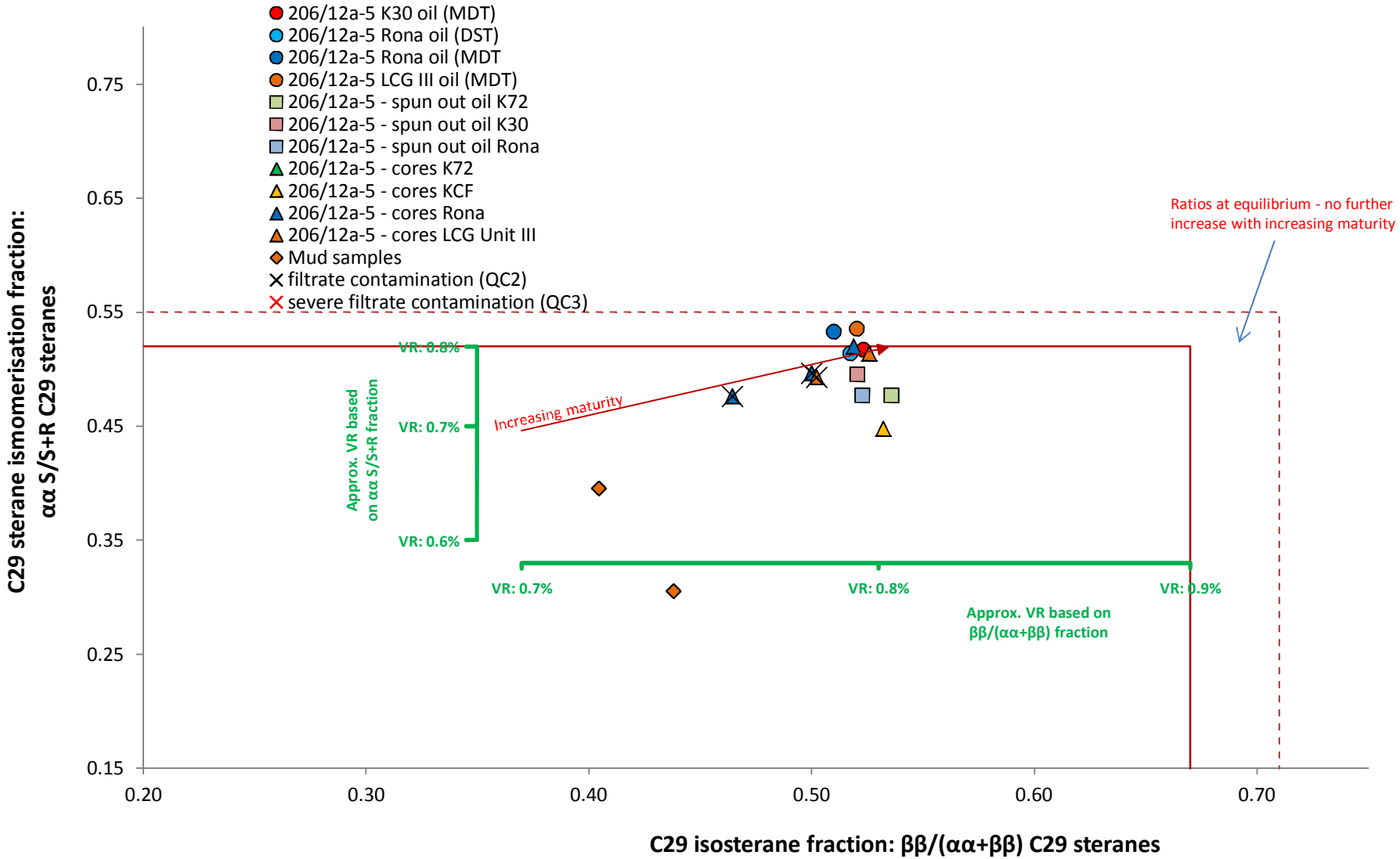


FIGURE 5.16 Cross plot of C29 sterane isomerisation against C29 isosterane fraction

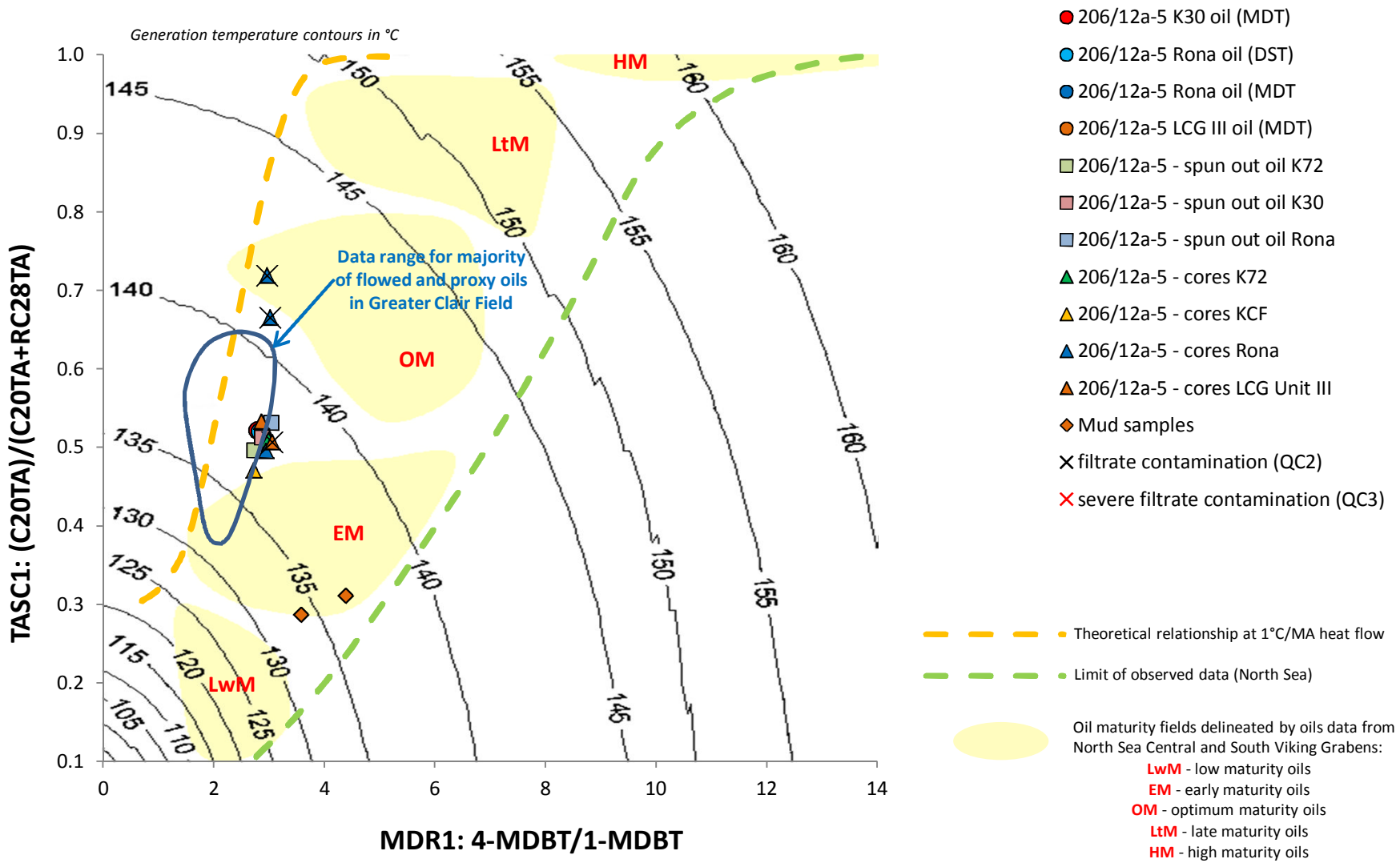


FIGURE 5.17 Cross plot of MDR1 against TASC1 with interpreted temperature of oil generation

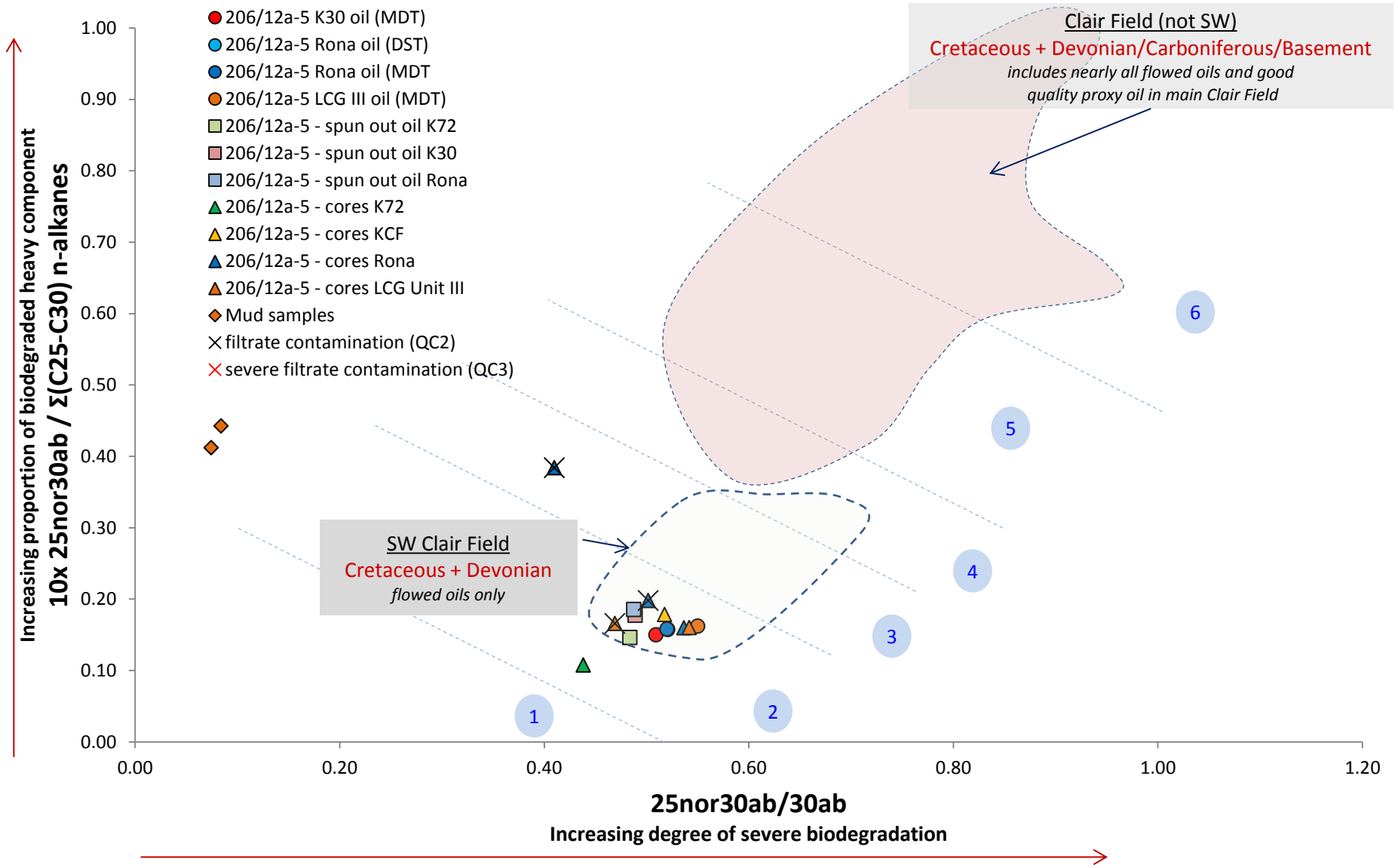


FIGURE 5.18 Cross plot of demethylated hopane/Σ(C25-C30) n-alkanes against demethylated hopane/hopane biomarker ratios

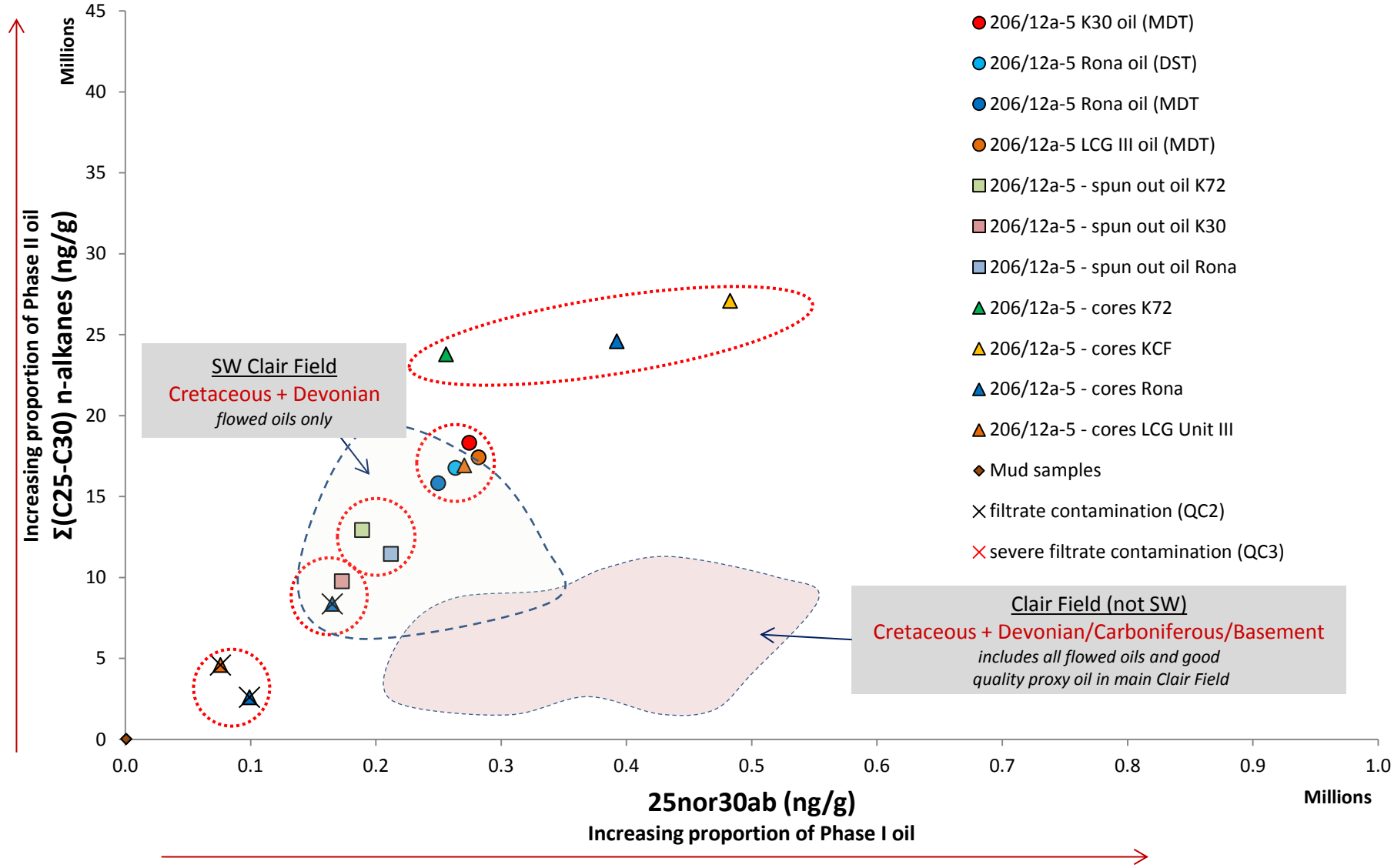
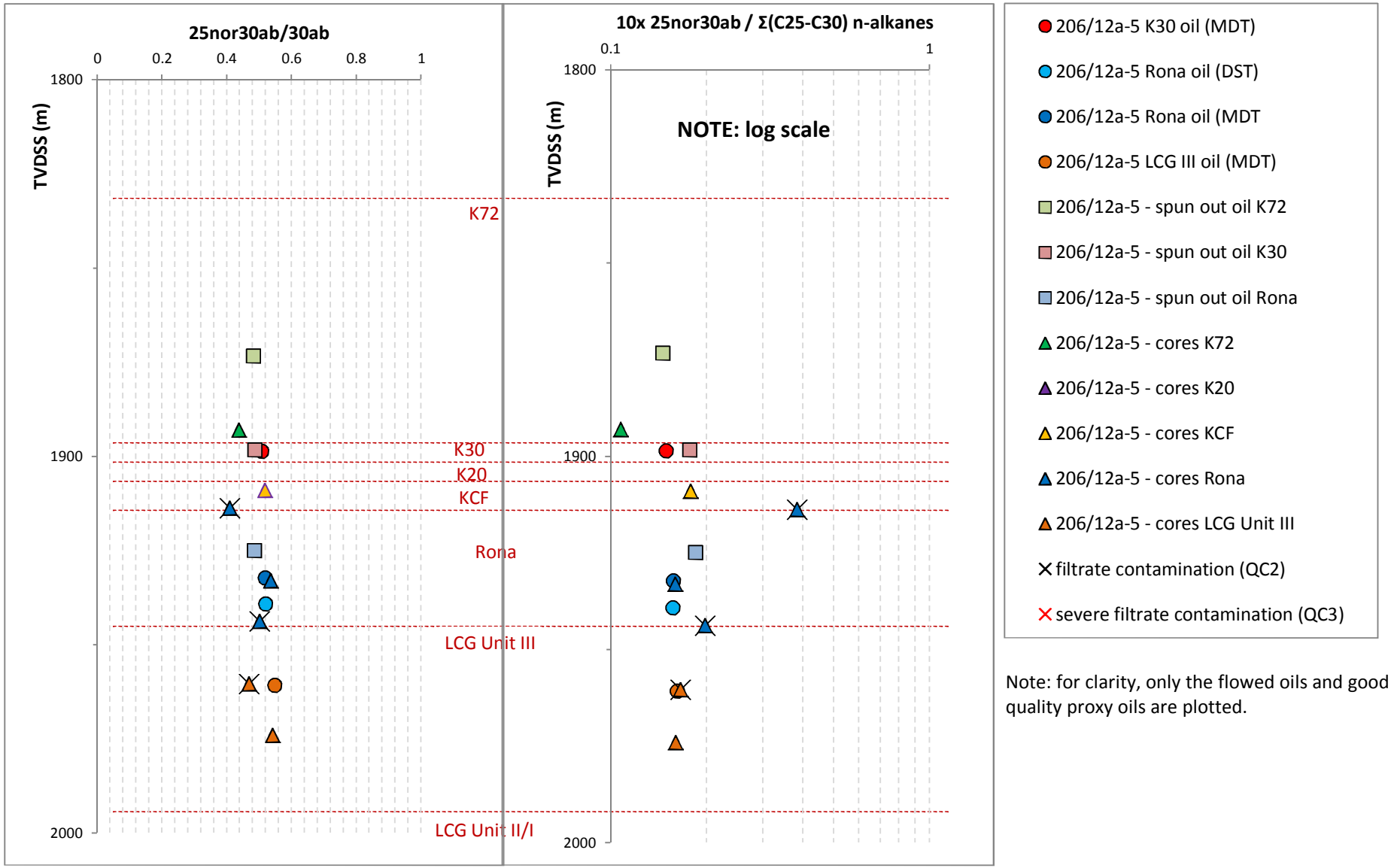
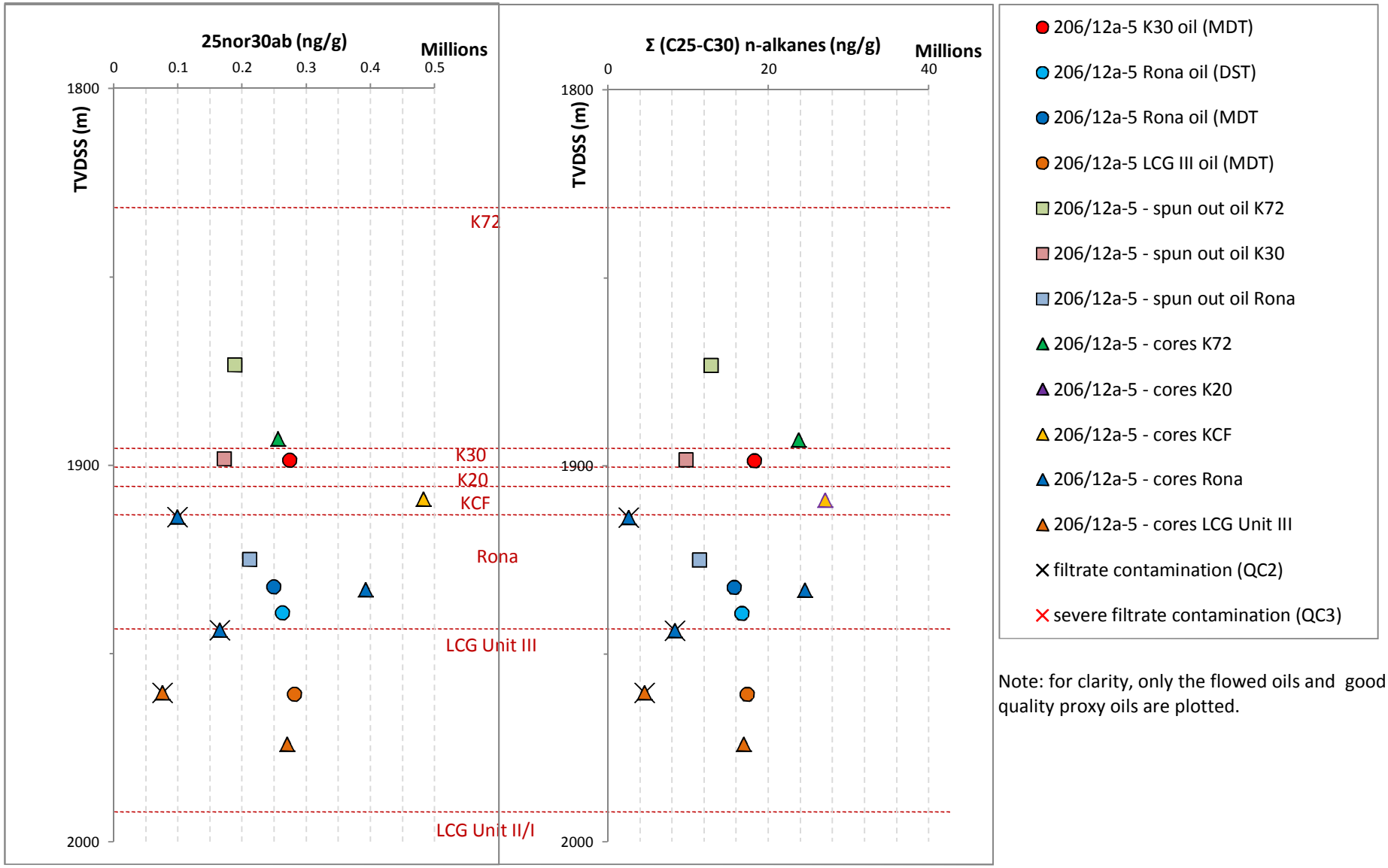


FIGURE 5.19 Cross plot of absolute concentrations of demethylated hopane against Σ(C25-C30) n-alkanes



Note: for clarity, only the flowed oils and good quality proxy oils are plotted.

FIGURE 5.20 Depth plots of demethylated hopane/Σ(C25-C30) n-alkanes and demethylated hopane/hopane biomarker ratios



Note: for clarity, only the flowed oils and good quality proxy oils are plotted.

FIGURE 5.21 Depth plots of absolute concentrations of demethylated hopane and Σ(C25-C30) n-alkanes

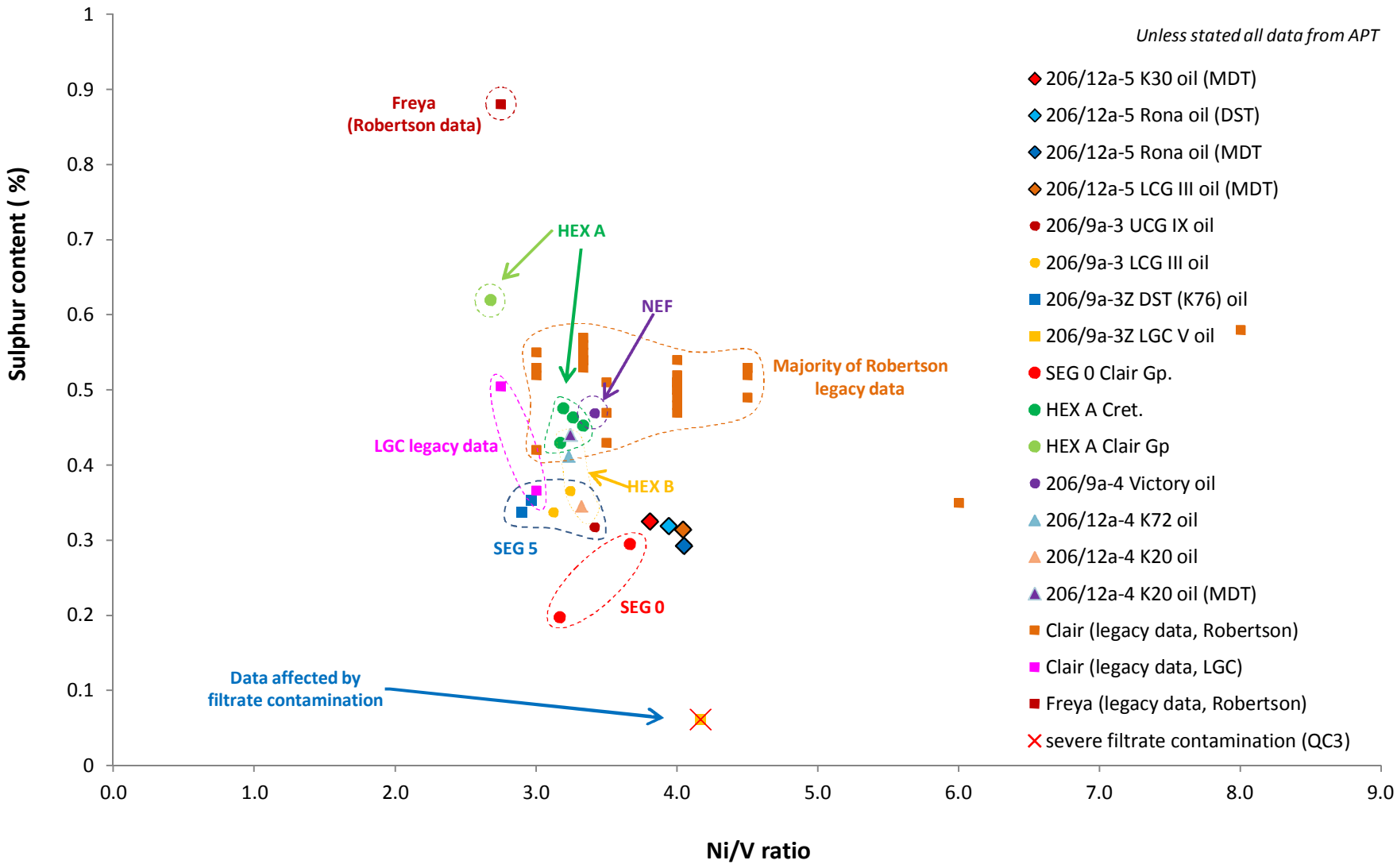


FIGURE 5.22 Cross plot of Ni/V ratios against sulphur contents

CHAPTER 6
FLOWED, ISOTUBE AND MUD (ISOJAR) GASES

6.1 Samples and analyses

Bulk composition and carbon and deuterium isotopic analyses have been undertaken on gas samples received as follows:

- 5 samples of flowed DST and MDT gases from 1922.4m MDRT (Turonian (K72)), 1936 to 1962 m MDRT (Rona), 1956 m MDRT (Rona) and 1984.5 m MDRT (LCG);
- 57 Isotube gas samples from 950m MDRT to 2340m MDRT;
- 15 Isojar gas samples from 1120m MDRT to 2352m MDRT.

Gas composition and carbon and deuterium isotope analyses have been undertaken on the flowed gases, Isotube gases and Isojar gases. A full list of the samples received is given in Tables 2.5. A list of analysed samples along with the analytical programme is tabulated in Table 2.6, and the distribution of samples throughout the well section is shown diagrammatically on Figures 2.2.

The data are fully reported in Appendix 11.

6.2 Discussion of results

6.2.1 Depth relationships

Figures 6.1 to 6.5 are depth plots of the gas concentration data, gas index data and carbon isotope data.

Gas concentrations & indices

Isotube gas concentrations measured in the Upper Cretaceous are generally similar and constant with depth, the methane concentration over this interval averages 710ppm. The methane concentration increases and reaches a maximum in the Jurassic and Devonian intervals (6700ppm and 4522 ppm respectively) coincident with the tested gases. The methane concentrations decrease below about 2120m (equivalent to the supposed field wide fwl) into the basal conglomerates (average 660ppm and a maximum 1500ppm). Interestingly the abundances of gases pick up significantly into the basal part of the LCG and into the Precambrian (average 1376ppm and a maximum of 2200ppm).

In general, the abundance data measured from the Isotube samples broadly match the data obtained from the Isojar samples (Figure 6.1).

Gas fraction data (Figures 6.2 and 6.3) indicate the gases throughout the Upper Cretaceous interval are dry, typically 94% methane or greater. The section becomes increasingly wetter into the Lower Cretaceous and deeper, peaking at 2096m MDRT with a wetness of 16% at the top of the Basal interval. The separator gas sample from the interval 1936m to 1963m MDRT has a wet gas content of 8%, concurrent with the Isotube results in the same tested interval. The other four tested gases have much wetter contents (14 to 15%) presumably because of association with liquid hydrocarbons. Isojar data also show elevated wet gas contents over the

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same interval. Below about 2200m in the LCG and the Basement the wet gas contents pick up significantly (10% and more) similarly to the absolute gas abundances.

Isotope ratios

Carbon isotope (CI) ratios against depth are shown on Figures 6.4 and 6.5. It is noted that the measured ratios from the Isotube samples are lighter than the data recorded from the Isojar samples. The Isotube data are, however, consistent with the data measured for the MDT samples. The same effect has been noted in previous studies (e.g. the Clair Field) and no reason for this discrepancy is apparent but casts some doubt on the reliability of the data measured from the Isojar samples.

For the Isotube and MDT samples, methane CI ratios average get heavier with depth, with an increase from around -80.1‰ in the Balder interval to around -40‰ at the base of the oil leg in the LCG Unit II/I interval (Figure 6.5). The value of -40‰ matches well with that seen in previous wells from the Clair Group and it is assumed that progressive dilution with isotopically lighter (biogenic) gases is occurring moving up the section.

In common with the increasing total gas and wet gas abundances, the carbon isotopes in the basal LCG and Basement (well below the apparent fwl) become markedly heavier at -36 to -38‰ for methane while the ethane and propane surprisingly become isotopically lighter. The reason for these observations are not at all clear at this time.

6.2.2 Gas provenance and maturity

Three interpretative plots relating the gas data to provenance and source rock maturity are shown as Figures 6.6 to 6.8.

Wet gas and methane carbon isotope ratio

A cross plot of wet gas content against carbon isotope ratio of methane is shown as Figures 6.6. The flowed gas data are considered the bench mark data on these plots and indicate that the reservoired gases are oil to condensate associated, i.e. generated from a source rock within the main to late oil window. Compared to previous studies the methane in the 206/12a-5 (E-1) well at -42.6‰ is consistent with 206/13a-3 (HEX A) (-42.5‰), 206/12a-4 (HEX B) (-43.4‰), 206/8-16 (SEG 0) (-40.5‰) and 206/9a-3 & -3Z (SEG 5) (-40.5‰) values, but is heavier than the gas in 206/9a-4 (NEF) (45.5‰).

Most of the Isotube and Isojar gases from the reservoir intervals have wet gas contents lower than the tested gases; those that don't are primarily Isojars of various stratigraphies and Isotubes of Basal gas samples. The reason for the differences between the tested samples and the Isotubes/Isojars is uncertain, but is probably a result of the relative ease of exsolving methane from particularly oil based mud compared to the heavier compounds. For this reason, the interpretation of the data based on this plot is biased to the benchmark flowed gas samples which are considered the most reliable.

The strong dilution with biogenic gas noted in previous wells is noted here, but on a lower scale in only the shallower Cretaceous gases. A clear depth trend to lighter gases with shallower depths noted above is clearly seen.

The five tested gases split into 3 distinct groups separated by their wet gas contents (Figure 6.6). The drier of the tested gases is the separator gas from the Rona, which plots with a tight

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cluster of Isotubes gases also from the Rona. The other DST (bottom hole) gas from the Rona plots with the Devonian and Cretaceous gases close to the 206/13a-3 (HEX A) Cretaceous and Devonian tested gases. The third and wet group is the MDT gas from the Rona which plots in between the 206/13a-3 (HEX A) gases and very close to the 206/8-16 (SEG 0) and 206/9a-3 & -3Z (SEG 5) flowed gases. The splitting of the Rona gases is most probably down to the different sampling types, (MDT, separator and bottom hole).

Methane and ethane carbon isotope ratios

The methane and ethane carbon isotope ratios are cross plotted on Figures 6.7. Again the flowed gas is considered the benchmark. The five tested gas all plot in a tight cluster similar to the 206/13a-3 (HEX A) tested gases and close to the 206/12a-4 (HEX B) tested gas, but differently from the 206/9a-4 (NEF) and 206/8-16 (SEG 0) and 206/9a-3 & -3Z (SEG 5) gases. The data match well with the Isotube data apart from some shallower Cretaceous and Basal samples. The inferred vitrinite reflectivity (VR) at the time of generation based on the tight cluster of tested and Isotube gases is around 1.1-1.2% VRE. This compares to suggested values of around 1.2% in 206/13a-3 (HEX A) and 1.25% in 206/8-16 (SEG 0) and 206/9a-3 & -3Z (SEG 5).

Methane carbon isotope and deuterium isotope ratios

The methane carbon and deuterium isotope (DI) ratios are plotted on Figure 6.8. The benchmark tested gas data suggest that this is an oil associated gas with the source rock at the upper end of the oil window. The tested gases are in a tight cluster with the analysed Isotubes, excluding some shallower Cretaceous and Basal samples. The tested gases plot within the tested 206/13a-3 (HEX A) and 206/12a-4 (HEX B) gas area.

Of some interest are the results from the Basal LCG and Basement section where isotopically much heavier deuterium values have been encountered; heavier than seen anywhere else in the Greater Clair Field area and occurring apparently well below the free water level. The significance of these results is not known.

6.2.3 Homogeneity of the tested gases

Figure 6.9 shows Galimov-style plots of gas concentrations, gas fractions and gas isotope ratios against compound for all the tested gases. Data from all previously studied wells (206/13a-3 (HEX A), 206/12a-4 (HEX B), 206/8-16 (SEG 0) and 206/9a-3 & -3Z (SEG 5) and 206/9a-4 (NEF)) plus the gas from this study (206/12a-5 - E-1) are shown for comparison.

It is clear that all five newly tested gases plot differently from the other Clair Field tested gases. The Lower Cretaceous, Jurassic and Devonian MDT gases are generally present in lower concentrations than most of the other Clair wells but are similar to the 206/13a-3 (HEX A) and 206/12a-4 (HEX B) gases in terms of carbon isotope ratios. The DST separator gas sample from the Jurassic interval is clearly very different from the bottom hole DST sample taken over the same interval. The differences are mostly likely down to the different sampling types rather than differences in the gases themselves as they were taken over the same interval.

The C2+ isotopes of all the analysed gases are all essentially the same in all wells analysed. Differences occur in the methane CI ratio with most of the Cretaceous gases lighter than the Palaeozoic gases. This is not the case for the 206/12a-5 (E-1) well gases. The Palaeozoic gas has identical carbon isotope ratios for methane (-42.6‰) as for the Cretaceous and Jurassic

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gases. The average carbon isotope ratio for methane and ethane in the 206/13a-3 (HEX A) and 206/12a-4 (HEX B) wells is the same as the results from the 206/12a-5 (E-1) well.

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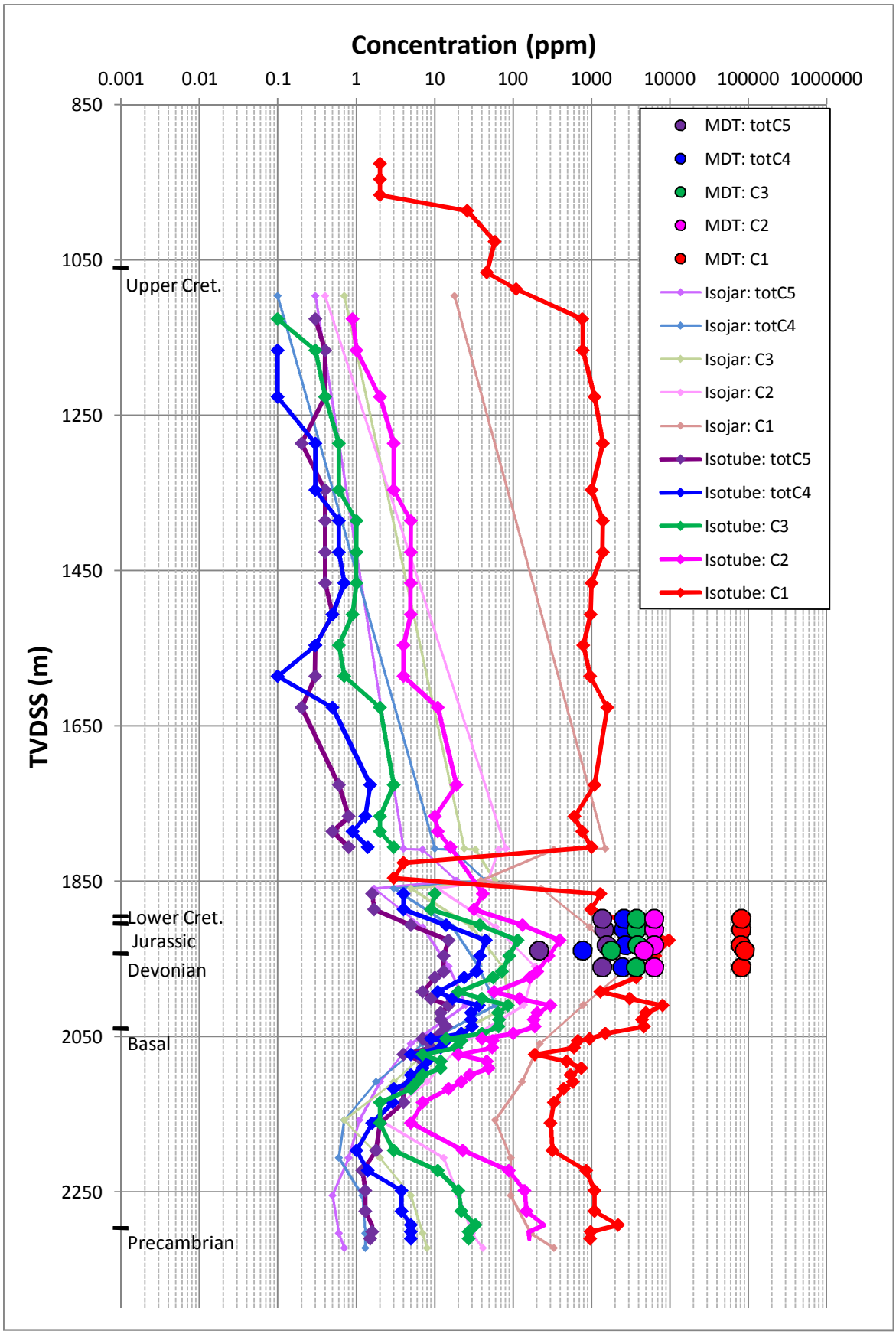

FIGURE 6.1 Depth plot of gas concentration data

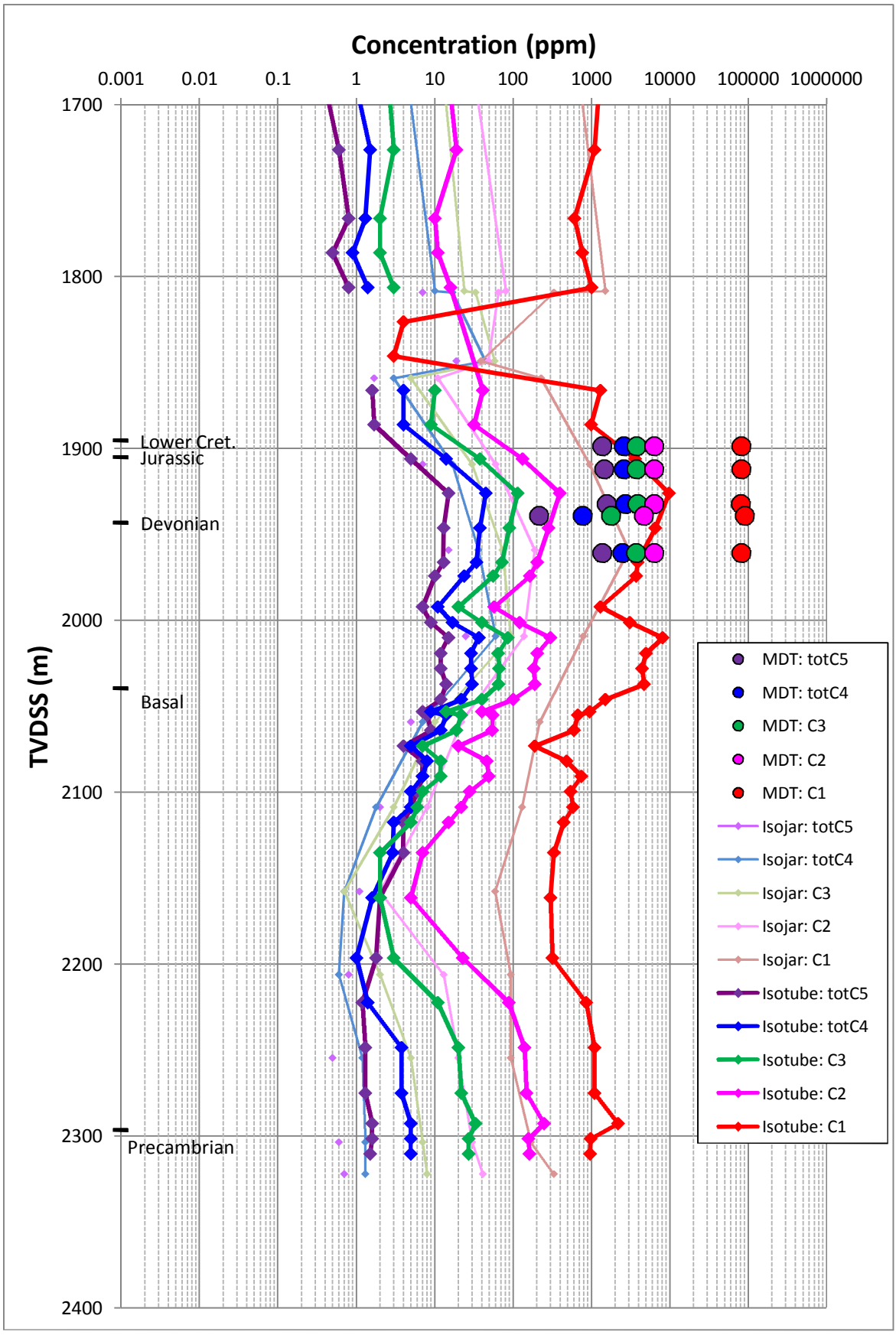

FIGURE 6.1 Depth plot of gas concentration data - expanded plot

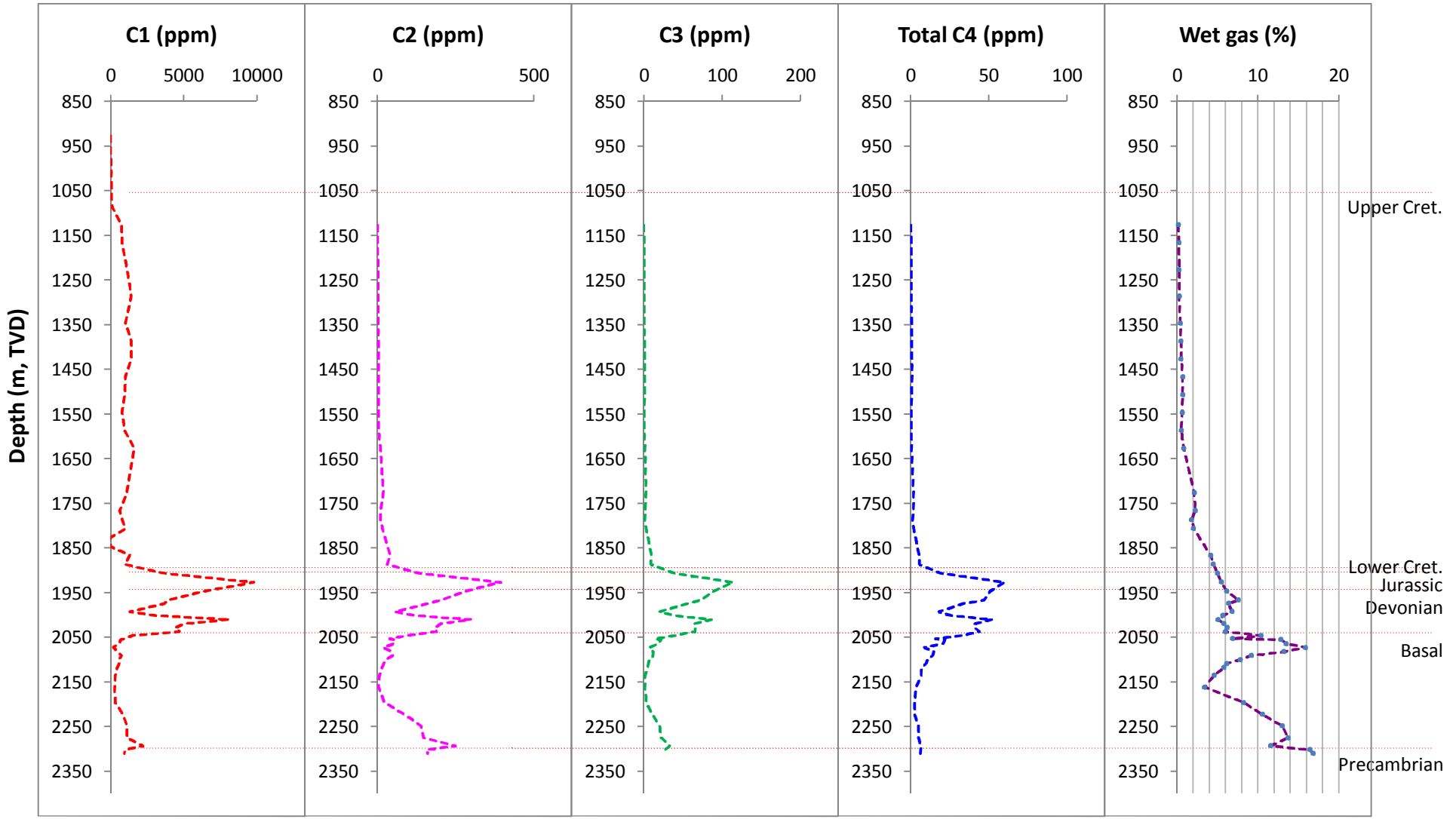

FIGURE 6.2 Depth plots of Isotube gas concentration data and wet gas percentage

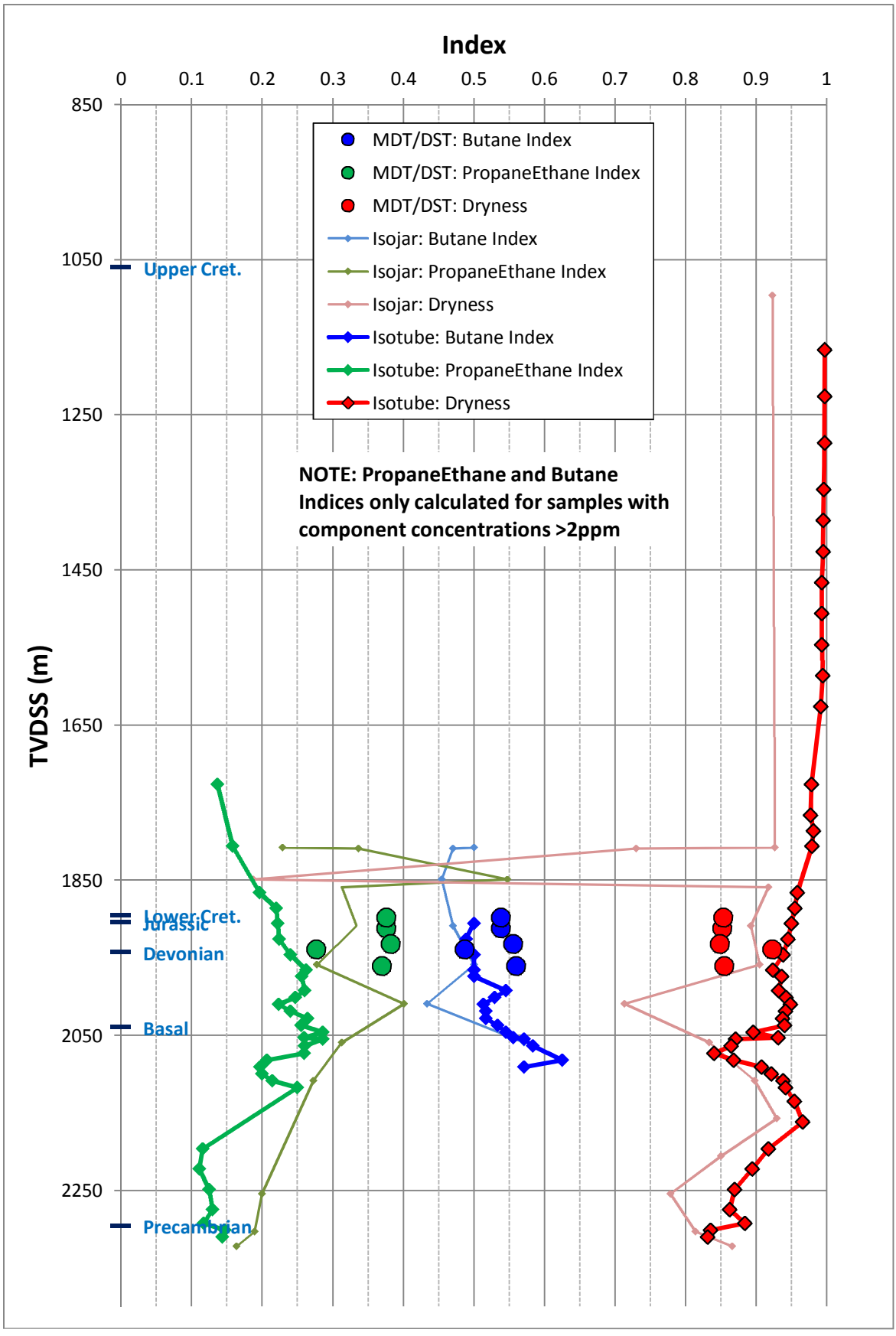

FIGURE 6.3 Depth plot of gas index data

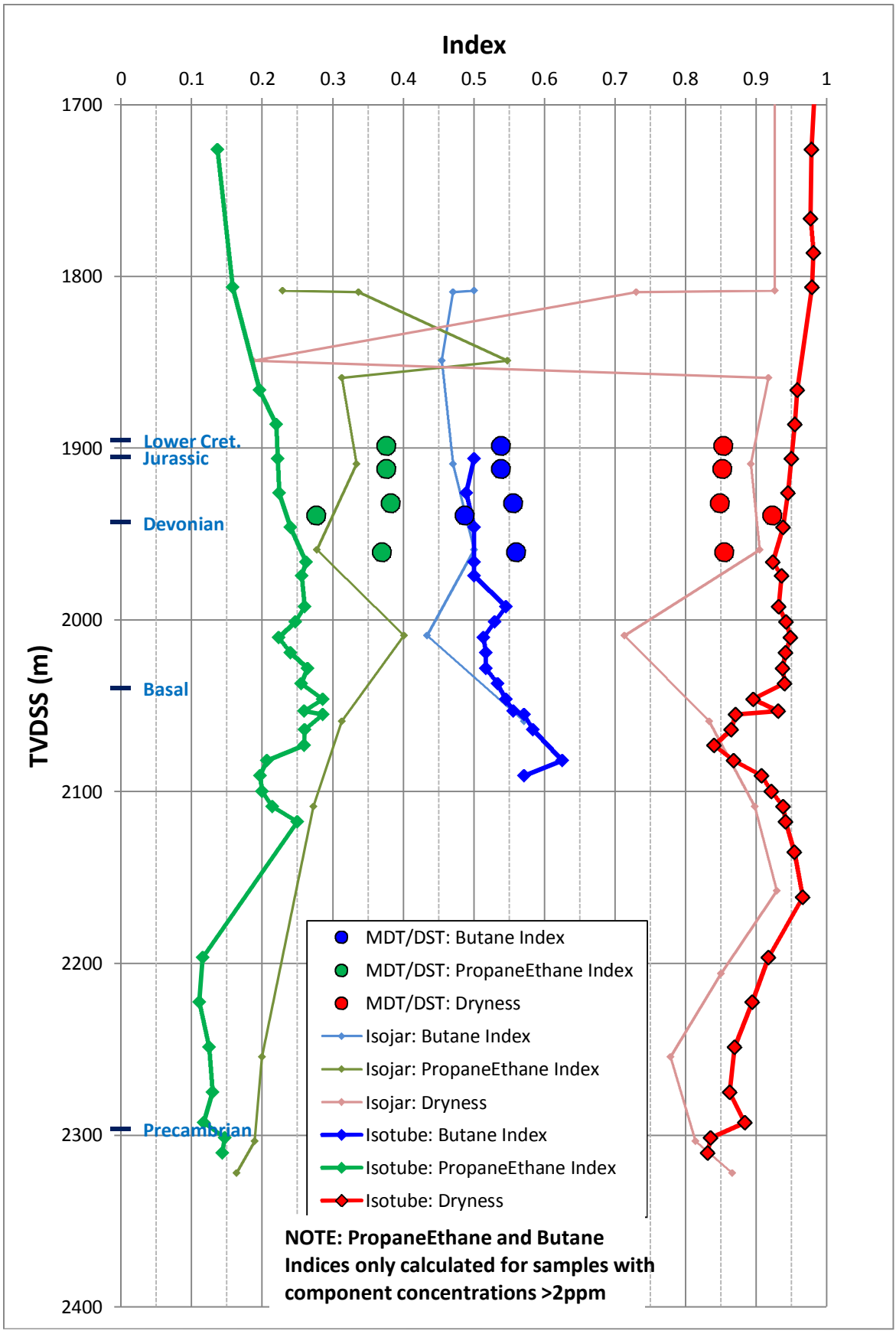

FIGURE 6.3 Depth plot of gas index data - expanded plot

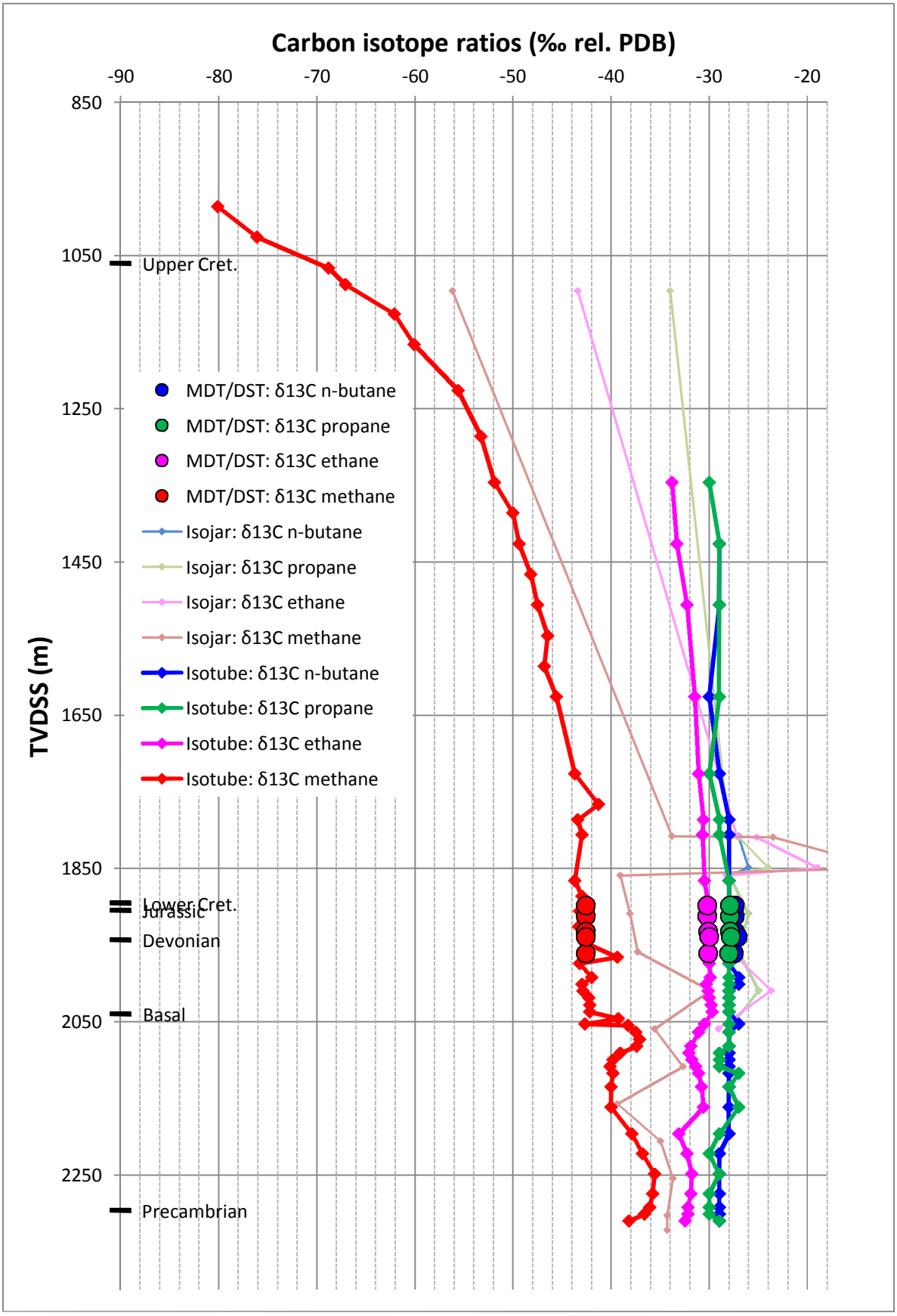

FIGURE 6.4 Depth plot of gas carbon isotope data

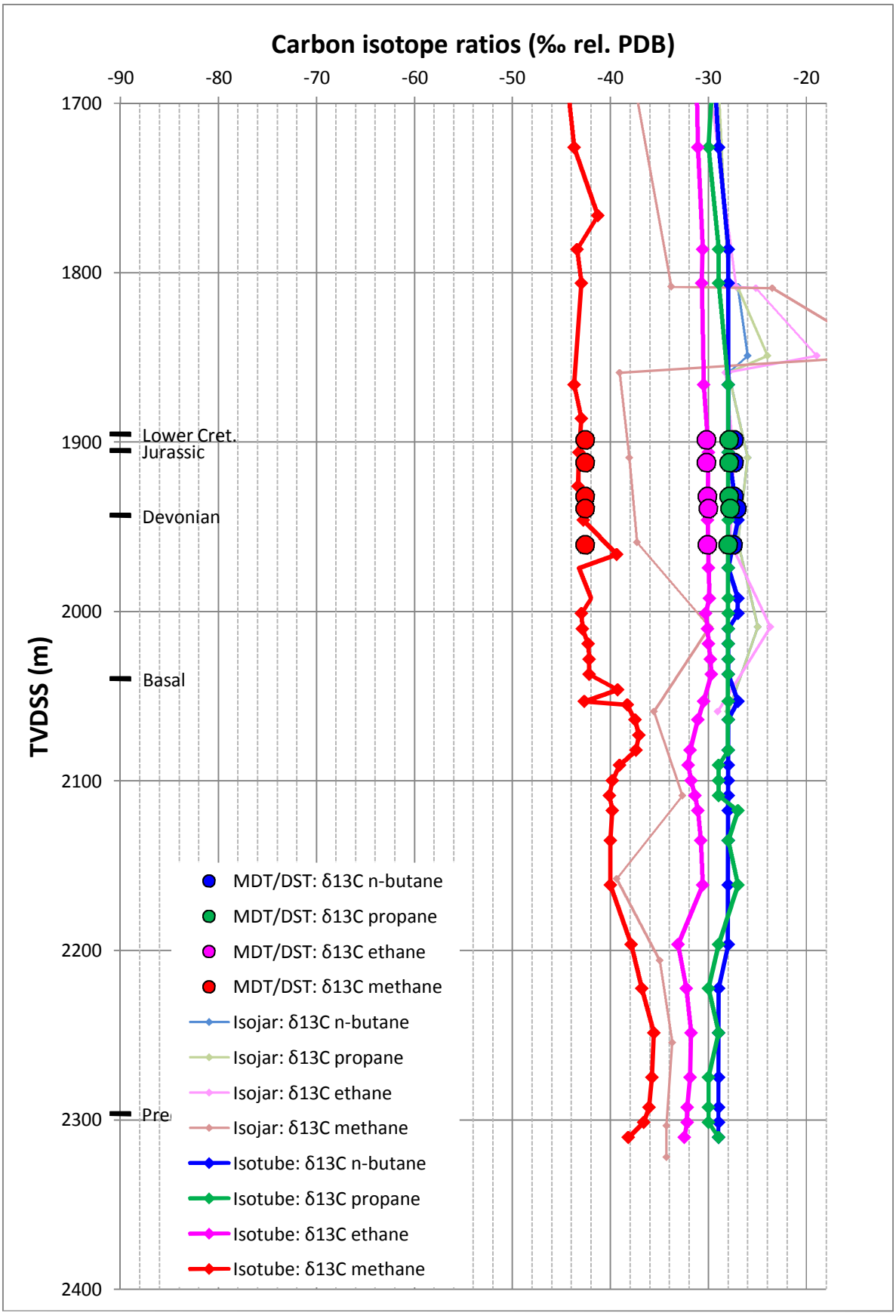

FIGURE 6.4 Depth plot of gas carbon isotope data - expanded plot

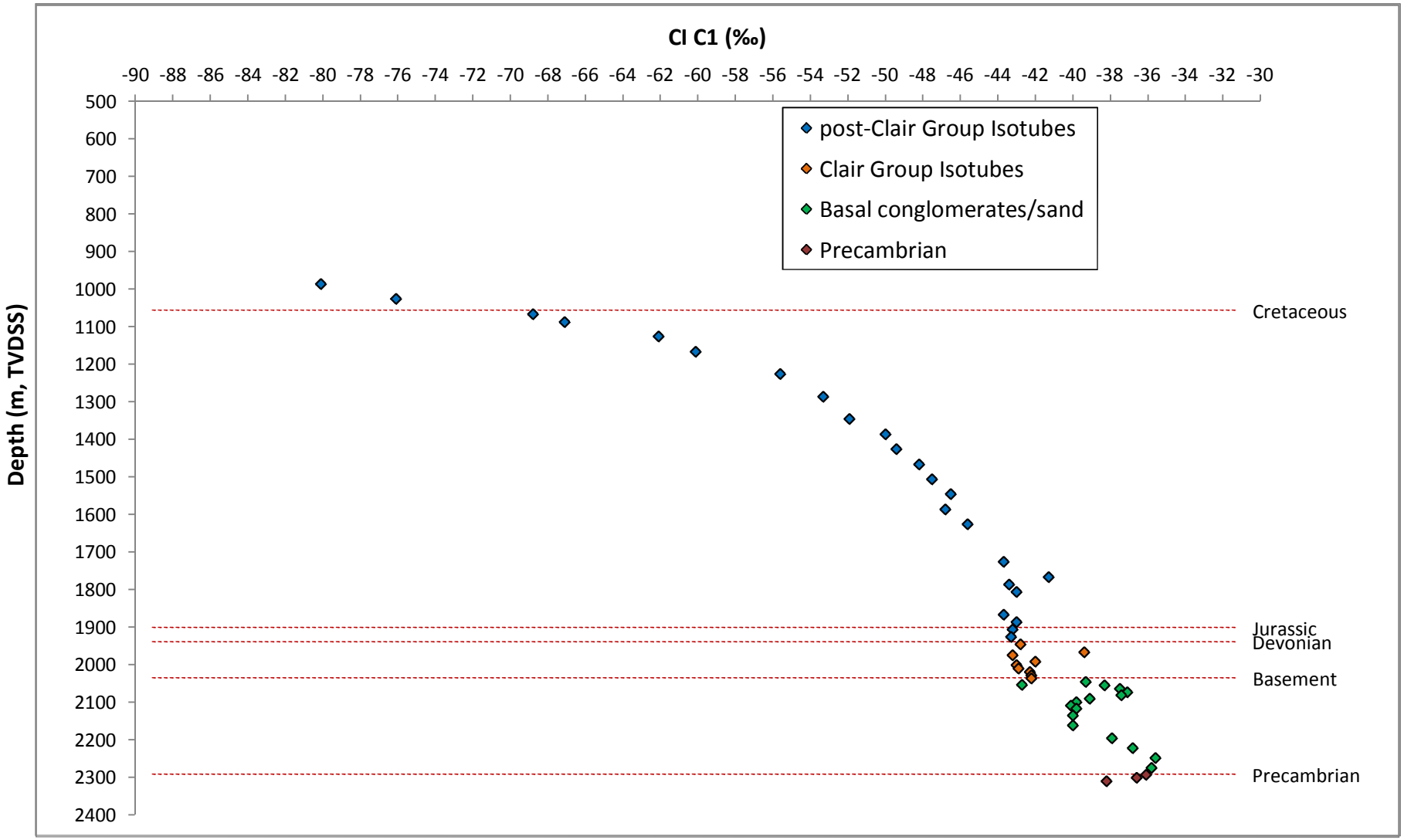

FIGURE 6.5 Depth plots of Isotube gas methane carbon isotope data

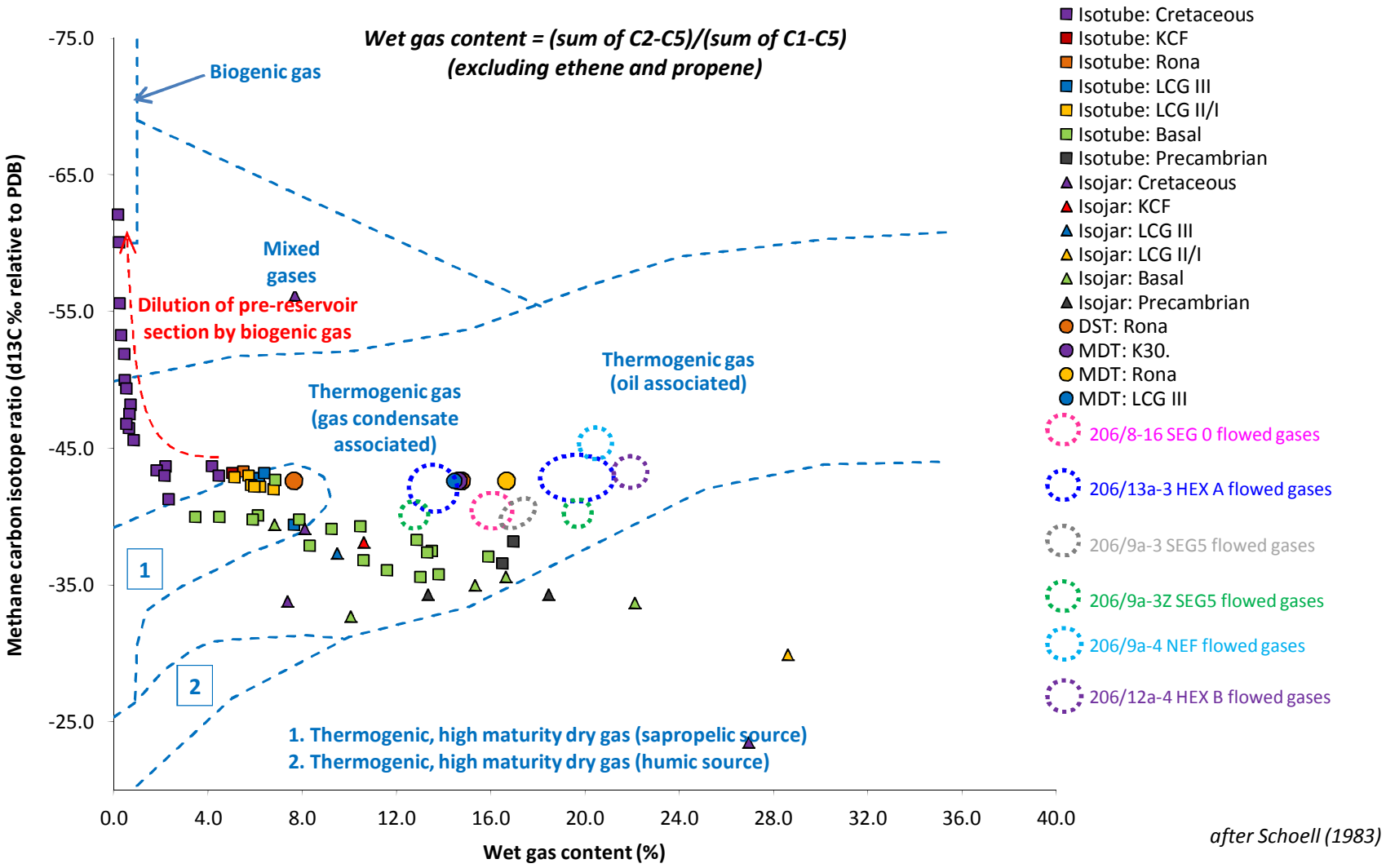

FIGURE 6.6 Cross plot of methane carbon isotope ratio data against wet gas content

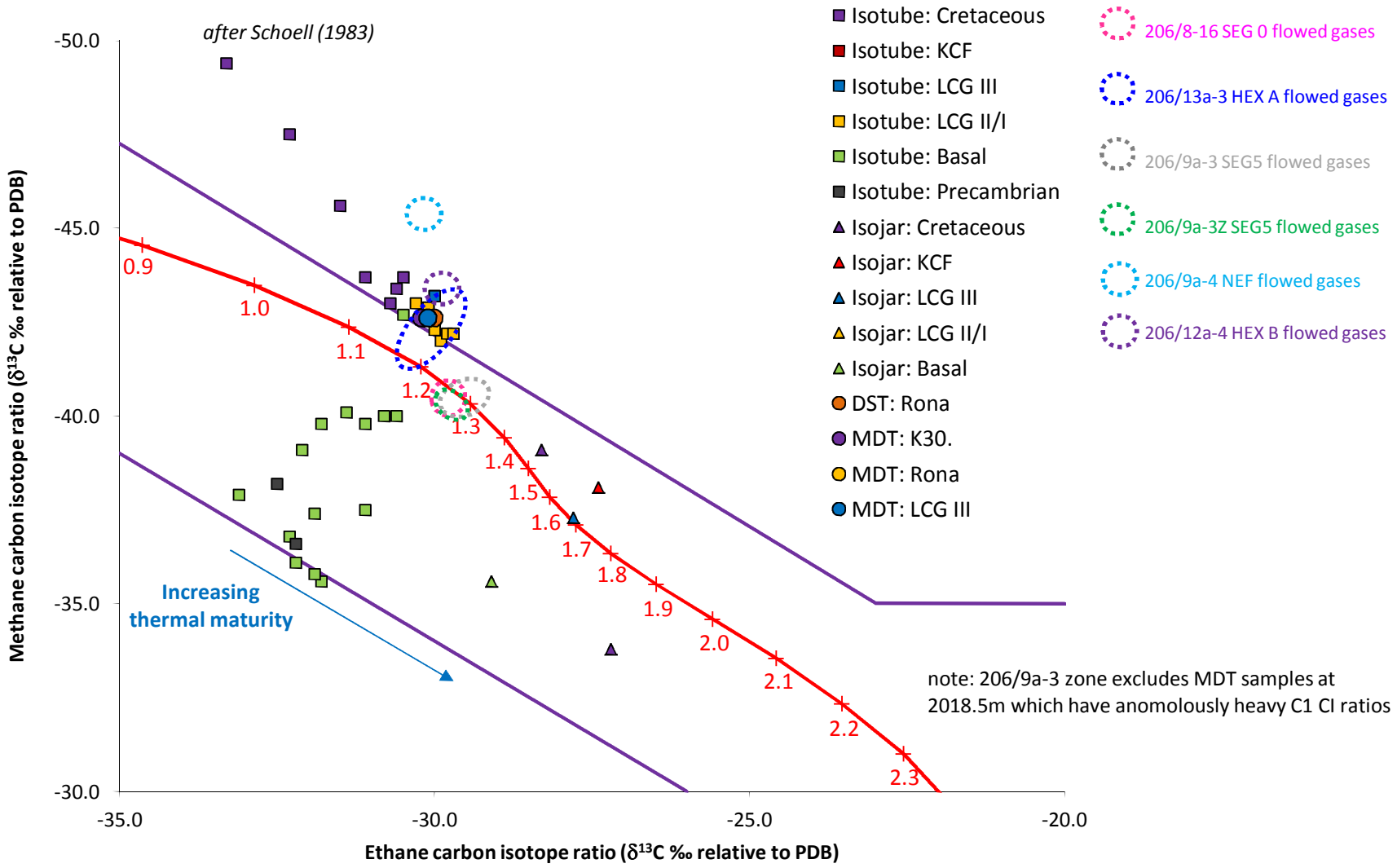

FIGURE 6.7 Cross plot of methane against ethane carbon isotope ratio data

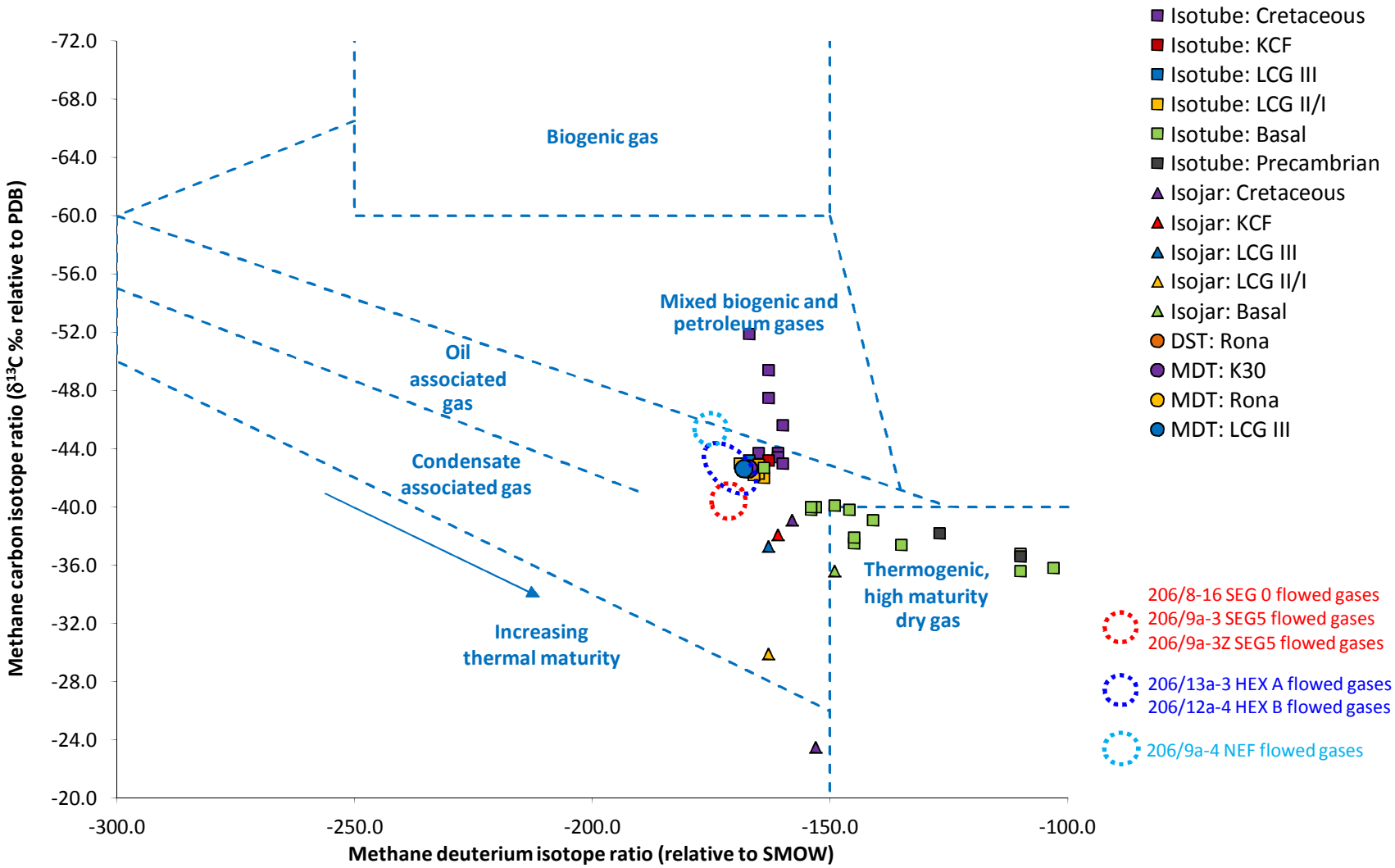

FIGURE 6.8 Cross plot of methane carbon against methane deuterium isotope ratio data

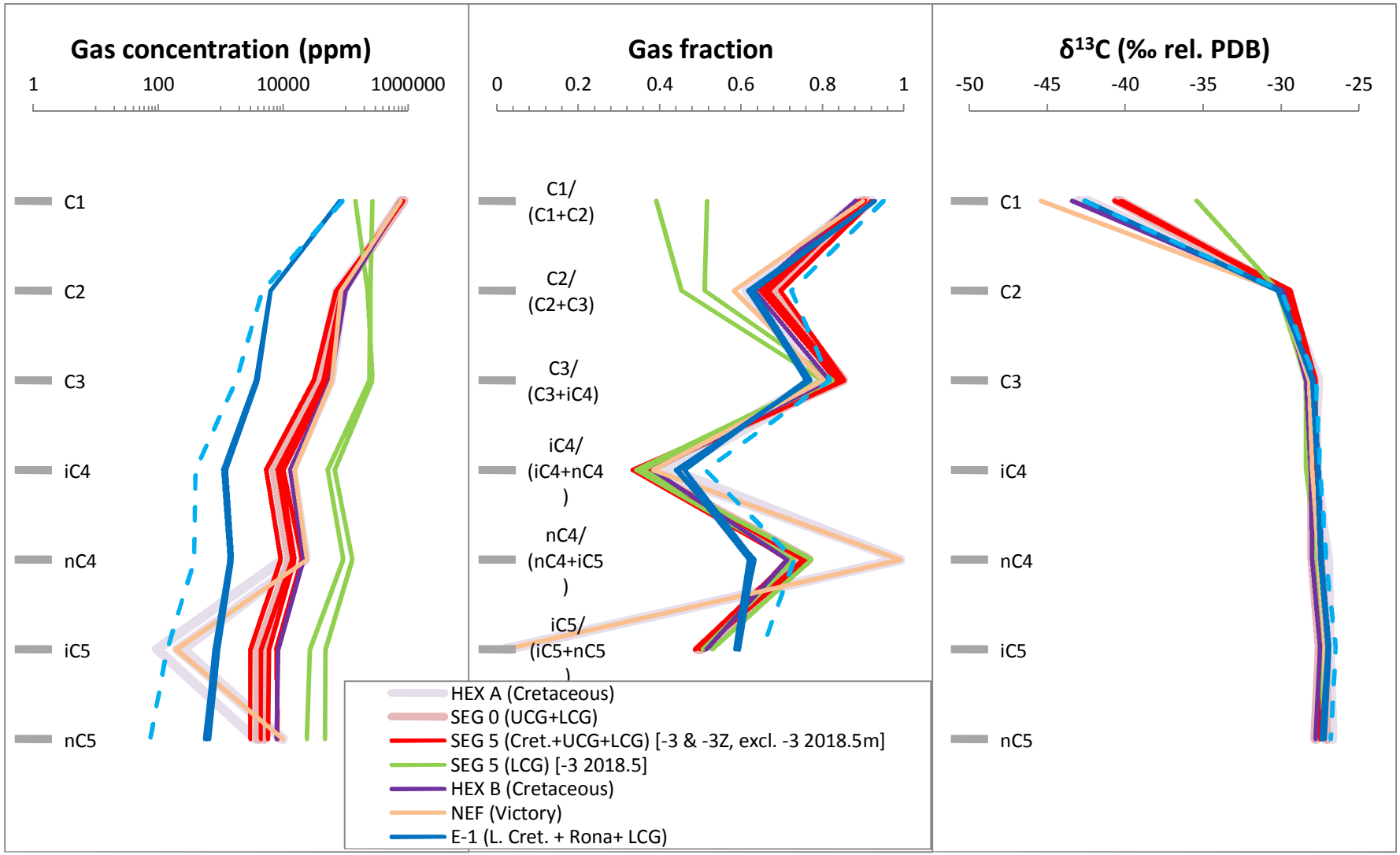

FIGURE 6.9 Galimov-style plots of gas concentrations, gas fractions and gas isotopes: 206/12a-5 (all Clair Field tested gases)

CHAPTER 7
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APPENDIX 1
VITRINITE REFLECTIVITY DATA

Tables

Ap1.1 Vitrinite reflectivity data

Figures

Ap1.1 Histograms, data and statistics for vitrinite reflectivity: 206/12a-5

Well	MD (m)	TVDSS (m)	Sample type	Vitrinite Reflectivity										Comments
				Sample prep.	Indigenous				Other populations					
					No.	Ave. %Ro	Min % Ro	Max %Ro	Type	No.	Ave. %Ro	Min % Ro	Max %Ro	
206/12a-5	270	246.5	cuttings	Kerogen	40	0.19	0.12	0.26						AOM dominates, but occasional fragments of peat are present, sometimes weathered.
	360	336.5	cuttings	Kerogen	40	0.18	0.12	0.27						AOM dominates with occasional peat like material present. Structured kerogen including inertinite more common.
	480	456.5	cuttings	Kerogen	40	0.22	0.15	0.30						AOM dominates with occasional peat like material present. Structured kerogen including inertinite more common.
	540	516.5	cuttings	Kerogen	40	0.21	0.15	0.26						AOM dominates with occasional peat like material present. Structured kerogen including inertinite more common.
	570	546.5	cuttings	Kerogen	40	0.23	0.15	0.31						AOM dominates with occasional peat like material present. Structured kerogen including inertinite more common.
	600	576.5	cuttings	Whole rock	55	0.18	0.12	0.26						Peat sample. Vascular, telinite-like tissue with %Ro < 0.1 the primary component. Only brighter corpocollinite-like particles have been measured.
	720	696.5	cuttings	Kerogen	40	0.21	0.15	0.31						Sclerotinite at 0.4% sets upper limit. Super abundant pyrite - vitrinite may be largely perhydrous.
	760	736.5	cuttings	Kerogen	40	0.27	0.16	0.39						Good sample. Huminite increasingly abundant and generally well-preserved.

TABLE Ap1.1 Vitrinite reflectivity data

Well	MD (m)	TVDSS (m)	Sample type	Vitrinite Reflectivity										Comments
				Sample prep.	Indigenous				Other populations					
					No.	Ave. %Ro	Min % Ro	Max %Ro	Type	No.	Ave. %Ro	Min % Ro	Max %Ro	
206/12a-5	1050	1026.5	cuttings	Kerogen	40	0.30	0.20	0.37						Abundant well-preserved huminite. Very good sample.
	1100	1076.5	cuttings	Kerogen	47	0.24	0.17	0.30						Vitrinite rare but sufficient for reliable data; morphology is variable. Distinction from other macerals by reflectivity is clear.
	1150	1126.5	cuttings	Kerogen	55	0.28	0.18	0.37						Vitrinite rare and morphologically similar to semifusinite, but easily distinguished by reflectivity.
	1210	1186.4	cuttings	Kerogen	49	0.30	0.19	0.40	R	6	0.53	0.50	0.64	Vitrinite rare, wispy to coaly.
	1250	1226.4	cuttings	Kerogen	27	0.34	0.25	0.44						Relatively low recovery and vitrinite very rare. Generally wispy.
	1300	1276.4	cuttings	Kerogen	55	0.34	0.26	0.46						Increased AOM, but occasional good wispy and blocky vitrinite present.
	1360	1336.4	cuttings	Kerogen	38	0.32	0.25	0.42	L R	4 13	0.18 0.50	0.16 0.44	0.21 0.65	Increased AOM, but occasional good wispy and blocky vitrinite present.
	1400	1376.4	cuttings	Kerogen	46	0.32	0.23	0.43	R	9	0.50	0.49	0.56	Vitrinite slightly more abundant than above, generally wispy to irregular.

TABLE Ap1.1 Vitrinite reflectivity data

Well	MD (m)	TVDSS (m)	Sample type	Vitrinite Reflectivity										Comments
				Sample prep.	Indigenous				Other populations					
					No.	Ave. %Ro	Min % Ro	Max %Ro	Type	No.	Ave. %Ro	Min % Ro	Max %Ro	
206/12a-5	1450	1426.4	DC	Kerogen	55	0.35	0.24	0.46						Vitrinite slightly more abundant than above, generally wispy to irregular.
	1500	1476.4	DC	Kerogen	40	0.36	0.25	0.48						Poor recovery but otherwise as above.
	1550	1526.4	DC	Kerogen	5	0.51	0.42	0.59						Little structured organic matter.
	1600	1576.4	DC	Kerogen	20	0.57	0.49	0.68	R	13	0.79	0.70	0.96	Very lean sample. Particles quite small.
	1650	1626.4	DC	Kerogen	26	0.59	0.50	0.66	R	29	0.83	0.70	1.11	Particles in both populations are similar, of moderate size, equidimensional and sub-rounded.
	1700	1676.4	DC	Kerogen	12	0.67	0.58	0.72	L	5	0.47	0.39	0.52	Poor recovery.
									R	2	0.84	0.82	0.85	
	1749	1725.4	DC	Kerogen	28	0.50	0.42	0.57	L	8	0.37	0.34	0.40	All particles sub-rounded, wispy to blocky.
									R	19	0.69	0.60	0.84	
	1806	1782.4	DC	Kerogen					R	3	0.69	0.65	0.73	
									R	3	0.90	0.87	1.01	

TABLE Ap1.1 Vitrinite reflectivity data

Well	MD (m)	TVDSS (m)	Sample type	Vitrinite Reflectivity										Comments
				Sample prep.	Indigenous				Other populations					
					No.	Ave. %Ro	Min % Ro	Max %Ro	Type	No.	Ave. %Ro	Min % Ro	Max %Ro	
206/12a-5	1884.58	1860.9	COCH	Kerogen	-	-	-	-						Inertinite is abundant and trace amounts of exinite are present, but vitrinite is entirely absent.
	1892.66	1869.0	COCH	Kerogen	20	0.42	0.33	0.51						Low recovery. Inertinite dominates the sample with only moderate amounts of vitrinite, which may be oxidised and probably overstates maturity.
	1902.70	1879.0	COCH	Kerogen	3	0.35	0.27	0.47						Low recovery. Inertinite common but vitrinite nearly absent.
	1911.17	1887.5	COCH	Kerogen	-	-	-	-						100% AOM with trace amounts of inertinite and no vitrinite. Pyrite is abundant.
	1928.61	1905.0	COCH	Kerogen	-	-	-	-						Very lean. Pyrite dominant with traces of very badly preserved inertinite. Vitrinite absent.
	1932.67	1909.0	COCH	Kerogen	55	0.42	0.34	0.52						Vitrinite abundant for KCF. Some particles are large, blocky and coaly.
	1944.05	1920.4	COCH	Whole rock	-	-	-	-	B	5	0.58	0.56	0.59	Organic matter is monomacerallic, with slightly granular surfaces. Interpreted to be semi bituminite. Abundant framboidal pyrite. AOM has brown fluorescence.
	1948.68	1925.0	COCH	Whole rock	-	-	-	-	B	40	0.62	0.57	0.66	Organic matter is rather soft looking. Clay and framboidal pyrite inclusions are common. No fluorescence. Under UV exposure fluid diffuses out of pore spaces.

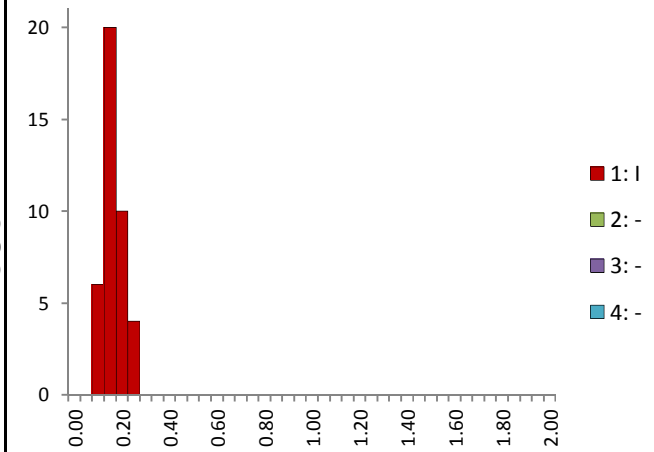
TABLE Ap1.1 Vitrinite reflectivity data

Well	MD (m)	TVDSS (m)	Sample type	Vitrinite Reflectivity										Comments
				Sample prep.	Indigenous				Other populations					
					No.	Ave. %Ro	Min % Ro	Max %Ro	Type	No.	Ave. %Ro	Min % Ro	Max %Ro	
206/12a-5	1960.35	1936.7	COCH	Whole rock	40	0.43	0.38	0.49						Large 'coaly' particles showing weakly laminar texture, probably eutextinite. Under UV exposure fluid diffuses out of pore spaces.
	1966.70	1943.0	COCH	Whole rock	40	0.35	0.32	0.41						Homogenous eutextinite (or bituminite). Under UV exposure fluid diffuses out of pore spaces.
	1967.35	1943.7	COCH	Whole rock	40	0.40	0.34	0.43						Organic matter heterogeneous with a relatively bright circular to curved smooth particles (textinite & phlobaphinite, possibly rootlets) in a dull granular densinite matrix.
	1984	1960.3	SWC	Kerogen										Sample was barren.
	2001	1977.3	SWC	Whole rock	-	-	-	-	R	10	0.70	0.61	0.79	
									R	6	1.10	0.88	1.24	
	2022	1998.3	SWC	Kerogen	-	-	-	-						Sample was barren.
	2048	2024.3	SWC	Kerogen	-	-	-	-						Sample was barren.
	2202	2176.2	DC	Kerogen	-	-	-	-						Sample was barren.

TABLE Ap1.1 Vitrinite reflectivity data

SAMPLE DATA
DEPTH/ORDER: 270m
SAMPLE-ID: Westray Gp.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 16-305
APT-ID.: 0

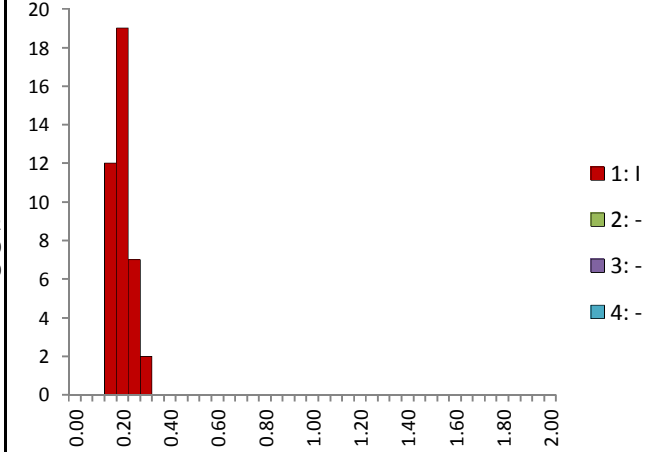
STATISTICS
POPN 1 - Indigenous
Mean value: 0.19
No. of values: 40
Range: 0.12 to 0.29
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.12	0.14	0.16	0.17	0.18	0.19	0.22	0.24
0.12	0.15	0.17	0.17	0.19	0.20	0.22	0.25
0.12	0.15	0.17	0.17	0.19	0.20	0.22	0.25
0.12	0.16	0.17	0.17	0.19	0.20	0.24	0.26
0.14	0.16	0.17	0.17	0.19	0.20	0.24	0.26

SAMPLE DATA
DEPTH/ORDER: 480m
SAMPLE-ID: Westray Gp.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 16-307
APT-ID.: 0

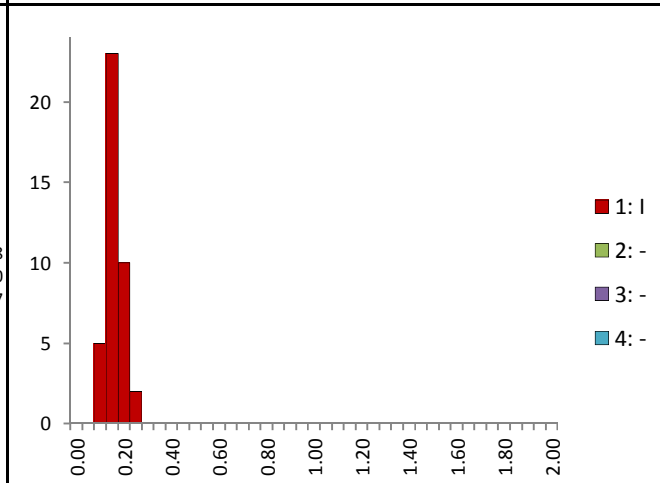
STATISTICS
POPN 1 - Indigenous
Mean value: 0.22
No. of values: 40
Range: 0.10 to 0.30
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.15	0.18	0.19	0.20	0.21	0.23	0.24	0.26
0.16	0.18	0.19	0.21	0.22	0.23	0.25	0.28
0.17	0.18	0.20	0.21	0.22	0.24	0.26	0.28
0.17	0.19	0.20	0.21	0.22	0.24	0.26	0.30
0.17	0.19	0.20	0.21	0.22	0.24	0.26	0.30

SAMPLE DATA
DEPTH/ORDER: 360m
SAMPLE-ID: Westray Gp.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 16-306
APT-ID.: 0

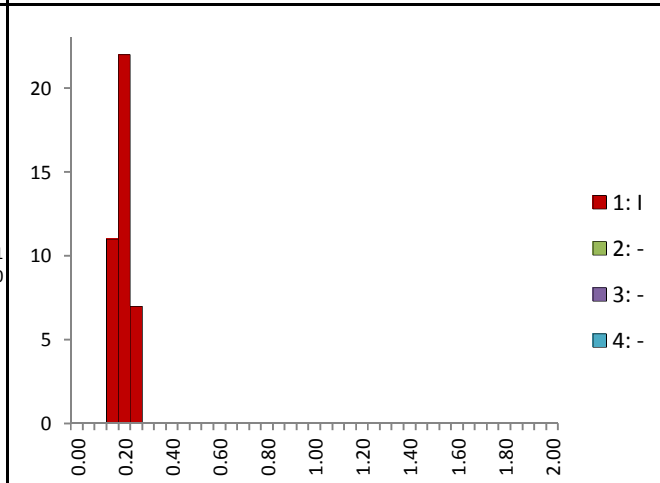
STATISTICS
POPN 1 - Indigenous
Mean value: 0.18
No. of values: 40
Range: 0.12 to 0.27
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.12	0.15	0.16	0.17	0.18	0.18	0.21	0.24
0.12	0.15	0.16	0.17	0.18	0.18	0.21	0.24
0.13	0.15	0.16	0.17	0.18	0.19	0.21	0.24
0.13	0.15	0.17	0.18	0.18	0.20	0.23	0.25
0.14	0.16	0.17	0.18	0.18	0.21	0.23	0.27

SAMPLE DATA
DEPTH/ORDER: 540m
SAMPLE-ID: Stronsay Gp
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 16-308
APT-ID.: 0

STATISTICS
POPN 1 - Indigenous
Mean value: 0.21
No. of values: 40
Range: 0.15 to
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a

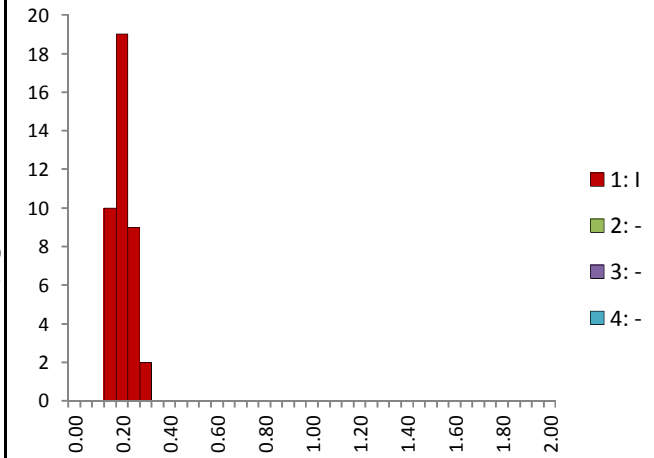


0.15	0.17	0.19	0.21	0.22	0.23	0.24	0.25
0.16	0.18	0.20	0.21	0.22	0.23	0.24	0.25
0.17	0.18	0.20	0.21	0.22	0.23	0.24	0.26
0.17	0.18	0.20	0.21	0.22	0.23	0.25	0.26
0.17	0.19	0.20	0.22	0.23	0.23	0.25	0.26

APPENDIX 1.1 Histograms, data and statistics for vitrinite reflectivity: 206/12a-5

SAMPLE DATA
DEPTH/ORDER: 570m
SAMPLE-ID: Stronsay Gp
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 16-309
APT-ID.: 0

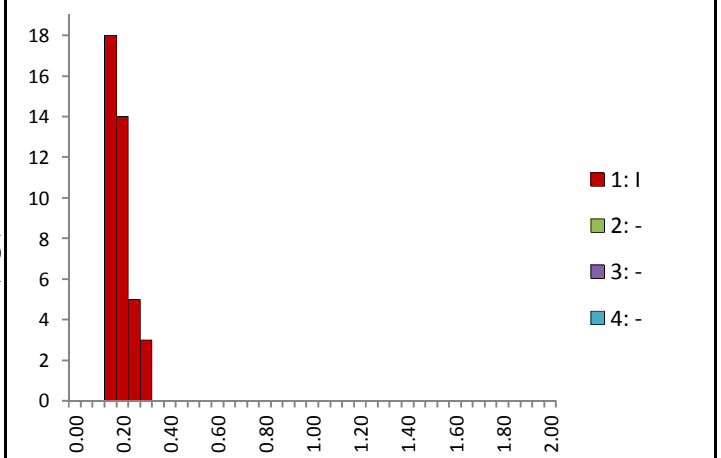
STATISTICS
POPN 1 - Indigenous
Mean value: 0.23
No. of values: 40
Range: 0.15 to 0.31
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.15	0.18	0.20	0.21	0.22	0.24	0.26	0.28
0.15	0.18	0.21	0.21	0.23	0.24	0.28	0.28
0.16	0.19	0.21	0.22	0.23	0.24	0.28	0.29
0.17	0.19	0.21	0.22	0.23	0.24	0.28	0.30
0.18	0.19	0.21	0.22	0.24	0.25	0.28	0.31

SAMPLE DATA
DEPTH/ORDER: 720m
SAMPLE-ID: Stronsay Gp
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 16-310
APT-ID.: 0

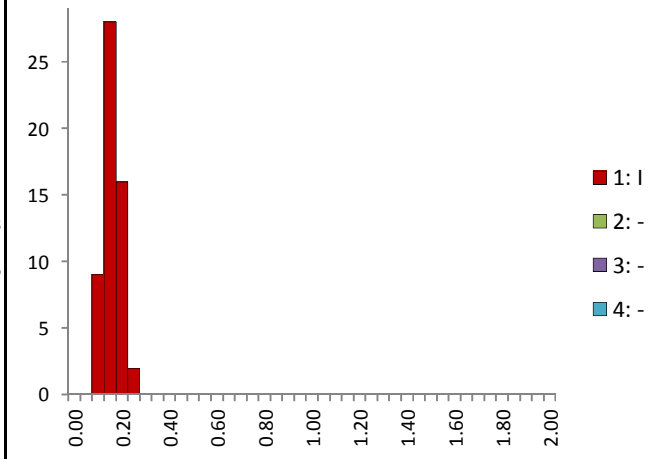
STATISTICS
POPN 1 - Indigenous
Mean value: 0.21
No. of values: 40
Range: 0.15 to 0.31
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.15	0.17	0.18	0.19	0.20	0.22	0.24	0.28
0.15	0.17	0.18	0.19	0.20	0.22	0.24	0.28
0.16	0.18	0.18	0.19	0.20	0.22	0.26	0.30
0.16	0.18	0.19	0.20	0.20	0.23	0.27	0.31
0.16	0.18	0.19	0.20	0.21	0.24	0.27	0.31

SAMPLE DATA
DEPTH/ORDER: 600m
SAMPLE-ID: Stronsay Gp
SAMPLE TYPE: DC
BLOCK TYPE.: Whole rock
BLOCK No..: 15-695
APT-ID.: 0

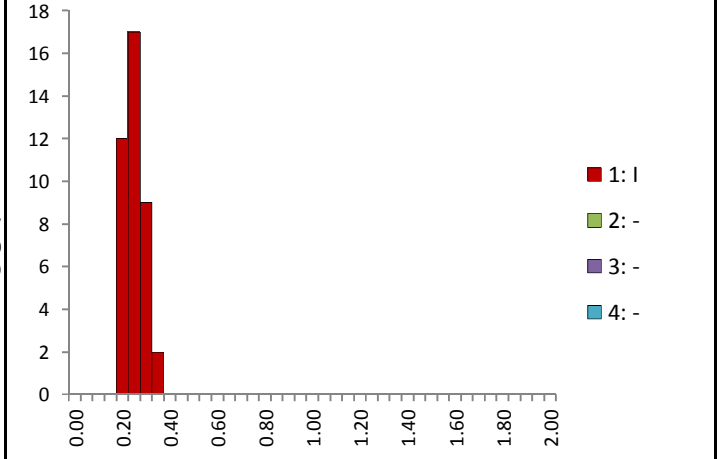
STATISTICS
POPN 1 - Indigenous
Mean value: 0.18
No. of values: 55
Range: 0.12 to 0.26
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.12	0.13	0.15	0.16	0.17	0.17	0.18	0.19	0.20	0.20	0.22
0.12	0.13	0.15	0.16	0.17	0.17	0.18	0.19	0.20	0.21	0.23
0.12	0.13	0.15	0.16	0.17	0.18	0.18	0.20	0.20	0.22	0.24
0.12	0.14	0.16	0.16	0.17	0.18	0.18	0.20	0.20	0.22	0.25
0.13	0.15	0.16	0.16	0.17	0.18	0.18	0.20	0.20	0.22	0.26

SAMPLE DATA
DEPTH/ORDER:760m
SAMPLE-ID: Stronsay Gp
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 16-311
APT-ID.: 0

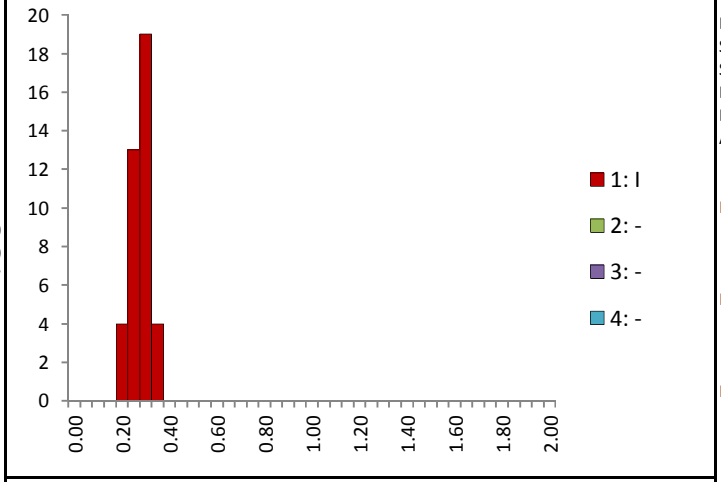
STATISTICS
POPN 1 - Indigenous
Mean value: 0.27
No. of values: 40
Range: 0.16 to 0.39
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.20	0.22	0.24	0.27	0.28	0.28	0.30	0.33
0.21	0.23	0.24	0.27	0.28	0.29	0.31	0.33
0.22	0.24	0.25	0.27	0.28	0.29	0.31	0.34
0.22	0.24	0.26	0.27	0.28	0.29	0.31	0.35
0.22	0.24	0.26	0.27	0.28	0.30	0.32	0.35

SAMPLE DATA
DEPTH/ORDER: 1050m
SAMPLE-ID: Balder
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 16-312
APT-ID.: 0

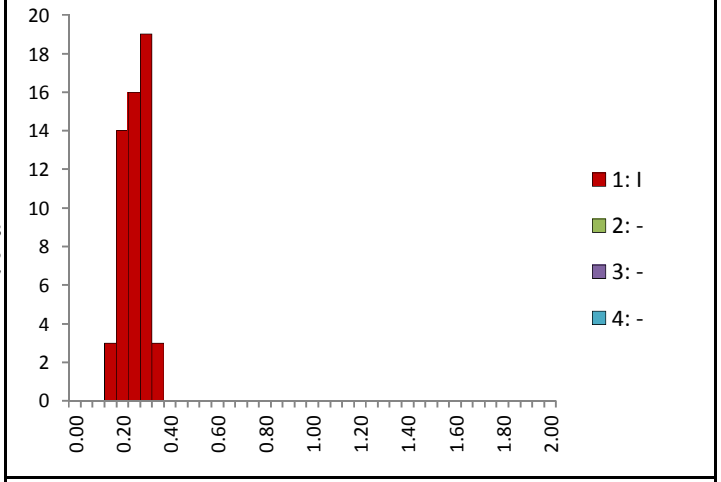
STATISTICS
POPN 1 - Indigenous
Mean value: 0.30
No. of values: 40
Range: 0.20 to 0.37
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.20	0.25	0.27	0.29	0.30	0.32	0.32	0.34
0.21	0.25	0.28	0.29	0.31	0.32	0.33	0.35
0.21	0.26	0.29	0.30	0.31	0.32	0.33	0.36
0.22	0.27	0.29	0.30	0.31	0.32	0.33	0.36
0.25	0.27	0.29	0.30	0.31	0.32	0.34	0.37

SAMPLE DATA
DEPTH/ORDER: 1150m
SAMPLE-ID: Late Maast.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-697
APT-ID.: 0

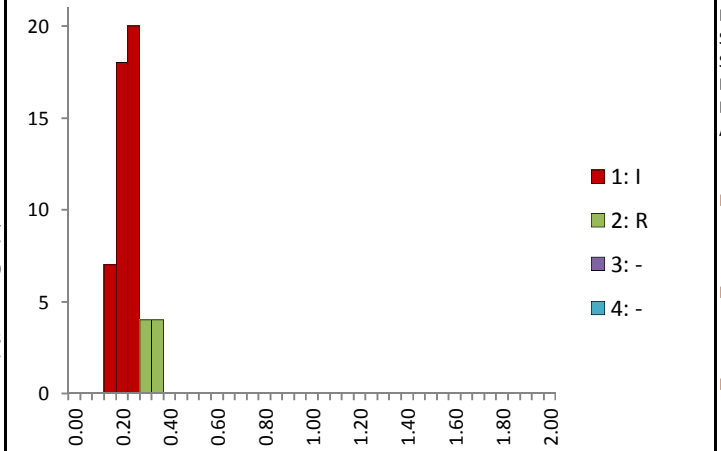
STATISTICS
POPN 1 - Indigenous
Mean value: 0.28
No. of values: 55
Range: 0.18 to 0.37
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.18	0.21	0.23	0.23	0.26	0.28	0.29	0.30	0.31	0.32	0.34
0.19	0.21	0.23	0.24	0.27	0.28	0.29	0.30	0.32	0.32	0.34
0.19	0.22	0.23	0.25	0.27	0.28	0.29	0.30	0.32	0.33	0.35
0.20	0.22	0.23	0.25	0.28	0.28	0.30	0.31	0.32	0.33	0.35
0.20	0.22	0.23	0.26	0.28	0.28	0.30	0.31	0.32	0.34	0.37

SAMPLE DATA
DEPTH/ORDER: 1100m
SAMPLE-ID: Late Maast.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-696
APT-ID.: 0

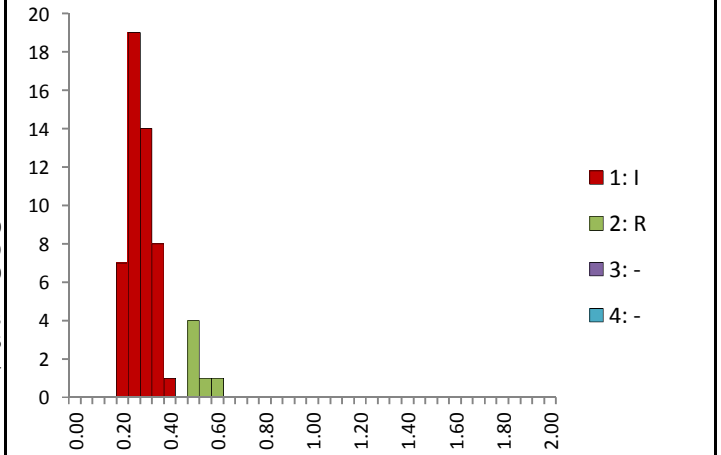
STATISTICS
POPN 1 - Indigenous
Mean value: 0.24
No. of values: 47
Range: 0.17 to 0.30
POPN 2 - Reworked
Mean value: 0.35
No. of values: 8
Range: 0.33 to 0.37
POPN 3 - n/a
POPN 4 - n/a



0.17	0.19	0.21	0.21	0.23	0.25	0.25	0.26	0.27	0.30	0.34
0.18	0.19	0.21	0.21	0.23	0.25	0.26	0.26	0.27	0.30	0.35
0.18	0.20	0.21	0.22	0.23	0.25	0.26	0.27	0.27	0.33	0.36
0.19	0.20	0.21	0.22	0.24	0.25	0.26	0.27	0.28	0.34	0.37
0.19	0.20	0.21	0.23	0.24	0.25	0.26	0.27	0.29	0.34	0.37

SAMPLE DATA
DEPTH/ORDER: 1210m
SAMPLE-ID: Late Maast.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-698
APT-ID.: 0

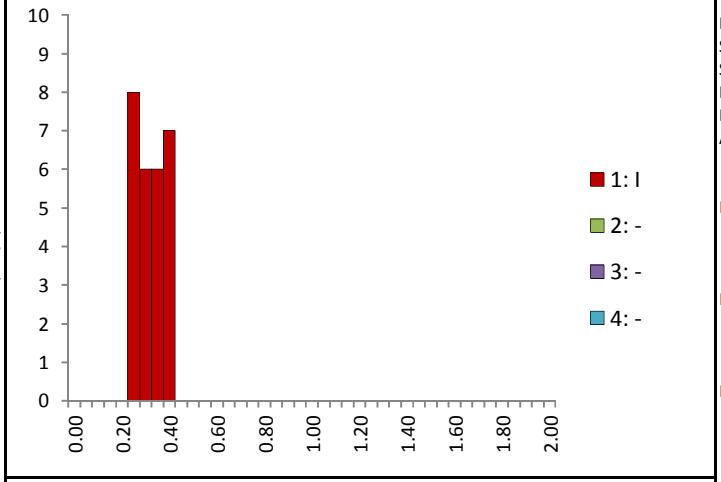
STATISTICS
POPN 1 - Indigenous
Mean value: 0.30
No. of values: 49
Range: 0.19 to 0.40
POPN 2 - Reworked
Mean value: 0.53
No. of values: 6
Range: 0.50 to 0.64
POPN 3 - n/a
POPN 4 - n/a



0.22	0.24	0.26	0.27	0.28	0.29	0.31	0.32	0.35	0.38	0.50
0.23	0.24	0.26	0.28	0.29	0.30	0.32	0.33	0.35	0.38	0.50
0.23	0.25	0.27	0.28	0.29	0.31	0.32	0.33	0.36	0.39	0.51
0.23	0.25	0.27	0.28	0.29	0.31	0.32	0.33	0.36	0.40	0.55
0.24	0.25	0.27	0.28	0.29	0.31	0.32	0.34	0.37	0.50	0.64

SAMPLE DATA
DEPTH/ORDER: 1250m
SAMPLE-ID: Late Maast.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-699
APT-ID.: 0

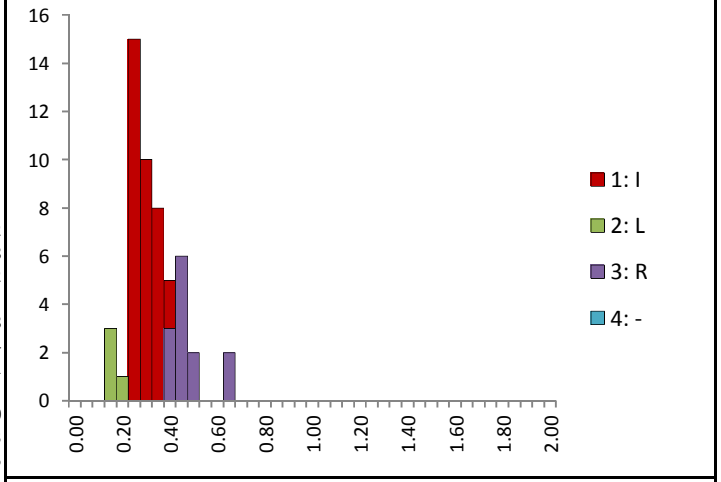
STATISTICS
POPN 1 - Indigenous
Mean value: 0.34
No. of values: 27
Range: 0.25 to 0.44
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.25	0.27	0.31	0.35	0.40	0.44	
0.26	0.28	0.32	0.36	0.42	0.44	
0.26	0.29	0.32	0.37	0.43		
0.27	0.30	0.34	0.38	0.43		
0.27	0.31	0.35	0.38	0.43		

SAMPLE DATA
DEPTH/ORDER: 1360m
SAMPLE-ID: Ear. Maast.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-701
APT-ID.: 0

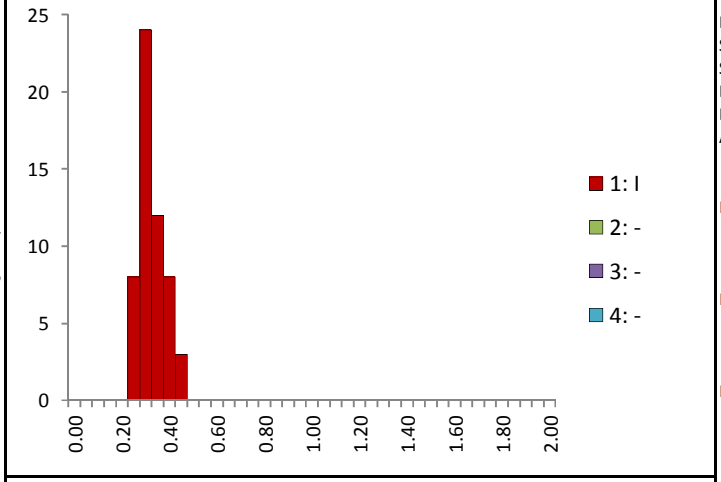
STATISTICS
POPN 1 - Indigenous
Mean value: 0.32
No. of values: 38
Range: 0.25 to 0.42
POPN 2 - Low reflecting
Mean value: 0.18
No. of values: 4
Range: 0.16 to 0.21
POPN 3 - Reworked
Mean value: 0.50
No. of values: 13
Range: 0.44 to 0.65
POPN 4 - n/a



0.16	0.25	0.26	0.28	0.30	0.31	0.35	0.39	0.41	0.45	0.48
0.16	0.25	0.27	0.29	0.30	0.31	0.36	0.39	0.42	0.45	0.51
0.19	0.26	0.28	0.29	0.30	0.31	0.36	0.40	0.44	0.46	0.52
0.21	0.26	0.28	0.29	0.30	0.34	0.37	0.40	0.44	0.47	0.65
0.25	0.26	0.28	0.30	0.31	0.35	0.39	0.40	0.44	0.48	0.65

SAMPLE DATA
DEPTH/ORDER: 1300m
SAMPLE-ID: Late Maast.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-700
APT-ID.: 0

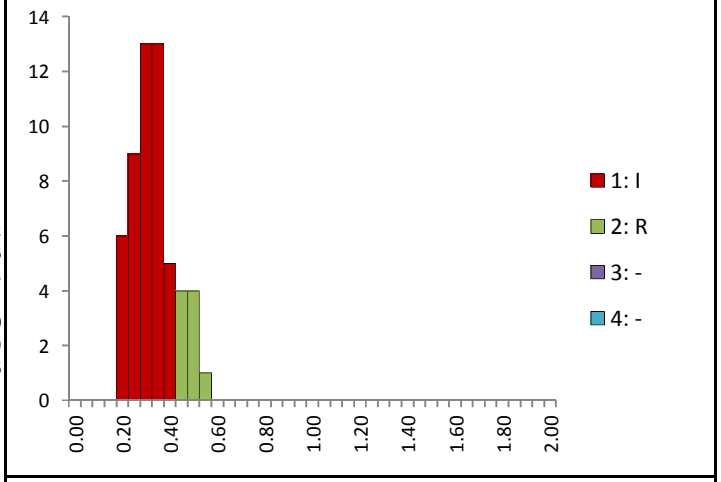
STATISTICS
POPN 1 - Indigenous
Mean value: 0.34
No. of values: 55
Range: 0.26 to 0.46
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.26	0.29	0.30	0.31	0.32	0.33	0.34	0.35	0.37	0.40	0.44
0.27	0.29	0.30	0.31	0.32	0.33	0.34	0.36	0.37	0.40	0.44
0.27	0.29	0.31	0.31	0.32	0.33	0.35	0.36	0.38	0.41	0.45
0.27	0.30	0.31	0.32	0.33	0.33	0.35	0.36	0.38	0.41	0.45
0.28	0.30	0.31	0.32	0.33	0.33	0.35	0.36	0.40	0.43	0.46

SAMPLE DATA
DEPTH/ORDER: 1400m
SAMPLE-ID: Ear. Maast.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-702
APT-ID.: 0

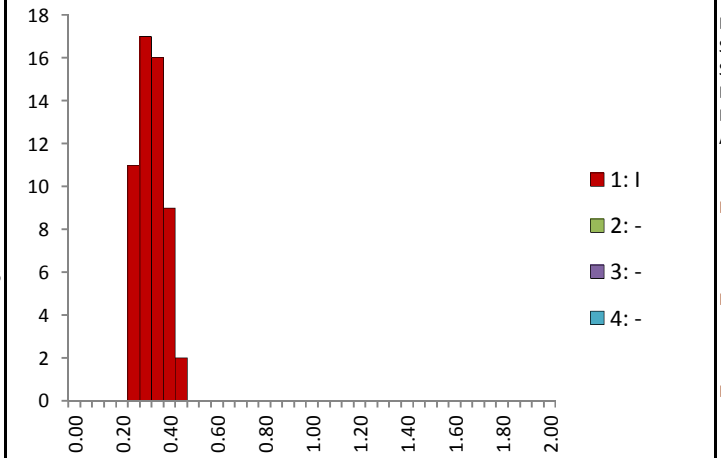
STATISTICS
POPN 1 - Indigenous
Mean value: 0.32
No. of values: 46
Range: 0.23 to 0.43
POPN 2 - Reworked
Mean value: 0.50
No. of values: 9
Range: 0.46 to 0.56
POPN 3 - n/a
POPN 4 - n/a



0.23	0.24	0.27	0.30	0.32	0.34	0.35	0.37	0.39	0.43	0.50
0.23	0.25	0.28	0.30	0.32	0.34	0.35	0.37	0.40	0.46	0.52
0.23	0.26	0.29	0.30	0.32	0.34	0.35	0.38	0.41	0.46	0.53
0.24	0.27	0.29	0.31	0.33	0.35	0.36	0.38	0.41	0.46	0.54
0.24	0.27	0.29	0.31	0.33	0.35	0.36	0.39	0.41	0.49	0.56

SAMPLE DATA
DEPTH/ORDER: 1450m
SAMPLE-ID: Ear. Maast.
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-703
APT-ID.: 0

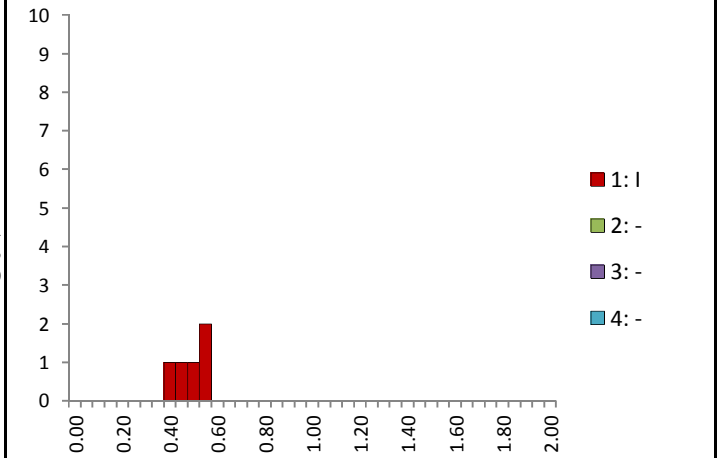
STATISTICS
POPN 1 - Indigenous
Mean value: 0.35
No. of values: 55
Range: 0.24 to 0.46
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.27	0.29	0.29	0.30	0.32	0.33	0.36	0.37	0.39	0.41	0.43
0.28	0.29	0.30	0.30	0.32	0.34	0.36	0.37	0.39	0.42	0.44
0.28	0.29	0.30	0.31	0.32	0.34	0.36	0.37	0.39	0.43	0.44
0.28	0.29	0.30	0.31	0.33	0.35	0.36	0.37	0.39	0.43	0.45
0.28	0.29	0.30	0.31	0.33	0.36	0.36	0.38	0.40	0.43	0.46

SAMPLE DATA
DEPTH/ORDER: 1550m
SAMPLE-ID: Campanian
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-705
APT-ID.: 0

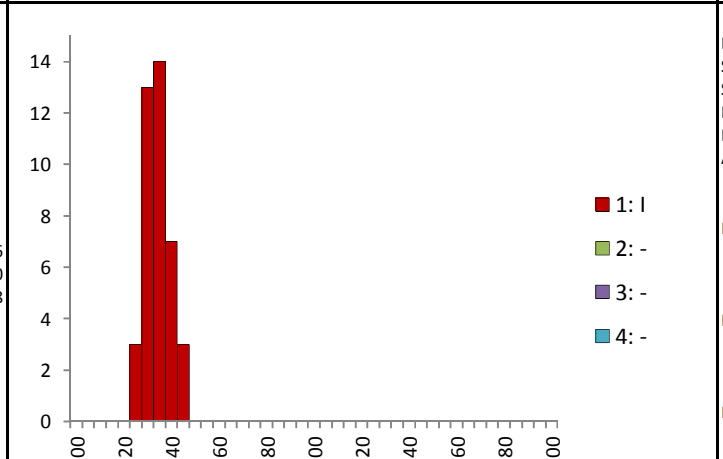
STATISTICS
POPN 1 - Indigenous
Mean value: 0.51
No. of values: 5
Range: 0.42 to 0.59
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.42
0.49
0.52
0.55
0.59

SAMPLE DATA
DEPTH/ORDER: 1500m
SAMPLE-ID: Ear. Maast.
SAMPLE TYPE: DC
BLOCK TYPE.: 0
BLOCK No..: 15-704
APT-ID.: 0

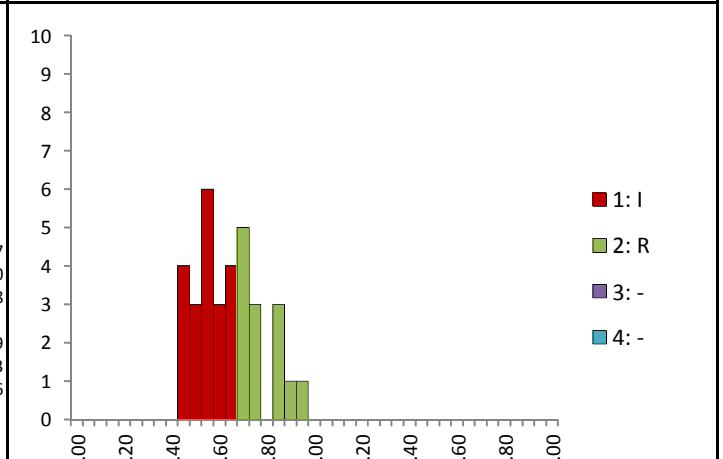
STATISTICS
POPN 1 - Indigenous
Mean value: 0.36
No. of values: 40
Range: 0.25 to 0.48
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.25	0.32	0.33	0.34	0.37	0.38	0.40	0.42
0.28	0.32	0.33	0.35	0.37	0.38	0.40	0.44
0.29	0.32	0.33	0.35	0.37	0.38	0.40	0.46
0.30	0.32	0.34	0.36	0.38	0.39	0.40	0.46
0.32	0.32	0.34	0.36	0.38	0.39	0.41	0.48

SAMPLE DATA
DEPTH/ORDER: 1600m
SAMPLE-ID: Campanian
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-706
APT-ID.: 0

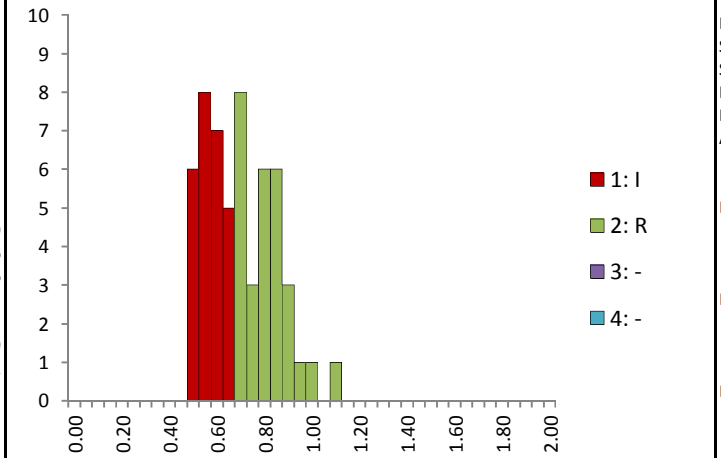
STATISTICS
POPN 1 - Indigenous
Mean value: 0.57
No. of values: 20
Range: 0.46 to 0.68
POPN 2 - Reworked
Mean value: 0.79
No. of values: 13
Range: 0.70 to 0.96
POPN 3 - n/a
POPN 4 - n/a



0.46	0.52	0.57	0.64	0.70	0.75	0.88
0.46	0.54	0.57	0.66	0.72	0.76	0.90
0.48	0.55	0.58	0.67	0.72	0.76	0.96
0.49	0.55	0.60	0.67	0.73	0.85	
0.51	0.55	0.63	0.68	0.73	0.87	

SAMPLE DATA
DEPTH/ORDER: 1650m
SAMPLE-ID: Campanian
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-707
APT-ID.: 0

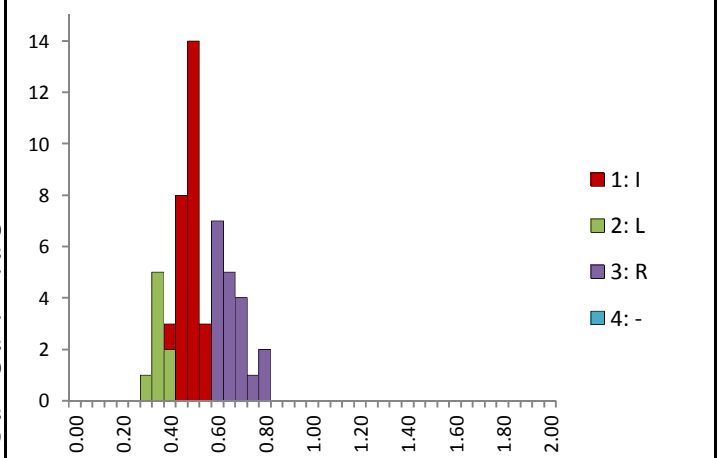
STATISTICS
POPN 1 - Indigenous
Mean value: 0.59
No. of values: 26
Range: 0.50 to 0.66
POPN 2 - Reworked
Mean value: 0.83
No. of values: 29
Range: 0.70 to 1.11
POPN 3 - n/a
POPN 4 - n/a



0.50	0.54	0.57	0.60	0.64	0.66	0.73	0.77	0.81	0.85	0.92
0.50	0.55	0.58	0.62	0.65	0.70	0.73	0.79	0.82	0.87	0.93
0.51	0.55	0.59	0.63	0.65	0.71	0.73	0.80	0.83	0.88	0.95
0.52	0.55	0.59	0.63	0.65	0.71	0.74	0.80	0.85	0.89	1.03
0.53	0.56	0.60	0.64	0.65	0.72	0.77	0.81	0.85	0.91	1.11

SAMPLE DATA
DEPTH/ORDER: 1749m
SAMPLE-ID: Santonian
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-709
APT-ID.: 0

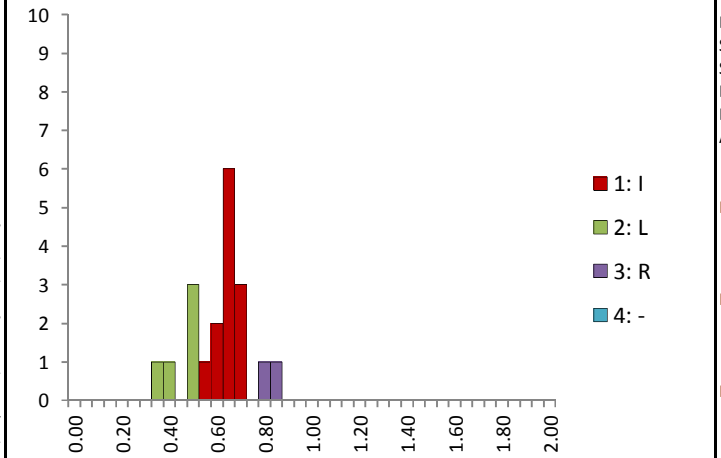
STATISTICS
POPN 1 - Indigenous
Mean value: 0.50
No. of values: 28
Range: 0.42 to 0.57
POPN 2 - Low reflecting
Mean value: 0.37
No. of values: 8
Range: 0.34 to 0.40
POPN 3 - Reworked
Mean value: 0.68
No. of values: 19
Range: 0.60 to 0.84
POPN 4 - n/a



0.34	0.39	0.44	0.47	0.50	0.51	0.53	0.57	0.63	0.65	0.73
0.35	0.40	0.45	0.47	0.50	0.52	0.53	0.60	0.63	0.67	0.74
0.35	0.40	0.46	0.48	0.50	0.52	0.54	0.61	0.64	0.69	0.77
0.37	0.42	0.47	0.49	0.51	0.52	0.56	0.61	0.65	0.72	0.81
0.37	0.42	0.47	0.50	0.51	0.52	0.56	0.62	0.65	0.72	0.84

SAMPLE DATA
DEPTH/ORDER: 1700m
SAMPLE-ID: Santonian
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-708
APT-ID.: 0

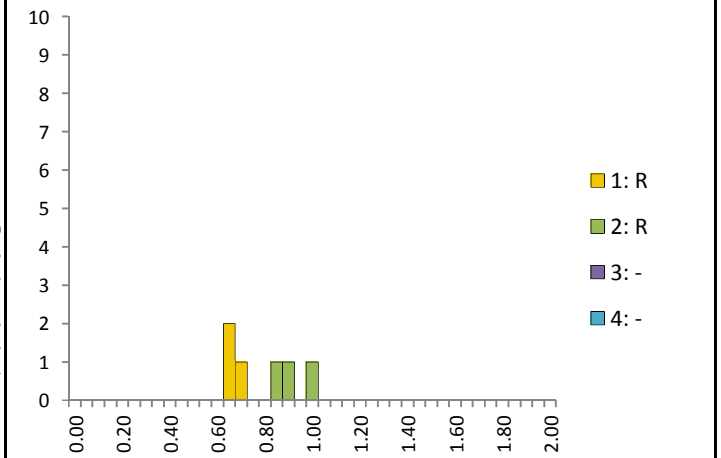
STATISTICS
POPN 1 - Indigenous
Mean value: 0.67
No. of values: 12
Range: 0.58 to 0.72
POPN 2 - Low reflecting
Mean value: 0.47
No. of values: 5
Range: 0.39 to 0.52
POPN 3 - Reworked
Mean value: 0.84
No. of values: 2
Range: 0.82 to 0.85
POPN 4 - n/a



0.39	0.58	0.69	0.72
0.43	0.61	0.69	0.72
0.50	0.62	0.69	0.82
0.52	0.65	0.69	0.85
0.52	0.66	0.71	

SAMPLE DATA
DEPTH/ORDER: 1806m
SAMPLE-ID: Coniacian
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-710
APT-ID.: 0

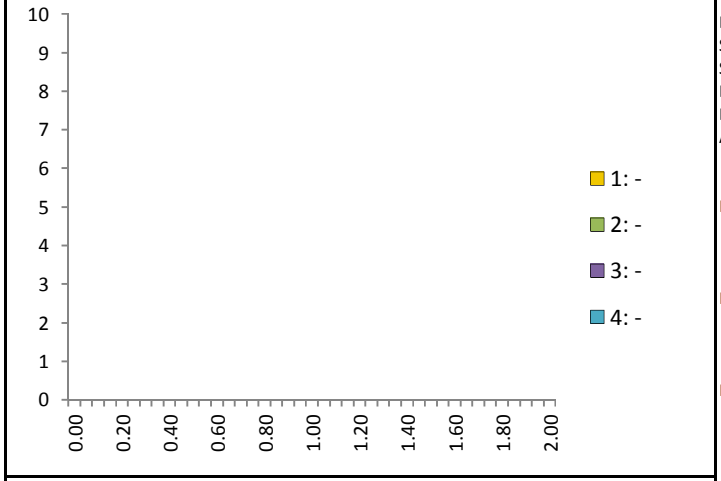
STATISTICS
POPN 1 - Reworked
Mean value: 0.69
No. of values: 3
Range: 0.65 to 0.73
POPN 2 - Reworked
Mean value: 0.93
No. of values: 3
Range: 0.87 to 1.01
POPN 3 - n/a
POPN 4 - n/a



0.65	1.01
0.68	
0.73	
0.87	
0.91	

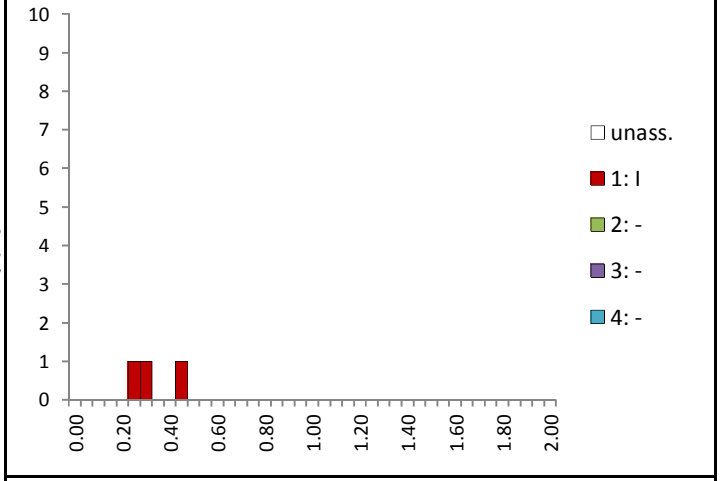
SAMPLE DATA
DEPTH/ORDER: 1884.58m
SAMPLE-ID: Turonian
SAMPLE TYPE: SWC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-284
APT-ID.: 0

STATISTICS
POPN 1 - n/a
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



SAMPLE DATA
DEPTH/ORDER: 1902.7m
SAMPLE-ID: Turonian
SAMPLE TYPE: SWC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-286
APT-ID.: 0

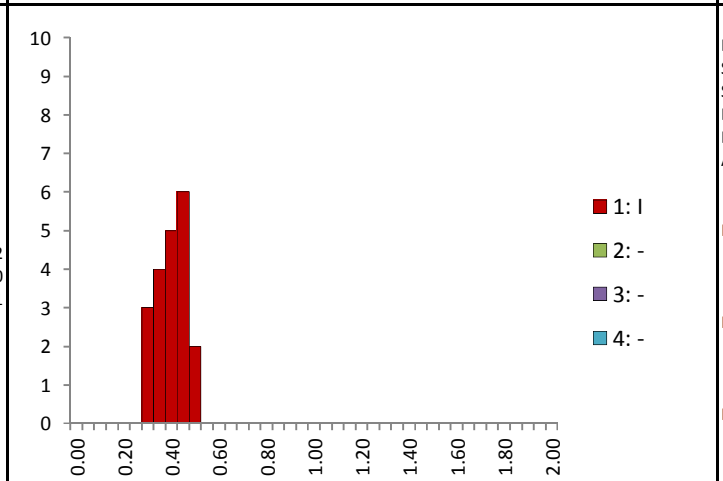
STATISTICS
POPN 1 - Indigenous
Mean value: 0.35
No. of values: 3
Range: 0.27 to 0.47
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.27
0.32
0.47

SAMPLE DATA
DEPTH/ORDER: 1892.66m
SAMPLE-ID: Turonian
SAMPLE TYPE: SWC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-285
APT-ID.: 0

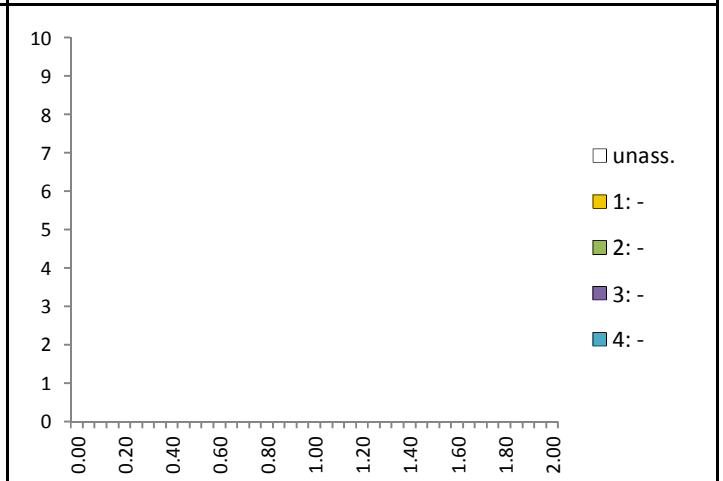
STATISTICS
POPN 1 - Indigenous
Mean value: 0.42
No. of values: 20
Range: 0.33 to 0.51
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.33 0.39 0.43 0.47
0.33 0.39 0.44 0.48
0.34 0.41 0.45 0.48
0.35 0.42 0.46 0.50
0.37 0.43 0.47 0.51

SAMPLE DATA
DEPTH/ORDER: 1911.17m
SAMPLE-ID: Turonian
SAMPLE TYPE: SWC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-287
APT-ID.: 0

STATISTICS
POPN 1 - n/a
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



APPENDIX 1.1 Histograms, data and statistics for vitrinite reflectivity: 206/12a-5

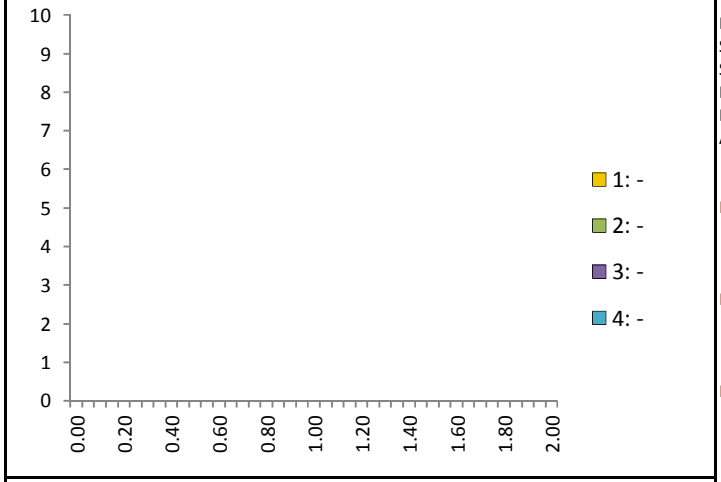
SAMPLE DATA
DEPTH/ORDER: 1928.61m
SAMPLE-ID: Cret K20
SAMPLE TYPE: COCH
BLOCK TYPE.: Kerogen
BLOCK No..: 15-239
APT-ID.: 0

STATISTICS
POPN 1 - n/a

POPN 2 - n/a

POPN 3 - n/a

POPN 4 - n/a



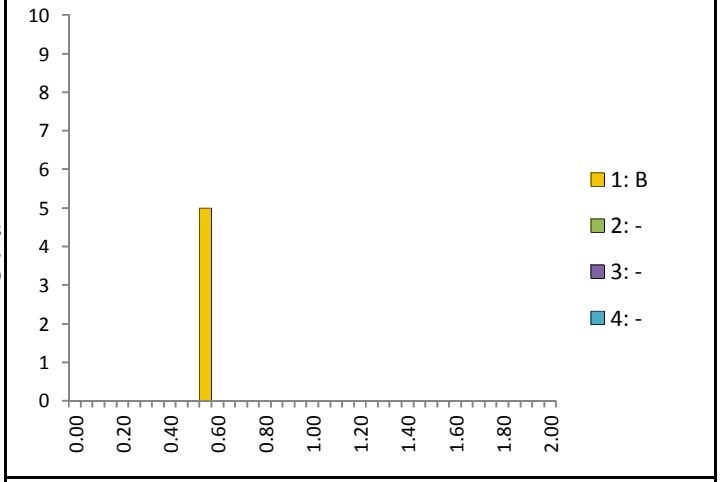
SAMPLE DATA
DEPTH/ORDER: 1944.05m
SAMPLE-ID: Rona
SAMPLE TYPE: COCH
BLOCK TYPE.: Whole rock
BLOCK No..: 15-316
APT-ID.: 0

STATISTICS
POPN 1 - Bitumen
Mean value: 0.58
No. of values: 5
Range: 0.56 to 0.59

POPN 2 - n/a

POPN 3 - n/a

POPN 4 - n/a



0.56 0.57 0.58 0.59 0.59

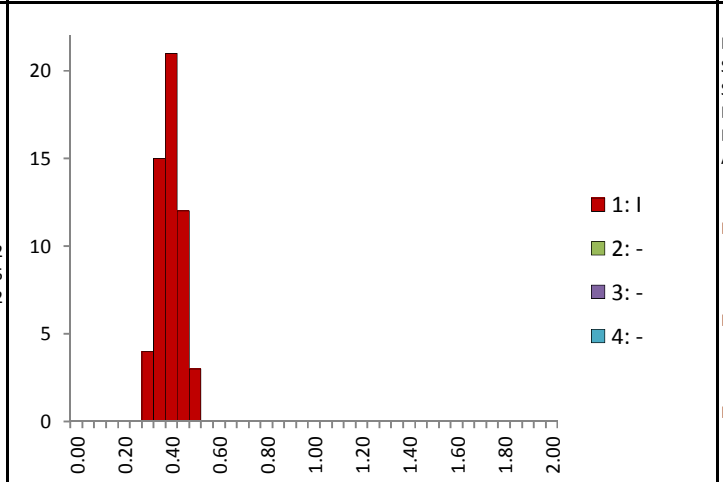
SAMPLE DATA
DEPTH/ORDER: 1932.67m
SAMPLE-ID: KCF
SAMPLE TYPE: COCH
BLOCK TYPE.: Kerogen
BLOCK No..: 15-240
APT-ID.: 0

STATISTICS
POPN 1 - Indigenous
Mean value: 0.42
No. of values: 55
Range: 0.34 to 0.52

POPN 2 - n/a

POPN 3 - n/a

POPN 4 - n/a



0.34	0.36	0.37	0.39	0.40	0.41	0.41	0.44	0.45	0.47	0.49
0.34	0.36	0.38	0.39	0.40	0.41	0.42	0.44	0.45	0.47	0.49
0.34	0.36	0.38	0.39	0.40	0.41	0.42	0.44	0.45	0.48	0.50
0.34	0.37	0.38	0.39	0.40	0.41	0.43	0.44	0.46	0.48	0.51
0.35	0.37	0.38	0.40	0.40	0.41	0.43	0.44	0.46	0.49	0.52

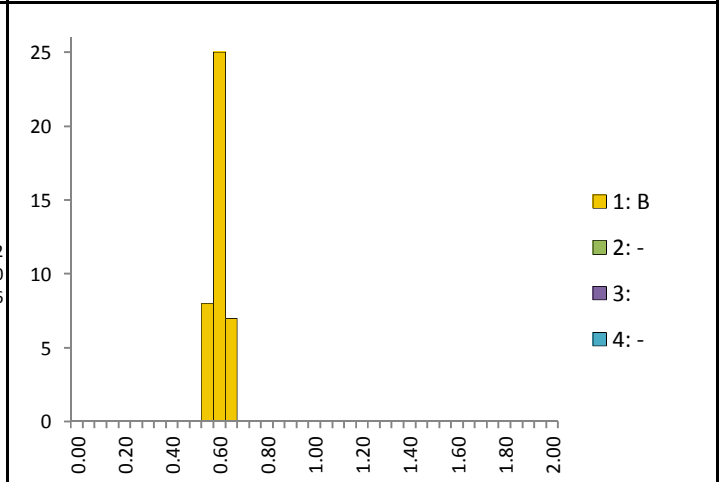
SAMPLE DATA
DEPTH/ORDER: 1948.68m
SAMPLE-ID: Rona
SAMPLE TYPE: COCH
BLOCK TYPE.: Whole rock
BLOCK No..: 15-317
APT-ID.: 0

STATISTICS
POPN 1 - Bitumen
Mean value: 0.62
No. of values: 40
Range: 0.57 to 0.66

POPN 2 - n/a

POPN 3 - n/a

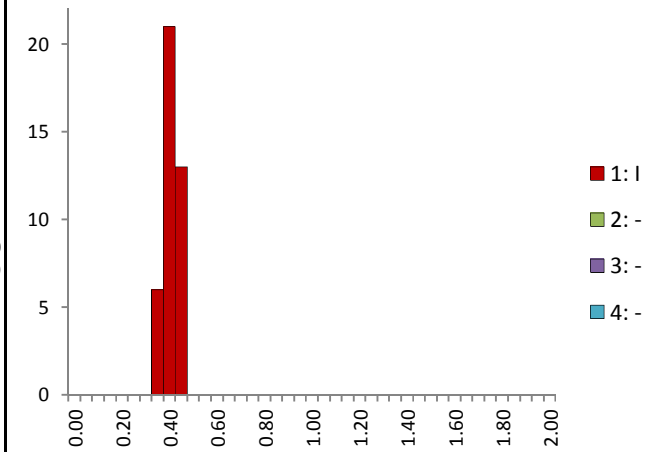
POPN 4 - n/a



0.57	0.59	0.60	0.62	0.62	0.63	0.64	0.65
0.57	0.59	0.61	0.62	0.62	0.63	0.64	0.65
0.57	0.59	0.61	0.62	0.62	0.63	0.64	0.65
0.58	0.60	0.61	0.62	0.63	0.63	0.65	0.65
0.58	0.60	0.62	0.62	0.63	0.63	0.65	0.66

SAMPLE DATA
DEPTH/ORDER: 1960.35m
SAMPLE-ID: Rona
SAMPLE TYPE: COCH
BLOCK TYPE.: Whole rock
BLOCK No..: 15-318
APT-ID.: 0

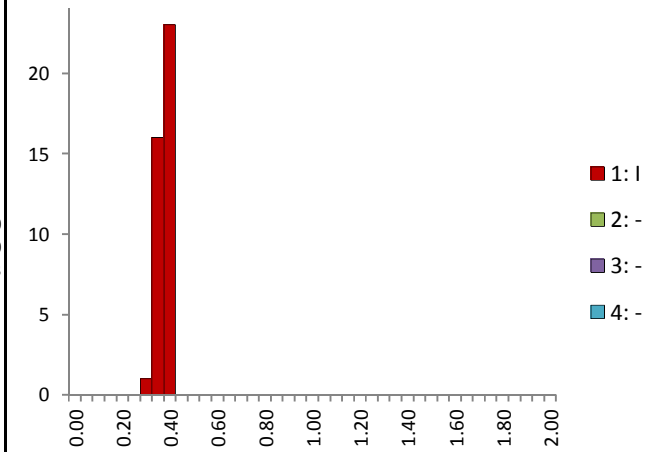
STATISTICS
POPN 1 - Indigenous
Mean value: 0.43
No. of values: 40
Range: 0.38 to 0.49
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.38	0.39	0.42	0.43	0.44	0.44	0.45	0.46
0.38	0.40	0.42	0.44	0.44	0.44	0.46	0.47
0.38	0.40	0.43	0.44	0.44	0.45	0.46	0.47
0.39	0.41	0.43	0.44	0.44	0.45	0.46	0.48
0.39	0.42	0.43	0.44	0.44	0.45	0.46	0.49

SAMPLE DATA
DEPTH/ORDER: 1967.35m
SAMPLE-ID: Rona
SAMPLE TYPE: COCH
BLOCK TYPE.: Whole rock
BLOCK No..: 15-319
APT-ID.: 0

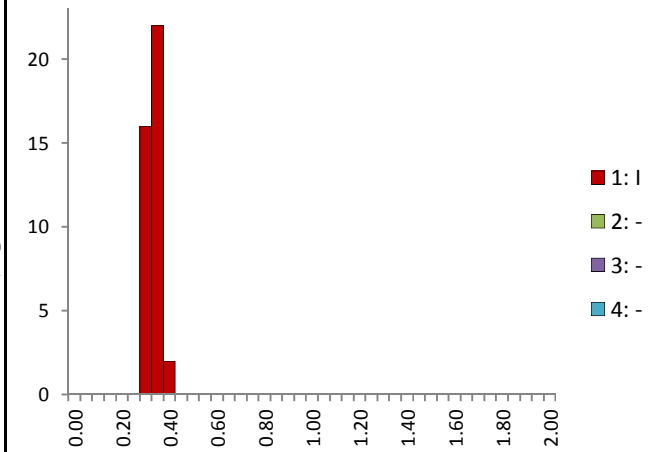
STATISTICS
POPN 1 - Indigenous
Mean value: 0.40
No. of values: 40
Range: 0.34 to 0.43
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.34	0.37	0.38	0.39	0.40	0.41	0.41	0.42
0.36	0.37	0.39	0.39	0.40	0.41	0.41	0.42
0.36	0.38	0.39	0.40	0.40	0.41	0.41	0.43
0.37	0.38	0.39	0.40	0.40	0.41	0.42	0.43
0.37	0.38	0.39	0.40	0.40	0.41	0.42	0.43

SAMPLE DATA
DEPTH/ORDER: 1966.7m
SAMPLE-ID: Rona
SAMPLE TYPE: COCH
BLOCK TYPE.: Whole rock
BLOCK No..: 15-320
APT-ID.: 0

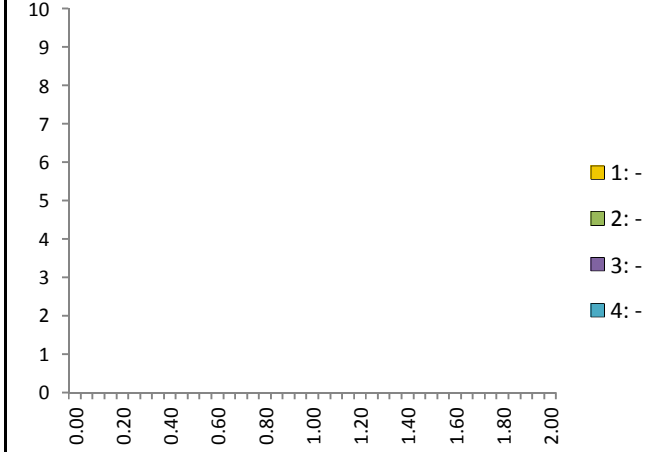
STATISTICS
POPN 1 - Indigenous
Mean value: 0.35
No. of values: 40
Range: 0.32 to 0.41
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



0.32	0.33	0.34	0.34	0.35	0.36	0.36	0.38
0.32	0.33	0.34	0.35	0.35	0.36	0.36	0.38
0.32	0.34	0.34	0.35	0.35	0.36	0.37	0.39
0.33	0.34	0.34	0.35	0.35	0.36	0.37	0.40
0.33	0.34	0.34	0.35	0.35	0.36	0.37	0.41

SAMPLE DATA
DEPTH/ORDER: 1984m
SAMPLE-ID: Lwr. Clair Gp.
SAMPLE TYPE: COCH
BLOCK TYPE.: Kerogen
BLOCK No..: 15-711
APT-ID.: 0

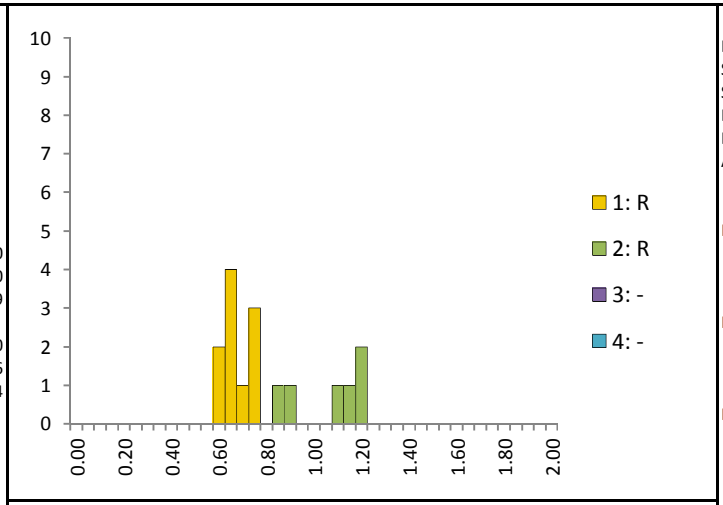
STATISTICS
POPN 1 - n/a
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



APPENDIX 1.1 Histograms, data and statistics for vitrinite reflectivity: 206/12a-5

SAMPLE DATA
DEPTH/ORDER: 2001m
SAMPLE-ID: Lwr. Clair Gp.
SAMPLE TYPE: COCH
BLOCK TYPE.: Whole rock
BLOCK No..: 15-315
APT-ID.: 0

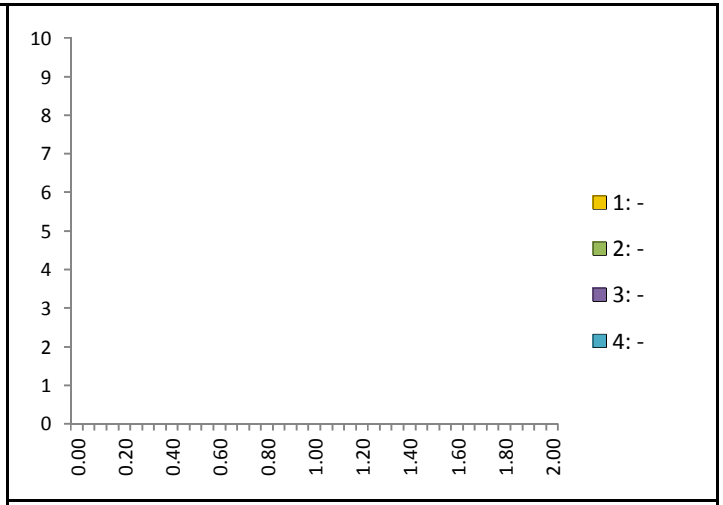
STATISTICS
POPN 1 - Reworked
Mean value: 0.70
No. of values: 10
Range: 0.61 to 0.79
POPN 2 - Reworked
Mean value: 1.10
No. of values: 6
Range: 0.88 to 1.24
POPN 3 - n/a
POPN 4 - n/a



0.61	0.69	0.88	1.24
0.63	0.71	0.93	
0.66	0.77	1.14	
0.67	0.79	1.17	
0.67	0.79	1.22	

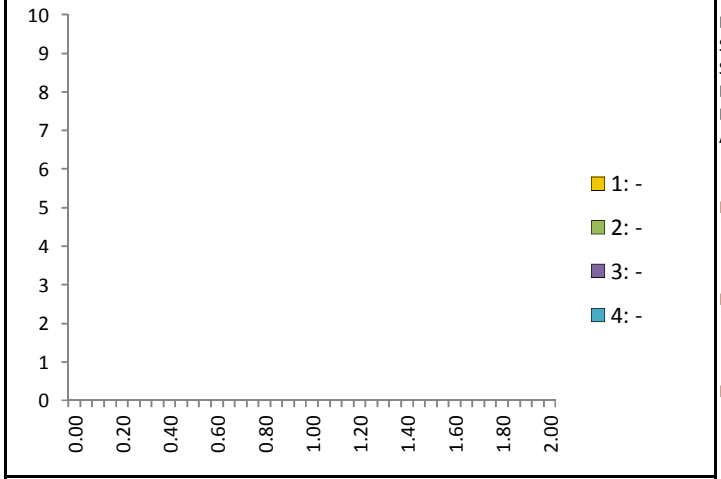
SAMPLE DATA
DEPTH/ORDER: 2048m
SAMPLE-ID: Lwr. Clair Gp.
SAMPLE TYPE: COCH
BLOCK TYPE.: Kerogen
BLOCK No..: 15-713
APT-ID.: 0

STATISTICS
POPN 1 - n/a
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



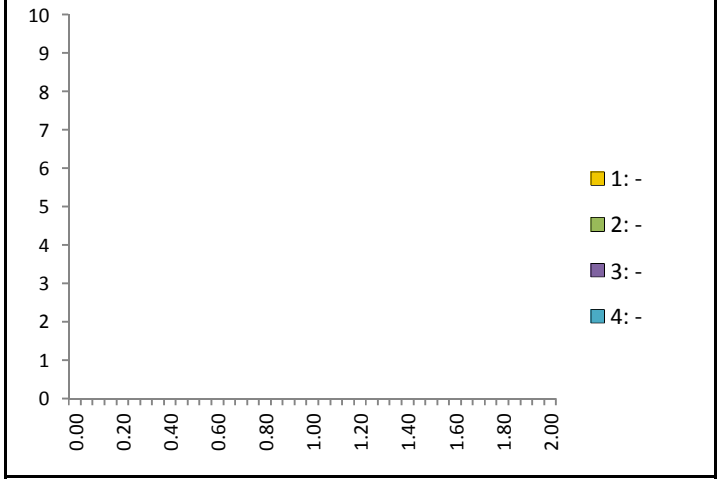
SAMPLE DATA
DEPTH/ORDER: 2022m
SAMPLE-ID: Lwr. Clair Gp.
SAMPLE TYPE: COCH
BLOCK TYPE.: Kerogen
BLOCK No..: 15-712
APT-ID.: 0

STATISTICS
POPN 1 - n/a
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



SAMPLE DATA
DEPTH/ORDER: 2202m
SAMPLE-ID: Basal
SAMPLE TYPE: DC
BLOCK TYPE.: Kerogen
BLOCK No..: 15-714
APT-ID.: 0

STATISTICS
POPN 1 - n/a
POPN 2 - n/a
POPN 3 - n/a
POPN 4 - n/a



APPENDIX 2
SPORE COLOUR AND KEROGEN COMPOSITION DATA

Tables

Ap2.1 Spore colour and kerogen composition – interpreted data
Ap2.2 Spore colour index and kerogen composition – microscopists data sheets

Well	MD (m)	TVDSS (m)	Sample type	Spore Colour Index (SCI)					Six Component Composition (%)						Comments
				Indigenous		Other populations									
				SCI	No.	SCI	No.	Type	FA	HA	AL	HE	WO	CO	
206/12a-5	270	246.5	DC	2.0	7				-	10	-	tr	80	10	Very low recovery, humic additive (?) abundant.
	360	336.5	DC	2.0	3	8.0	2	R	80	-	-	tr	10	10	Reworked miospores and woody material common.
	480	456.5	DC	2.5	23	6.0	3	R	30	-	tr	tr	60	10	Most of the darker miospores are stained. One Classopollis suggests Jurassic reworking and an SCI of 3.5 sets an upper SCI limit.
	540	516.5	DC	3.0	14				mnr	mnr	-	tr	95	5	Should be good for VR.
	570	546.5	DC	3.0	20				10	-	-	5	75	10	Mainly very large translucent, fibrous woody OM. Should be good for VR.
	720	696.5	DC	3.5	9				-	75	-	5	10	10	OM degraded with orange-brown stain. Sapropel, humics & pyrite. Minor fibrous woody material. Herbaceous material mostly cuticle.
	760	736.5	DC	3.0	19	4.0-8.0	5	R	-	80	-	mnr	10	10	More marine than above. Reworked Classopollis at 3.5 sets an upper limit for SCI. Callialasporites (Jurassic) at 6.5.
	1050	1026.5	DC	3.0	26	4.5	2	R	-	15	-	tr	70	15	Callialasporites (Jurassic) at 4.0. Exinite fluorescence unusually weak given the low SCI. Should be good for VR.
	1100	1076.5	DC	2.5	16	6.0	6	R	-	10	5	5	5	75	Mostly terrestrial organic matter, dinos are present. Whole kerogen fraction is dominated by opaque, equidimensional
	1150	1126.5	DC	3.0	28	5.0	4	R	-	30	5	5	5	55	As above, but preservation is not as good and miospores are darker.
	1210	1186.4	DC	3.5	24	5.0	8	R	-	70	5	tr	tr	25	Abundant opaques with secondary woody debris, but dinocysts are more abundant than miospores. Aquillapollenites at 3.0-3.5.
	1250	1226.4	DC	3.5	18	6.0	7	R	-	-	-	-	-	-	Low recovery. Organic matter darker - possibly due to oil or mud staining.
	1300	1276.4	DC	3.0	21	6.0	4	R	-	60	5	mnr	5	30	Aquillapollenites at 4.5 noted.
	1360	1336.4	DC	3.5	19	7.0	3	R	-	55	5	5	5	30	2-2.5 SCI miospores poorly preserved and thinned. Vitrinite is more abundant than above in the sieved fraction.

TABLE Ap2.1 Spore colour and kerogen composition - interpreted data

Well	MD (m)	TVDSS (m)	Sample type	Spore Colour Index (SCI)					Six Component Composition (%)						Comments
				Indigenous		Other populations									
				SCI	No.	SCI	No.	Type	FA	HA	AL	HE	WO	CO	
206/12a-5	1400	1376.4	DC	3.5	21	7.5	2	R	-	75	mnr	mnr	5	20	
	1450	1426.4	DC	4.0	17	6.5	2	R	-	90	tr	tr	mnr	10	
	1500	1476.4	DC	3.5	21	6.0	2	R	-	85	mnr	5	5	5	
	1550	1526.4	DC	4.0	8	2.5 6.5	1 10	C R	-	90	mnr	mnr	mnr	10	Low recovery. Organic matter generally fairly dark. Aquillapollenites present. Evidence of localised heating event.
	1600	1576.4	DC	4.5	7	6.5	6	R	-	70	tr	tr	tr	30	Little sieved material recovered.
	1650	1626.4	DC	4.0	6	2.5 6.5	3 11	C R	-	70	tr	tr	tr	30	Wide range of SCI from *in situ* spores and dinocysts suggests a highly localised heating event.
	1700	1676.4	DC	-	-	6.5	3	R	-	80	tr	tr	-	20	Mostly ragged brown FDOM. Palynomorphs rare nothing suggest SCI lower than 5.0. Palynomorph identification uncertain.
	1749	1725.4	DC	4.5	25				-	80	5	mnr	-	15	Abundant opaques and common well-preserved dinocysts.
	1806	1782.4	DC	-	-	-	-	-	-	60	tr	-	-	40	Low recovery - mostly a mix of opaques and ragged brown finely disseminated organic matter.
	1884.58	1860.9	COCH	3.5	14	5.5	5	R	-	15	10	tr	5	70	Sieved fraction nearly barren. Mostly small blocky opaques. A reworked Carboniferous *Densosporites* spp. present.
	1892.66	1869.0	COCH	3.0	18	6.0	4	R	-	65	5	tr	10	20	Dinos more common than miospores
	1902.70	1879.0	COCH	4.0	6	7.0	2	R	-	50	10	tr	-	40	Some staining. Abundant dinos in sieved fraction. Reworked Permo-Triassic striate bisaccate present.
	1911.17	1887.5	COCH	3.0	1				-	65	15	-	-	20	Dominated by pyrite rich non fluorescing AOM.
	1928.61	1905.0	COCH	-	-	-	-	-	-	95	-	mnr	5	-	Barren sample.

TABLE Ap2.1 Spore colour and kerogen composition - interpreted data

Well	MD (m)	TVDSS (m)	Sample type	Spore Colour Index (SCI)					Six Component Composition (%)						Comments
				Indigenous		Other populations									
				SCI	No.	SCI	No.	Type	FA	HA	AL	HE	WO	CO	
206/12a-5	1932.67	1909.0	COCH	4.5	2				-	90	5	mnr	mnr	5	Few miospores and algae present. Possible exine (tissue) present at 4.5. Inertinite laths are common.
	1984.00	1960.3	COCH	-	-	-	-	-	-	tr	-	-	-	-	Nearly barren, with only traces of dark brown finely disseminated organic matter.
	2022.00	1998.3	COCH	-	-	-	-	-	-	-	-	-	-	tr	Nearly barren. Only trace amounts of small opaques.
	2048.00	2024.3	COCH	-	-	-	-	-	-	-	-	-	-	-	Low recovery - mostly light coloured plant tissue and light orange AOM, most of which is probably drilling additive.
	2202.0	2176.2	DC	-	-	-	-	-	-	-	-	-	-	-	Barren sample.

TABLE Ap2.1 Spore colour and kerogen composition - interpreted data

Depth:	270m	APT ID	Spores/ang. pollen	2.0 3.0 2.5 1.5 2.5 2.0 1.5
Fm:	Pliocence	-	Saccates	
Age:	Tertiary		Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	10	-	tr	80	10	yel	-	-	-	-	-	-	poor	rare	2.0	-	-	-

AOM Descr.	Additional comments:	
-		Very low revovery, humic additive (?) abundant

Depth:	360m	APT ID	Spores/ang. pollen	2.0 1.5 7.5 8.0
Fm:	Pliocence	-	Saccates	2.5
Age:	Tertiary		Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
80	-	-	tr	10	10	yel	-	-	-	-	-	yel to orng, occ	pooor	rare	2.0	-	8.0	4

AOM Descr.	Additional comments:	
-		Reworked miospores and woody material common.

Depth:	480m	APT ID	Spores/ang. pollen	3.5 2.0 3.0 2.0 1.5 3.0 2.5 3.5 3.0 2.0 5.0 6.0 7.0
Fm:	Pliocence	-	Saccates	2.0 2.0 3.0 2.0 3.5 2.5 2.5 4.0 3.0 2.0 1.5 3.0 3.5
Age:	Tertiary		Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
30	-	tr	tr	60	10	yel	yel-orng	0	0	0	0	orng-yel	fair to poor	abundant	2.5	-	6.0	4

AOM Descr.	Additional comments:	
-		Most of the darker miospores are stained, but one appears to be Jurassic reworking. A reworked Classopollis at 3.5 sets an upper limit on SCI.

Depth:	540m	APT ID	Spores/ang. pollen	3.0 1.5 2.5 3.5 3.0 3.0 3.0 6.5
Fm:	Eocene	-	Saccates	3.0 2.5 3.5 3.0 3.0 2.5 2.0
Age:	Tertiary		Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
mnr	mnr	-	tr	95	5	yel	grnish yel	0	0	0	0	brn to nf	fair	rare	3.0	-	-	3

AOM Descr.	Additional comments:	
-		

TABLE Ap2.2 Spore colour index and kerogen composition - microscopists data sheets

Depth:	570m	APT ID	**Spores/ang. pollen**	3.5 2.5 3.5 2.5 3.5 3.5 3.0 3.5 2.5 3.0
Fm:	Eocene	-	**Saccates**	
Age:	Tertiary		**Dinos/algae**	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
10	-	-	5	75	10	yel + yel-orng	orng-yel	0	0	0	0	yel to brn	fair to poor	common	3.0	-	-	42463

AOM Descr.	Additional comments:	
-		Mainly very large translucent, fibrous woody organic.

Depth:	720m	APT ID	**Spores/ang. pollen**	3.5 3.0 4.0 3.5 3.0 4.0 4.0
Fm:	Eocene	-	**Saccates**	3.5 3.0
Age:	Tertiary		**Dinos/algae**	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	75	-	5	10	10	orng	orng-yel	0	0	0	0	yel-orng	poor	abundant	2.5?	0	0.0	0

AOM Descr.	Additional comments:	
0		Organic matter more degraded than above. Abundant nearly opaque, small equidimensional amorphous particles, probably mixed sapropel, humic and pyrite. Some fibrous woody material as above, but less well preserved. All organic matter has an orange-brown stain. Most herbaceous material is cuticle.

Depth:	760m	APT ID	**Spores/ang. pollen**	3.0 3.5 3.0 3.0 3.0 3.0 3.0 3.5 2.5 2.5 3.0 3.0 3.5 2.5 2.5 3.5 4.5 4.5 6.5 7.0 8.0
Fm:	Eocene	-	**Saccates**	3.0 3.0 1.5
Age:	Tertiary		**Dinos/algae**	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	80	-	mnr	10	10	yel, occ brn	yel	0	0	0	0	0	poor	abundant	3.0	-	4.0-8.0	0

AOM Descr.	Additional comments:	
orng to yel - brn fdom		More marine than above. Reworked Classopollis at 3.5 sets an upper limit SCI. Callialasporites (Jurassic) at 6.5.

Depth:	1050m	APT ID	**Spores/ang. pollen**	2.0 2.0 1.5 3.0 3.0 3.5 3.0 2.0 2.5 3.0 3.5 2.5 3.0 2.5 3.0 3.0 3.0 ## 2.5 3.0 4.5 4.0
Fm:	Balder	-	**Saccates**	2.5 3.0 3.0 3.0 3.0 4.0
Age:	Tertiary		**Dinos/algae**	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	15	-	tr	70	15	yel-orng	0	0	0	0	0	nf	fair to poor	abundant	3.0	-	4.5	0

AOM Descr.	Additional comments:	
0		Callialasporites (Jurassic) at 4.0. Exinitte flourescence unusually weak given the low SCI.

TABLE Ap2.2 Spore colour index and kerogen composition - microscopists data sheets

Depth:	1100m	APT ID	Spores/ang. pollen	2.5 1.5 1.5 2.5 2.5 1.5 2.5 2.5 2.5 3.0 3.0 2.5 3.5 3.5 8.0 6.5 4.0 4.5 6.0 7.5
Fm:	Maastrichtian	-	Saccates	3.5 3.0
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	10	5	5	5	75	yel	-	-	-	-	-	-	good	occasional	2.5	-	6.0	0

AOM Descr.	Additional comments:	
-		Mostly terrestrial organic matter but dinos are presesnt. Whole kerogen fraction is dominated by opaque, equidimensional particles.

Depth:	1150m	APT ID	Spores/ang. pollen	3.0 3.0 3.0 2.0 2.0 2.0 2.5 3.5 3.5 2.5 3.5 2.0 3.0 3.0 3.0 1.5 3.5 5.0 5.0
Fm:	Maastrichtian	-	Saccates	3.0 3.5 2.5 2.5 3.0 3.5 3.5 2.5 3.5 3.0 3.0 4.0 5.0
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	30	5	5	5	55	yel to brn	brn	-	-	-	-	-	fair	common	3.0	-	5.0	0

AOM Descr.	Additional comments:	
-		As above, but perservation is not as good and miospores are darker.

Depth:	1210m	APT ID	Spores/ang. pollen	3.0 4.5 3.0 3.0 4.5 3.5 3.5 4.0 3.0 3.5 3.0 3.0 2.5 3.5 6.0 5.0 8.0 5.0 5.0
Fm:	Maastrichtian	-	Saccates	3.0 2.5 4.0 3.0 2.5 3.0 3.5 3.5 3.5 3.5 5.0 6.0 6.0
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	70	5	tr	tr	25	-	-	-	-	-	-	-	fair	occasional	3.5	-	5.0	0

AOM Descr.	Additional comments:	
brn, granular, wk		Abunant opaques with secondary woody debris, but dinocysts are more abundant than miospores. Aquillapollenites at 3.0-3.5. Marked increase in no-fluorescing AOM.

Depth:	1250m	APT ID	Spores/ang. pollen	4.0 4.0 4.5 3.5 3.0 3.5 2.5 3.0 3.0 3.0 3.5 3.5 3.5 2.5 6.5 6.0 7.0 6.0 5.0
Fm:	Maastrichtian	-	Saccates	3.5 3.5 4.0 3.0 4.5 5.0
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	-	-	-	-	-	-	-	-	-	-	-	-	fair	common	3.5	-	6.0	-

AOM Descr.	Additional comments:	
-		Low recovery. OM darker - possibly due to oil or mud staining.

TABLE Ap2.2 Spore colour index and kerogen composition - microscopists data sheets

Depth:	1300m	APT ID	Spores/ang. pollen	4.0 3.0 3.5 3.5 3.5 2.5 3.0 3.0 3.0 4.5 3.0 2.5 3.5 3.0 3.0 2.5 3.0 ## 7.5
Fm:	Maastrichtian	-	Saccates	3.5 3.5 3.0 5.0 7.0 5.0
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	60	5	mnr	5	30	-	-	-	-	-	-	-	fair	occasional	3.0	-	6.0	0

AOM Descr.	Additional comments:	
-		Aquillapollenites at 4.5 noted.

Depth:	1360m	APT ID	Spores/ang. pollen	4.0 3.5 3.5 4.0 ## ## ##
Fm:	Maastrichtian	-	Saccates	-
Age:	Upper Cretaceous		Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	55	5	5	5	30	-	-	-	-	-	-	-	fair to good	occasional	2.5?	-	7.0	0

AOM Descr.	Additional comments:	
brown ragged		2-2.5 SCI miospores poorly preserved and thinned. Mitrinite ias more abundant than above in the sieved fraction.

Depth:	1400m	APT ID	Spores/ang. pollen	3.5 4.0 4.0 3.0 3.5 2.5 3.0 2.5 3.0 3.5 3.5 3.5 3.0 3.0 4.0 4.0 3.0 ## 5.0 7.5 7.0
Fm:	Maastrichtian	-	Saccates	3.0 3.0
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	75	mnr	mnr	5	20	-	-	-	-	-	-	-	fair	common	3.5	-	7.5	0

AOM Descr.	Additional comments:	
brn, granular		

Depth:	1450m	APT ID	Spores/ang. pollen	2.5 3.0 4.5 4.0 4.0 4.0 3.5 2.5 4.5 4.5 3.5 4.0 3.0 3.5 4.5 5.5 7.5
Fm:	Maastrichtian	-	Saccates	3.5 4.5
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	90	tr	tr	mnr	10	-	-	-	-	-	-	-	-	common	4.0	-	6.5	0

AOM Descr.	Additional comments:	
orng-brn, more coh		

TABLE Ap2.2 Spore colour index and kerogen composition - microscopists data sheets

Depth:	1500m	APT ID	Spores/ang. pollen	4.5 5.0 5.0 3.5 3.5 2.5 3.5 3.0 3.0 3.5 3.5 3.0 3.0 4.0 4.0 3.5 3.5 ## 6.0 6.0
Fm:	Maastrichtian	-	Saccates	3.5 3.5 3.5
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	85	mnr	5	5	5	-	-	-	-	-	-	-	fair to good	occasional	3.5	-	6.0	-

AOM Descr.	Additional comments:
orng-brn, coh	

Depth:	1550m	APT ID	Spores/ang. pollen	3.5 3.5 4.0 4.0 3.0 5.0 5.0 5.0 2.5 6.0 7.5 6.0 6.0 7.0 6.0 7.5 6.0 6.0
Fm:	Maastrichtian	-	Saccates	- 6.0
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	90	mnr	mnr	mnr	10	-	-	-	-	-	-	-	fair to poor	common	4.0	2.5	6.5	-

AOM Descr.	Additional comments:
dk brn FDOM	Low recovery. Organic matter fairly dark - localised heating event? Aquillapollenites.

Depth:	1600m	APT ID	Spores/ang. pollen	3.0 4.0 4.0 4.5 6.0 6.0 6.0 7.0 7.5
Fm:	Campanian	-	Saccates	5.0 5.0 7.5
Age:	Upper Cretaceous		Dinos/algae	6.0 4.5 7.0

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	70	tr	tr	tr	30	-	-	-	-	-	-	-	poor	occasional	4.5	-	6.5	-

AOM Descr.	Additional comments:
dk brn ragged	Little sieved material recovered.

Depth:	1650m	APT ID	Spores/ang. pollen	3.0 3.0 4.0 4.0 4.5 5.0 3.0 2.5 ## 6.0 6.0 6.0 6.0 6.5 7.0 7.0 7.0 7.5 8.0
Fm:	Campanian	-	Saccates	-
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	70	tr	tr	tr	30	-	-	-	-	-	-	-	fair to poor	occasional	4.0	2.5	6.5	-

AOM Descr.	Additional comments:
-	The wide range of SCI from both in situ spore and dinocysts suggest the possibility of a significat but highly localised heating event. Aquillapollenites are present. Two specimens of Inapaturopollinites at 2.5 and 3.0 also present.

TABLE Ap2.2 Spore colour index and kerogen composition - microscopists data sheets

Depth:	1700m	APT ID	Spores/ang. pollen	5.0 8.0 7.0
Fm:	Campanian	-	Saccates	6.0
Age:	Upper Cretaceous		Dinos/algae	6.0

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	80	tr	tr	-	20	nf	nf	-	-	-	-	-	poor	occasional	-	-	6.5	-

AOM Descr.	Additional comments:
ragged brn FDOM	Low recovery. Mostly ragged brown finely disseminated organic matter. Palynomorphs very rare but there is nothing present suggesting SCI lower than 5.0 (this may be entirely reworked). Palynomorph identification uncertain due to poor preservation.

Depth:	1749m	APT ID	Spores/ang. pollen	4.5 4.0 4.5 4.0 6.0 3.5 4.5 6.0 5.5 5.0
Fm:	Santonian	-	Saccates	5.0 4.0 5.0 4.5 5.0 5.0 5.0 3.5 4.5 4.0 4.5 4.5 4.5 3.5 4.5
Age:	Upper Cretaceous		Dinos/algae	0-7.

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	80	5	mnr	-	15	strong orng	-	-	-	-	-	-	fair to good	-	4.5	-	-	-

AOM Descr.	Additional comments:
-	Whole kerogen as above but sieved fraction has abundant opaques and common well-preserved dinocysts. A well preserved Glechiniidetes spp. specimen is present and considered in situ. Maturity appears lower than previous samples but SCI data seems reliable. Possible Jurassic reworking is possible, given the presence of mature looking specimens tentatively identified as Mendicoinium spp. and Chytroesphaeridia spp.

Depth:	1806m	APT ID	Spores/ang. pollen	
Fm:	Coniacian	-	Saccates	
Age:	Upper Cretaceous		Dinos/algae	6.0

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	60	tr	-	-	40	-	-	-	-	-	-	-	poor	common	-	-	-	-

AOM Descr.	Additional comments:
-	Low recovery - mostly a mix of opaques and ragged brown finely disseminated organic matter

Depth:	1884.58m	APT ID	Spores/ang. pollen	2.0 3.0 3.0 3.5 3.0 3.5 3.5 3.0 4.0 3.5 3.5 7.0 6.0 6.0 5.0
Fm:	Turonian	-	Saccates	3.5 3.0 4.0
Age:	Upper Cretaceous		Dinos/algae	3.0

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	15	10	tr	5	70	yel to yel-orng	-	-	-	-	-	wk orng-brn	fair-good	rare	3.5	-	5.5	3

AOM Descr.	Additional comments:
-	Sieved fraction nearly barren. Mostly small blocky opaques. A reworked Carboniferous Densosporites spp. present. * Beneath the Santonian.

TABLE Ap2.2 Spore colour index and kerogen composition - microscopists data sheets

Depth:	1892.66m	APT ID	Spores/ang. pollen	3.0 3.0 3.5 2.0 2.5 2.5 3.5 3.0 3.0 3.0 3.5 3.0 3.0 2.0 2.5 5.0 5.5 7.0
Fm:	Turonian	-	Saccates	3.5 3.5 4.0
Age:	Upper Cretaceous		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	65	5	tr	10	20	yel to yel orng	yellow	-	-	-	-	-	fair	rare	3.0	-	6.0	2

AOM Descr.	Additional comments:	
-		More miospores present than dinos.

Depth:	1902.7m	APT ID	Spores/ang. pollen	3.0 4.0 4.0 4.0 4.0 .
Fm:	Turonian	-	Saccates	3.5
Age:	Upper Cretaceous		Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	50	10	tr	-	40	yel-orng	yel to yel orng	-	yel	-	-	nf	fair to poor	occ	4.0	-	7.0	2

AOM Descr.	Additional comments:	
-		Some staining. Abundant dinos in sieved fraction. Reworked Permo-Triassic striate bisaccate present.

Depth:	1911.17m	APT ID	Spores/ang. pollen	-
Fm:	Turonian	-	Saccates	3.0
Age:	Upper Cretaceous		Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	65	15	-	-	20	yel	yel	-	yel, occ. grn	-	-	nf	fair to poor	abn	3.0	-	-	2

AOM Descr.	Additional comments:	
-		Lots of juicy pyrite-rich but non-fluorescent AOM.

Depth:	1928.61m	APT ID	Spores/ang. pollen	
Fm:	Cromer Knoll	-	Saccates	
Age:	Lower Cretaceous		Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	95	-	mnr	5	-	pl yel	-	-	-	-	-	yel/ wk yel	poor	common	-	-	-	3

AOM Descr.	Additional comments:	
-		Fairly lean sample - sieved fraction almost barren.

TABLE Ap2.2 Spore colour index and kerogen composition - microscopists data sheets

Depth:	1932.67m	APT ID	Spores/ang. pollen	
Fm:	KCF	-	Saccates	4.0 4.5
Age:	Jurassic		Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	90	5	mnr	mnr	5	wk yel to orng brn	-	-	-	-	-	v wk brn, yel	fair to poor	moderate	4.5	-	-	3

AOM Descr.	Additional comments:	
-		Few miospores and / or algae (yellow fluorescing). Some tissue present that may be exine at ~4.5. Inertinite laths common.

Depth:	1984m	APT ID	Spores/ang. pollen	-
Fm:	LCG	-	Saccates	-
Age:	Devonian		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	tr	-	-	-	-	-	-	-	-	-	-	-	poor	common	-	-	-	-

AOM Descr.	Additional comments:	
-		Nearly barren, with only traces of dark brown finely disseminated organic matter.

Depth:	2022m	APT ID	Spores/ang. pollen	-
Fm:	LCG	-	Saccates	-
Age:	Devonian		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

AOM Descr.	Additional comments:	
-		Nearly barren. Only trace amounts of small opaques.

Depth:	2048m	APT ID	Spores/ang. pollen	-
Fm:	LCG	-	Saccates	-
Age:	Devonian		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

AOM Descr.	Additional comments:	
-		Low recovery - mostly light coloured plant tissue and light orange AOM, most of which is probably drilling additive.

TABLE Ap2.2 Spore colour index and kerogen composition - microscopists data sheets

Depth:	2202m	APT ID	Spores/ang. pollen	-
Fm:	LCG	-	Saccates	-
Age:	Devonian		Dinos/algae	-

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-

AOM Descr.	Additional comments:	Barren.
-		

Depth:		APT ID	Spores/ang. pollen	
Fm:		-	Saccates	
Age:			Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0	0.0	0

AOM Descr.	Additional comments:	#VALUE!
0		

Depth:		APT ID	Spores/ang. pollen	
Fm:		-	Saccates	
Age:			Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	
0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0.0	0	0.0	0

AOM Descr.	Additional comments:	#VALUE!
0		

Depth:		APT ID	Spores/ang. pollen	
Fm:			Saccates	
Age:			Dinos/algae	

Kerogen composition (%)						Fluorescence							Pres.	Pyrite	Mean SCI			FPI
F-Am	H-Am	Algae	Herb	Woody	Coaly	Miosp.	Dinos	Acrit.	Sph.mphs	Botry.	Pediastr.	AOM			Indig.	Caved	Rwk.	

AOM Descr.	Additional comments:	

TABLE Ap2.2 Spore colour index and kerogen composition - microscopists data sheets

APPENDIX 3
SOURCE ROCK SCREENING (TOC AND ROCK-EVAL PYROLYSIS) DATA

Tables

Ap3.1 TOC and Rock-Eval pyrolysis data

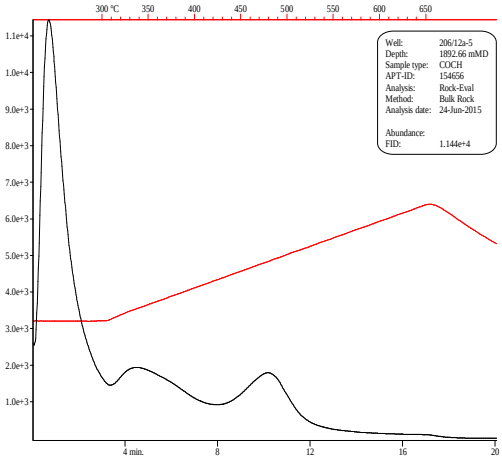
Figures

Ap3.1 Rock-Eval pyrograms

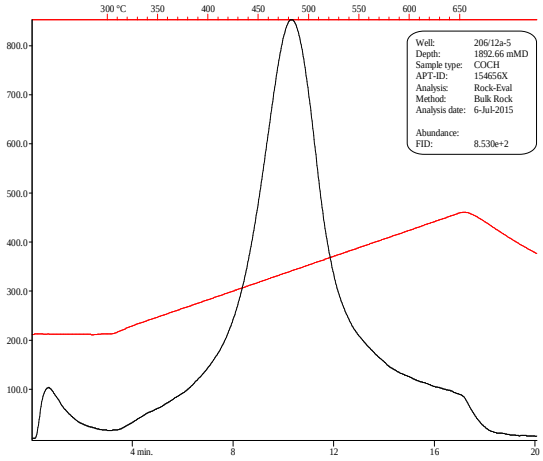
Well	MD (m)	TVDSS (m)	Sample type	APTID	TOC (%)	Rock-Eval pyrolysis data				Rock-Eval pyrolysis ratios			
						S1 (mg/g)	S2 (mg/g)	S3 (mg/g)	Tmax (°C)	HI (mg HC/ g TOC)	OI (mg CO2/	PI	PP (mg/g)
206/12a-5	1884.58	1860.9	Core	154655	0.64	-	-	-	-	-	-	-	-
				154655X	0.49								
	1892.66	1869.0	Core	154656	0.81	1.28	1.03	0.54	290	127	67	0.55	2.31
				154656X	0.61	0.01	0.26	0.44	434	43	73	0.04	0.27
	1902.70	1879.0	Core	154657	0.64	-	-	-	-	-	-	-	-
				154657X	0.47	-	-	-	-	-	-	-	-
	1911.17	1887.5	Core	154658	0.86	1.41	1.60	0.70	434	185	81	0.47	3.01
				154658X	0.74	0.02	0.90	0.86	438	121	116	0.02	0.92
	1928.61	1905.0	Core plug	146975X	0.05	0	0.02	0.16	438	39	314	0.00	0.02
	1932.42	1908.8	Core	154659	3.84	3.84	13.31	0.91	436	347	24	0.22	17.15
				154659X	3.35	0.02	9.56	0.39	438	285	12	0.00	9.58
	1932.67	1909.0	Core plug	146976	2.45	0.05	8.26	0.24	436	337	10	0.01	8.31
				146976X	2.11	0.04	6.84	0.59	437	324	28	0.01	6.88
	1937.14	1913.5	Core	154660	2.06	6.53	4.11	0.38	425	200	18	0.61	10.64
				154660X	1.50	0.01	1.67	0.24	427	111	16	0.01	1.68
	1937.32	1913.7	Core	154661	8.21	4.56	28.78	0.16	433	351	2	0.14	33.34
				154661X	7.21	0.03	23.31	0.13	433	323	2	0.00	23.34
	1944.05	1920.4	Core	154662	4.99	33.50	23.96	1.50	428	480	30	0.58	57.46
				154662X	2.83	0.07	6.21	0.12	438	219	4	0.01	6.28
	1948.68	1925.0	Core	154663	4.72	33.19	19.15	0.91	421	406	19	0.63	52.34
				154663X	3.51	0.27	7.80	0.20	434	222	6	0.03	8.07
	1960.35	1936.7	Core	154664	3.10	20.90	14.01	2.33	422	452	75	0.60	34.91
				154664X	0.88	0.05	1.29	0.15	418	147	17	0.04	1.34
	1966.70	1943.0	Core	154665	2.96	14.12	12.34	1.00	411	417	34	0.53	26.46
				154665X	1.15	0.03	2.58	0.23	421	224	20	0.01	2.61
	1967.35	1943.7	Core	154666	23.20	47.40	113.05	2.23	424	487	10	0.30	160.45
				154666X	49.40	0.25	101.44	1.50	430	205	3	0.00	101.69

TABLE Ap3.1 TOC and Rock-Eval pyrolysis data

Before extraction: APTID - 154656

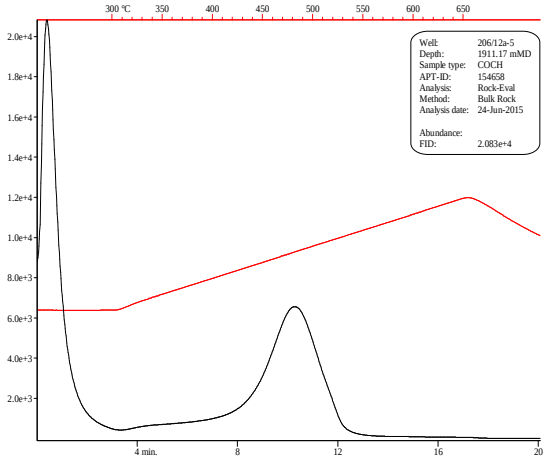


After extraction: APTID - 154656X

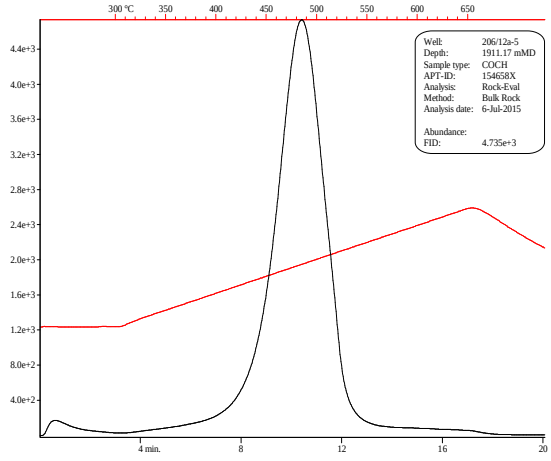


206/12a-5, 1892.66m MD, Core (U. Cret., Turonian K72)

Before extraction: APTID - 154658



After extraction: APTID - 154658X

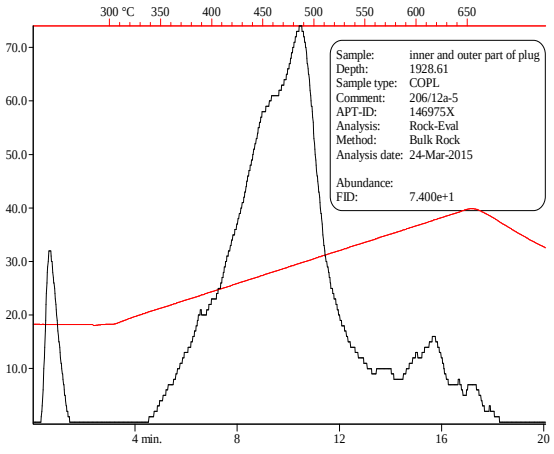


206/12a-5, 1911.17m MD, Core (U. Cret., Turonian K72)

Before extraction: APTID - 146975

Not analysed

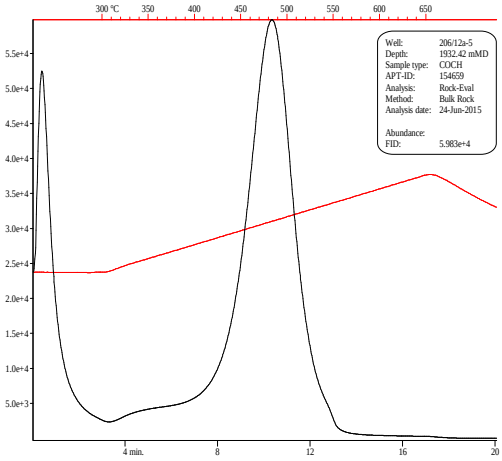
After extraction: APTID - 146975X



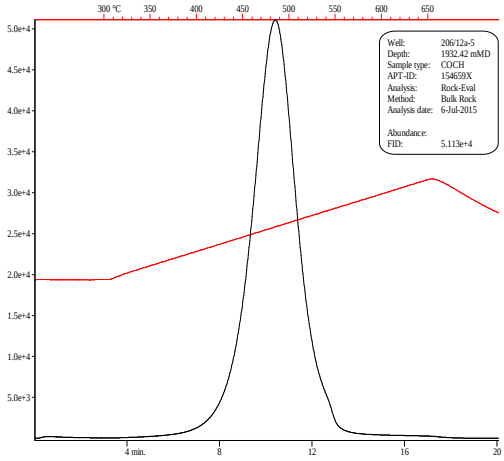
206/12a-5, 1928.61m MD, Core Plug (L. Cret. Cromer Knoll K20)

FIGURE Ap3.1 Rock-Eval pyrograms

Before extraction: APTID - 154659

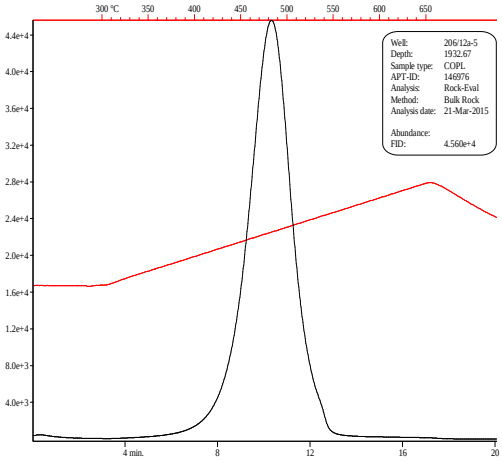


After extraction: APTID - 154659X

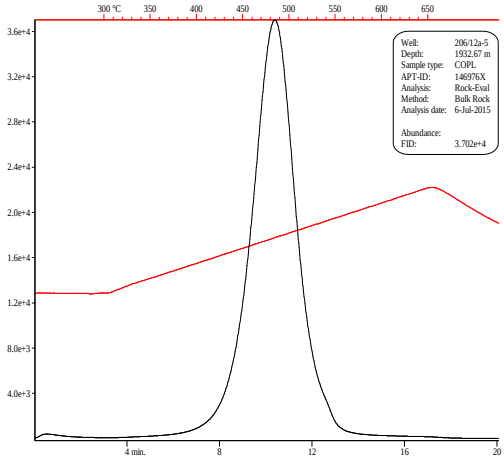


206/12a-5, 1932.42m MD, Core (KCF)

Before extraction: APTID - 146976

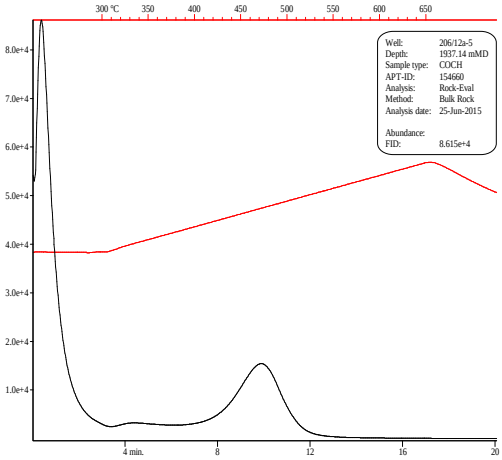


After extraction: APTID - 146976X

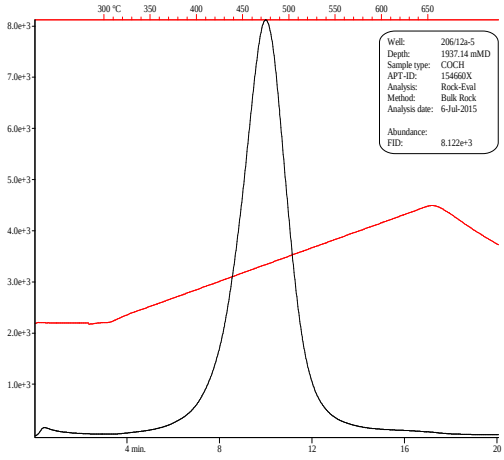


206/12a-5, 1932.67m MD, Core plug (KCF)

Before extraction: APTID - 154660



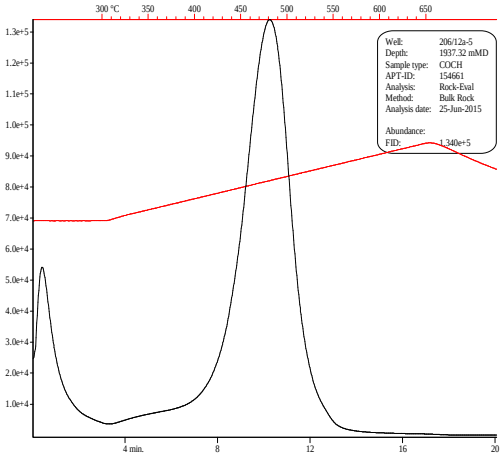
After extraction: APTID - 154660X



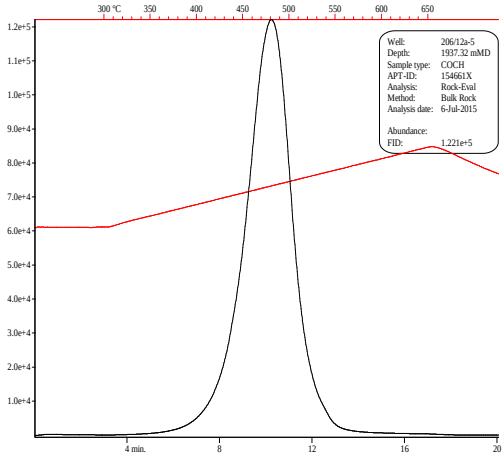
206/12a-5, 1937.14m MD, Core (Rona)

FIGURE Ap3.1 Rock-Eval pyrograms

Before extraction: APTID - 154661

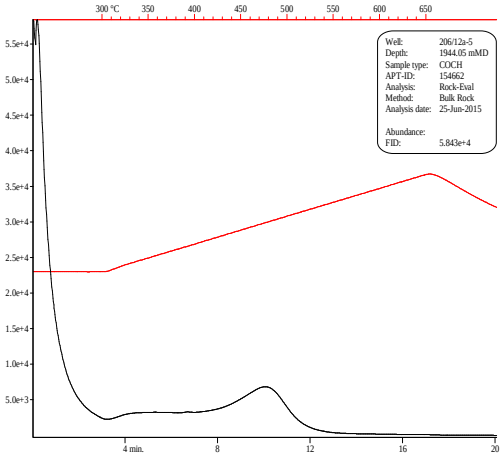


After extraction: APTID - 154661X

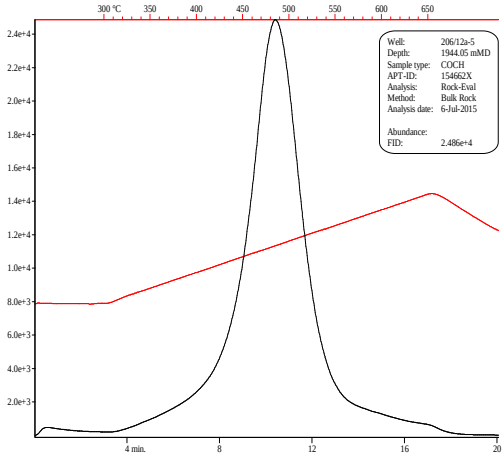


206/12a-5, 1937.32m MD, Core (Rona)

Before extraction: APTID - 154662

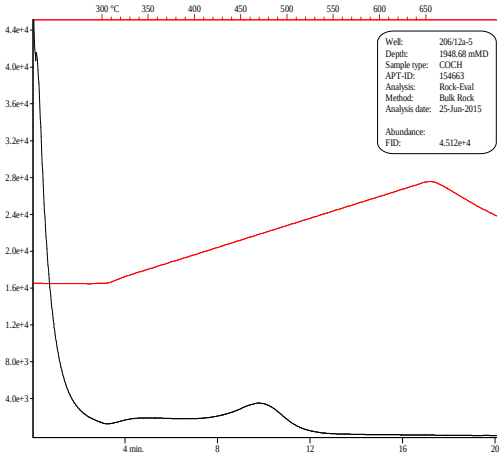


After extraction: APTID - 154662X

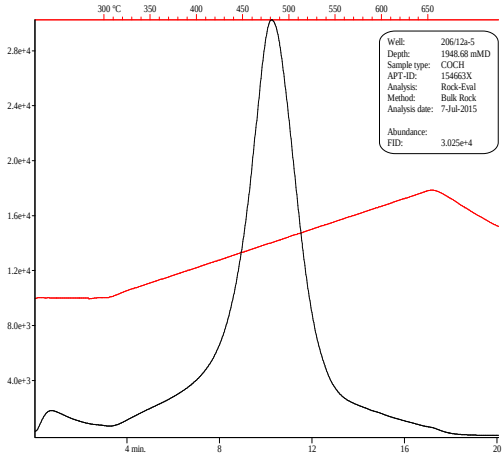


206/12a-5, 1944.05m MD, Core (Rona)

Before extraction: APTID - 154663



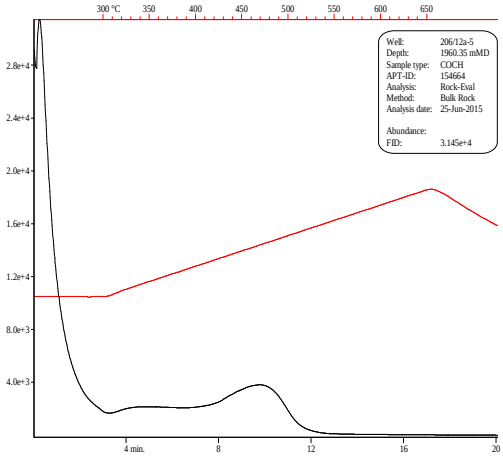
After extraction: APTID - 154663X



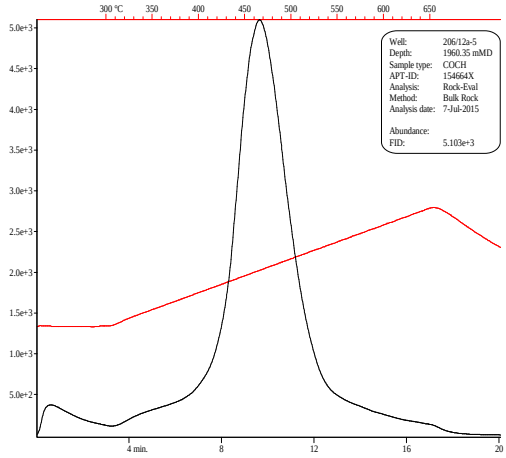
206/12a-5, 1948.68m MD, Core (Rona)

FIGURE Ap3.1 Rock-Eval pyrograms

Before extraction: APTID - 154664

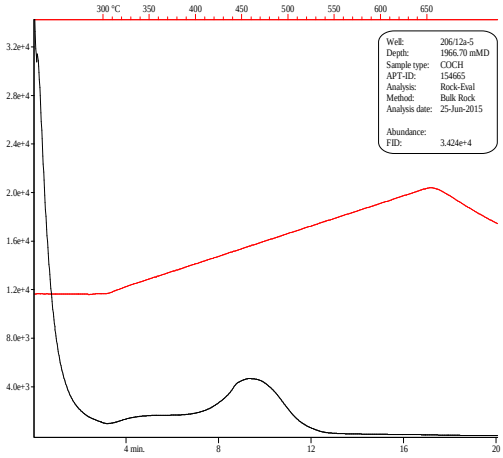


After extraction: APTID - 154664X

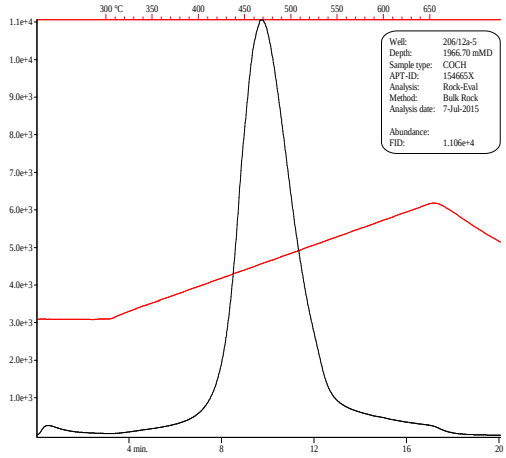


206/12a-5, 1960.35m MD, Core (Rona)

Before extraction: APTID - 154665

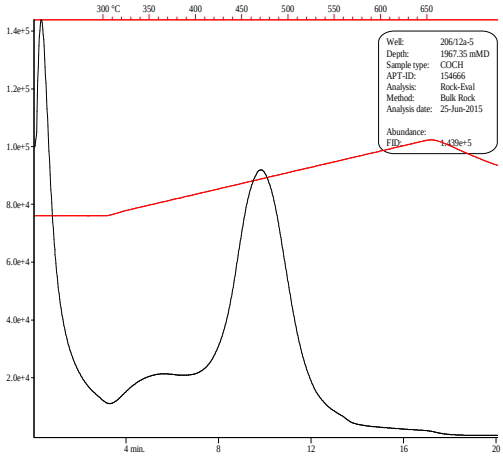


After extraction: APTID - 154665X

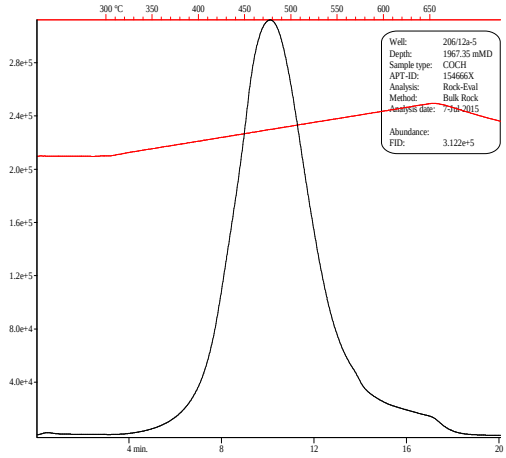


206/12a-5, 1966.7m MD, Core (Rona)

Before extraction: APTID - 154666



After extraction: APTID - 154666X



206/12a-5, 1967.35m MD, Core (Rona)

FIGURE Ap3.1 Rock-Eval pyrograms

APPENDIX 4
Solvent extraction, extract fractionation and volatile content data

Tables

Ap.4 Solvent extraction, volatiles and fractionation data

Well	MD (m)	TVDSS (m)	Sample type	APTID	Extraction data			Volatiles (%)	MPLC & asphaltene precipitation data					
					Rock wt. (g)	EOM (mg)	EOM (mg/kg rock)		Fraction (wt% of EOM/oil)				Ratios	
									Alks.	Aroms.	Pols.	Asph.	Alks/ aroms	HC/ non-HC
206/12a-5	1884.58	1860.9	Core	154655	10.166	11	1043							
	1884.67	1861.0	Core plug	146965	19.634	5	250		34.8	22.3	18.6	24.3	1.6	1.3
	1892.66	1869.0	Core	154656	9.642	24	2499							
	1897.03	1873.4	Spun out oil	146645					50.6	20.3	22.4	6.7	2.5	2.4
	1900.67	1877.0	Core plug	146969	20.578	65	3173		95.7	0.6	2.8	0.9	159.5	26.0
	1902.70	1879.0	Core	154657	12.995	27	2047							
	1911.17	1887.5	Core	154658	10.486	25	2384							
	1916.67	1893.0	Core plug	146972	20.157	318	15766		53.3	20.8	19.2	6.7	2.6	2.9
	1922.08	1898.4	Spun out oil	146643					54.1	20.2	21.8	3.8	2.7	2.9
	1922.40	1898.7	Oil	160302				11.6	52.3	28.5	14.9	4.3	1.8	4.2
	1924.67	1901.0	Core plug	146974	20.031	0	20		7.0	6.2	9.1	77.8	1.1	0.2
	1932.42	1908.8	Core	154659	13.230	106	7982							
	1932.67	1909.0	Core plug	146976	18.890	93	4902		44.0	20.9	17.5	17.6	2.1	1.9
	1937.14	1913.5	Core	154660	14.479	128	8868							
	1937.32	1913.7	Core	154661	7.723	87	11317		27.5	28.5	31.4	12.6	1.0	1.3
	1938.00	1914.3	Mud	163301		107			95.4	0.1	3.5	1.0	954.0	21.2
	1944.05	1920.4	Core	154662	12.747	830	65128							
	1948.67	1925.0	Spun out oil	146644					53.6	20.1	21.6	4.8	2.7	2.8
	1948.67	1925.0	Core plug	146980	13.850	959	69263		47.0	23.3	21.4	8.3	2.0	2.4
	1948.68	1925.0	Core	154663	10.821	710	65640							
	1956.00	1932.3	Oil	160303				7.2	54.9	25.6	14.5	5.1	2.1	4.1
	1956.67	1933.0	Core plug	146982	10.385	765	73633		56.2	26.7	7.4	9.7	2.1	4.9
	1960.35	1936.7	Core	154664	11.073	514	46448							
	1936-1963	1912.3-1939.3	Oil	160301				7.4	50.5	26.0	18.3	5.3	1.9	3.2
	1966.70	1943.0	Core	154665	15.056	457	30354							
	1967.35	1943.7	Core	154666	9.669	994	102786		35.4	20.2	22.7	21.8	1.8	1.3
	1984.00	1960.3	SWC	153498	15.663	178	11345		77.4	7.7	12.0	2.8	10.1	5.8

TABLE Ap4.1 Solvent extraction, volatiles and fractionation data

Well	MD (m)	TVDSS (m)	Sample type	APTID	Extraction data			Volatiles (%)	MPLC & asphaltene precipitation data					
					Rock wt. (g)	EOM (mg)	EOM (mg/kg rock)		Fraction (wt% of EOM/oil)				Ratios	
									Alks.	Aroms.	Pols.	Asph.	Alks/ aroms	HC/ non-HC
206/12a-5	1984.50	1960.8	Oil	160304				6.7	52.4	26.6	15.4	5.7	2.0	3.7
	1988.67	1965.0	Core plug	146990	15.254	259	17006		33.6	26.8	32.6	7.0	1.3	1.5
	1997.67	1974.0	Core plug	146992	13.230	287	21656		50.1	20.8	23.0	6.1	2.4	2.4
	2001.00	1977.3	SWC	153499	18.386	303	16458		87.1	3.2	8.0	1.7	27.2	9.3
	2022.00	1998.3	SWC	153500	17.910	691	38571		89.4	2.1	7.4	1.0	42.6	10.9
	2048.00	2024.3	SWC	153501	17.211	221	12823		89.2	1.4	8.1	1.3	63.7	9.6
	2352.00	2336.6	Mud	163302		116			95.7	0.5	2.9	1.0	191.4	24.7

TABLE Ap4.1 Solvent extraction, volatiles and fractionation data

APPENDIX 5
Whole oil gas chromatography data

Tables

Ap5.1 Whole oil gas chromatography parameters
Ap5.2 Whole oil gas chromatography peak areas
Ap5.3 Whole oil gas chromatography amounts in ng/g

Figures

Ap5.1 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72))
Ap5.2 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret)
Ap5.3 Whole oil gas chromatogram: 206/12a-5, OIL, 1922.4m (K30 Sand Cret)
Ap5.4 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona)
Ap5.5 Whole oil gas chromatogram: 206/12a-5, OIL, 1956m (Rona)
Ap5.6 Whole oil gas chromatogram: 206/12a-5, OIL, 1936-1963m (Rona)
Ap5.7 Whole oil gas chromatogram: 206/12a-5, OIL, 1984.5m (LCG Unit III)

Well	MD (m)	TVDSS (m)	Sample type	APTID	A	B	X	W	C	I	F	H	U	R
206/12a-5	1897.03	1873.4	Spun out oil	146645	0.30	0.72	0.63	1.24	0.37	1.19	0.36	17.32	2.27	3.14
	1922.08	1898.4	Spun out oil	146643	0.31	0.92	1.04	2.99	0.64	1.42	0.53	20.91	1.62	2.99
	1922.40	1898.7	Oil	160302	0.27	0.87	0.82	2.46	0.62	1.36	0.52	20.50	1.60	2.99
	1948.67	1925.0	Spun out oil	146644	0.32	0.96	0.92	3.22	0.65	1.42	0.53	20.94	1.61	2.99
	1956.00	1932.3	Oil	160303	0.30	0.86	0.78	2.76	0.62	1.35	0.51	20.57	1.63	3.06
	1936-1963	1912.3-1939.3	Oil	160301	0.31	0.85	0.79	2.81	0.62	1.36	0.51	20.55	1.63	3.03
	1984.50	1960.8	Oil	160304	0.29	0.79	0.77	2.73	0.62	1.35	0.52	20.67	1.64	3.07

Well	MD (m)	TVDSS (m)	Sample type	APTID	S	Late Mature Index	Heptane Index	Kerogen Type Index	CPI-1	Pr / n-C17	Ph / n-C18	(Pr/ n-C17) / (Ph/ n-C18)	Pr / Ph	n-C17 / (n-C17+ n-C27)
206/12a-5	1897.03	1873.4	Spun out oil	146645	10.60	0.03	17.01	10.89	1.12	0.39	0.25	1.54	1.60	0.68
	1922.08	1898.4	Spun out oil	146643	12.72	0.11	20.56	11.66	1.12	0.42	0.30	1.41	1.66	0.71
	1922.40	1898.7	Oil	160302	11.47	0.09	20.15	11.85	1.09	0.45	0.27	1.65	1.77	0.69
	1948.67	1925.0	Spun out oil	146644	12.36	0.11	20.58	11.71	1.13	0.41	0.27	1.52	1.64	0.70
	1956.00	1932.3	Oil	160303	11.03	0.10	20.21	11.99	1.11	0.45	0.29	1.58	1.95	0.74
	1936-1963	1912.3-1939.3	Oil	160301	11.53	0.10	20.19	11.92	1.08	0.45	0.28	1.61	1.72	0.69
	1984.50	1960.8	Oil	160304	11.38	0.10	20.30	12.03	1.07	0.45	0.27	1.64	1.81	0.70

see last page of table for key to ratios

TABLE Ap5.1 Whole oil gas chromatography parameters

KEY: Whole oil GC ratios

A :	Benz/n-C6
B :	Tol/n-C7 - Aromaticity Index
X :	m+p-Xyl/n-C8
W :	Benz*10/CyC6
C :	(n-C6+n-C7)/(CyC6+MCyC6)
I :	(2-MC6+3-MC6)/(c1,3-DMCyC5+t1,3-DMCyC5+t1,2-DMCyC5) - Isoheptane Index
F :	n-C7/MCyC6 - Paraffinicity Index
H :	n-C7*100/(CyC6+2-MC6+3-MC6+c1,3-DMCyC5+t1,3-DMCyC5+t1,2-DMCyC5+n-C7+MCyC6)
U :	CyC6/MCyC5
R :	n-C7/2-MC6
S :	n-C6/2,2-DMC4
Late Mature Index :	Benz/[MCyC6+c-1,2-DMCyC5]
Heptane Index :	n-C7*100/(CyC6+2-MC6+1,1-DMCyC5+3-MC6+c-1,3-DMCyC5+t-1,3-DMCyC5+t-1,2-DMCyC5+n-C7+MCyC6)
Kerogen Type Index :	(Heptane Index-4)/Isoheptane Index
CPI-1 :	Carbon Preference Index-1: 0.5 * [{ (C25+C27+C29+C31)/(C24+C26+C28+C30) } + { (C25+C27+C29+C31)/(C26+C28+C30+C32) }]
Pr /n-C17 :	Pristane / normal C17
Ph /n-C18 :	Phytane / normal C18
(Pr/n-C17) /(Ph/n-C18)) :	(Pristane / normal C17) / (Phytane / normal C18)
Pr /Ph :	Pristane / phytane
n-C17 /(n-C17+n-C27) :	normal C17 / (C17+C27) fraction

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C3	i-C4	n-C4	i-C5	n-C5	2,2-DMC4	CyC5	2,3-DMC4
206/12a-5	1897.03	1873.4	Spun out oil	146645		20.416		8.415	5.055	2.064	1.648	4.526
	1922.08	1898.4	Spun out oil	146643	15.744	91.491	164.449	442.903	382.482	53.205	54.033	103.679
	1922.40	1898.7	Oil	160302	189.317	215.885	382.696	701.207	590.432	81.657	63.932	166.514
	1948.67	1925.0	Spun out oil	146644	33.633	118.633	218.063	542.114	457.685	62.88	33.522	128.216
	1956.00	1932.3	Oil	160303	141.387	162.563	295.35	566.115	484.444	71.916	50.797	147.323
	1936-1963	1912.3-1939.3	Oil	160301	163.965	187.9	336.047	637.072	541.714	77.246	53.518	154.782
	1984.50	1960.8	Oil	160304	153.317	176.43	321.476	606.931	524.611	75.444	40.364	177.884

Well	MD (m)	TVDSS (m)	Sample type	APTID	2-MC5	3-MC5	n-C6	2,2-DMC5	MCyC5	2,4-DMC5	2,2,3-TMC4	Benz
206/12a-5	1897.03	1873.4	Spun out oil	146645	13.303	19.472	21.881	3.168	23.004	4.867	1.408	6.505
	1922.08	1898.4	Spun out oil	146643	378.761	315.402	676.917	39.878	430.311	69.422	17.933	208.057
	1922.40	1898.7	Oil	160302	546.809	450.499	936.225	46.978	630.386	101.777	25.938	248.179
	1948.67	1925.0	Spun out oil	146644	439.568	364.113	777.152	44.262	485.277	78.336	19.959	251.756
	1956.00	1932.3	Oil	160303	459.234	380.795	793.013	41.286	533.697	87.571	22.307	240.008
	1936-1963	1912.3-1939.3	Oil	160301	514.923	426.726	890.613	45.146	599.545	98.389	26.046	274.017
	1984.50	1960.8	Oil	160304	494.463	408.288	858.233	42.683	563.343	94.18	23.81	251.826

TABLE Ap5.2 Whole oil gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	3,3-DMC5	CyC6	2-MC6	2,3-DMC5	1,1-DMCyC5	3-MC6	c-1,3-DMCyC5	t-1,3-DMCyC5
206/12a-5	1897.03	1873.4	Spun out oil	146645	3.038	52.304	25.497	12.596	8.356	32.499	13.648	12.288
	1922.08	1898.4	Spun out oil	146643	31.967	695.468	348.316	137.007	85.188	392.088	143.906	141.12
	1922.40	1898.7	Oil	160302	58.939	1010.176	479.717	194.933	124.413	542.81	204.328	200.354
	1948.67	1925.0	Spun out oil	146644	36.034	781.045	387.898	155.931	96.124	437.854	161.678	157.38
	1956.00	1932.3	Oil	160303	52.176	869.24	409.163	172.411	107.616	468.842	176.767	174.034
	1936-1963	1912.3-1939.3	Oil	160301	58.323	974.709	462.438	188.996	120.537	525.365	198.153	193.945
	1984.50	1960.8	Oil	160304	55.11	923.172	438.502	183.482	116.999	502.393	188.463	186.862

Well	MD (m)	TVDSS (m)	Sample type	APTID	3-EC5	t-1,2-DMCyC5	n-C7	c-1,2-DMCyC5	MCyC6	1,1,3-TMCyC5	ECyC5	2,5-DMC6
206/12a-5	1897.03	1873.4	Spun out oil	146645	2.842	22.618	80.125		223.618	14.275	10.032	9.229
	1922.08	1898.4	Spun out oil	146643	26.25	236.106	1040.059		1975.911	106.097	90.195	75.965
	1922.40	1898.7	Oil	160302	25.398	345.585	1436.414		2786.069	154.568	117.145	111.343
	1948.67	1925.0	Spun out oil	146644	29.641	263.933	1160.676		2191.989	118.15	99.069	84.884
	1956.00	1932.3	Oil	160303	21.98	298.361	1251.399		2435.442	135.97	103.323	95.988
	1936-1963	1912.3-1939.3	Oil	160301	24.568	335.035	1401.839		2731.485	156.375	116.219	109.638
	1984.50	1960.8	Oil	160304	24.008	319.221	1344.668		2602.561	147.54	109.634	103.974

TABLE Ap5.2 Whole oil gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	2,2,3-TMC5/2,4-DMC6	c,t-1,2,4-TMCyC5	3,3-DMC6	t,c-1,2,3-TMCyC5	2,3,4-TMC5	Tol	2,3-DMC6	2-MC7
206/12a-5	1897.03	1873.4	Spun out oil	146645	13.155	11.157	5.103	12.415	2.544	57.803	10.49	63.212
	1922.08	1898.4	Spun out oil	146643	102.858	81.302	33.455	83.989	15.162	961.82	75.697	463.03
	1922.40	1898.7	Oil	160302	138.852	109.583	47.128	114.538	20.204	1256.639	140.882	609.807
	1948.67	1925.0	Spun out oil	146644	113.039	89.522	37.14	93.007	16.474	1110.31	81.984	506.14
	1956.00	1932.3	Oil	160303	122.175	96.71	41.906	102.014	18.408	1072.186	124.689	535.143
	1936-1963	1912.3-1939.3	Oil	160301	137.087	109.087	46.401	113.965	20.321	1194.495	140.034	600.468
	1984.50	1960.8	Oil	160304	130.742	104.517	44.776	110.088	19.89	1068.609	123.033	573.844

Well	MD (m)	TVDSS (m)	Sample type	APTID	4-MC7	3-MC7	c-1,3-DMCyC6	t-1,4-DMCyC6	1,1-DMCyC6	t-1,2-DMCyC6	n-C8	E-CyC6
206/12a-5	1897.03	1873.4	Spun out oil	146645	21.283	51.799	91.551	35.05	12.824	47.015	190.988	105.146
	1922.08	1898.4	Spun out oil	146643	153.476	366.853	578.283	234.522	75.65	264.491	1366.017	540.747
	1922.40	1898.7	Oil	160302	217.144	474.493	805.012	328.668	104.131	358.215	1791.039	691.489
	1948.67	1925.0	Spun out oil	146644	167.424	400.378	632.918	257.088	83.495	289.673	1483.519	588.312
	1956.00	1932.3	Oil	160303	191.855	420.184	711.989	290.766	93.01	320.165	1578.022	618.238
	1936-1963	1912.3-1939.3	Oil	160301	214.316	470.993	795.759	323.736	104.728	357.858	1771.092	693.001
	1984.50	1960.8	Oil	160304	203.583	445.946	765.317	312.522	100.975	345.611	1669.909	664.098

TABLE Ap5.2 Whole oil gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	i-C9	E-Benz	m-Xyl	p-Xyl	4-MC8	2-MC8	3-MC8	o-Xyl
206/12a-5	1897.03	1873.4	Spun out oil	146645	43.194	20.326	100.433	19.146	44.941	61.172	74.037	53.182
	1922.08	1898.4	Spun out oil	146643	183.668	184.086	994.971	420.725	227.225	309.704	356.176	395.96
	1922.40	1898.7	Oil	160302	241.258	246.72	1173.646	297.738	286.352	409.421	440.103	484.18
	1948.67	1925.0	Spun out oil	146644	198.314	203.236	1086.202	274.449	239.174	330.56	378.008	433.625
	1956.00	1932.3	Oil	160303	217.987	212.994	1007.226	228.175	254.788	365.444	393.03	423.878
	1936-1963	1912.3-1939.3	Oil	160301	244.07	242.975	1133.341	260.103	284.349	408.527	438.423	475.527
	1984.50	1960.8	Oil	160304	236.301	223.426	1011.663	279.112	270.46	388.69	418.16	434.724

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C9	i-C10	n-C10	i-C11	n-C11	n-C12	i-C13	i-C14
206/12a-5	1897.03	1873.4	Spun out oil	146645	346.812	73.362	499.18	94.501	657.91	816.436	152.527	186.006
	1922.08	1898.4	Spun out oil	146643	1671.001	279.342	1890.523	310.137	2167.056	2584.984	536.995	658.481
	1922.40	1898.7	Oil	160302	2082.658	405.958	2264.929	397.21	2456.276	2729.635	479.398	471.599
	1948.67	1925.0	Spun out oil	146644	1764.041	294.383	1955.268	319.341	2186.864	2386.397	449.219	534.128
	1956.00	1932.3	Oil	160303	1865.44	368.152	2076.252	369.662	2365.616	3228.502	657.939	752.551
	1936-1963	1912.3-1939.3	Oil	160301	2078.949	409.358	2274.91	404.087	2471.999	2757.686	486.753	479.038
	1984.50	1960.8	Oil	160304	1982.181	393.537	2180.389	389.491	2392.487	2777.911	510.259	520.914

TABLE Ap5.2 Whole oil gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C13	i-C15	n-C14	i-C16	n-C15	n-C16	i-C18	n-C17
206/12a-5	1897.03	1873.4	Spun out oil	146645	994.739	206.959	1104.721	367.55	1262.769	1257.866	283.073	1302.75
	1922.08	1898.4	Spun out oil	146643	3323.219	771.2	3254.401	1198.116	3406.611	3037.886	769.847	2820.373
	1922.40	1898.7	Oil	160302	2675.33	550.538	2597.008	939.108	2861.42	2779.98	695.651	2850.604
	1948.67	1925.0	Spun out oil	146644	2639.592	569.117	2612.511	962.028	2777.214	2624.615	662.073	2559.789
	1956.00	1932.3	Oil	160303	4235.319	1037.841	4243.832	1522.761	4262.663	3675.781	954.862	3281.866
	1936-1963	1912.3-1939.3	Oil	160301	2701.961	554.689	2611.185	947.728	2869.673	2812.227	696.048	2837.86
	1984.50	1960.8	Oil	160304	2959.172	640.16	2891.586	1043.548	3104.543	2936.635	739.878	2915.28

Well	MD (m)	TVDSS (m)	Sample type	APTID	Pr	n-C18	Ph	n-C19	n-C20	n-C21	n-C22	n-C23
206/12a-5	1897.03	1873.4	Spun out oil	146645	507.182	1250.465	316.442	1387.571	1223.778	1202.726	1110.233	1107.897
	1922.08	1898.4	Spun out oil	146643	1198.103	2400.161	723.441	2592.535	2142.915	2119.013	1957.287	1939.486
	1922.40	1898.7	Oil	160302	1273.248	2658.987	718.501	2981.269	2566.627	2496.621	2289.88	2249.83
	1948.67	1925.0	Spun out oil	146644	1044.103	2368.479	635.424	2594.005	2179.306	2168.621	1983.005	1960.989
	1956.00	1932.3	Oil	160303	1483.538	2661.176	762.21	2768.892	2369.93	2248.35	2061.772	2027.974
	1936-1963	1912.3-1939.3	Oil	160301	1270.004	2652.195	737.965	2959.55	2530.446	2455.414	2262.293	2221.274
	1984.50	1960.8	Oil	160304	1306.929	2644.325	723.039	2926.105	2509.563	2431.189	2237.815	2199.491

TABLE Ap5.2 Whole oil gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C24	n-C25	n-C26	n-C27	n-C28	n-C29	n-C30	n-C31
206/12a-5	1897.03	1873.4	Spun out oil	146645	938.735	906.512	715.878	626.691	454.41	427.139	329.5	248.08
	1922.08	1898.4	Spun out oil	146643	1637.173	1604.883	1261.496	1142.462	826.367	750.32	582.827	431.946
	1922.40	1898.7	Oil	160302	1890.359	1792.062	1468.769	1299.837	951.59	831.107	668.201	512.509
	1948.67	1925.0	Spun out oil	146644	1655.872	1585.602	1271.006	1121.998	766.851	749.194	575.831	411.427
	1956.00	1932.3	Oil	160303	1691.613	1611.753	1310.632	1146.256	840.989	803.257	585.426	459.842
	1936-1963	1912.3-1939.3	Oil	160301	1863.279	1754.07	1429.189	1259.201	925.772	814.701	686.386	495.259
	1984.50	1960.8	Oil	160304	1842.726	1701.748	1414.13	1237.046	900.649	761.981	633.089	451.615

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C32	n-C33	n-C34	n-C35	n-C36
206/12a-5	1897.03	1873.4	Spun out oil	146645	144.518	126.599	85.454	41.21	
	1922.08	1898.4	Spun out oil	146643	269.982	223.425	133.788	95.069	79.272
	1922.40	1898.7	Oil	160302	358.95	309.238	195.011	184.122	125.984
	1948.67	1925.0	Spun out oil	146644	228.711	185.85	100.444	44.094	34.038
	1956.00	1932.3	Oil	160303	335.032	283.296	176.627	176.553	118.826
	1936-1963	1912.3-1939.3	Oil	160301	356.895	303.382	186.015	189.457	133.301
	1984.50	1960.8	Oil	160304	311.645	260.805	142.598	130.953	78.205

TABLE Ap5.2 Whole oil gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C3	i-C4	n-C4	i-C5	n-C5	2,2-DMC4	CyC5	2,3-DMC4
206/12a-5	1897.03	1873.4	Spun out oil	146645		102856		42096	25289	10284	8017	22550
	1922.08	1898.4	Spun out oil	146643	35265	202640	364231	974032	841154	116536	115562	227092
	1922.40	1898.7	Oil	160302	433571	488885	866639	1576698	1327615	182872	139799	372908
	1948.67	1925.0	Spun out oil	146644	119106	415422	763600	1884925	1591368	217752	113349	444012
	1956.00	1932.3	Oil	160303	317061	360468	654910	1246432	1066614	157702	108764	323060
	1936-1963	1912.3-1939.3	Oil	160301	427849	484820	867067	1632147	1387844	197103	133339	394949
	1984.50	1960.8	Oil	160304	350480	398802	726666	1362211	1177449	168647	88101	397639

Well	MD (m)	TVDSS (m)	Sample type	APTID	2-MC5	3-MC5	n-C6	2,2-DMC5	MCyC5	2,4-DMC5	2,2,3-TMC4	Benz
206/12a-5	1897.03	1873.4	Spun out oil	146645	66277	97015	109018	15721	111909	24151	6985	29391
	1922.08	1898.4	Spun out oil	146643	829618	690840	1482685	86997	920309	151448	39123	413306
	1922.40	1898.7	Oil	160302	1224579	1008893	2096676	104786	1378466	227016	57855	504071
	1948.67	1925.0	Spun out oil	146644	1522222	1260921	2691276	152664	1640896	270192	68842	790693
	1956.00	1932.3	Oil	160303	1007039	835033	1738973	90172	1142735	191261	48720	477325
	1936-1963	1912.3-1939.3	Oil	160301	1313900	1088851	2272526	114735	1493758	250048	66193	634122
	1984.50	1960.8	Oil	160304	1105317	912683	1918485	95031	1229603	209686	53011	510539

TABLE Ap5.3 Whole oil gas chromatography amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	3,3-DMC5	CyC6	2-MC6	2,3-DMC5	1,1-DMCyC5	3-MC6	c-1,3-DMCyC5	t-1,3-DMCyC5
206/12a-5	1897.03	1873.4	Spun out oil	146645	15073	254447	126524	62505	40650	161267	66397	59777
	1922.08	1898.4	Spun out oil	146643	69739	1487405	759875	298891	182193	855367	307773	301814
	1922.40	1898.7	Oil	160302	131465	2208954	1070021	434802	272054	1210752	446805	438114
	1948.67	1925.0	Spun out oil	146644	124285	2640992	1337907	537825	325029	1510210	546690	532157
	1956.00	1932.3	Oil	160303	113957	1861190	893645	376560	230425	1023989	378488	372637
	1936-1963	1912.3-1939.3	Oil	160301	148224	2428473	1175247	480317	300315	1335172	493694	483212
	1984.50	1960.8	Oil	160304	122698	2014998	976293	408510	255372	1118544	411356	407862

Well	MD (m)	TVDSS (m)	Sample type	APTID	3-EC5	t-1,2-DMCyC5	n-C7	c-1,2-DMCyC5	MCyC6	1,1,3-TMCyC5	ECyC5	2,5-DMC6
206/12a-5	1897.03	1873.4	Spun out oil	146645	14104	110031	397600		1087853	69444	48801	45706
	1922.08	1898.4	Spun out oil	146643	57267	504962	2268959		4225898	226910	192900	165391
	1922.40	1898.7	Oil	160302	56651	755693	3203956		6092304	337994	256160	247857
	1948.67	1925.0	Spun out oil	146644	102236	892451	4003309		7411894	399507	334986	292188
	1956.00	1932.3	Oil	160303	48007	638841	2733155		5214695	291134	221231	209227
	1936-1963	1912.3-1939.3	Oil	160301	62438	834736	3562655		6805457	389607	289557	278078
	1984.50	1960.8	Oil	160304	53452	696760	2993810		5680582	322033	239296	231028

TABLE Ap5.3 Whole oil gas chromatography amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	2,2,3-TMC5/2,4-DMC6	c,t-1,2,4-TMCyC5	3,3-DMC6	t,c-1,2,3-TMCyC5	2,3,4-TMC5	Tol	2,3-DMC6	2-MC7
206/12a-5	1897.03	1873.4	Spun out oil	146645	65149	54276	25270	60396	12601	264079	51951	313049
	1922.08	1898.4	Spun out oil	146643	223943	173881	72839	179628	33011	1931807	164809	1008112
	1922.40	1898.7	Oil	160302	309092	239625	104910	250461	44976	2580588	313612	1357469
	1948.67	1925.0	Spun out oil	146644	389106	302706	127843	314491	56705	3525765	282206	1742244
	1956.00	1932.3	Oil	160303	266306	207072	91342	218428	40123	2155957	271786	1166457
	1936-1963	1912.3-1939.3	Oil	160301	347698	271788	117688	283942	51542	2794868	355172	1522985
	1984.50	1960.8	Oil	160304	290507	228129	99490	240288	44196	2190430	273377	1275068

Well	MD (m)	TVDSS (m)	Sample type	APTID	4-MC7	3-MC7	c-1,3-DMCyC6	t-1,4-DMCyC6	1,1-DMCyC6	t-1,2-DMCyC6	n-C8	E-CyC6
206/12a-5	1897.03	1873.4	Spun out oil	146645	105400	256529	445376	170509	62387	228717	945842	511512
	1922.08	1898.4	Spun out oil	146643	334149	798714	1236780	501575	161794	565669	2974100	1156499
	1922.40	1898.7	Oil	160302	483377	1056252	1760321	718699	227704	783309	3986966	1512081
	1948.67	1925.0	Spun out oil	146644	576311	1378188	2140121	869306	282326	979486	5106600	1989291
	1956.00	1932.3	Oil	160303	418188	915881	1524489	622580	199150	685528	3439632	1323752
	1936-1963	1912.3-1939.3	Oil	160301	543576	1194596	1982621	806582	260929	891599	4492078	1726603
	1984.50	1960.8	Oil	160304	452358	990882	1670449	682139	220397	754362	3710500	1449520

TABLE Ap5.3 Whole oil gas chromatography amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	i-C9	E-Benz	m-Xyl	p-Xyl	4-MC8	2-MC8	3-MC8	o-Xyl
206/12a-5	1897.03	1873.4	Spun out oil	146645	213487	93640	462679	88204	222119	302339	365926	244999
	1922.08	1898.4	Spun out oil	146643	399084	372831	2015123	852097	493728	672944	773920	801941
	1922.40	1898.7	Oil	160302	535985	510896	2430336	616542	636166	909577	977743	1002620
	1948.67	1925.0	Spun out oil	146644	681276	650774	3478088	878803	821647	1135590	1298590	1388496
	1956.00	1932.3	Oil	160303	474201	431874	2042290	462657	554255	794972	854983	859470
	1936-1963	1912.3-1939.3	Oil	160301	617807	573269	2673982	613682	719764	1034092	1109767	1121949
	1984.50	1960.8	Oil	160304	524007	461813	2091062	576913	599756	861937	927288	898555

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C9	i-C10	n-C10	i-C11	n-C11	n-C12	i-C13	i-C14
206/12a-5	1897.03	1873.4	Spun out oil	146645	1714110	361868	2462270	465674	3241997	4019167	750115	913859
	1922.08	1898.4	Spun out oil	146643	3630852	605762	4099658	671872	4694653	5594475	1161022	1422270
	1922.40	1898.7	Oil	160302	4626875	900087	5021788	879815	5440623	6040100	1059751	1041478
	1948.67	1925.0	Spun out oil	146644	6060096	1009294	6703647	1093775	7490218	8165510	1535560	1823993
	1956.00	1932.3	Oil	160303	4058004	799268	4507600	801749	5130707	6995231	1424149	1627324
	1936-1963	1912.3-1939.3	Oil	160301	5262380	1034131	5746940	1019801	6238617	6952693	1225987	1205356
	1984.50	1960.8	Oil	160304	4395571	870947	4825474	861136	5289607	6135644	1125903	1148274

TABLE Ap5.3 Whole oil gas chromatography amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C13	i-C15	n-C14	i-C16	n-C15	n-C16	i-C18	n-C17
206/12a-5	1897.03	1873.4	Spun out oil	146645	4892060	1016799	5427551	1804001	6204050	6173836	1387996	6394139
	1922.08	1898.4	Spun out oil	146643	7185036	1665736	7029268	2585277	7358030	6555108	1659519	6085761
	1922.40	1898.7	Oil	160302	5914056	1215805	5735223	2071867	6319148	6133213	1533230	6289024
	1948.67	1925.0	Spun out oil	146644	9022895	1943477	8921465	3281975	9483910	8953918	2256435	8732763
	1956.00	1932.3	Oil	160303	9167599	2244241	9176913	3289572	9217635	7940675	2060716	7089712
	1936-1963	1912.3-1939.3	Oil	160301	6805433	1395710	6570272	2382314	7220679	7069122	1747930	7133555
	1984.50	1960.8	Oil	160304	6529509	1411133	6374050	2298058	6843479	6466937	1627715	6419909

Well	MD (m)	TVDSS (m)	Sample type	APTID	Pr	n-C18	Ph	n-C19	n-C20	n-C21	n-C22	n-C23
206/12a-5	1897.03	1873.4	Spun out oil	146645	2486876	6131436	1550084	6803711	5994647	5891522	5438447	5421645
	1922.08	1898.4	Spun out oil	146643	2582689	5173905	1557943	5588597	4614804	4563330	4215051	4172588
	1922.40	1898.7	Oil	160302	2806267	5860469	1582027	6570786	5651309	5497167	5041955	4948877
	1948.67	1925.0	Spun out oil	146644	3558449	8072106	2163472	8840727	7420032	7383651	6751670	6670115
	1956.00	1932.3	Oil	160303	3201668	5743163	1643321	5975630	5109558	4847432	4445171	4367981
	1936-1963	1912.3-1939.3	Oil	160301	3189260	6660247	1851361	7432083	6348222	6159986	5675497	5567086
	1984.50	1960.8	Oil	160304	2875215	5817459	1589098	6437368	5515524	5343273	4918275	4829270

TABLE Ap5.3 Whole oil gas chromatography amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C24	n-C25	n-C26	n-C27	n-C28	n-C29	n-C30	n-C31
206/12a-5	1897.03	1873.4	Spun out oil	146645	4593825	4436139	3501862	3064977	2221962	2088197	1610700	1212454
	1922.08	1898.4	Spun out oil	146643	3522195	3452727	2712895	2456425	1776433	1612637	1252527	928092
	1922.40	1898.7	Oil	160302	4158162	3941940	3229526	2857516	2091529	1826354	1468224	1125903
	1948.67	1925.0	Spun out oil	146644	5632287	5393273	4321497	3814105	2606308	2545795	1956507	1397634
	1956.00	1932.3	Oil	160303	3643507	3471499	2821810	2467419	1809946	1728398	1259559	989167
	1936-1963	1912.3-1939.3	Oil	160301	4669858	4396150	3580502	3154013	2318390	2039835	1718392	1239654
	1984.50	1960.8	Oil	160304	4045946	3736410	3103681	2714487	1975930	1671377	1388520	990308

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C32	n-C33	n-C34	n-C35	n-C36
206/12a-5	1897.03	1873.4	Spun out oil	146645	706239	618553	417477	201310	
	1922.08	1898.4	Spun out oil	146643	580033	479916	287346	204166	170225
	1922.40	1898.7	Oil	160302	788480	679146	428241	404289	276603
	1948.67	1925.0	Spun out oil	146644	776865	631152	341078	149713	115559
	1956.00	1932.3	Oil	160303	720616	609218	379794	379597	255454
	1936-1963	1912.3-1939.3	Oil	160301	893236	759153	465420	473985	333461
	1984.50	1960.8	Oil	160304	683312	571728	312568	287013	171387

TABLE Ap5.3 Whole oil gas chromatography amounts in ng/g

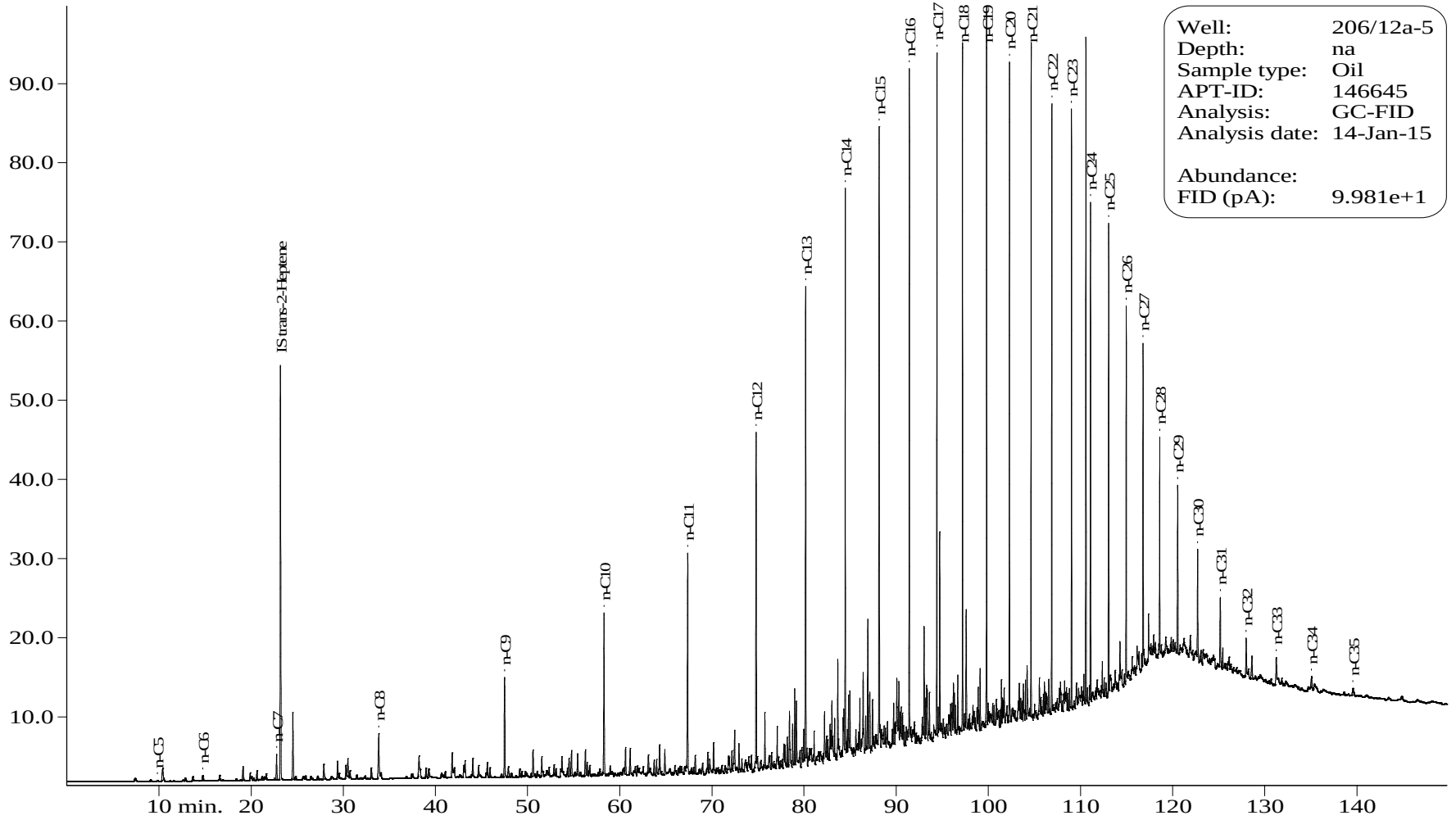


FIGURE Ap5.1 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72)) (full-scale)

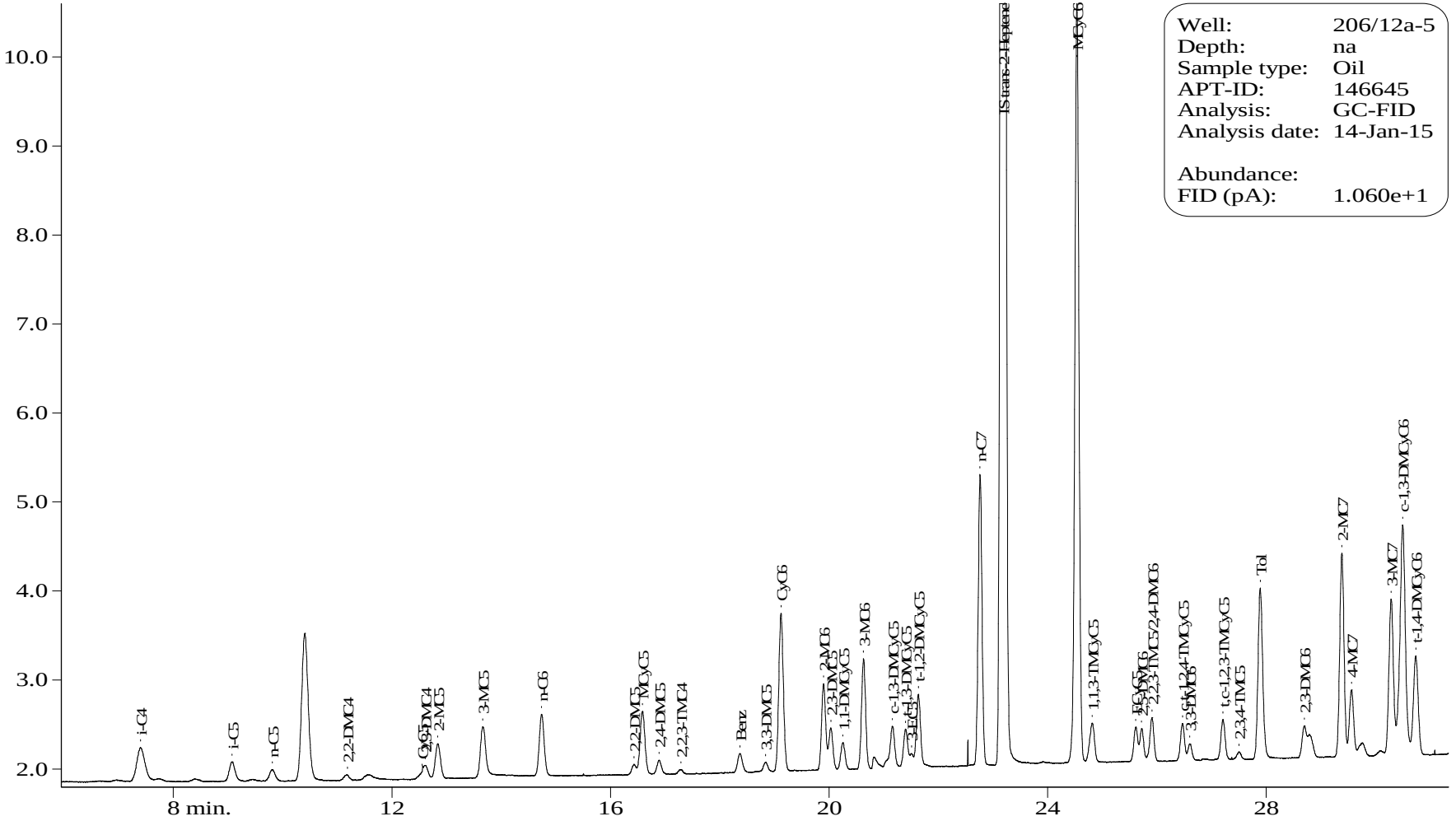


FIGURE Ap5.1 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72)) (C3-C7 range)

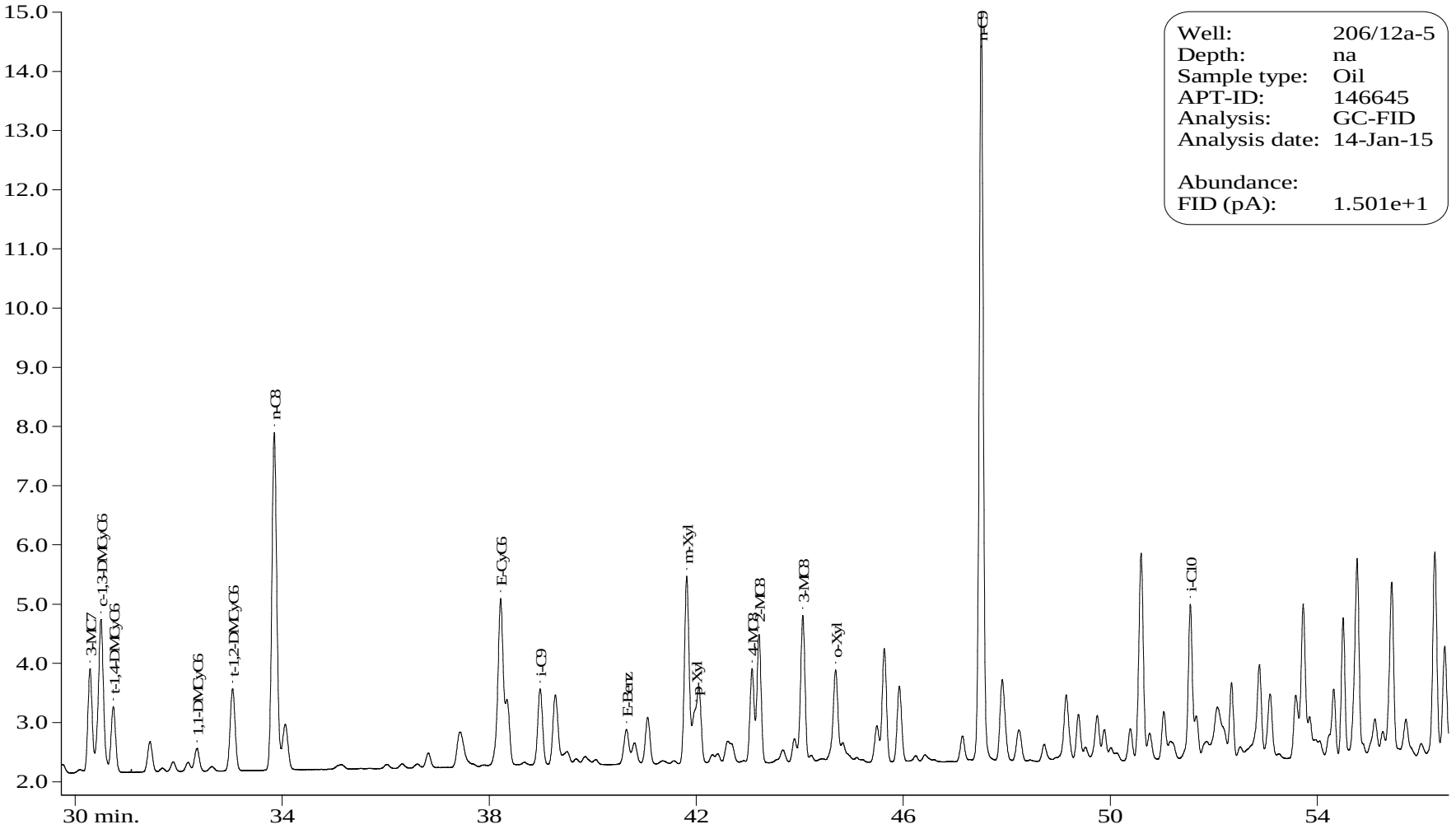


FIGURE Ap5.1 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72)) (C8-C10 range)

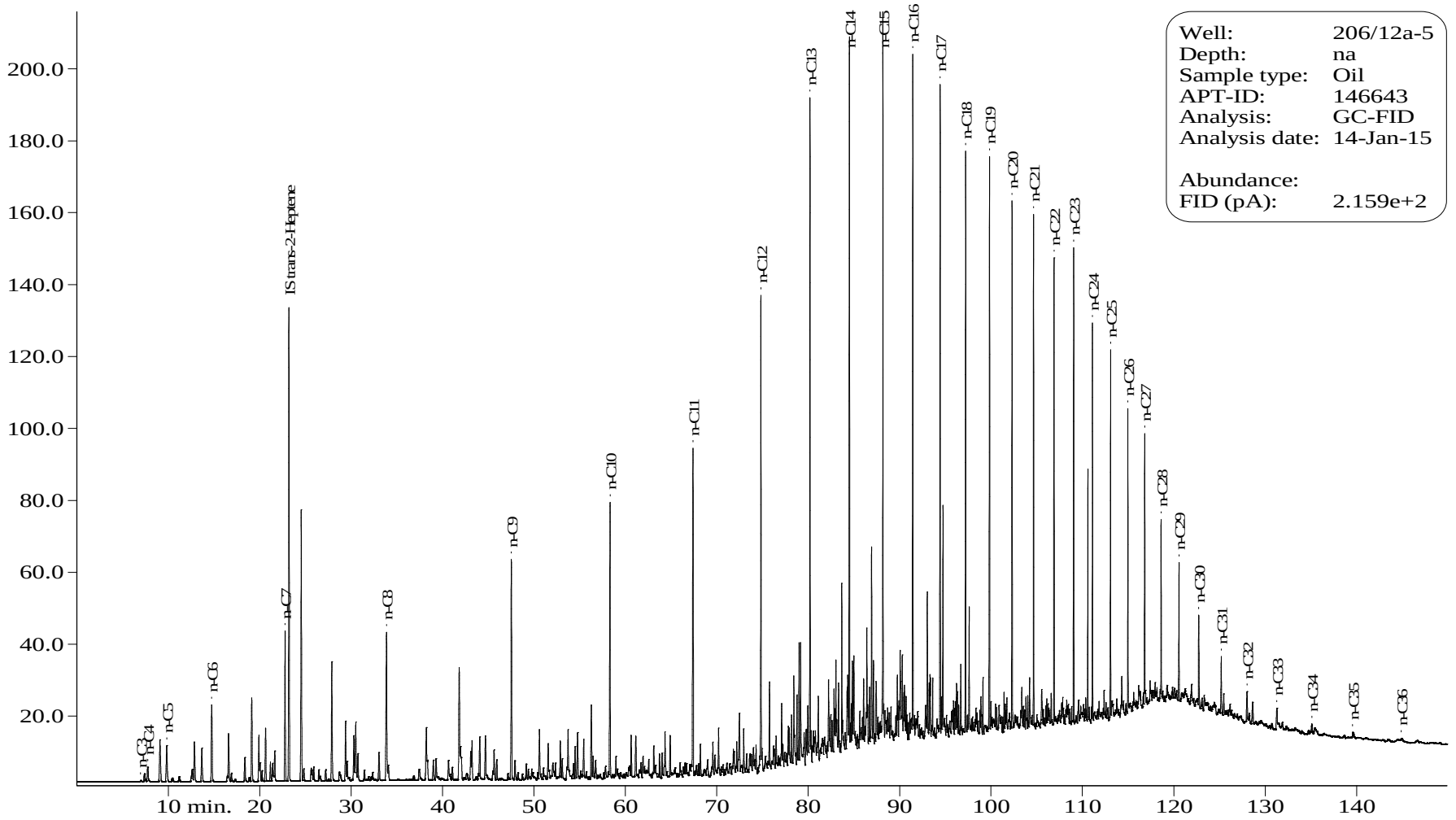


FIGURE Ap5.2 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret) (full-scale)

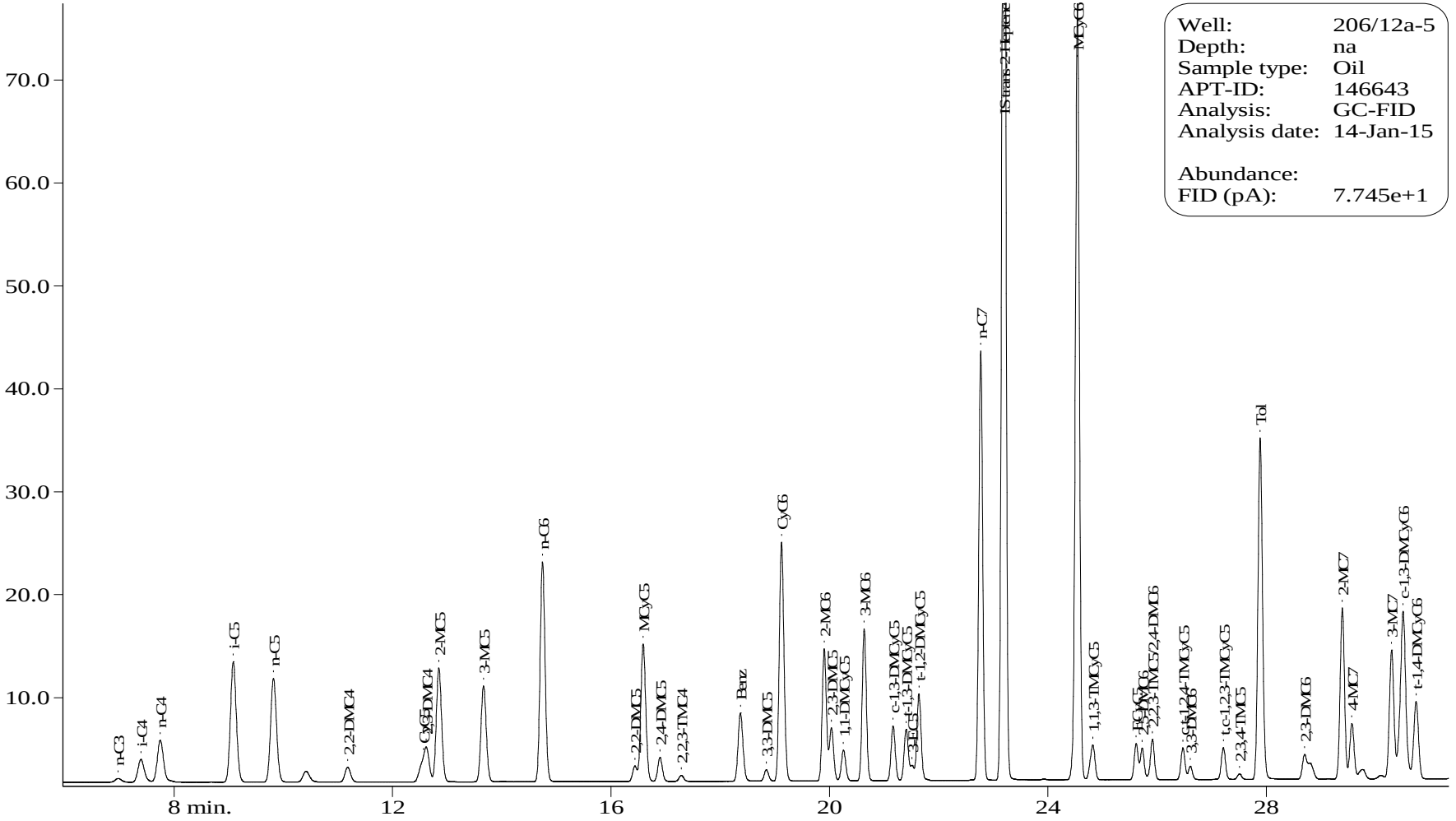


FIGURE Ap5.2 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret) (C3-C7 range)

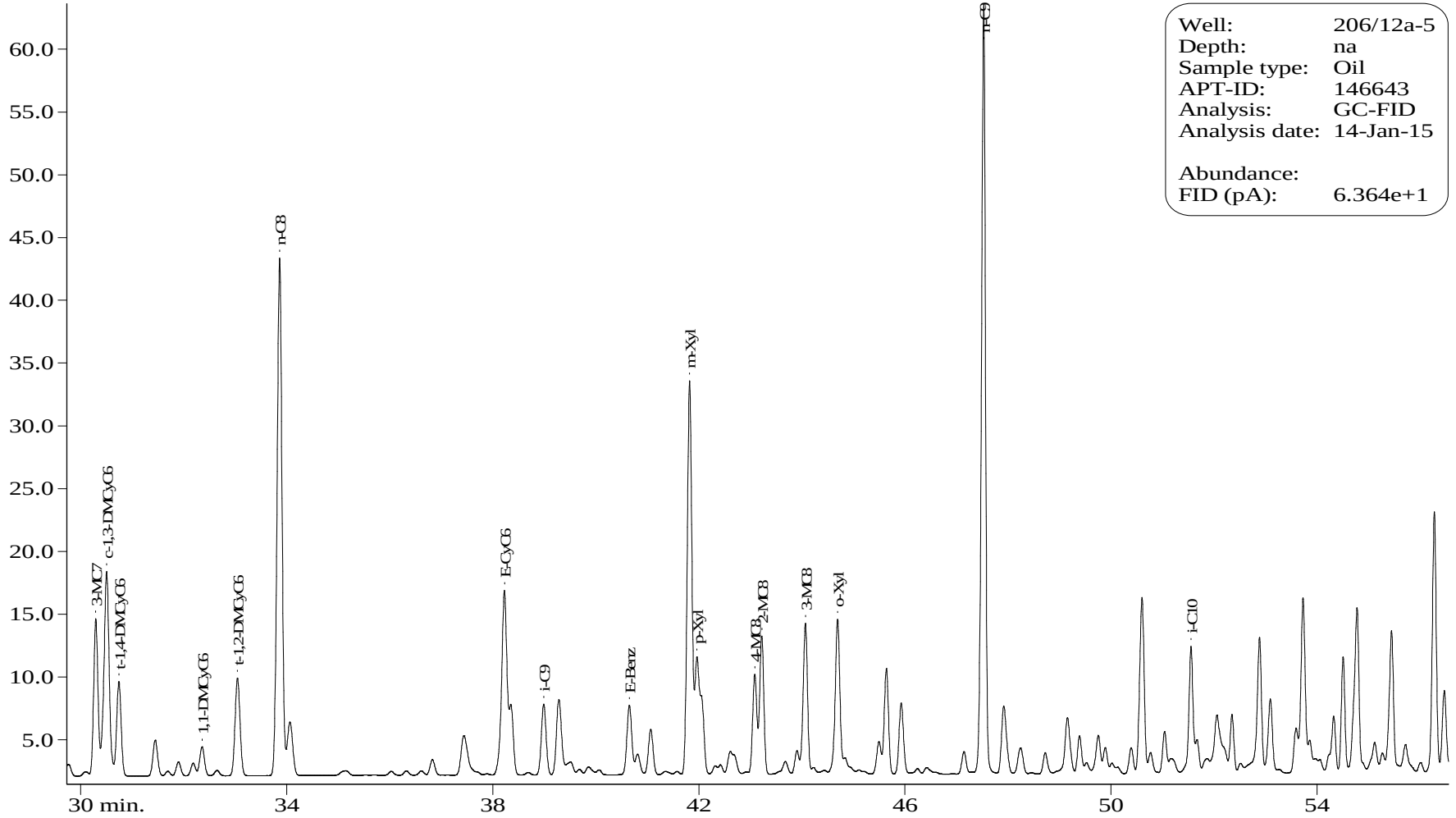


FIGURE Ap5.2 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret) (C8-C10 range)

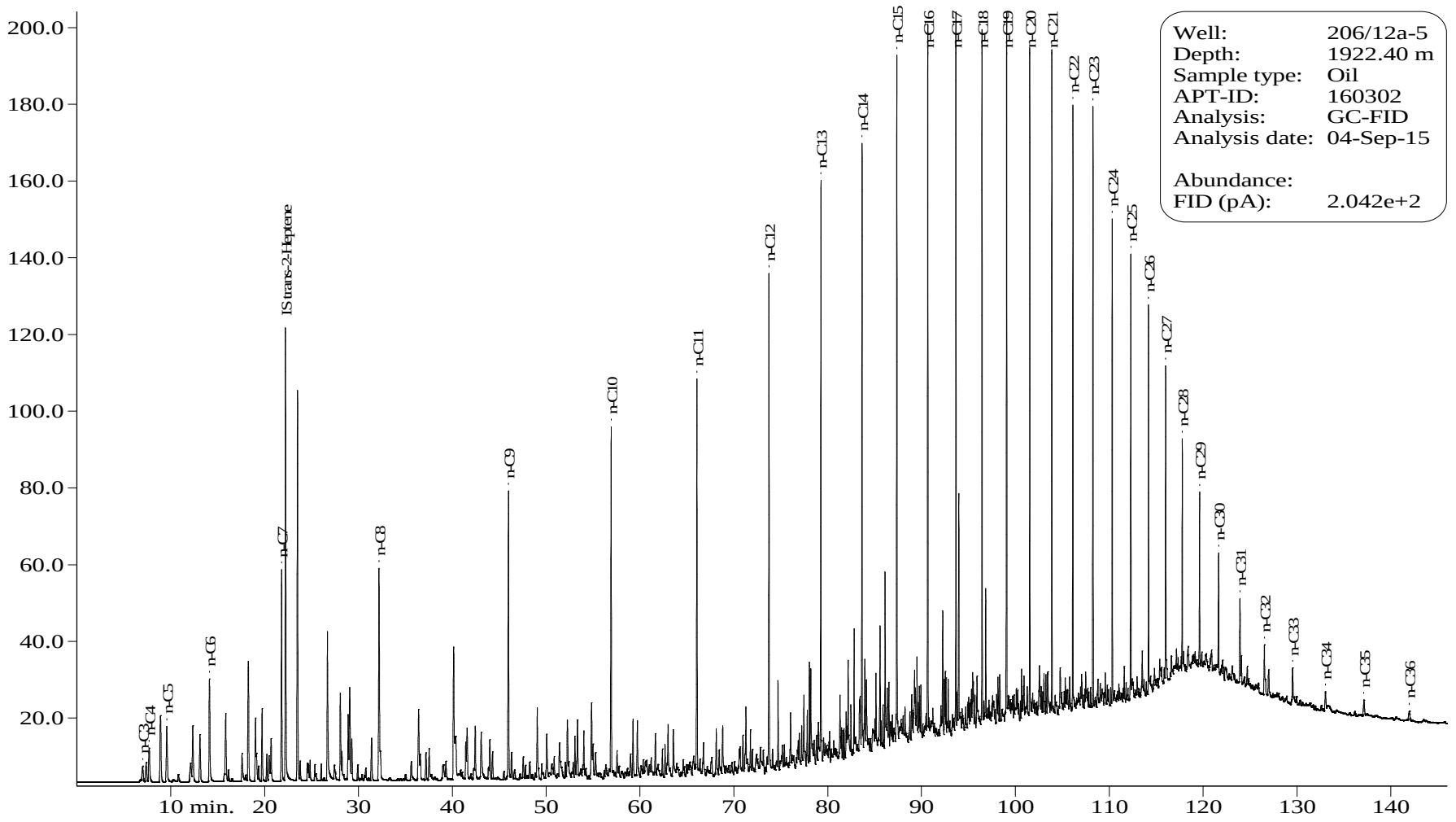


FIGURE Ap5.3 Whole oil gas chromatogram: 206/12a-5, OIL, 1922.4m (K30 Sand Cret) (full-scale)

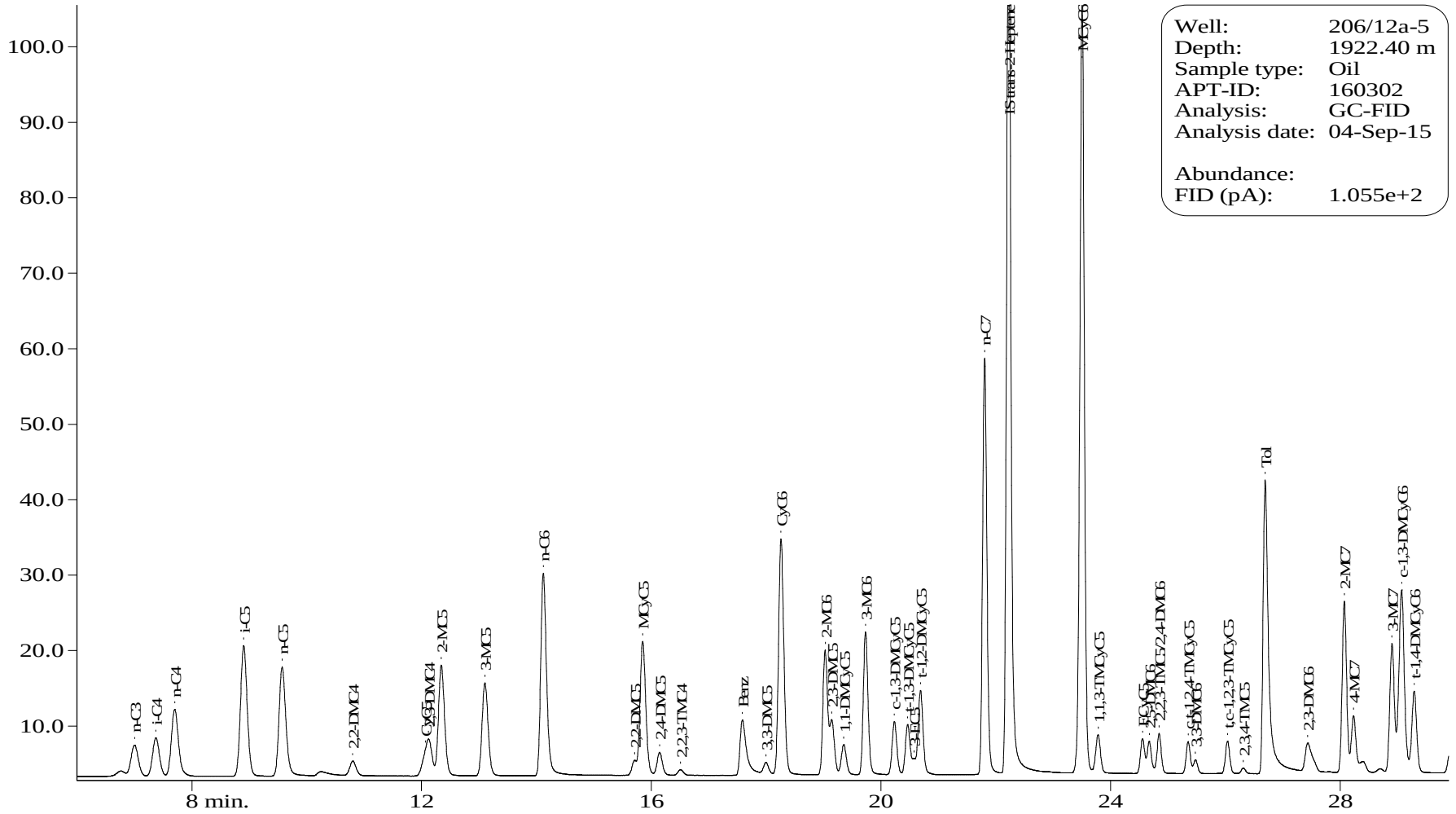


FIGURE Ap5.3 Whole oil gas chromatogram: 206/12a-5, OIL, 1922.4m (K30 Sand Cret) (C3-C7 range)

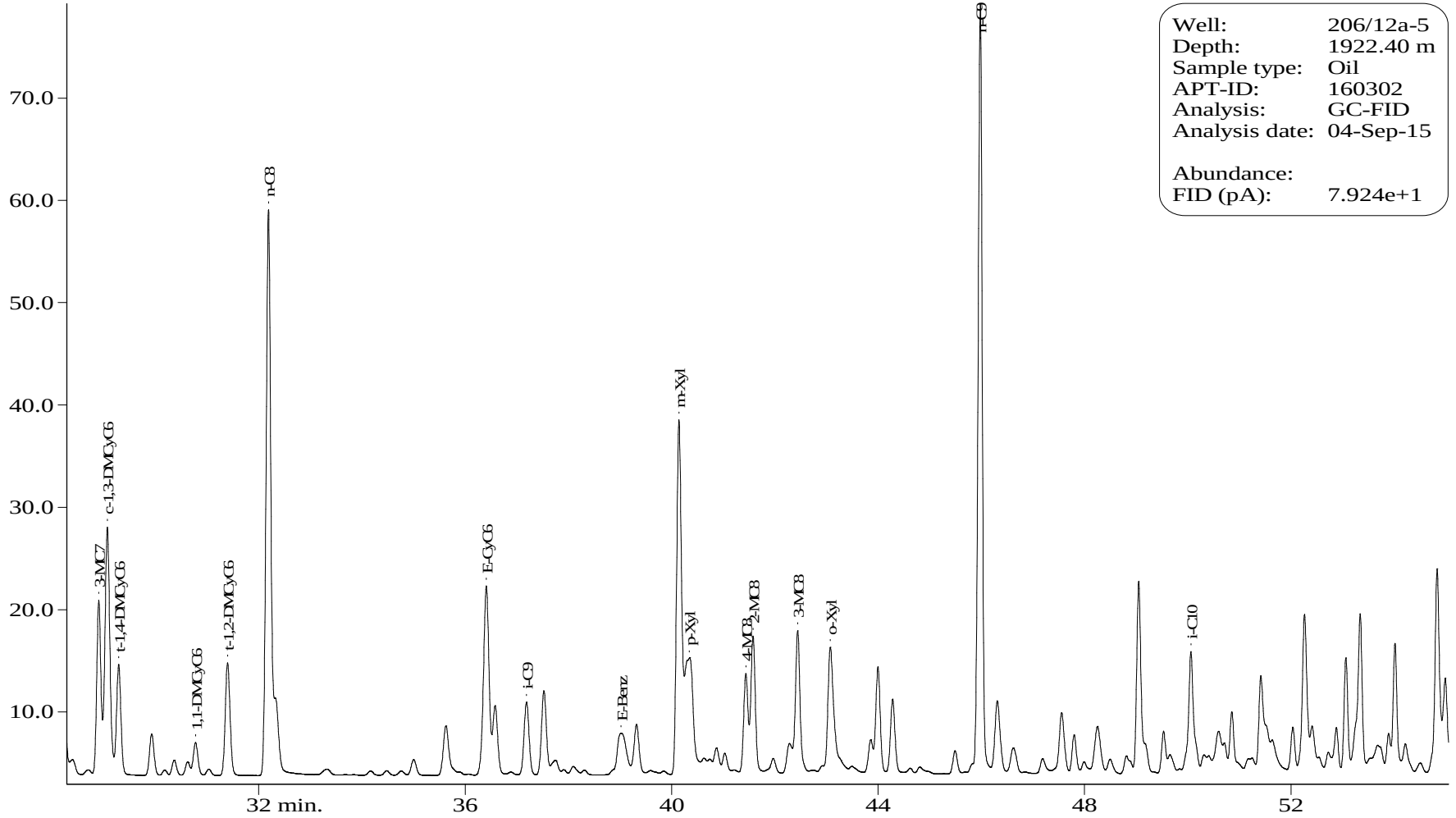


FIGURE Ap5.3 Whole oil gas chromatogram: 206/12a-5, OIL, 1922.4m (K30 Sand Cret) (C8-C10 range)

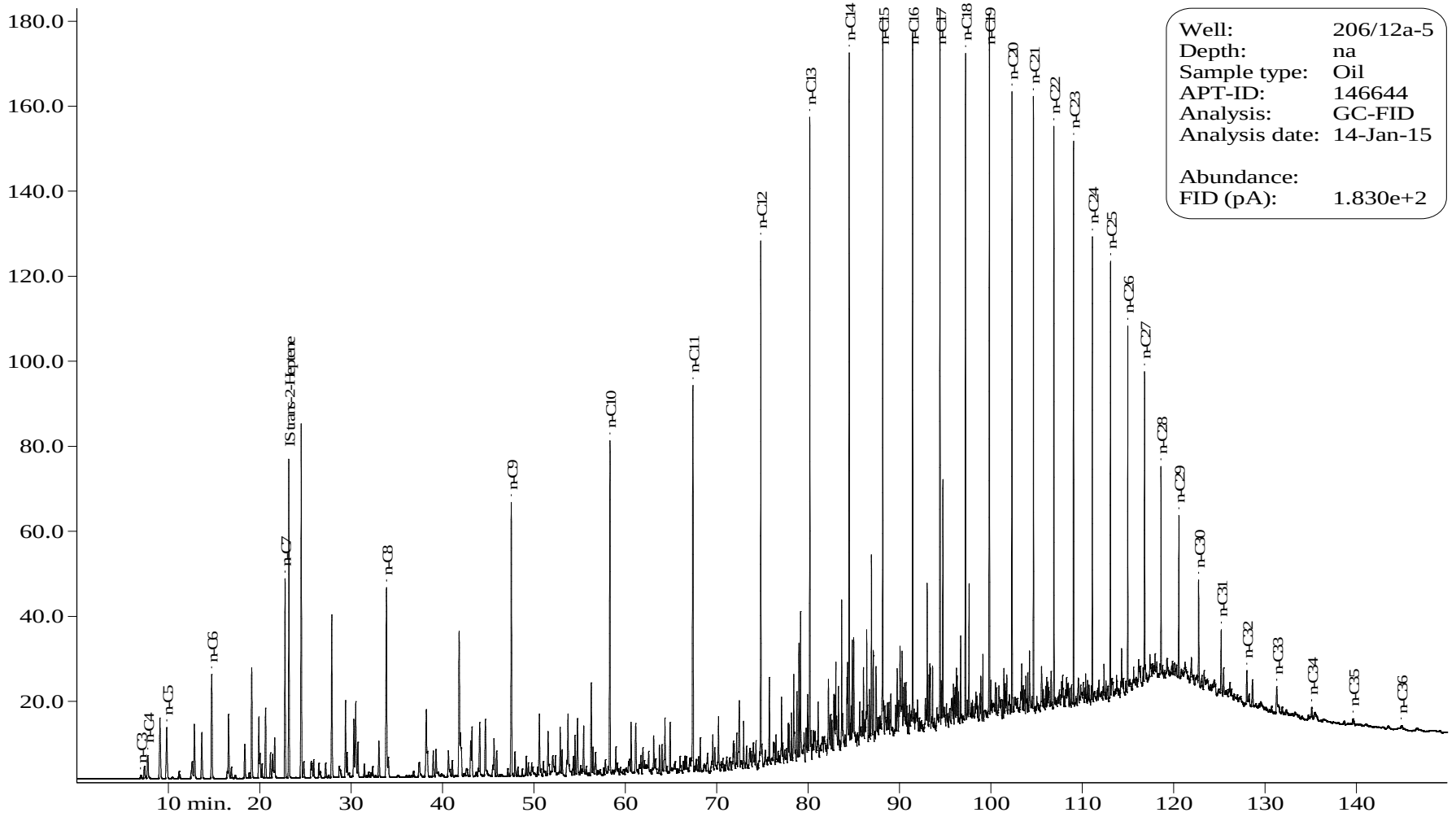


FIGURE Ap5.4 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona) (full-scale)

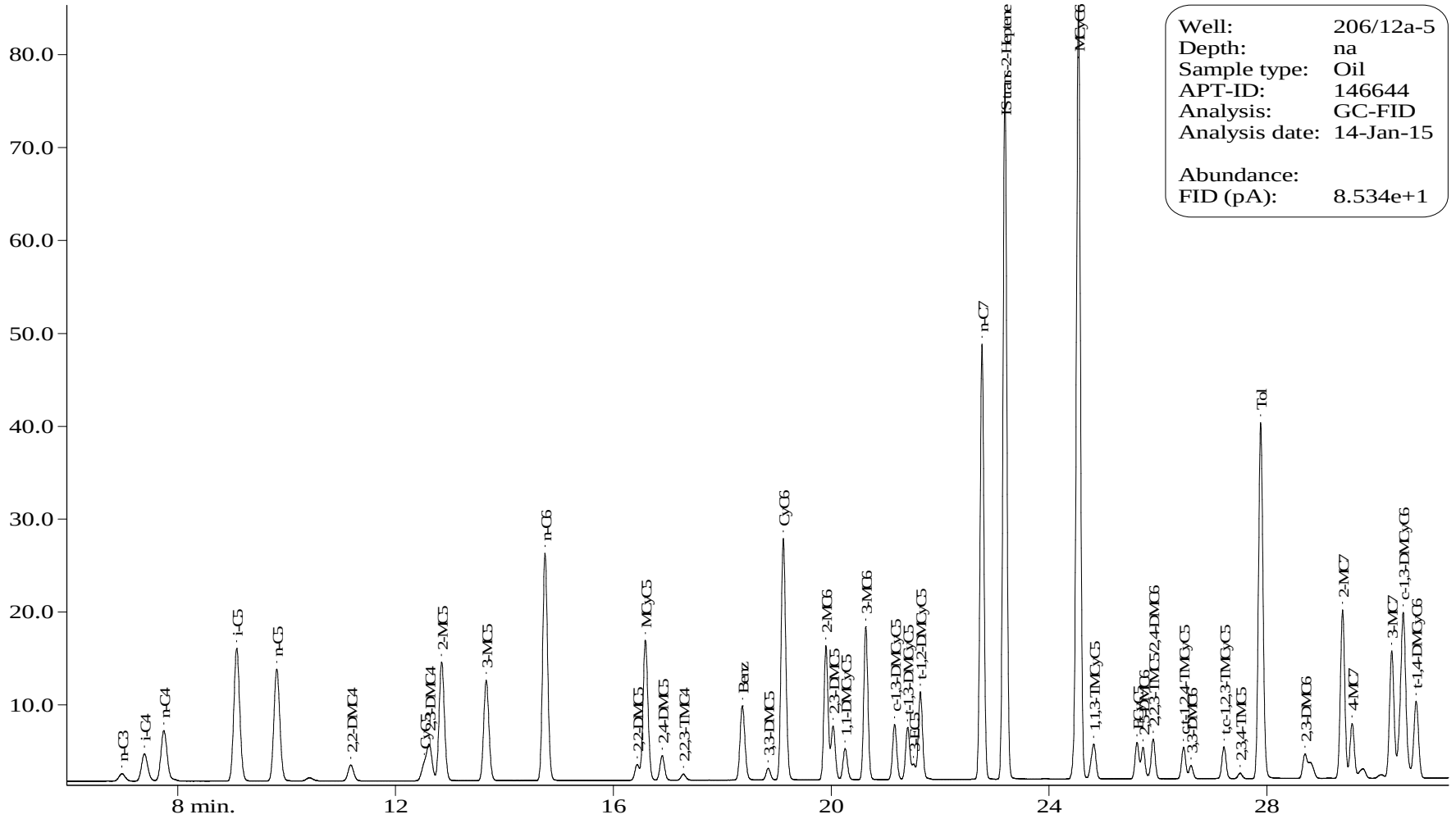


FIGURE Ap5.4 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona) (C3-C7 range)

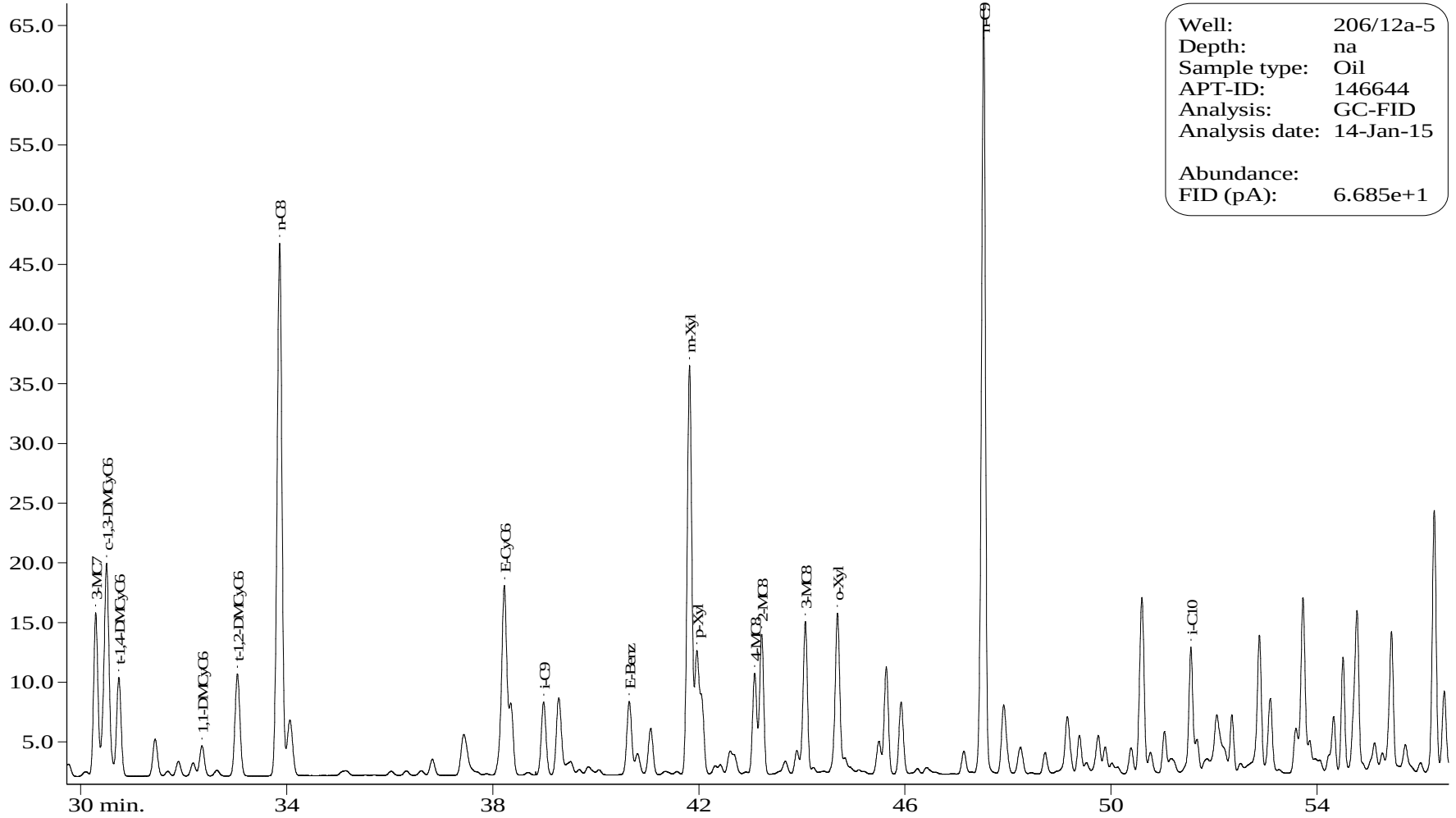


FIGURE Ap5.4 Whole oil gas chromatogram: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona) (C8-C10 range)

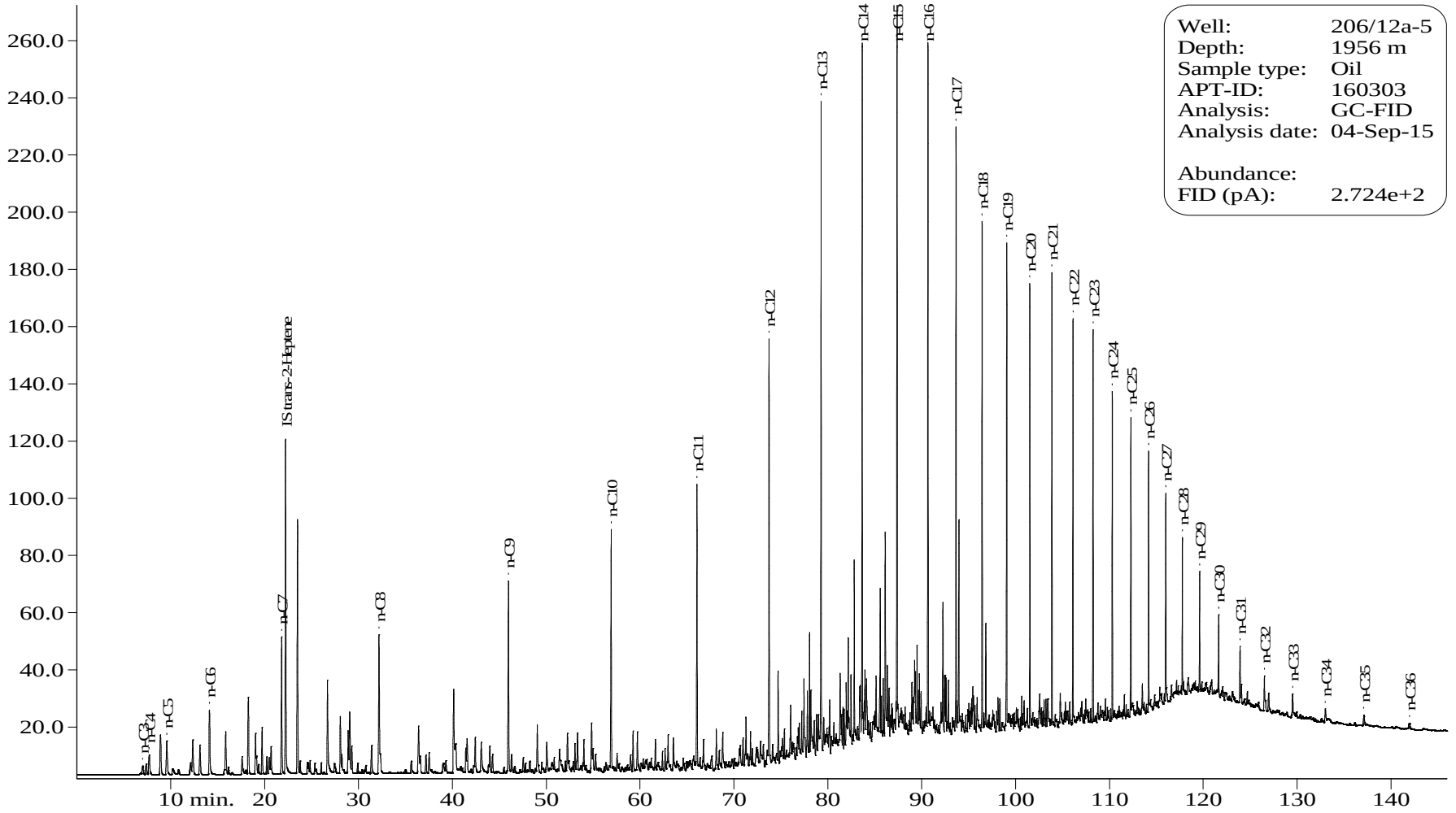


FIGURE Ap5.5 Whole oil gas chromatogram: 206/12a-5, OIL, 1956m (Rona) (full-scale)

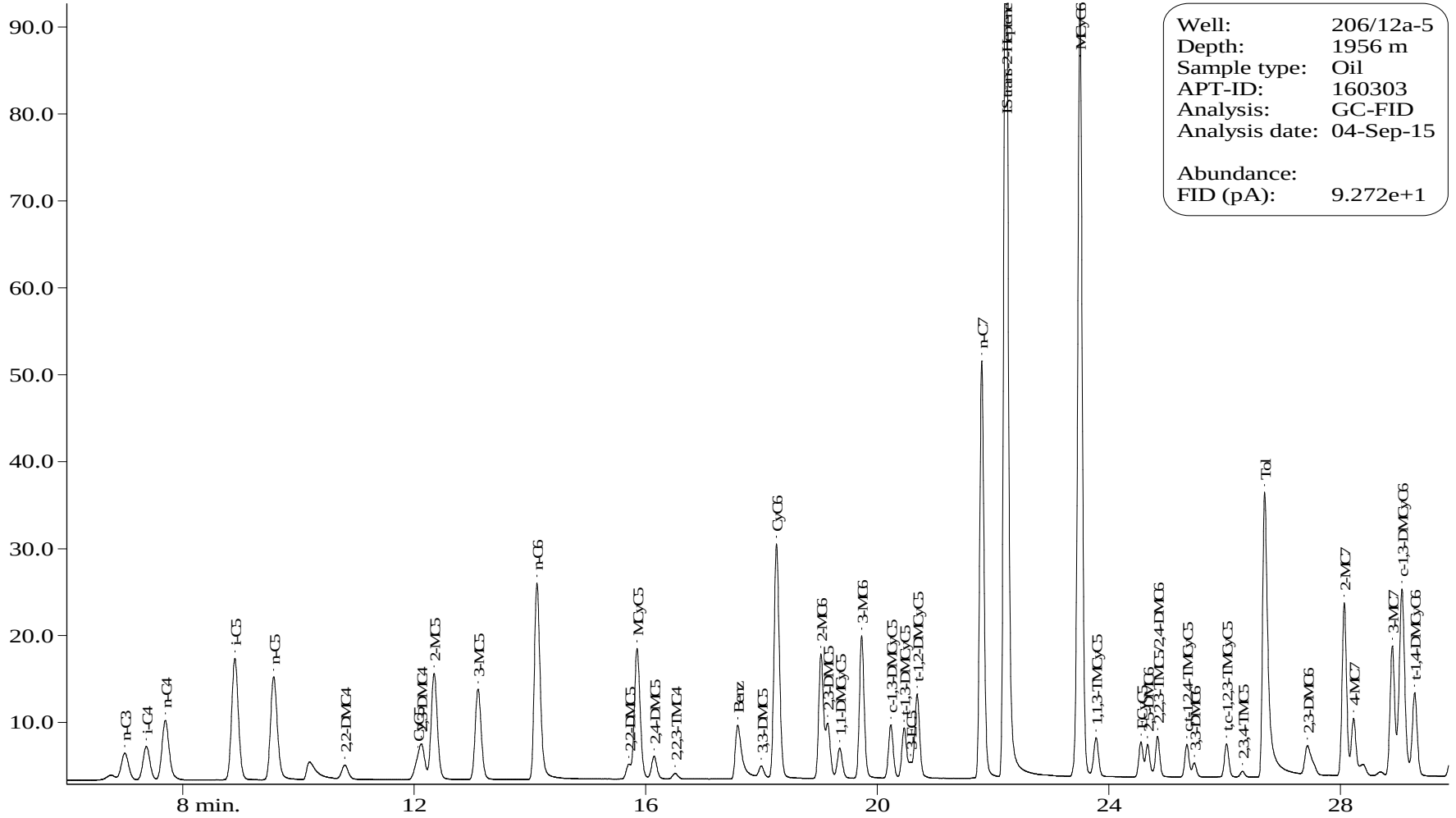


FIGURE Ap5.5 Whole oil gas chromatogram: 206/12a-5, OIL, 1956m (Rona) (C3-C7 range)

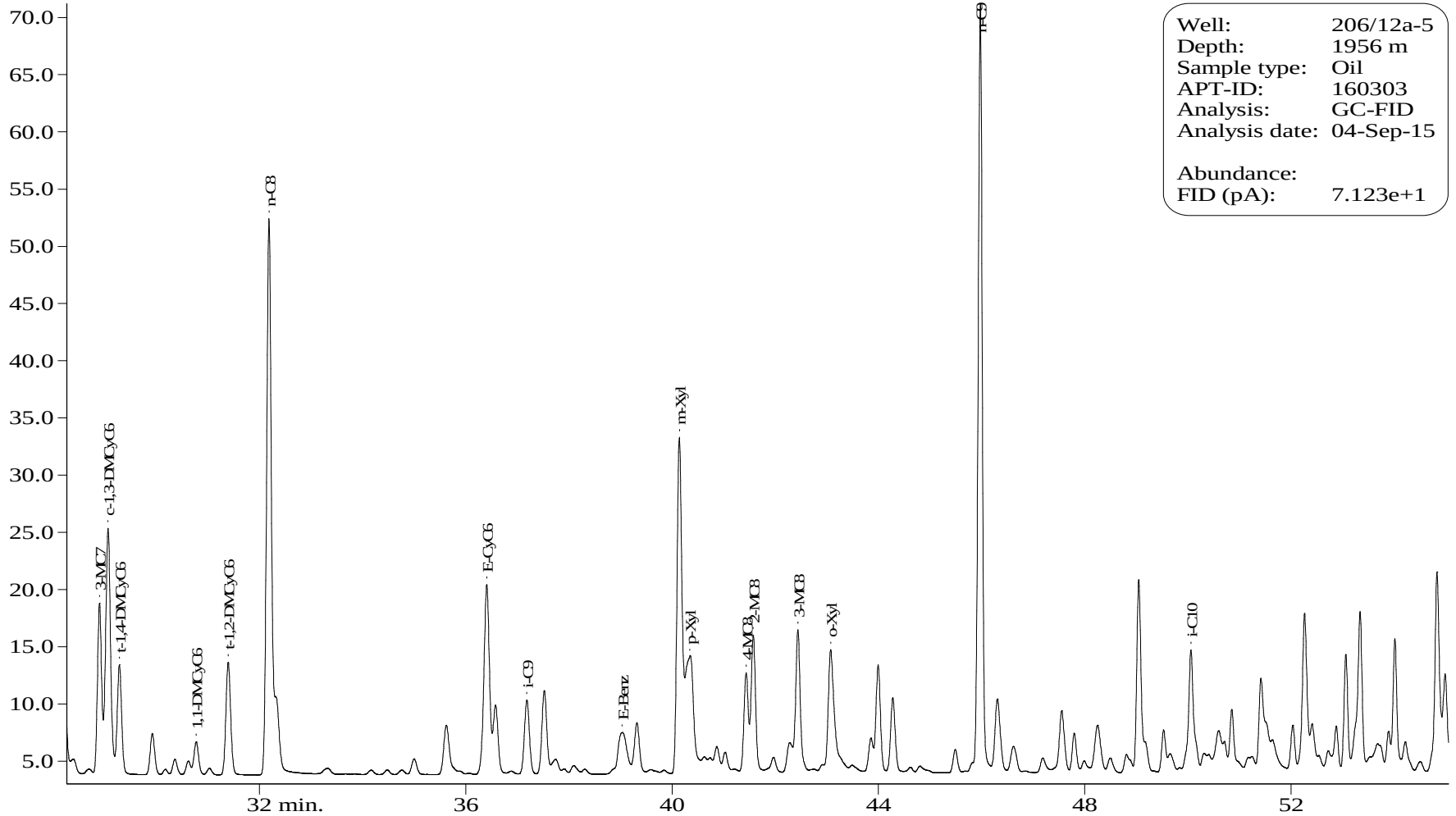


FIGURE Ap5.5 Whole oil gas chromatogram: 206/12a-5, OIL, 1956m (Rona) (C8-C10 range)

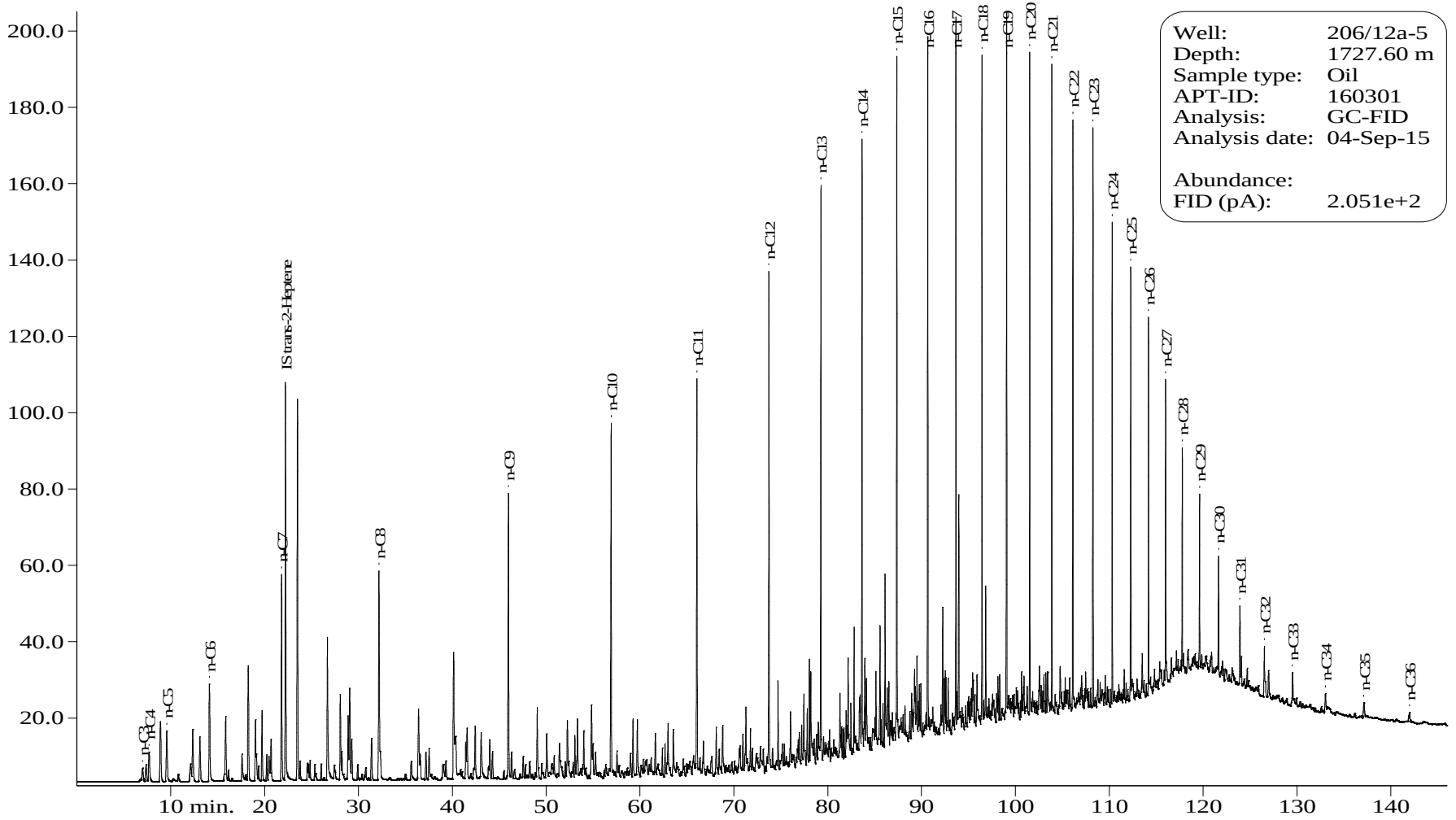


FIGURE Ap5.6 Whole oil gas chromatogram: 206/12a-5, OIL, 1936-1963m (Rona) (full-scale)

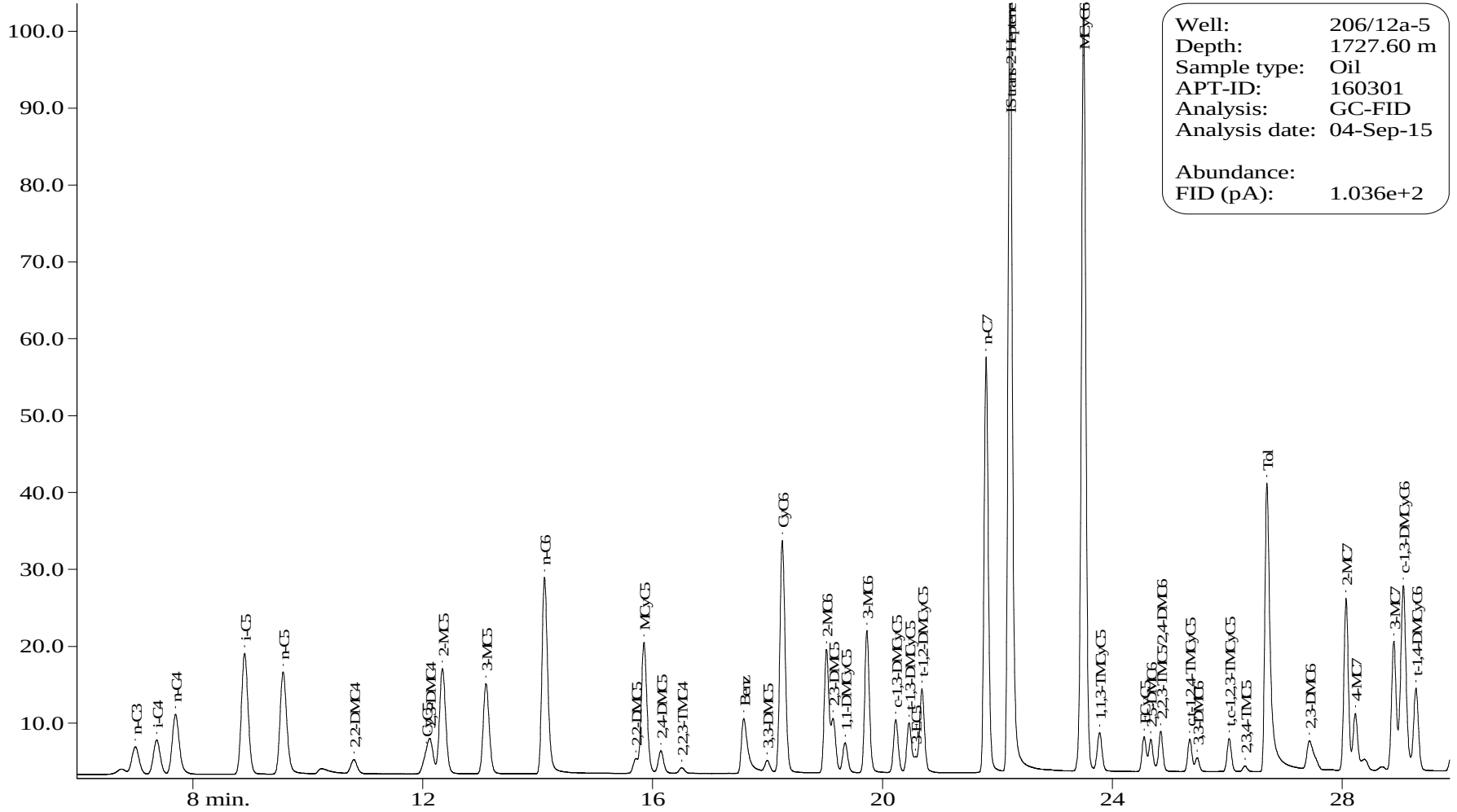


FIGURE Ap5.6 Whole oil gas chromatogram: 206/12a-5, OIL, 1936-1963m (Rona) (C3-C7 range)

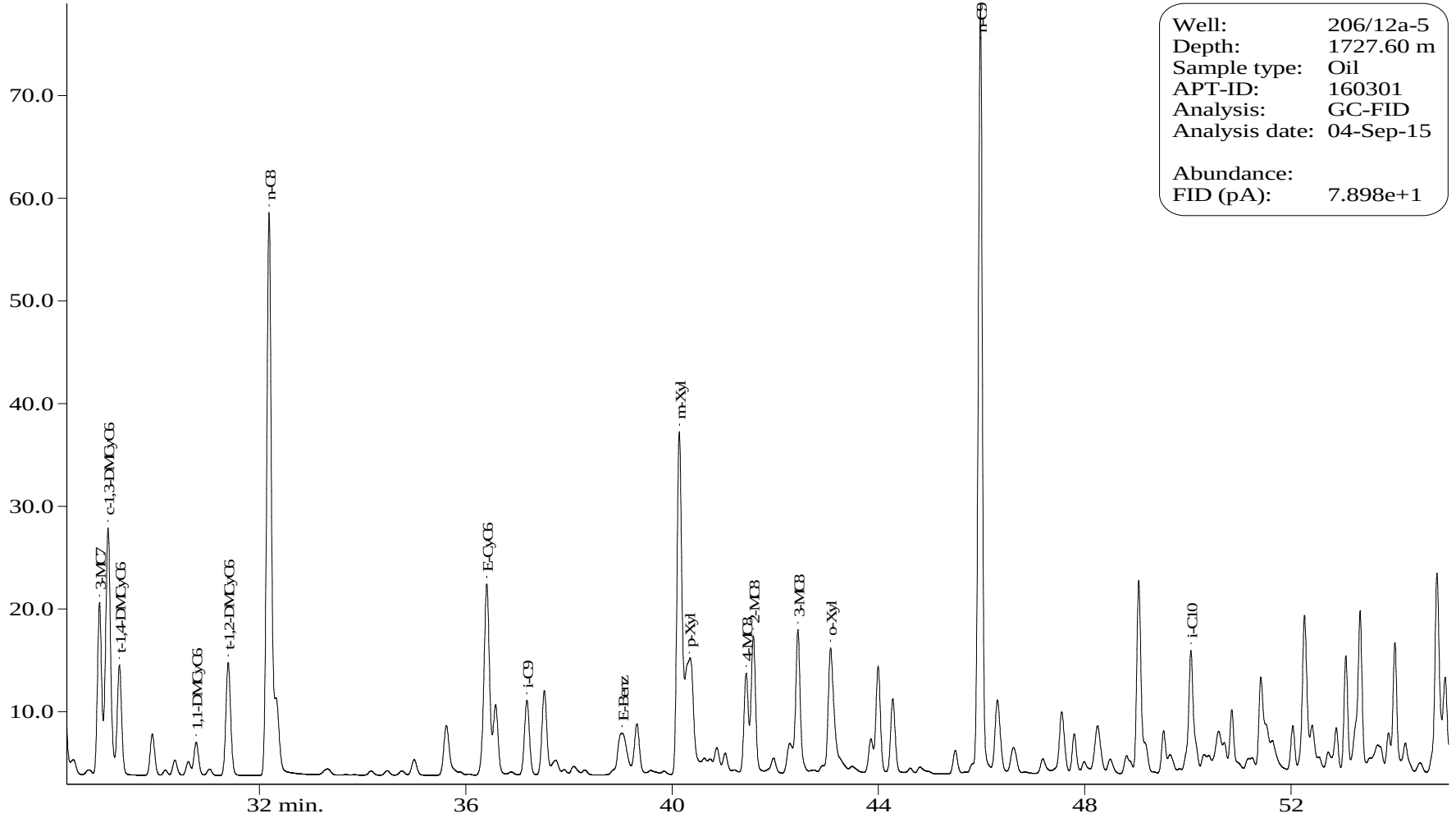


FIGURE Ap5.6 Whole oil gas chromatogram: 206/12a-5, OIL, 1936-1963m (Rona) (C8-C10 range)

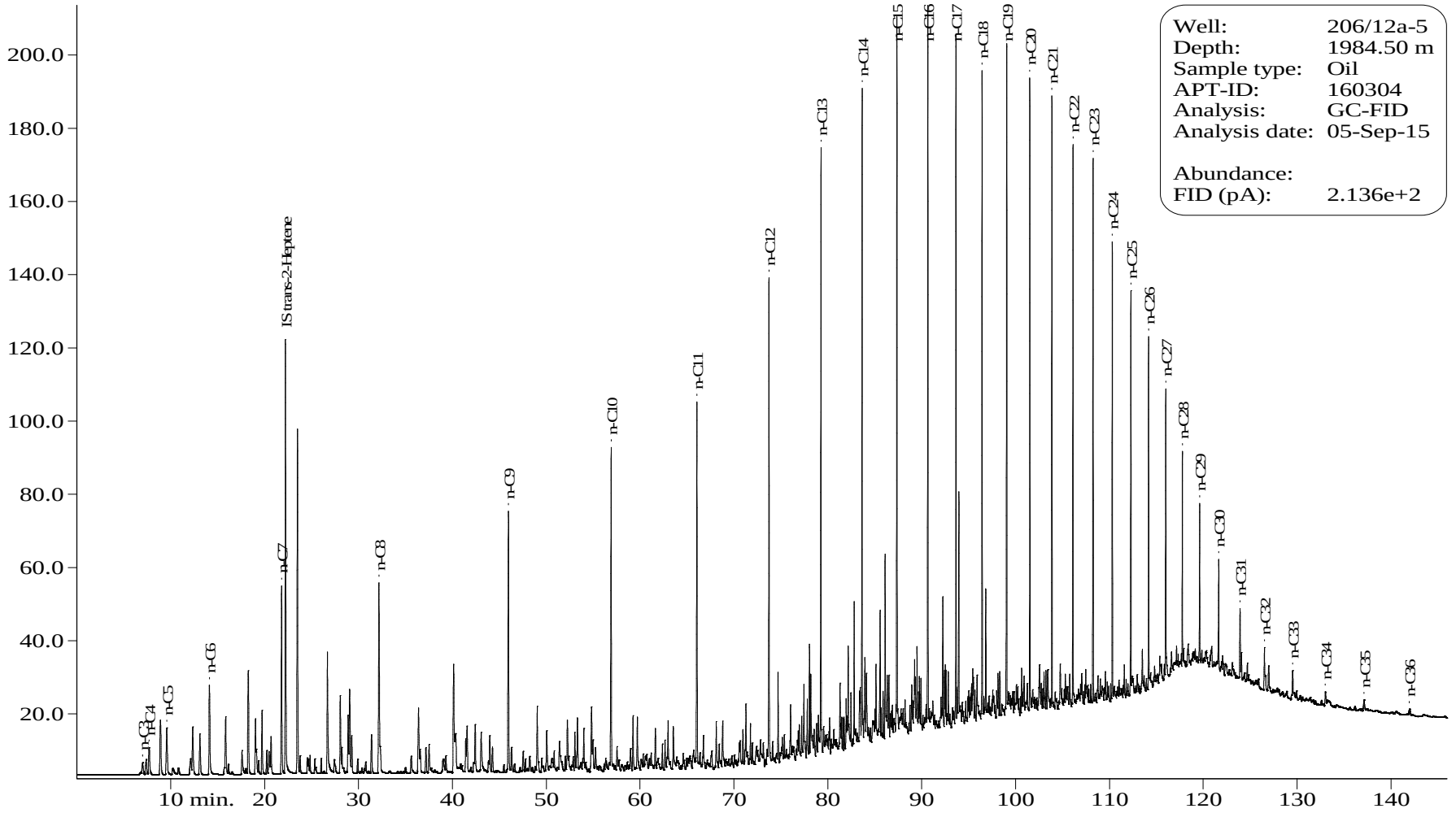


FIGURE Ap5.7 Whole oil gas chromatogram: 206/12a-5, OIL, 1984.5m (LCG Unit III) (full-scale)

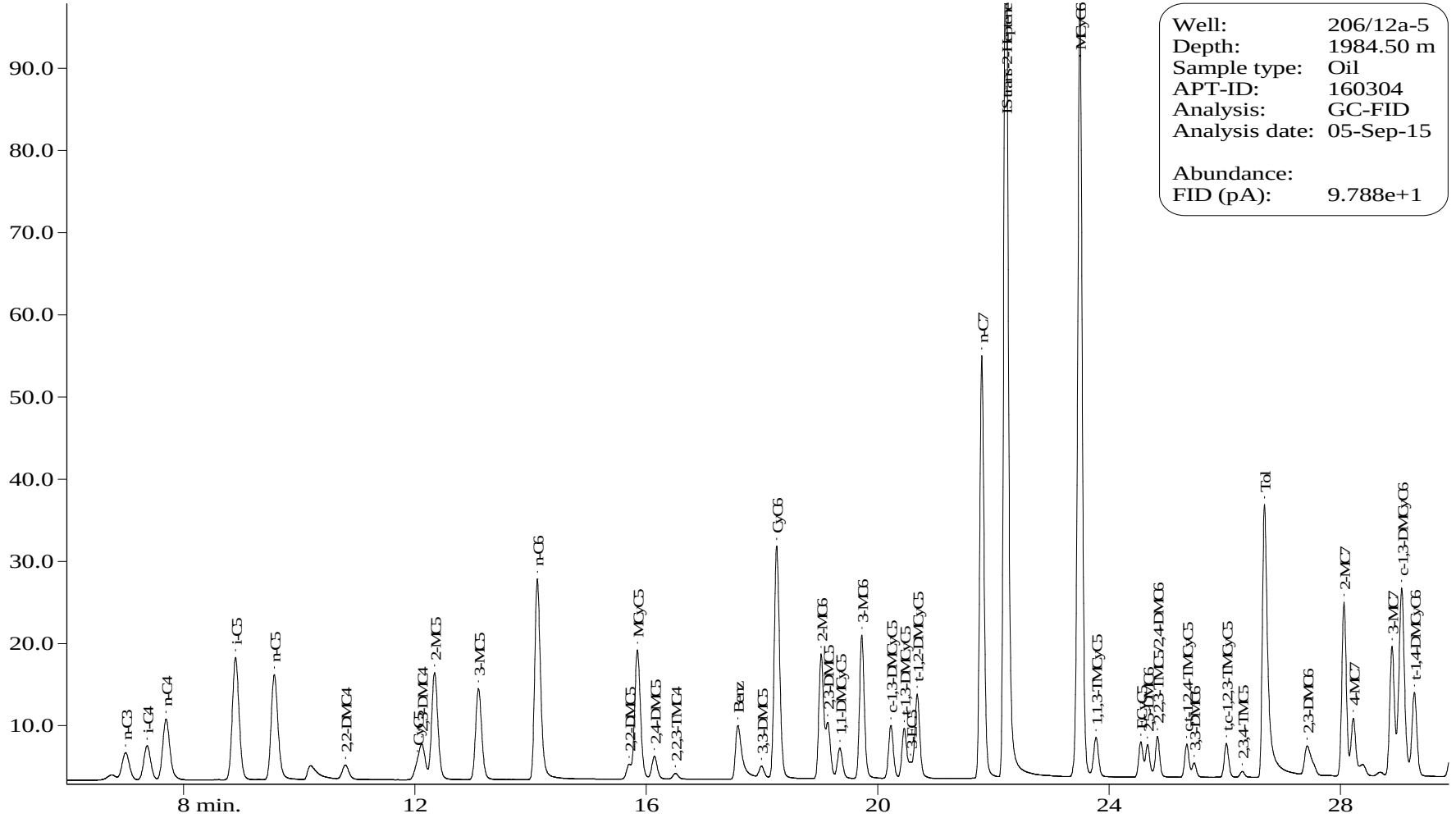


FIGURE Ap5.7 Whole oil gas chromatogram: 206/12a-5, OIL, 1984.5m (LCG Unit III) (C3-C7 range)

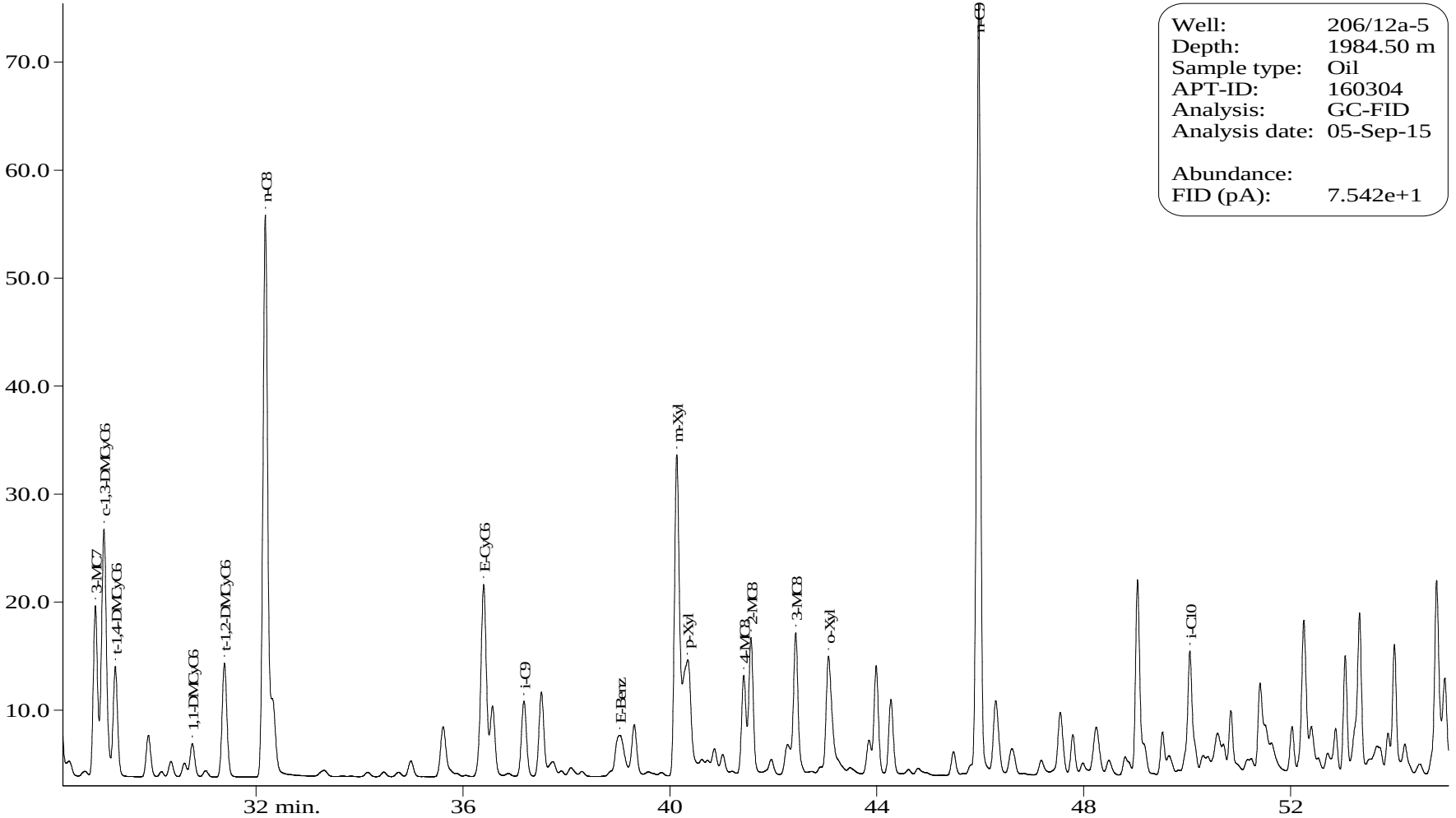


FIGURE Ap5.7 Whole oil gas chromatogram: 206/12a-5, OIL, 1984.5m (LCG Unit III) (C8-C10 range)

APPENDIX 6
Whole extract gas chromatography data

Tables

Ap6.1 Whole extract gas chromatography parameters
Ap6.2 Whole extract gas chromatography peak areas
Ap6.3 Whole extract gas chromatography amounts in ng/g

Figures

Ap6.1 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1884.67m (Turonian (K72))
Ap6.2 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1900.67m (Turonian (K72))
Ap6.3 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))
Ap6.4 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1924.67m (K20 Sand Cret)
Ap6.5 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1932.67m (Kimmeridge Clay Fm.)
Ap6.6 Whole extract gas chromatogram: 206/12a-5, CORE, 1937.32m (Rona)
Ap6.7 Whole extract gas chromatogram: 206/12a-5, MUD, 1938m (Rona)
Ap6.8 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1948.67m (Rona)
Ap6.9 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1956.67m (Rona)
Ap6.10 Whole extract gas chromatogram: 206/12a-5, CORE, 1967.35m (Rona)
Ap6.11 Whole extract gas chromatogram: 206/12a-5, SWC, 1984m (LCG Unit III)
Ap6.12 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1988.67m (LCG Unit III)
Ap6.13 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)
Ap6.14 Whole extract gas chromatogram: 206/12a-5, SWC, 2001m (LCG Unit III)
Ap6.15 Whole extract gas chromatogram: 206/12a-5, SWC, 2022m (LCG Unit II /I)
Ap6.16 Whole extract gas chromatogram: 206/12a-5, SWC, 2048m (LCG Unit II /I)
Ap6.17 Whole extract gas chromatogram: 206/12a-5, MUD, 2352m (Basement)

Well	MD (m)	TVDSS (m)	Sample type	APTID	CPI-1	Pr / n-C17	Ph / n-C18	(Pr/ n-C17) / (Ph/ n-C18)	Pr / Ph	n-C17 / (n-C17+ n-C27)
206/12a-5	1884.67	1861.0	Core plug	146965	1.13	0.36	0.26	1.37	1.42	0.79
	1900.67	1877.0	Core plug	146969	1.13	0.48	0.51	0.95	2.69	1.00
	1916.67	1893.0	Core plug	146972	1.09	0.37	0.24	1.52	1.58	0.69
	1924.67	1901.0	Core plug	146974	1.26	0.57	0.64	0.89	1.04	0.83
	1932.67	1909.0	Core plug	146976	1.09	0.41	0.27	1.52	1.58	0.69
	1937.32	1913.7	Core	154661	1.15	0.43	0.35	1.23	2.10	0.89
	1938.00	1914.3	Mud	163301	0.95	0.50	0.55	0.91	2.58	1.00
	1948.67	1925.0	Core plug	146980	1.10	0.42	0.27	1.54	1.58	0.69
	1956.67	1933.0	Core plug	146982	1.10	0.42	0.27	1.55	1.59	0.70
	1967.35	1943.7	Core	154666	1.13	0.44	0.31	1.40	1.97	0.79
	1984.00	1960.3	SWC	153498	1.09	0.50	0.51	0.97	2.23	0.98
	1988.67	1965.0	Core plug	146990	1.10	0.39	0.25	1.55	1.63	0.74
	1997.67	1974.0	Core plug	146992	1.10	0.41	0.26	1.56	1.63	0.69
	2001.00	1977.3	SWC	153499	1.10	0.49	0.53	0.94	2.58	0.99
	2022.00	1998.3	SWC	153500	1.12	0.48	0.52	0.91	2.57	0.99
	2048.00	2024.3	SWC	153501	1.03	0.51	0.54	0.95	2.45	1.00
	2352.00	2336.6	Mud	163302	0.92	0.51	0.57	0.90	2.72	1.00

CPI-1 : Carbon Preference Index-1: 0.5 * [{ (C25+C27+C29+C31)/(C24+C26+C28+C30) } + { (C25+C27+C29+C31)/(C26+C28+C30+C32) }]

TABLE AP6.1 Whole extract gas chromatography parameters

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C10	n-C11	n-C12	i-C13	i-C14	n-C13	i-C15	n-C14	i-C16
206/12a-5	1884.67	1861.0	Core plug	146965	70	102	164	27	369	298	97	668	395
	1900.67	1877.0	Core plug	146969	29	495	3686	1481	2507	14447	4505	15823	6164
	1916.67	1893.0	Core plug	146972	531	1126	1689	326	418	2433	504	3056	1234
	1924.67	1901.0	Core plug	146974	9	3	2	1	1	8	5	28	21
	1932.67	1909.0	Core plug	146976	914	2172	3636	700	1041	4882	1175	6200	2715
	1937.32	1913.7	Core	154661	939	1074	1760	389	2641	3511	1145	6847	3115
	1938.00	1914.3	Mud	163301	362	2163	9559	3991	2847	28376	8766	30133	12084
	1948.67	1925.0	Core plug	146980	873	1407	1962	371	500	2404	514	2961	1307
	1956.67	1933.0	Core plug	146982	1151	1901	2618	492	920	3184	757	3977	1748
	1967.35	1943.7	Core	154666	94	212	999	258	2427	4520	1962	11119	4956
	1984.00	1960.3	SWC	153498	5931	1	77	51	409	2966	2537	11260	7493
	1988.67	1965.0	Core plug	146990	63	126	238	46	370	425	128	794	378
	1997.67	1974.0	Core plug	146992	942	1483	1828	344	519	2085	421	2412	1024
	2001.00	1977.3	SWC	153499	5825	1	179	129	976	7505	5880	24837	13888
	2022.00	1998.3	SWC	153500	2200	1	134	101	737	5962	4805	19719	10764
	2048.00	2024.3	SWC	153501	6035	1	118	83	646	5749	4981	22193	12830
	2352.00	2336.6	Mud	163302	305	1476	8607	3806	2751	26584	8429	28675	11399

TABLE AP6.2 Whole extract gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C15	n-C16	i-C18	n-C17	Pr	n-C18	Ph	n-C19	n-C20
206/12a-5	1884.67	1861.0	Core plug	146965	660	967	240	1022	369	985	260	747	673
	1900.67	1877.0	Core plug	146969	14522	10279	4580	6477	3118	2272	1157	596	227
	1916.67	1893.0	Core plug	146972	3233	3472	826	3517	1305	3396	827	3392	3365
	1924.67	1901.0	Core plug	146974	62	78	31	88	50	75	48	61	46
	1932.67	1909.0	Core plug	146976	6294	6643	1807	6689	2762	6407	1744	6309	6175
	1937.32	1913.7	Core	154661	5789	4806	2180	3587	1557	2102	741	1408	1315
	1938.00	1914.3	Mud	163301	26249	18725	8468	11994	6026	4225	2338	1223	512
	1948.67	1925.0	Core plug	146980	3021	3283	846	3322	1391	3248	881	3174	3162
	1956.67	1933.0	Core plug	146982	3974	4303	1150	4447	1872	4325	1175	4240	4191
	1967.35	1943.7	Core	154666	9423	7468	3205	5918	2587	4209	1312	3517	3441
	1984.00	1960.3	SWC	153498	22874	31789	16026	27557	13708	11960	6139	4097	2340
	1988.67	1965.0	Core plug	146990	692	777	204	704	277	669	170	602	599
	1997.67	1974.0	Core plug	146992	2413	2522	622	2514	1037	2407	637	2372	2355
	2001.00	1977.3	SWC	153499	34681	33504	14685	22409	11076	8160	4300	2420	1150
	2022.00	1998.3	SWC	153500	26471	21154	9156	14140	6751	5024	2622	1388	565
	2048.00	2024.3	SWC	153501	34847	35182	16025	23866	12256	9226	5007	2540	1028
	2352.00	2336.6	Mud	163302	24583	18521	8488	12162	6184	4001	2272	982	345

TABLE AP6.2 Whole extract gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C21	n-C22	n-C23	n-C24	n-C25	n-C26	n-C27	n-C28	n-C29
206/12a-5	1884.67	1861.0	Core plug	146965	595	544	511	418	421	310	277	235	209
	1900.67	1877.0	Core plug	146969	141	65	36	23	16	14	10	7	11
	1916.67	1893.0	Core plug	146972	3194	2933	2979	2559	2436	1856	1616	1387	1251
	1924.67	1901.0	Core plug	146974	32	30	23	25	36	21	18	16	15
	1932.67	1909.0	Core plug	146976	5846	5310	5344	4576	4402	3331	2939	2601	2402
	1937.32	1913.7	Core	154661	1256	955	961	673	668	490	424	344	274
	1938.00	1914.3	Mud	163301	265	128	54	36	20	19	10	6	6
	1948.67	1925.0	Core plug	146980	2956	2692	2719	2336	2234	1682	1474	1261	1158
	1956.67	1933.0	Core plug	146982	3929	3575	3610	3093	2937	2235	1951	1699	1563
	1967.35	1943.7	Core	154666	3393	2956	3010	2411	2516	1815	1581	1325	1097
	1984.00	1960.3	SWC	153498	1769	1416	1207	1013	928	728	625	529	505
	1988.67	1965.0	Core plug	146990	570	501	490	403	373	280	243	202	170
	1997.67	1974.0	Core plug	146992	2207	2023	2034	1748	1682	1275	1116	970	890
	2001.00	1977.3	SWC	153499	680	566	402	323	289	229	193	149	153
	2022.00	1998.3	SWC	153500	366	258	173	131	114	92	72	52	59
	2048.00	2024.3	SWC	153501	514	341	187	140	117	107	81	64	66
	2352.00	2336.6	Mud	163302	175	82	39	27	17	15	9	5	5

TABLE AP6.2 Whole extract gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C30	n-C31	n-C32	n-C33	n-C34	n-C35	n-C36
206/12a-5	1884.67	1861.0	Core plug	146965	146	130	92	83	97	66	31
	1900.67	1877.0	Core plug	146969	6	7	5	3	3	2	
	1916.67	1893.0	Core plug	146972	902	764	590	507	562	348	259
	1924.67	1901.0	Core plug	146974	13	16	11	10	9	7	25
	1932.67	1909.0	Core plug	146976	1738	1466	1128	1034	1053	909	549
	1937.32	1913.7	Core	154661	167	179	115	97	116	42	43
	1938.00	1914.3	Mud	163301	5	4	1	21	2	50	1
	1948.67	1925.0	Core plug	146980	842	693	534	485	499	340	237
	1956.67	1933.0	Core plug	146982	1103	949	731	656	656	467	329
	1967.35	1943.7	Core	154666	801	718	509	456	489	279	179
	1984.00	1960.3	SWC	153498	351	273	199	413	182	1258	64
	1988.67	1965.0	Core plug	146990	116	99	73	66	54	38	27
	1997.67	1974.0	Core plug	146992	644	547	410	370	385	244	188
	2001.00	1977.3	SWC	153499	107	83	59	348	56	1031	48
	2022.00	1998.3	SWC	153500	37	26	18	186	26	543	24
	2048.00	2024.3	SWC	153501	48	34	25	347	41	1021	47
	2352.00	2336.6	Mud	163302	7	3	1	32	2	80	2

TABLE AP6.2 Whole extract gas chromatography peak areas

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C10	n-C11	n-C12	i-C13	i-C14	n-C13	i-C15	n-C14	i-C16
206/12a-5	1884.67	1861.0	Core plug	146965	491319	715249	1146379	186882	2572932	2082226	676792	4663442	2755935
	1900.67	1877.0	Core plug	146969	167587	2898710	2.2E+07	8670454	1.5E+07	8.5E+07	2.6E+07	9.3E+07	3.6E+07
	1916.67	1893.0	Core plug	146972	1629556	3456921	5184353	1002289	1283531	7469398	1547760	9383768	3789763
	1924.67	1901.0	Core plug	146974	423786	166315	109701	24964	66832	395559	266394	1412993	1056778
	1932.67	1909.0	Core plug	146976	1976647	4695011	7859823	1512586	2249176	1.1E+07	2539526	1.3E+07	5869678
	1937.32	1913.7	Core	154661	2102359	2403887	3940881	871003	5913026	7861624	2564005	1.5E+07	6973420
	1938.00	1914.3	Mud	163301	443434	2650771	1.2E+07	4890528	3488754	3.5E+07	1.1E+07	3.7E+07	1.5E+07
	1948.67	1925.0	Core plug	146980	1921364	3096012	4317893	816390	1099719	5291390	1130485	6517181	2876939
	1956.67	1933.0	Core plug	146982	1994328	3294232	4536188	852378	1594621	5516870	1312137	6889938	3027841
	1967.35	1943.7	Core	154666	98832	221992	1046876	270361	2542788	4734790	2055628	1.2E+07	5191803
	1984.00	1960.3	SWC	153498	7678772	1545	99107	65609	529294	3839580	3284883	1.5E+07	9700408
	1988.67	1965.0	Core plug	146990	159726	321015	604844	117443	938761	1078594	325451	2016022	960635
	1997.67	1974.0	Core plug	146992	1960023	3086632	3803747	715003	1080359	4338909	876954	5019094	2131563
	2001.00	1977.3	SWC	153499	1.7E+07	3171	515880	371188	2812661	2.2E+07	1.7E+07	7.2E+07	4E+07
	2022.00	1998.3	SWC	153500	5001876	1293	305702	229367	1674560	1.4E+07	1.1E+07	4.5E+07	2.4E+07
	2048.00	2024.3	SWC	153501	1.1E+07	1711	221317	155721	1215298	1.1E+07	9363915	4.2E+07	2.4E+07
	2352.00	2336.6	Mud	163302	413588	2002196	1.2E+07	5163939	3732897	3.6E+07	1.1E+07	3.9E+07	1.5E+07

TABLE AP6.3 Whole extract gas chromatography amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C15	n-C16	i-C18	n-C17	Pr	n-C18	Ph	n-C19	n-C20
206/12a-5	1884.67	1861.0	Core plug	146965	4609381	6752573	1678756	7136887	2576694	6875948	1813311	5214643	4696794
	1900.67	1877.0	Core plug	146969	8.5E+07	6E+07	2.7E+07	3.8E+07	1.8E+07	1.3E+07	6771368	3488203	1328223
	1916.67	1893.0	Core plug	146972	9926120	1.1E+07	2534896	1.1E+07	4005508	1E+07	2539238	1E+07	1E+07
	1924.67	1901.0	Core plug	146974	3053951	3878434	1547639	4346643	2476874	3718488	2357426	3045955	2281637
	1932.67	1909.0	Core plug	146976	1.4E+07	1.4E+07	3906217	1.4E+07	5970038	1.4E+07	3768981	1.4E+07	1.3E+07
	1937.32	1913.7	Core	154661	1.3E+07	1.1E+07	4881970	8031344	3486071	4707364	1659283	3152930	2943958
	1938.00	1914.3	Mud	163301	3.2E+07	2.3E+07	1E+07	1.5E+07	7384371	5177018	2865206	1498382	627335
	1948.67	1925.0	Core plug	146980	6648870	7225643	1861265	7311898	3062396	7148177	1940001	6985729	6960416
	1956.67	1933.0	Core plug	146982	6884668	7455019	1992895	7704961	3243934	7494187	2035804	7346075	7261776
	1967.35	1943.7	Core	154666	9871352	7823586	3358107	6199443	2710174	4408935	1374056	3684471	3605344
	1984.00	1960.3	SWC	153498	3E+07	4.1E+07	2.1E+07	3.6E+07	1.8E+07	1.5E+07	7947611	5304272	3029916
	1988.67	1965.0	Core plug	146990	1757572	1973546	518545	1787340	702991	1700224	430990	1528554	1522556
	1997.67	1974.0	Core plug	146992	5020794	5248119	1294142	5230567	2157715	5009175	1325999	4936196	4900337
	2001.00	1977.3	SWC	153499	1E+08	9.7E+07	4.2E+07	6.5E+07	3.2E+07	2.4E+07	1.2E+07	6974573	3313486
	2022.00	1998.3	SWC	153500	6E+07	4.8E+07	2.1E+07	3.2E+07	1.5E+07	1.1E+07	5962266	3155433	1283561
	2048.00	2024.3	SWC	153501	6.6E+07	6.6E+07	3E+07	4.5E+07	2.3E+07	1.7E+07	9411913	4774680	1932629
	2352.00	2336.6	Mud	163302	3.3E+07	2.5E+07	1.2E+07	1.7E+07	8390964	5428376	3083430	1332491	467956

TABLE AP6.3 Whole extract gas chromatography amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C21	n-C22	n-C23	n-C24	n-C25	n-C26	n-C27	n-C28	n-C29
206/12a-5	1884.67	1861.0	Core plug	146965	4150871	3799108	3566540	2920354	2938566	2164300	1931332	1642156	1460136
	1900.67	1877.0	Core plug	146969	827439	377695	212753	135872	96040	79371	56076	41180	63560
	1916.67	1893.0	Core plug	146972	9806222	9005728	9147243	7855074	7478396	5699114	4962718	4258536	3840743
	1924.67	1901.0	Core plug	146974	1587909	1470599	1163072	1258681	1806598	1064362	892565	805591	731051
	1932.67	1909.0	Core plug	146976	1.3E+07	1.1E+07	1.2E+07	9891891	9515143	7199961	6352207	5622406	5192602
	1937.32	1913.7	Core	154661	2811127	2139178	2152651	1506944	1496542	1097193	949421	771107	612979
	1938.00	1914.3	Mud	163301	325062	157366	66737	43504	24445	23528	12412	7134	6867
	1948.67	1925.0	Core plug	146980	6505019	5925769	5984975	5140618	4917747	3702134	3244175	2776293	2548916
	1956.67	1933.0	Core plug	146982	6806959	6194128	6253987	5359304	5089312	3872675	3380368	2943277	2707468
	1967.35	1943.7	Core	154666	3555055	3096559	3153683	2525305	2635913	1901409	1656035	1388187	1148881
	1984.00	1960.3	SWC	153498	2290683	1832678	1563105	1311577	1201245	942607	808745	685295	653208
	1988.67	1965.0	Core plug	146990	1448766	1272352	1244333	1023641	948233	711609	618071	513202	431975
	1997.67	1974.0	Core plug	146992	4591843	4210392	4232748	3636833	3498939	2653669	2322850	2019148	1851450
	2001.00	1977.3	SWC	153499	1960263	1631067	1157959	930632	833430	661423	556879	428926	441901
	2022.00	1998.3	SWC	153500	831629	587291	392526	298770	260275	209881	163137	118560	134324
	2048.00	2024.3	SWC	153501	966444	640335	351706	264103	220579	200715	151971	119796	123897
	2352.00	2336.6	Mud	163302	238097	111664	52757	36901	22659	20347	12472	6850	6170

TABLE AP6.3 Whole extract gas chromatography amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	n-C30	n-C31	n-C32	n-C33	n-C34	n-C35	n-C36
206/12a-5	1884.67	1861.0	Core plug	146965	1017868	906750	640300	581516	673712	457295	218314
	1900.67	1877.0	Core plug	146969	33068	41598	26377	17242	20364	11546	
	1916.67	1893.0	Core plug	146972	2769962	2344612	1811553	1556711	1726310	1069435	796631
	1924.67	1901.0	Core plug	146974	650376	796901	548482	513097	453854	353364	1258776
	1932.67	1909.0	Core plug	146976	3756346	3169946	2437236	2234462	2275499	1964299	1187551
	1937.32	1913.7	Core	154661	373717	399840	258263	217878	260381	94471	96512
	1938.00	1914.3	Mud	163301	5684	5114	1355	25760	1956	60875	1262
	1948.67	1925.0	Core plug	146980	1853356	1525819	1175781	1067705	1097433	747479	522629
	1956.67	1933.0	Core plug	146982	1910574	1643836	1267381	1136622	1137342	808558	569632
	1967.35	1943.7	Core	154666	839364	752480	532720	477889	511840	291880	187755
	1984.00	1960.3	SWC	153498	454683	353382	257860	535088	235967	1628966	82779
	1988.67	1965.0	Core plug	146990	294342	250687	185118	168513	137403	96770	68888
	1997.67	1974.0	Core plug	146992	1339605	1137201	852430	768938	802030	508479	392012
	2001.00	1977.3	SWC	153499	309604	238398	171031	1002033	162084	2970418	139255
	2022.00	1998.3	SWC	153500	83528	58424	40959	423738	58129	1235306	54113
	2048.00	2024.3	SWC	153501	89989	63168	47051	653259	77049	1919587	88772
	2352.00	2336.6	Mud	163302	8923	4511	1421	43951	3139	108889	2782

TABLE AP6.3 Whole extract gas chromatography amounts in ng/g

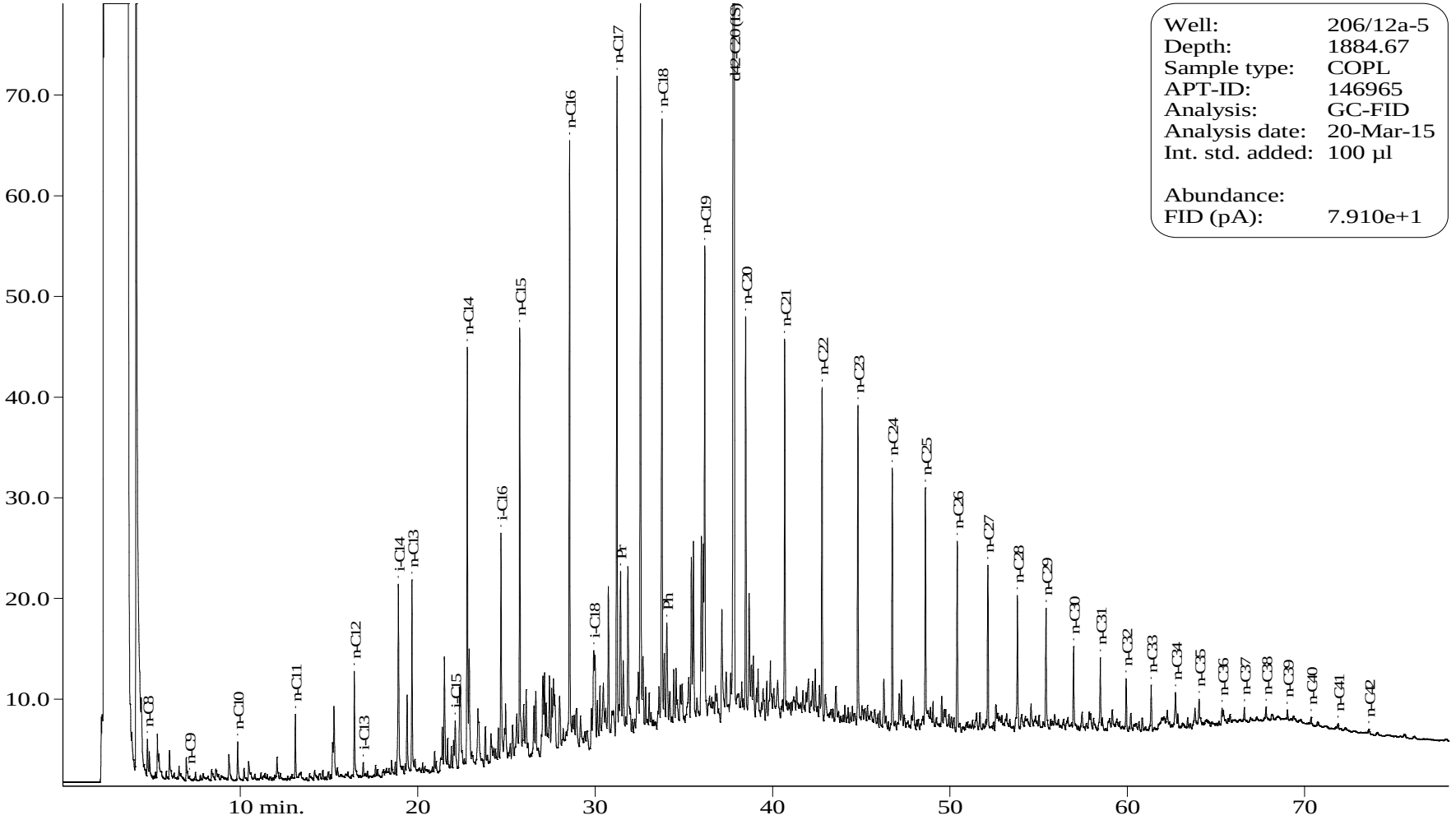


FIGURE AP6.1 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1884.67m (Turonian (K72))

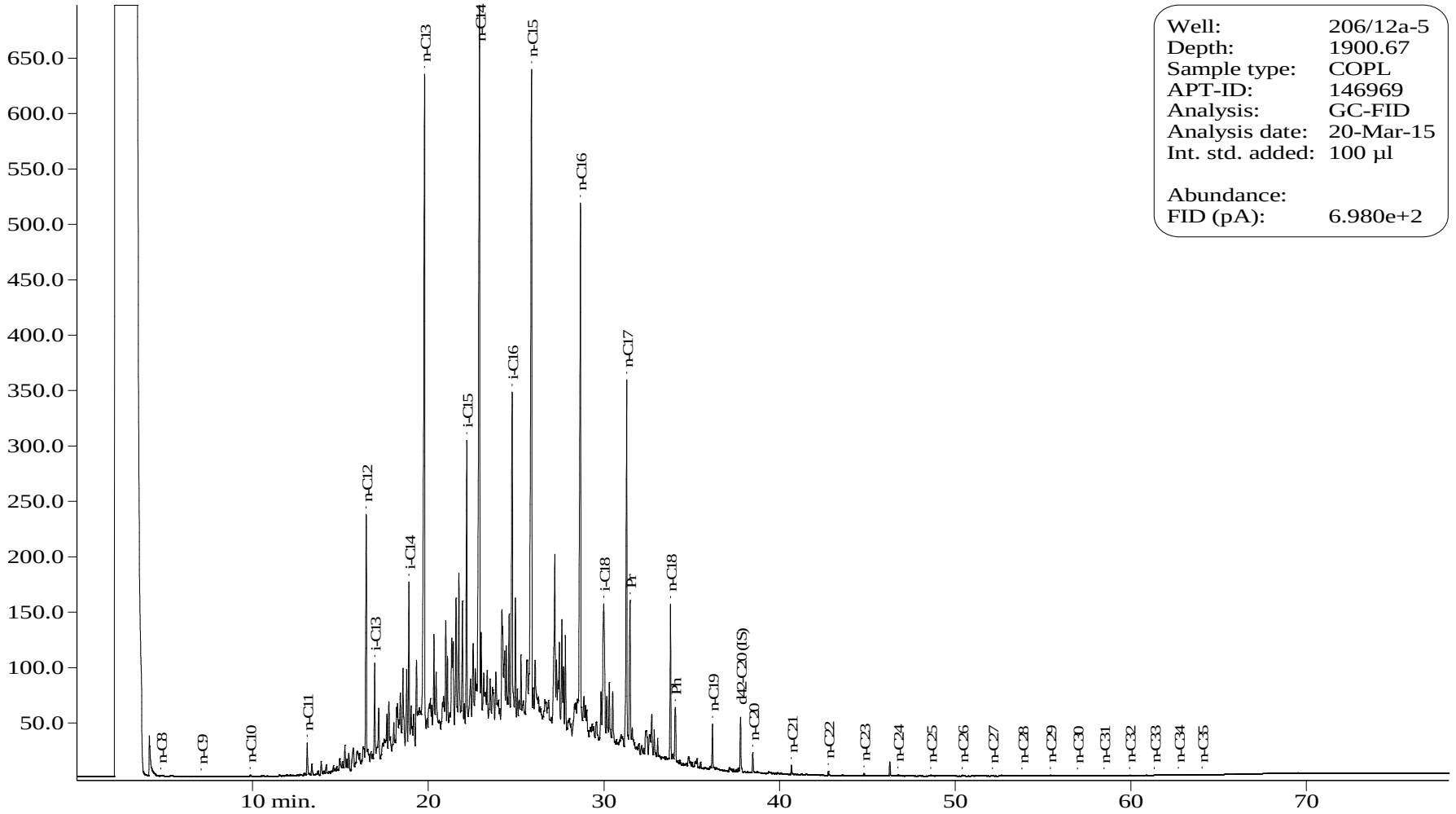


FIGURE AP6.2 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1900.67m (Turonian (K72))

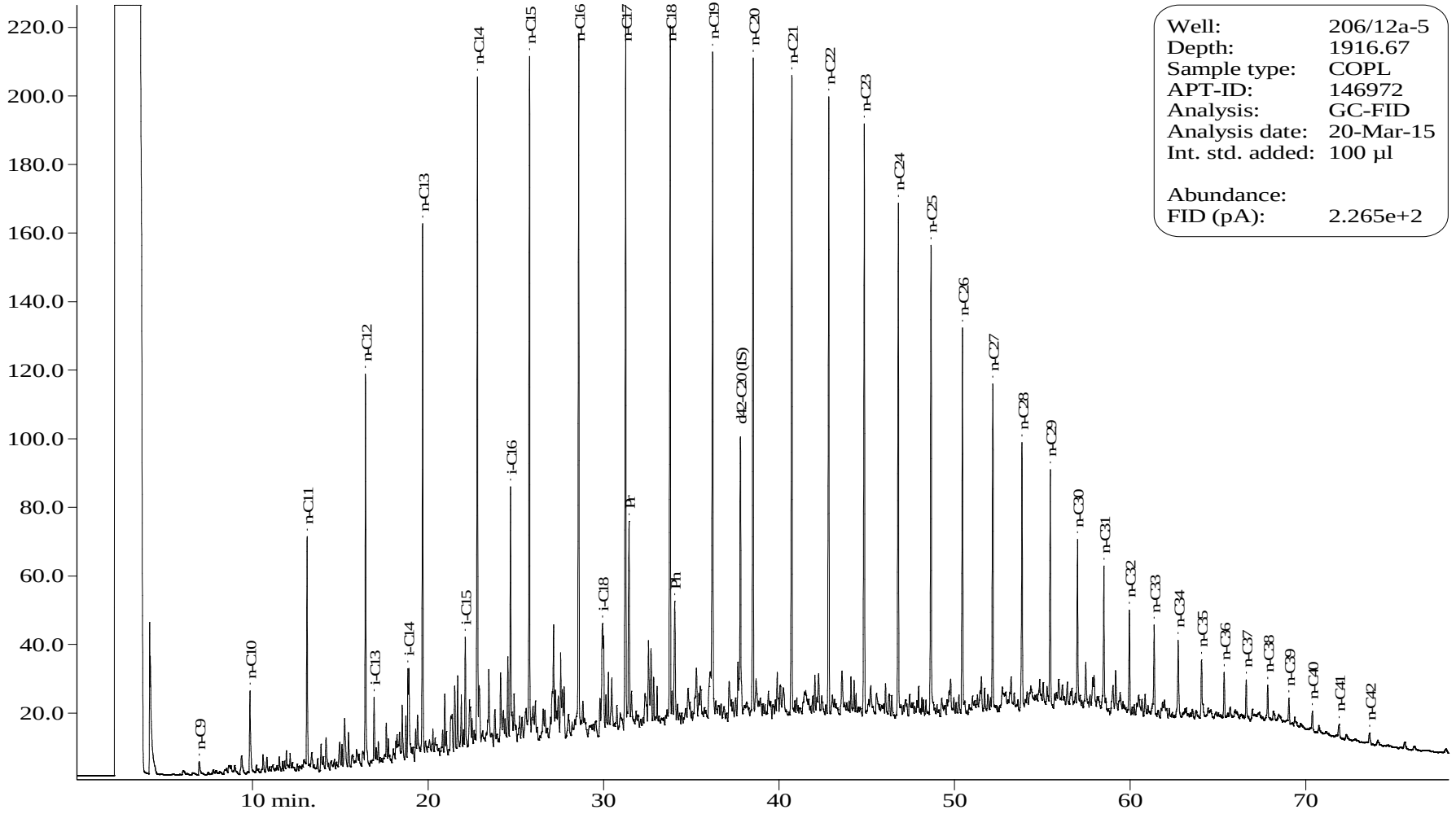


FIGURE AP6.3 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))

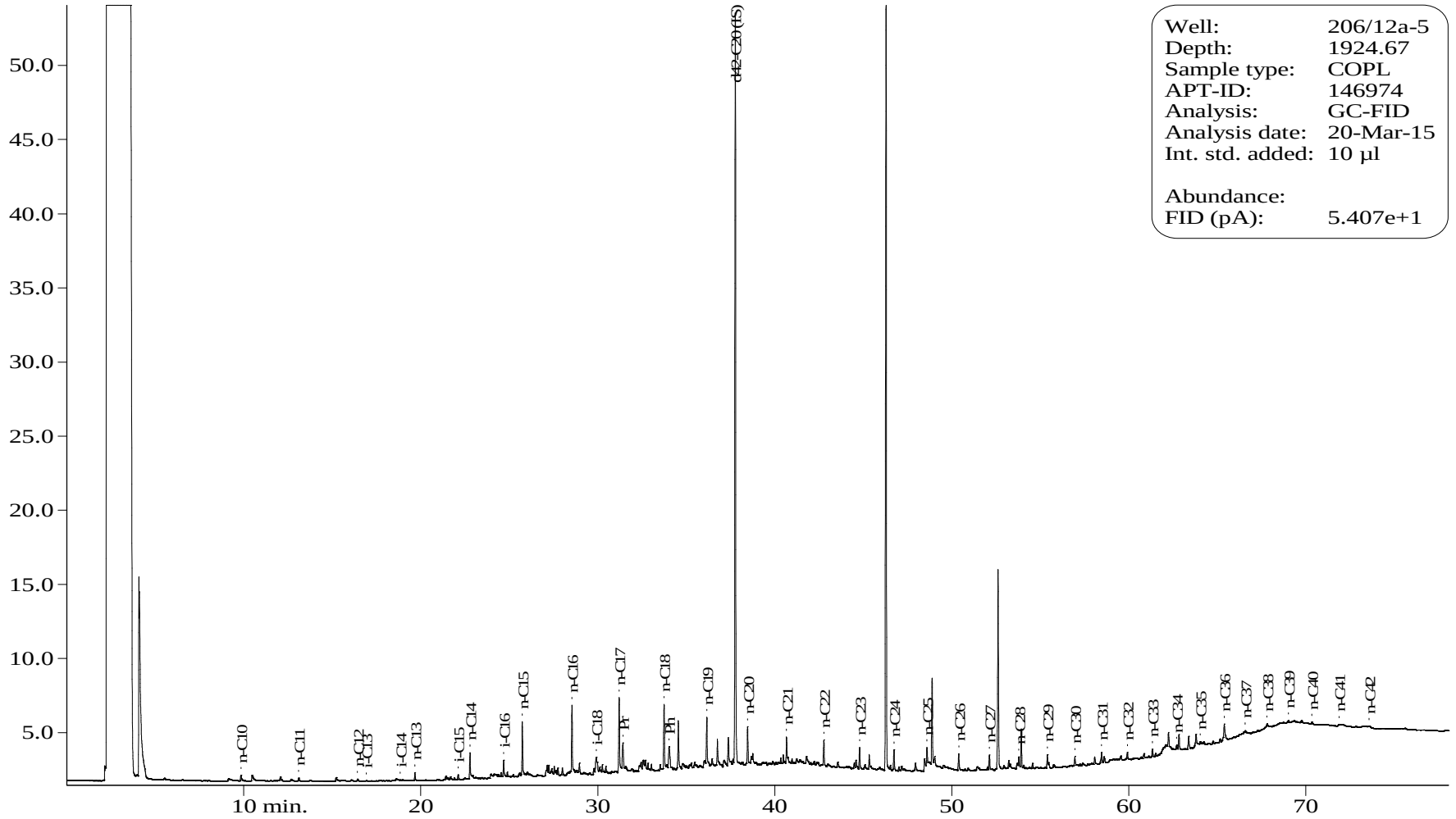


FIGURE AP6.4 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1924.67m (K20 Sand Cret)

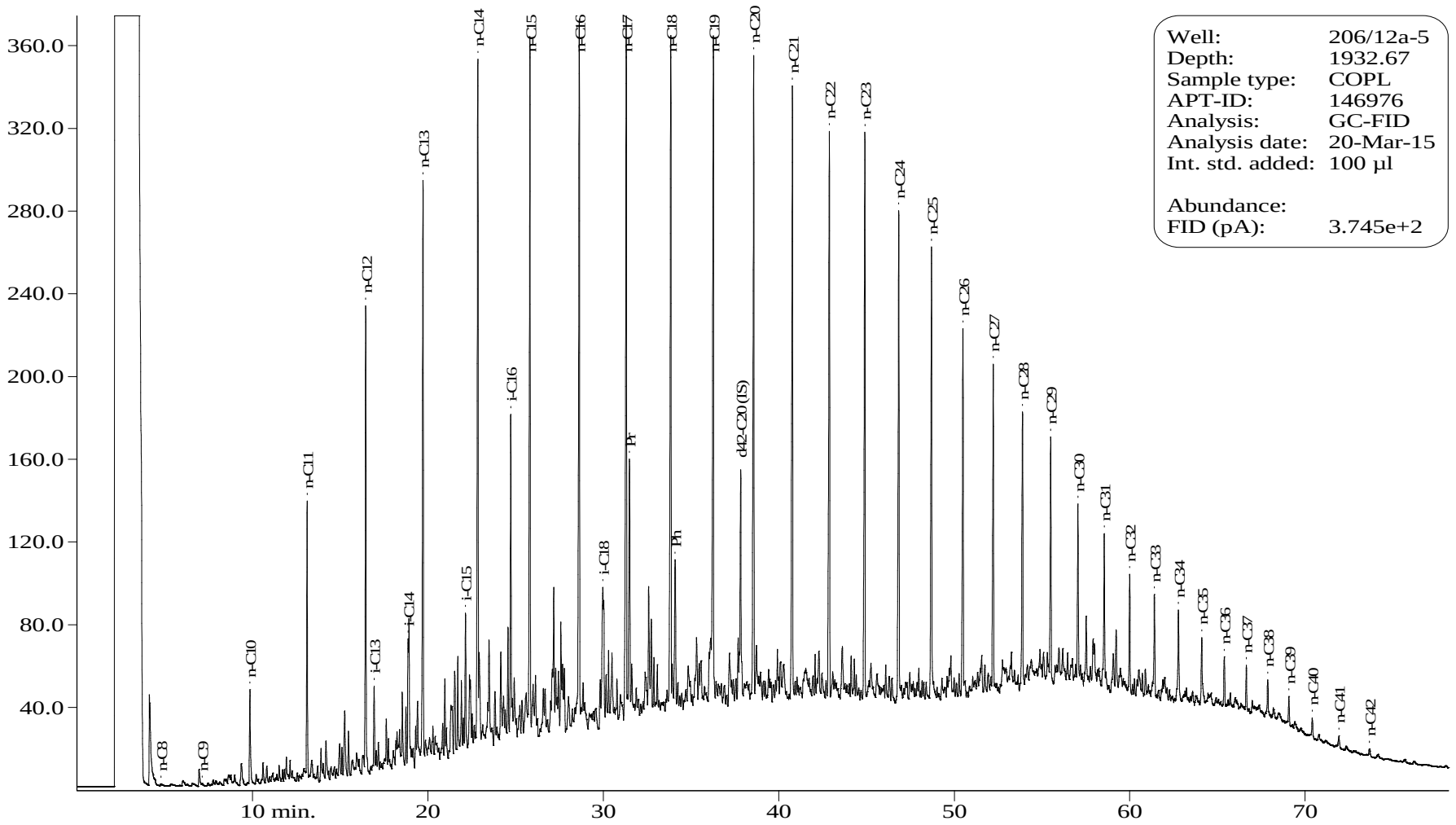


FIGURE AP6.5 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1932.67m (Kimmeridge Clay Fm.)

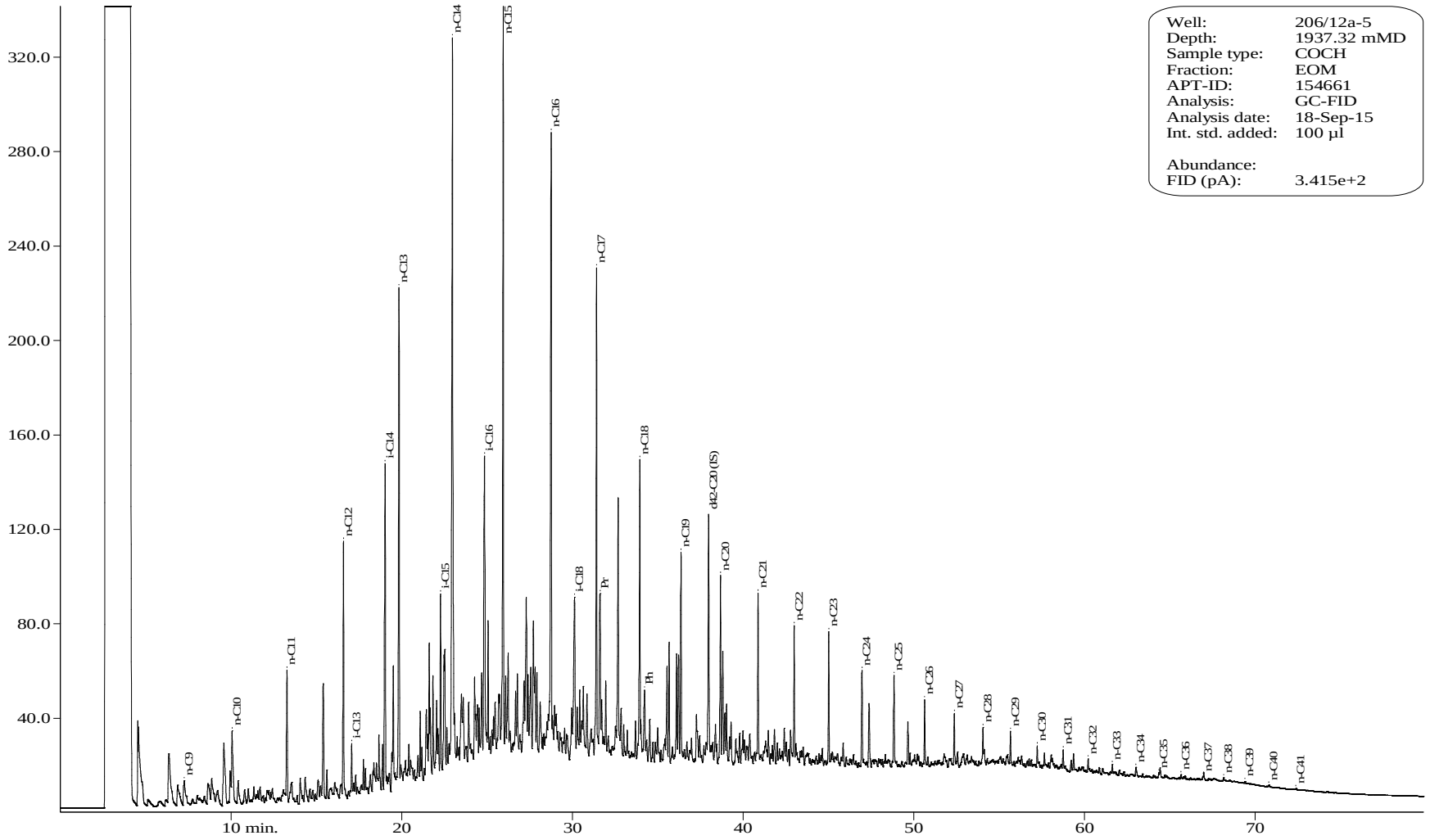


FIGURE AP6.6 Whole extract gas chromatogram: 206/12a-5, CORE, 1937.32m (Rona)

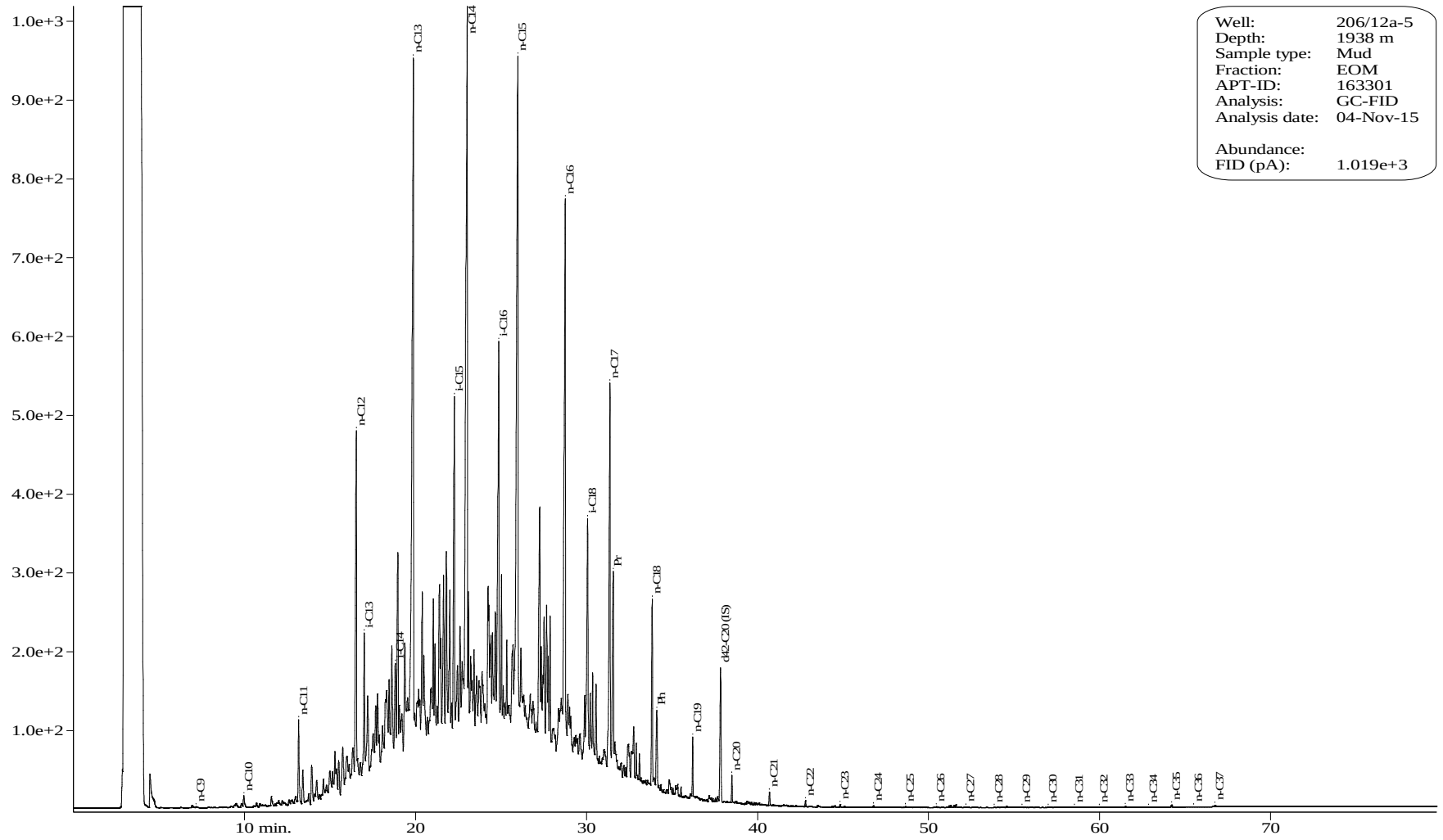


FIGURE AP6.7 Whole extract gas chromatogram: 206/12a-5, MUD, 1938m (Rona)

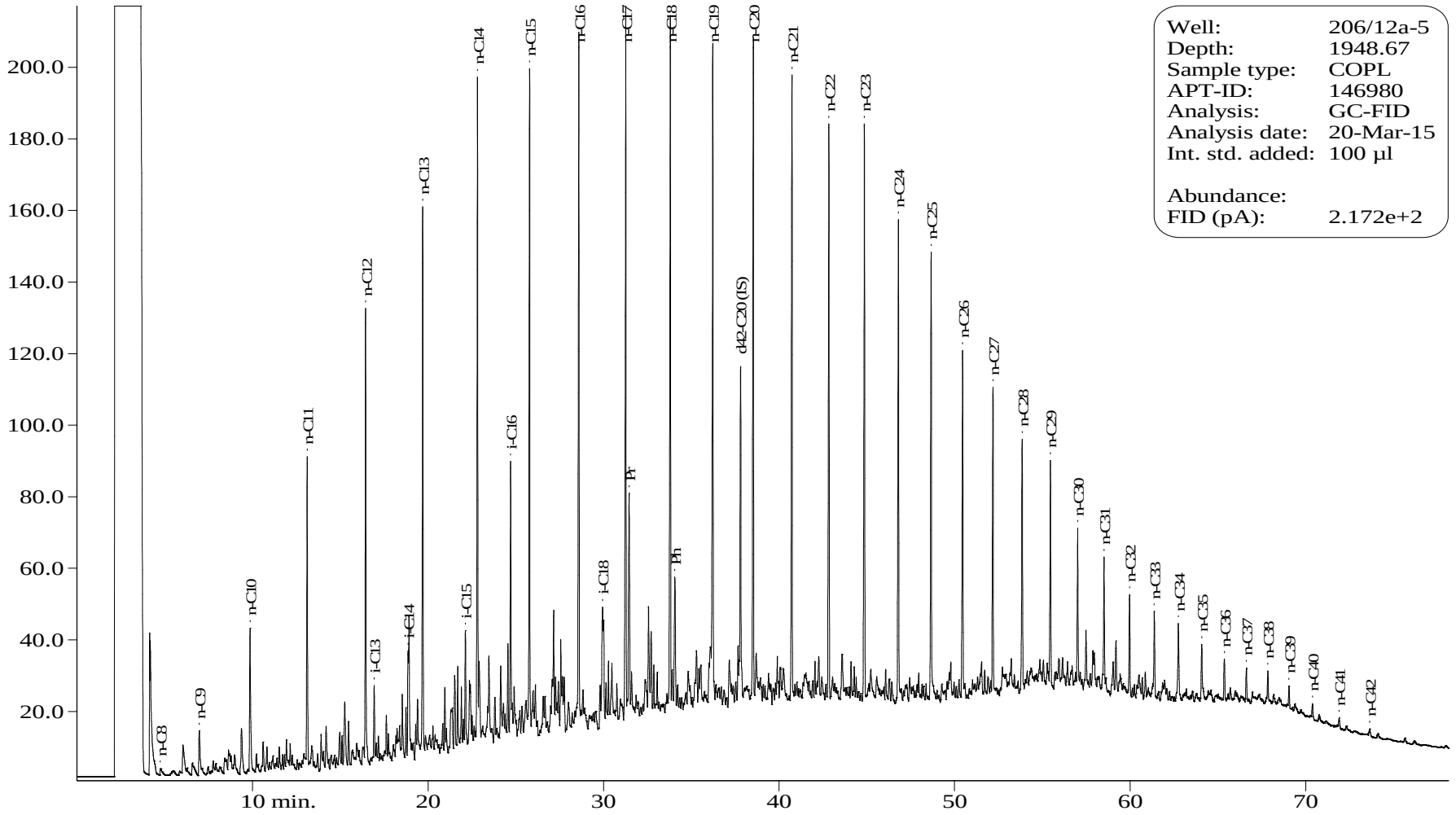


FIGURE AP6.8 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1948.67m (Rona)

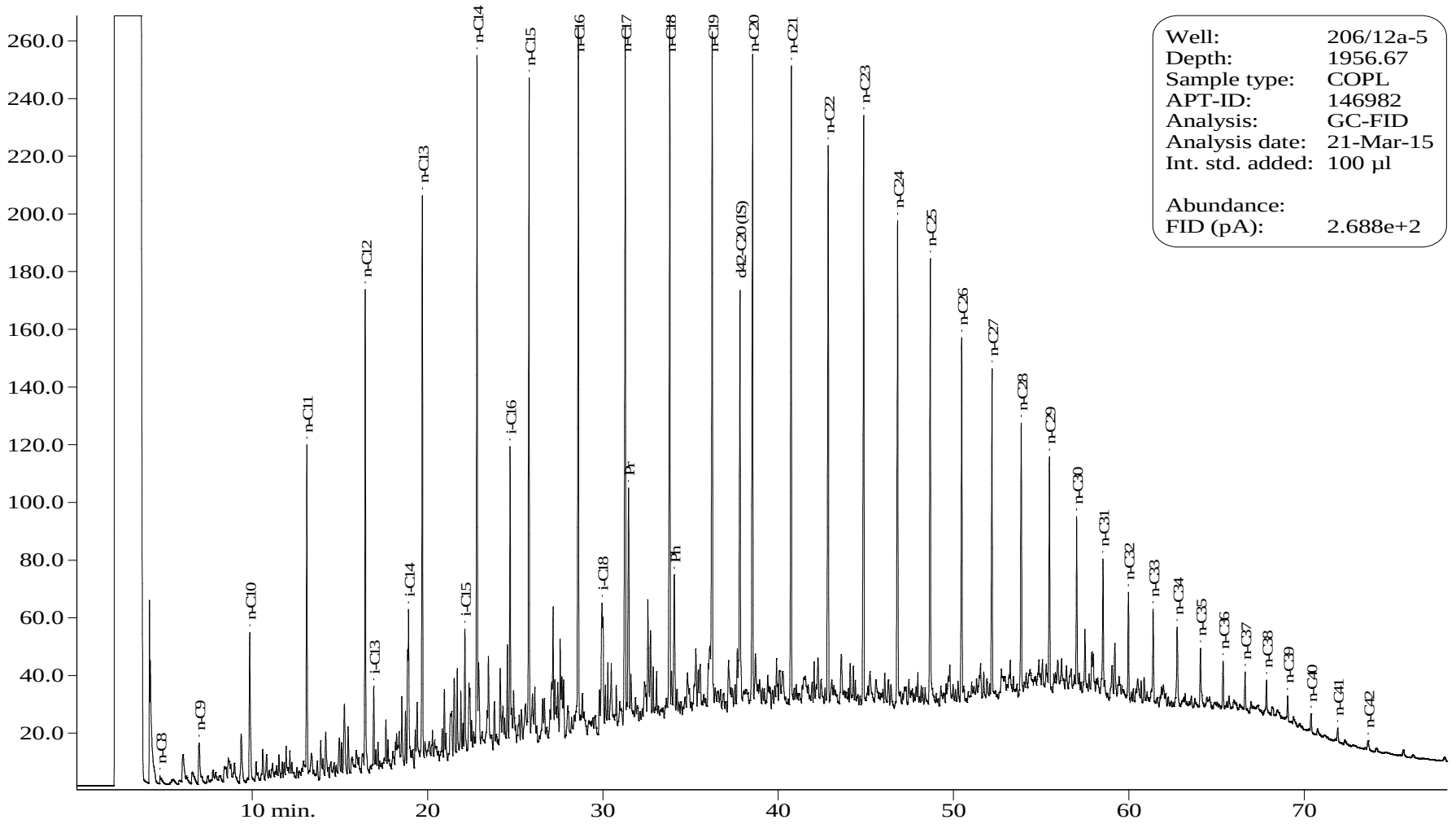


FIGURE AP6.9 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1956.67m (Rona)

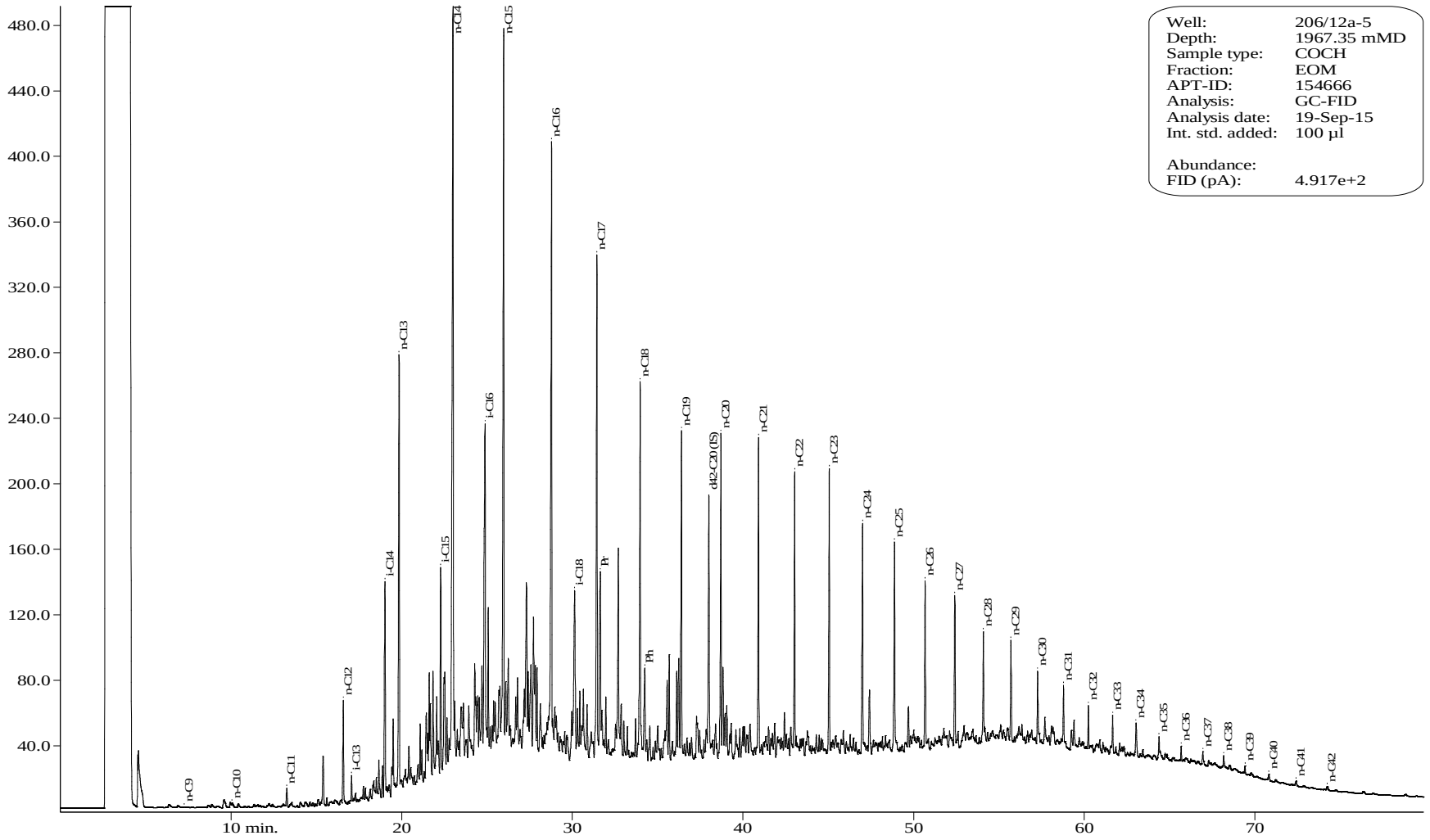


FIGURE AP6.10 Whole extract gas chromatogram: 206/12a-5, CORE, 1967.35m (Rona)

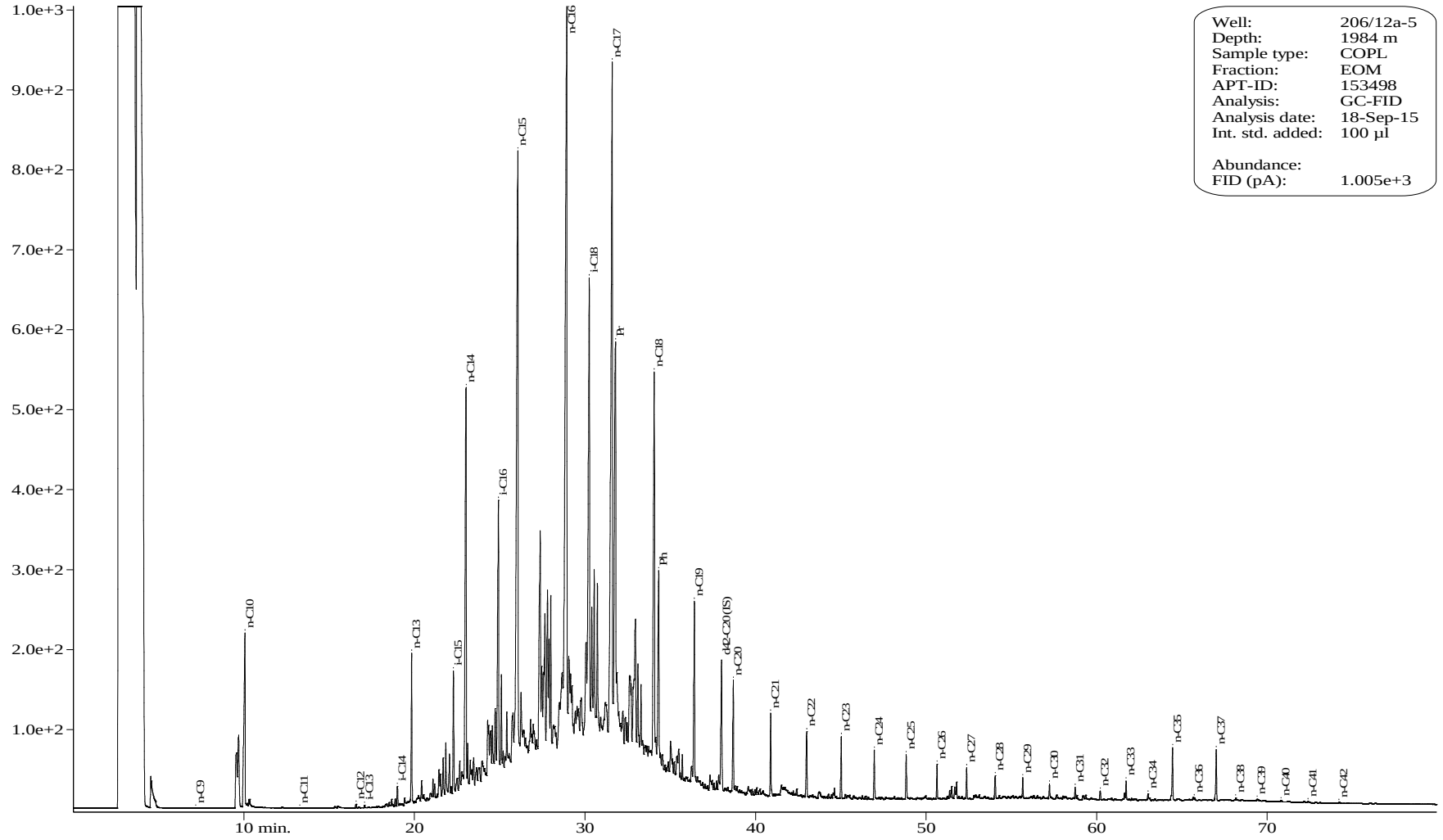


FIGURE AP6.11 Whole extract gas chromatogram: 206/12a-5, SWC, 1984m (LCG Unit III)

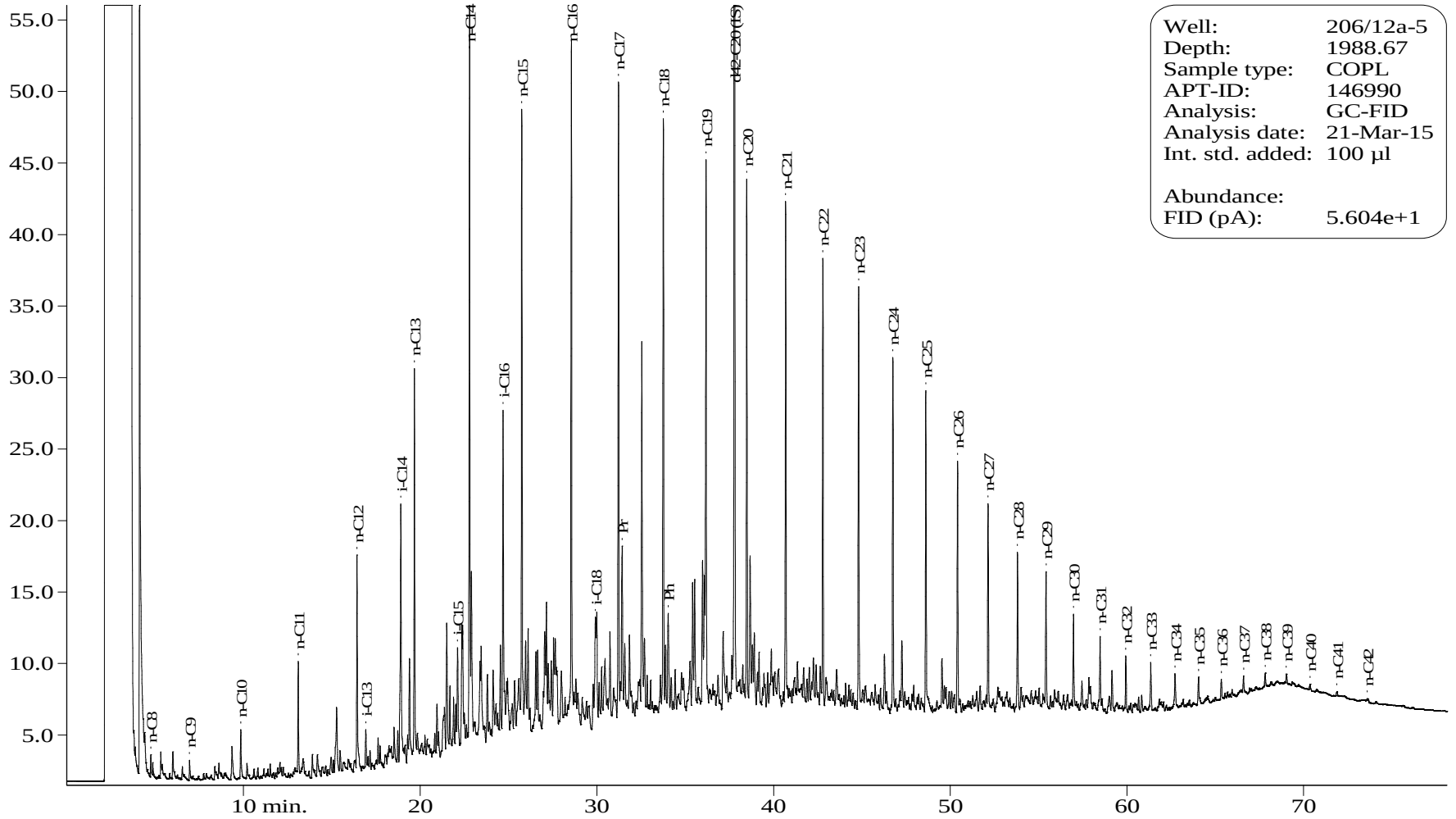


FIGURE AP6.12 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1988.67m (LCG Unit III)

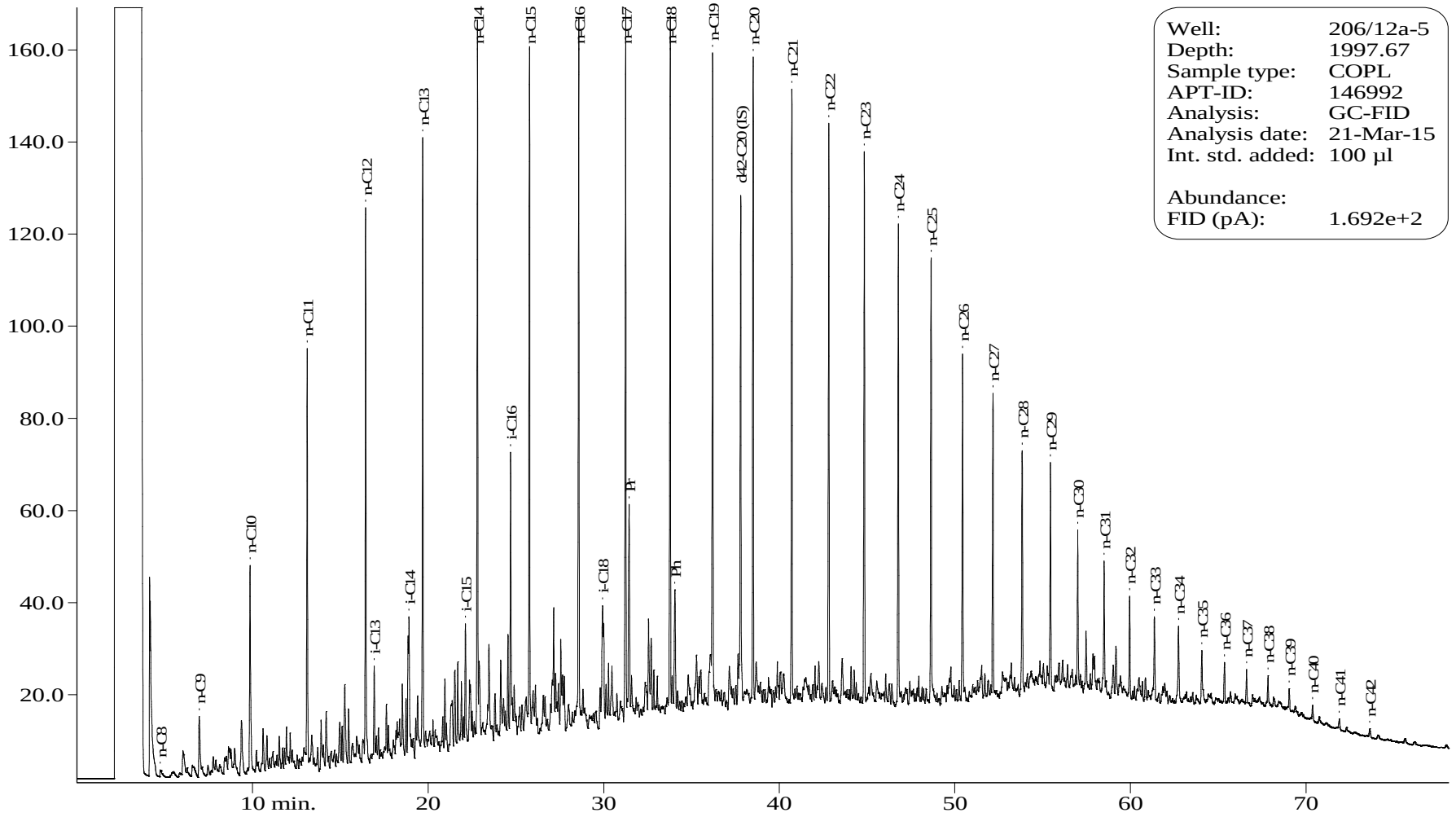


FIGURE AP6.13 Whole extract gas chromatogram: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)

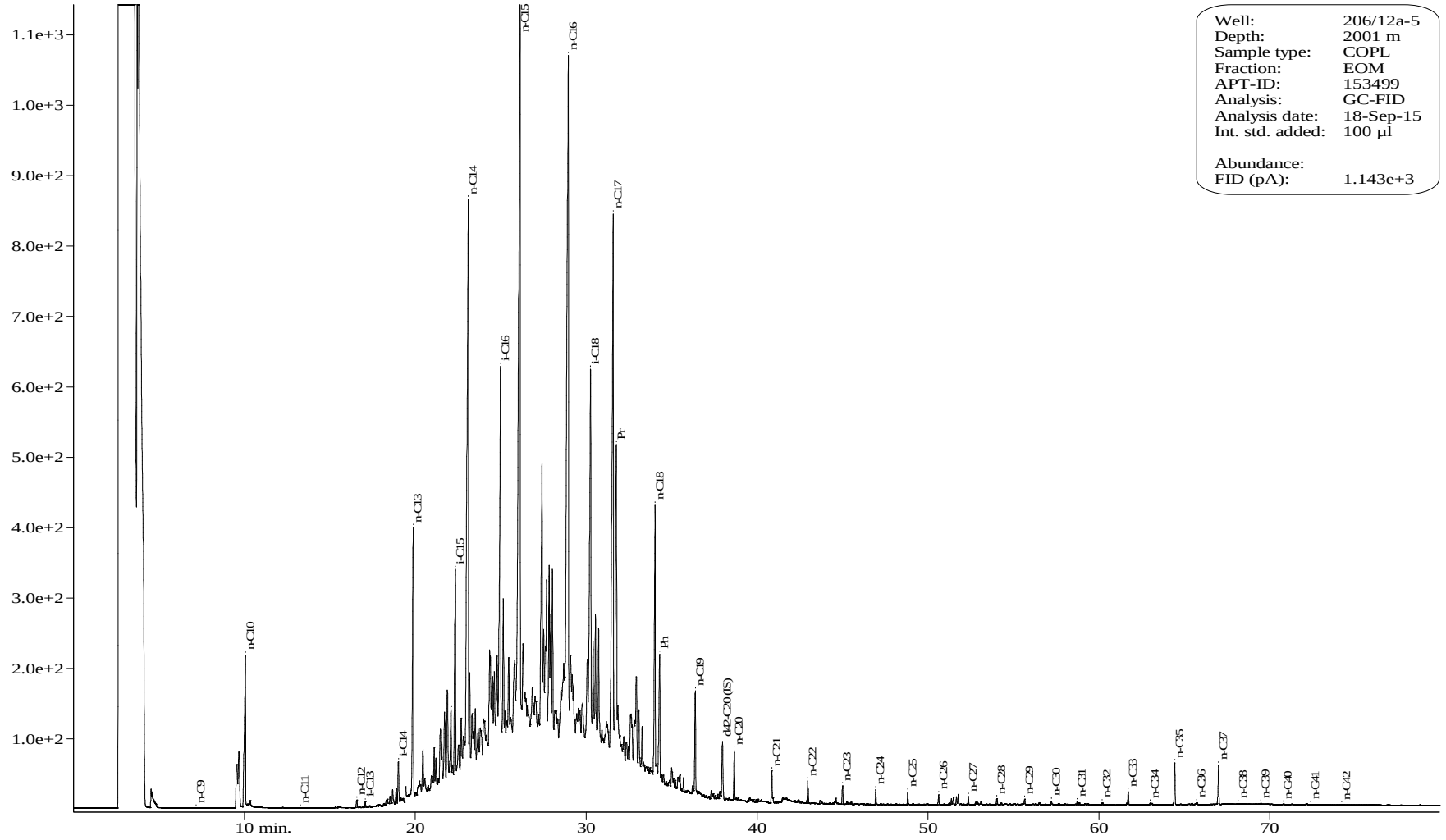


FIGURE AP6.14 Whole extract gas chromatogram: 206/12a-5, SWC, 2001m (LCG Unit III)

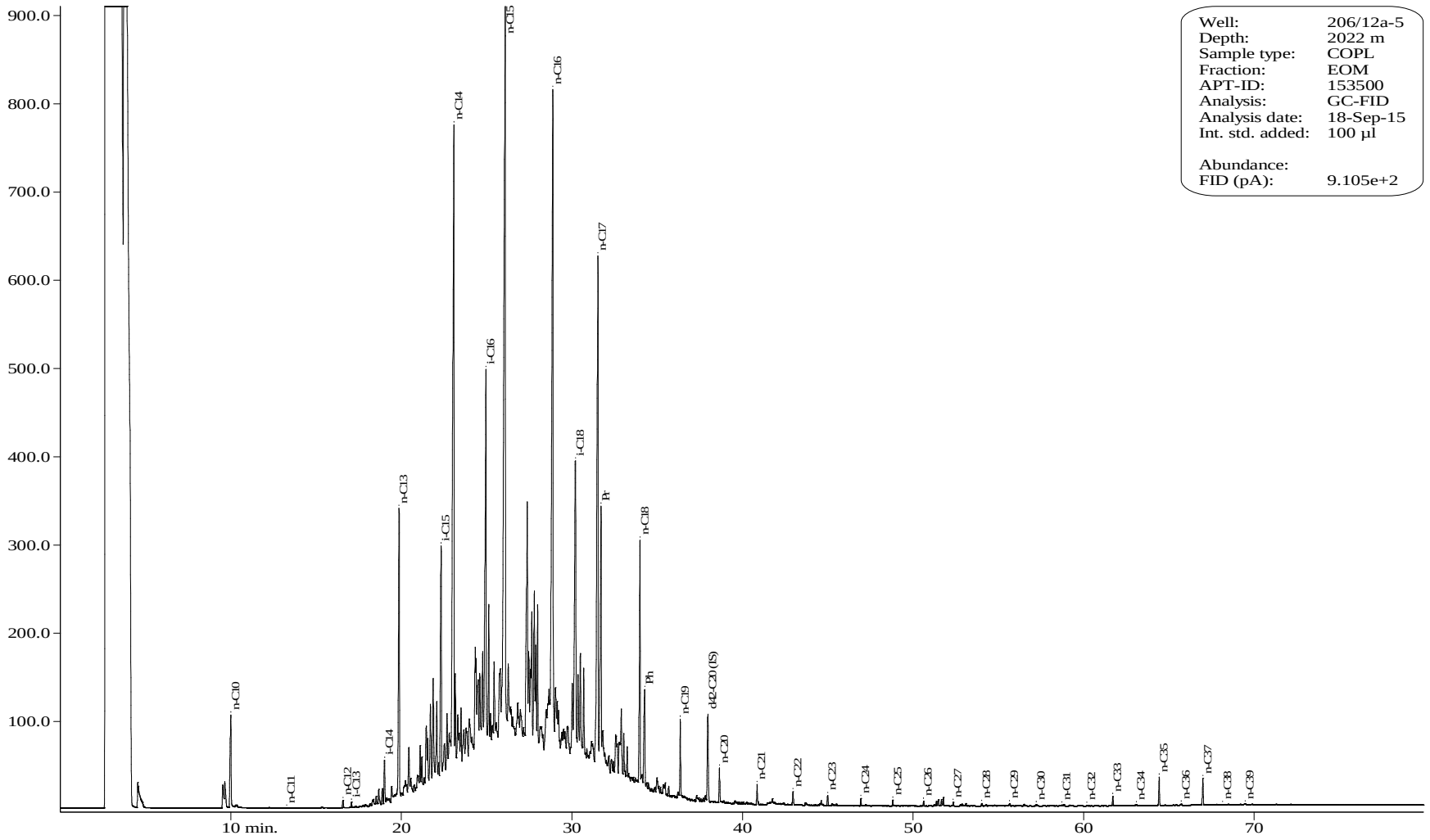


FIGURE AP6.15 Whole extract gas chromatogram: 206/12a-5, SWC, 2022m (LCG Unit II /I)

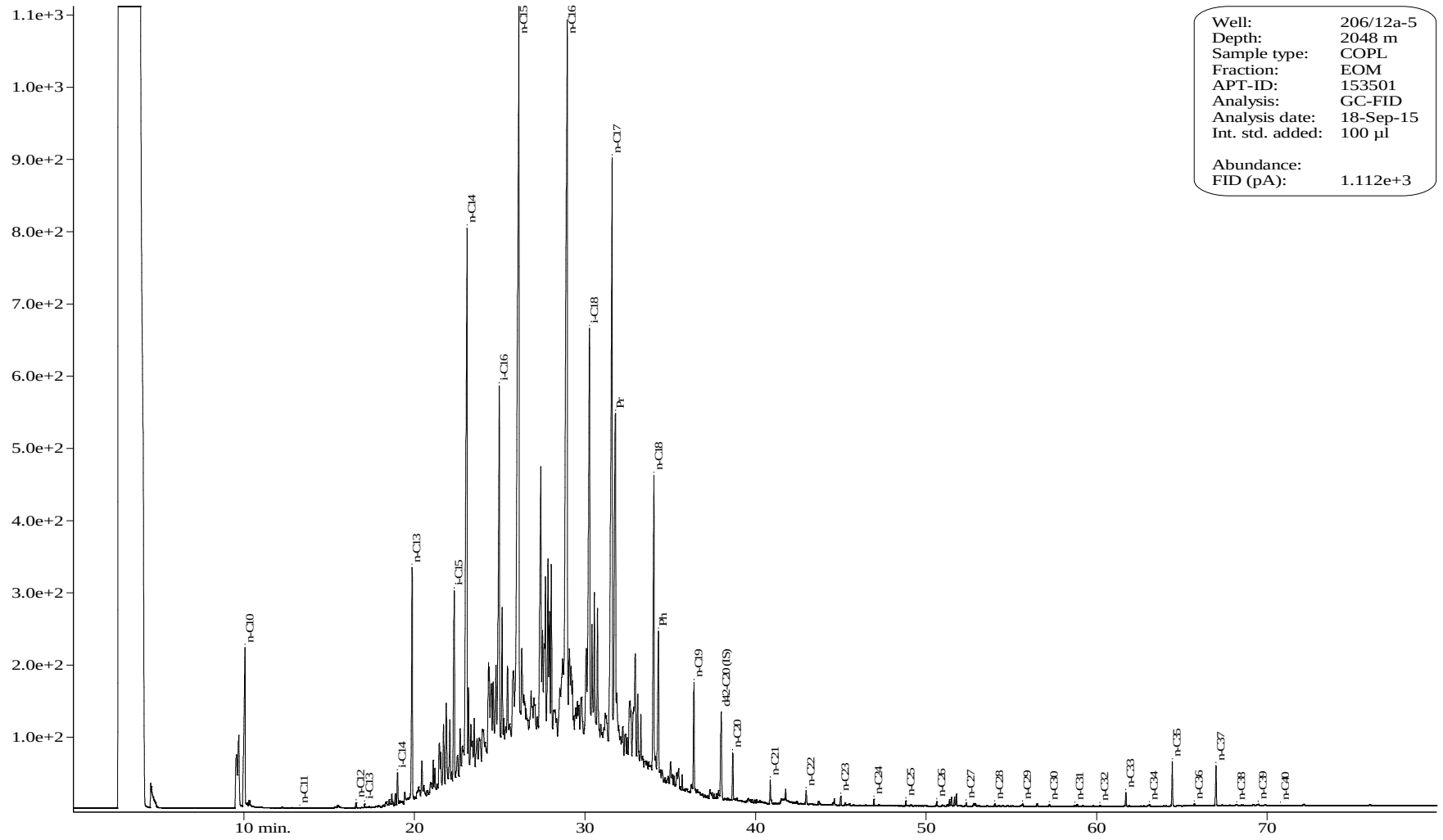


FIGURE AP6.16 Whole extract gas chromatogram: 206/12a-5, SWC, 2048m (LCG Unit II /I)

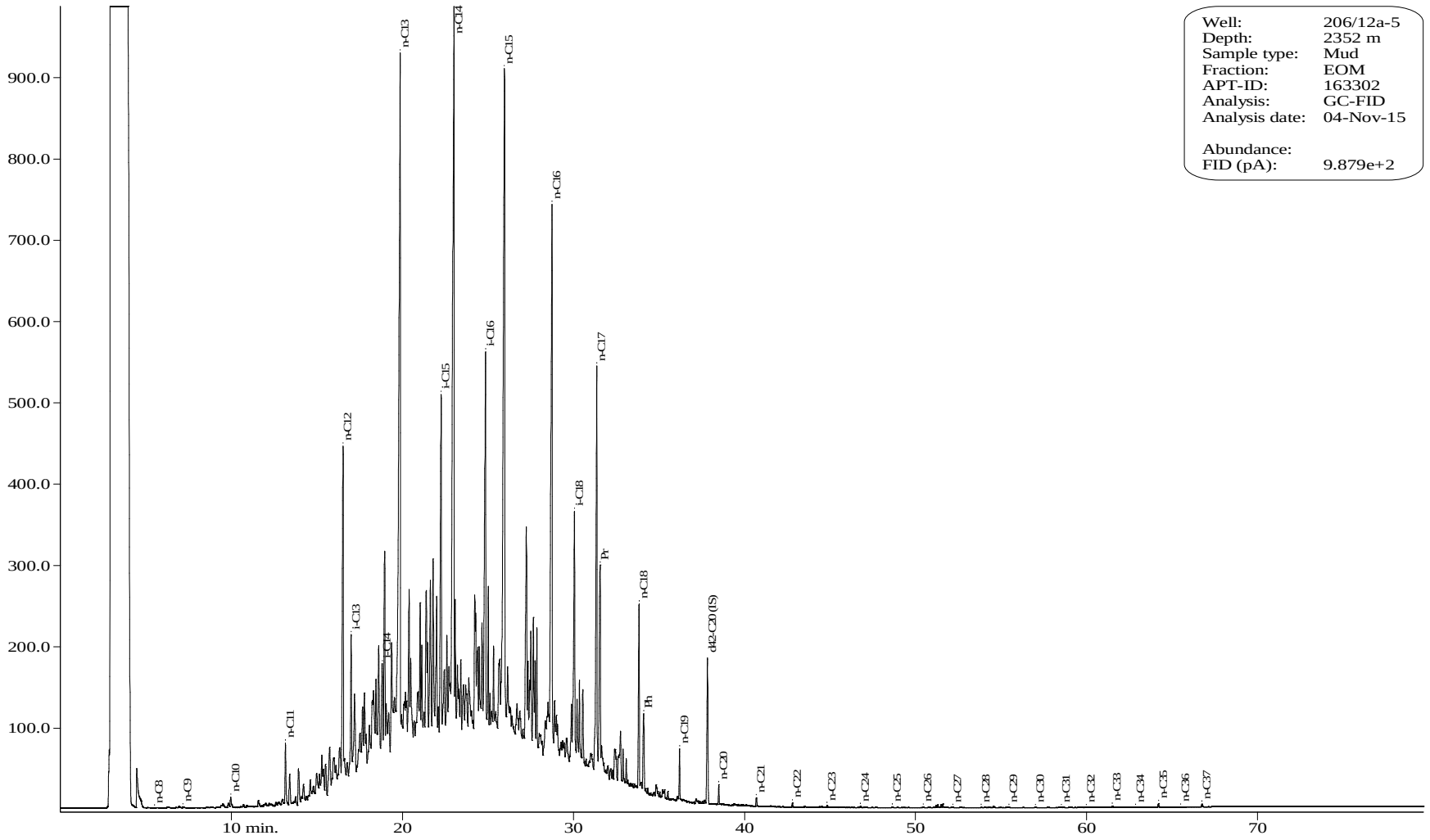


FIGURE AP6.17 Whole extract gas chromatogram: 206/12a-5, MUD, 2352m (Basement)

APPENDIX 7
Alkane fraction GC-MS data

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Ap7.14 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1984.5m (LCG Unit III)
Ap7.15 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)
Ap7.16 Alkane fraction GC-MS fragmentograms: 206/12a-5, MUD, 2352m (Basement)

Well	MD (m)	TVDSS (m)	Sample type	APTID	Ts/Tm	C30 mor/ C30 hop	C29 mor/ C29 hop	C31 S/R hop	C32 S/R hop	C35S+R/ C31S+R hop	C35S+R/ C34S+R hop	C29/ C29+C30 hop	Bnh/hop
206/12a-5	1897.03	1873.4	Spun out oil	146645	1.01	0.14	0.15	1.32	1.22	0.26	0.79	0.34	0.09
	1916.67	1893.0	Core plug	146972	1.09	0.14	0.12	1.30	1.31	0.25	0.64	0.33	0.09
	1922.08	1898.4	Spun out oil	146643	1.05	0.13	0.15	1.33	1.35	0.26	0.79	0.34	0.09
	1922.40	1898.7	Oil	160302	1.09	0.16	0.14	1.33	1.29	0.28	0.66	0.34	0.10
	1932.67	1909.0	Core plug	146976	1.12	0.15	0.17	1.23	1.24	0.29	0.76	0.34	0.10
	1937.32	1913.7	Core	154661	1.03	0.15	0.18	1.45	1.26	0.26	0.90	0.34	0.10
	1938.00	1914.3	Mud	163301	0.68	0.14		1.21	1.17	0.52	1.12	0.28	0.04
	1948.67	1925.0	Spun out oil	146644	0.94	0.14	0.14	1.28	1.28	0.27	0.75	0.33	0.09
	1956.00	1932.3	Oil	160303	1.16	0.16	0.12	1.37	1.29	0.26	0.65	0.35	0.11
	1956.67	1933.0	Core plug	146982	1.11	0.15	0.13	1.35	1.27	0.26	0.65	0.35	0.11

Well	MD (m)	TVDSS (m)	Sample type	APTID	G/hop	O/hop	Demeth hop/hop	Diahop/ hop	28+29 tricy/ hop	23 tricy/ hop	24 tetracy/ 26 tricy	24 tetracy/ hop	hop/ster
206/12a-5	1897.03	1873.4	Spun out oil	146645	0.09		0.48	0.16	0.11	0.07	1.57	0.08	4.98
	1916.67	1893.0	Core plug	146972	0.11		0.44	0.16	0.10	0.09	1.54	0.08	5.61
	1922.08	1898.4	Spun out oil	146643	0.09		0.49	0.16	0.11	0.08	1.48	0.07	5.27
	1922.40	1898.7	Oil	160302	0.11		0.51	0.17	0.11	0.09	1.49	0.08	5.34
	1932.67	1909.0	Core plug	146976	0.11		0.52	0.17	0.14	0.09	1.47	0.08	4.93
	1937.32	1913.7	Core	154661	0.10		0.41	0.15	0.08	0.10	2.29	0.10	5.43
	1938.00	1914.3	Mud	163301	0.12		0.15	0.06		0.02		0.01	12.66
	1948.67	1925.0	Spun out oil	146644	0.10		0.49	0.15	0.10	0.09	1.61	0.08	5.86
	1956.00	1932.3	Oil	160303	0.12		0.52	0.17	0.11	0.09	1.57	0.09	5.61
	1956.67	1933.0	Core plug	146982	0.11		0.54	0.16	0.11	0.10	1.43	0.08	5.17

TABLE Ap7.1 Alkane fraction GC-MS parameters: terpane and sterane ratios

Well	MD (m)	TVDSS (m)	Sample type	APTID	αα S/S+R C29 ster	ββ/ αα+ββ C29 ster	Sterane composition (ααR)			Iso-sterane composition (ββ S+R)			28/29
							C27 (%)	C28 (%)	C29 (%)	C27 (%)	C28 (%)	C29 (%)	
206/12a-5	1897.03	1873.4	Spun out oil	146645	0.48	0.54	39.8	22.7	37.6	34.7	27.9	37.4	0.61
	1916.67	1893.0	Core plug	146972	0.51	0.52	43.6	22.7	33.7	36.9	28.3	34.8	0.68
	1922.08	1898.4	Spun out oil	146643	0.50	0.52	39.3	22.8	37.8	35.7	28.8	35.4	0.72
	1922.40	1898.7	Oil	160302	0.52	0.52	43.7	22.6	33.7	35.2	28.0	36.8	0.68
	1932.67	1909.0	Core plug	146976	0.45	0.53	40.0	23.7	36.3	35.4	28.3	36.3	0.72
	1937.32	1913.7	Core	154661	0.48	0.46	42.5	22.4	35.2	37.7	27.7	34.6	0.64
	1938.00	1914.3	Mud	163301	0.31	0.44	68.1	11.9	20.0	18.7	34.1	47.2	0.55
	1948.67	1925.0	Spun out oil	146644	0.48	0.52	41.8	21.2	37.0	36.2	27.9	35.9	0.67
	1956.00	1932.3	Oil	160303	0.53	0.51	44.3	24.0	31.7	37.0	27.9	35.1	0.72
	1956.67	1933.0	Core plug	146982	0.52	0.52	42.4	23.7	33.9	35.5	28.3	36.2	0.72

Ts/Tm : C27 trisnorhopane ratio - 27Ts/27Tm

C30 mor/hop : C30 moretane/hopane ratio - 30βα/30αβ

C29 mor/hop : C29 normoretane/norhopane ratio - 29βα/29αβ

C31 S/R hop : C31 hopane isomer ratio - 31αβS/31αβR

C32 S/R hop : C32 hopane isomer ratio - 32αβS/32αβR

35S+R/C31S+R hop : C35/C31 hopane ratio - 35αβS+R/31αβS+R

35S+R/C34S+R hop : C35/C34 hopane ratio - 35αβS+R/34αβS+R

C29/C29+C30 hop : Norhopane/hopane fraction - 29αβ/(29αβ+30αβ)

Bnh/hop : Bisnorhopane/hopane ratio - 28αβ/30αβ

G/hop : Gammacerane/hopane ratio - 30G/30αβ

O/hop : Oleanane/hopane ratio - 30O/30αβ

Demeth hop/hop : Demethylated hopane/hopane ratio - 25nor30αβ/30αβ

Diahop/hop : C30 diahopane/hopane ratio - 30d/30αβ

28+29 tricy/hop : Tricyclic terpane/hopane ratio - (28/3R+S +29/3R+S)/30αβ

23 tricy/hop : Tricyclic terpane/hopane ratio - 23/3 / 30αβ

24 tetracy/26 tricy : Tetracyclic/tricyclic terpane ratio - 24/4 / 26/3R+S

24 tetracy/hop : Tetracyclic terpane/hopane ratio - 24/3 / 30αβ

hop/ster : Hopane/sterane ratio - 30αβ/(29ααS+29ββR+29ββS+29αααR)

αα S/S+R C29 ster : Sterane isomerisation fraction - 29ααS/(29ααS+29αααR)

ββ/αα+ββ C29 ster : Isosterane fraction - (29ββR+29ββS) / (29ααS+29ββR+29ββS+29αααR)

Sterane composition (ααR)
- **C27 (%) :** 27αααR/(27αααR+28αααR+29αααR)*100
- **C28 (%) :** 28αααR/(27αααR+28αααR+29αααR)*100
- **C29 (%) :** 29αααR/(27αααR+28αααR+29αααR)*100

Iso-sterane composition (ββ S+R)
- **C27 (%) :** (27ββR+27ββS)/(27ββR+27ββS+28ββR+28ββS+29ββR+29ββS)*100
- **C28 (%) :** (28ββR+28ββS)/(27ββR+27ββS+28ββR+28ββS+29ββR+29ββS)*100
- **C29 (%) :** (29ββR+29ββS)/(27ββR+27ββS+28ββR+28ββS+29ββR+29ββS)*100

28/29 : (28αα(R+S)+28ββ(R+S))/(29αα(R+S)+29ββ(R+S))

TABLE Ap7.1 Alkane fraction GC-MS parameters: terpane and sterane ratios

Well	MD (m)	TVDSS (m)	Sample type	APTID	Ts/Tm	C30 mor/ C30 hop	C29 mor/ C29 hop	C31 S/R hop	C32 S/R hop	C35S+R/ C31S+R hop	C35S+R/ C34S+R hop	C29/ C29+C30 hop	Bnh/hop
206/12a-5	1936-1963	1912.3-1939.3	Oil	160301	1.12	0.16	0.14	1.33	1.29	0.27	0.67	0.34	0.11
	1967.35	1943.7	Core	154666	1.02	0.15	0.14	1.37	1.23	0.26	0.74	0.33	0.10
	1984.00	1960.3	SWC	153498	0.95	0.14	0.13	1.34	1.25	0.28	0.75	0.32	0.09
	1984.50	1960.8	Oil	160304	1.03	0.15	0.15	1.35	1.22	0.26	0.64	0.34	0.11
	1997.67	1974.0	Core plug	146992	1.05	0.16	0.14	1.42	1.26	0.26	0.69	0.35	0.11
	2352.00	2336.6	Mud	163302	0.79	0.08	0.05	1.63	1.24	0.36	0.89	0.34	0.04

Well	MD (m)	TVDSS (m)	Sample type	APTID	G/hop	O/hop	Demeth hop/hop	Diahop/ hop	28+29 tricy/ hop	23 tricy/ hop	24 tetracy/ 26 tricy	24 tetracy/ hop	hop/ster
206/12a-5	1936-1963	1912.3-1939.3	Oil	160301	0.11		0.52	0.17	0.11	0.09	1.42	0.08	5.32
	1967.35	1943.7	Core	154666	0.10		0.50	0.16	0.10	0.09	1.75	0.09	5.37
	1984.00	1960.3	SWC	153498	0.10		0.47	0.15	0.10	0.08	1.69	0.07	5.71
	1984.50	1960.8	Oil	160304	0.11		0.55	0.17	0.11	0.10	1.54	0.09	5.25
	1997.67	1974.0	Core plug	146992	0.11		0.54	0.17	0.11	0.10	1.49	0.09	5.41
	2352.00	2336.6	Mud	163302	0.08		0.18	0.04		0.03		0.01	13.47

TABLE Ap7.1 Alkane fraction GC-MS parameters: terpane and sterane ratios

Well	MD (m)	TVDSS (m)	Sample type	APTID	αα S/S+R C29 ster	ββ/ αα+ββ C29 ster	Sterane composition (ααR)			Iso-sterane composition (ββ S+R)			28/29
							C27 (%)	C28 (%)	C29 (%)	C27 (%)	C28 (%)	C29 (%)	
206/12a-5	1936-1963	1912.3-1939.3	Oil	160301	0.51	0.52	44.3	21.5	34.2	35.3	28.3	36.4	0.64
	1967.35	1943.7	Core	154666	0.50	0.50	44.5	22.2	33.3	36.5	28.1	35.4	0.69
	1984.00	1960.3	SWC	153498	0.49	0.50	50.5	19.8	29.7	35.3	27.3	37.4	0.72
	1984.50	1960.8	Oil	160304	0.54	0.52	43.3	23.0	33.7	36.0	27.8	36.2	0.64
	1997.67	1974.0	Core plug	146992	0.51	0.53	44.8	22.3	33.0	35.4	29.2	35.4	0.74
	2352.00	2336.6	Mud	163302	0.40	0.40	71.7	9.3	19.0	28.4	29.8	41.8	0.46

Ts/Tm : C27 trisnorhopane ratio - 27Ts/27Tm

C30 mor/hop : C30 moretane/hopane ratio - 30βα/30αβ

C29 mor/hop : C29 normoretane/norhopane ratio - 29βα/29αβ

C31 S/R hop : C31 hopane isomer ratio - 31αβS/31αβR

C32 S/R hop : C32 hopane isomer ratio - 32αβS/32αβR

35S+R/C31S+R hop : C35/C31 hopane ratio - 35αβS+R/31αβS+R

35S+R/C34S+R hop : C35/C34 hopane ratio - 35αβS+R/34αβS+R

C29/C29+C30 hop : Norhopane/hopane fraction - 29αβ/(29αβ+30αβ)

Bnh/hop : Bisnorhopane/hopane ratio - 28αβ/30αβ

G/hop : Gammacerane/hopane ratio - 30G/30αβ

O/hop : Oleanane/hopane ratio - 30O/30αβ

Demeth hop/hop : Demethylated hopane/hopane ratio - 25nor30αβ/30αβ

Diahop/hop : C30 diahopane/hopane ratio - 30d/30αβ

28+29 tricy/hop : Tricyclic terpane/hopane ratio - (28/3R+S +29/3R+S)/30αβ

23 tricy/hop : Tricyclic terpane/hopane ratio - 23/3 / 30αβ

24 tetracy/26 tricy : Tetracyclic/tricyclic terpane ratio - 24/4 / 26/3R+S

24 tetracy/hop : Tetracyclic terpane/hopane ratio - 24/3 / 30αβ

hop/ster : Hopane/sterane ratio - 30αβ/(29ααS+29ββR+29ββS+29αααR)

αα S/S+R C29 ster : Sterane isomerisation fraction - 29ααS/(29ααS+29αααR)

ββ/αα+ββ C29 ster : Isosterane fraction - (29ββR+29ββS) / (29ααS+29ββR+29ββS+29αααR)

Sterane composition (ααR)
- **C27 (%) :** 27αααR/(27αααR+28αααR+29αααR)*100
- **C28 (%) :** 28αααR/(27αααR+28αααR+29αααR)*100
- **C29 (%) :** 29αααR/(27αααR+28αααR+29αααR)*100

Iso-sterane composition (ββ S+R)
- **C27 (%) :** (27ββR+27ββS)/(27ββR+27ββS+28ββR+28ββS+29ββR+29ββS)*100
- **C28 (%) :** (28ββR+28ββS)/(27ββR+27ββS+28ββR+28ββS+29ββR+29ββS)*100
- **C29 (%) :** (29ββR+29ββS)/(27ββR+27ββS+28ββR+28ββS+29ββR+29ββS)*100

28/29 : (28αα(R+S)+28ββ(R+S))/(29αα(R+S)+29ββ(R+S))

TABLE Ap7.1 Alkane fraction GC-MS parameters: terpane and sterane ratios

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 177				m/e 191			
					25-nor-28ab	25-nor-29ab	25-nor-30ab	25-nor-31ab	19/3	20/3	21/3	22/3
206/12a-5	1897.03	1873.4	Spun out oil	146645	2559611	10556697	7868687	3585060	1010197	1150180	1260213	561902
	1916.67	1893.0	Core plug	146972	1673727	6662121	4760610	2482059	678573	844800	971860	278819
	1922.08	1898.4	Spun out oil	146643	2346690	10084224	7824663	3142596	990964	1096783	1279822	559522
	1922.40	1898.7	Oil	160302	1698025	6779587	4792195	2429477	730951	872410	939423	282869
	1932.67	1909.0	Core plug	146976	9407961	40302305	27713254	12682415	4137402	4327721	4606668	1662393
	1937.32	1913.7	Core	154661	1112758	3781267	2483130	1220804	669555	723978	708617	278010
	1938.00	1914.3	Mud	163301	4647	16216	10964	7609	5554	6261	5775	
	1948.67	1925.0	Spun out oil	146644	3841841	15488362	10747317	4763552	1461720	1661542	1976397	785043
	1956.00	1932.3	Oil	160303	1574766	6388260	4612131	2235687	690844	779543	897595	287642
	1956.67	1933.0	Core plug	146982	2467493	10805310	7407293	3595916	1099512	1332991	1450145	442550
	1936-1963	1912.3-1939.3	Oil	160301	1580118	7263717	4731036	2227065	760963	813362	960574	261003
	1967.35	1943.7	Core	154666	1696552	6588884	4702393	2038601	716160	779802	847734	354999
	1984.00	1960.3	SWC	153498	699310	2644665	2006537	913949	242484	324904	378962	128491
	1984.50	1960.8	Oil	160304	1442485	6537479	4837222	2134555	662357	792778	866644	258715
	1997.67	1974.0	Core plug	146992	1908761	8191816	5376506	2532082	772092	896316	1020864	308229
	2352.00	2336.6	Mud	163302	2840	12939	9046	5592	6618	5639	3808	856

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191							
					23/3	24/3	25/3R	25/3S	24/4	26/3R	26/3S	28/3R
206/12a-5	1897.03	1873.4	Spun out oil	146645	1751554	1347205	574731	628442	1782962	578750	556089	529329
	1916.67	1893.0	Core plug	146972	1284571	954172	425493	411808	1131180	357325	374862	408626
	1922.08	1898.4	Spun out oil	146643	1695895	1249048	523640	538981	1578048	563232	501758	485045
	1922.40	1898.7	Oil	160302	1208230	1004757	442764	390696	1119483	380511	369478	372580
	1932.67	1909.0	Core plug	146976	7069640	4472393	2241477	2271453	5866187	1947885	2047935	2617792
	1937.32	1913.7	Core	154661	789287	478910	192019	204947	772055	174673	162045	163247
	1938.00	1914.3	Mud	163301	3003	1410			1997			
	1948.67	1925.0	Spun out oil	146644	2718997	1857751	812692	828884	2543420	779668	803265	757536
	1956.00	1932.3	Oil	160303	1146791	830475	366156	350881	1120225	348671	363060	375785
	1956.67	1933.0	Core plug	146982	1926868	1471166	648328	616722	1696315	588550	594791	610231
	1936-1963	1912.3-1939.3	Oil	160301	1168487	833141	391346	380474	1047723	369461	370132	346454
	1967.35	1943.7	Core	154666	1047291	797868	337017	342739	1066417	304760	303553	302818
	1984.00	1960.3	SWC	153498	465931	331761	123583	131132	405002	120886	118963	140122
	1984.50	1960.8	Oil	160304	1146839	882473	405517	362484	1061330	339713	348711	365857
	1997.67	1974.0	Core plug	146992	1382066	1054964	448562	429846	1243387	416844	416985	428788
	2352.00	2336.6	Mud	163302	2949	1924	702	1204	1461			

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191							
					28/3S	29/3R	29/3S	27Ts	27Tm	30/3R	30/3S	28ab
206/12a-5	1897.03	1873.4	Spun out oil	146645	578395	733170	652132	4972787	4930230	738651	1369866	2083070
	1916.67	1893.0	Core plug	146972	323697	412077	301364	3204014	2948366	190180	722218	1322853
	1922.08	1898.4	Spun out oil	146643	537015	649290	566218	4584156	4355537	662214	1249647	1871043
	1922.40	1898.7	Oil	160302	334053	482267	233661	3169142	2903288	178383	615966	1398921
	1932.67	1909.0	Core plug	146976	2188289	4080737	2262123	17337434	15423414	3014836	4328587	7980282
	1937.32	1913.7	Core	154661	148169	154290	154938	2176631	2123206	288414	474713	766782
	1938.00	1914.3	Mud	163301				8099	11910		1632	5134
	1948.67	1925.0	Spun out oil	146644	716480	903476	865507	6485500	6891899	982766	2430770	2921579
	1956.00	1932.3	Oil	160303	309255	436185	231383	2952440	2543460	163665	585537	1325995
	1956.67	1933.0	Core plug	146982	555801	683418	424398	4664271	4192153	199486	891644	2283154
	1936-1963	1912.3-1939.3	Oil	160301	341033	393297	241494	2924696	2615630	195474	471078	1350778
	1967.35	1943.7	Core	154666	252159	332611	298435	2898979	2850101	473322	952370	1199575
	1984.00	1960.3	SWC	153498	105706	151613	137432	1116839	1176772	182871	375325	499258
	1984.50	1960.8	Oil	160304	322169	380607	199662	2682202	2615746	118631	653649	1300121
	1997.67	1974.0	Core plug	146992	397606	488184	195603	3297829	3155187	192860	660259	1514661
	2352.00	2336.6	Mud	163302				6550	8244		3258	4044

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191							
					25nor-30ab	29ab	29Ts	30d	29ba	30O	30ab	30ba
206/12a-5	1897.03	1873.4	Spun out oil	146645	11388219	12141883	4774977	3736695	1766330		23540021	3293942
	1916.67	1893.0	Core plug	146972	6458979	7325371	2999120	2289155	908463		14741442	2112485
	1922.08	1898.4	Spun out oil	146643	10389059	10817777	4700874	3301414	1618650		21247528	2859807
	1922.40	1898.7	Oil	160302	6824185	7015215	2940857	2243792	972323		13404162	2199384
	1932.67	1909.0	Core plug	146976	39822387	39594898	17497177	13313419	6594920		76920903	11431160
	1937.32	1913.7	Core	154661	3204839	4013237	1637575	1137277	710952		7819446	1184676
	1938.00	1914.3	Mud	163301	19935	52659	10740	7666			136668	19651
	1948.67	1925.0	Spun out oil	146644	15214994	15089495	6526162	4773874	2056996		31214339	4516069
	1956.00	1932.3	Oil	160303	6369091	6647625	2691054	2122507	791576		12245277	1987602
	1956.67	1933.0	Core plug	146982	10867688	10904443	4619697	3238695	1392397		20252184	3094163
	1936-1963	1912.3-1939.3	Oil	160301	6460875	6413110	2915622	2111880	902288		12407955	1982736
	1967.35	1943.7	Core	154666	5972086	5952485	2715335	1911202	852498		11906836	1824187
	1984.00	1960.3	SWC	153498	2611445	2662476	1054403	816990	356285		5567222	795434
	1984.50	1960.8	Oil	160304	6467855	5962716	2578660	1943483	881397		11759326	1822027
	1997.67	1974.0	Core plug	146992	7433942	7283761	3263768	2372146	1035118		13722017	2241876
	2352.00	2336.6	Mud	163302	17322	49730	6862	3576	2338		98560	7718

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191							
					31abS	31abR	30G	31ba	32abS	32abR	33abS	33abR
206/12a-5	1897.03	1873.4	Spun out oil	146645	7475457	5673629	2008774	1245108	5346038	4399083	4240044	2606425
	1916.67	1893.0	Core plug	146972	4447146	3418904	1578764	766809	3308805	2521729	2530197	1634531
	1922.08	1898.4	Spun out oil	146643	6941348	5222585	1855363	1279462	5005567	3713010	3911601	2514160
	1922.40	1898.7	Oil	160302	4289210	3217409	1502873	767650	2998971	2330912	2466775	1748845
	1932.67	1909.0	Core plug	146976	25440515	20700812	8153239	4494268	19361003	15577561	16374927	10228880
	1937.32	1913.7	Core	154661	2345577	1615829	757625	408966	1433206	1140271	1000083	740971
	1938.00	1914.3	Mud	163301	35402	29354	16330	7517	25373	21733	33290	21819
	1948.67	1925.0	Spun out oil	146644	9444965	7378847	3038556	1766492	7000236	5487396	5384236	3407785
	1956.00	1932.3	Oil	160303	4011062	2929137	1472520	652953	2960655	2298371	2221447	1526613
	1956.67	1933.0	Core plug	146982	6843513	5060108	2238915	1148930	4846843	3817491	3628340	2490210
	1936-1963	1912.3-1939.3	Oil	160301	4111247	3082236	1313302	741407	2951482	2296636	2218603	1571165
	1967.35	1943.7	Core	154666	3848080	2804493	1233455	578770	2684970	2174311	2054178	1332786
	1984.00	1960.3	SWC	153498	1775120	1321746	560181	251024	1241618	993938	1052896	684923
	1984.50	1960.8	Oil	160304	3986367	2953673	1288318	672495	2734443	2235717	2184420	1561674
	1997.67	1974.0	Core plug	146992	4887996	3437814	1521550	854684	3269069	2595651	2552247	1776525
	2352.00	2336.6	Mud	163302	30857	18901	8285	2921	15805	12705	16012	10253

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191				m/e 217			
					34abS	34abR	35abS	35abR	21aa	21bb	22aa	22bb
206/12a-5	1897.03	1873.4	Spun out oil	146645	2733281	1670618	2009357	1452756	1414100	1627804	1108217	981061
	1916.67	1893.0	Core plug	146972	2017107	1028419	1159161	793520	932117	1007494	736577	611814
	1922.08	1898.4	Spun out oil	146643	2508660	1555979	1895508	1325188	1369082	1612011	1063486	902330
	1922.40	1898.7	Oil	160302	2110376	1043915	1235103	847034	929926	1004085	717995	658072
	1932.67	1909.0	Core plug	146976	10771858	6693782	7749305	5473360	4961964	5783532	4133505	3576763
	1937.32	1913.7	Core	154661	702902	456049	620457	423059	699366	985274	471583	556787
	1938.00	1914.3	Mud	163301	16331	13656	18501	15164	1510	782		681
	1948.67	1925.0	Spun out oil	146644	3700788	2369908	2694186	1860682	1871407	2274797	1427716	1406511
	1956.00	1932.3	Oil	160303	1824240	952420	1075755	735130	836909	974131	685194	602358
	1956.67	1933.0	Core plug	146982	3229341	1557369	1812702	1306055	1340482	1522669	1104169	1005393
	1936-1963	1912.3-1939.3	Oil	160301	1959712	949990	1136626	812572	840926	942140	690967	600570
	1967.35	1943.7	Core	154666	1425487	887132	1035910	672632	946597	1133676	724541	677784
	1984.00	1960.3	SWC	153498	695312	449081	509333	346228	331380	386845	234446	215526
	1984.50	1960.8	Oil	160304	1924567	911291	1064778	755650	823795	919349	662288	589490
	1997.67	1974.0	Core plug	146992	2020207	1127389	1317159	869145	1013048	1169559	775914	742339
	2352.00	2336.6	Mud	163302	12570	7629	11641	6368	1258	1002	575	631

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 217							
					27dbS	27dbR	27daR	27daS	28dbS#1	28dbS#2	28dbR#1	28dbR#2
206/12a-5	1897.03	1873.4	Spun out oil	146645	2687217	1629168	755950	902640	1036549	1058405	559616	800389
	1916.67	1893.0	Core plug	146972	1637147	1071727	425476	584790	610588	658909	402301	470234
	1922.08	1898.4	Spun out oil	146643	2398193	1551194	640792	874841	949739	939435	556818	793282
	1922.40	1898.7	Oil	160302	1534445	1024655	393133	524936	590739	566794	374375	448660
	1932.67	1909.0	Core plug	146976	8291897	5603252	2280919	3105200	3360100	3886527	2227881	2974864
	1937.32	1913.7	Core	154661	650852	398629	174390	249574	242086	231475	151401	200423
	1938.00	1914.3	Mud	163301	2023	955			1117	914		
	1948.67	1925.0	Spun out oil	146644	3219961	1926830	880624	1114783	1241560	1300388	749976	1024352
	1956.00	1932.3	Oil	160303	1335282	899380	360826	481736	537289	538802	330889	397398
	1956.67	1933.0	Core plug	146982	2248131	1574945	611574	801024	881650	918707	545441	692083
	1936-1963	1912.3-1939.3	Oil	160301	1338412	900978	358289	504537	547153	526405	324715	408822
	1967.35	1943.7	Core	154666	1322759	884252	359986	499617	516572	495240	334452	409193
	1984.00	1960.3	SWC	153498	538457	327513	130967	187558	203590	200286	118142	157004
	1984.50	1960.8	Oil	160304	1335739	854384	374175	468070	514524	522408	326010	391781
	1997.67	1974.0	Core plug	146992	1588026	1051648	423896	554694	629396	622996	384829	448112
	2352.00	2336.6	Mud	163302	1787	1226			994	1144	606	845

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 217							
					28daR	27aaS	27bbR+ 29dbS	27bbS	28daS	27aaR	29dbR	29daR
206/12a-5	1897.03	1873.4	Spun out oil	146645	582329	1145336	1697370	1006378	347959	1214710	1337609	672552
	1916.67	1893.0	Core plug	146972	486520	682419	1023612	658801	247702	801160	791265	507619
	1922.08	1898.4	Spun out oil	146643	544157	1021130	1556786	928150	368608	1014681	1278531	638866
	1922.40	1898.7	Oil	160302	328015	647476	951169	605123	252399	748861	791785	496468
	1932.67	1909.0	Core plug	146976	1972954	4151035	6763380	3704754	1532163	4449459	5309251	2436755
	1937.32	1913.7	Core	154661	174889	402542	334451	311237	97887	488079	327619	283364
	1938.00	1914.3	Mud	163301		3147	1615			14339	937	
	1948.67	1925.0	Spun out oil	146644	739111	1248583	2061581	1394276	577463	1500654	1753666	1026027
	1956.00	1932.3	Oil	160303	294105	589291	842148	581230	214222	696361	702333	437764
	1956.67	1933.0	Core plug	146982	581704	1032121	1518548	970190	377147	1133371	1199899	820263
	1936-1963	1912.3-1939.3	Oil	160301	310750	583913	899576	612672	221199	708860	737972	456379
	1967.35	1943.7	Core	154666	284448	602337	828175	564219	242298	746484	666183	499644
	1984.00	1960.3	SWC	153498	129585	258188	328754	228607	82907	418078	294857	196156
	1984.50	1960.8	Oil	160304	292181	596689	829913	564962	224139	640796	703690	422289
	1997.67	1974.0	Core plug	146992	404981	675351	985805	641021	262285	792982	839363	529586
	2352.00	2336.6	Mud	163302		2288	1312			9940	1002	868

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 217							
					28aaS	29daS	28bbR	28bbS	28aaR	29aaS	29bbR	29bbS
206/12a-5	1897.03	1873.4	Spun out oil	146645	314785	591205	688078	1210609	692467	1046510	1408887	1125508
	1916.67	1893.0	Core plug	146972	128327	376816	529870	716014	416825	642162	658926	704737
	1922.08	1898.4	Spun out oil	146643	325083	585853	777775	1210472	589561	957818	1132523	968732
	1922.40	1898.7	Oil	160302	141607	372563	481262	684272	387817	618809	628476	685541
	1932.67	1909.0	Core plug	146976	949827	2244746	2765269	4931286	2632145	3274239	4720434	3588952
	1937.32	1913.7	Core	154661	92526	136811	248673	324864	257431	367560	342450	326828
	1938.00	1914.3	Mud	163301			1341	2042	2503	1851	2640	2089
	1948.67	1925.0	Spun out oil	146644	313652	756595	954781	1533625	762335	1212920	1483402	1302766
	1956.00	1932.3	Oil	160303	134423	326216	423787	638888	377195	569373	527737	584619
	1956.67	1933.0	Core plug	146982	286926	660973	794413	1091713	633732	981114	978811	1054781
	1936-1963	1912.3-1939.3	Oil	160301	115867	333911	422170	620102	343827	578034	587448	619883
	1967.35	1943.7	Core	154666	158250	289564	398498	605473	372023	550714	584403	524359
	1984.00	1960.3	SWC	153498	59027	138401	177504	297561	163540	239242	255869	233782
	1984.50	1960.8	Oil	160304	97229	340071	426413	579811	340601	574651	567190	597865
	1997.67	1974.0	Core plug	146992	184854	423358	551312	736579	394230	617446	610543	722322
	2352.00	2336.6	Mud	163302			1040	1035	1284	1723	1454	1507

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 217					m/e 218		
					29aaR	30aaS	30bbR	30bbS	30aaR	27bbR	27bbS	28bbR
206/12a-5	1897.03	1873.4	Spun out oil	146645	1147917	753322	443501	223481	314066	1346099	1234284	815697
	1916.67	1893.0	Core plug	146972	620040	348625	293002	186270	180888	1063382	977205	688026
	1922.08	1898.4	Spun out oil	146643	976061	663601	403166	213819	298419	1223045	1117575	824206
	1922.40	1898.7	Oil	160302	578000	372089	281368	197473	181992	930488	825874	600722
	1932.67	1909.0	Core plug	146976	4034562	2508399	1476253	789916	1350311	6731661	5772704	4134884
	1937.32	1913.7	Core	154661	404094	197837	126971	84194	119744	448244	418115	271376
	1938.00	1914.3	Mud	163301	4214	1640				1196	1075	2124
	1948.67	1925.0	Spun out oil	146644	1329682	735754	598808	315635	405038	2186778	2062849	1327856
	1956.00	1932.3	Oil	160303	499304	376870	265872	164670	158947	837922	809976	534734
	1956.67	1933.0	Core plug	146982	905521	610262	418006	304475	281288	1367497	1341780	940995
	1936-1963	1912.3-1939.3	Oil	160301	547515	317435	278501	184179	171567	797815	804047	553229
	1967.35	1943.7	Core	154666	558131	313018	243746	158450	159246	759118	703572	466280
	1984.00	1960.3	SWC	153498	245949	126006	101039	59634	64820	302298	300271	183255
	1984.50	1960.8	Oil	160304	499193	310998	244622	188081	159384	809645	763717	545785
	1997.67	1974.0	Core plug	146992	584124	375813	302294	214748	198267	932424	900733	651845
	2352.00	2336.6	Mud	163302	2635	937	620			1428	1542	1202

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 218					m/e 185		
					28bbS	29bbR	29bbS	30bbR	30bbS	n-C20	n-C21	n-C22
206/12a-5	1897.03	1873.4	Spun out oil	146645	1260117	1435834	1344123	303921	246251	15598324	15626613	15107268
	1916.67	1893.0	Core plug	146972	876534	970421	952794	260941	235423	15414274	15455852	13954276
	1922.08	1898.4	Spun out oil	146643	1063689	1152480	1166835	276984	268183	11721627	13065882	11535256
	1922.40	1898.7	Oil	160302	799964	927154	910863	267198	228842	11671198	11705218	10435417
	1932.67	1909.0	Core plug	146976	5883292	6640083	6194441	1511416	1146234	42342548	42987302	44264023
	1937.32	1913.7	Core	154661	364288	397216	395998	85532	65286	4394467	3876196	3303952
	1938.00	1914.3	Mud	163301	2024	3346	2403	1509		1393404	576081	174862
	1948.67	1925.0	Spun out oil	146644	1947932	2174130	2042507	533828	433654	18745846	19799271	18291552
	1956.00	1932.3	Oil	160303	710225	789495	775778	231442	201538	10665319	10729998	9854810
	1956.67	1933.0	Core plug	146982	1222833	1389838	1376956	387957	351329	17041400	17497220	16228530
	1936-1963	1912.3-1939.3	Oil	160301	732050	835715	816469	256770	208463	11962214	11342976	10196543
	1967.35	1943.7	Core	154666	659495	716777	701620	182878	156043	8805311	8429309	8269237
	1984.00	1960.3	SWC	153498	281683	315300	322571	56566	46008	8168106	6367579	4963842
	1984.50	1960.8	Oil	160304	666746	797259	783182	229412	192934	10114827	10641116	9765459
	1997.67	1974.0	Core plug	146992	862658	912313	919413	274589	221375	12179526	11373479	11107214
	2352.00	2336.6	Mud	163302	1905	2441	1924			1170198	405073	128999

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 185							
					n-C23	n-C24	n-C25	n-C26	n-C27	n-C28	n-C29	n-C30
206/12a-5	1897.03	1873.4	Spun out oil	146645	15910145	13981562	13800648	9964580	8913460	6190344	5051557	3483242
	1916.67	1893.0	Core plug	146972	14463483	11729013	11091960	8691818	6963216	5298635	4355857	3327296
	1922.08	1898.4	Spun out oil	146643	12192784	10672118	10241968	7540746	6648610	4892571	3911260	2465131
	1922.40	1898.7	Oil	160302	11229352	9101300	8460399	6259750	5494282	4252657	3436904	2270298
	1932.67	1909.0	Core plug	146976	44987158	38580716	38311124	30548139	26136150	21604149	17552947	13595676
	1937.32	1913.7	Core	154661	2933721	2249954	1922948	1413431	1080236	730922	536797	349628
	1938.00	1914.3	Mud	163301	60578	22668	11721	7481	4528	3468	4378	3354
	1948.67	1925.0	Spun out oil	146644	19113570	16226241	15045442	11418252	10183794	7064384	5761220	3812651
	1956.00	1932.3	Oil	160303	9646924	8379814	7647893	5730738	4776982	3586035	2936224	2061884
	1956.67	1933.0	Core plug	146982	16061397	13424025	12856726	9382246	7757869	5854015	5316549	3919814
	1936-1963	1912.3-1939.3	Oil	160301	10037516	8081448	7608410	5831950	5109041	3653320	2877893	2164756
	1967.35	1943.7	Core	154666	8308001	6976734	6109354	4824984	4022915	2720256	2372474	1690199
	1984.00	1960.3	SWC	153498	4448612	3487969	3312906	2399452	2008718	1483021	1212420	914700
	1984.50	1960.8	Oil	160304	9608681	7890944	7657165	5618548	4900801	3504954	2800676	1992652
	1997.67	1974.0	Core plug	146992	10825280	9521773	8782535	6411385	5257431	4336556	3421331	2578941
	2352.00	2336.6	Mud	163302	46774	17746	9567	6922	3304	3517	2964	2086

TABLE Ap7.2 Alkane fraction GC-MS peak heights (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 177				m/e 191			
					25-nor-28ab	25-nor-29ab	25-nor-30ab	25-nor-31ab	19/3	20/3	21/3	22/3
206/12a-5	1897.03	1873.4	Spun out oil	146645	42506	175307	130669	59534	16776	19100	20927	9331
	1916.67	1893.0	Core plug	146972	66375	264200	188792	98431	26910	33502	38541	11057
	1922.08	1898.4	Spun out oil	146643	39071	167895	130275	52322	16499	18261	21308	9316
	1922.40	1898.7	Oil	160302	68331	272820	192845	97766	29414	35107	37804	11383
	1932.67	1909.0	Core plug	146976	114080	488703	336049	153786	50170	52478	55860	20158
	1937.32	1913.7	Core	154661	34472	117139	76924	37819	20742	22428	21952	8612
	1938.00	1914.3	Mud	163301	188	657	444	308	225	254	234	
	1948.67	1925.0	Spun out oil	146644	53555	215908	149818	66404	20376	23162	27551	10944
	1956.00	1932.3	Oil	160303	61749	250493	180849	87665	27089	30567	35196	11279
	1956.67	1933.0	Core plug	146982	89085	390109	267429	129825	39696	48126	52355	15978
	1936-1963	1912.3-1939.3	Oil	160301	64469	296360	193027	90864	31047	33185	39191	10649
	1967.35	1943.7	Core	154666	46947	182327	130124	56412	19818	21579	23458	9824
	1984.00	1960.3	SWC	153498	20331	76888	58336	26571	7050	9446	11017	3736
	1984.50	1960.8	Oil	160304	62917	285144	210984	93103	28890	34578	37800	11284
	1997.67	1974.0	Core plug	146992	69475	298164	195693	92162	28102	32624	37157	11219
	2352.00	2336.6	Mud	163302	106	481	336	208	246	209	141	32

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191							
					23/3	24/3	25/3R	25/3S	24/4	26/3R	26/3S	28/3R
206/12a-5	1897.03	1873.4	Spun out oil	146645	29087	22372	9544	10436	29608	9611	9235	8790
	1916.67	1893.0	Core plug	146972	50942	37840	16874	16331	44859	14170	14866	16205
	1922.08	1898.4	Spun out oil	146643	28235	20796	8718	8974	26273	9377	8354	8076
	1922.40	1898.7	Oil	160302	48621	40433	17817	15722	45050	15312	14868	14993
	1932.67	1909.0	Core plug	146976	85726	54232	27180	27544	71133	23620	24833	31743
	1937.32	1913.7	Core	154661	24451	14836	5949	6349	23917	5411	5020	5057
	1938.00	1914.3	Mud	163301	122	57			81			
	1948.67	1925.0	Spun out oil	146644	37903	25897	11329	11555	35455	10869	11198	10560
	1956.00	1932.3	Oil	160303	44967	32564	14358	13759	43926	13672	14236	14735
	1956.67	1933.0	Core plug	146982	69567	53114	23407	22266	61243	21249	21474	22031
	1936-1963	1912.3-1939.3	Oil	160301	47674	33992	15967	15523	42747	15074	15101	14135
	1967.35	1943.7	Core	154666	28981	22079	9326	9484	29510	8433	8400	8380
	1984.00	1960.3	SWC	153498	13546	9645	3593	3812	11775	3514	3459	4074
	1984.50	1960.8	Oil	160304	50021	38491	17687	15810	46292	14817	15210	15958
	1997.67	1974.0	Core plug	146992	50304	38398	16327	15645	45257	15172	15177	15607
	2352.00	2336.6	Mud	163302	110	71	26	45	54			

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191							
					28/3S	29/3R	29/3S	27Ts	27Tm	30/3R	30/3S	28ab
206/12a-5	1897.03	1873.4	Spun out oil	146645	9605	12175	10829	82579	81873	12266	22748	34592
	1916.67	1893.0	Core plug	146972	12837	16342	11951	127062	116923	7542	28641	52460
	1922.08	1898.4	Spun out oil	146643	8941	10810	9427	76323	72516	11025	20806	31151
	1922.40	1898.7	Oil	160302	13443	19407	9403	127531	116832	7178	24787	56295
	1932.67	1909.0	Core plug	146976	26535	49483	27430	210233	187023	36558	52488	96768
	1937.32	1913.7	Core	154661	4590	4780	4800	67429	65774	8935	14706	23754
	1938.00	1914.3	Mud	163301				328	483		66	208
	1948.67	1925.0	Spun out oil	146644	9988	12595	12065	90408	96073	13700	33885	40727
	1956.00	1932.3	Oil	160303	12126	17103	9073	115770	99733	6418	22960	51994
	1956.67	1933.0	Core plug	146982	20066	24674	15322	168396	151351	7202	32191	82430
	1936-1963	1912.3-1939.3	Oil	160301	13914	16047	9853	119328	106718	7975	19220	55112
	1967.35	1943.7	Core	154666	6978	9204	8258	80220	78868	13098	26354	33195
	1984.00	1960.3	SWC	153498	3073	4408	3996	32470	34212	5317	10912	14515
	1984.50	1960.8	Oil	160304	14052	16601	8709	116989	114091	5174	28510	56707
	1997.67	1974.0	Core plug	146992	14472	17769	7120	120034	114842	7020	24032	55130
	2352.00	2336.6	Mud	163302				243	306		121	150

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191							
					25nor-30ab	29ab	29Ts	30d	29ba	30O	30ab	30ba
206/12a-5	1897.03	1873.4	Spun out oil	146645	189116	201631	79295	62053	29332		390912	54700
	1916.67	1893.0	Core plug	146972	256144	290503	118936	90781	36027		584602	83775
	1922.08	1898.4	Spun out oil	146643	172970	180108	78266	54966	26949		353755	47614
	1922.40	1898.7	Oil	160302	274615	282302	118344	90293	39128		539402	88506
	1932.67	1909.0	Core plug	146976	482884	480125	212170	161438	79970		932738	138614
	1937.32	1913.7	Core	154661	99282	124325	50730	35231	22024		242236	36700
	1938.00	1914.3	Mud	163301	808	2134	435	311			5539	796
	1948.67	1925.0	Spun out oil	146644	212098	210348	90975	66548	28675		435129	62954
	1956.00	1932.3	Oil	160303	249742	260664	105520	83227	31039		480156	77937
	1956.67	1933.0	Core plug	146982	392361	393688	166787	116928	50270		731174	111710
	1936-1963	1912.3-1939.3	Oil	160301	263604	261655	118958	86165	36813		506245	80896
	1967.35	1943.7	Core	154666	165259	164717	75139	52887	23590		329485	50479
	1984.00	1960.3	SWC	153498	75922	77406	30654	23752	10358		161855	23126
	1984.50	1960.8	Oil	160304	282107	260075	112473	84769	38444		512905	79471
	1997.67	1974.0	Core plug	146992	270579	265113	118794	86341	37676		499451	81599
	2352.00	2336.6	Mud	163302	644	1848	255	133	87		3662	287

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191							
					31abS	31abR	30G	31ba	32abS	32abR	33abS	33abR
206/12a-5	1897.03	1873.4	Spun out oil	146645	124139	94218	33358	20677	88778	73052	70411	43283
	1916.67	1893.0	Core plug	146972	176361	135584	62609	30409	131217	100004	100340	64821
	1922.08	1898.4	Spun out oil	146643	115568	86952	30890	21302	83339	61819	65125	41859
	1922.40	1898.7	Oil	160302	172604	129473	60478	30891	120683	93799	99266	70376
	1932.67	1909.0	Core plug	146976	308490	251017	98866	54497	234770	188893	198561	124035
	1937.32	1913.7	Core	154661	72663	50056	23470	12669	44399	35324	30981	22954
	1938.00	1914.3	Mud	163301	1435	1190	662	305	1028	881	1349	884
	1948.67	1925.0	Spun out oil	146644	131663	102861	42358	24625	97584	76495	75057	47505
	1956.00	1932.3	Oil	160303	157280	114856	57740	25603	116092	90123	87106	59861
	1956.67	1933.0	Core plug	146982	247075	182688	80833	41480	174988	137825	130996	89905
	1936-1963	1912.3-1939.3	Oil	160301	167739	125755	53583	30249	120421	93703	90519	64104
	1967.35	1943.7	Core	154666	106484	77606	34132	16016	74298	60167	56843	36881
	1984.00	1960.3	SWC	153498	51608	38427	16286	7298	36097	28897	30611	19913
	1984.50	1960.8	Oil	160304	173873	128830	56192	29332	119268	97515	95278	68115
	1997.67	1974.0	Core plug	146992	177912	125129	55381	31109	118987	94476	92896	64662
	2352.00	2336.6	Mud	163302	1146	702	308	109	587	472	595	381

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 191				m/e 217			
					34abS	34abR	35abS	35abR	21aa	21bb	22aa	22bb
206/12a-5	1897.03	1873.4	Spun out oil	146645	45390	27743	33368	24125	23483	27032	18403	16292
	1916.67	1893.0	Core plug	146972	79992	40784	45969	31469	36965	39954	29210	24263
	1922.08	1898.4	Spun out oil	146643	41767	25906	31559	22063	22794	26839	17706	15023
	1922.40	1898.7	Oil	160302	84925	42009	49702	34086	37422	40406	28893	26482
	1932.67	1909.0	Core plug	146976	130619	81168	93968	66370	60168	70131	50123	43372
	1937.32	1913.7	Core	154661	21775	14128	19221	13106	21665	30523	14609	17249
	1938.00	1914.3	Mud	163301	662	553	750	615	61	32		28
	1948.67	1925.0	Spun out oil	146644	51589	33037	37557	25938	26087	31711	19902	19607
	1956.00	1932.3	Oil	160303	71531	37346	42182	28826	32816	38197	26868	23619
	1956.67	1933.0	Core plug	146982	116590	56226	65445	47153	48396	54974	39864	36298
	1936-1963	1912.3-1939.3	Oil	160301	79956	38760	46374	33153	34310	38439	28192	24503
	1967.35	1943.7	Core	154666	39446	24549	28666	18613	26194	31371	20049	18756
	1984.00	1960.3	SWC	153498	20215	13056	14808	10066	9634	11247	6816	6266
	1984.50	1960.8	Oil	160304	83944	39748	46442	32959	35931	40099	28887	25712
	1997.67	1974.0	Core plug	146992	73531	41034	47942	31635	36873	42569	28242	27019
	2352.00	2336.6	Mud	163302	467	283	432	237	47	37	21	23

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 217							
					27dbS	27dbR	27daR	27daS	28dbS#1	28dbS#2	28dbR#1	28dbR#2
206/12a-5	1897.03	1873.4	Spun out oil	146645	44625	27054	12554	14989	17213	17576	9293	13291
	1916.67	1893.0	Core plug	146972	64924	42502	16873	23191	24214	26130	15954	18648
	1922.08	1898.4	Spun out oil	146643	39928	25826	10669	14565	15812	15641	9271	13208
	1922.40	1898.7	Oil	160302	61748	41234	15820	21124	23772	22809	15065	18055
	1932.67	1909.0	Core plug	146976	100547	67945	27658	37653	40744	47128	27015	36073
	1937.32	1913.7	Core	154661	20163	12349	5402	7731	7500	7171	4690	6209
	1938.00	1914.3	Mud	163301	82	39			45	37		
	1948.67	1925.0	Spun out oil	146644	44886	26860	12276	15540	17307	18127	10455	14280
	1956.00	1932.3	Oil	160303	52358	35266	14149	18890	21068	21127	12975	15583
	1956.67	1933.0	Core plug	146982	81165	56861	22080	28920	31831	33169	19692	24987
	1936-1963	1912.3-1939.3	Oil	160301	54607	36760	14618	20585	22324	21477	13248	16680
	1967.35	1943.7	Core	154666	36603	24469	9962	13825	14295	13704	9255	11323
	1984.00	1960.3	SWC	153498	15654	9522	3808	5453	5919	5823	3435	4565
	1984.50	1960.8	Oil	160304	58261	37266	16320	20416	22442	22786	14220	17088
	1997.67	1974.0	Core plug	146992	57801	38278	15429	20190	22909	22676	14007	16310
	2352.00	2336.6	Mud	163302	66	46			37	43	23	31

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 217							
					28daR	27aaS	27bbR+ 29dbS	27bbS	28daS	27aaR	29dbR	29daR
206/12a-5	1897.03	1873.4	Spun out oil	146645	9670	19020	28187	16712	5778	20172	22213	11169
	1916.67	1893.0	Core plug	146972	19294	27063	40593	26126	9823	31772	31379	20131
	1922.08	1898.4	Spun out oil	146643	9060	17001	25919	15453	6137	16894	21287	10637
	1922.40	1898.7	Oil	160302	13200	26055	38276	24351	10157	30135	31863	19979
	1932.67	1909.0	Core plug	146976	23924	50335	82012	44924	18579	53954	64380	29548
	1937.32	1913.7	Core	154661	5418	12470	10361	9642	3032	15120	10149	8778
	1938.00	1914.3	Mud	163301		128	65			581	38	
	1948.67	1925.0	Spun out oil	146644	10303	17405	28739	19436	8050	20919	24446	14303
	1956.00	1932.3	Oil	160303	11532	23107	33022	22791	8400	27305	27540	17165
	1956.67	1933.0	Core plug	146982	21002	37263	54825	35027	13616	40919	43321	29614
	1936-1963	1912.3-1939.3	Oil	160301	12679	23824	36703	24997	9025	28922	30109	18620
	1967.35	1943.7	Core	154666	7871	16668	22917	15613	6705	20657	18435	13826
	1984.00	1960.3	SWC	153498	3767	7506	9558	6646	2410	12155	8572	5703
	1984.50	1960.8	Oil	160304	12744	26026	36198	24642	9776	27950	30693	18419
	1997.67	1974.0	Core plug	146992	14740	24581	35881	23332	9547	28863	30551	19276
	2352.00	2336.6	Mud	163302		85	49			369	37	32

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 217							
					28aaS	29daS	28bbR	28bbS	28aaR	29aaS	29bbR	29bbS
206/12a-5	1897.03	1873.4	Spun out oil	146645	5227	9818	11426	20104	11499	17379	23396	18690
	1916.67	1893.0	Core plug	146972	5089	14943	21013	28395	16530	25466	26131	27948
	1922.08	1898.4	Spun out oil	146643	5412	9754	12949	20153	9816	15947	18856	16129
	1922.40	1898.7	Oil	160302	5698	14992	19367	27536	15606	24902	25291	27587
	1932.67	1909.0	Core plug	146976	11518	27220	33531	59796	31917	39703	57240	43519
	1937.32	1913.7	Core	154661	2866	4238	7704	10064	7975	11387	10609	10125
	1938.00	1914.3	Mud	163301			54	83	101	75	107	85
	1948.67	1925.0	Spun out oil	146644	4372	10547	13310	21379	10627	16908	20679	18161
	1956.00	1932.3	Oil	160303	5271	12791	16617	25052	14790	22326	20693	22924
	1956.67	1933.0	Core plug	146982	10359	23863	28681	39415	22880	35422	35338	38081
	1936-1963	1912.3-1939.3	Oil	160301	4727	13624	17225	25300	14028	23584	23968	25291
	1967.35	1943.7	Core	154666	4379	8013	11027	16755	10295	15239	16172	14510
	1984.00	1960.3	SWC	153498	1716	4024	5161	8651	4755	6955	7439	6797
	1984.50	1960.8	Oil	160304	4241	14833	18599	25290	14856	25064	24739	26077
	1997.67	1974.0	Core plug	146992	6728	15409	20067	26810	14349	22474	22222	26291
	2352.00	2336.6	Mud	163302			39	38	48	64	54	56

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 217					m/e 218		
					29aaR	30aaS	30bbR	30bbS	30aaR	27bbR	27bbS	28bbR
206/12a-5	1897.03	1873.4	Spun out oil	146645	19063	12510	7365	3711	5215	22354	20497	13546
	1916.67	1893.0	Core plug	146972	24589	13825	11620	7387	7173	42171	38753	27285
	1922.08	1898.4	Spun out oil	146643	16251	11048	6712	3560	4968	20363	18607	13722
	1922.40	1898.7	Oil	160302	23260	14973	11323	7947	7324	37444	33234	24174
	1932.67	1909.0	Core plug	146976	48923	30417	17901	9578	16374	81628	69999	50139
	1937.32	1913.7	Core	154661	12518	6129	3933	2608	3710	13886	12953	8407
	1938.00	1914.3	Mud	163301	171	66				48	44	86
	1948.67	1925.0	Spun out oil	146644	18536	10256	8347	4400	5646	30484	28756	18510
	1956.00	1932.3	Oil	160303	19578	14778	10425	6457	6233	32856	31760	20968
	1956.67	1933.0	Core plug	146982	32692	22033	15091	10993	10155	49371	48443	33973
	1936-1963	1912.3-1939.3	Oil	160301	22339	12951	11363	7514	7000	32551	32805	22572
	1967.35	1943.7	Core	154666	15445	8662	6745	4385	4407	21006	19469	12903
	1984.00	1960.3	SWC	153498	7150	3663	2937	1734	1884	8789	8730	5328
	1984.50	1960.8	Oil	160304	21773	13565	10670	8204	6952	35314	33311	23805
	1997.67	1974.0	Core plug	146992	21261	13679	11003	7816	7216	33938	32785	23726
	2352.00	2336.6	Mud	163302	98	35	23			53	57	45

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 218					m/e 185		
					28bbS	29bbR	29bbS	30bbR	30bbS	n-C20	n-C21	n-C22
206/12a-5	1897.03	1873.4	Spun out oil	146645	20926	23844	22321	5047	4089	5430851	4979556	4415574
	1916.67	1893.0	Core plug	146972	34761	38484	37785	10348	9336	10474374	9705991	8198300
	1922.08	1898.4	Spun out oil	146643	17710	19188	19427	4612	4465	4091671	4174336	3380269
	1922.40	1898.7	Oil	160302	32192	37310	36654	10752	9209	8047314	7458505	6220995
	1932.67	1909.0	Core plug	146976	71340	80517	75113	18327	13899	8797412	8253817	7951395
	1937.32	1913.7	Core	154661	11285	12305	12268	2650	2022	1965280	1607298	1345571
	1938.00	1914.3	Mud	163301	82	136	97	61		885297	348464	104870
	1948.67	1925.0	Spun out oil	146644	27154	30307	28473	7442	6045	4040166	3935700	3708280
	1956.00	1932.3	Oil	160303	27849	30957	30419	9075	7903	7165557	6662121	5724518
	1956.67	1933.0	Core plug	146982	44149	50178	49713	14007	12684	10541850	10002702	8679711
	1936-1963	1912.3-1939.3	Oil	160301	29868	34097	33312	10476	8505	8362902	7328488	6163252
	1967.35	1943.7	Core	154666	18250	19835	19415	5061	4318	3517543	3122187	3008255
	1984.00	1960.3	SWC	153498	8189	9167	9378	1645	1338	3428177	2477926	1897205
	1984.50	1960.8	Oil	160304	29081	34774	34160	10006	8415	7559187	7349209	6309920
	1997.67	1974.0	Core plug	146992	31399	33206	33465	9994	8058	7596086	6555333	5989301
	2352.00	2336.6	Mud	163302	71	91	71			681621	224636	70928

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

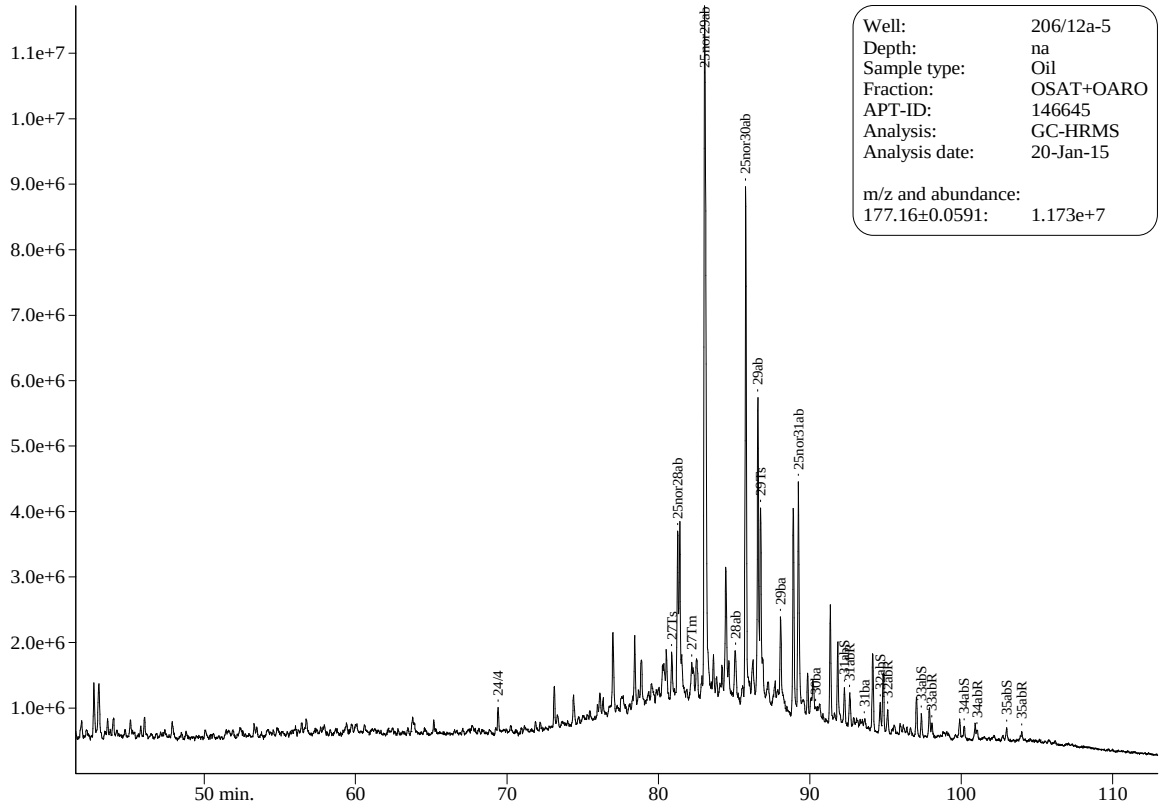
Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 185							
					n-C23	n-C24	n-C25	n-C26	n-C27	n-C28	n-C29	n-C30
206/12a-5	1897.03	1873.4	Spun out oil	146645	4374885	3690224	3699628	2656815	2492872	1634446	1449911	993792
	1916.67	1893.0	Core plug	146972	8395476	6462942	6406559	5268100	4128278	3077449	2793796	2107165
	1922.08	1898.4	Spun out oil	146643	3361384	2824037	2752738	2015763	1864264	1295137	1125526	705138
	1922.40	1898.7	Oil	160302	6614240	5088772	4958974	3849756	3305539	2506523	2236872	1459005
	1932.67	1909.0	Core plug	146976	7984656	6500133	6766565	5661134	4738222	3836993	3442440	2632796
	1937.32	1913.7	Core	154661	1166482	875505	803086	601716	455223	315618	240854	169034
	1938.00	1914.3	Mud	163301	33752	12510	6192	4186	2652	2006	2570	2003
	1948.67	1925.0	Spun out oil	146644	3496044	3086126	2877051	2437683	2166143	1493084	1460079	1007995
	1956.00	1932.3	Oil	160303	5536747	4565460	4368008	3434215	2800436	2059522	1862103	1291156
	1956.67	1933.0	Core plug	146982	8487594	6733926	6760944	5176773	4187452	3095574	3104415	2260036
	1936-1963	1912.3-1939.3	Oil	160301	5994309	4581402	4521170	3636617	3116297	2183006	1899049	1410446
	1967.35	1943.7	Core	154666	2950747	2425010	2279113	1834803	1514337	1049243	950871	729932
	1984.00	1960.3	SWC	153498	1659996	1273741	1298455	958634	794416	600981	510529	415021
	1984.50	1960.8	Oil	160304	6134373	4782111	4864633	3745256	3195800	2239107	1975686	1387991
	1997.67	1974.0	Core plug	146992	5767214	4815497	4655763	3566562	2860798	2311672	2014052	1499005
	2352.00	2336.6	Mud	163302	23893	8978	4633	3551	1774	1865	1595	1142

TABLE Ap7.3 Alkane fraction GC-MS amounts in ng/g (including m/e 183 ion)

Peak	Ion(s)	Compound
25nor28αβ	177	17α(H), 21β(H)-25,28,30-trisnorhopane
25nor29αβ	177	17α, 21β-25,30-bisnorhopane
25nor30αβ	177, 191	17α(H), 21β(H)-25-norhopane
25nor31αβ	177	17α, 21β, 22(R/S)-25-norhomohopane
19/3	191	C19H34 tricyclic terpane
20/3	191	C20H36 tricyclic terpane
21/3	191	C21H38 tricyclic terpane
22/3	191	C22H40 tricyclic terpane
23/3	191	C23H42 tricyclic terpane
24/3	191	C24H44 tricyclic terpane
25/3R	191	C25H46 tricyclic terpane
25/3S	191	C25H46 tricyclic terpane
24/4	191	C24H42 tetracyclic terpane
26/3R	191	C26H48 tricyclic terpane
26/3S	191	C26H48 tricyclic terpane
28/3R	191	C28H52 tricyclic terpane
28/3S	191	C28H52 tricyclic terpane
29/3R	191	C29H54 tricyclic terpane
29/3S	191	C29H54 tricyclic terpane
27Ts	191	18α(H)-22,29,30-trisnorneohopane
27Tm	191	17α(H)-22,29,30-trisnorhopane
30/3R	191	C30H56 tricyclic terpane
30/3S	191	C30H56 tricyclic terpane
28αβ	191	17α(H), 21β(H)-28,30-bisnorhopane
29αβ	191	17α(H), 21β(H)-30-norhopane
29Ts	191	18α(H)-30-norneohopane
30d	191	15α-methyl-17α(H)-27-norhopane (diahopane)
29βα	191	17β(H), 21α(H)-30-norhopane (normoretane)
30O	191	18α(H)-oleanane
30αβ	191	17α(H), 21β(H)-hopane
30βα	191	17β(H), 21α(H)-hopane (moretane)
31αβS	191	17α(H), 21β(H), 22(S)-homohopane
31αβR	191	17α(H), 21β(H), 22(R)-homohopane
30G	191	Gammacerane
31βα	191	17β(H), 21α(H)-homohopane
32αβS	191	17α(H), 21β(H), 22(S)-bishomohopane
32αβR	191	17α(H), 21β(H), 22(R)-bishomohopane
33αβS	191	17α(H), 21β(H), 22(S)-trishomohopane

Peak	Ion(s)	Compound
33αβR	191	17α(H), 21β(H), 22(R)-trishomohopane
34αβS	191	17α(H), 21β(H), 22(S)-tetrakishomohopane
34αβR	191	17α(H), 21β(H), 22(R)-tetrakishomohopane
35αβS	191	17α(H), 21β(H), 22(S)-pentakishomohopane
35αβR	191	17α(H), 21β(H), 22(R)-pentakishomohopane
21αα	217	C21-5α(H), 14α(H), 17α(H)-pregnane
21ββ	217	C21-5α(H), 14β(H), 17β(H)-pregnane
22αα	217	C22-5α(H), 14α(H), 17α(H)-pregnane
22ββ	217	C22-5α(H), 14β(H), 17β(H)-pregnane
27dβS	217	13β(H), 17α(H), 20(S)-cholestane (diasterane)
27dβR	217	13β(H), 17α(H), 20(R)-cholestane (diasterane)
27dαR	217	13α(H), 17β(H), 20(R)-cholestane (diasterane)
27dαS	217	13α(H), 17β(H), 20(S)-cholestane (diasterane)
28dβS	217	24-methyl-13β(H), 17α(H), 20(S)-cholestane (diasterane)
28dβR	217	24-methyl-13β(H), 17α(H), 20(R)-cholestane (diasterane)
28dαR	217	24-methyl-13α(H), 17β(H), 20(R)-cholestane (diasterane)
27ααS	217	5α(H), 14α(H), 17α(H), 20(S)-cholestane
27ββR	217, 218	5α(H), 14β(H), 17β(H), 20(R)-cholestane
29dβS	217	24-ethyl-13β(H), 17α(H), 20(S)-cholestane (diasterane)
27ββS	217, 218	5α(H), 14β(H), 17β(H), 20(S)-cholestane
28dαS	217	24-methyl-13α(H), 17β(H), 20(S)-cholestane (diasterane)
27ααR	217	5α(H), 14α(H), 17α(H), 20(R)-cholestane
29dβR	217	24-ethyl-13β(H), 17α(H), 20(R)-cholestane (diasterane)
29dαR	217	24-ethyl-13α(H), 17β(H), 20(R)-cholestane (diasterane)
28ααS	217	24-methyl-5α(H), 14α(H), 17α(H), 20(S)-cholestane
29dαS	217	24-ethyl-13α(H), 17β(H), 20(S)-cholestane (diasterane)
28ββR	217, 218	24-methyl-5α(H), 14β(H), 17β(H), 20(R)-cholestane
28ββS	217, 218	24-methyl-5α(H), 14β(H), 17β(H), 20(S)-cholestane
28ααR	217	24-methyl-5α(H), 14α(H), 17α(H), 20(R)-cholestane
29ααS	217	24-ethyl-5α(H), 14α(H), 17α(H), 20(S)-cholestane
29ββR	217, 218	24-ethyl-5α(H), 14β(H), 17β(H), 20(R)-cholestane
29ββS	217, 218	24-ethyl-5α(H), 14β(H), 17β(H), 20(S)-cholestane
29ααR	217	24-ethyl-5α(H), 14α(H), 17α(H), 20(R)-cholestane
30ααS	217	24-propyl-5α(H), 14α(H), 17α(H), 20(S)-cholestane
30ββR	217, 218	24-propyl-5α(H), 14β(H), 17β(H), 20(R)-cholestane
30ββS	217, 218	24-propyl-5α(H), 14β(H), 17β(H), 20(S)-cholestane
30ααR	217	24-propyl-5α(H), 14α(H), 17α(H), 20(R)-cholestane

TABLE Ap7KEY Abbreviations of alkane fraction biomarkers



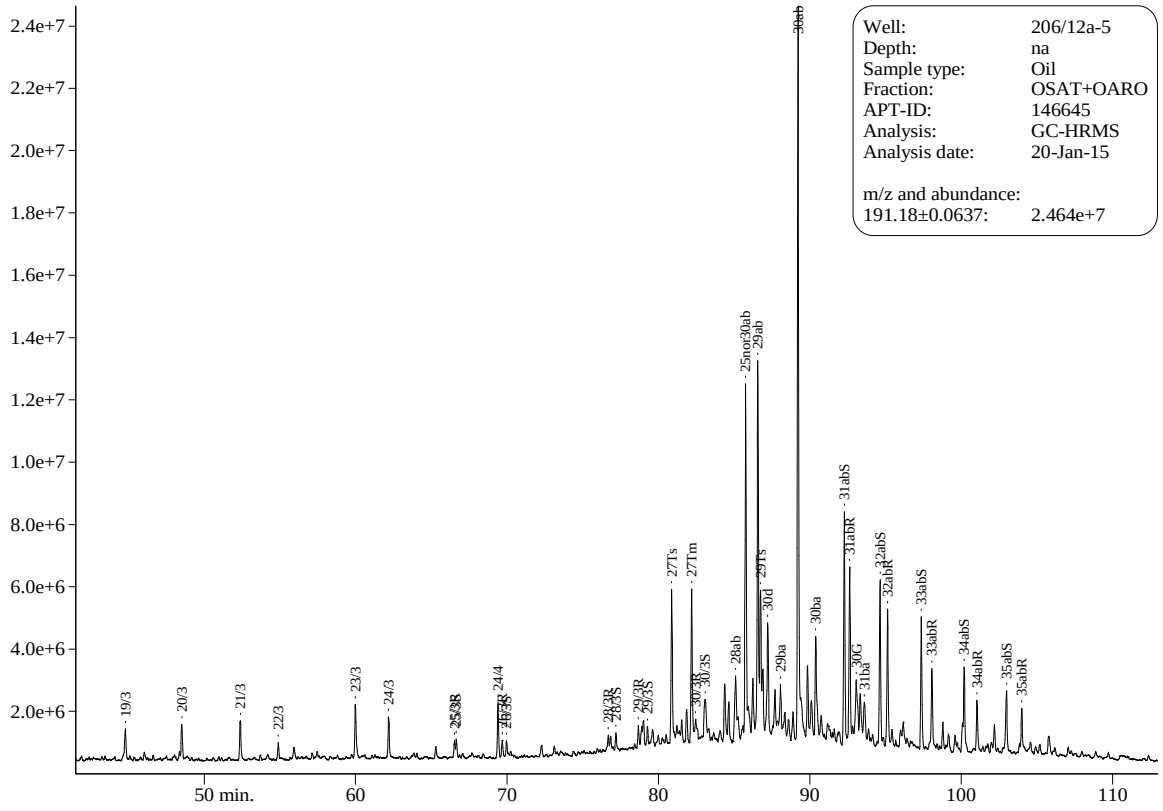
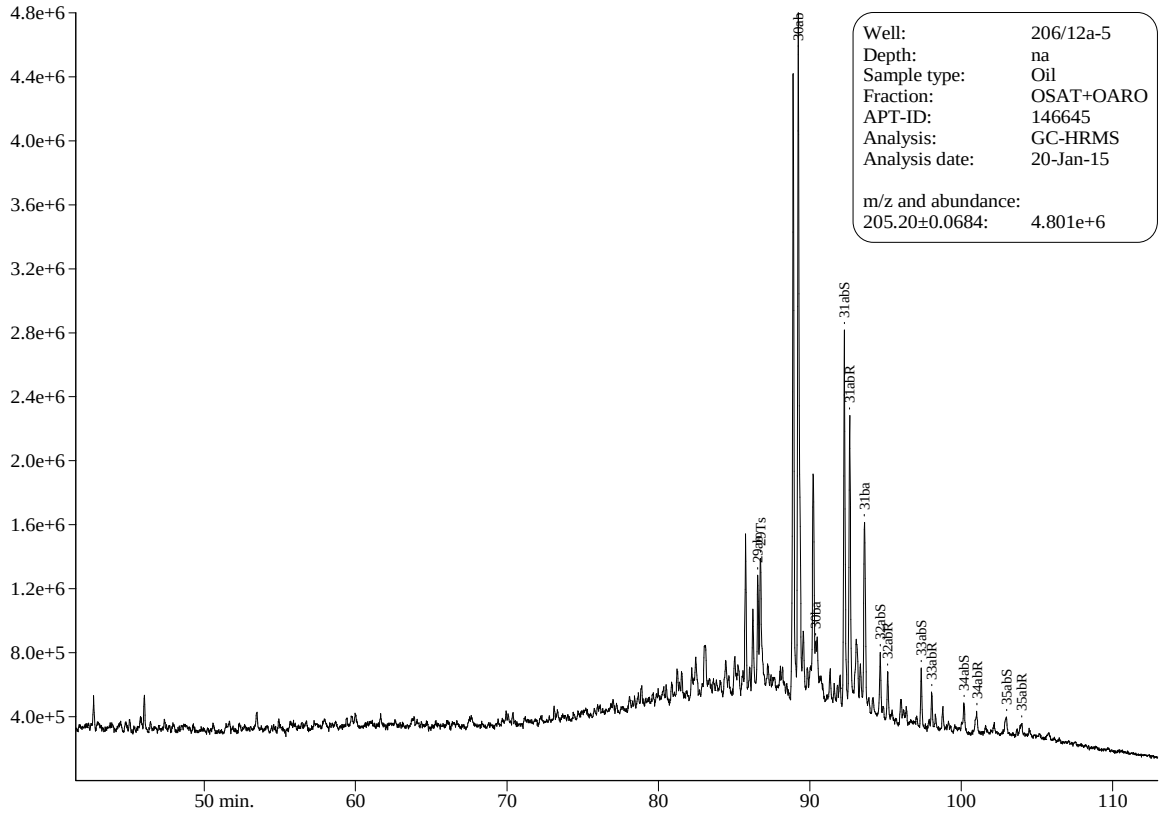


FIGURE Ap7.1 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72))



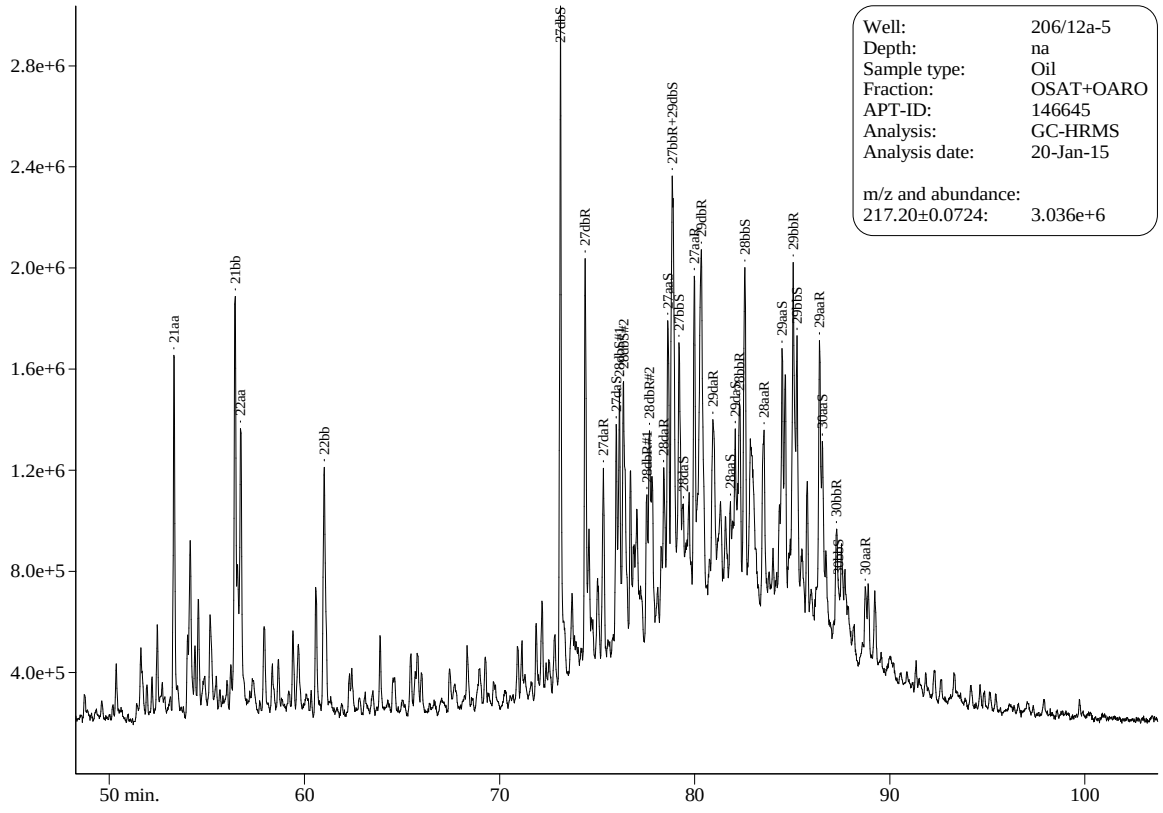
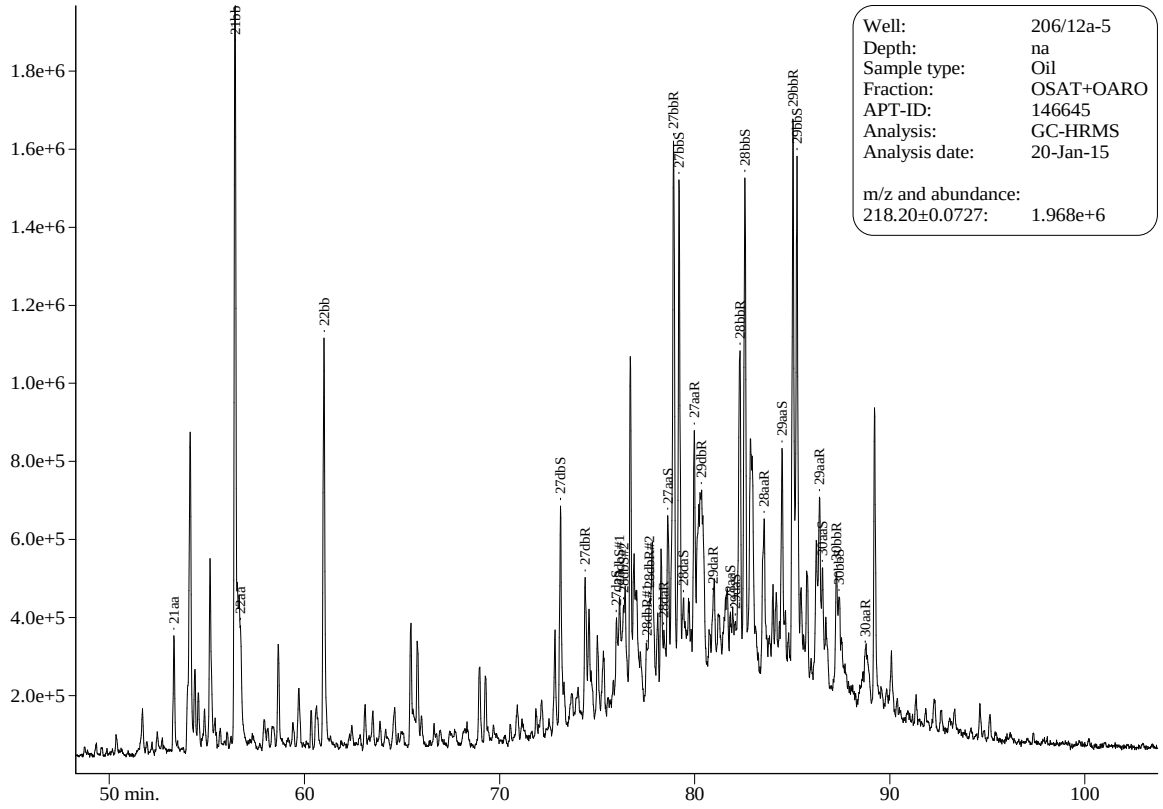


FIGURE Ap7.1 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72))



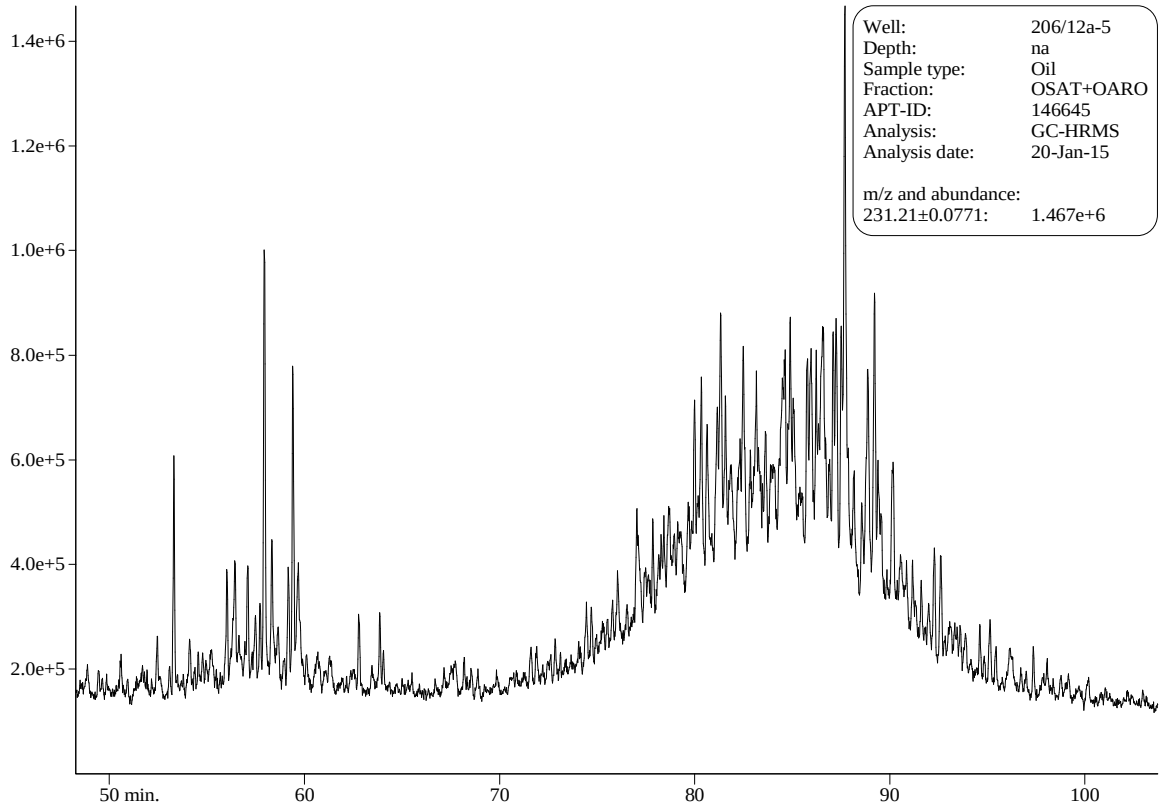
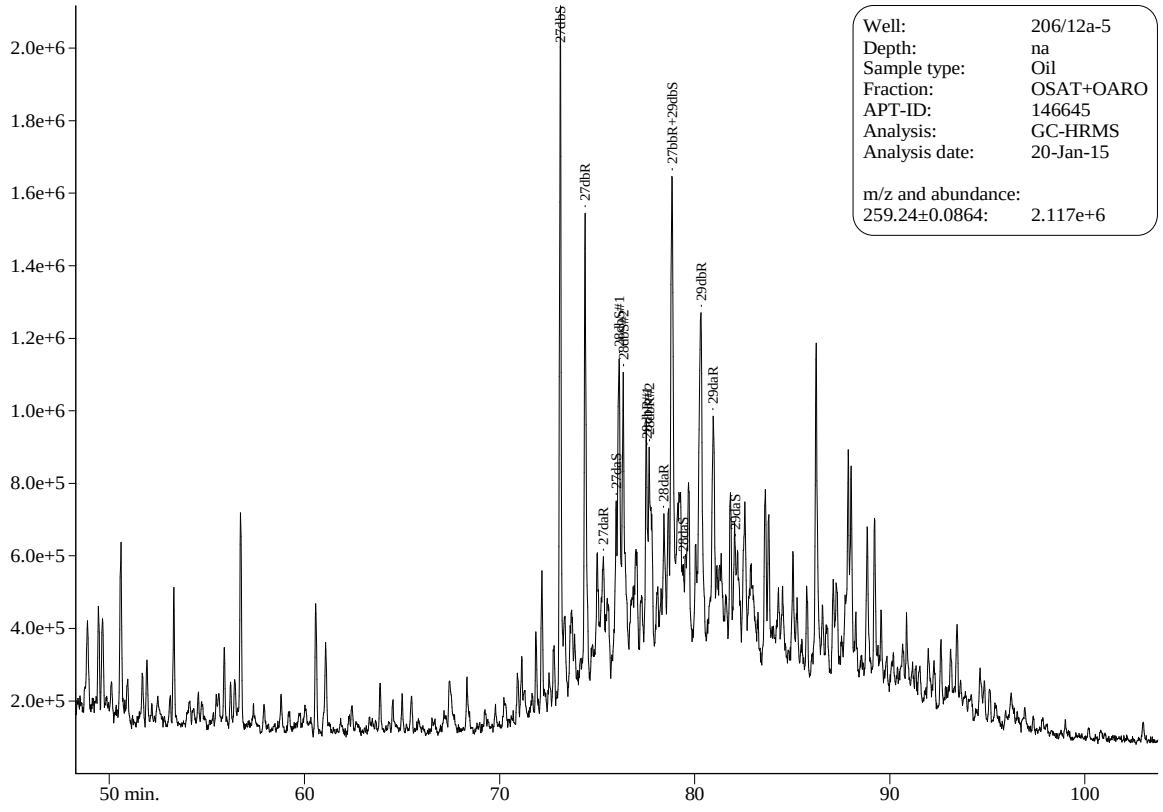


FIGURE Ap7.1 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72))



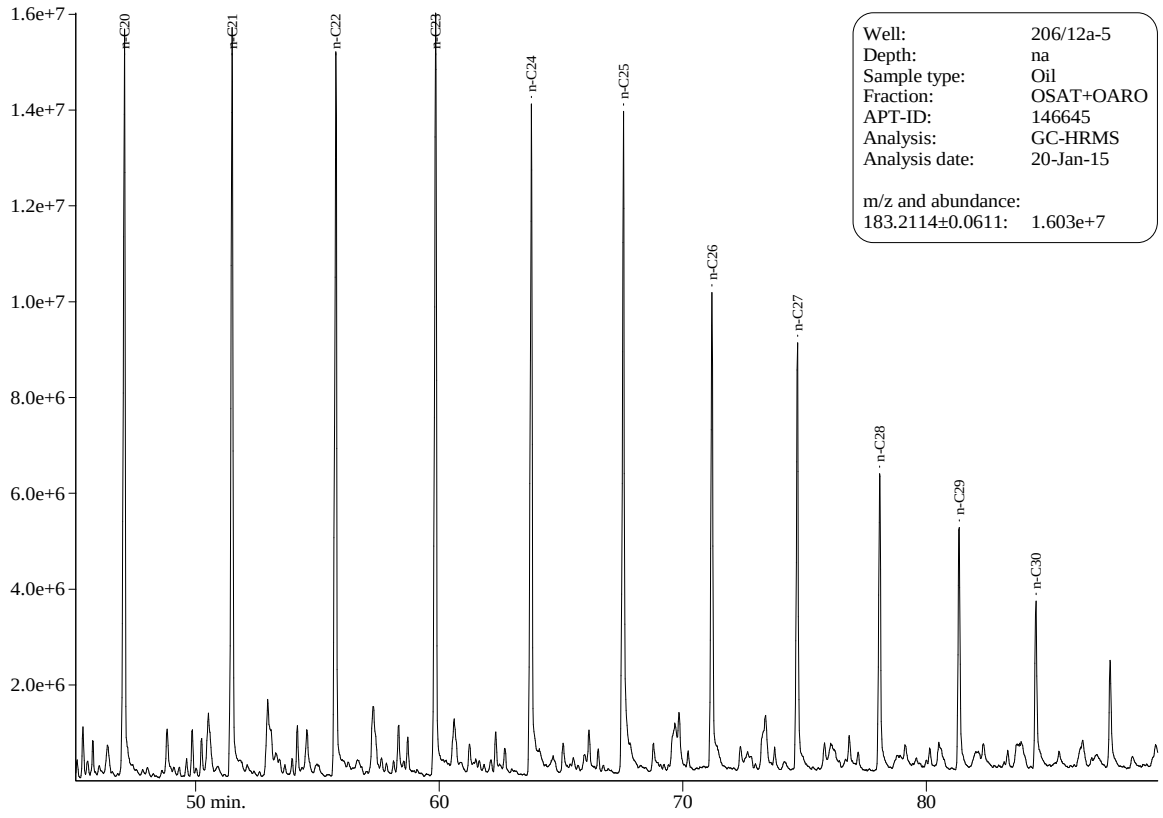
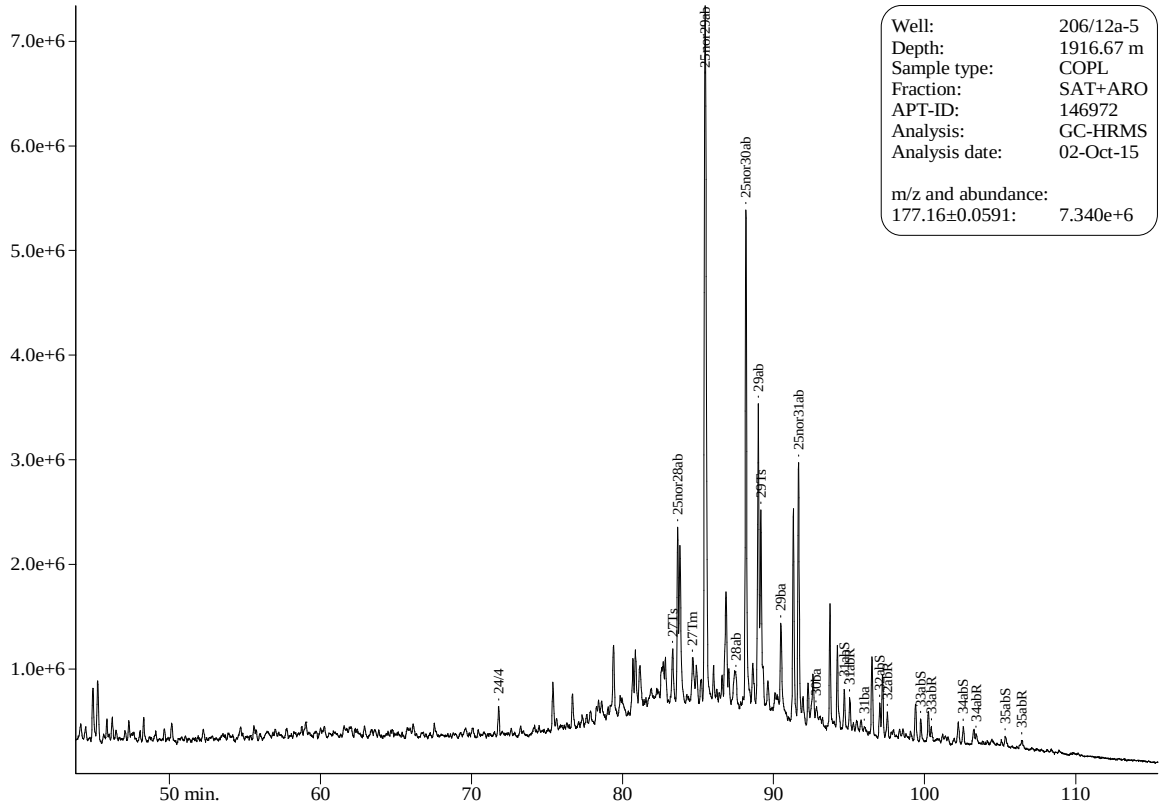


FIGURE Ap7.1 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72))



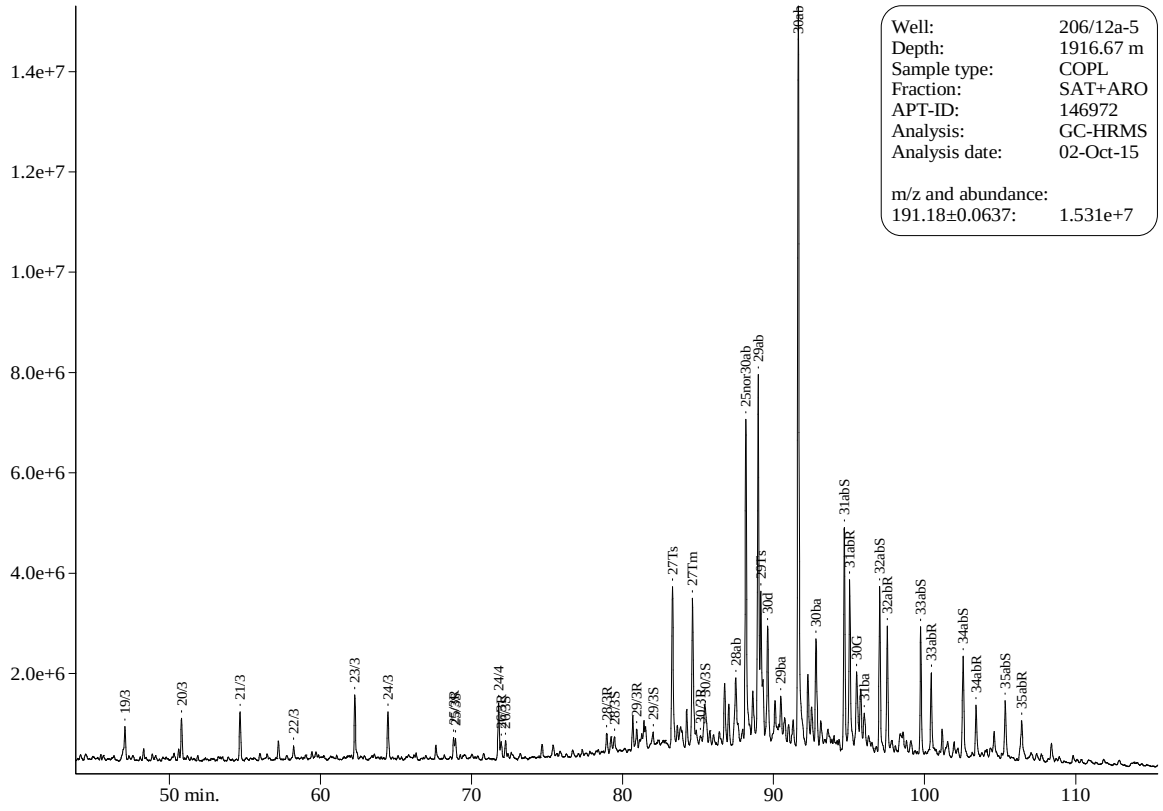
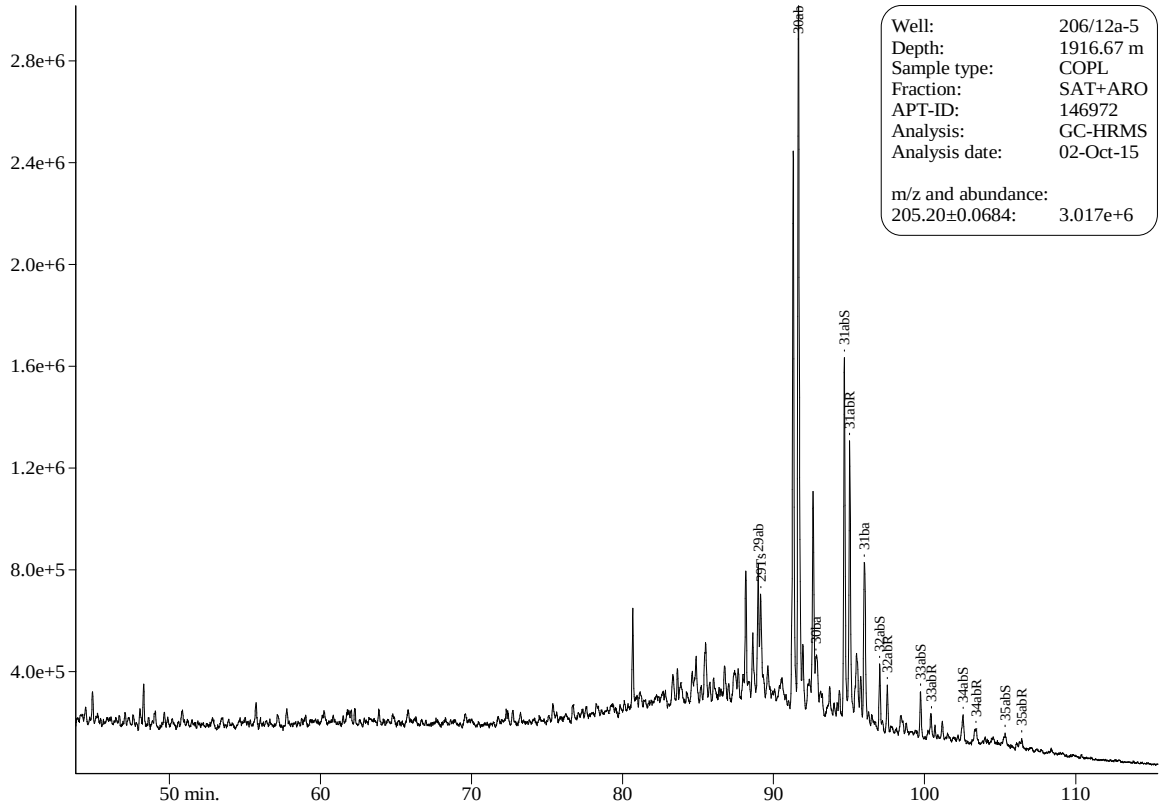


FIGURE Ap7.2 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))



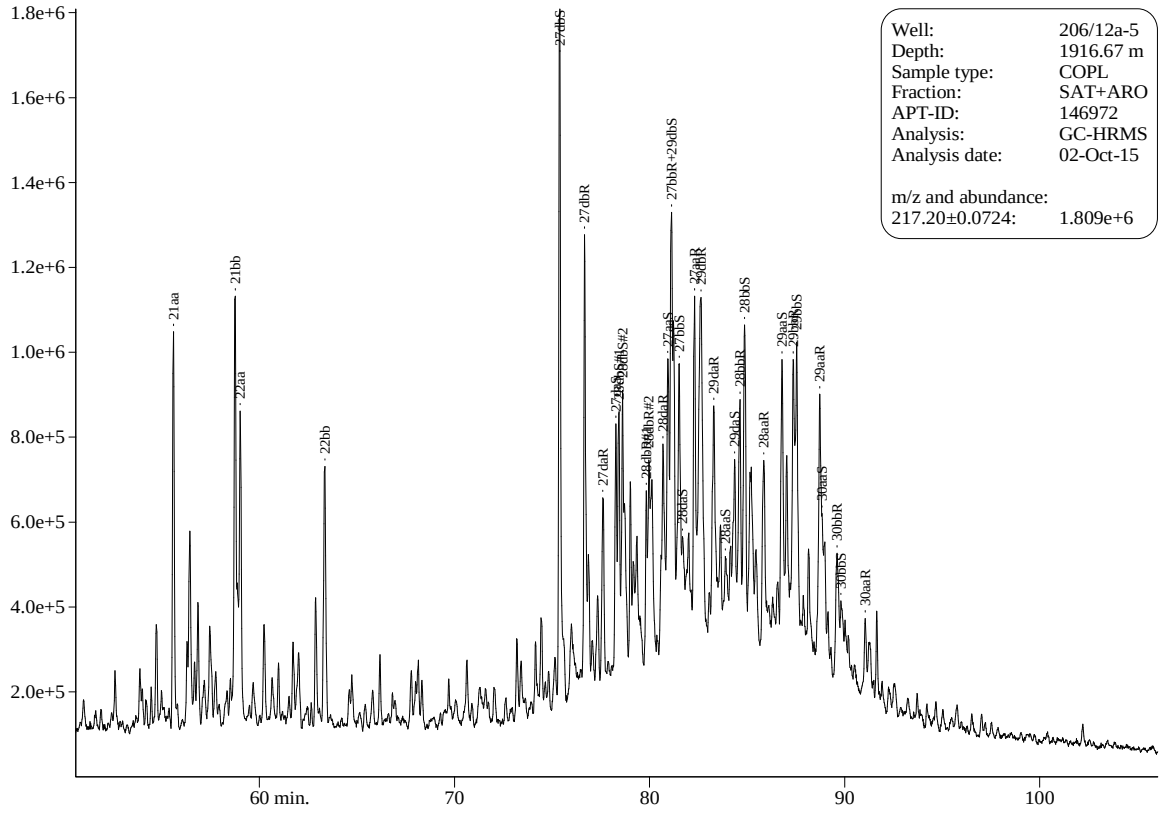
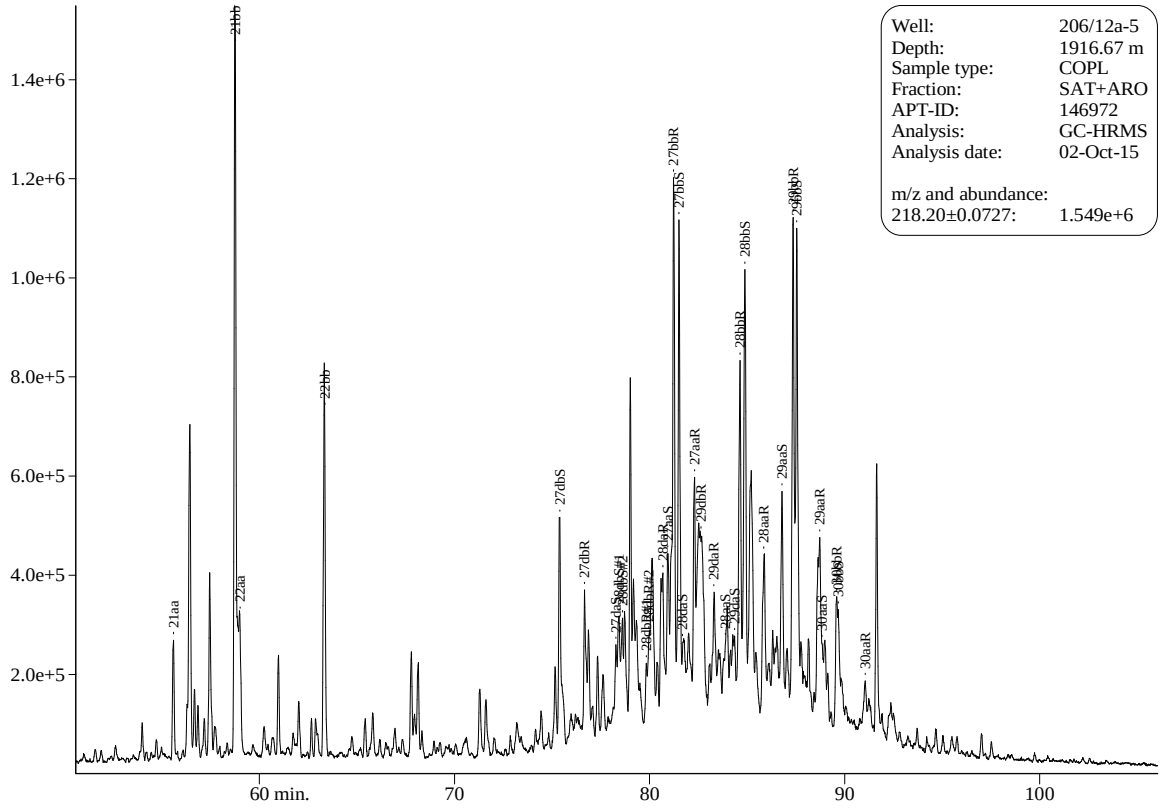


FIGURE Ap7.2 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))



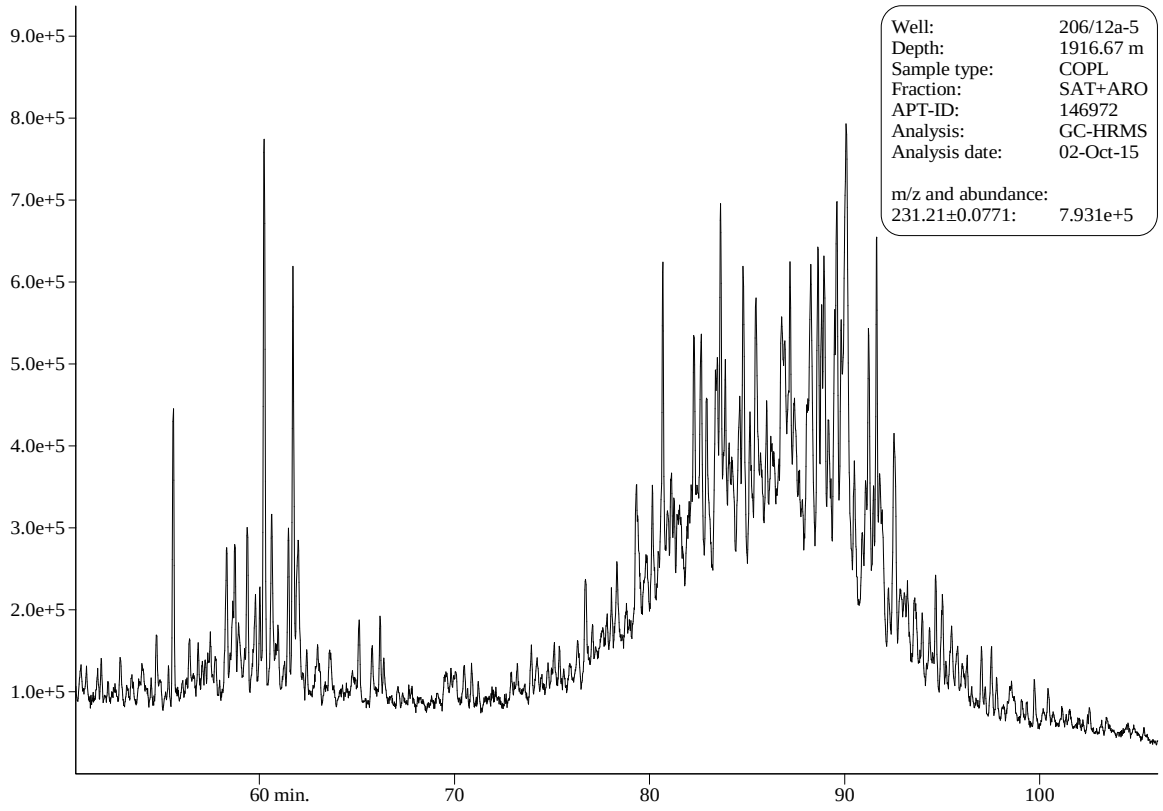
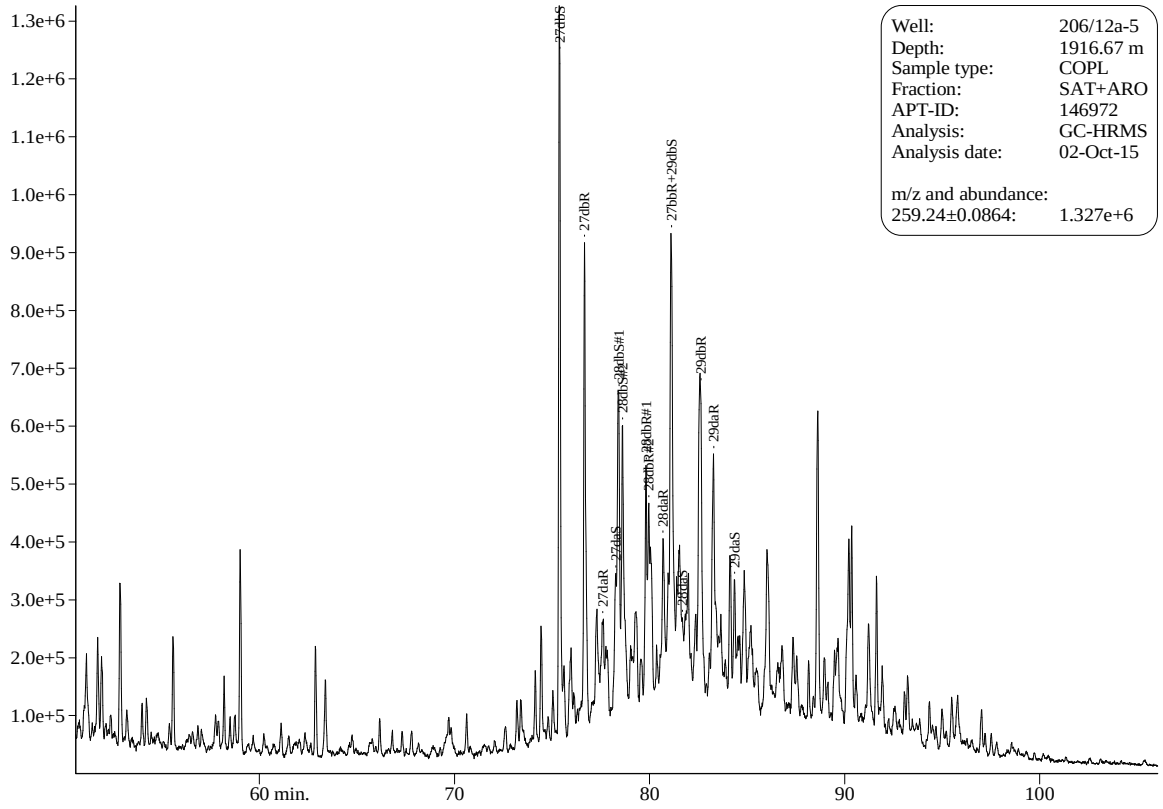


FIGURE Ap7.2 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))



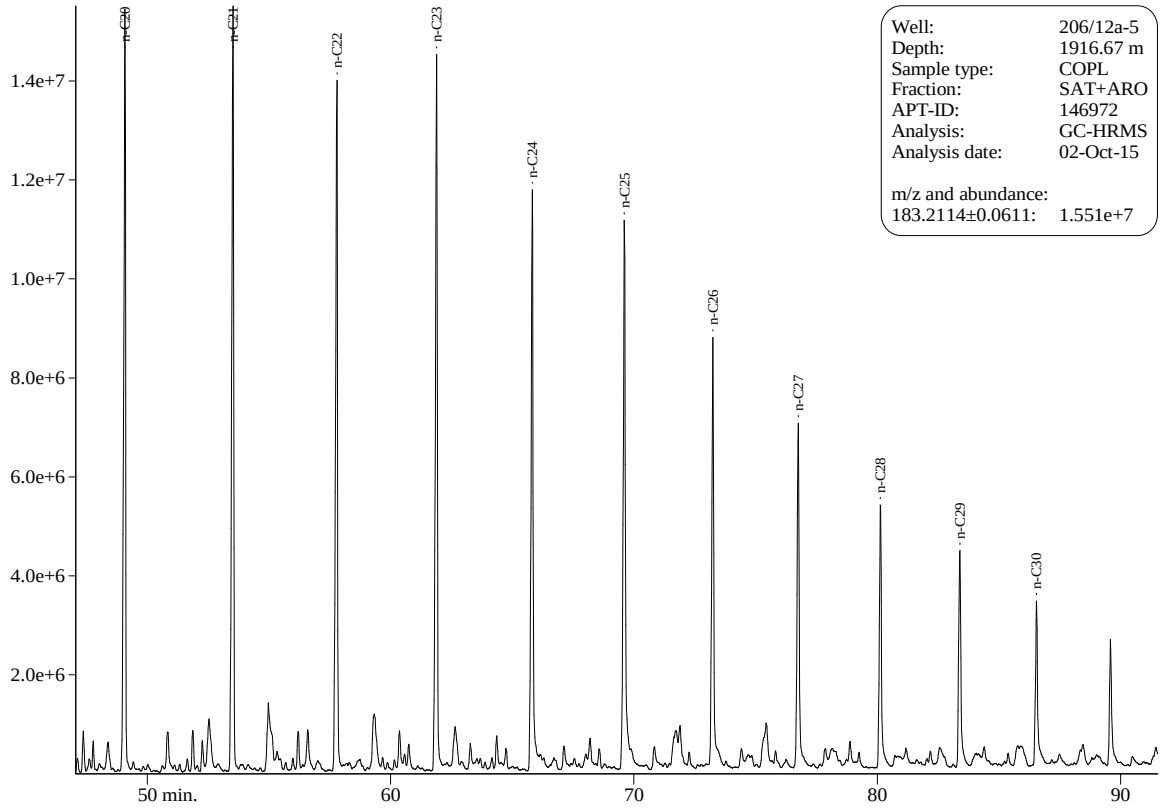
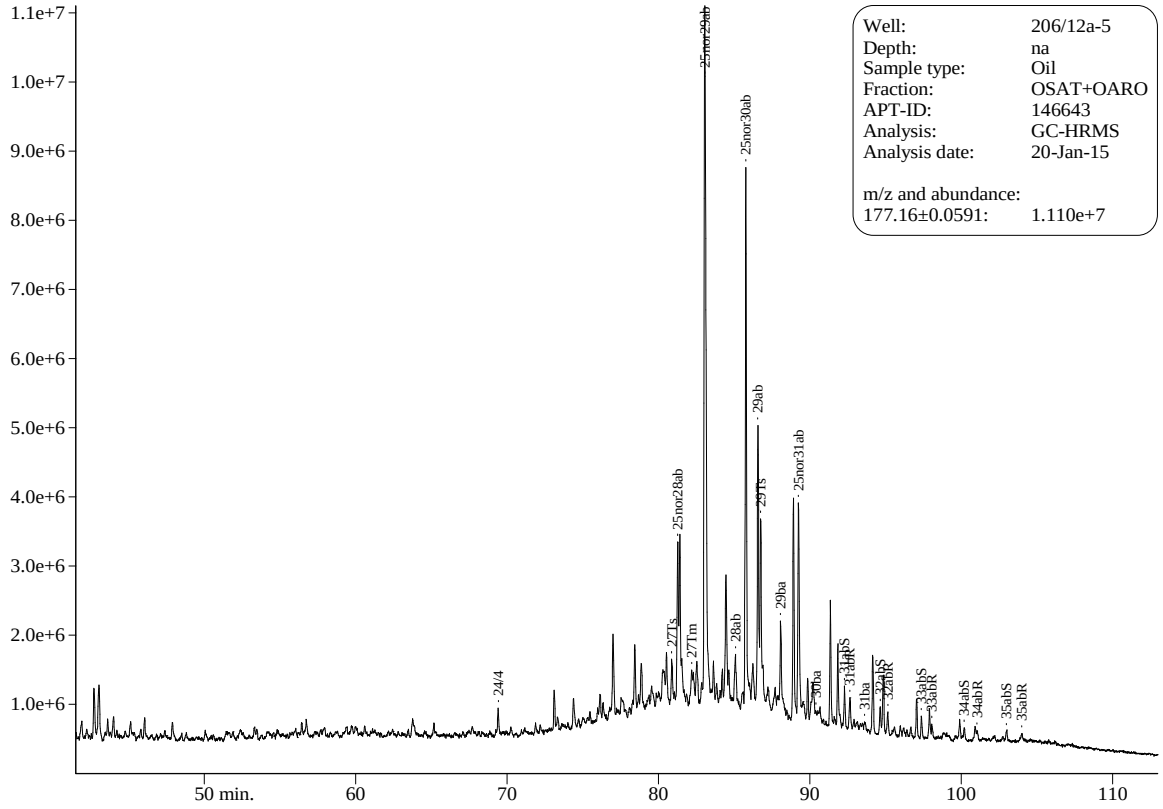


FIGURE Ap7.2 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))



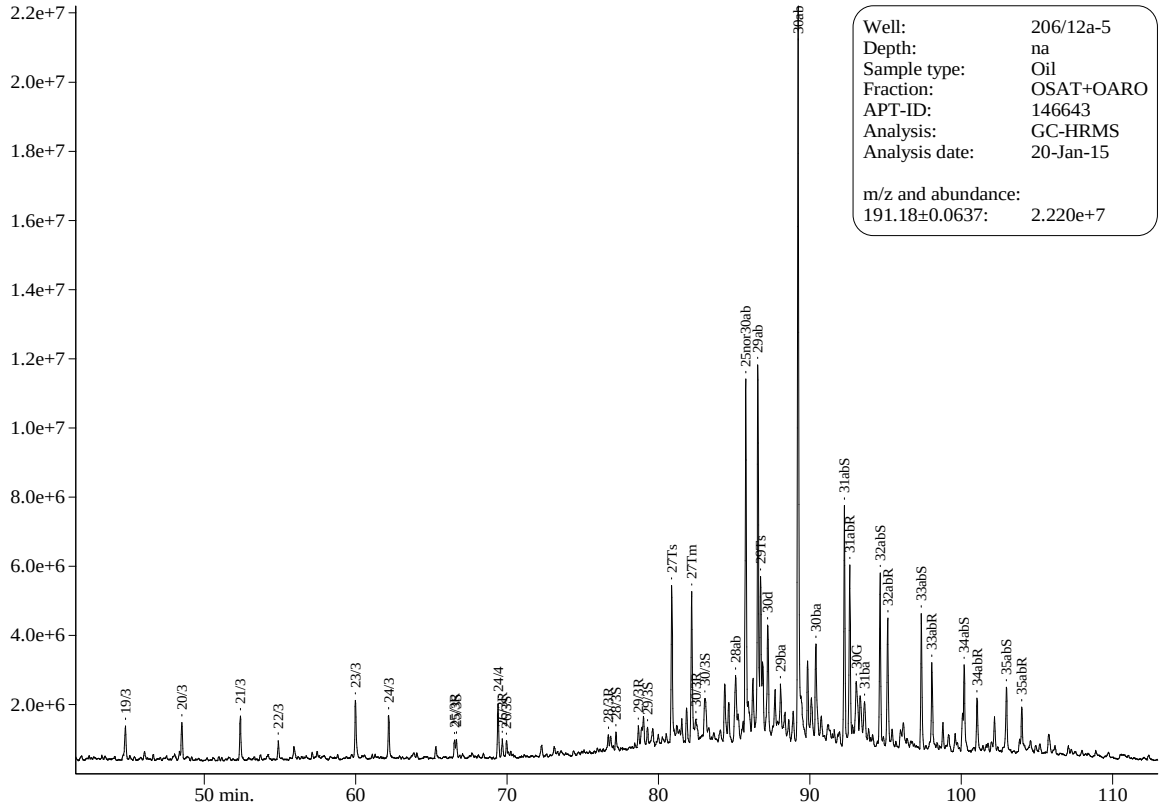
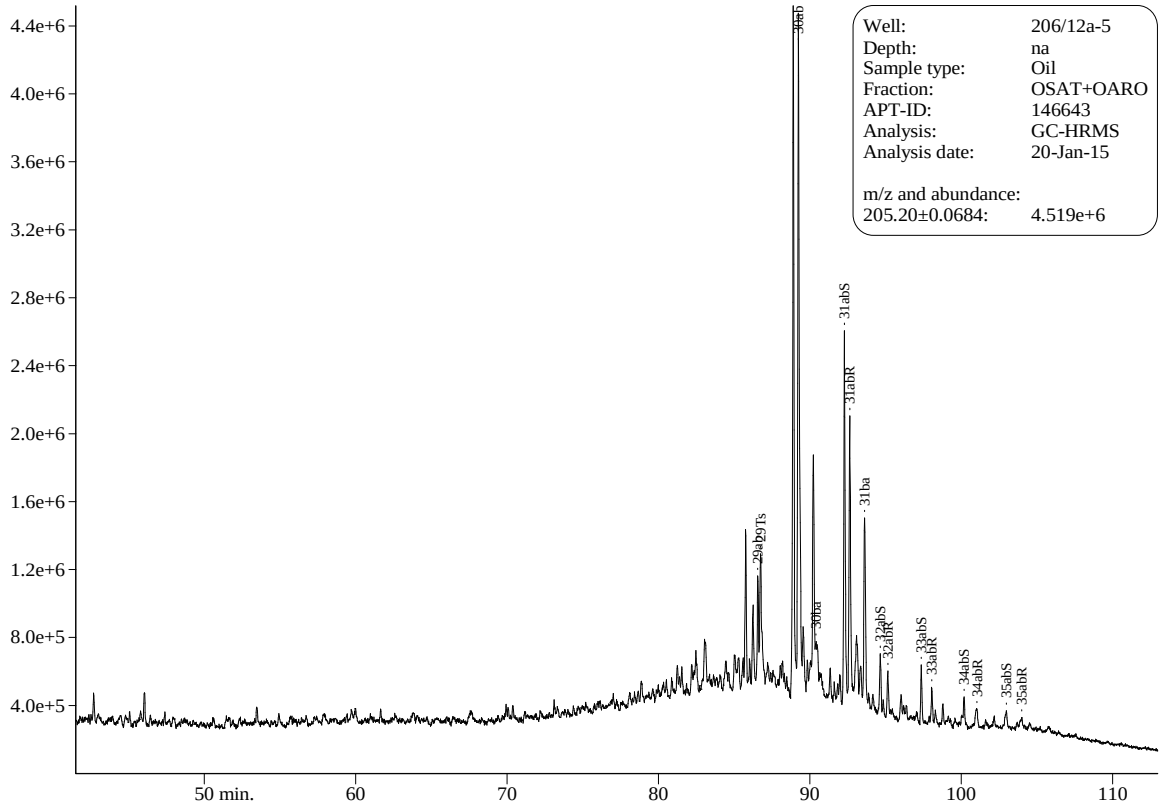


FIGURE Ap7.3 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret)



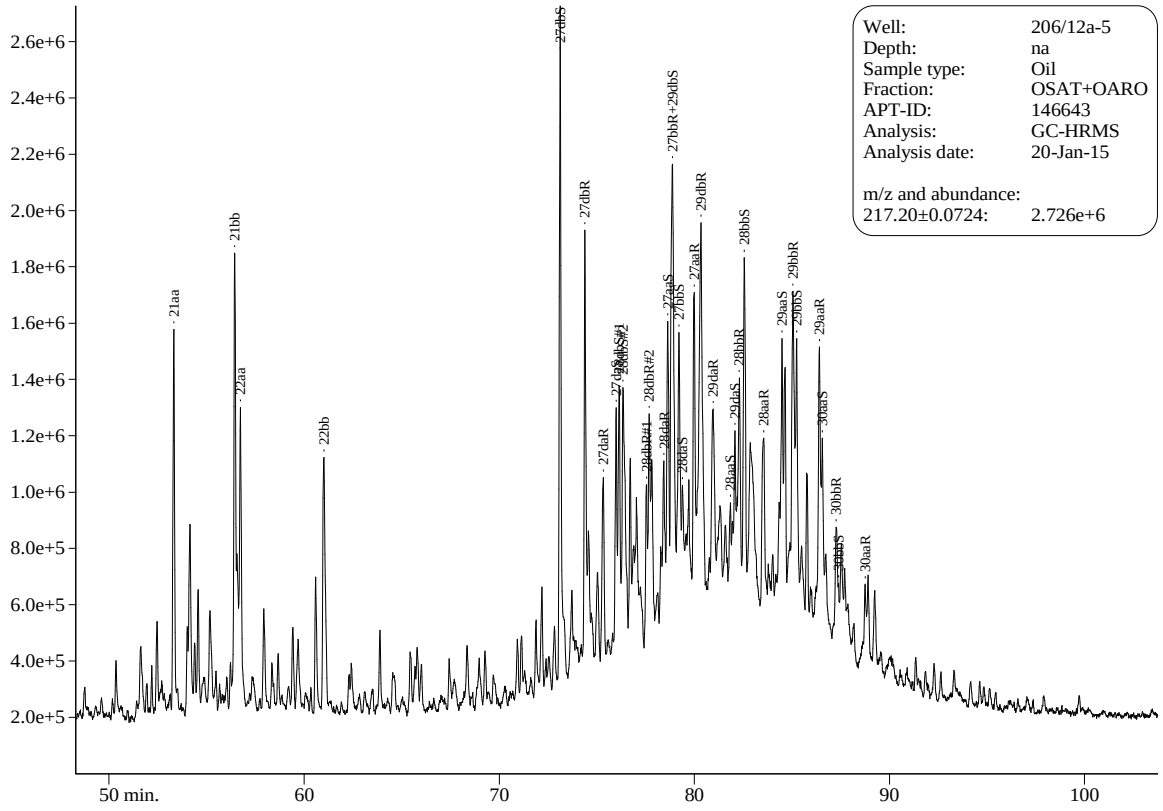
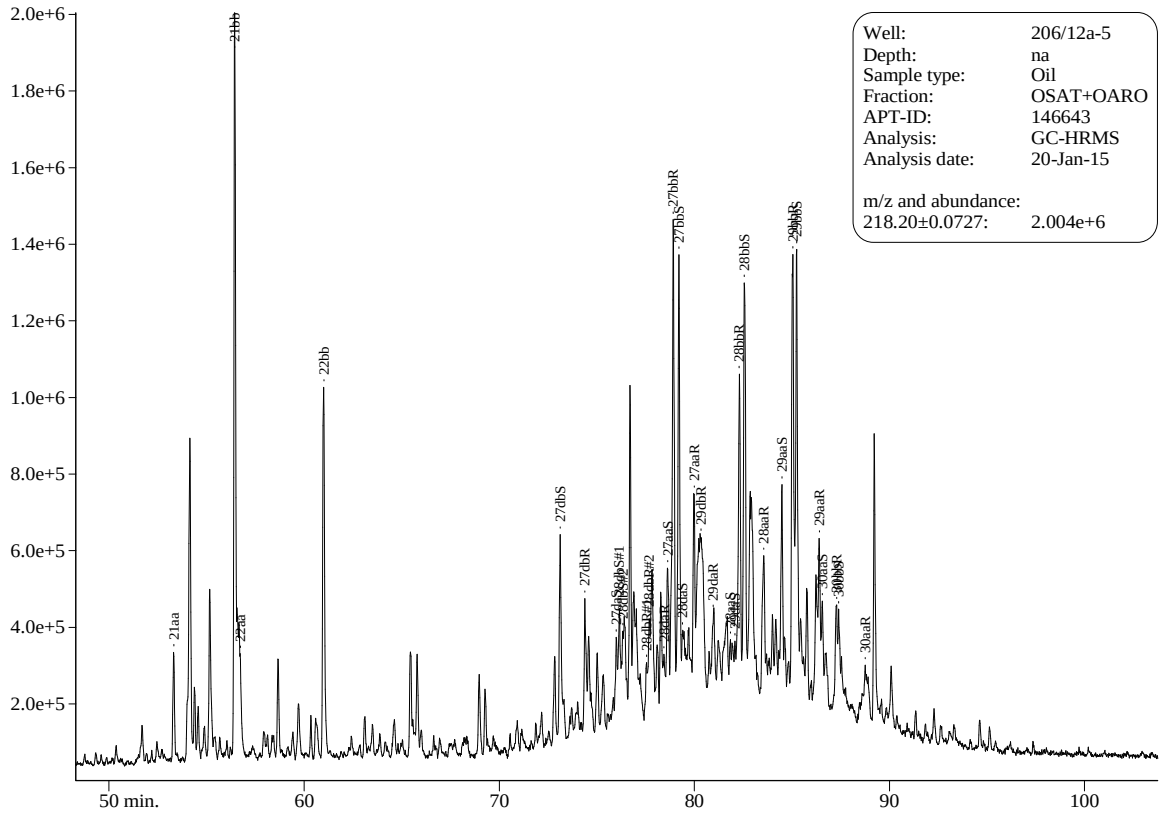


FIGURE Ap7.3 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret)



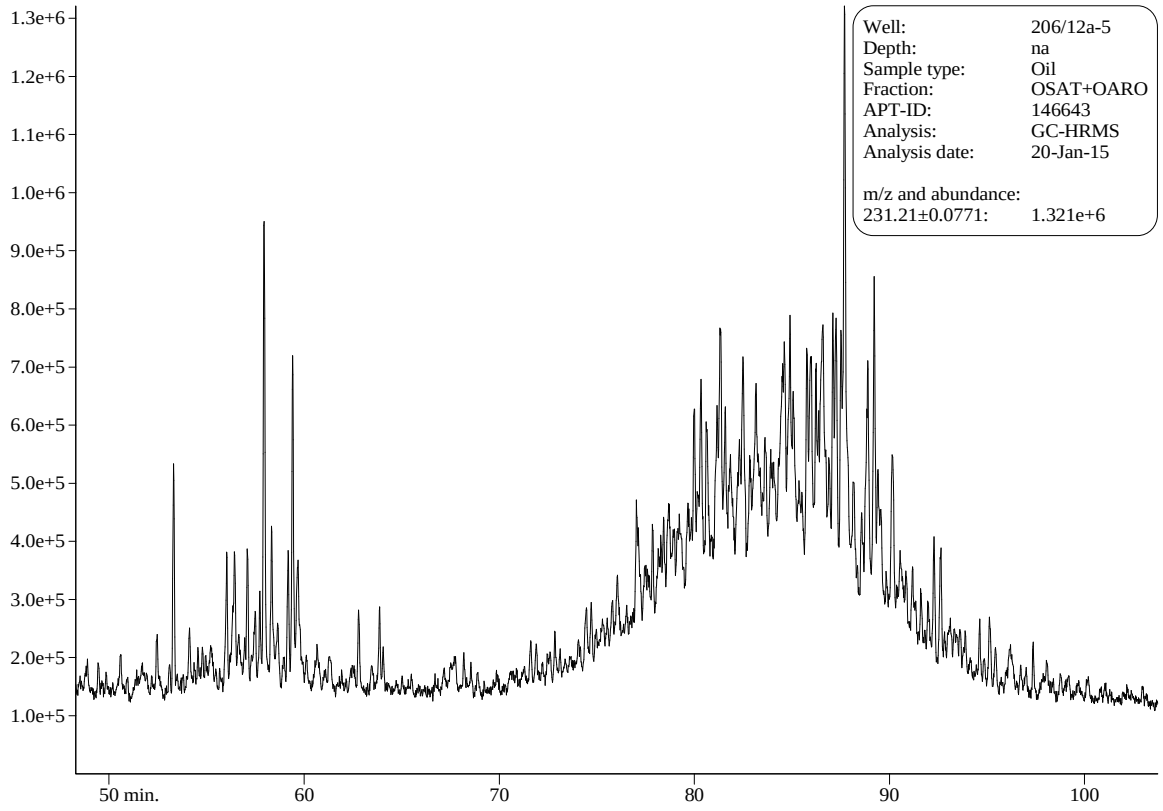
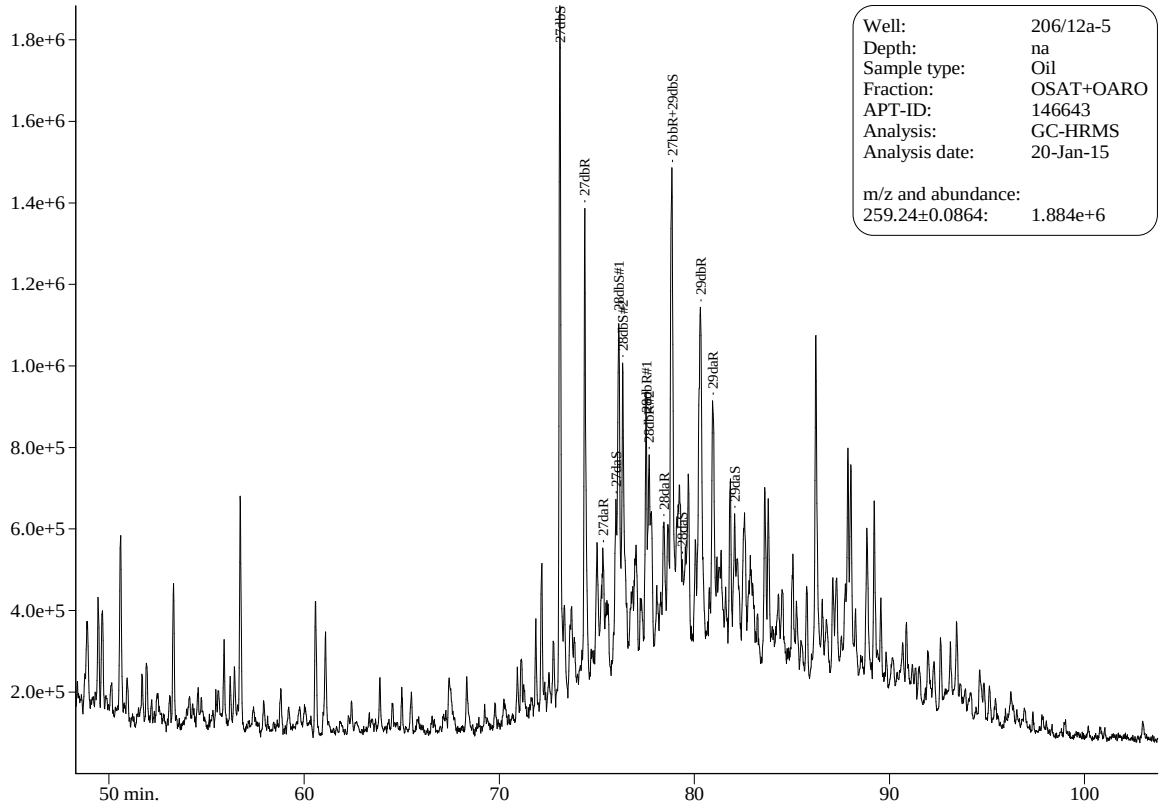


FIGURE Ap7.3 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret)



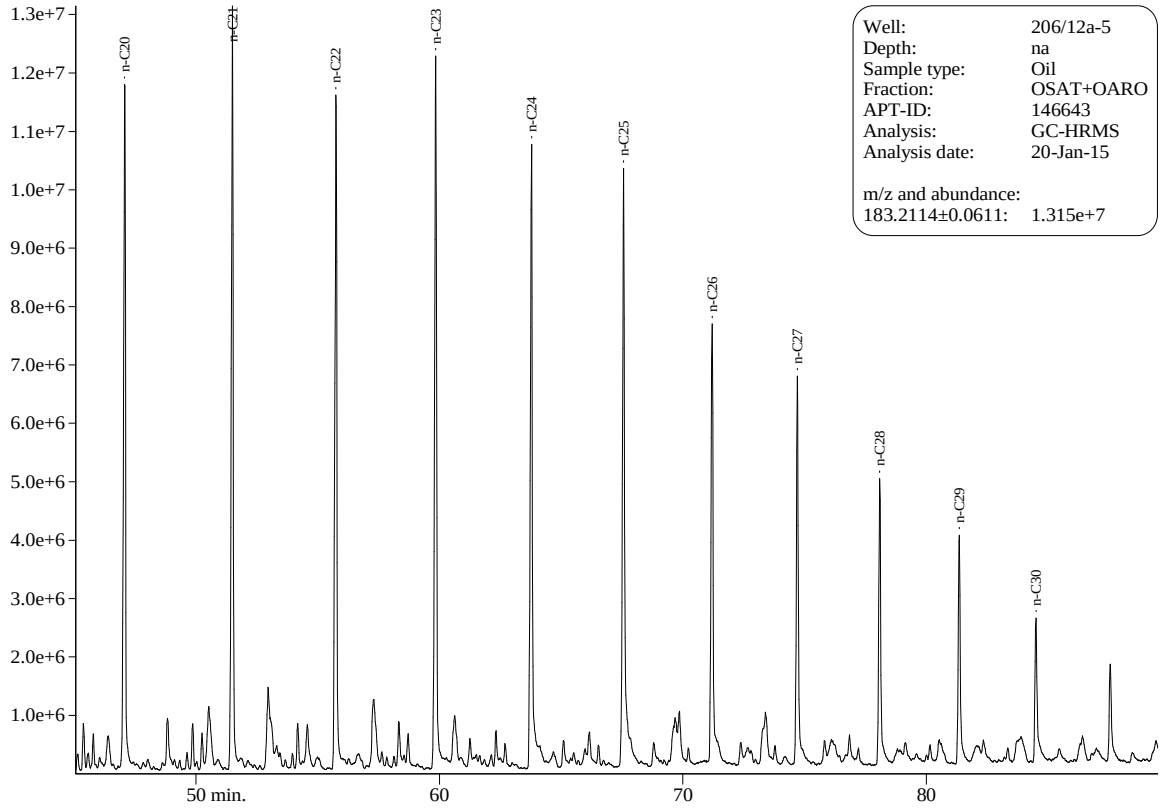
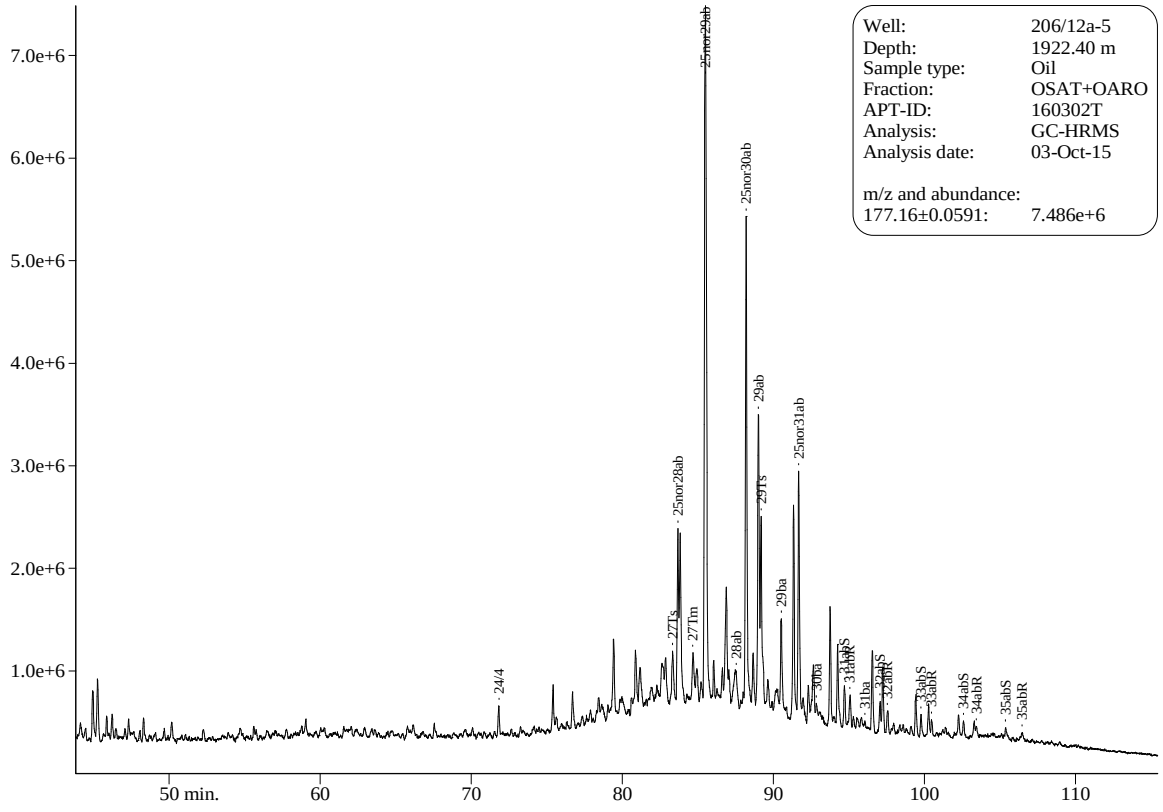


FIGURE Ap7.3 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret)



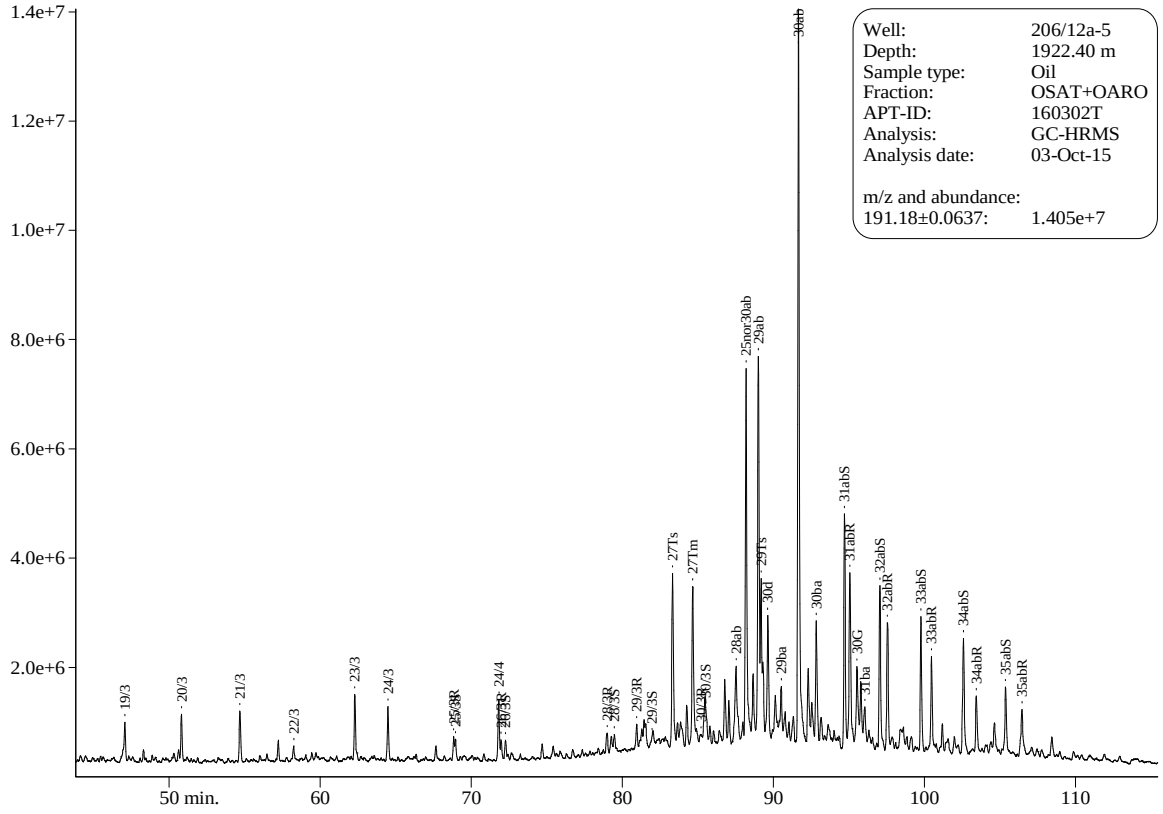
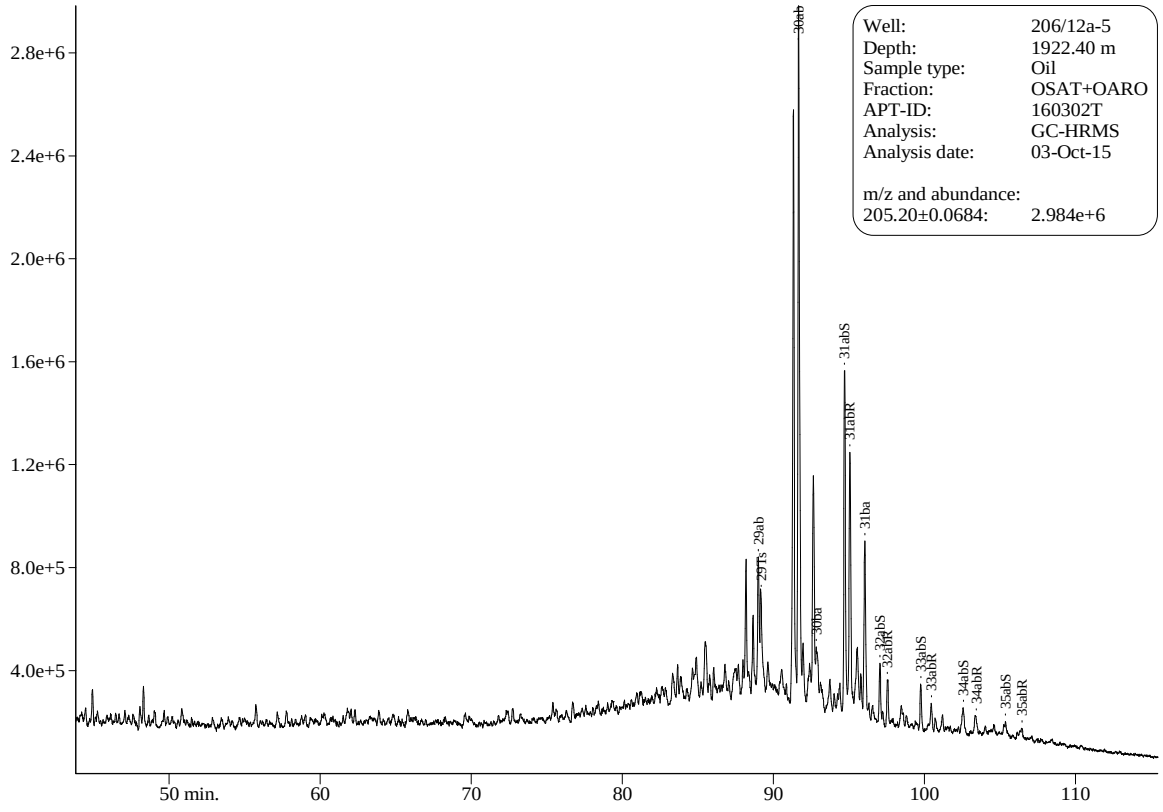


FIGURE Ap7.4 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1922.4m (K30 Sand Cret)



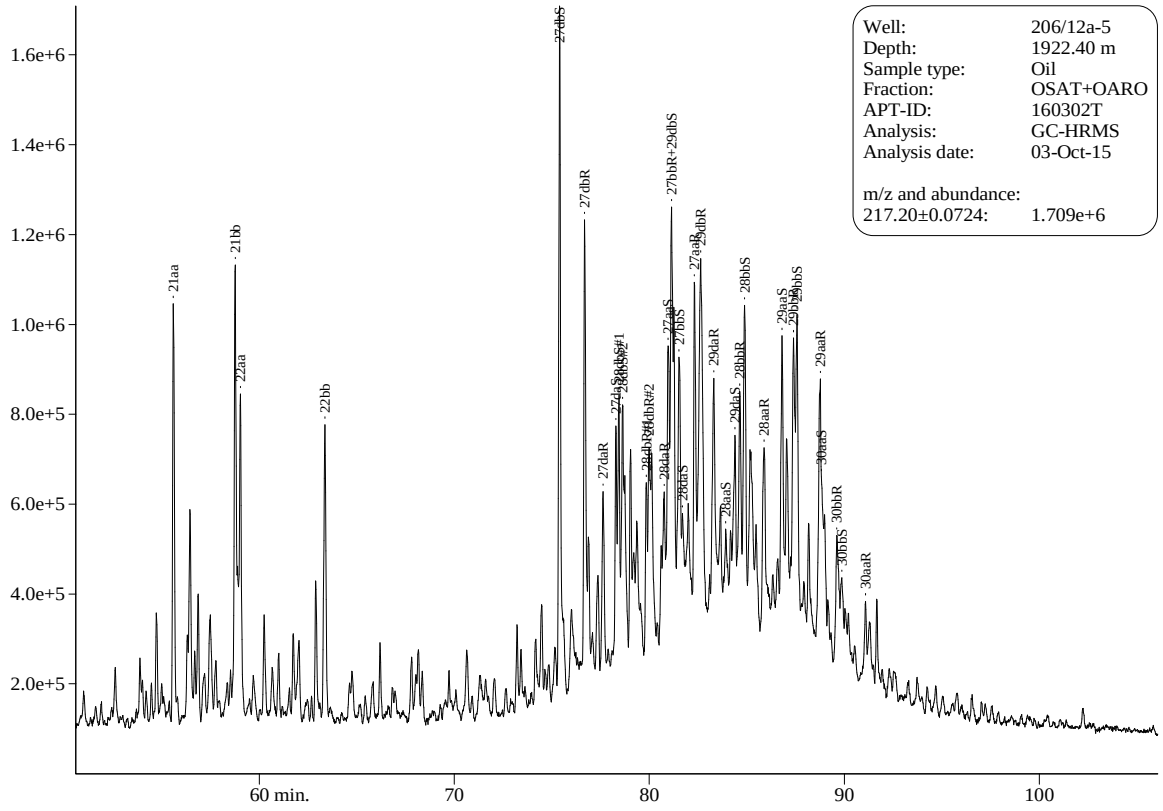
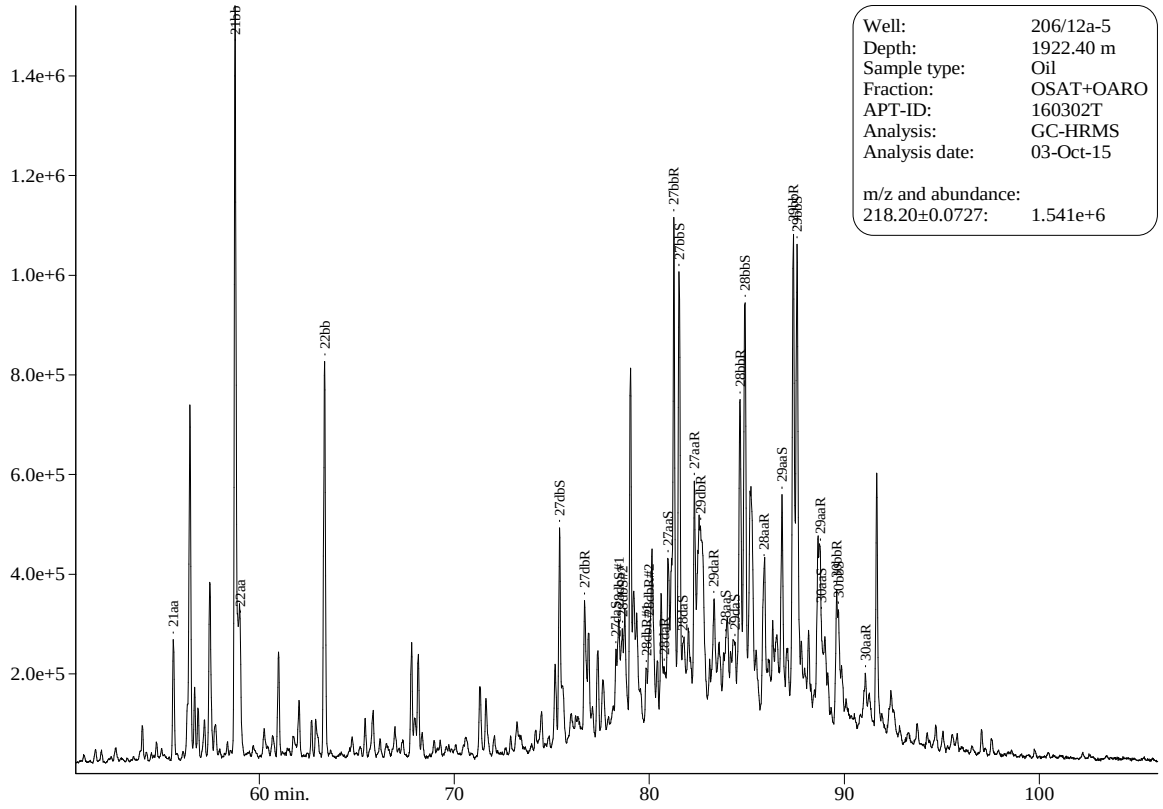


FIGURE Ap7.4 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1922.4m (K30 Sand Cret)



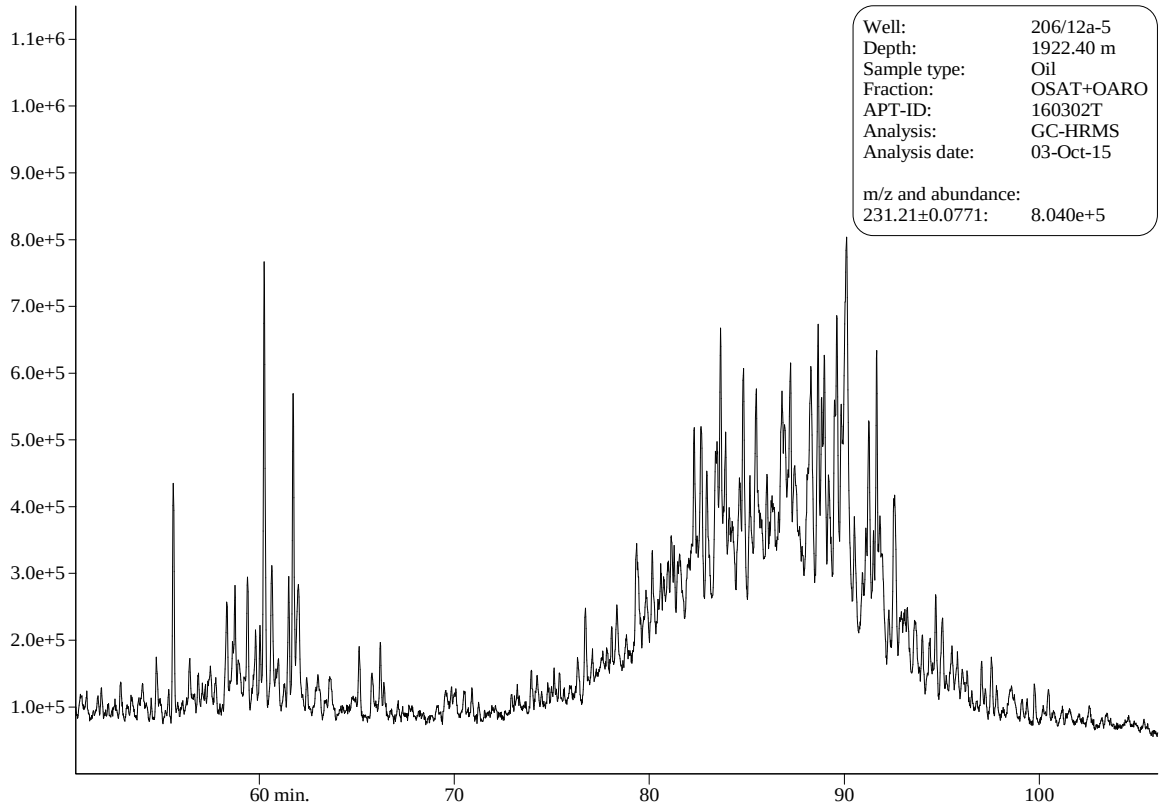
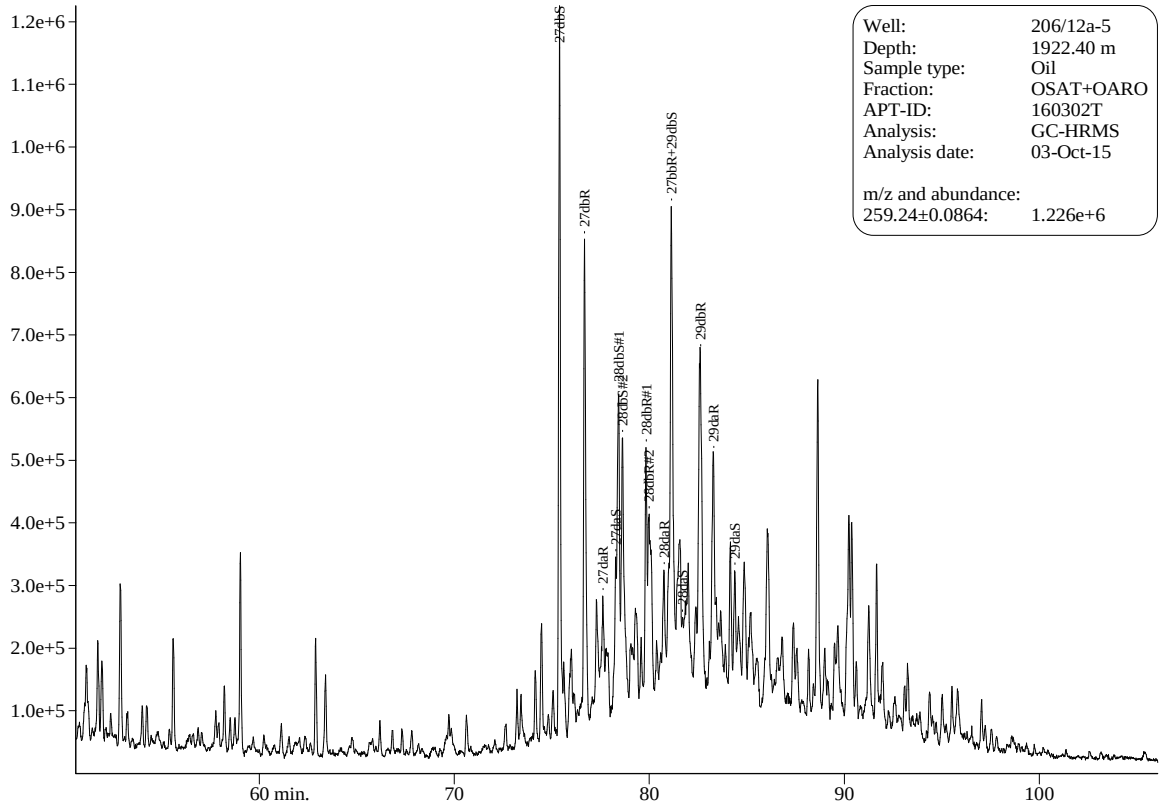


FIGURE Ap7.4 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1922.4m (K30 Sand Cret)



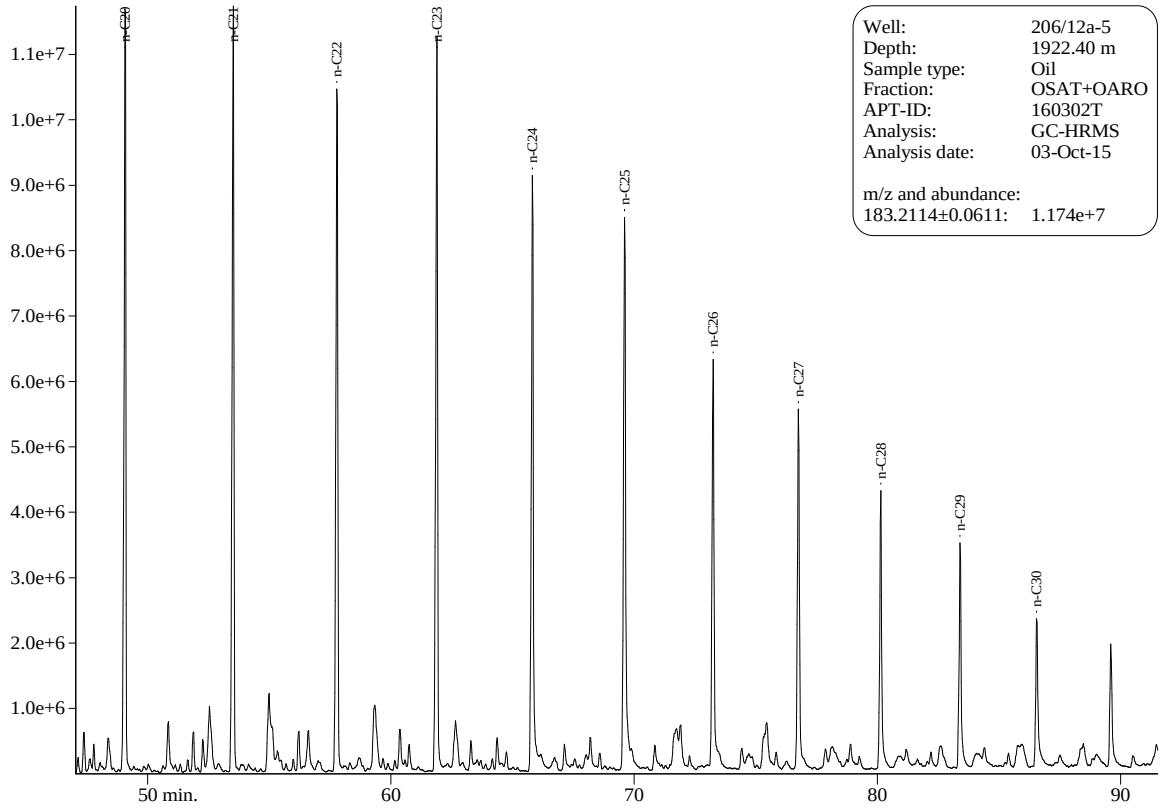
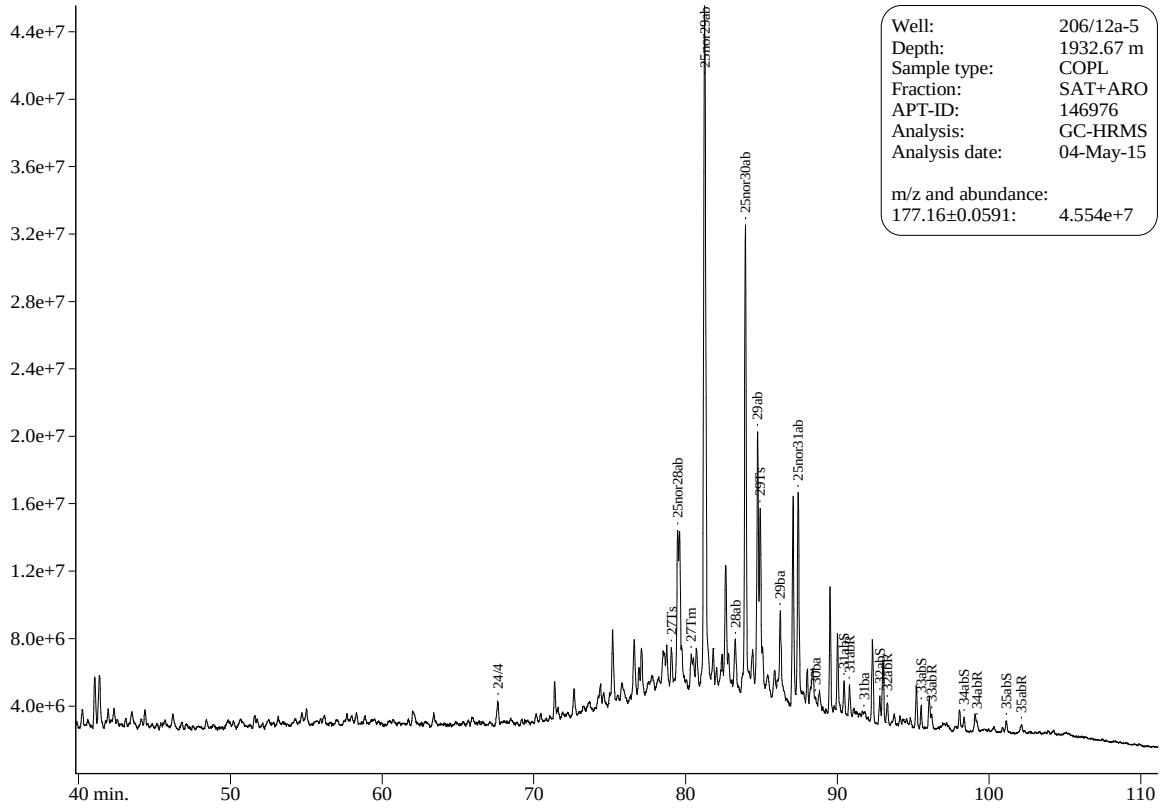


FIGURE Ap7.4 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1922.4m (K30 Sand Cret)



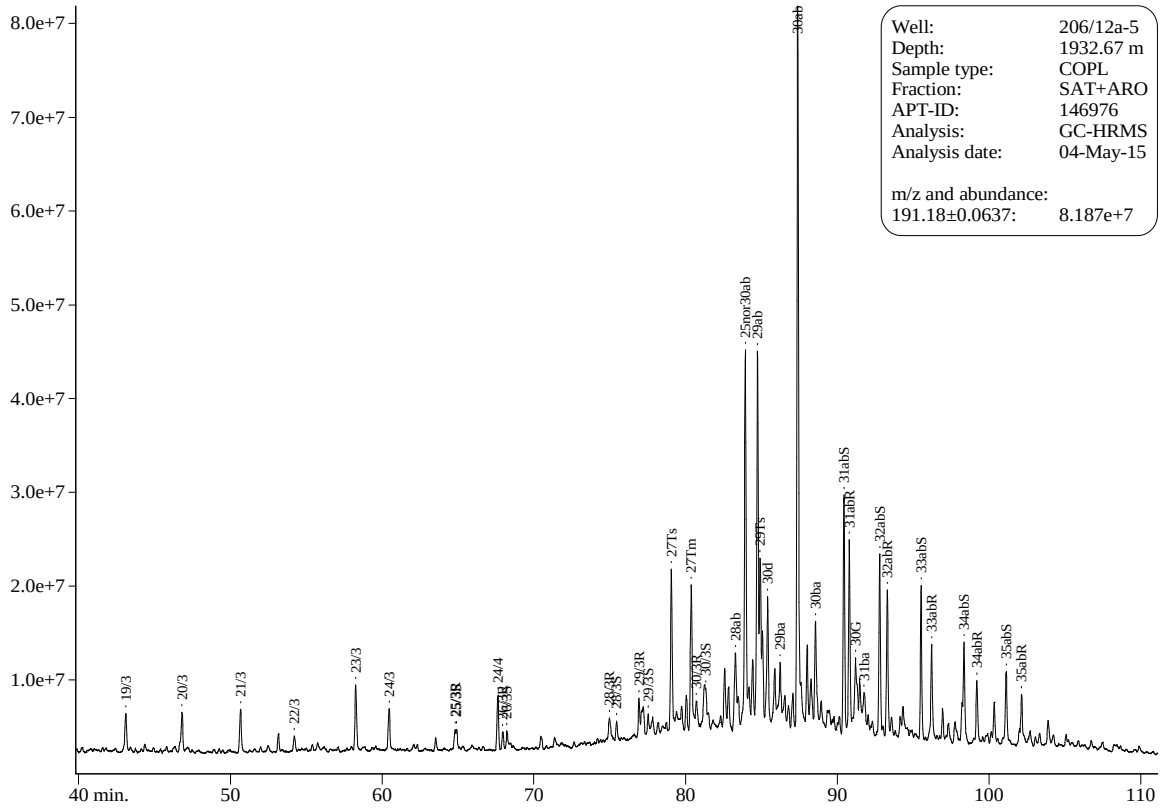
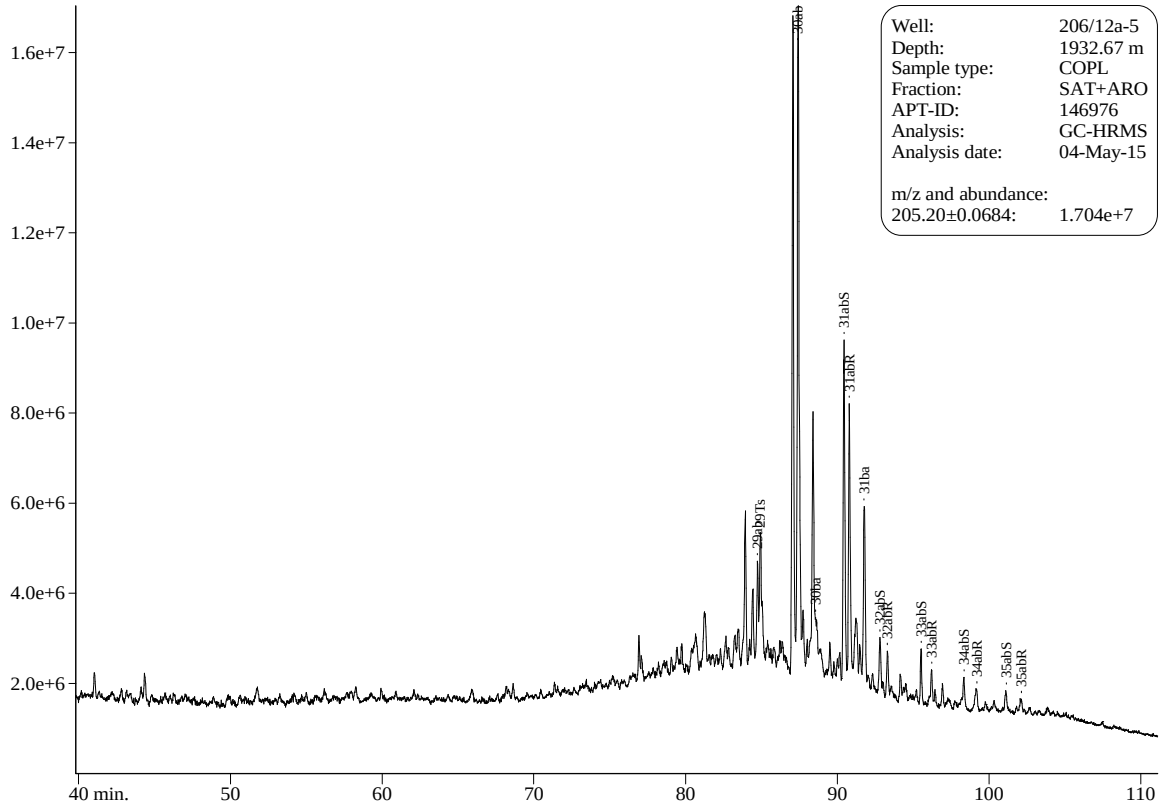


FIGURE Ap7.5 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1932.67m (Kimmeridge Clay Fm.)



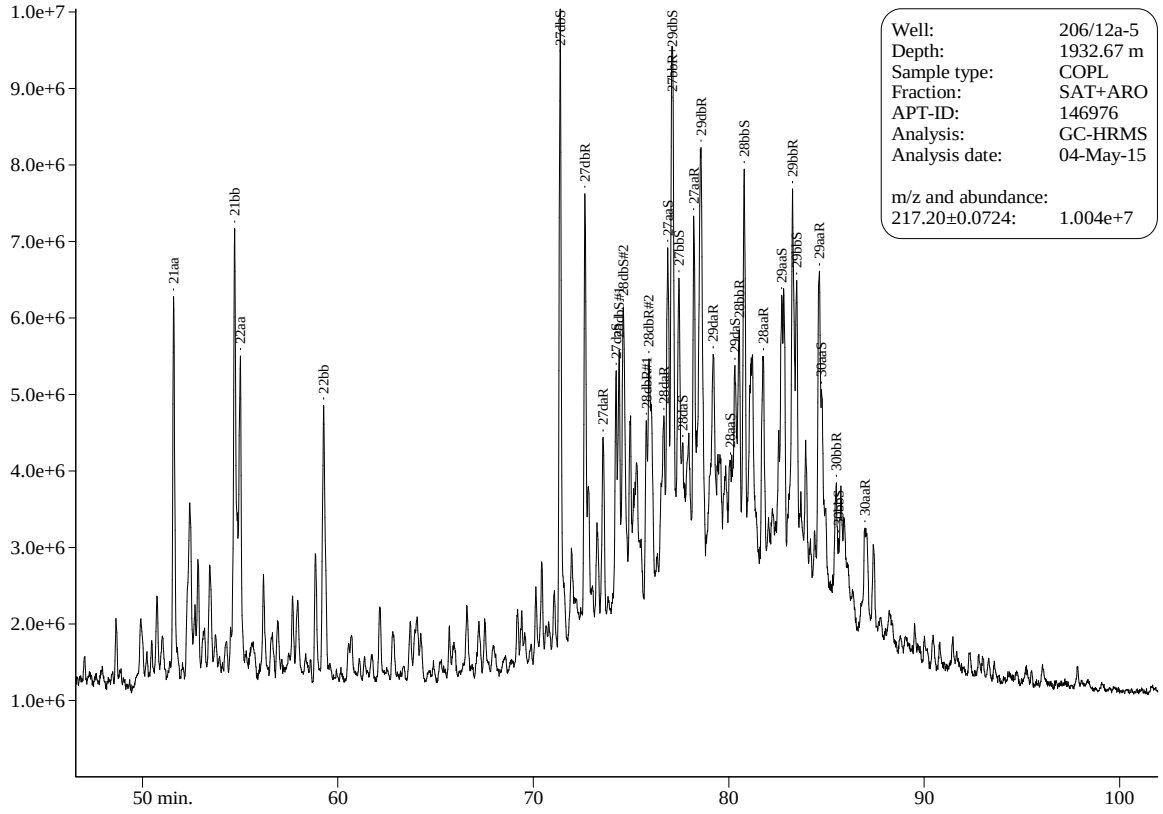
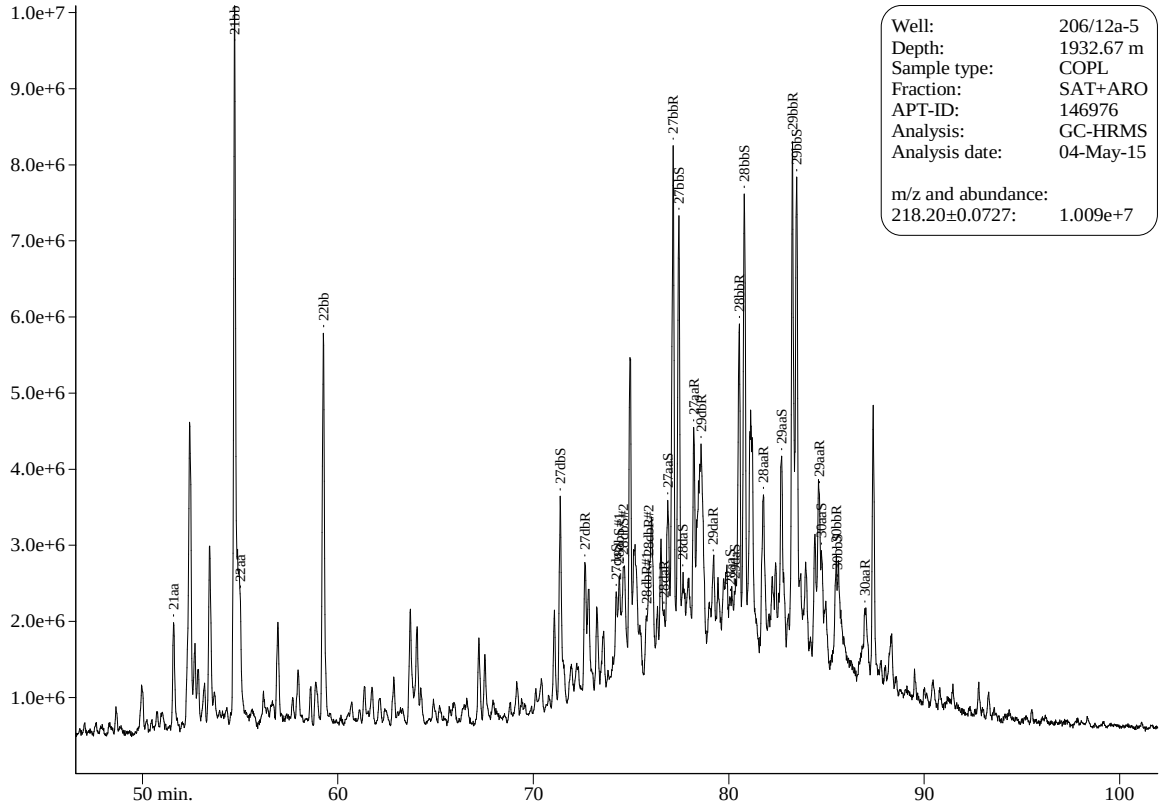


FIGURE Ap7.5 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1932.67m (Kimmeridge Clay Fm.)



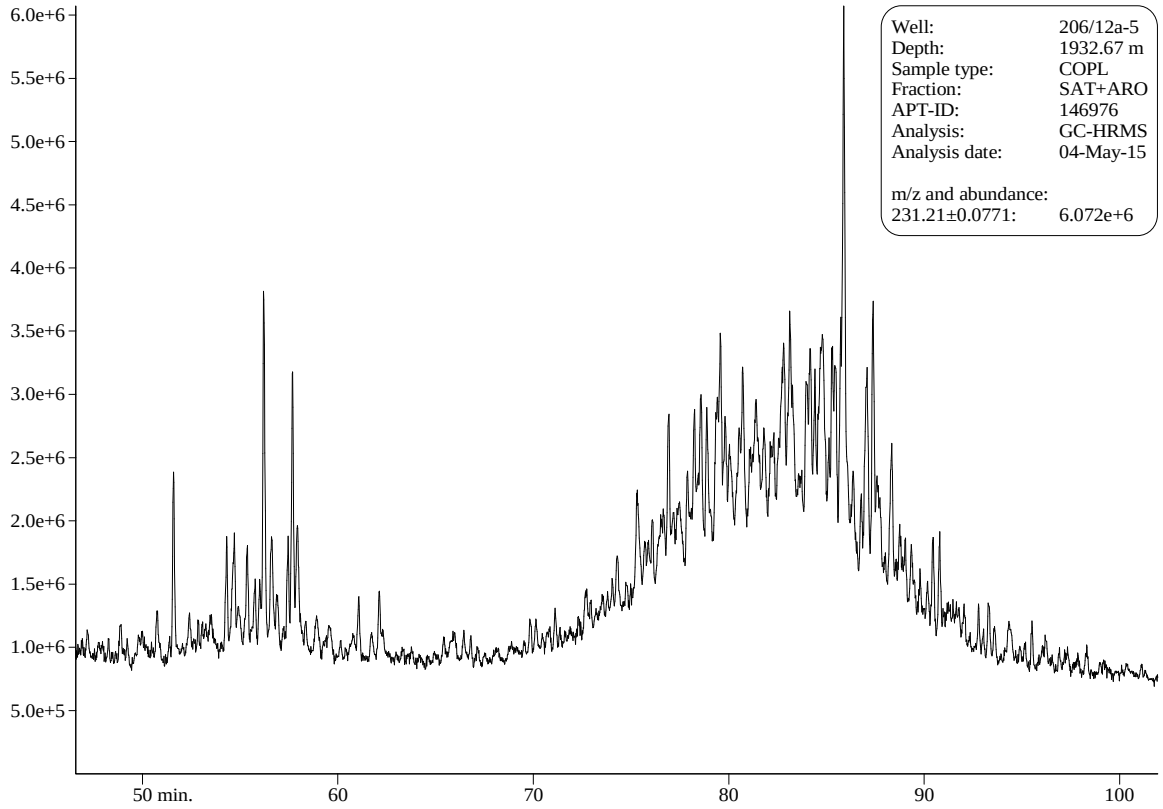
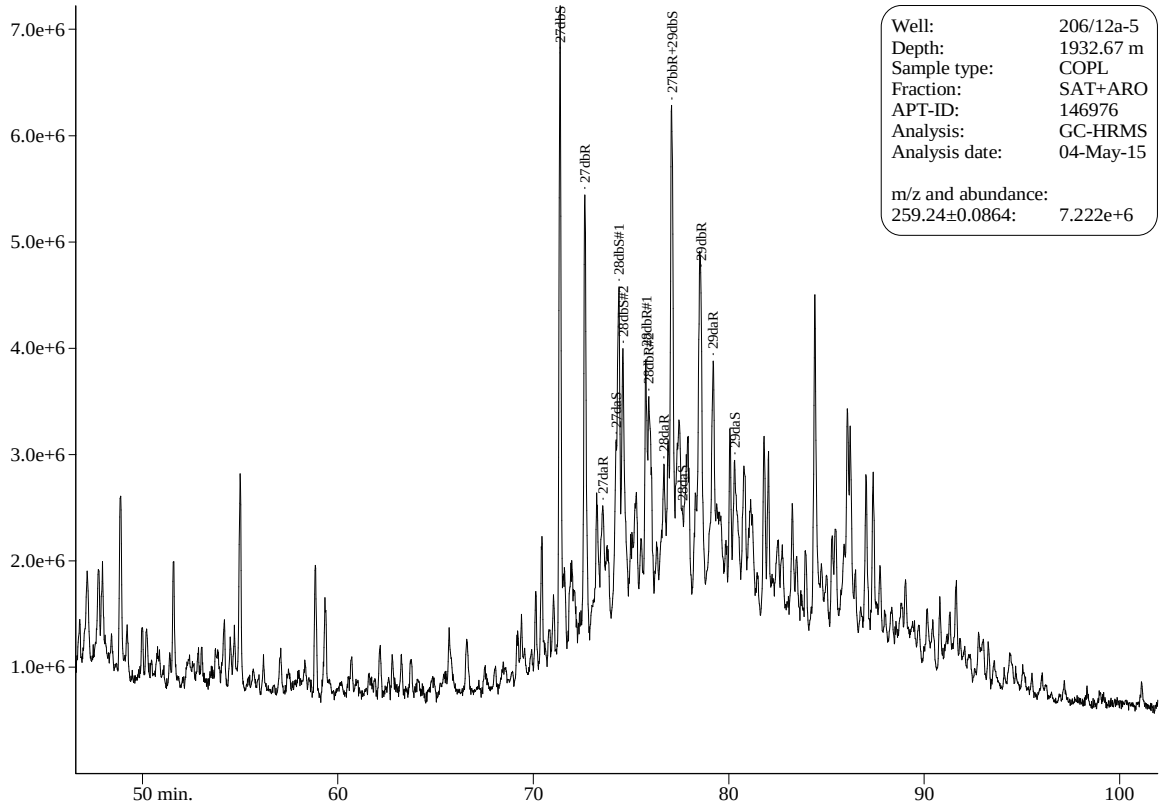


FIGURE Ap7.5 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1932.67m (Kimmeridge Clay Fm.)



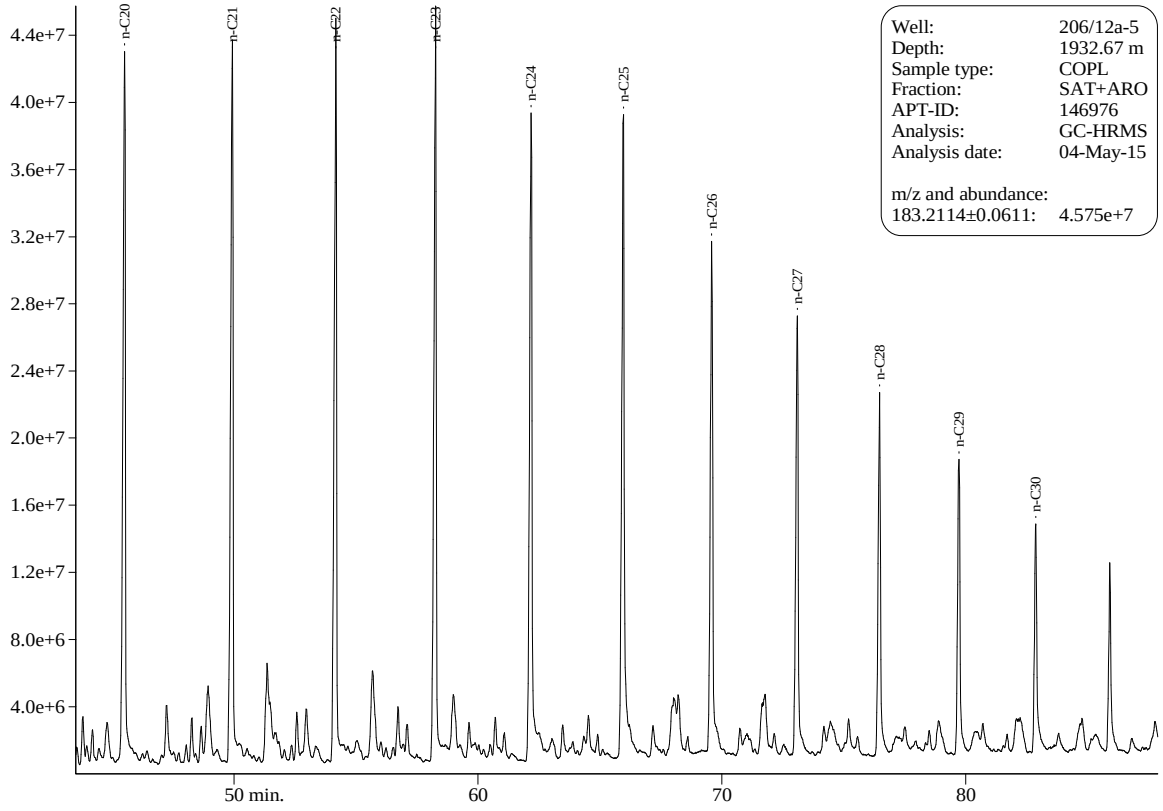
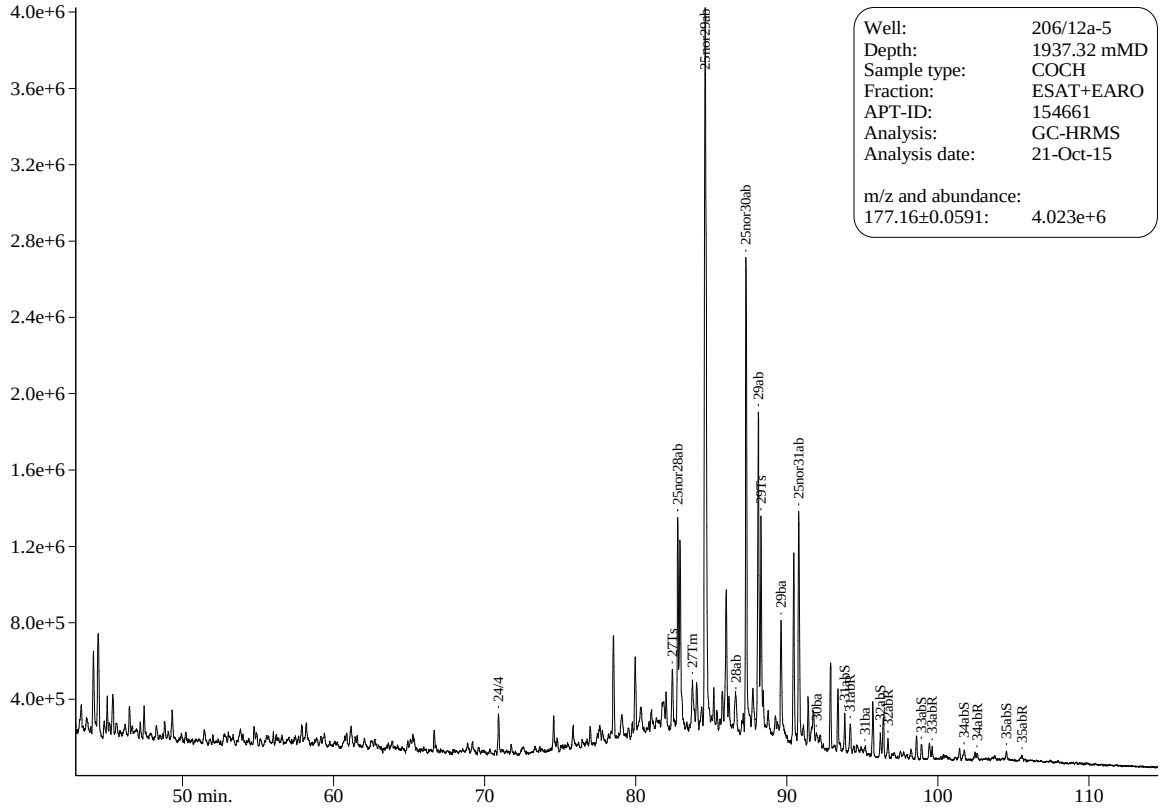


FIGURE Ap7.5 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1932.67m (Kimmeridge Clay Fm.)



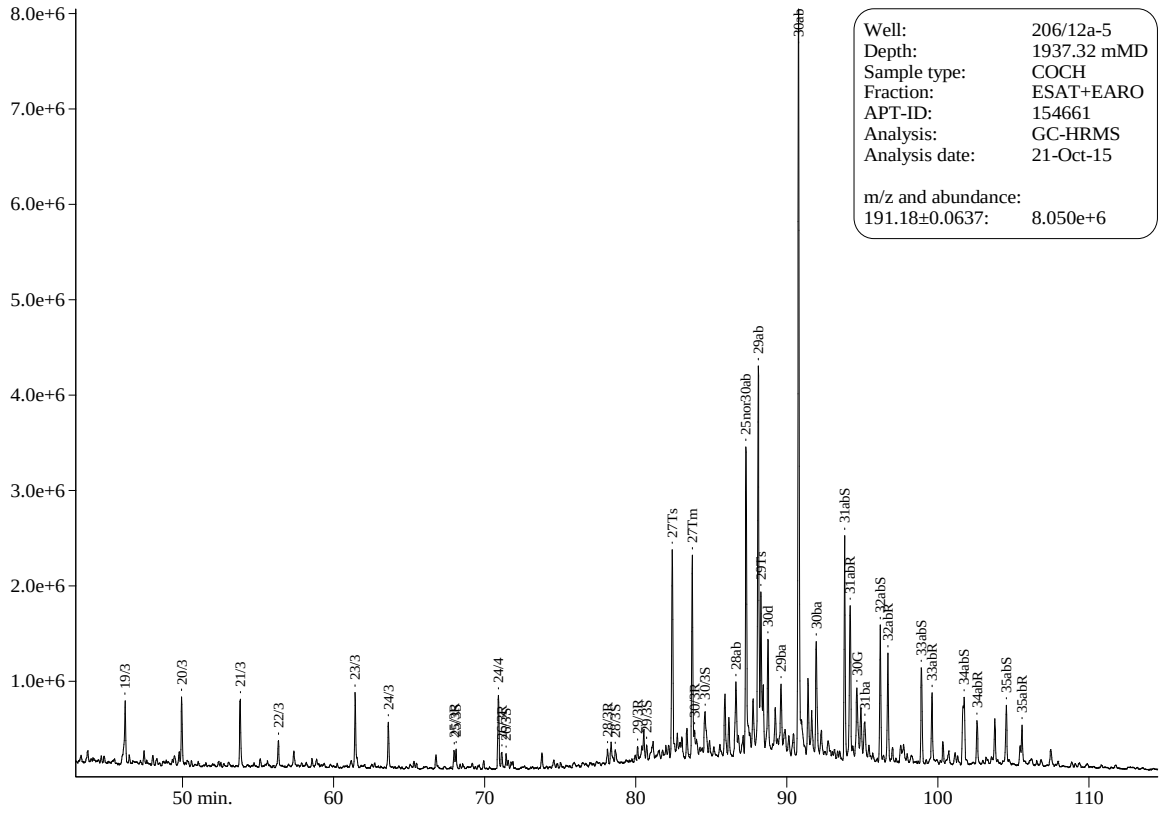
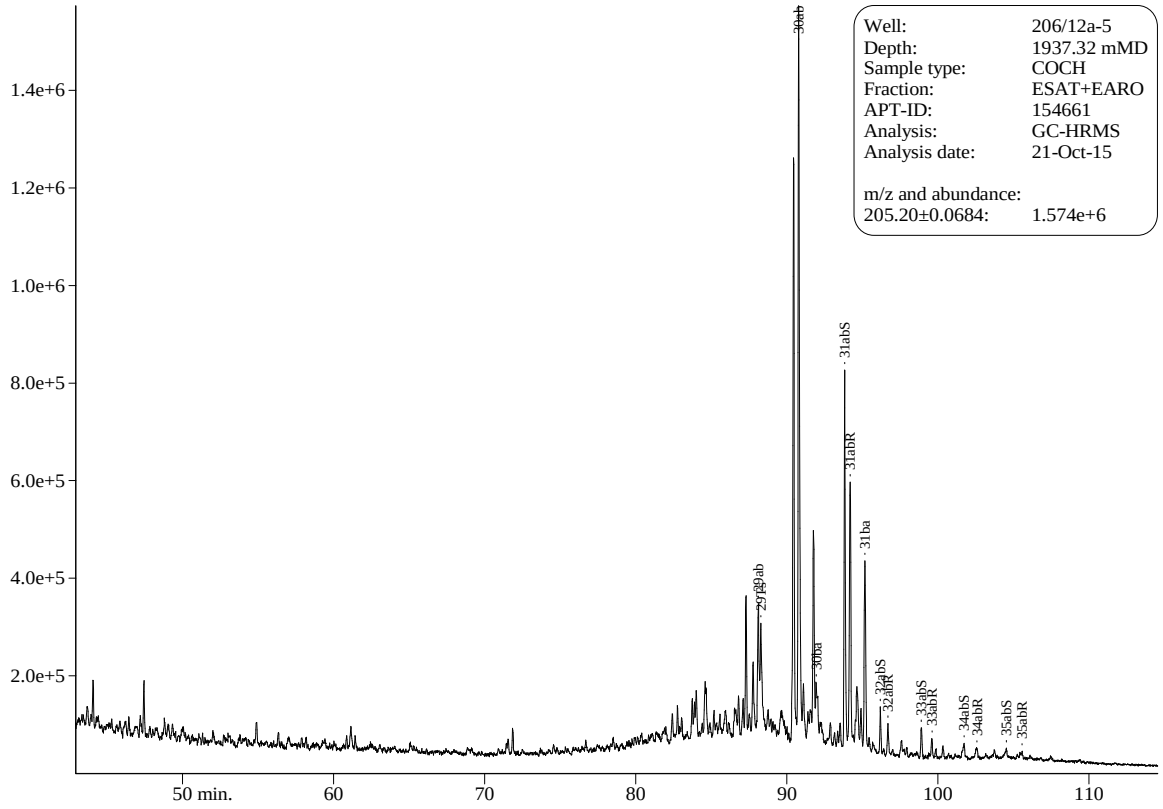


FIGURE Ap7.6 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE, 1937.32m (Rona)



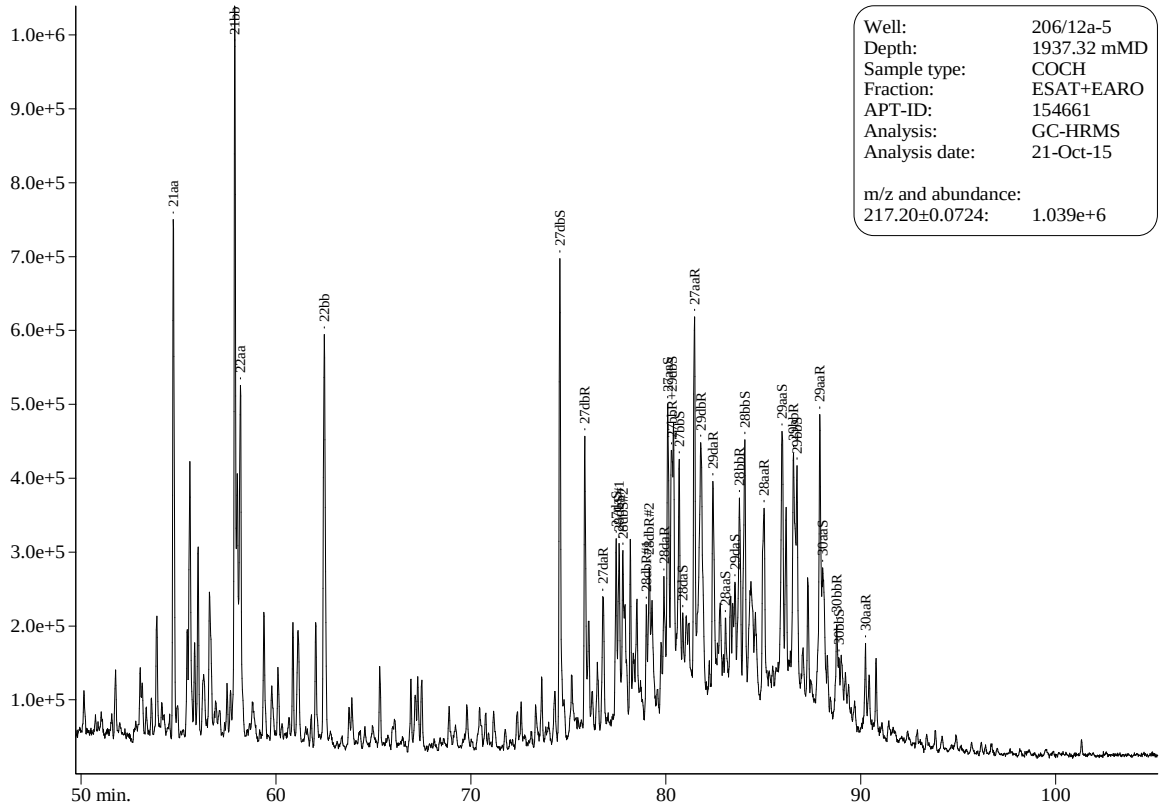
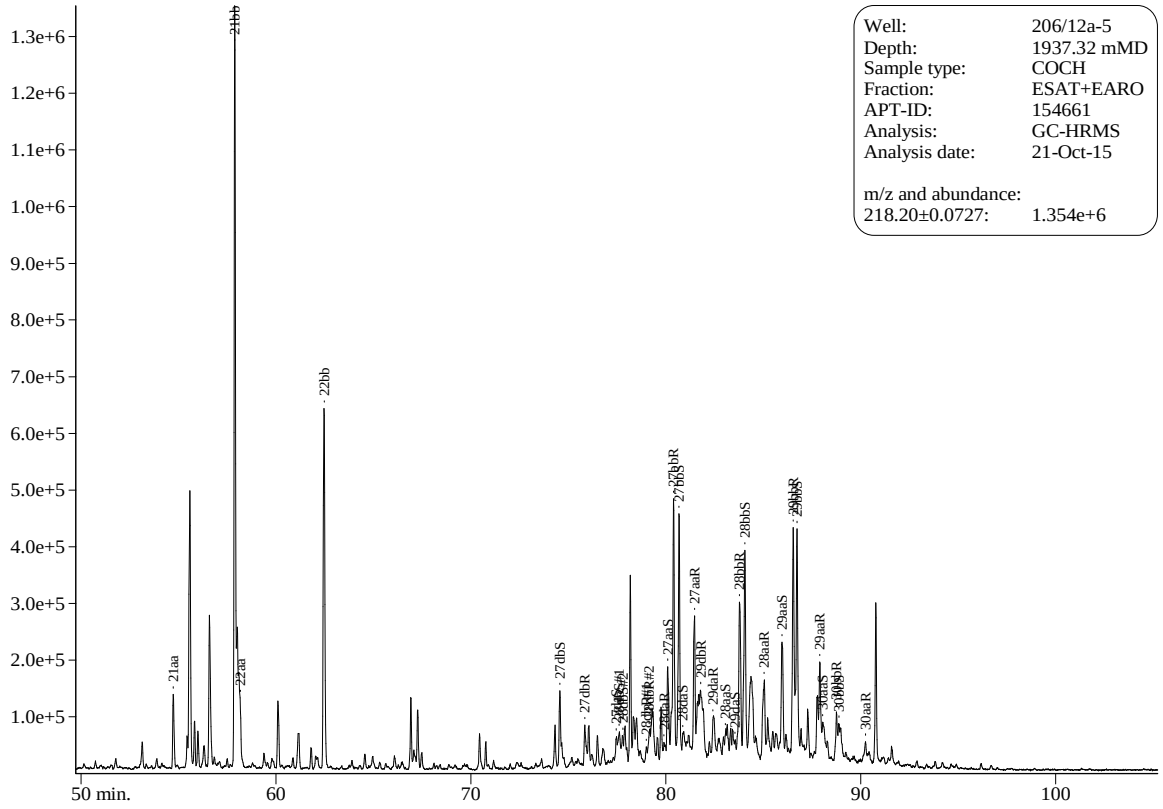


FIGURE Ap7.6 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE, 1937.32m (Rona)



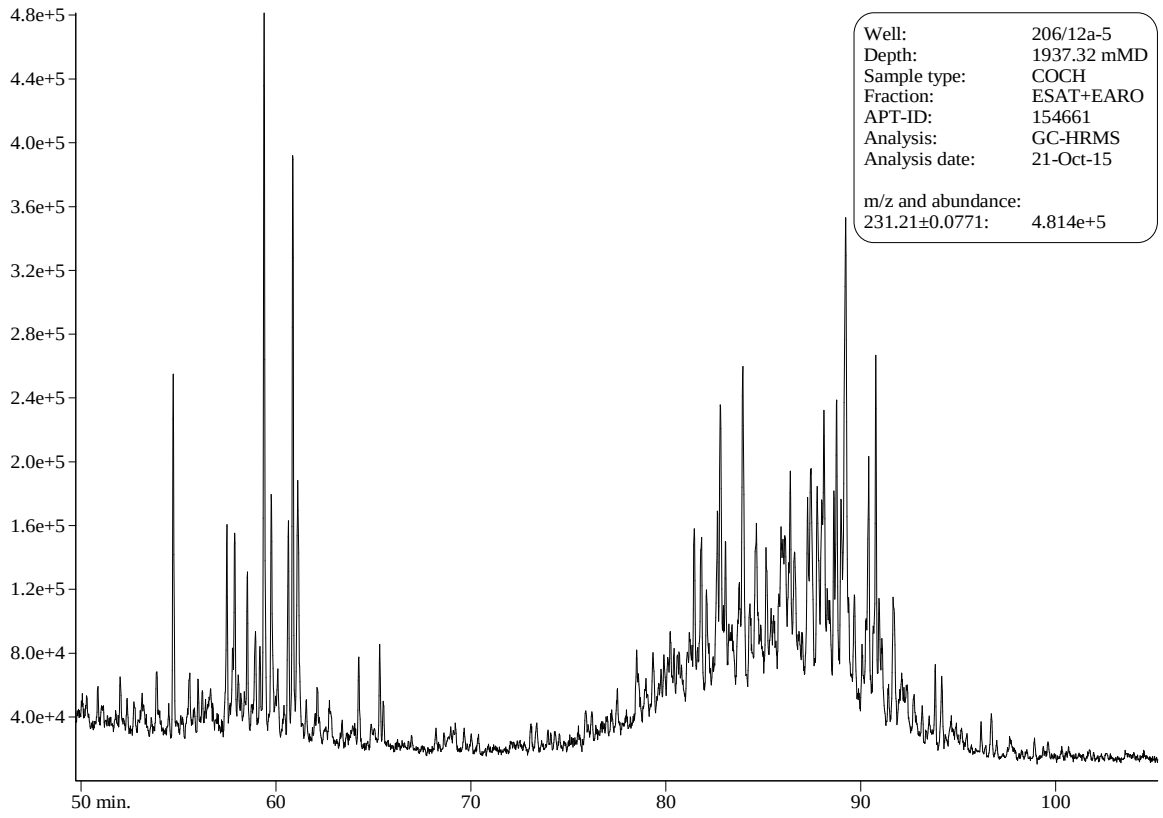
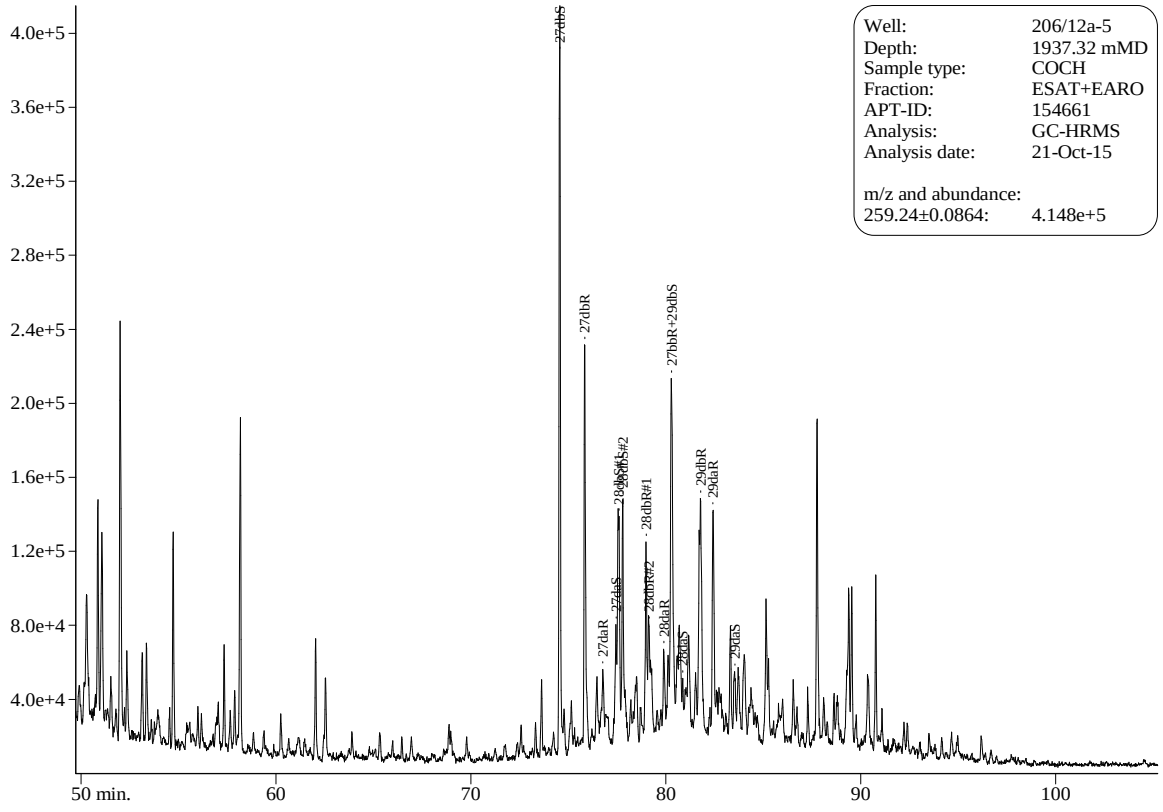


FIGURE Ap7.6 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE, 1937.32m (Rona)



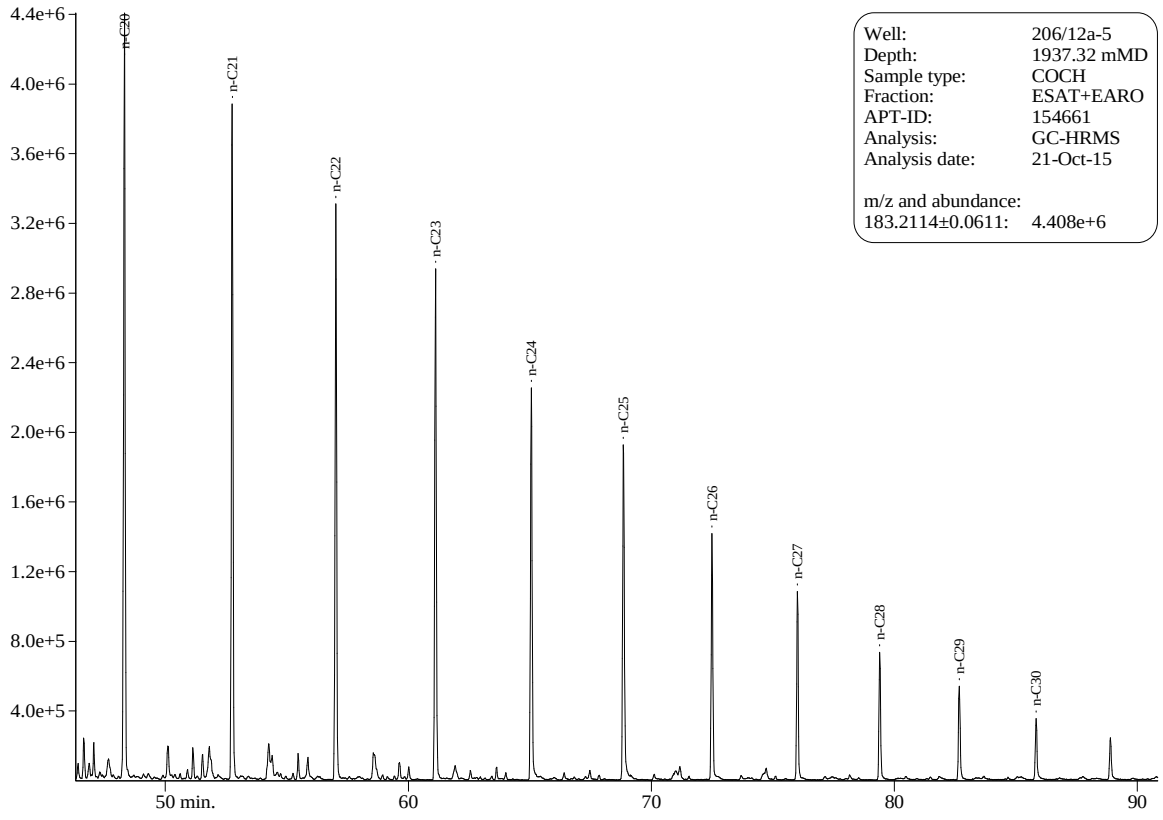
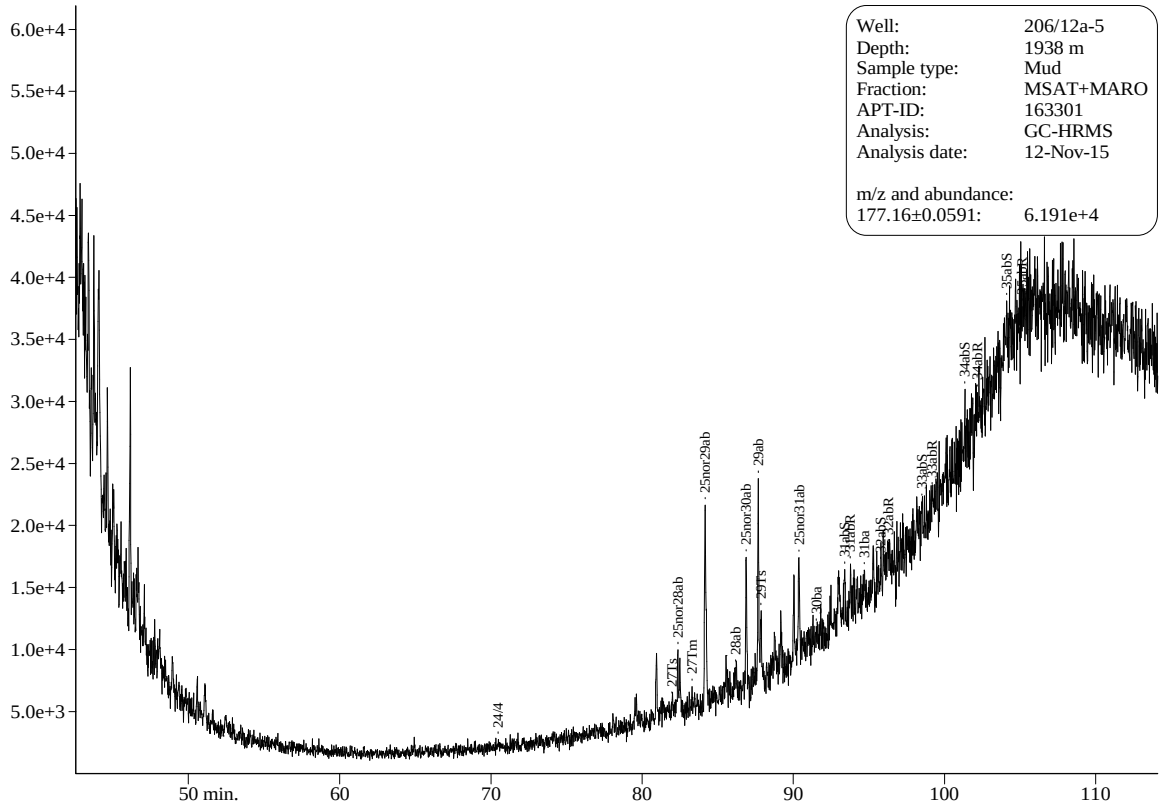


FIGURE Ap7.6 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE, 1937.32m (Rona)



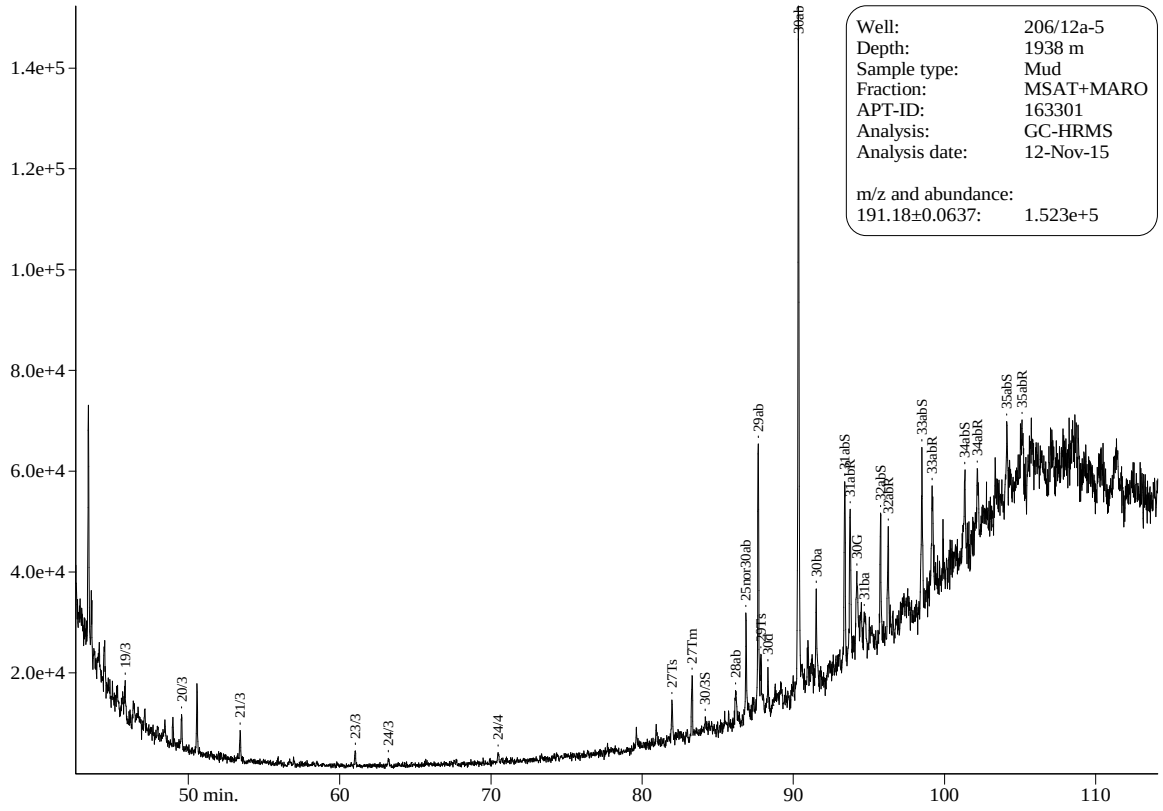
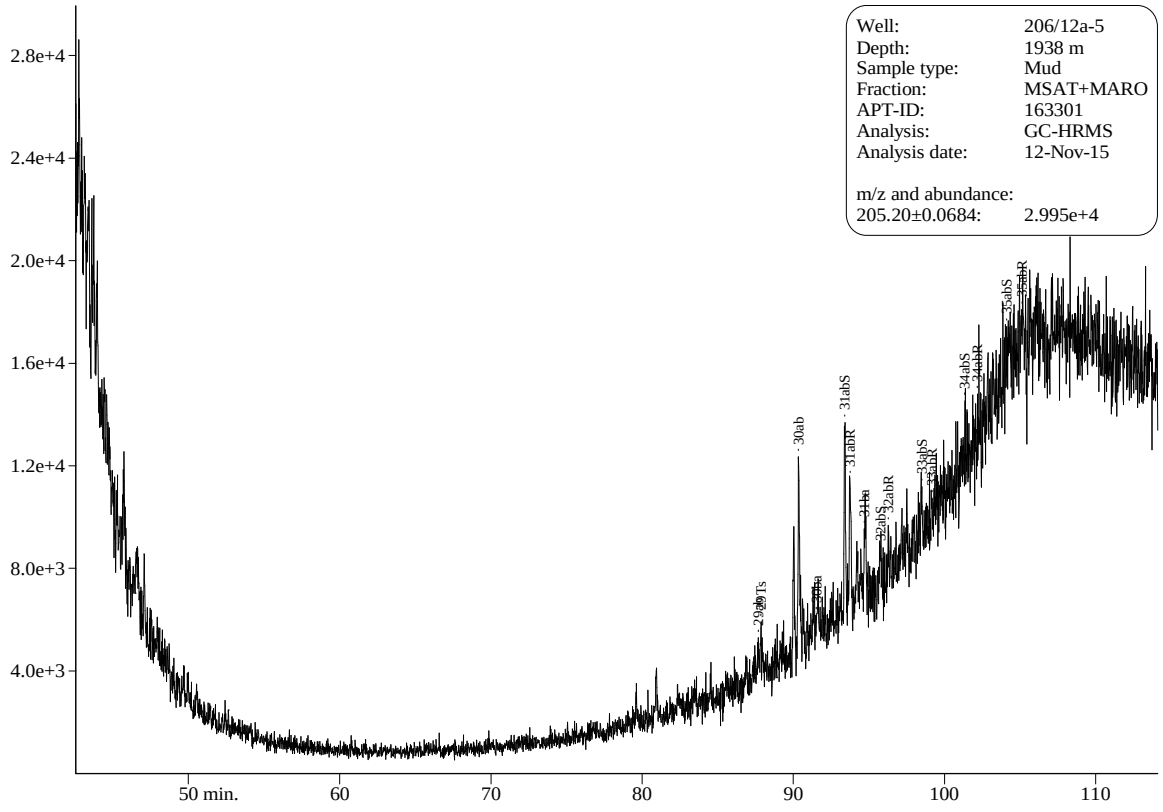


FIGURE Ap7.7 Alkane fraction GC-MS fragmentograms: 206/12a-5, MUD, 1938m (Rona)



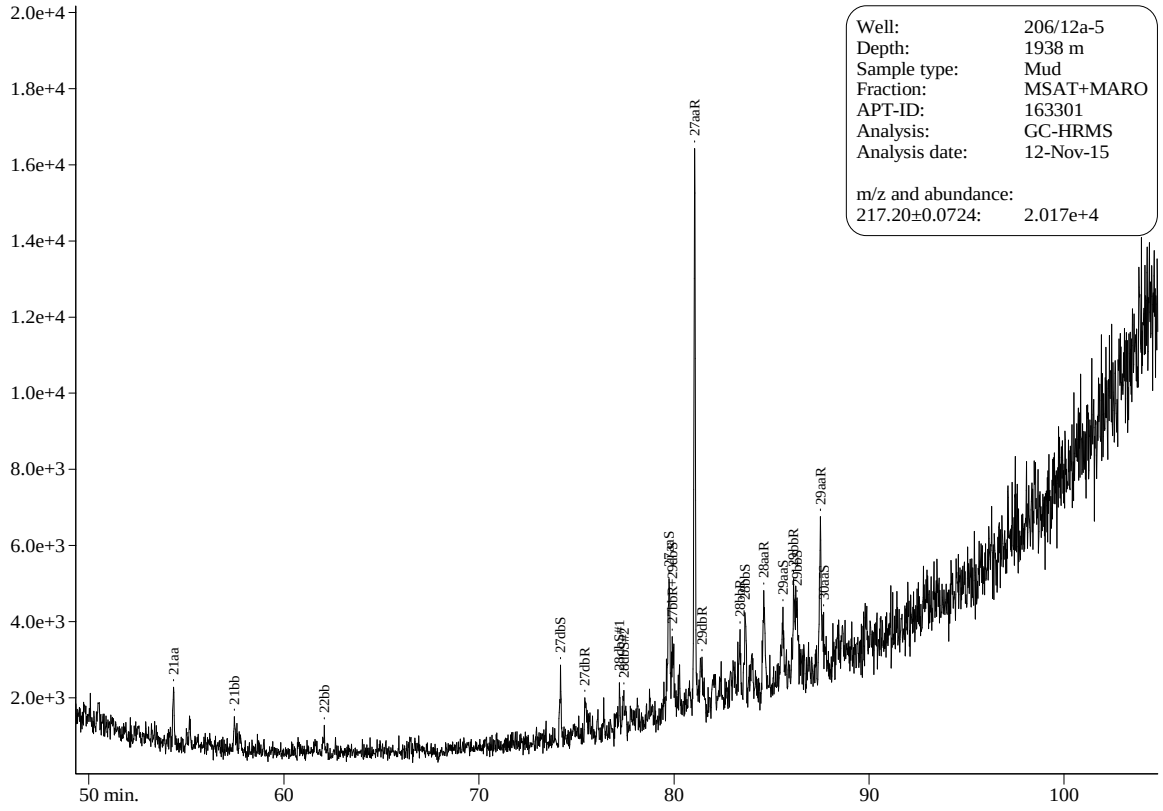
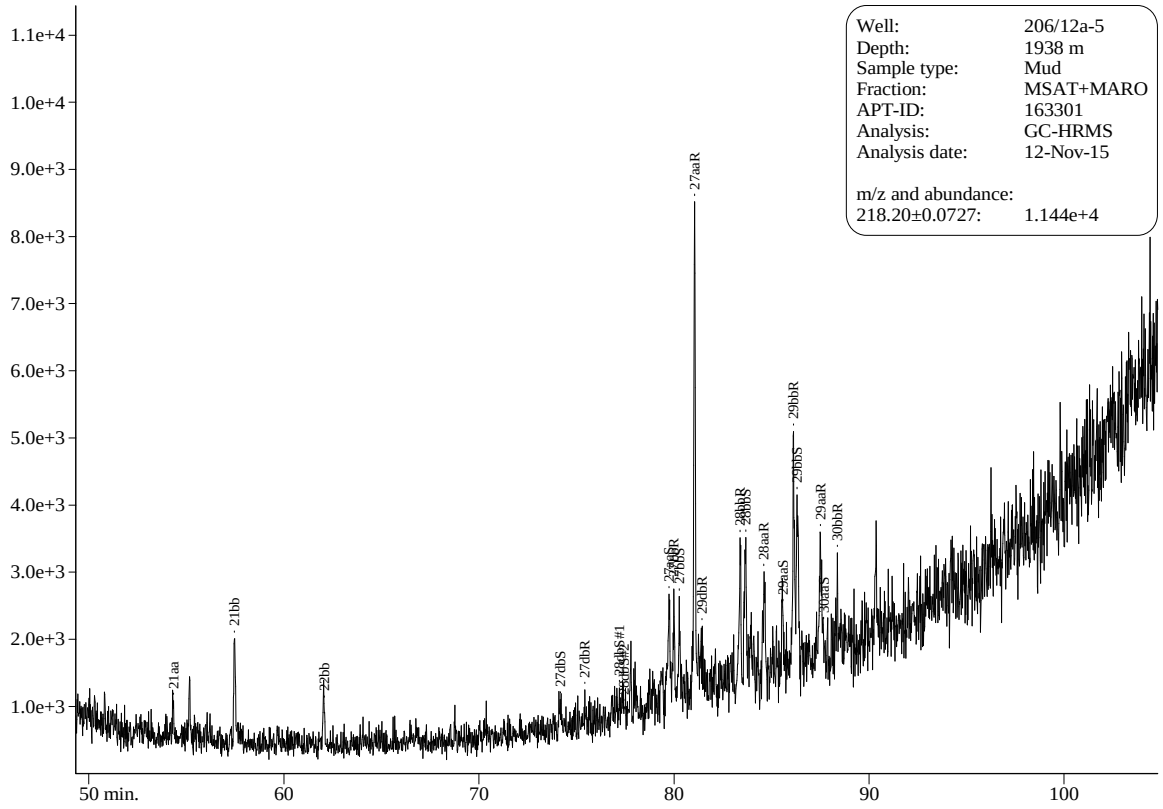


FIGURE Ap7.7 Alkane fraction GC-MS fragmentograms: 206/12a-5, MUD, 1938m (Rona)



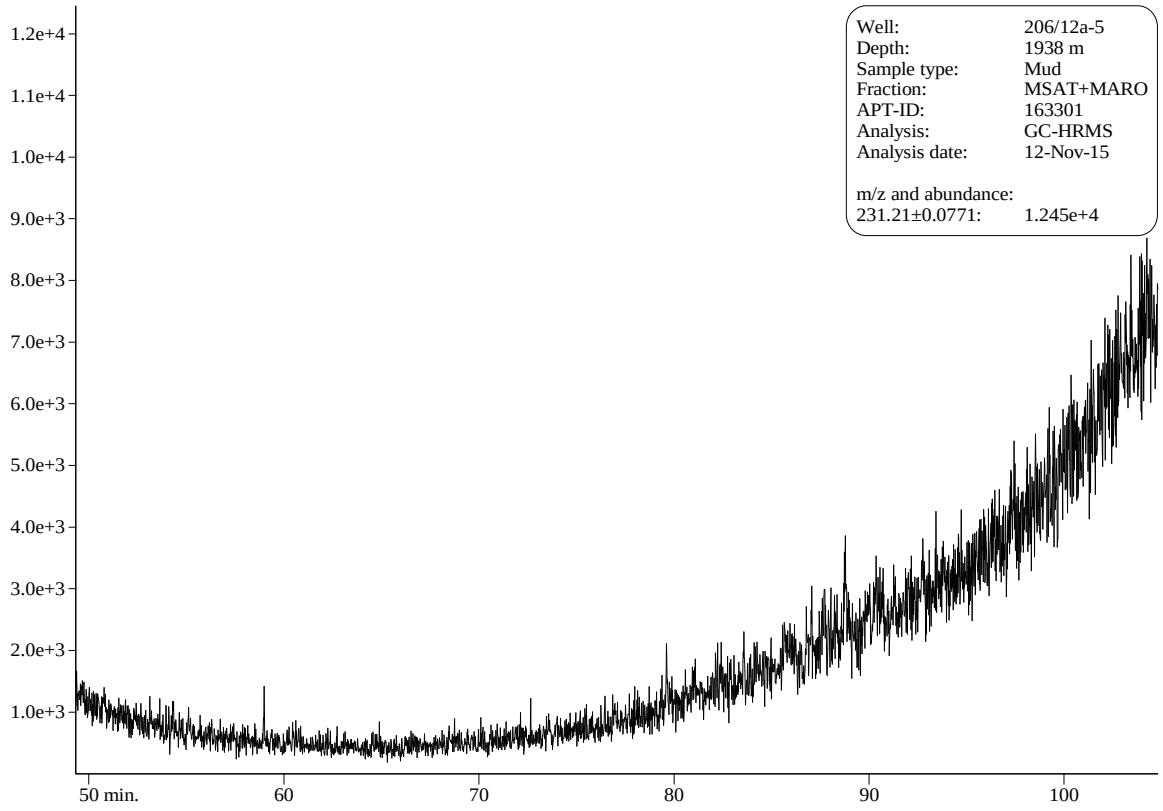
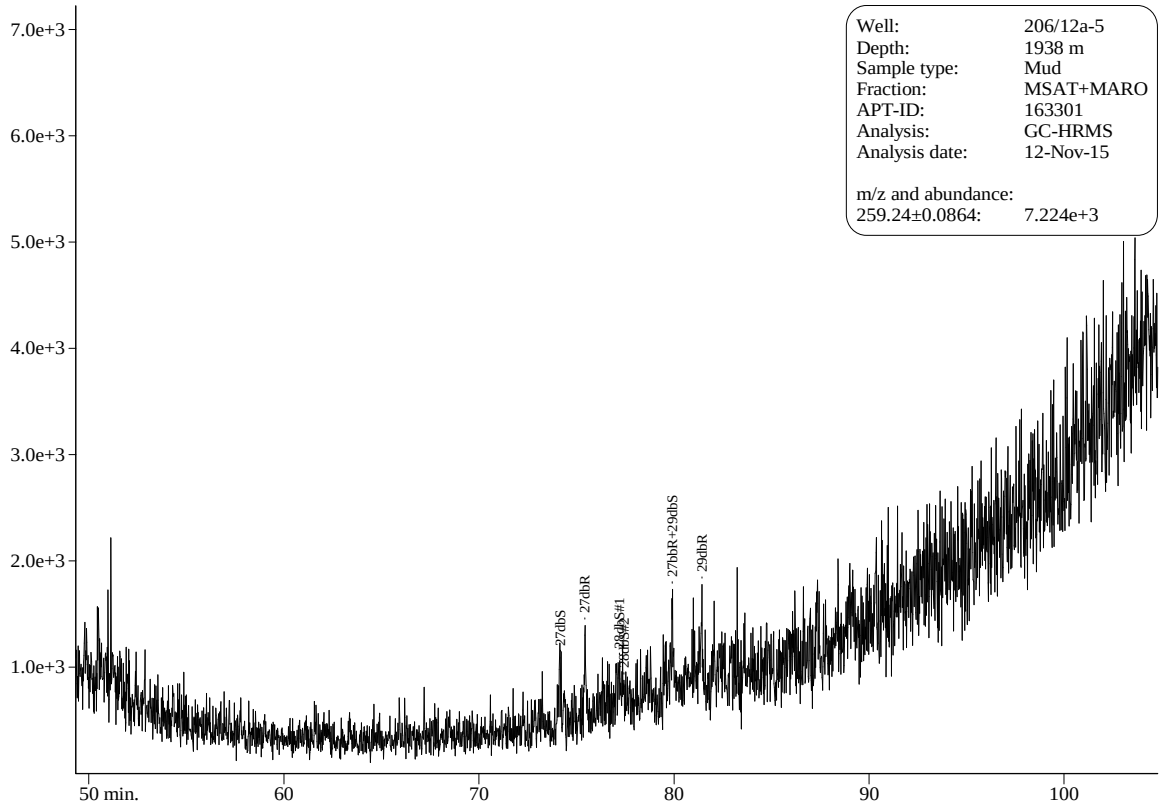


FIGURE Ap7.7 Alkane fraction GC-MS fragmentograms: 206/12a-5, MUD, 1938m (Rona)



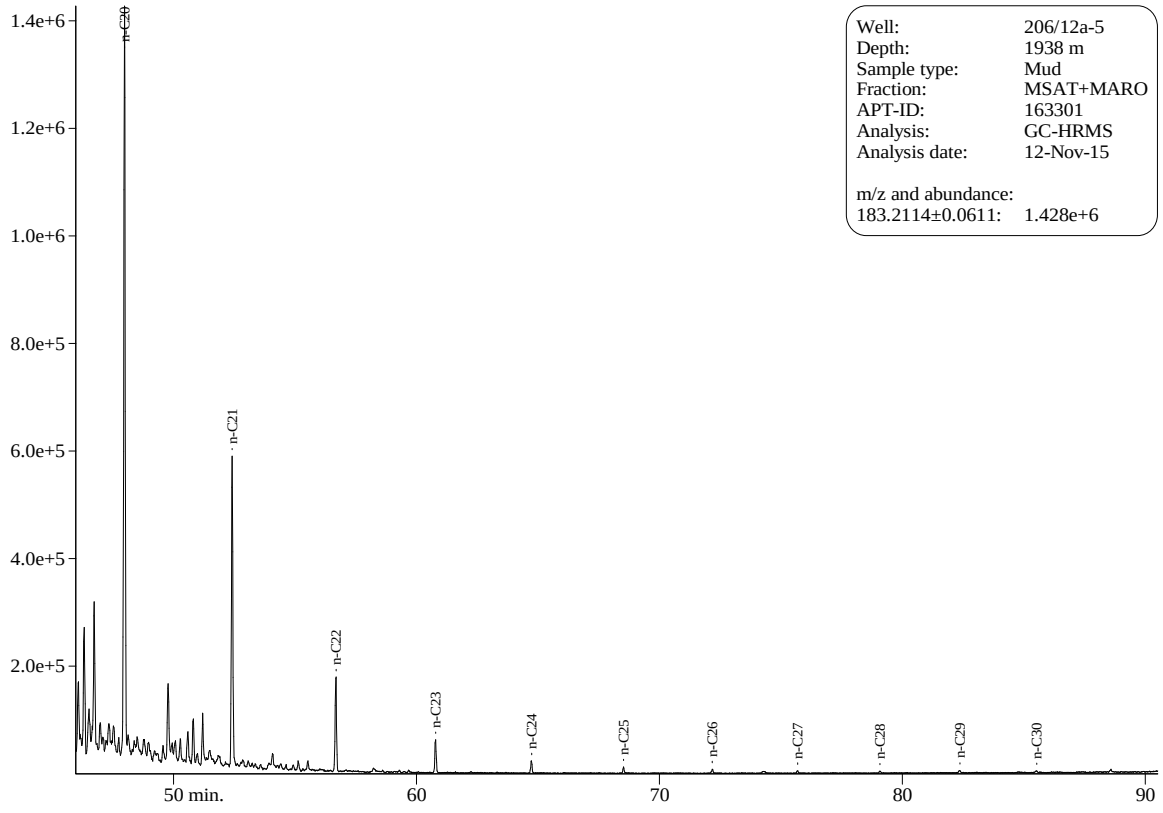
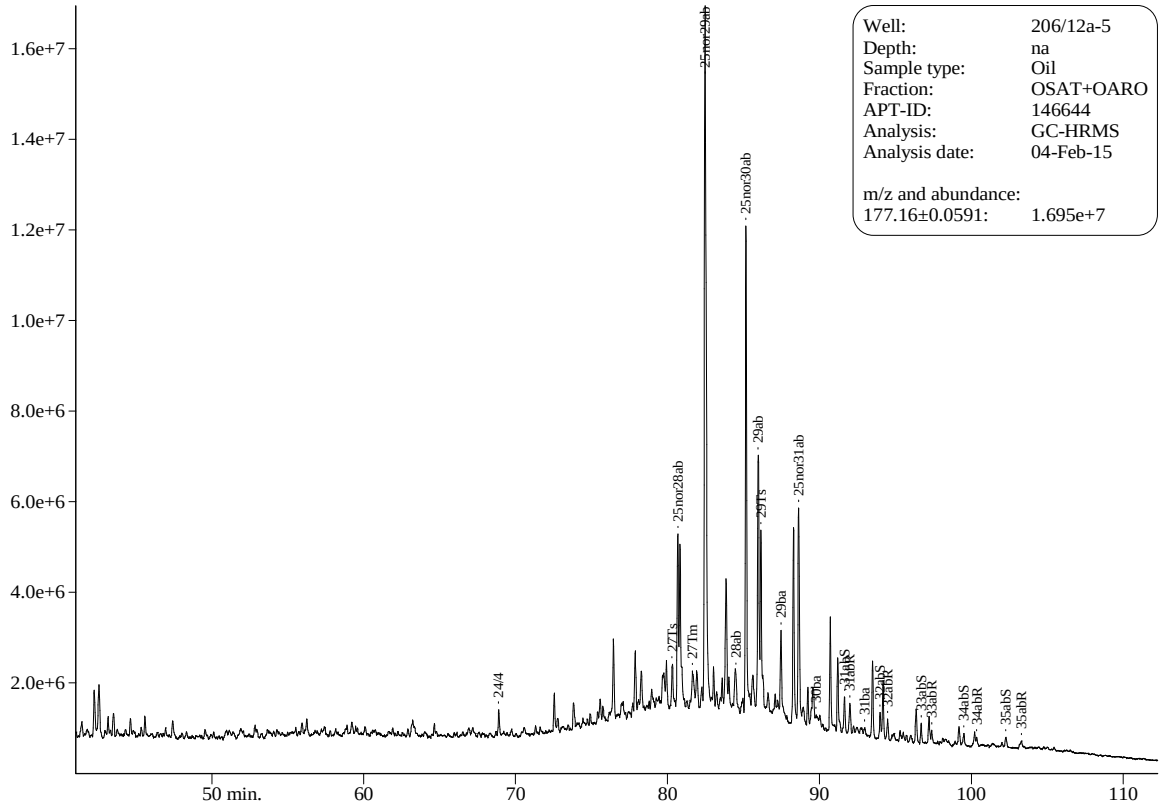


FIGURE Ap7.7 Alkane fraction GC-MS fragmentograms: 206/12a-5, MUD, 1938m (Rona)



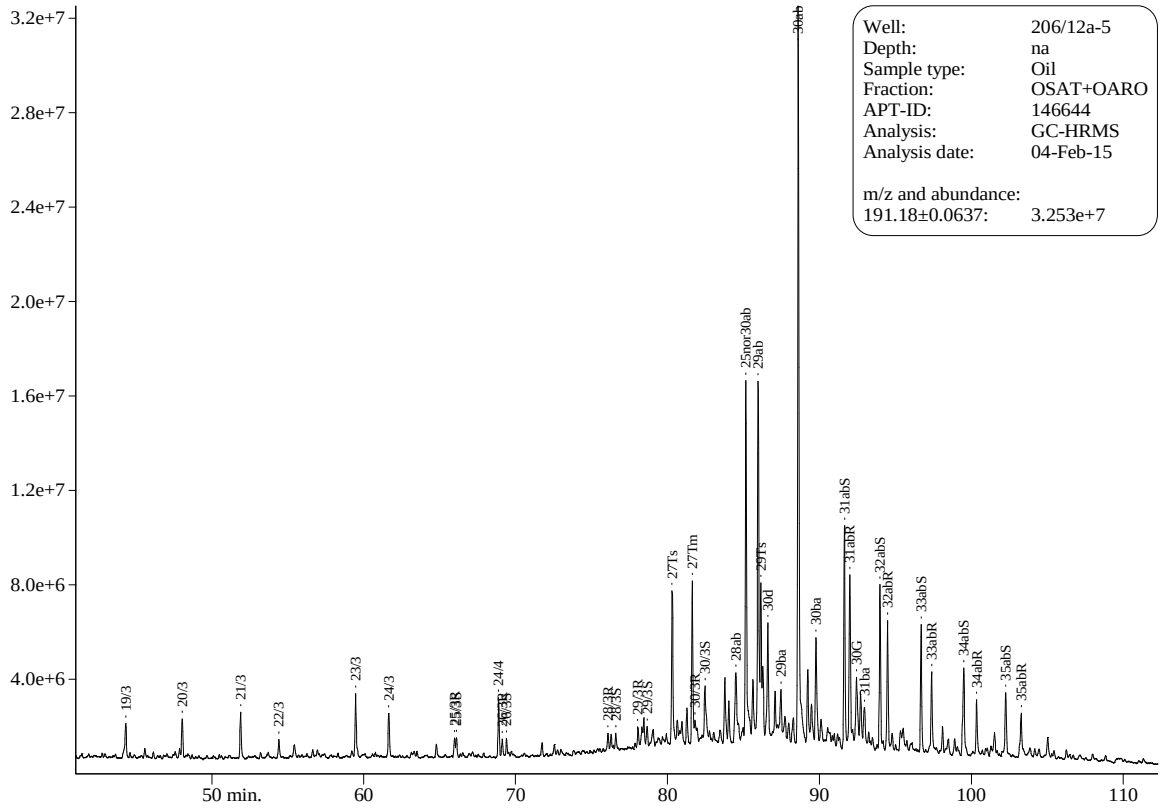
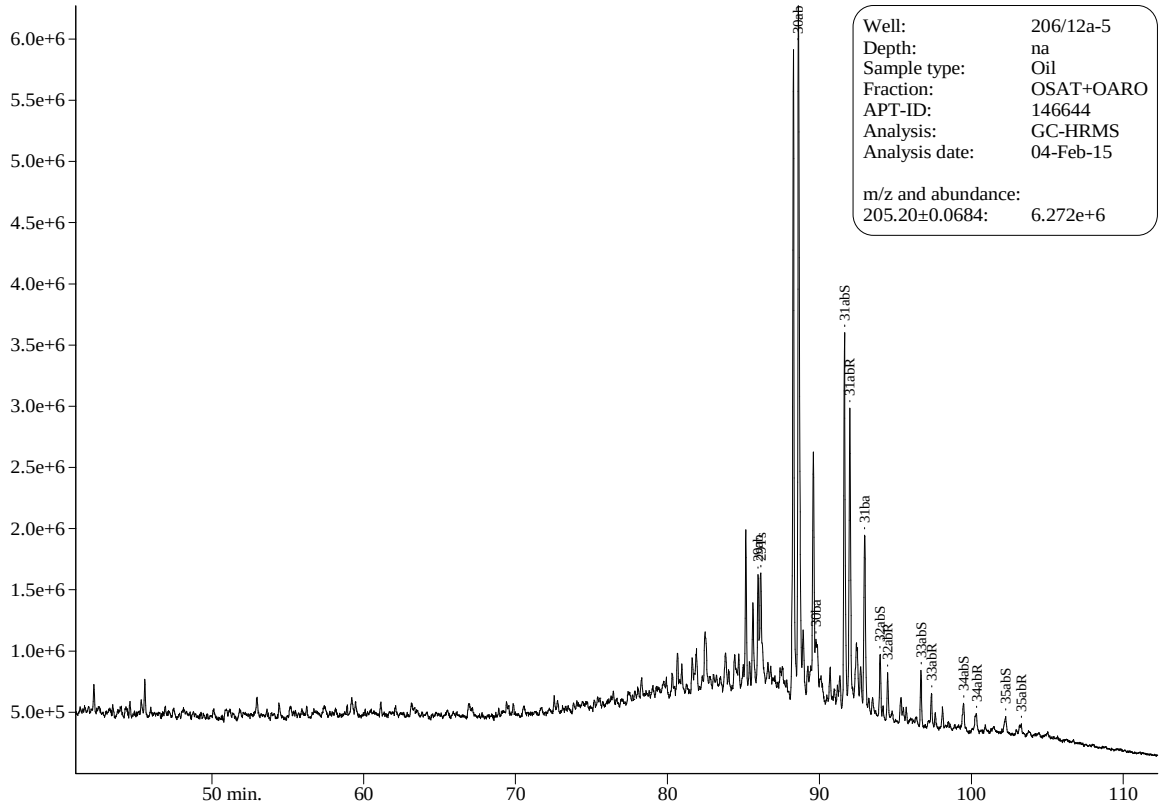


FIGURE Ap7.8 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona)



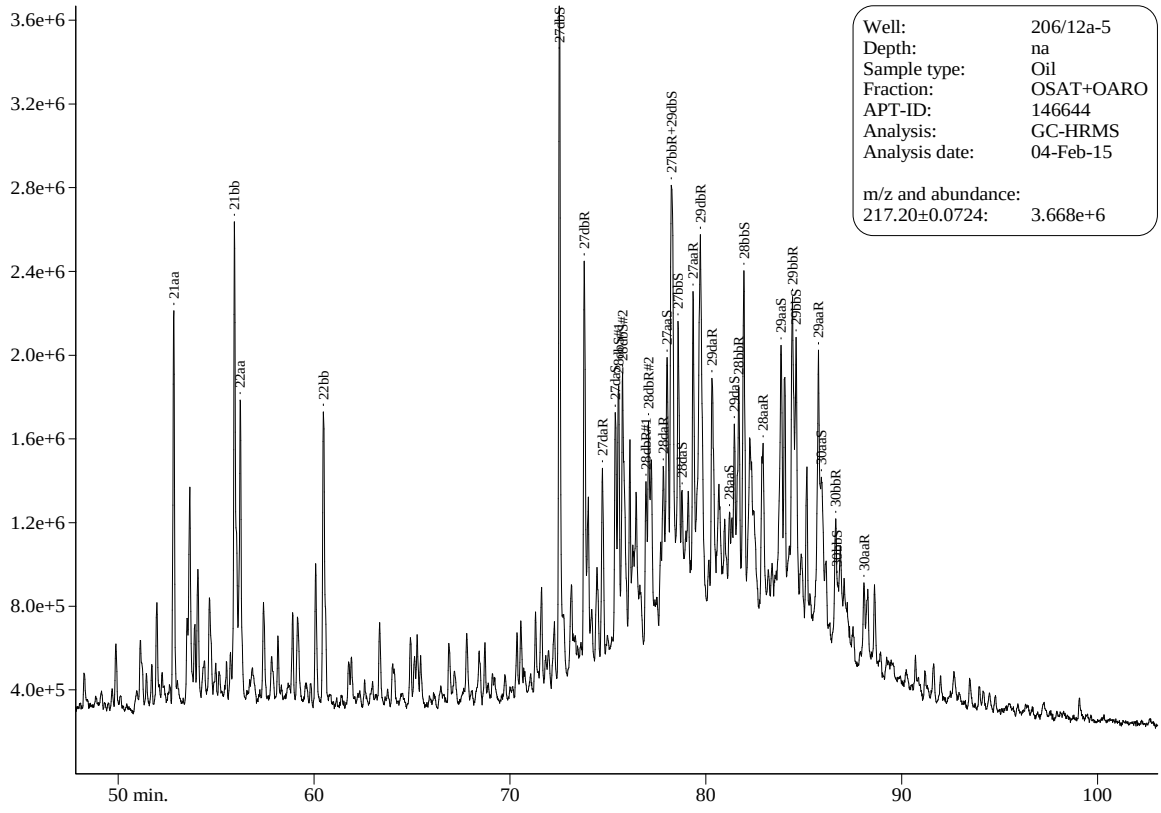
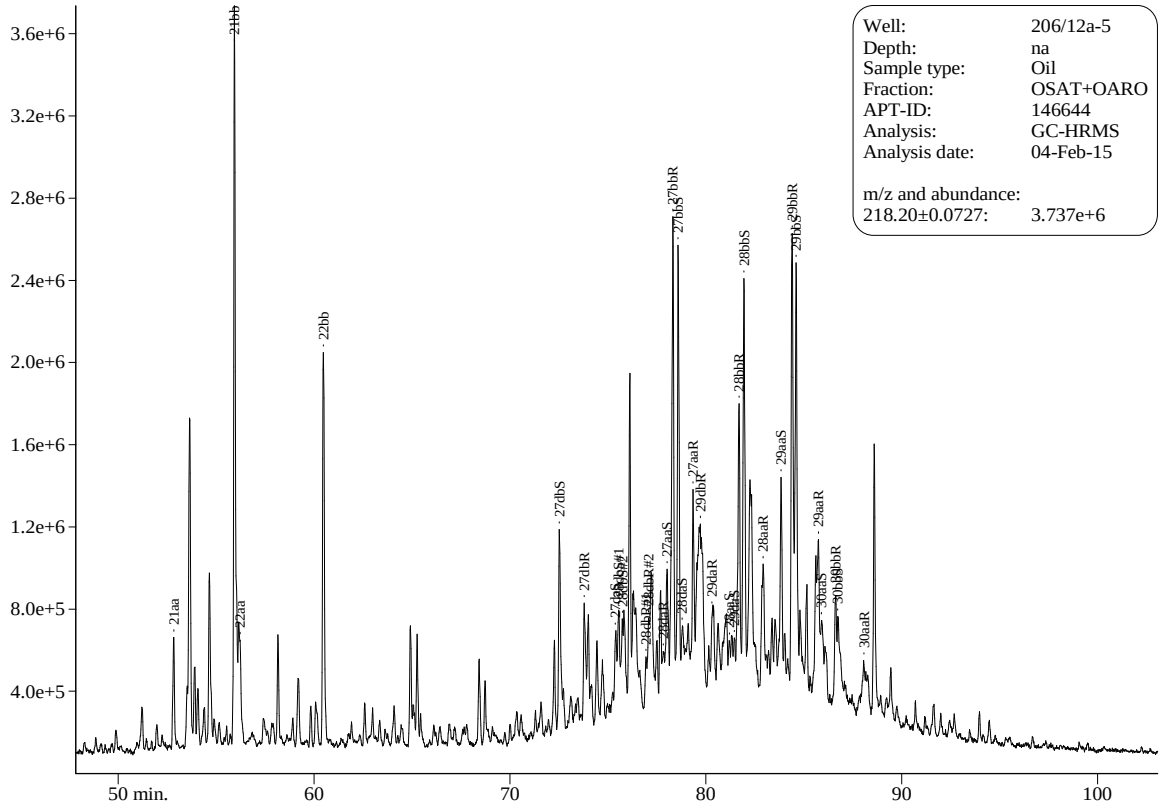


FIGURE Ap7.8 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona)



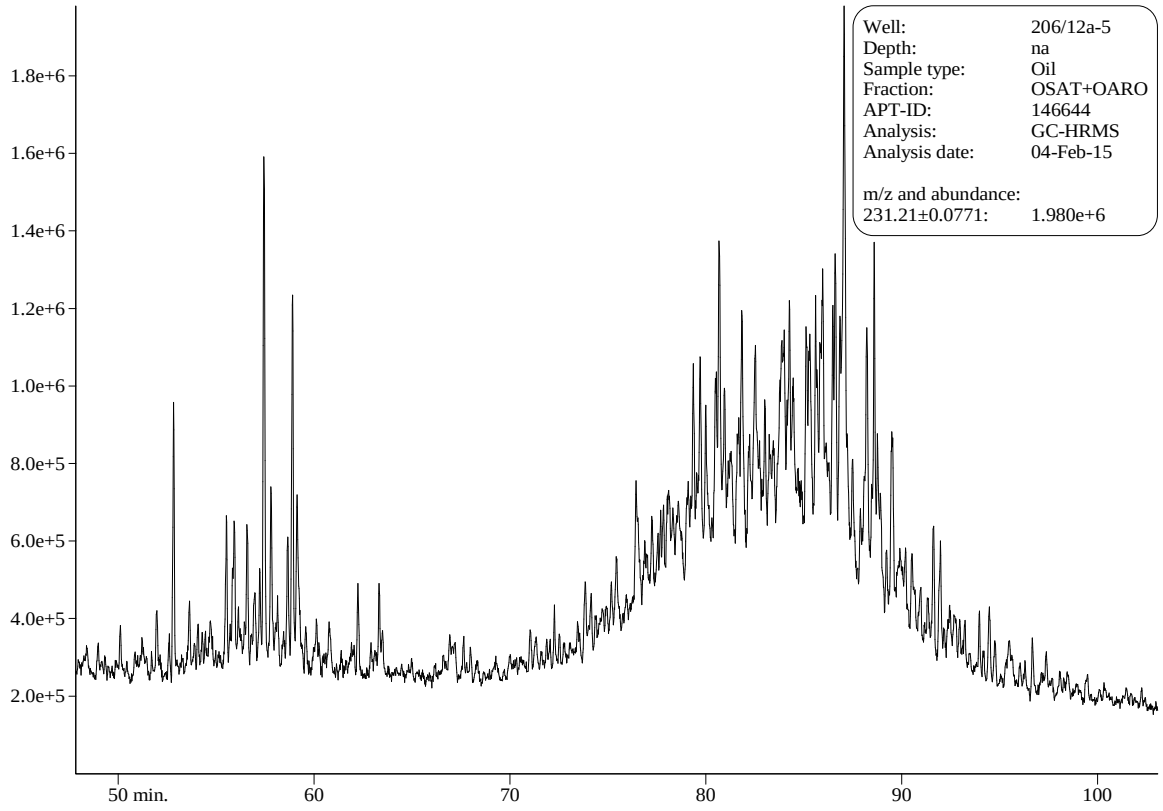
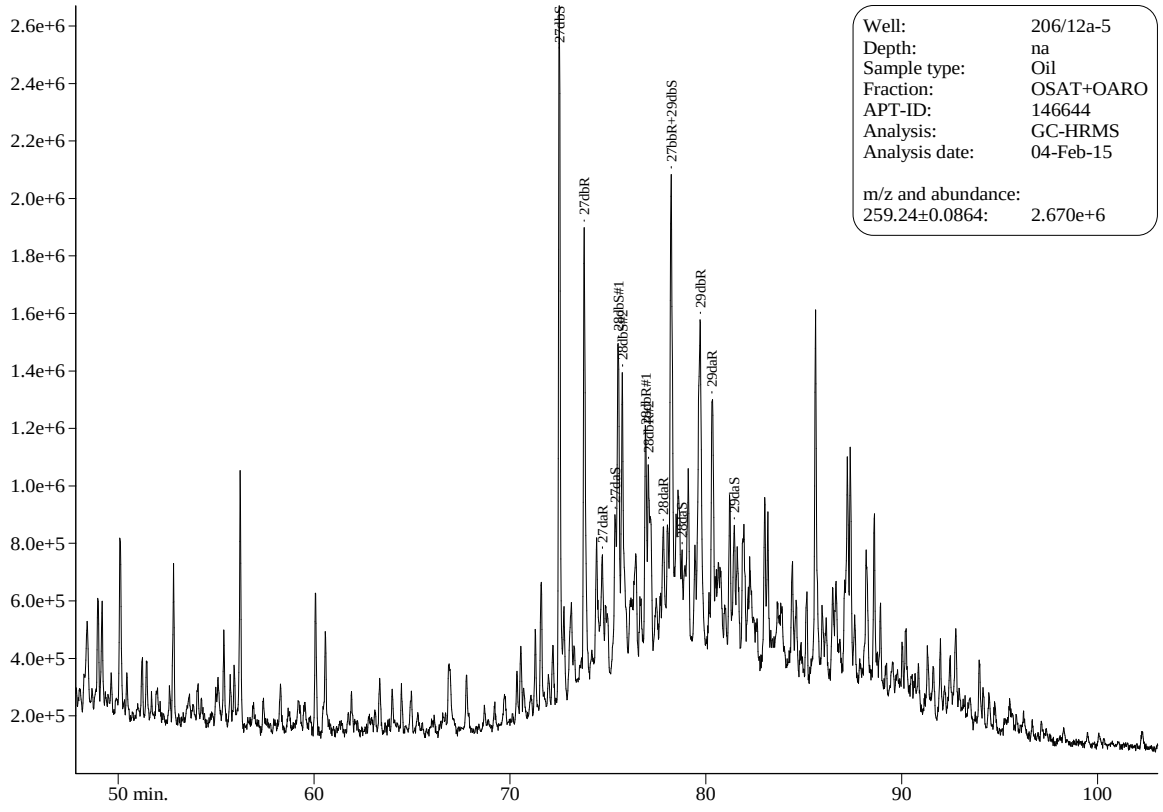


FIGURE Ap7.8 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona)



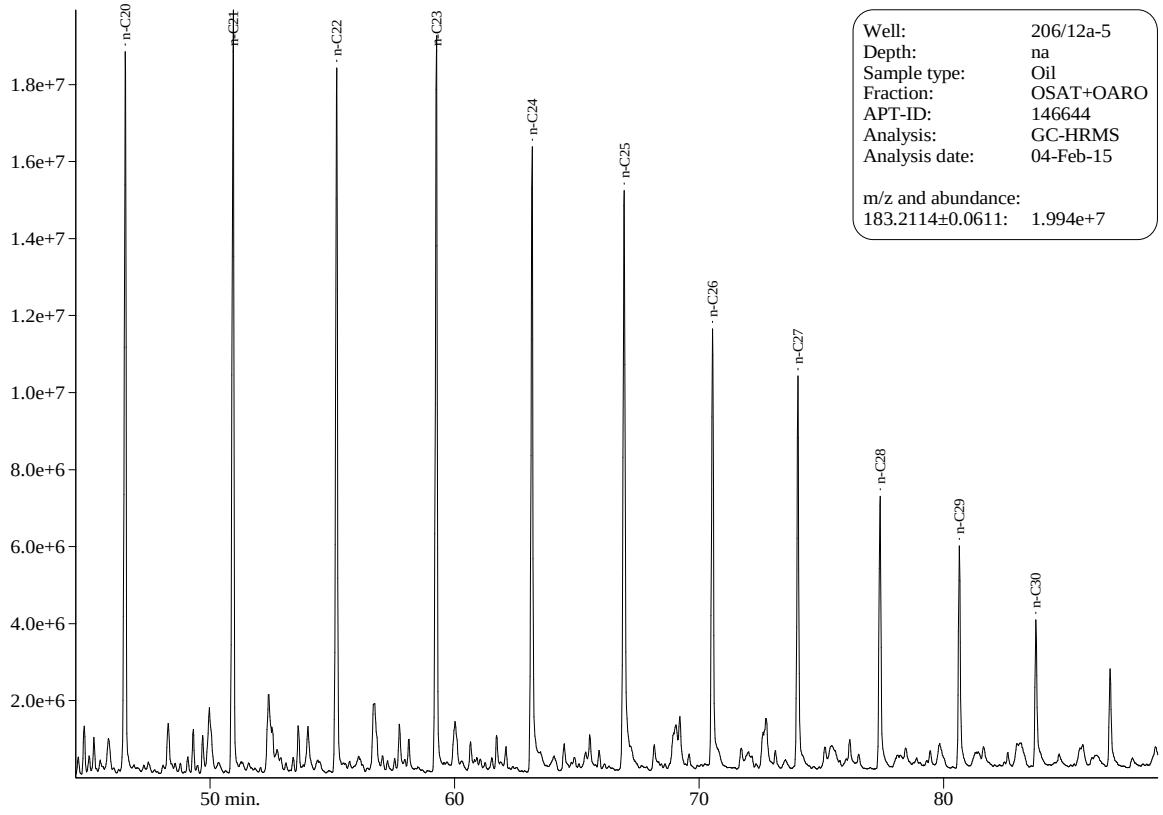
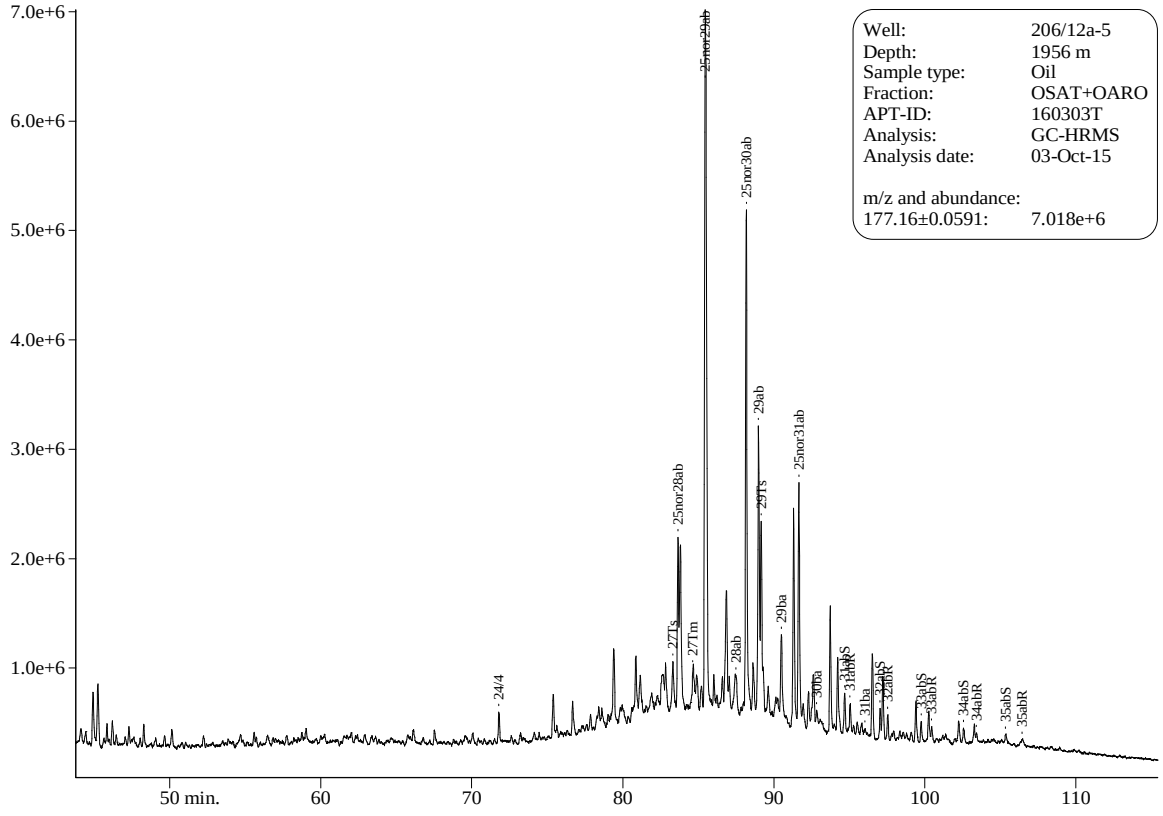


FIGURE Ap7.8 Alkane fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona)



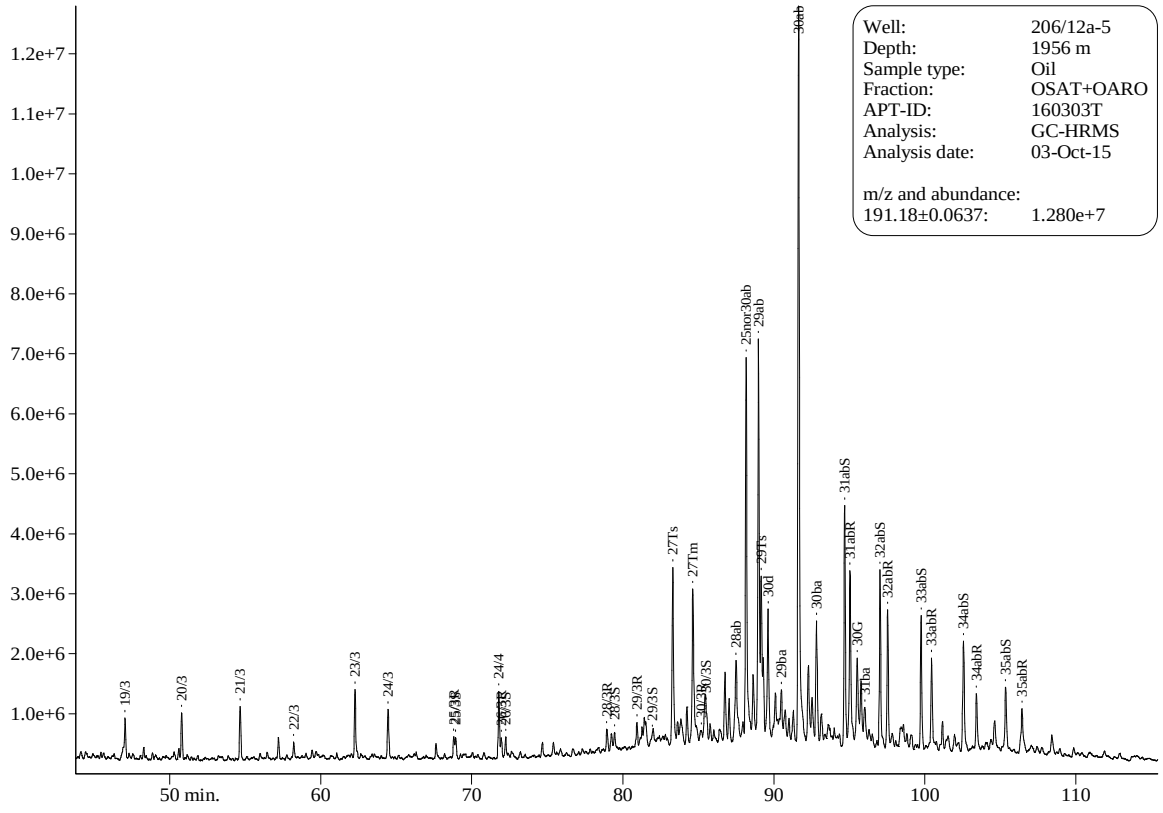
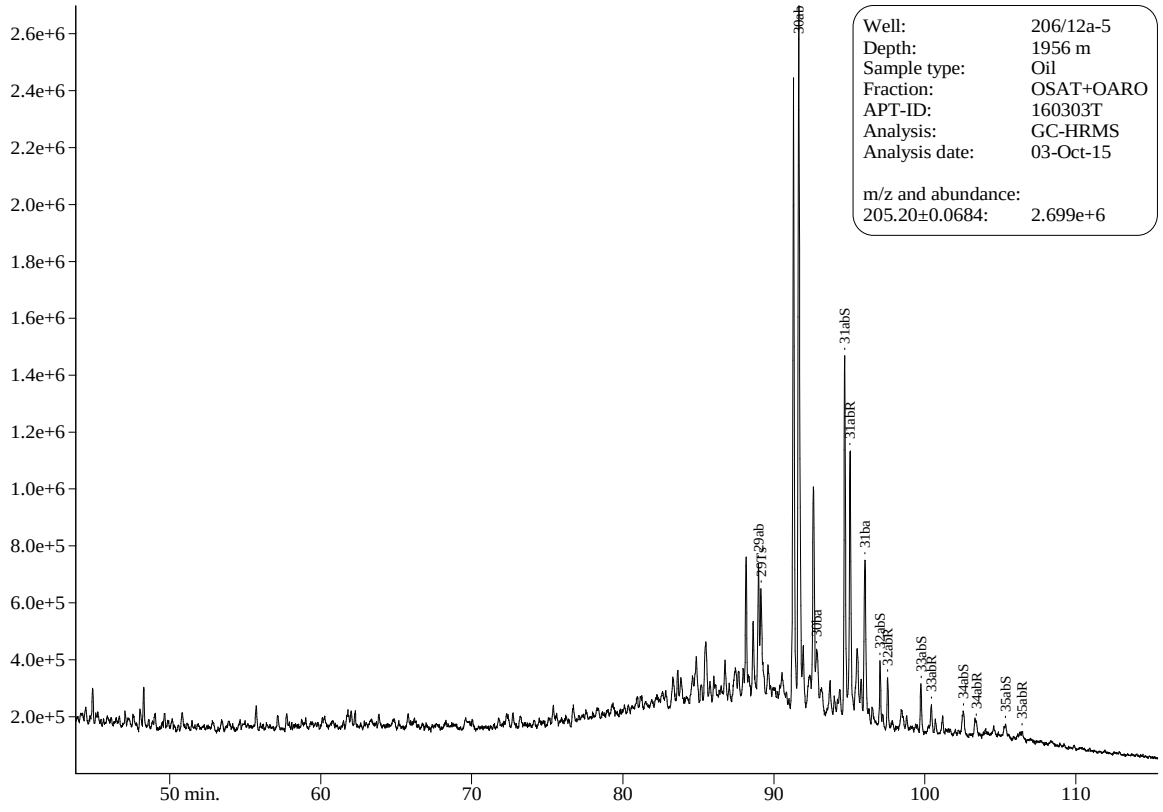


FIGURE Ap7.9 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1956m (Rona)



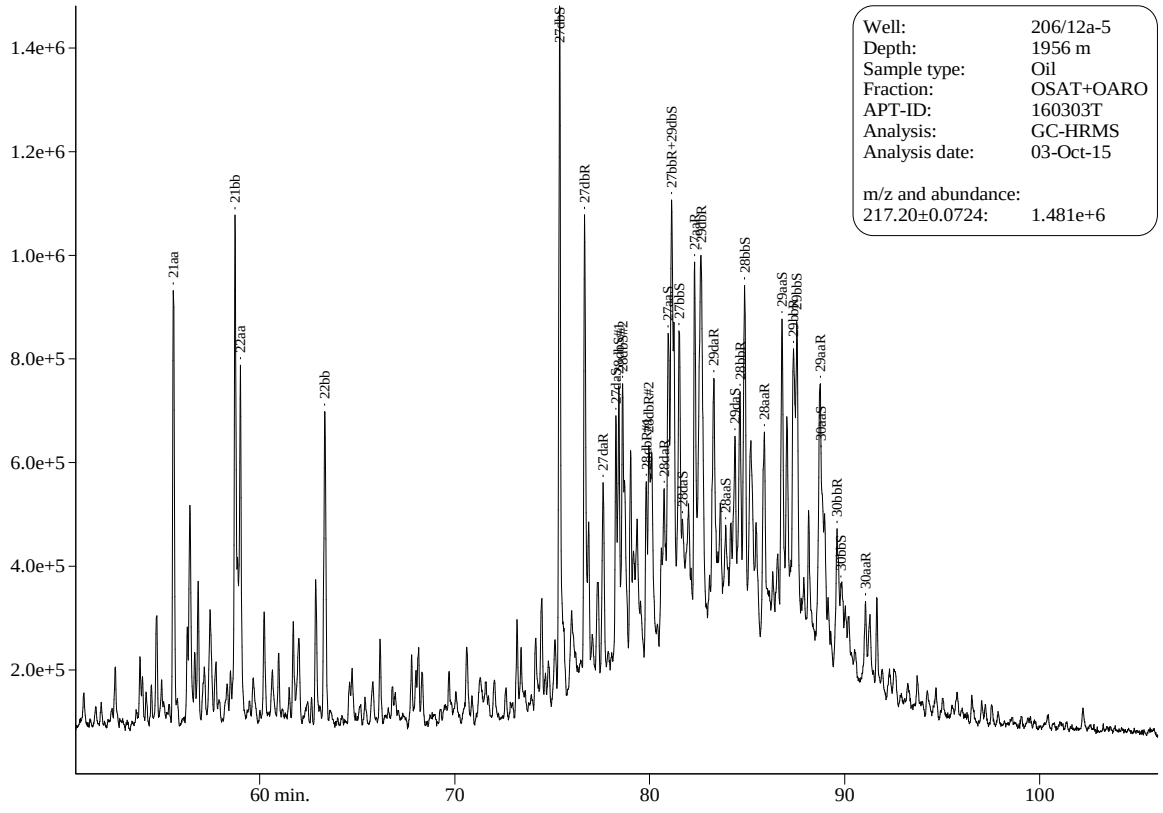
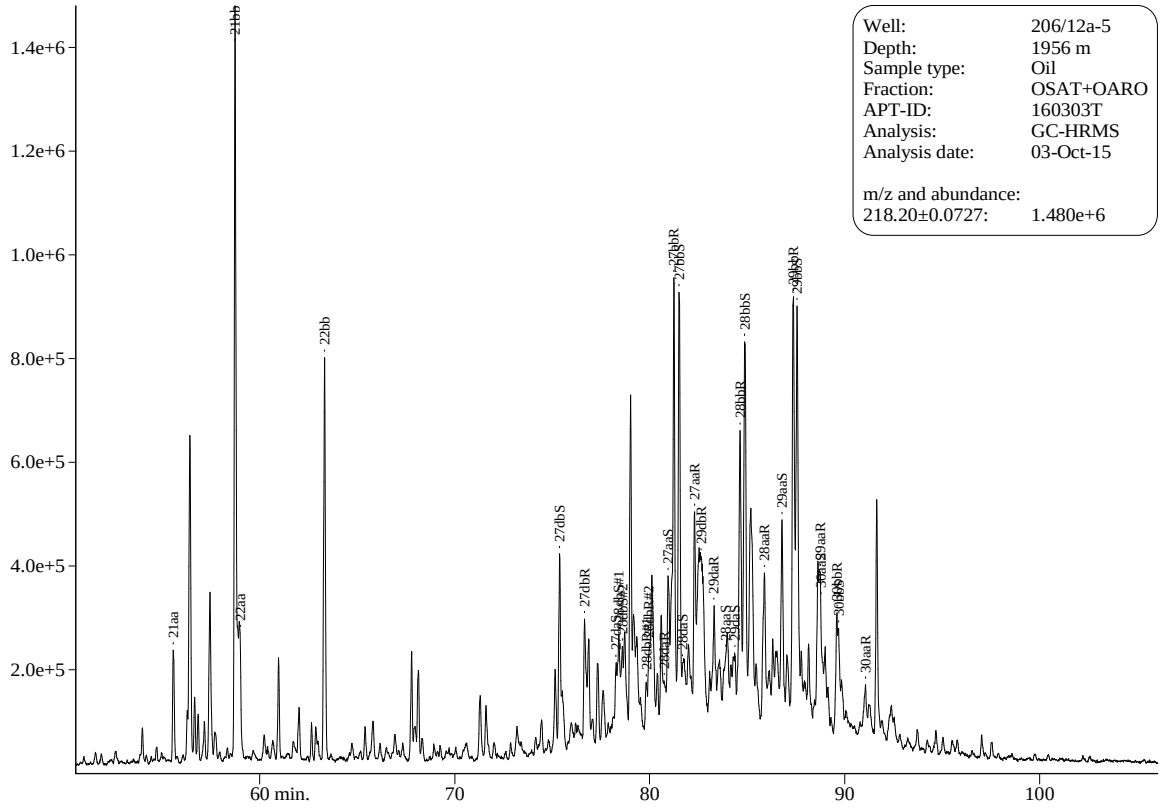


FIGURE Ap7.9 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1956m (Rona)



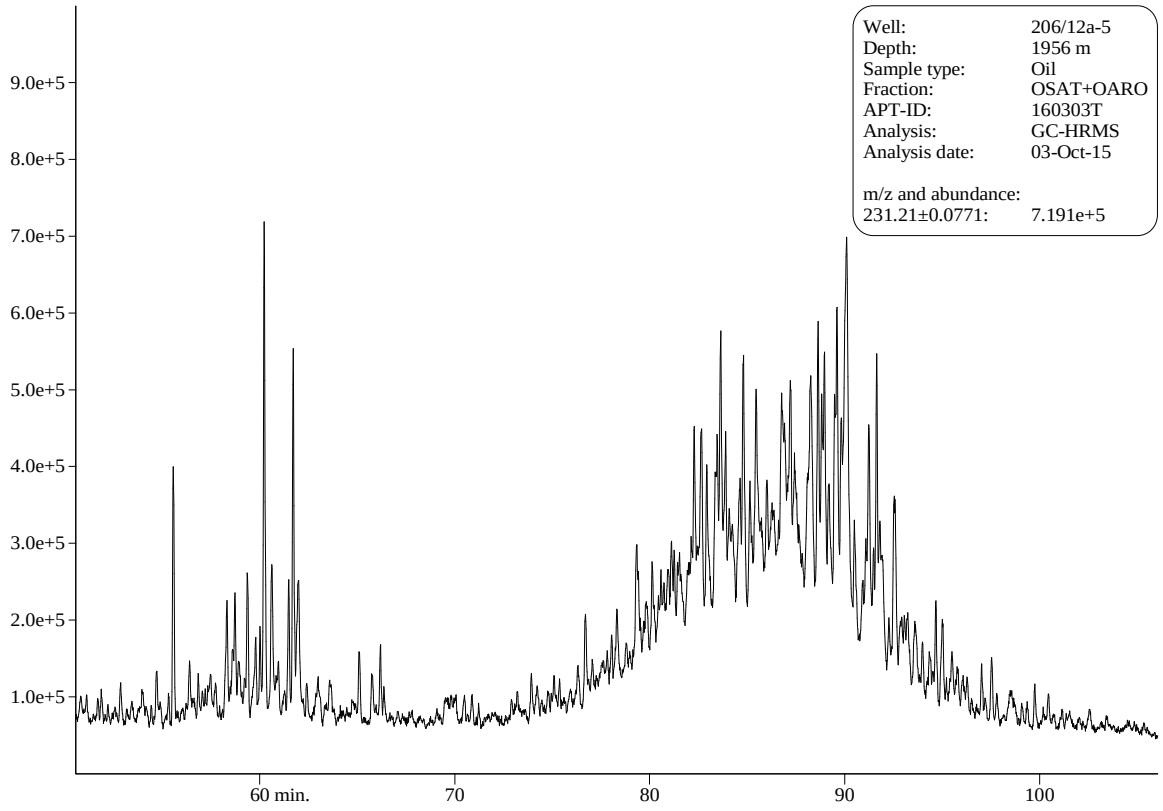
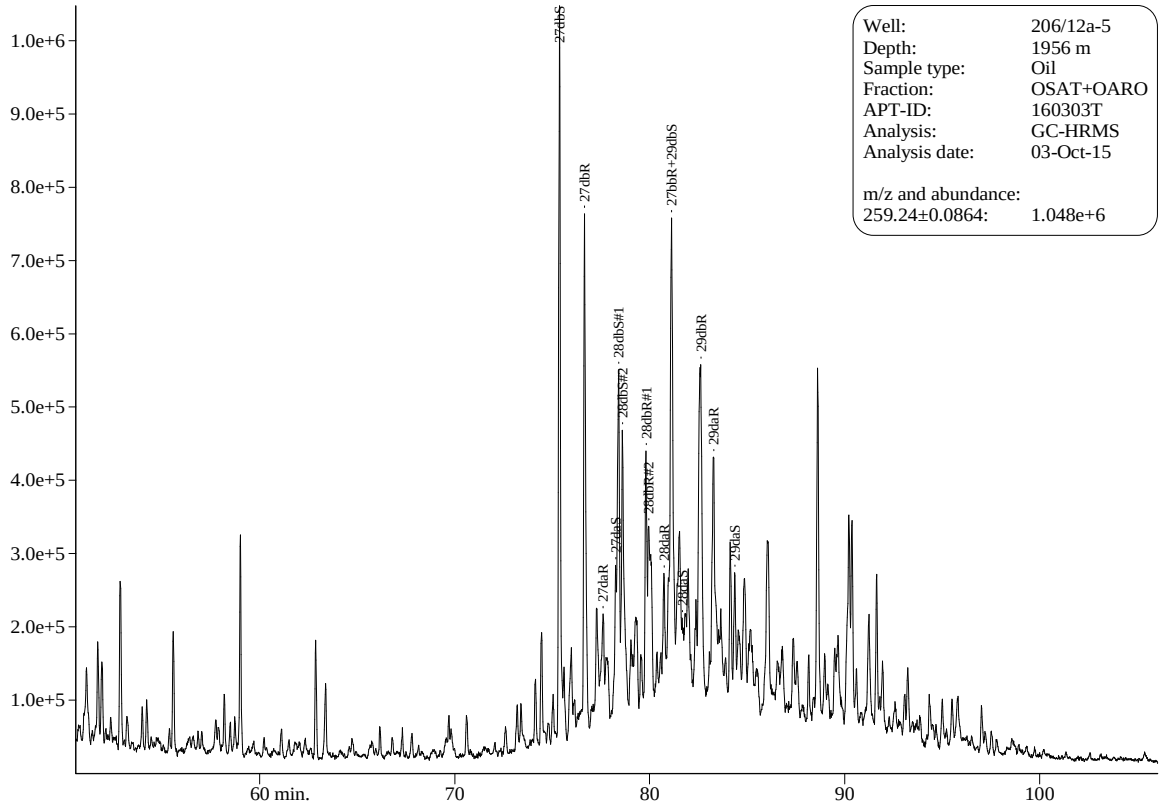


FIGURE Ap7.9 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1956m (Rona)



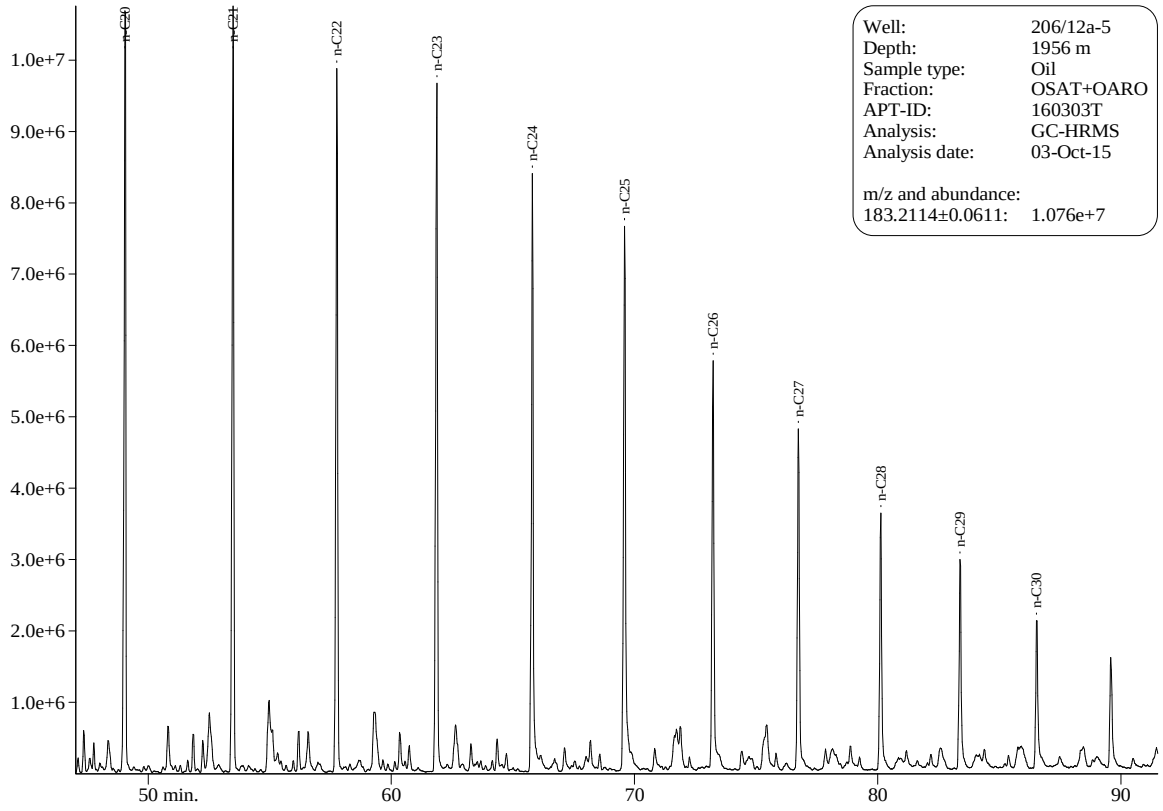
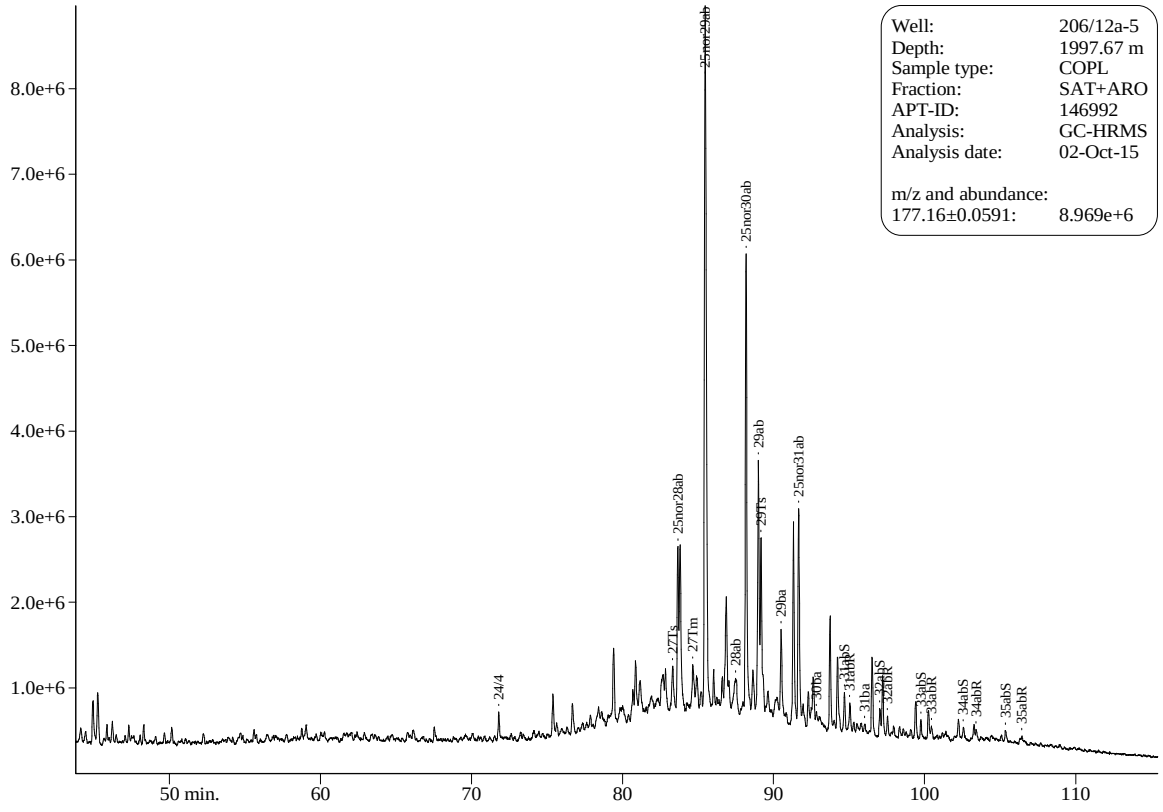


FIGURE Ap7.9 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1956m (Rona)



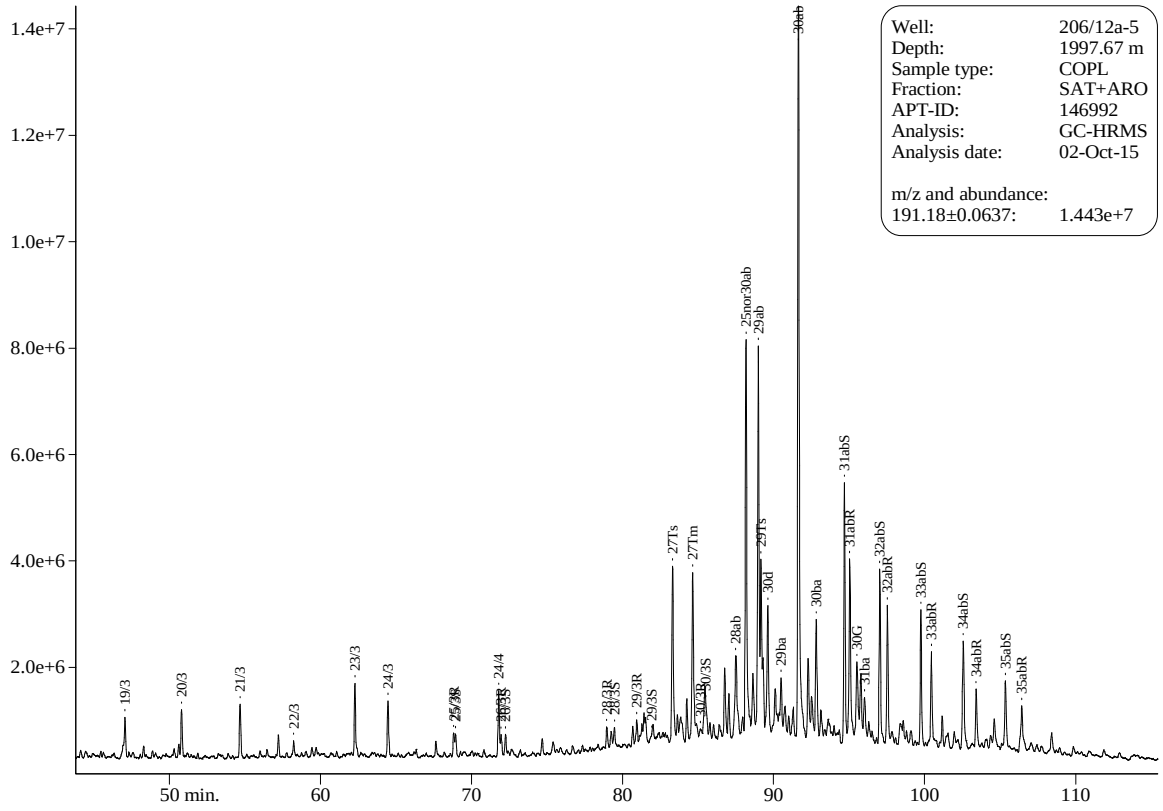
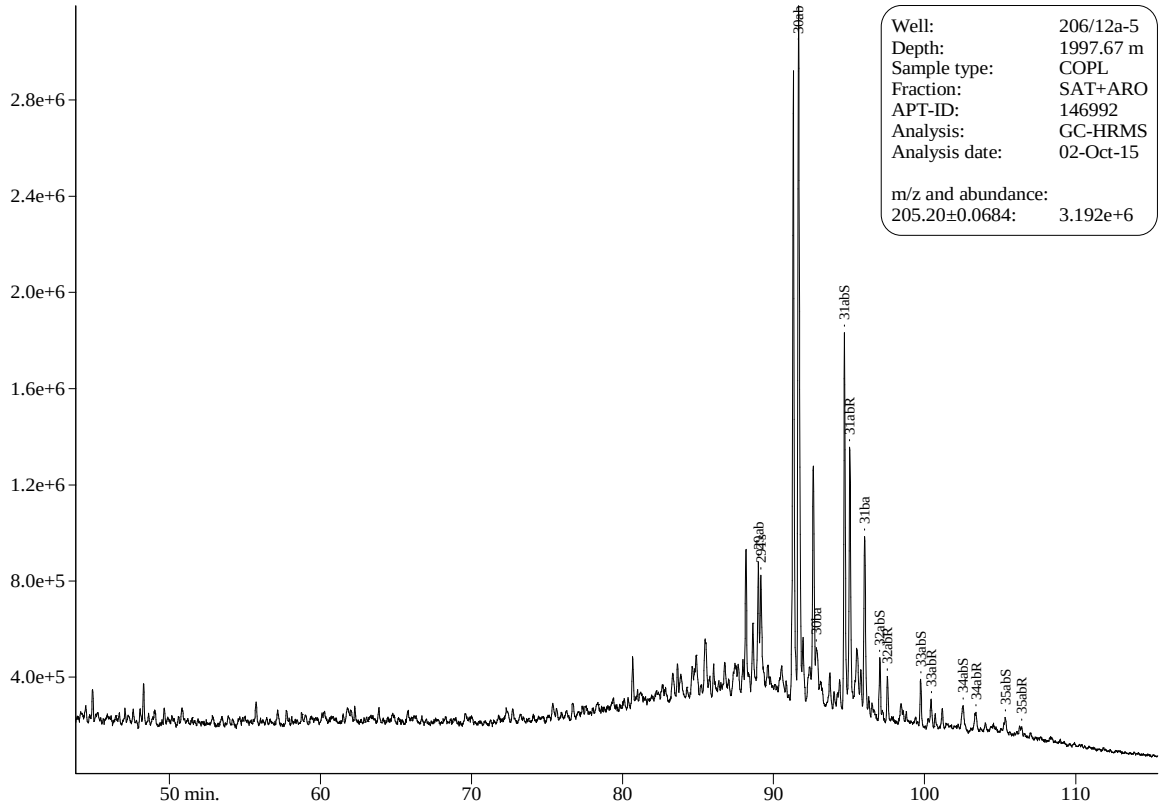


FIGURE Ap7.10 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1956.67m (Rona)



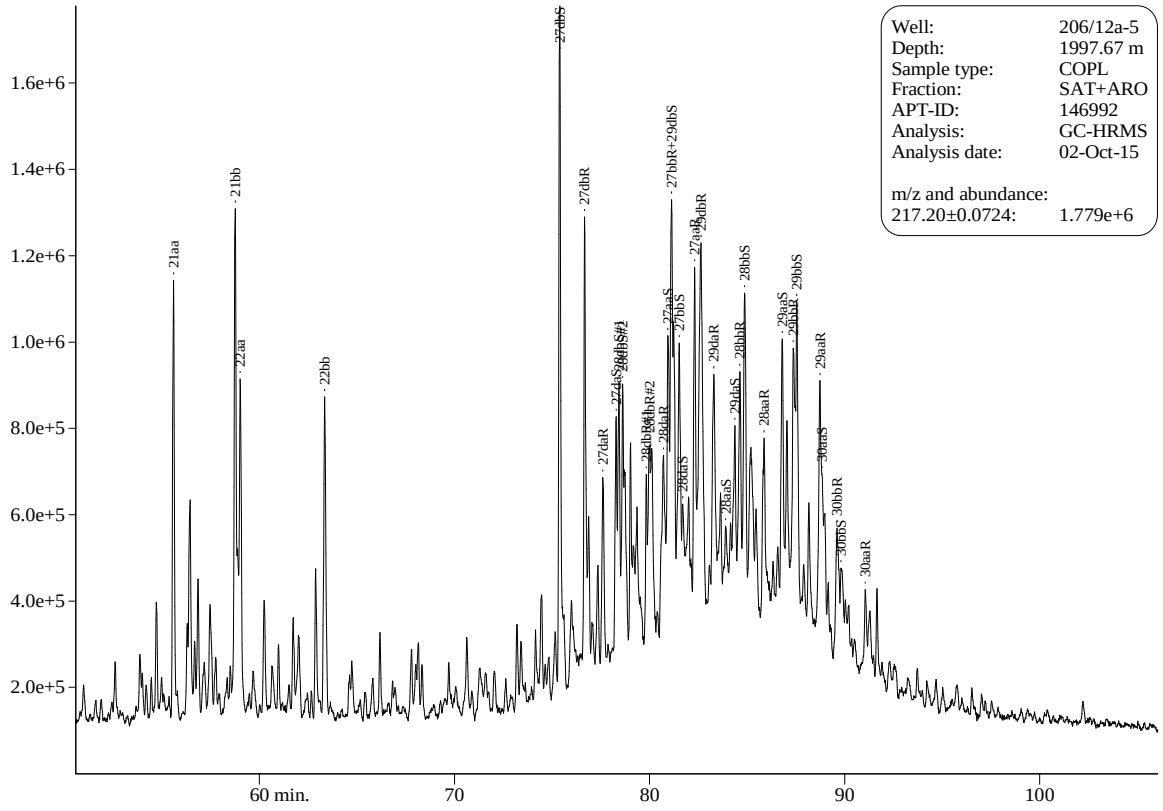
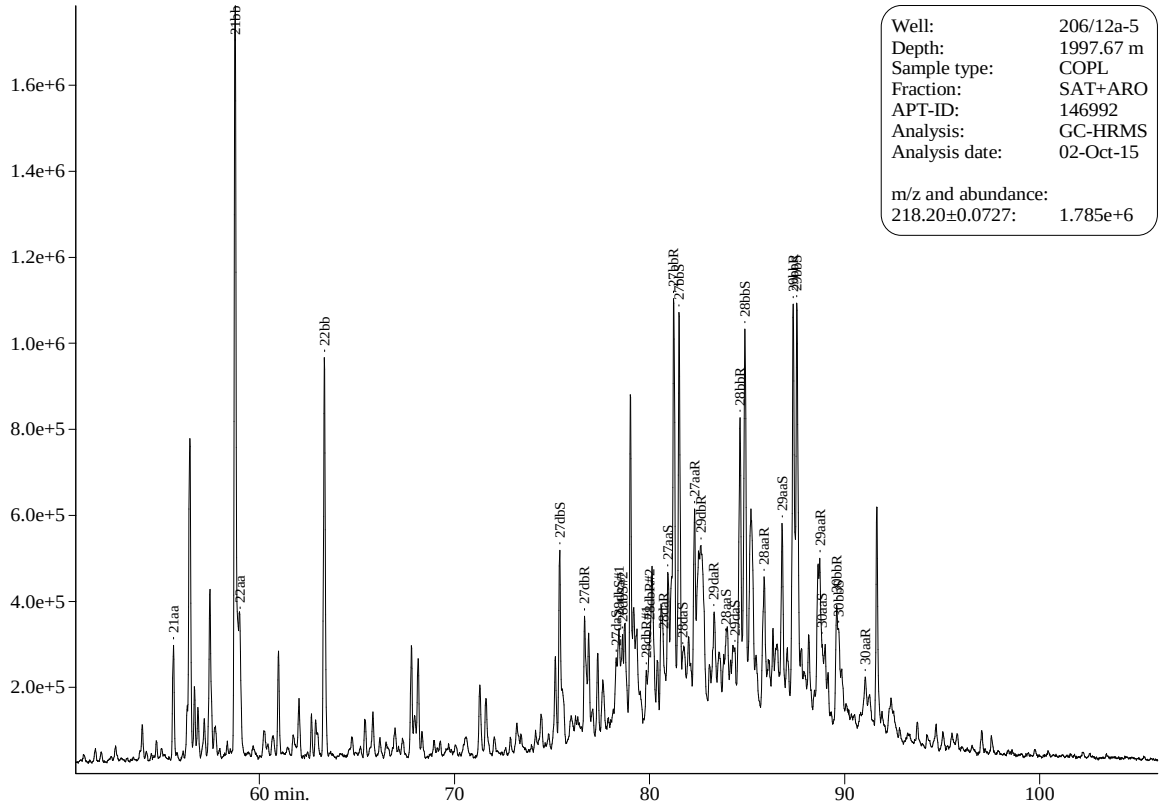


FIGURE Ap7.10 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1956.67m (Rona)



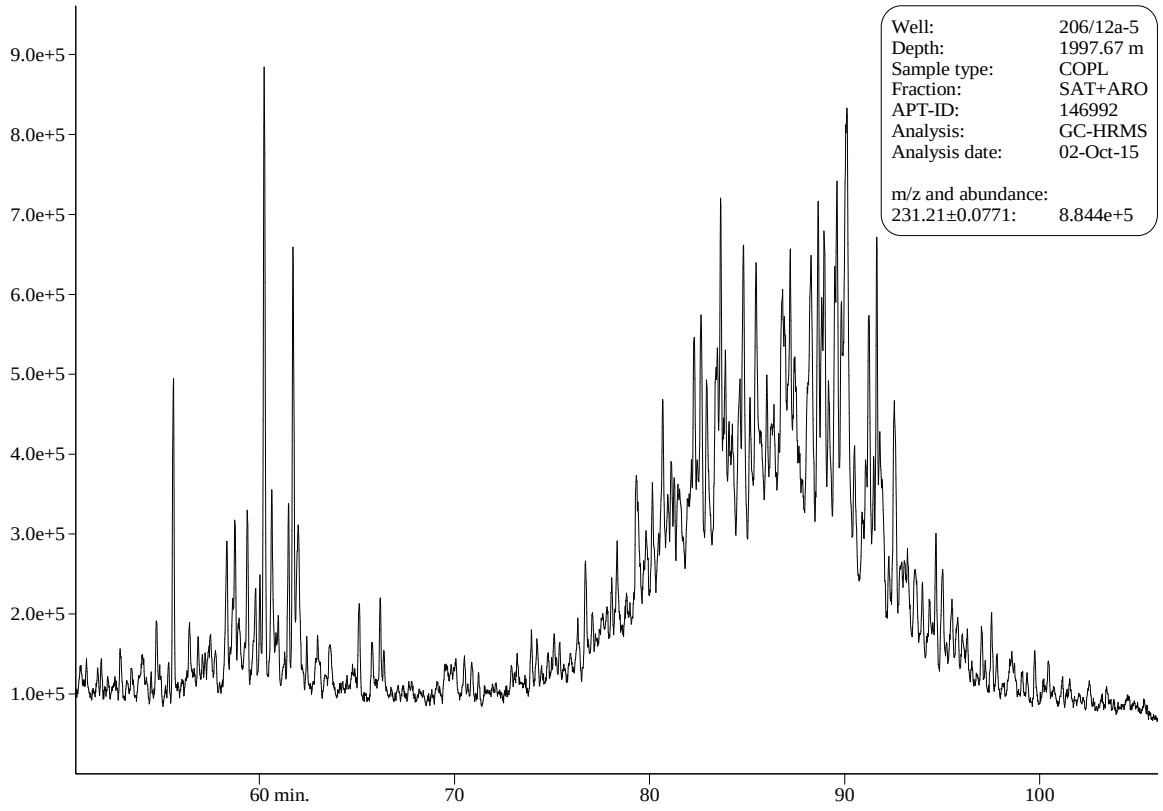
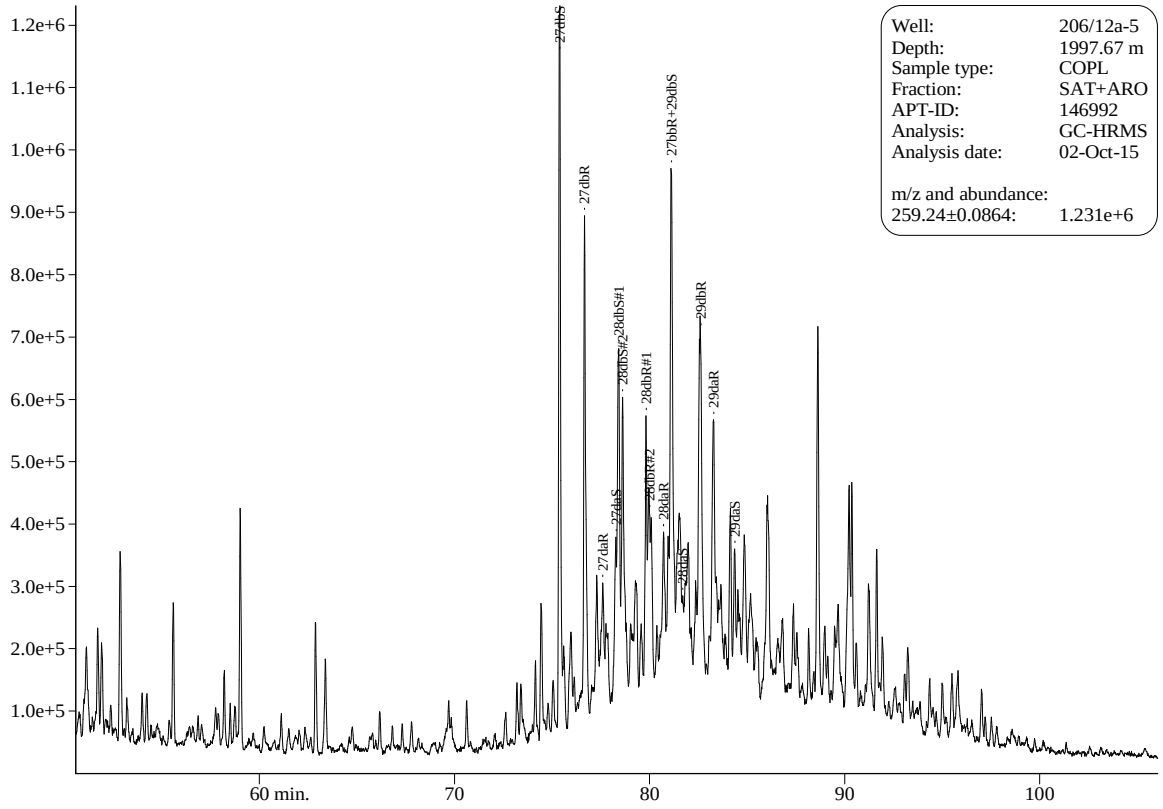


FIGURE Ap7.10 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1956.67m (Rona)



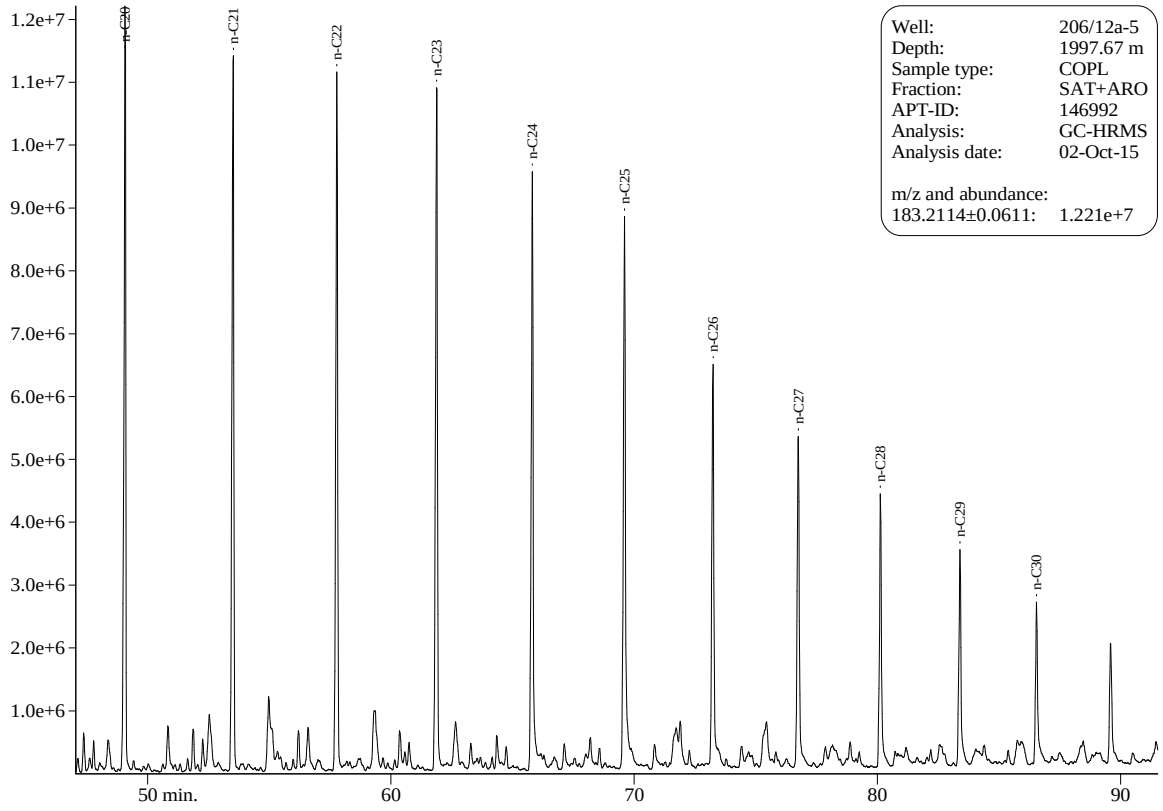
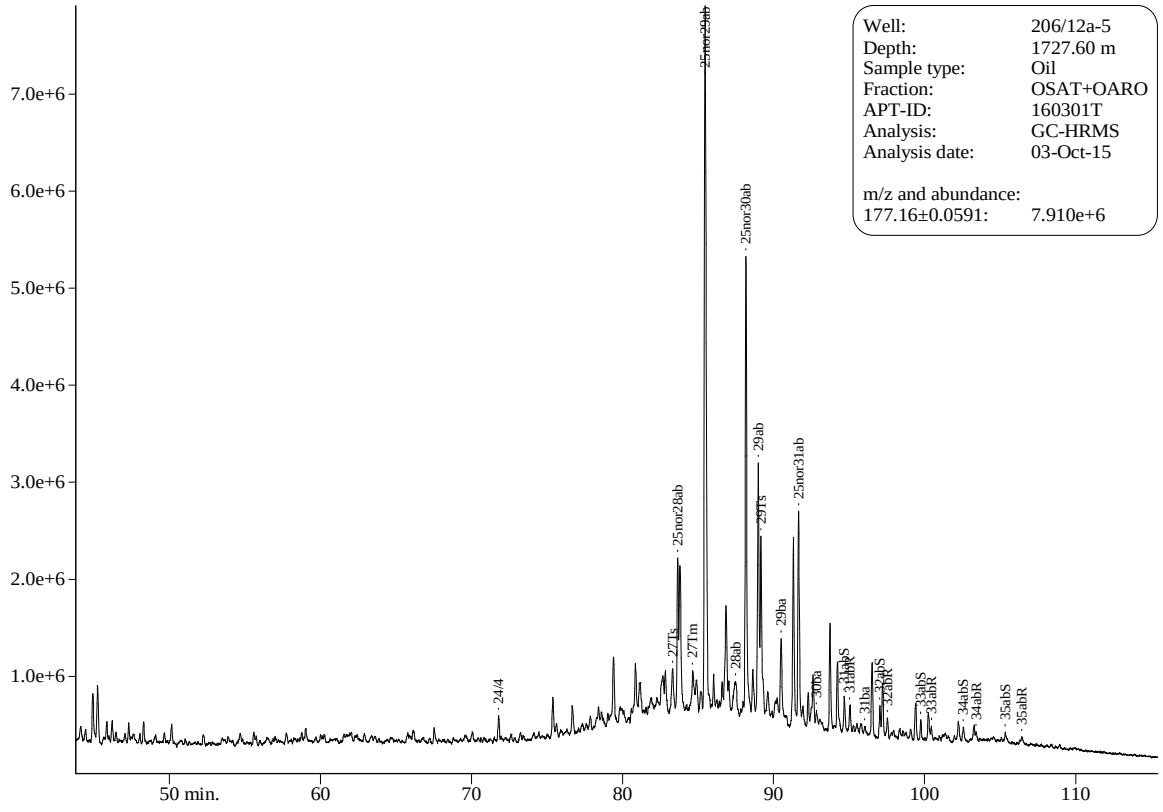


FIGURE Ap7.10 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1956.67m (Rona)



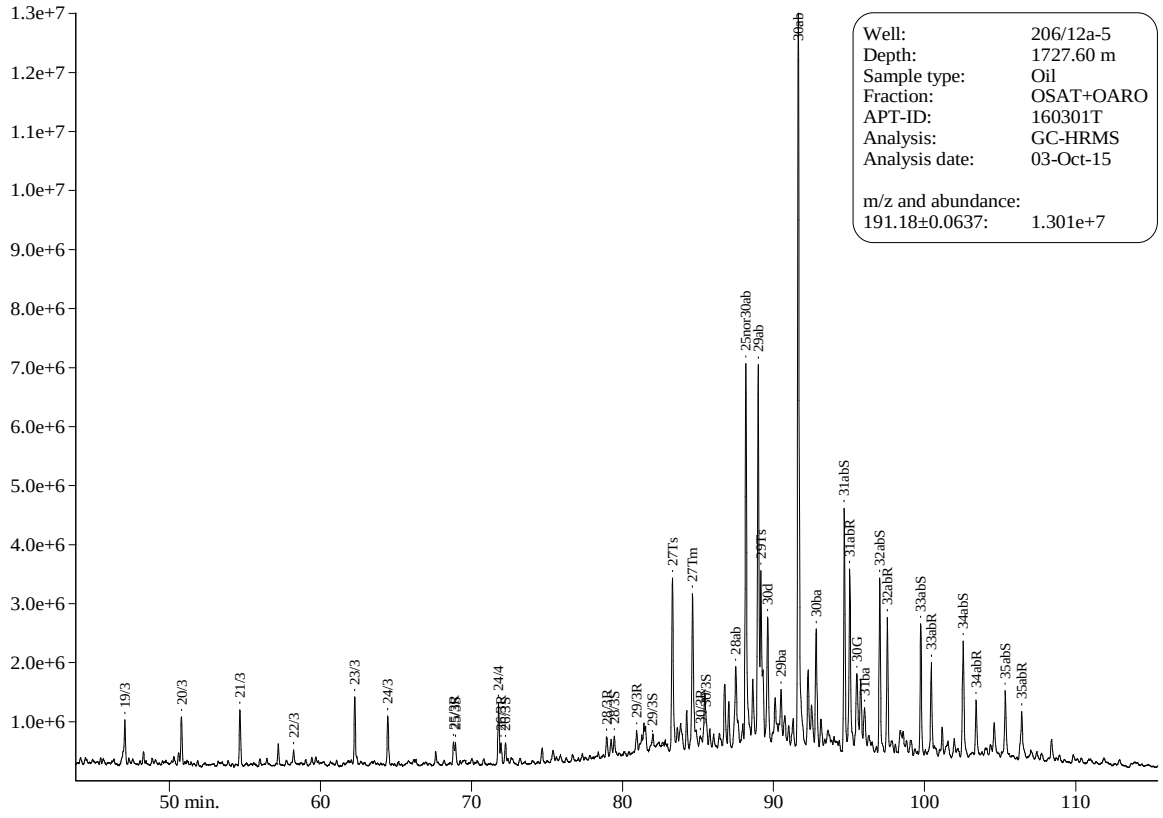
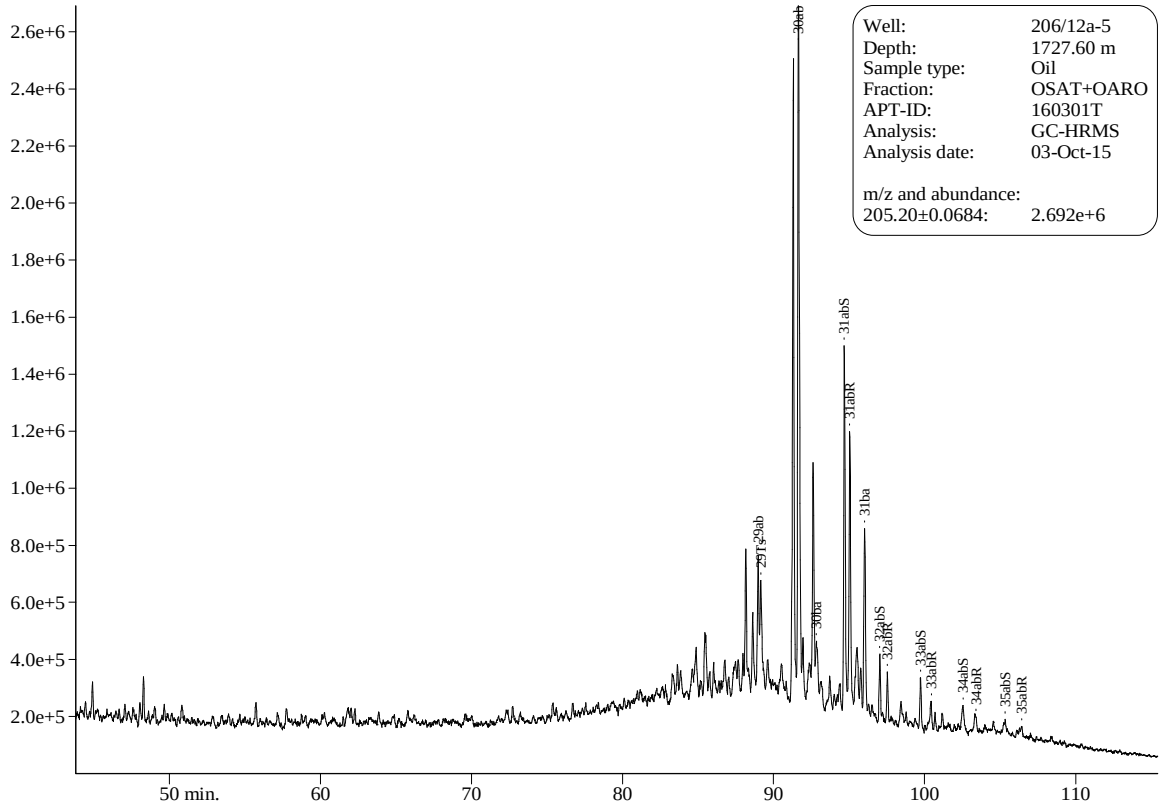


FIGURE Ap7.11 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1936-1963m (Rona)



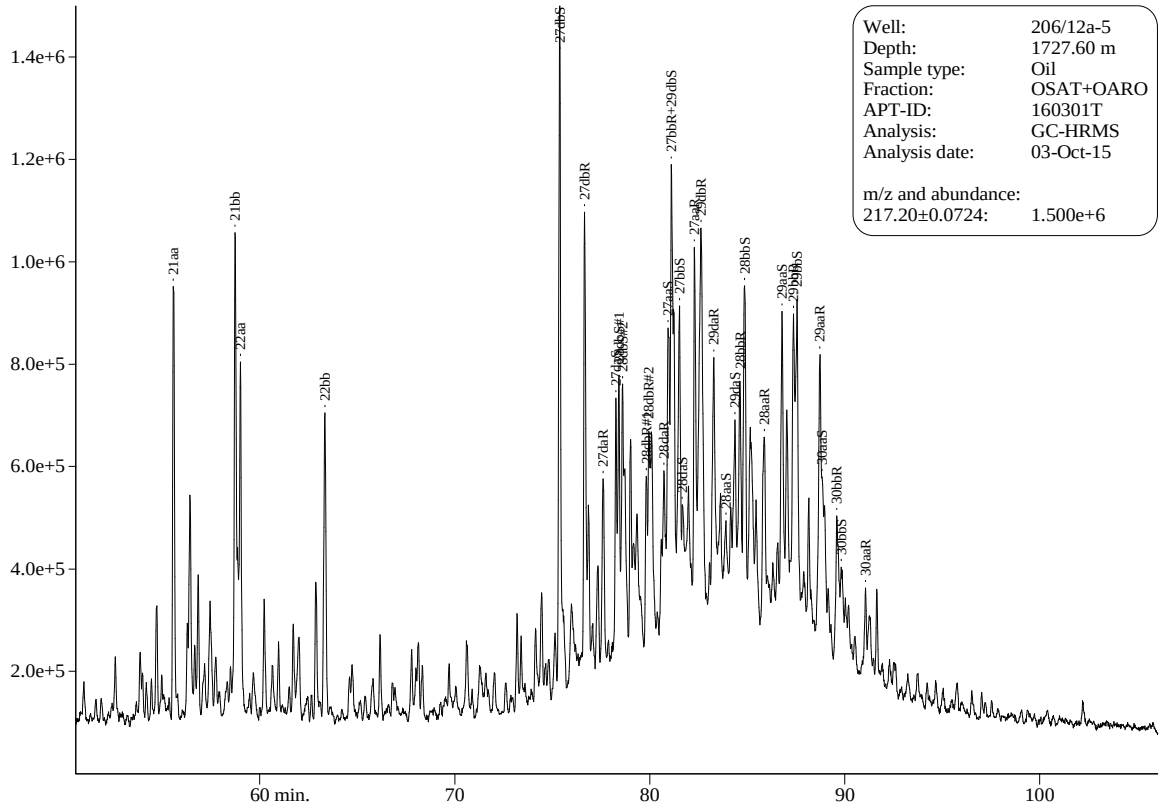
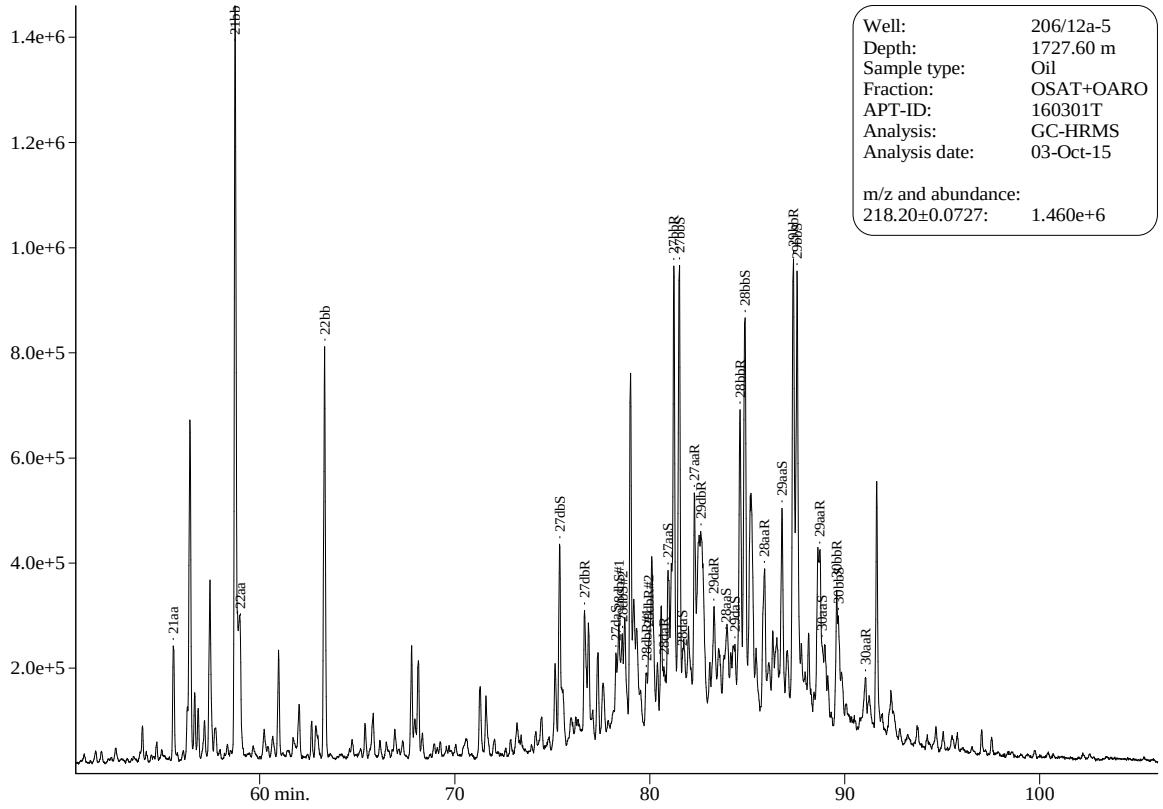


FIGURE Ap7.11 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1936-1963m (Rona)



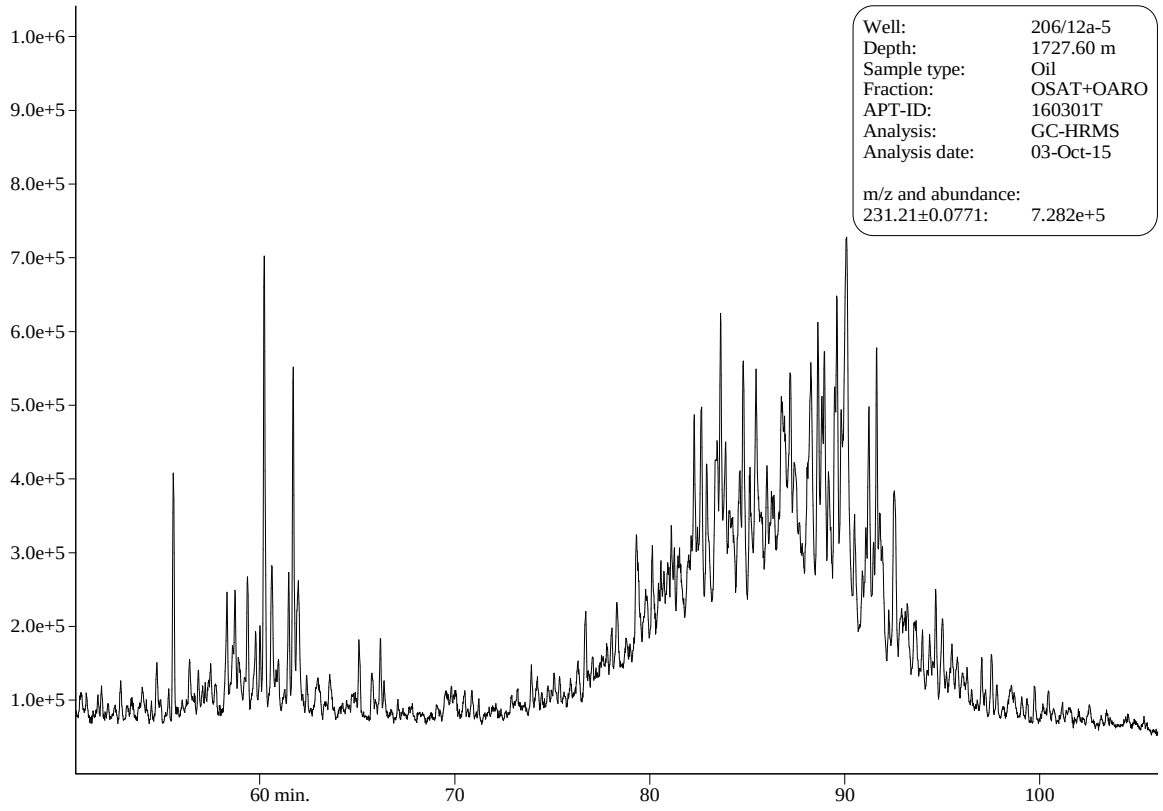
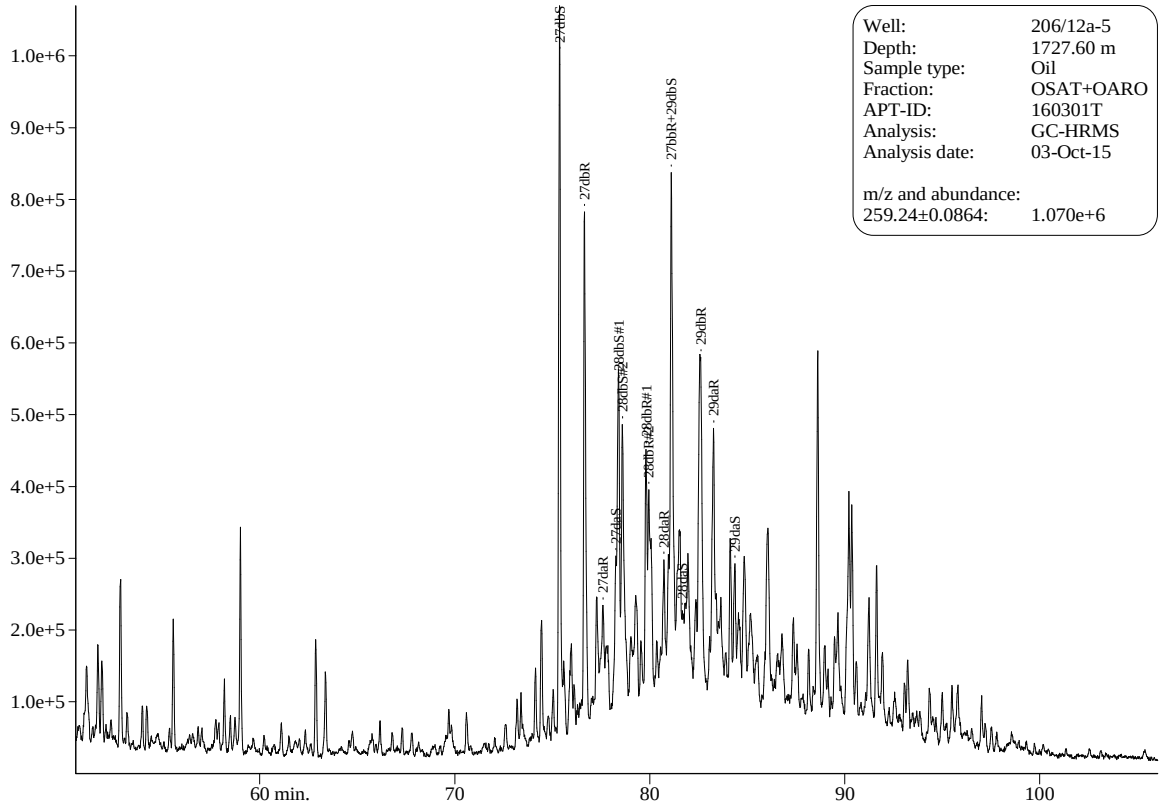


FIGURE Ap7.11 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1936-1963m (Rona)



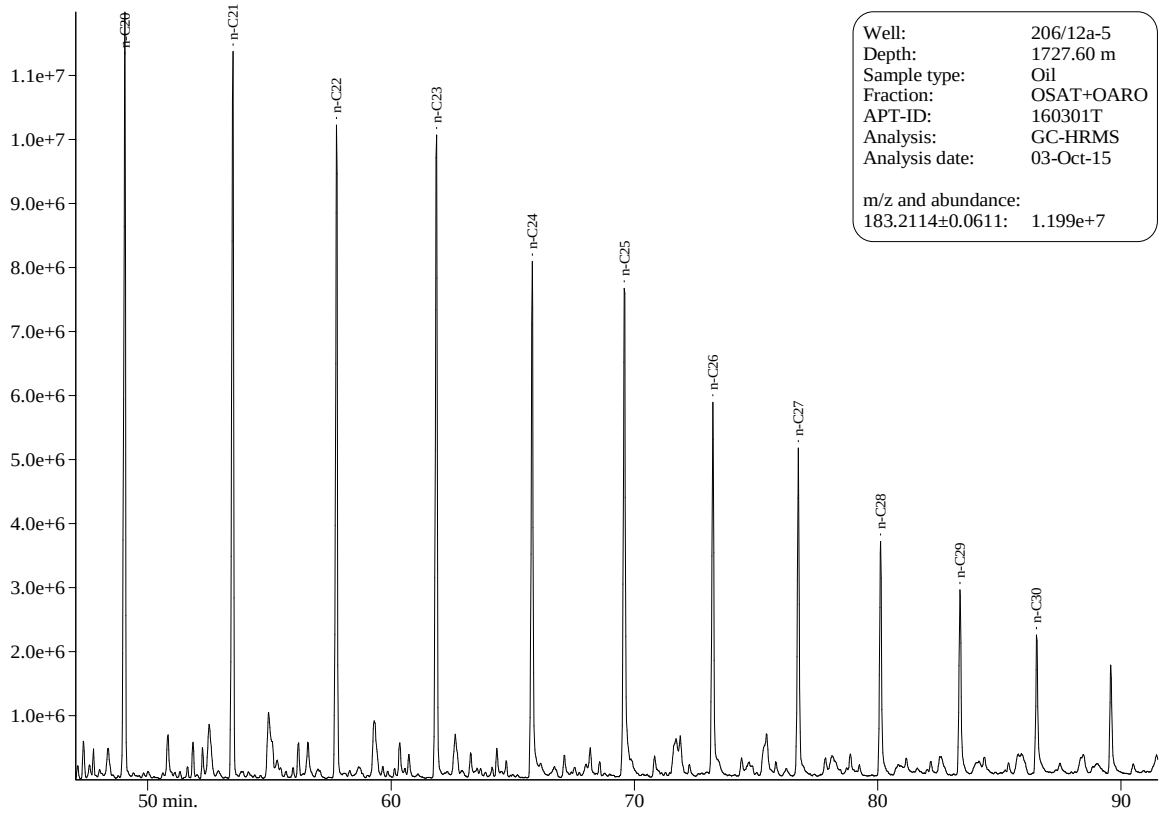
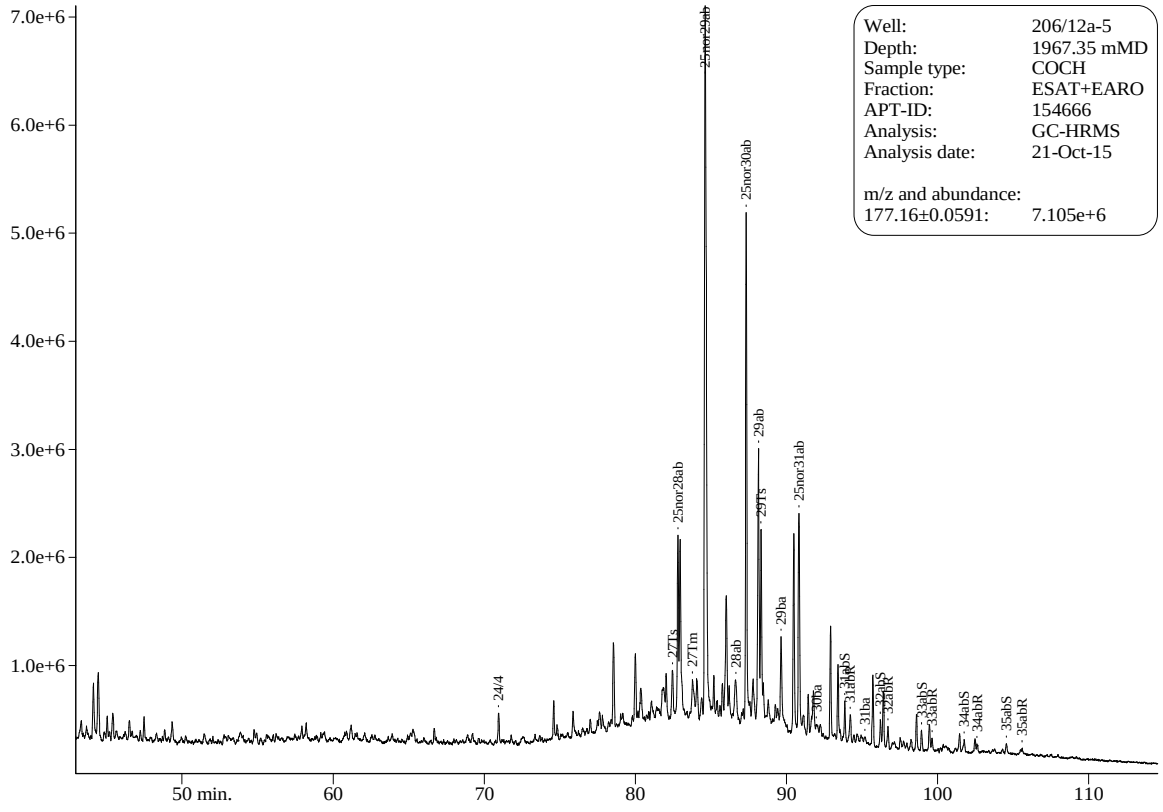


FIGURE Ap7.11 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1936-1963m (Rona)



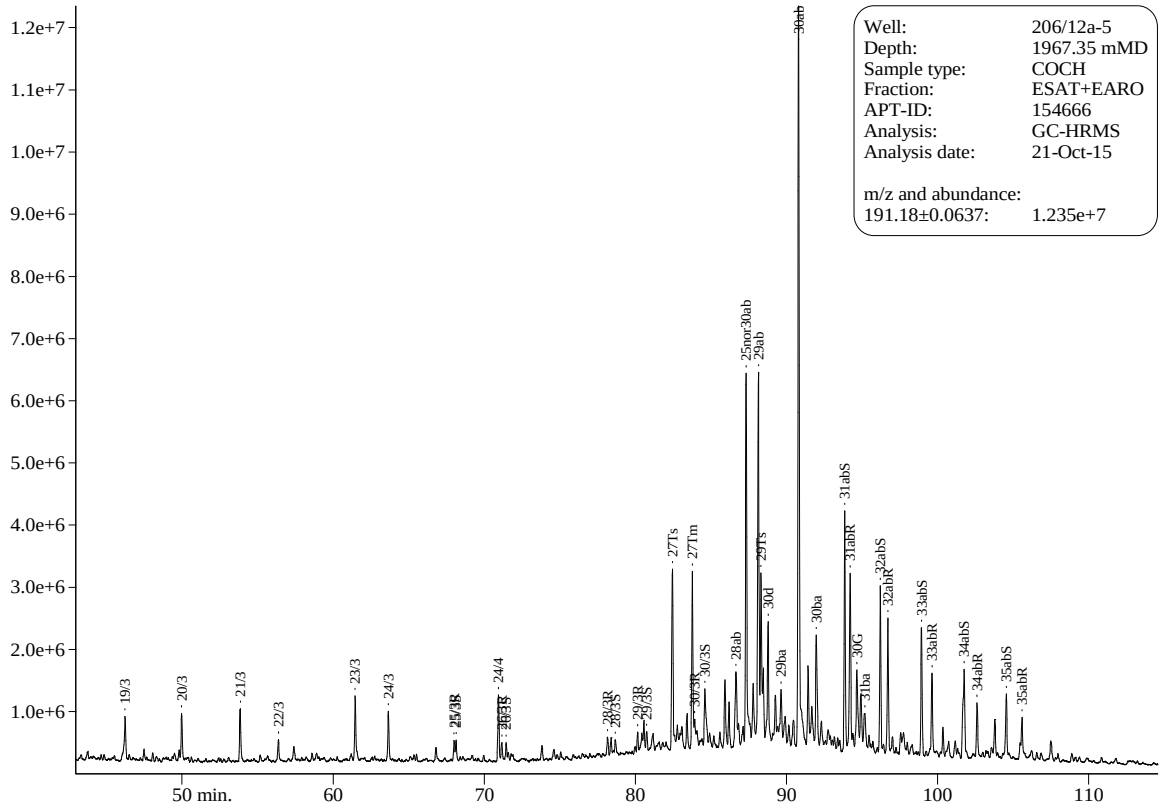
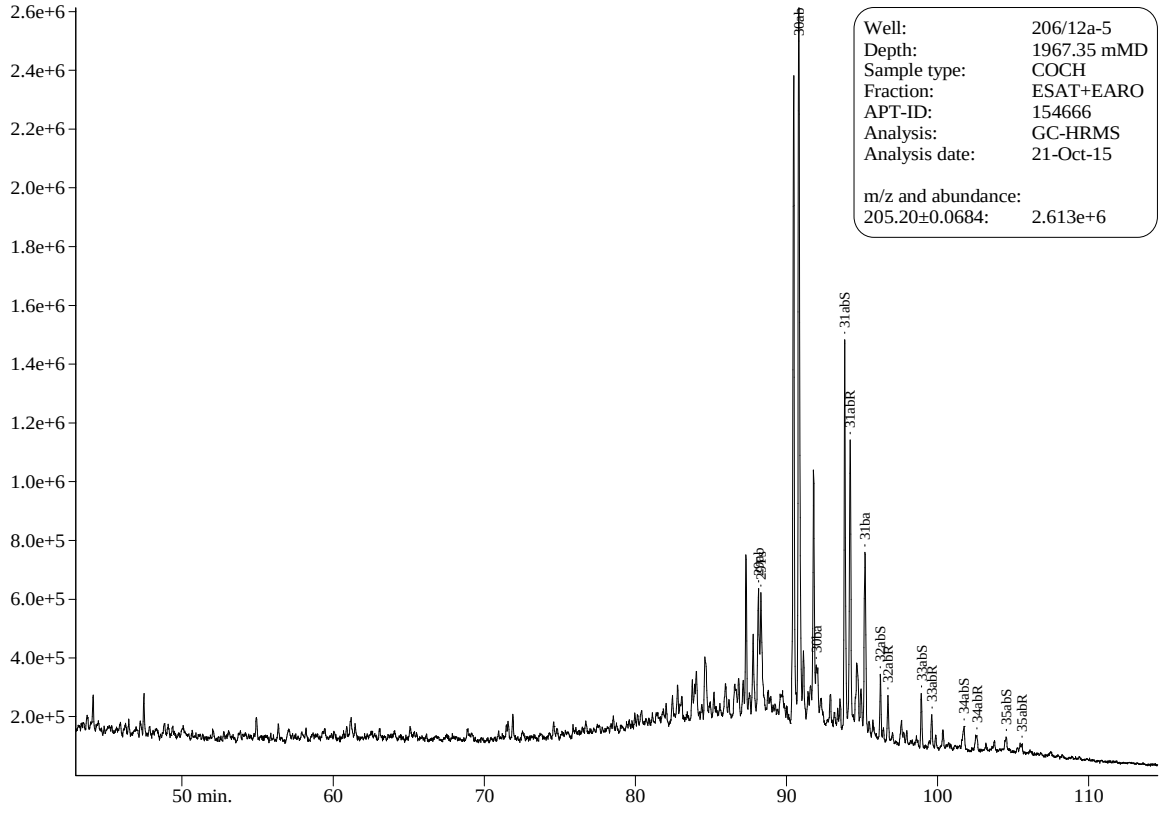


FIGURE Ap7.12 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE, 1967.35m (Rona)



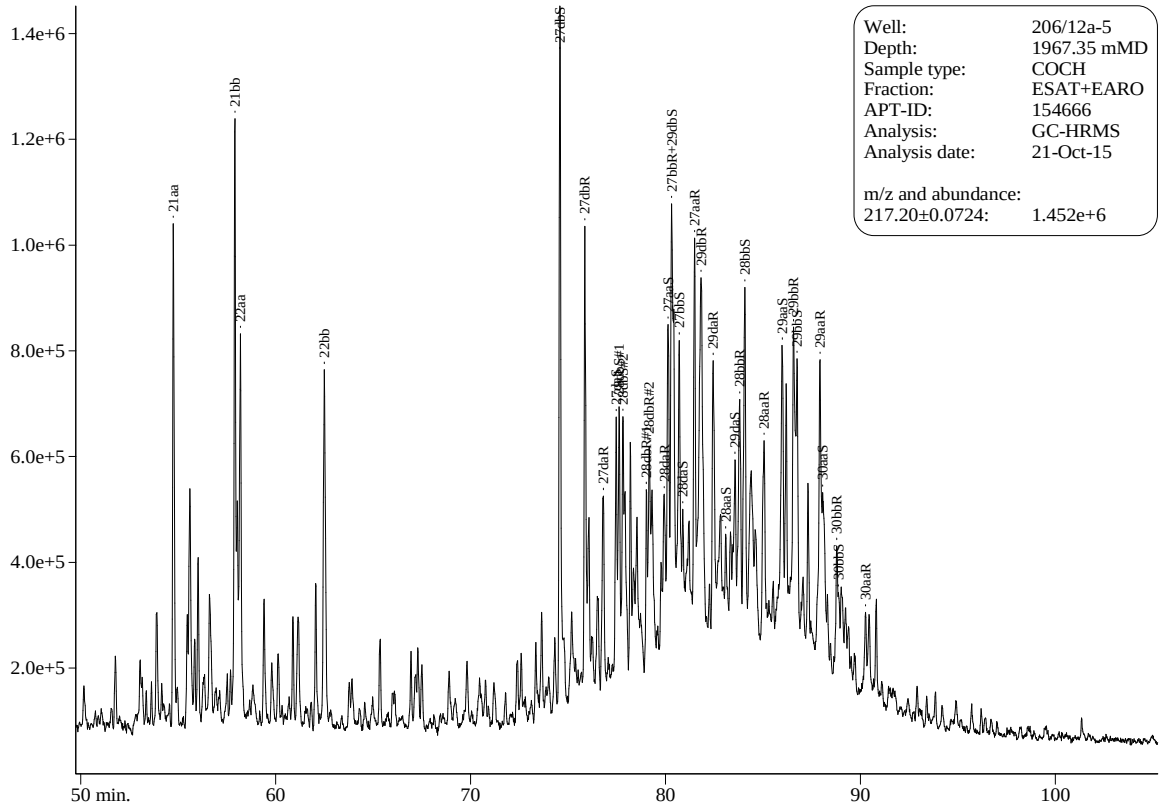
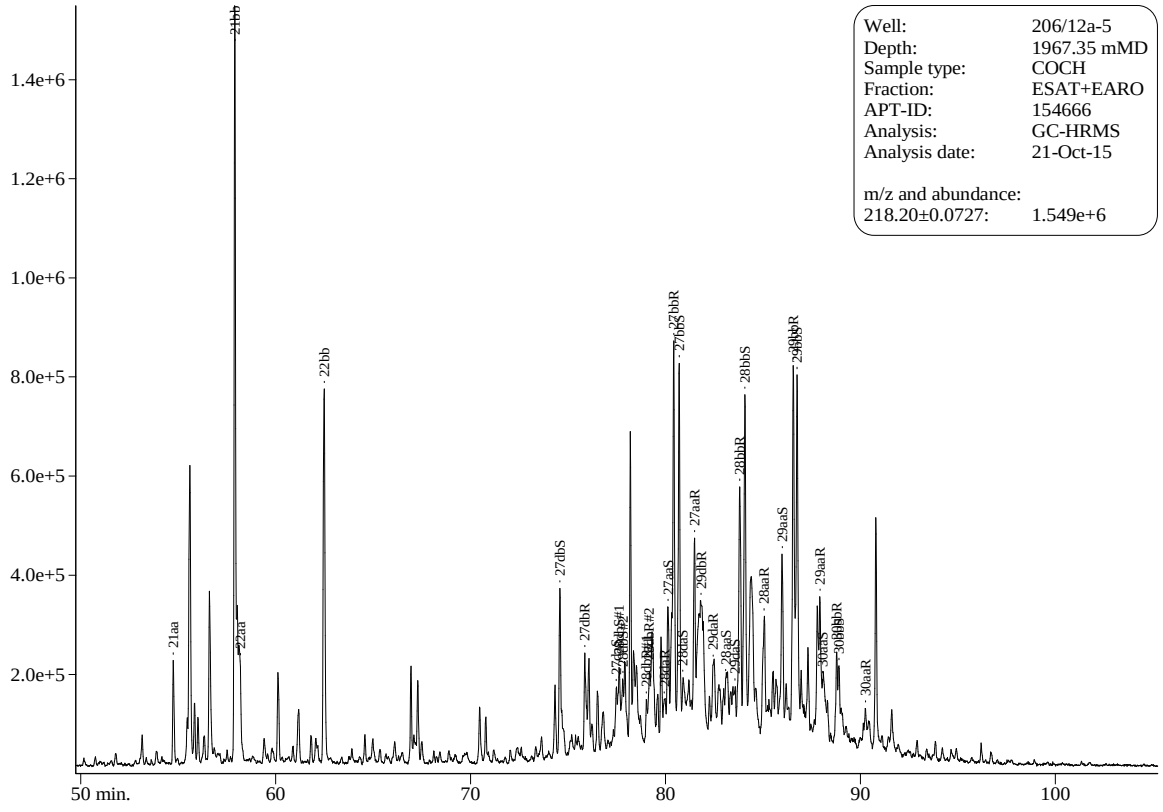


FIGURE Ap7.12 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE, 1967.35m (Rona)



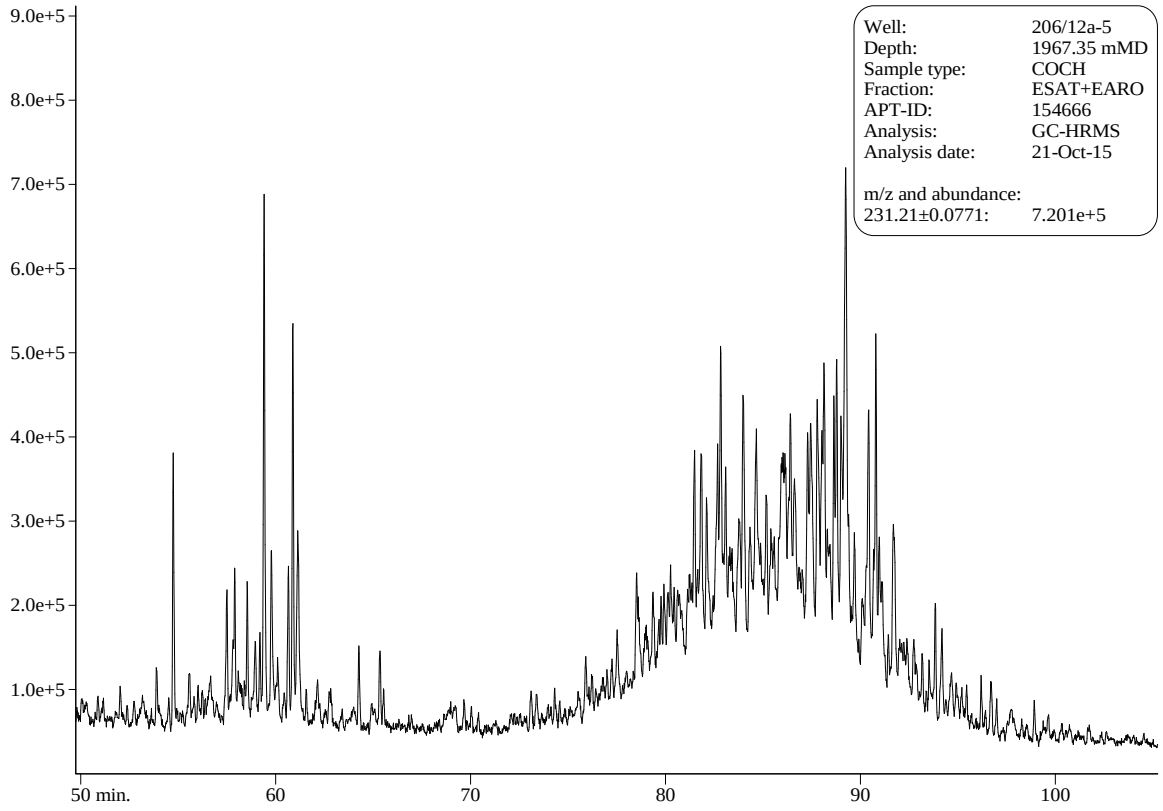
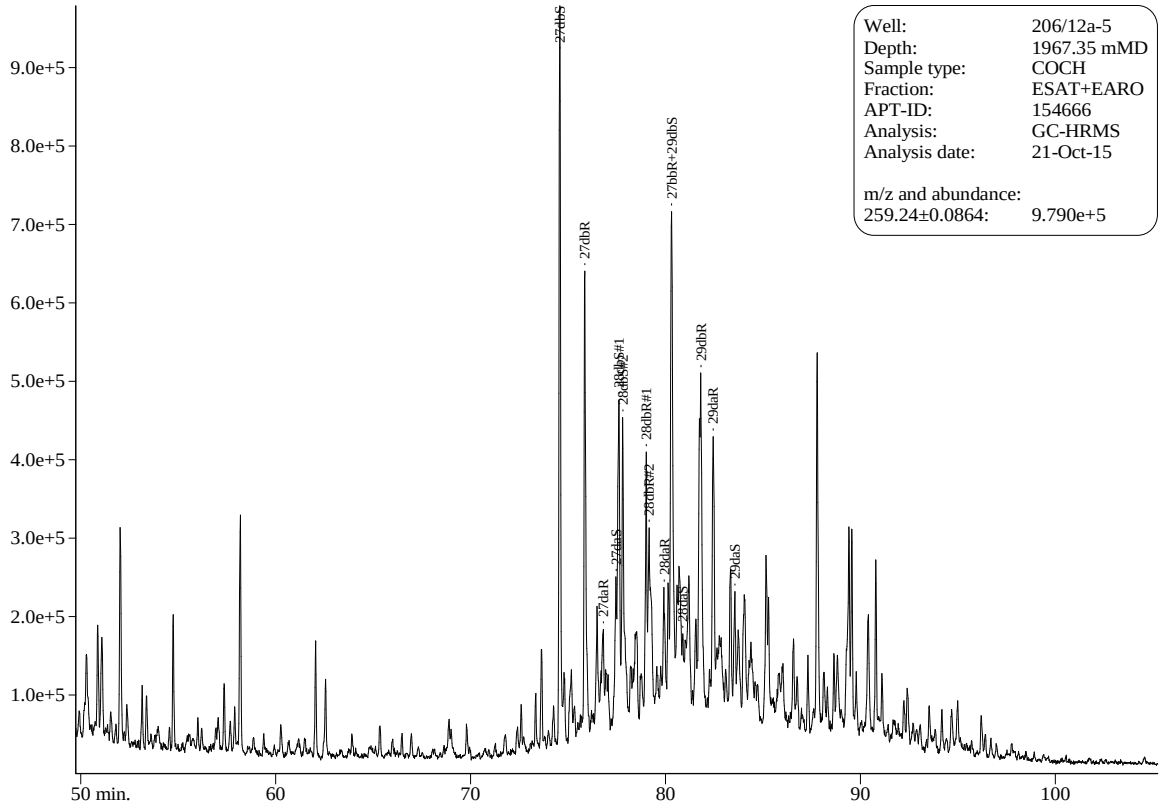


FIGURE Ap7.12 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE, 1967.35m (Rona)



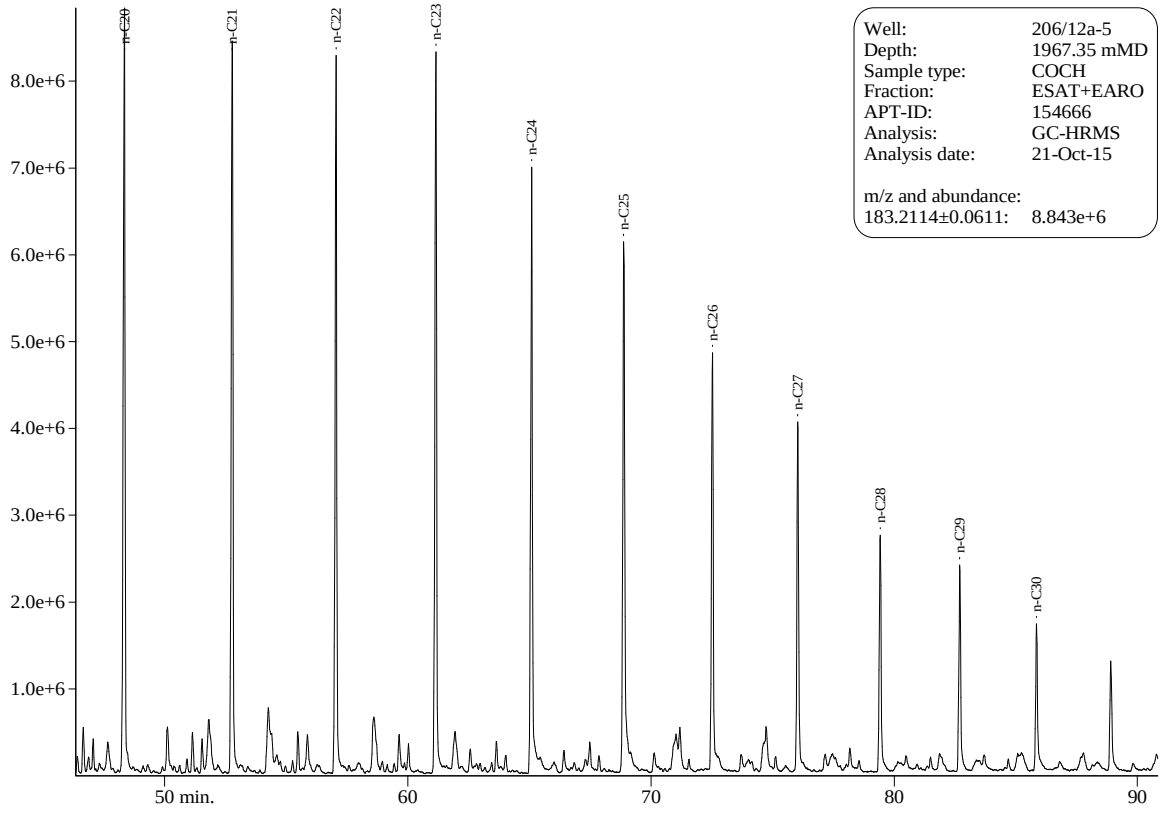
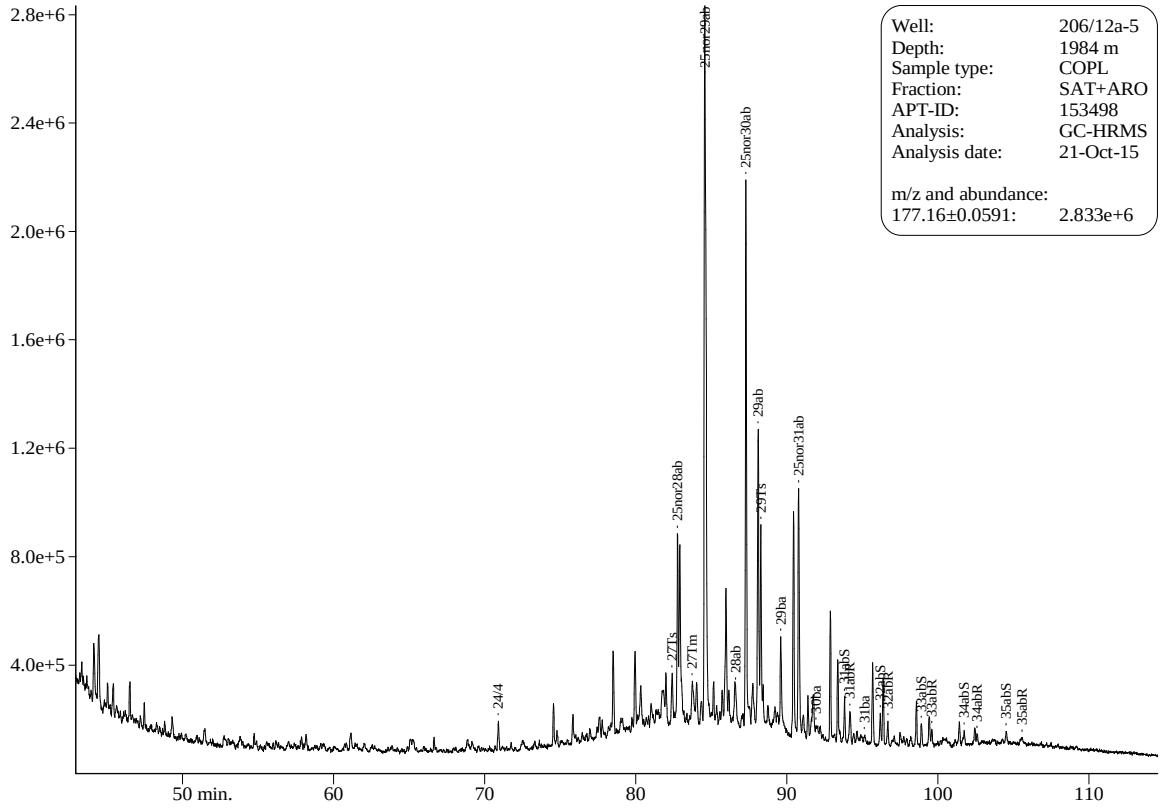


FIGURE Ap7.12 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE, 1967.35m (Rona)



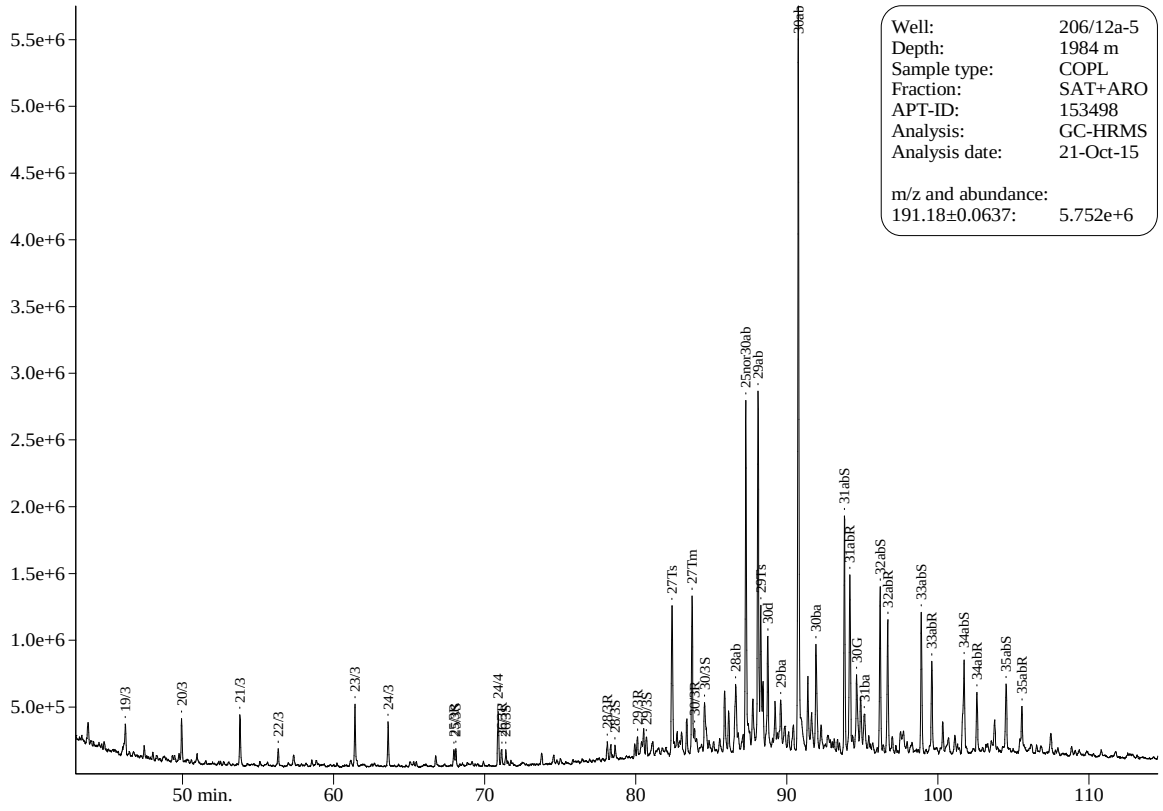
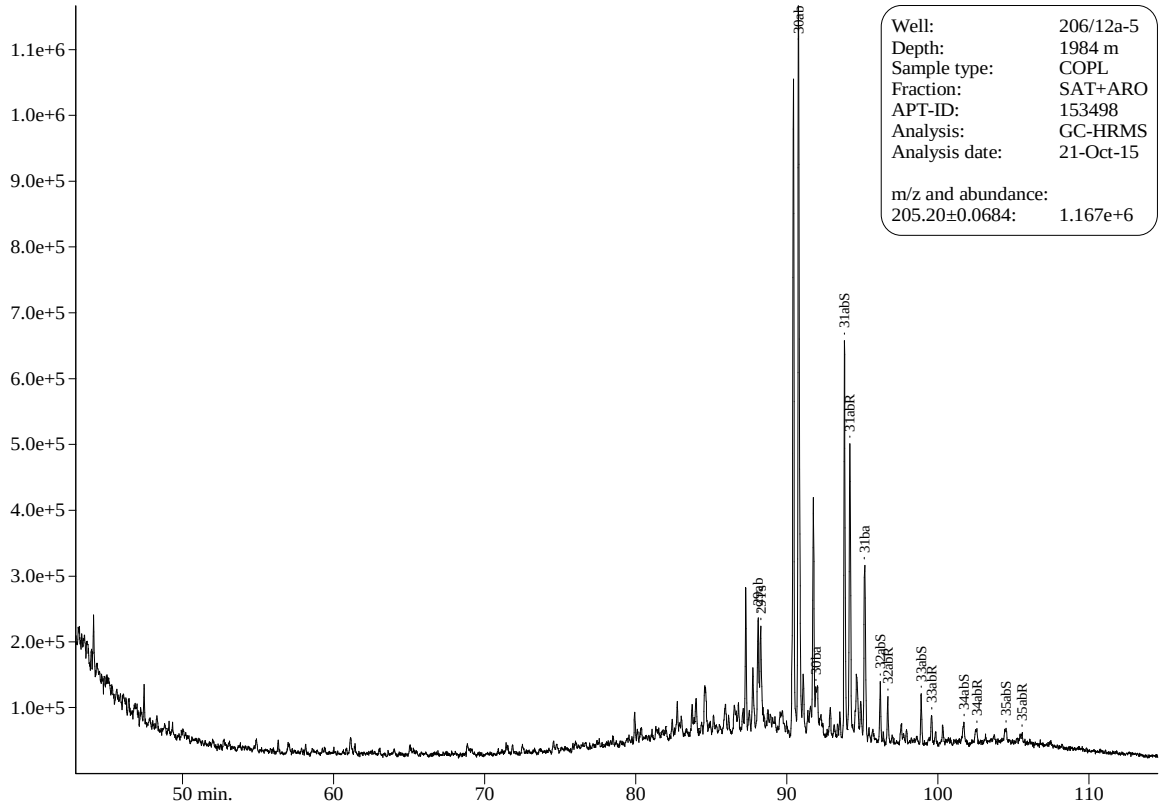


FIGURE Ap7.13 Alkane fraction GC-MS fragmentograms: 206/12a-5, SWC, 1984m (LCG Unit III)



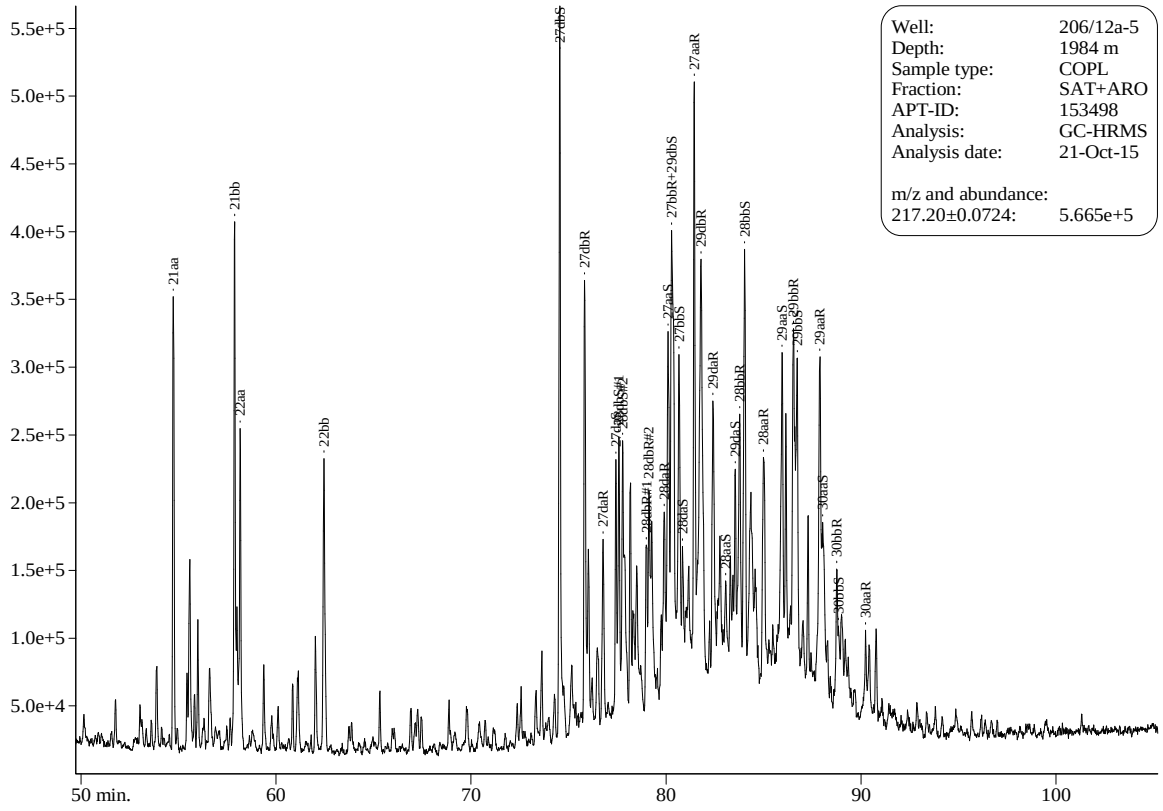
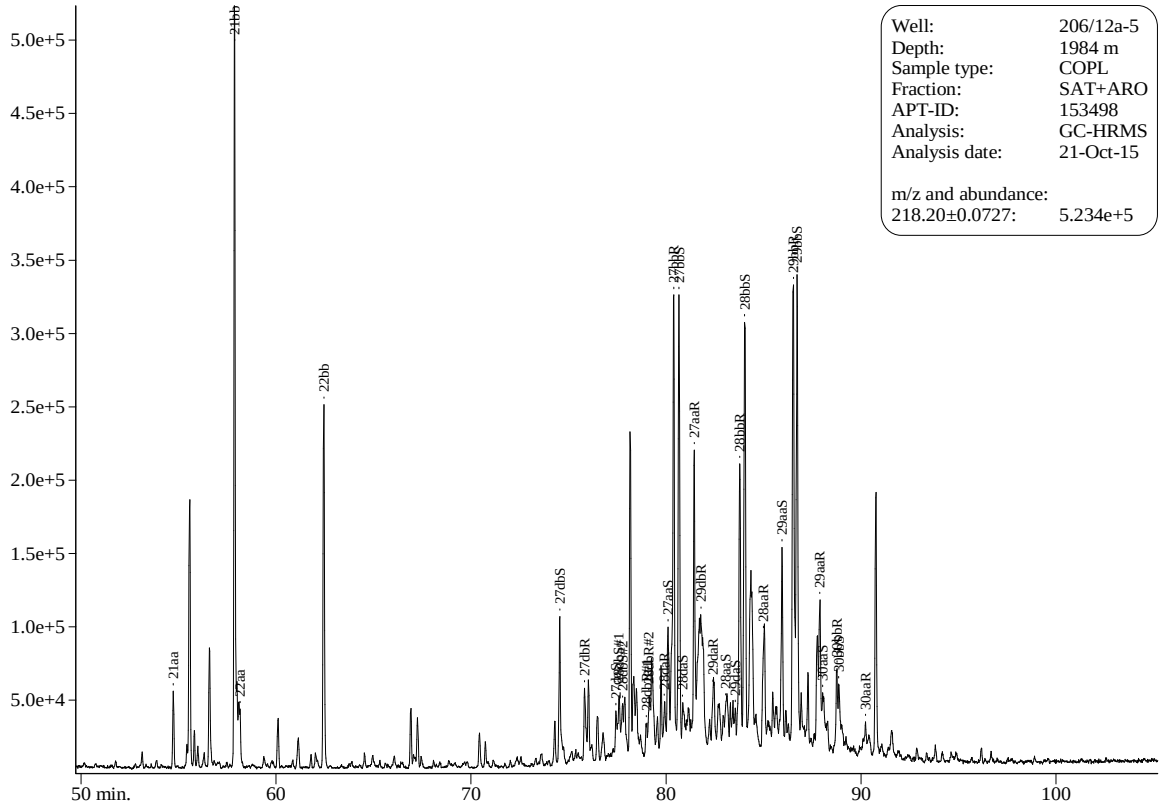


FIGURE Ap7.13 Alkane fraction GC-MS fragmentograms: 206/12a-5, SWC, 1984m (LCG Unit III)



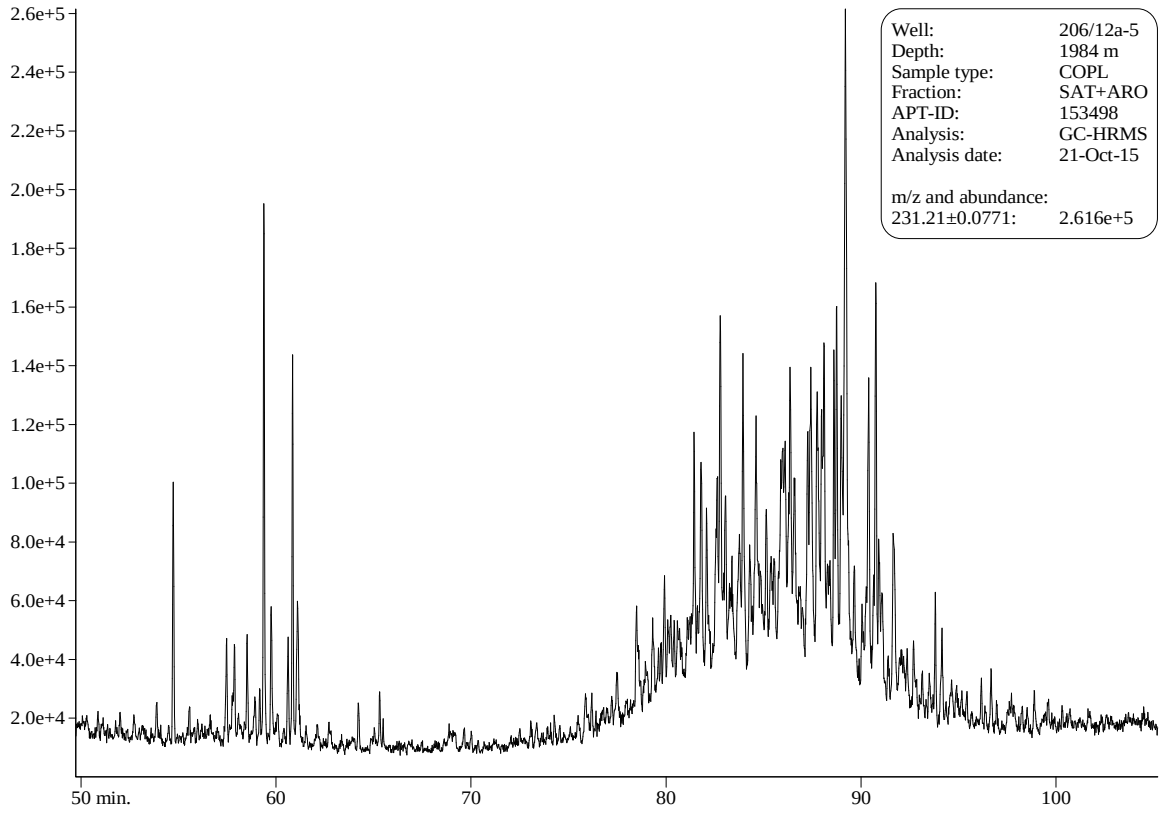
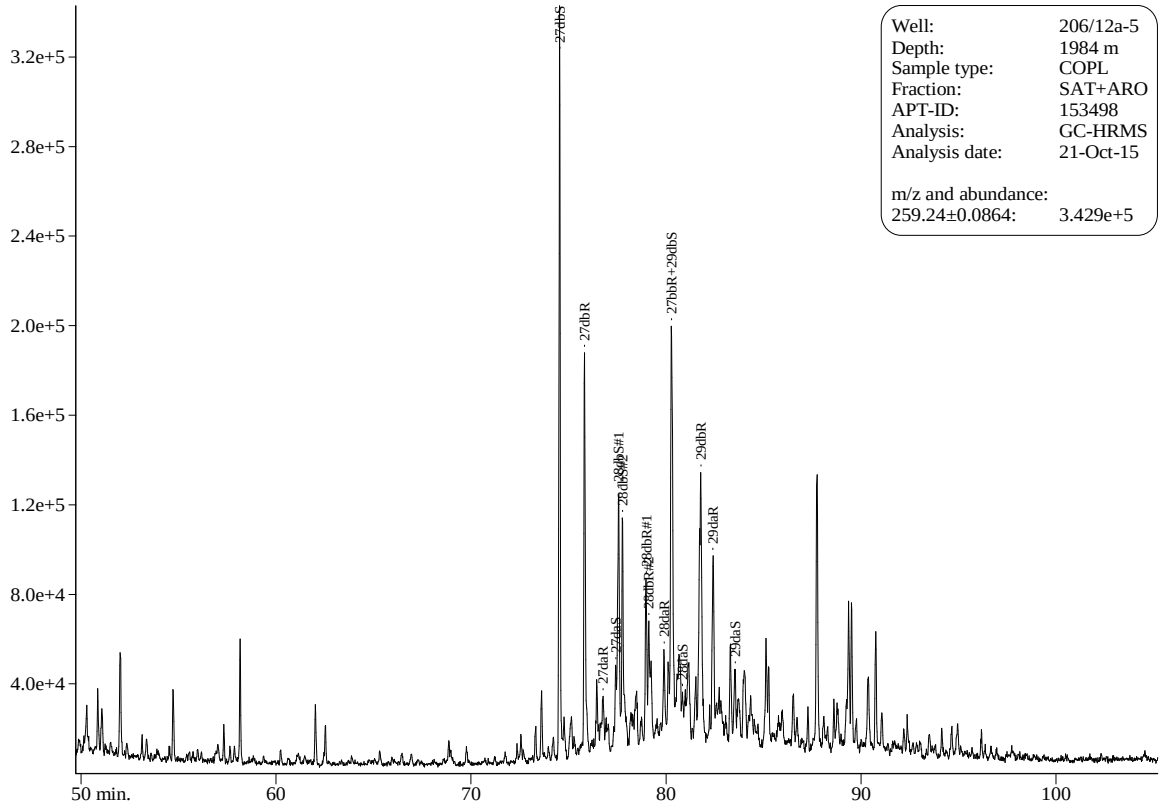


FIGURE Ap7.13 Alkane fraction GC-MS fragmentograms: 206/12a-5, SWC, 1984m (LCG Unit III)



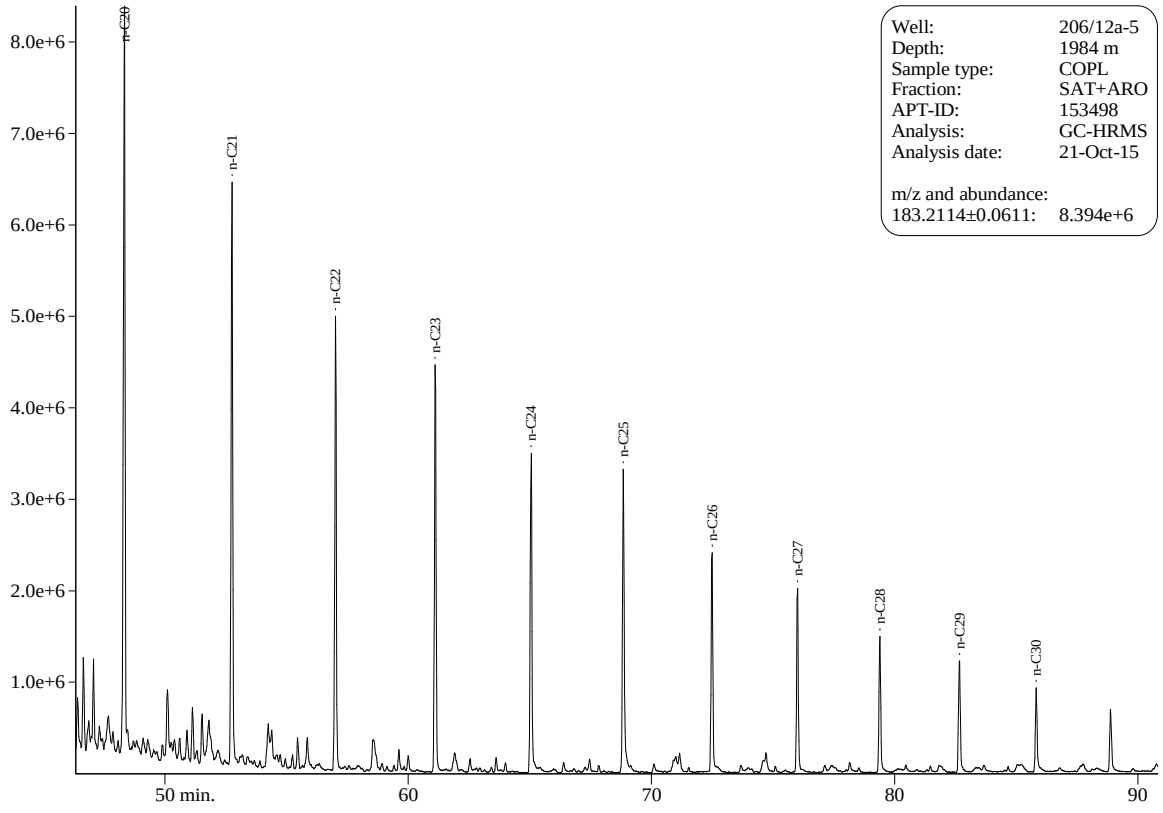
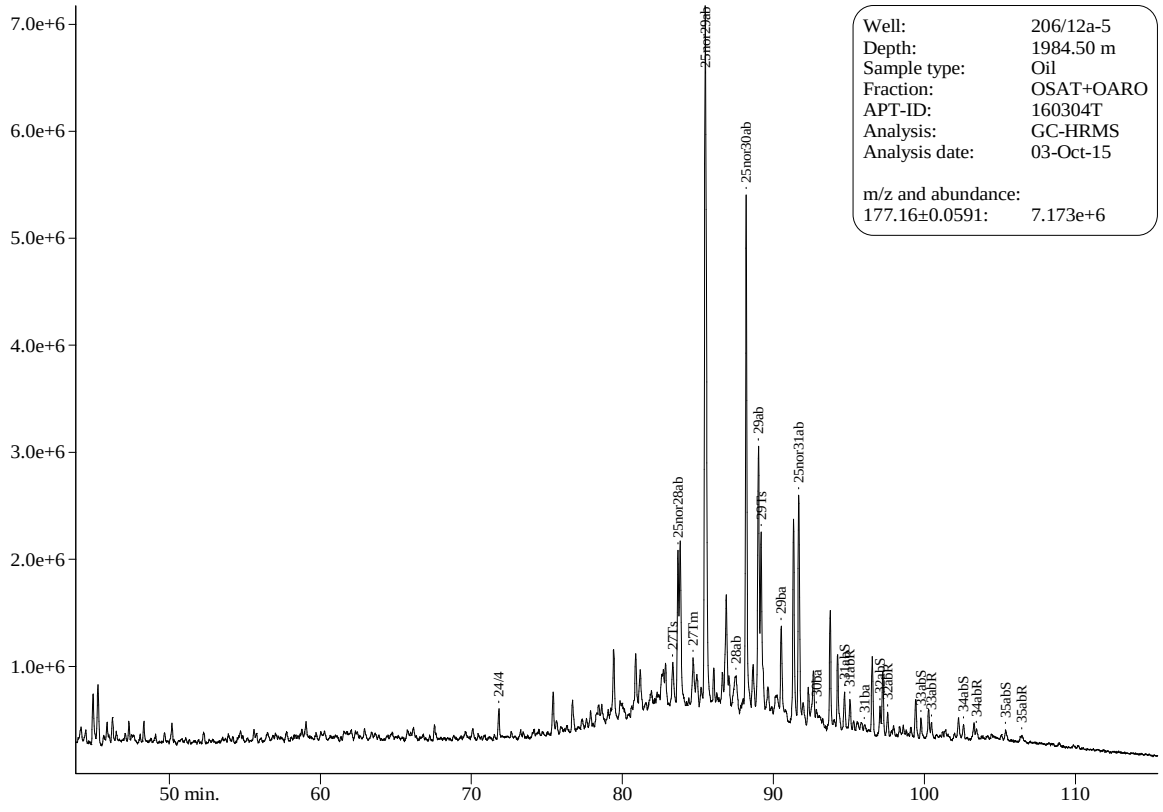


FIGURE Ap7.13 Alkane fraction GC-MS fragmentograms: 206/12a-5, SWC, 1984m (LCG Unit III)



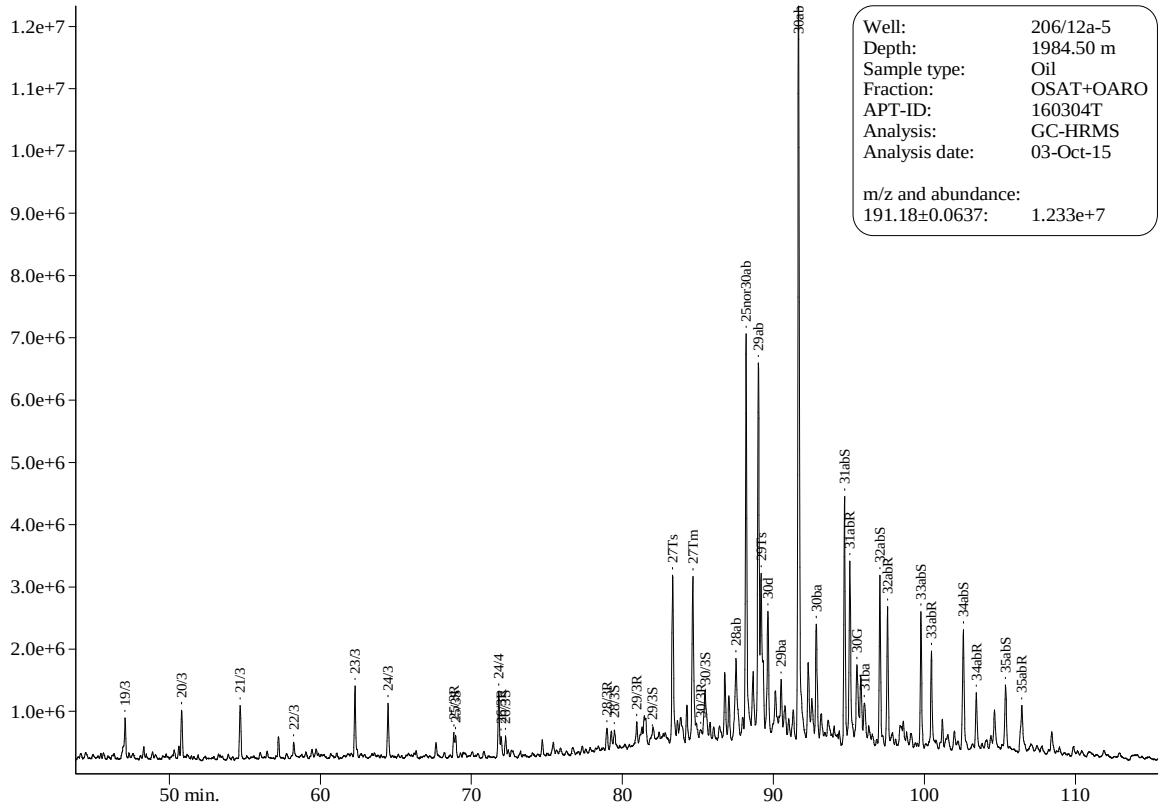
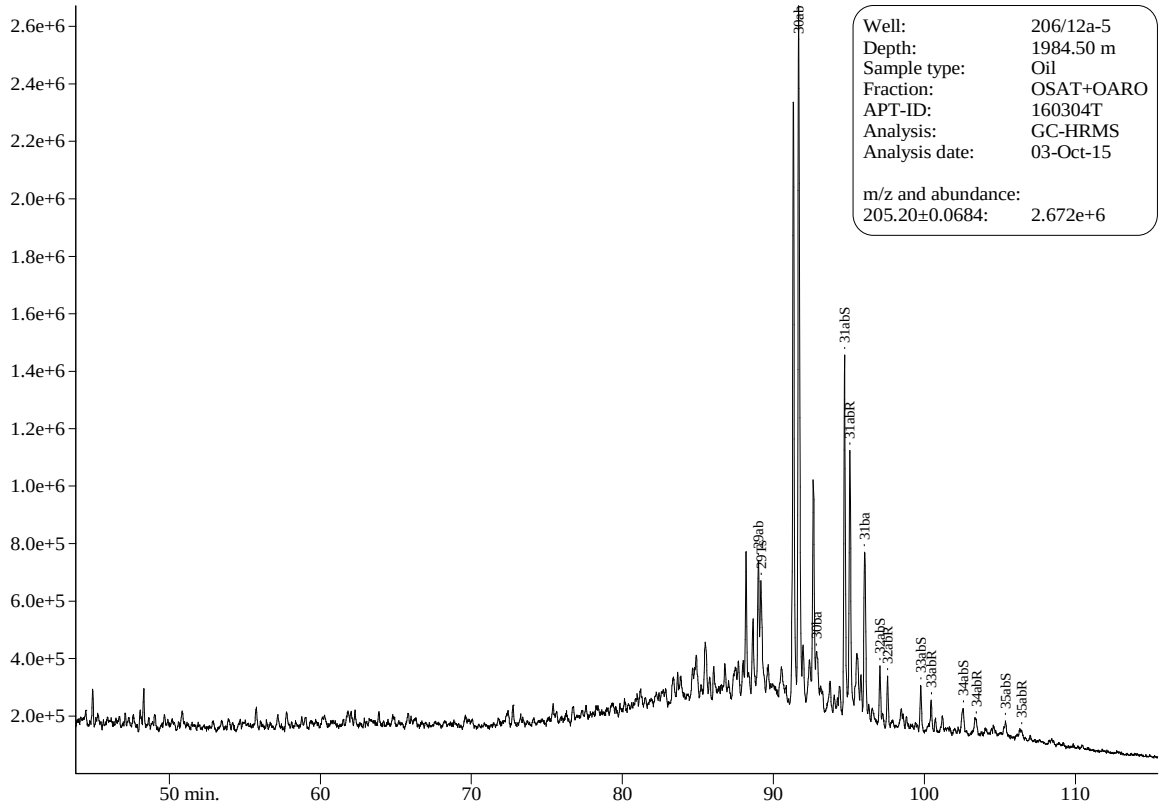


FIGURE Ap7.14 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1984.5m (LCG Unit III)



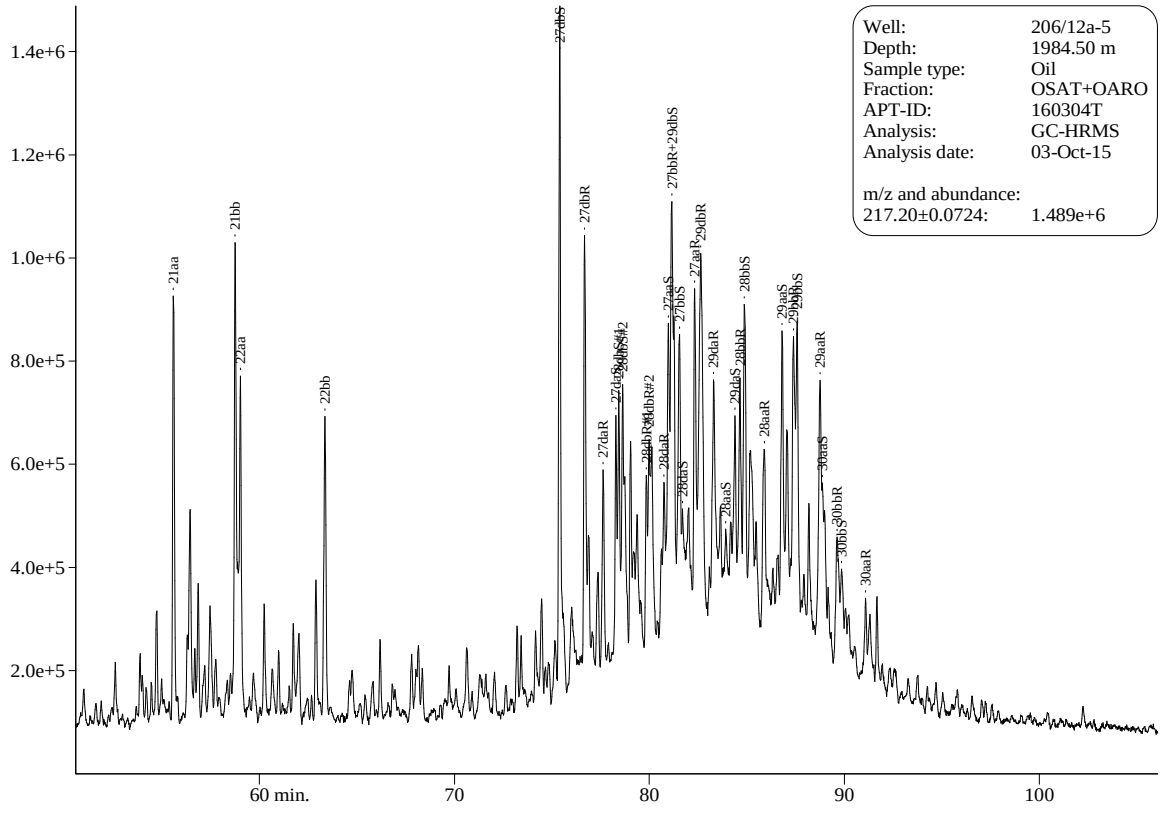
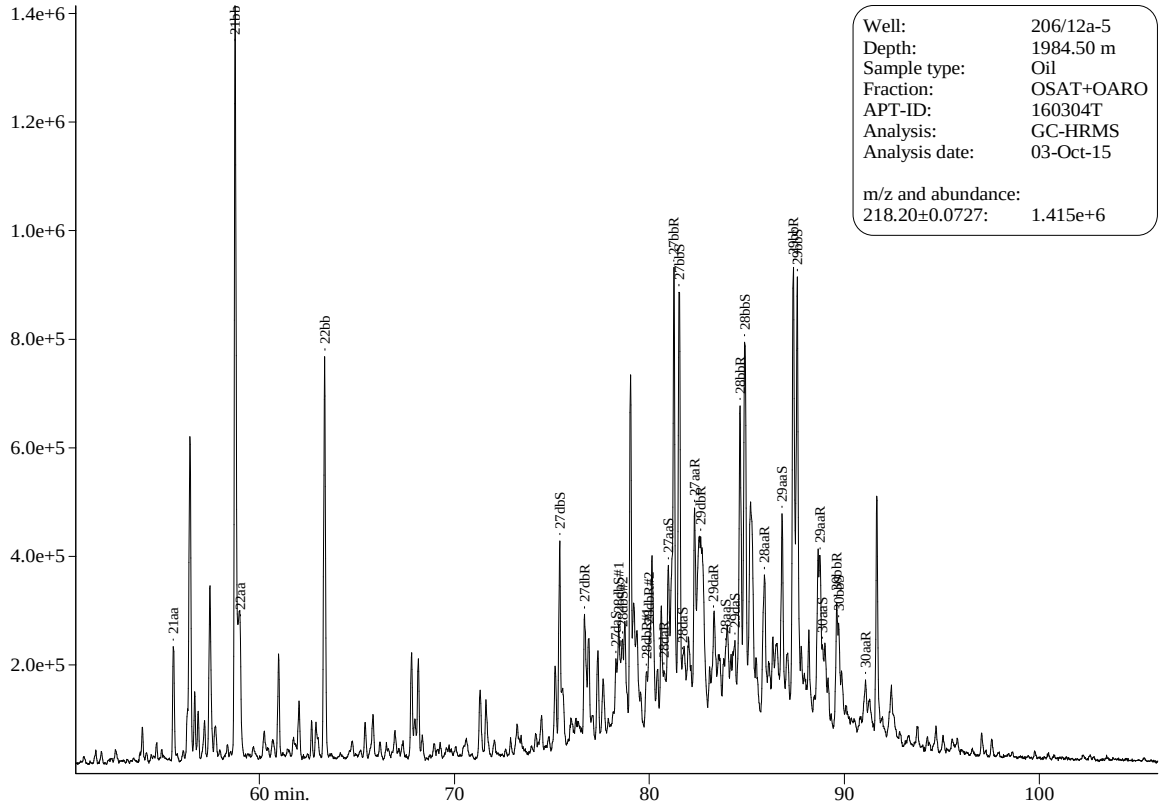


FIGURE Ap7.14 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1984.5m (LCG Unit III)



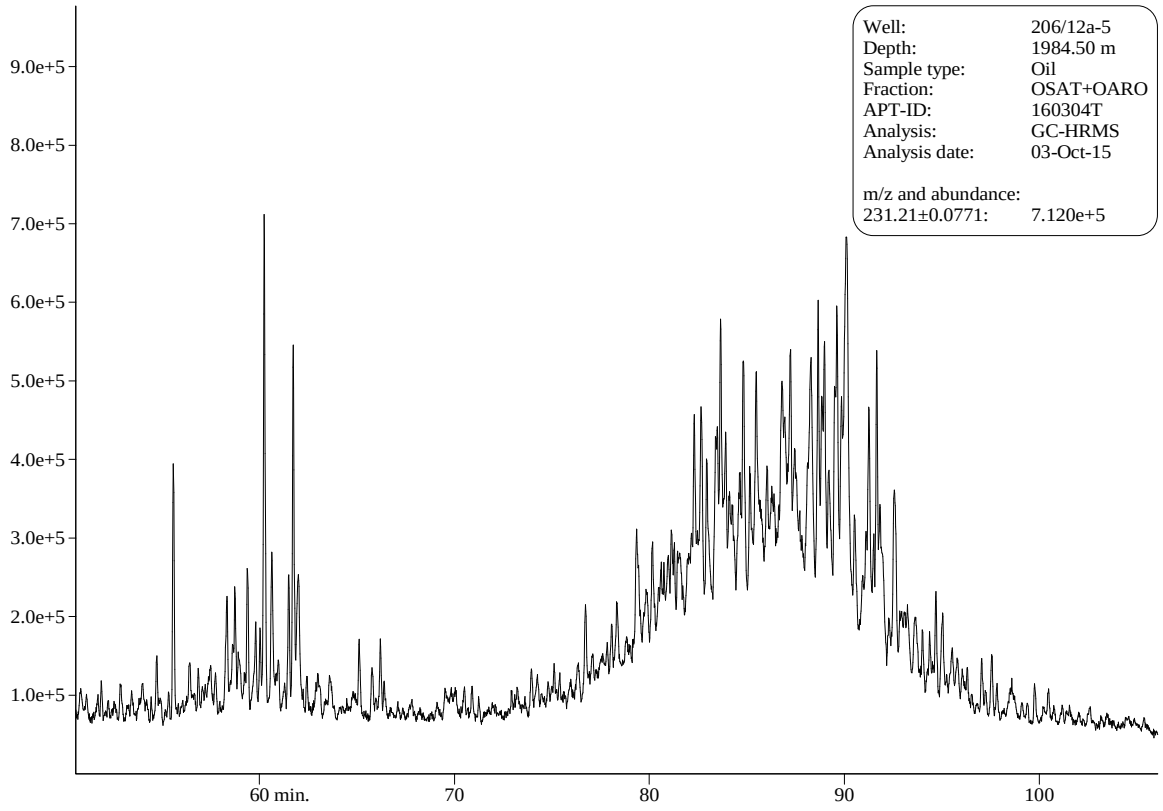
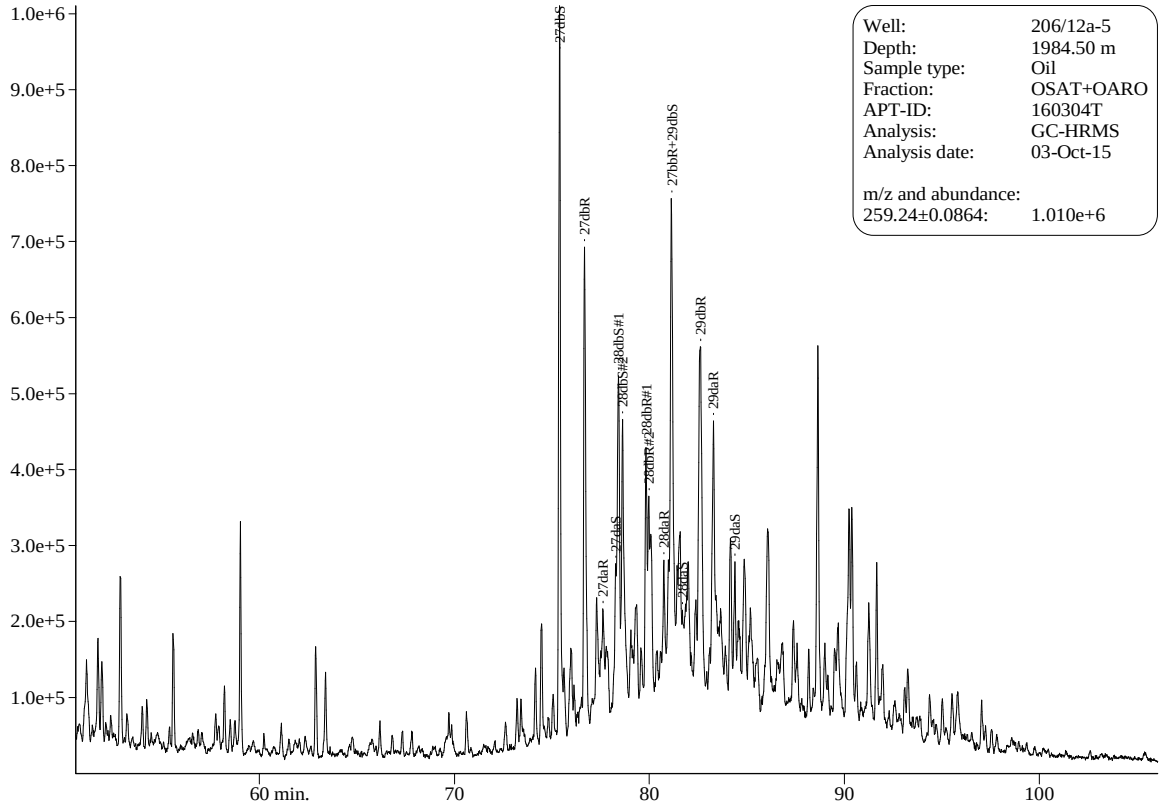


FIGURE Ap7.14 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1984.5m (LCG Unit III)



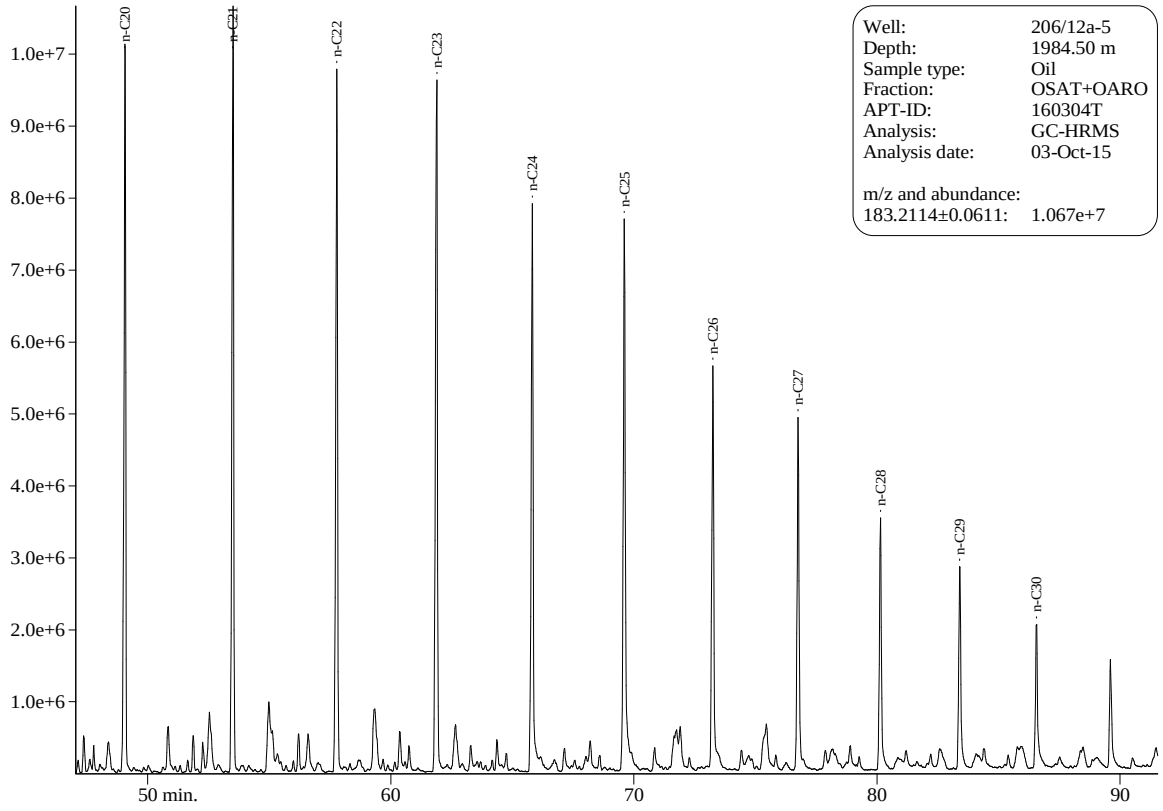
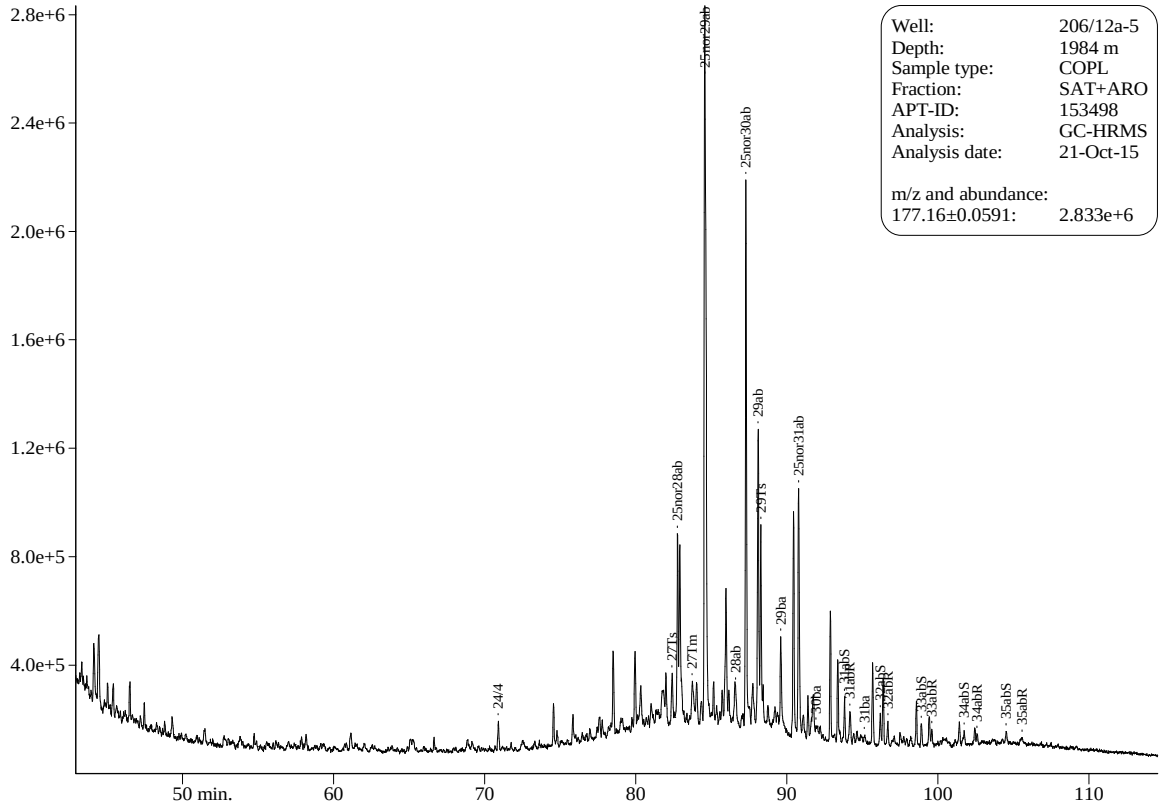


FIGURE Ap7.14 Alkane fraction GC-MS fragmentograms: 206/12a-5, OIL, 1984.5m (LCG Unit III)



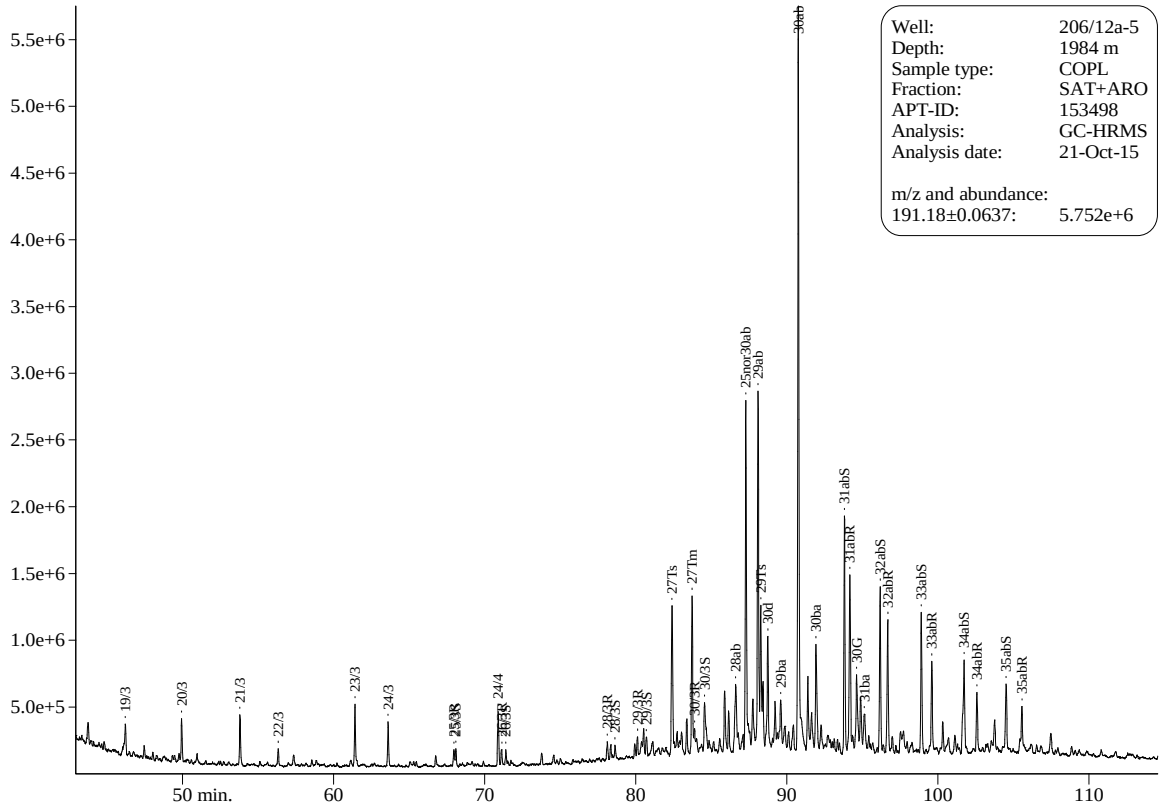
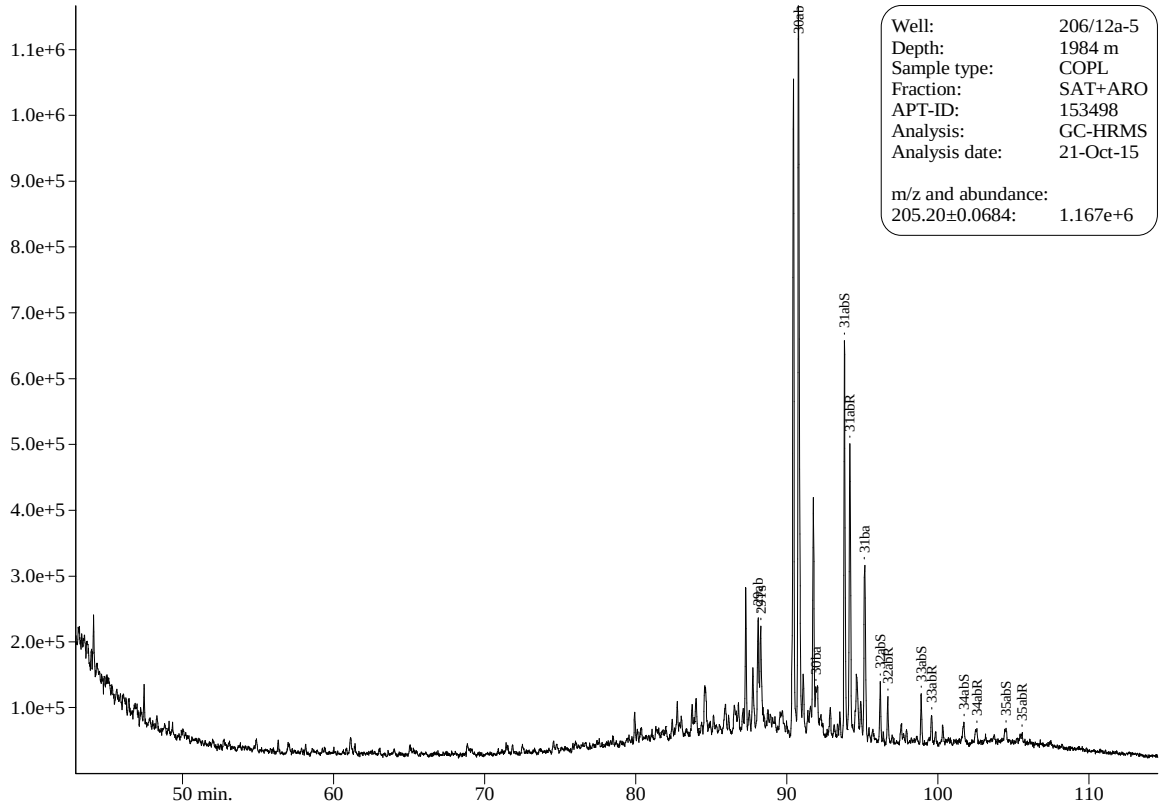


FIGURE Ap7.15 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)



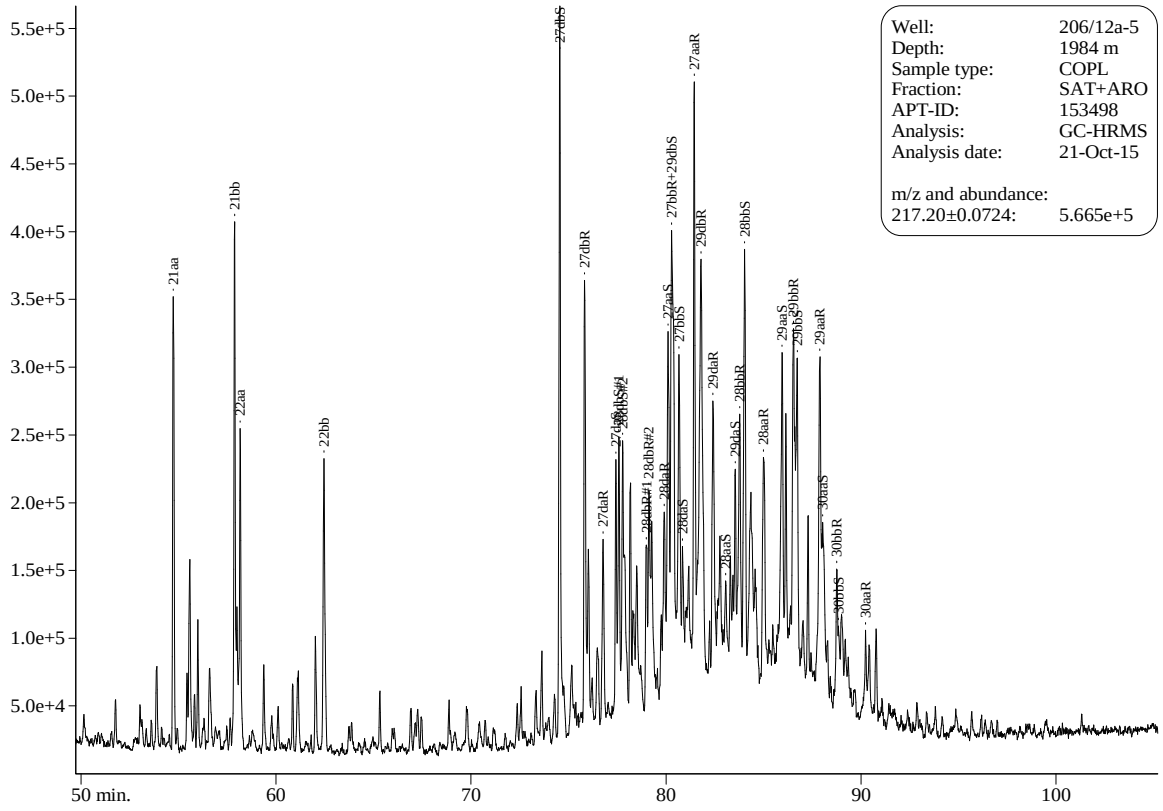
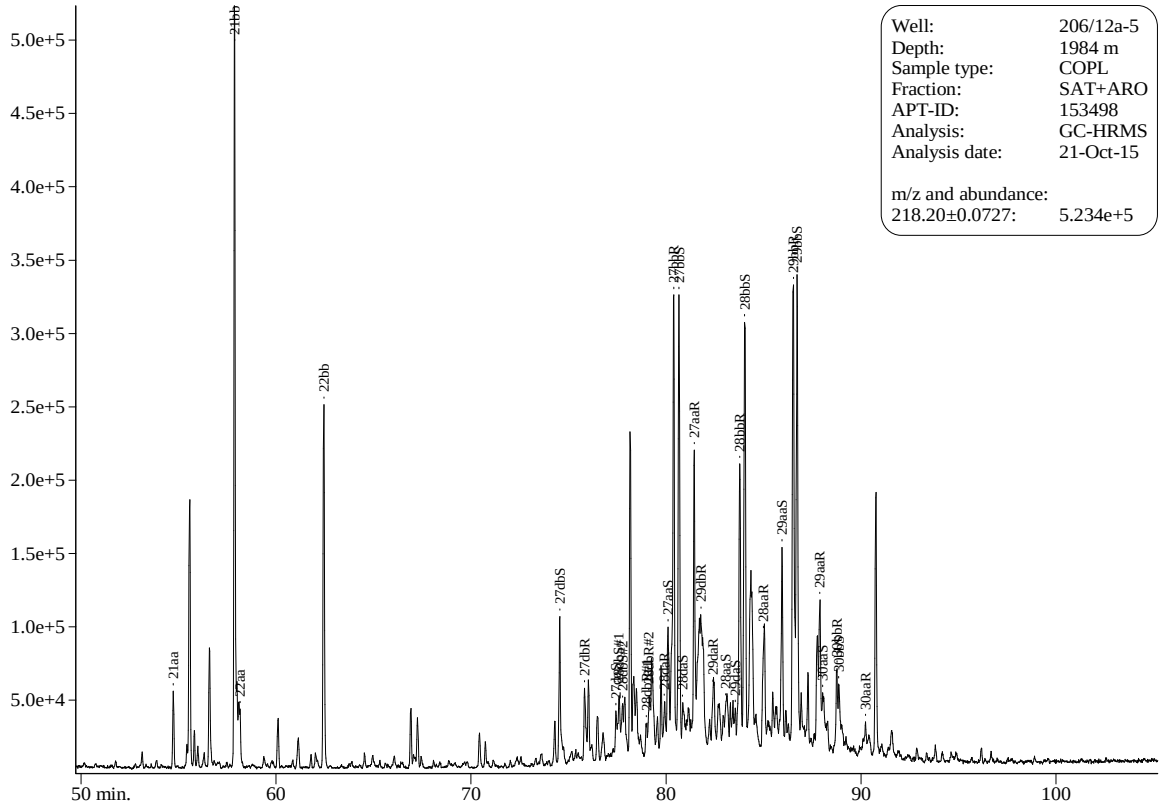


FIGURE Ap7.15 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)



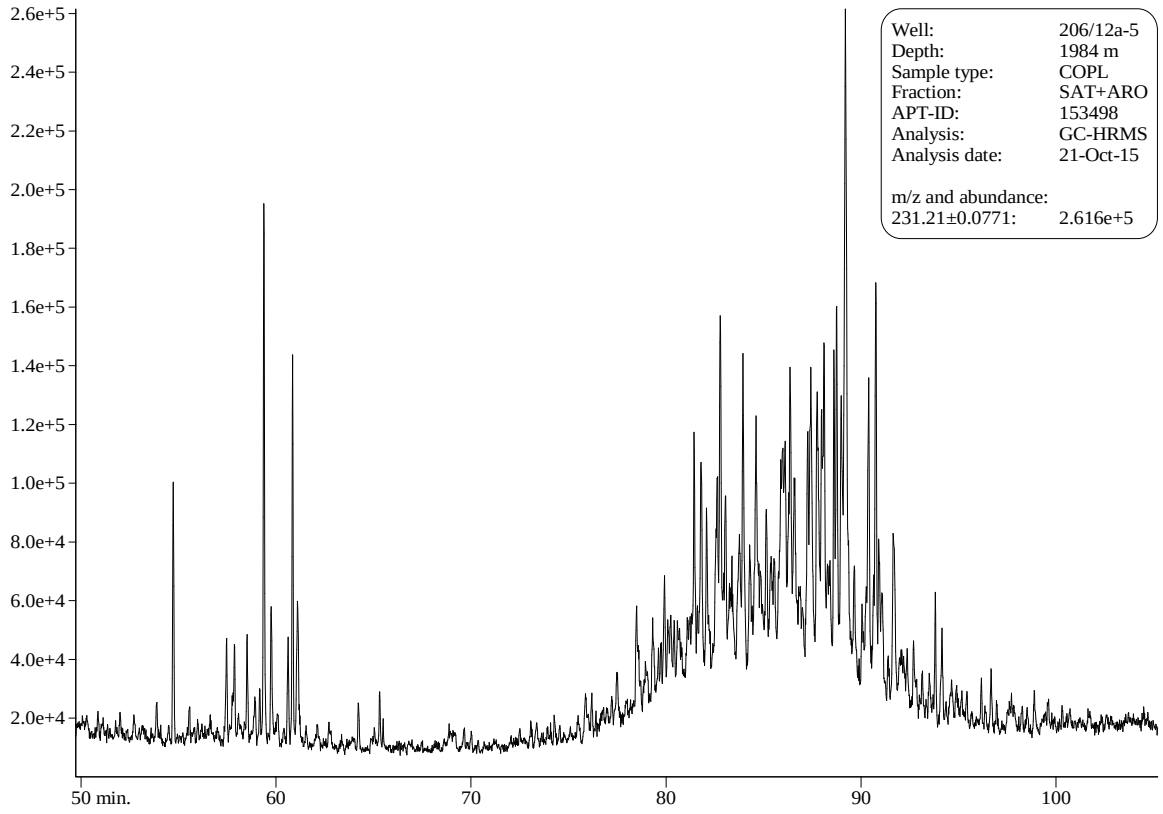
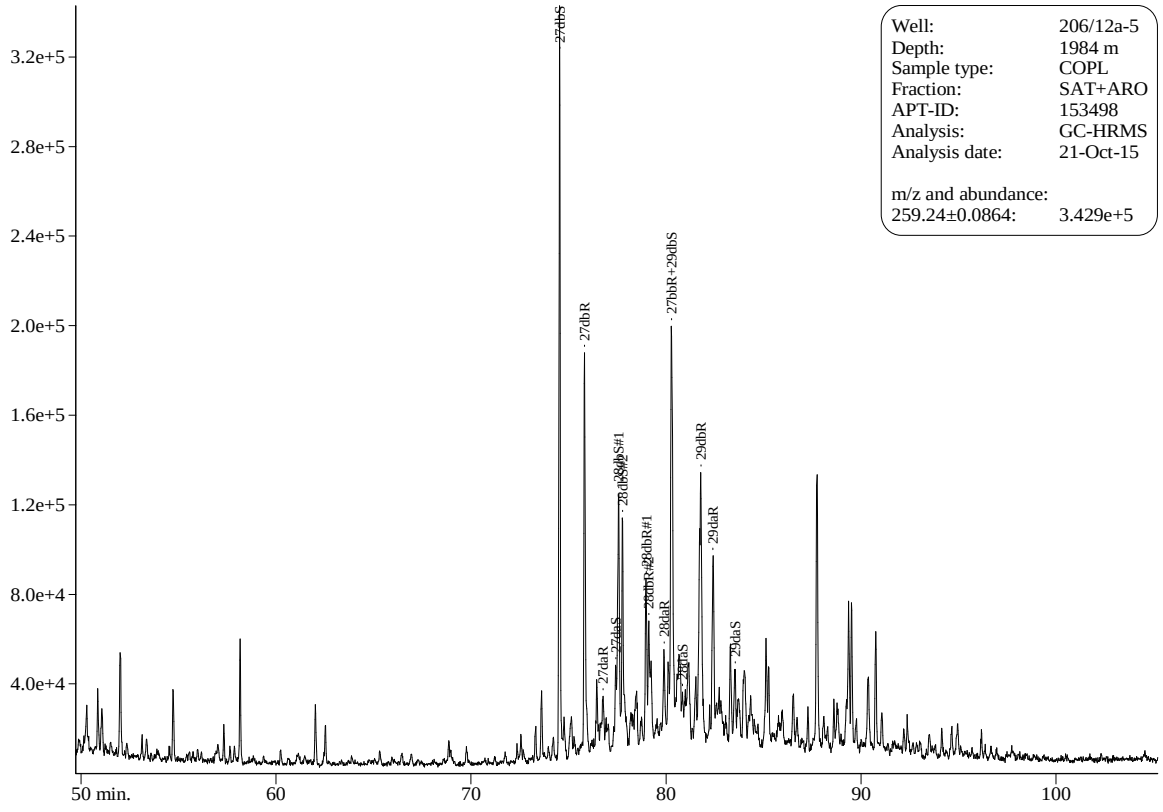


FIGURE Ap7.15 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)



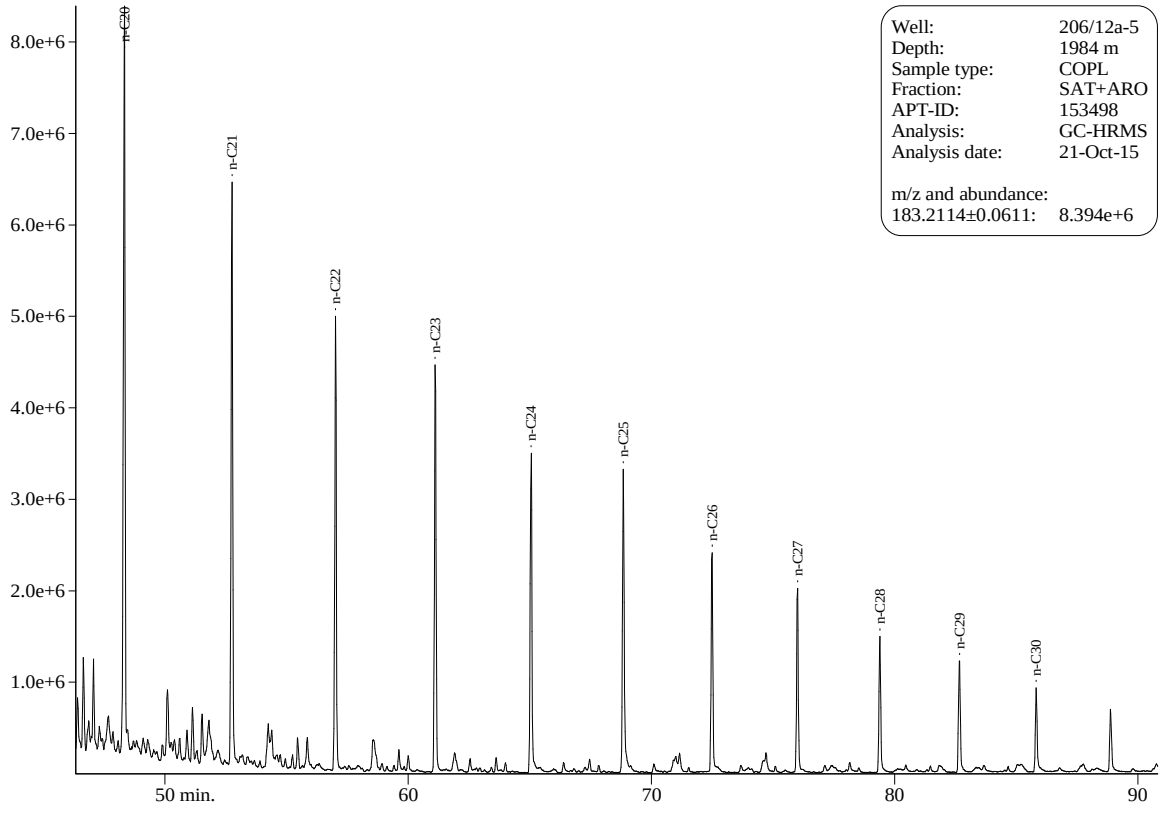
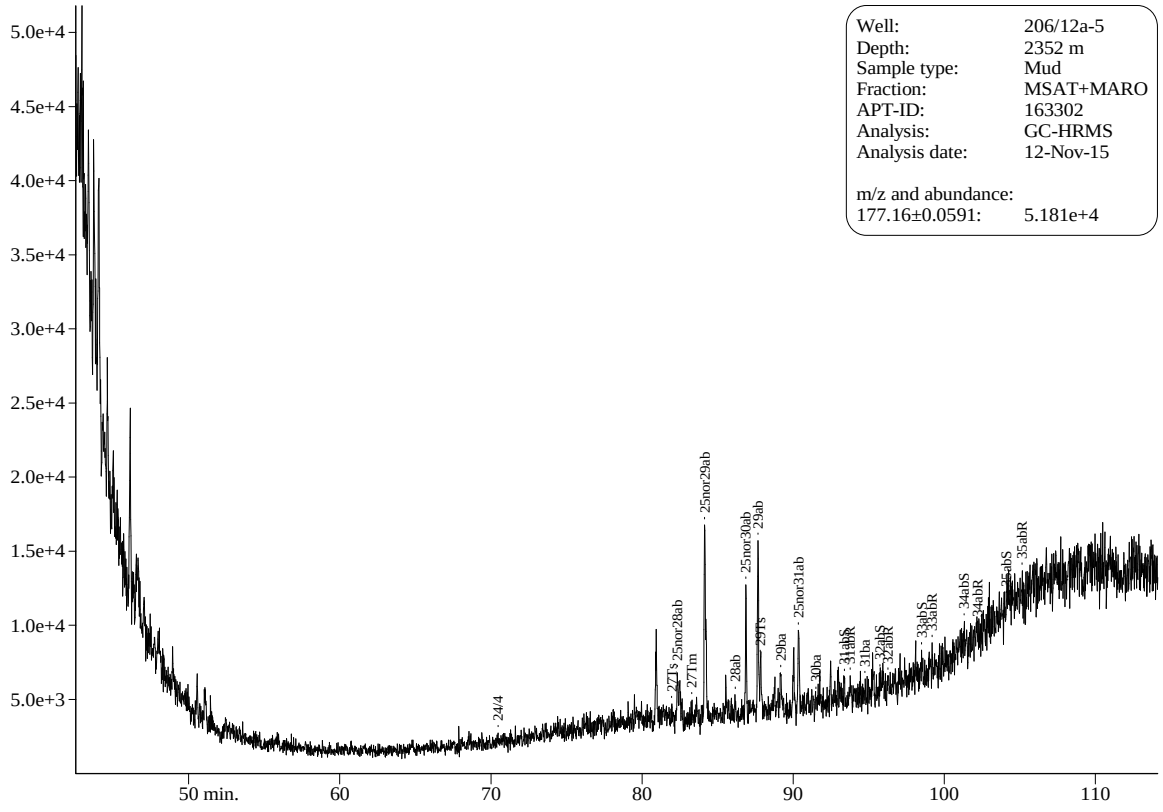


FIGURE Ap7.15 Alkane fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)



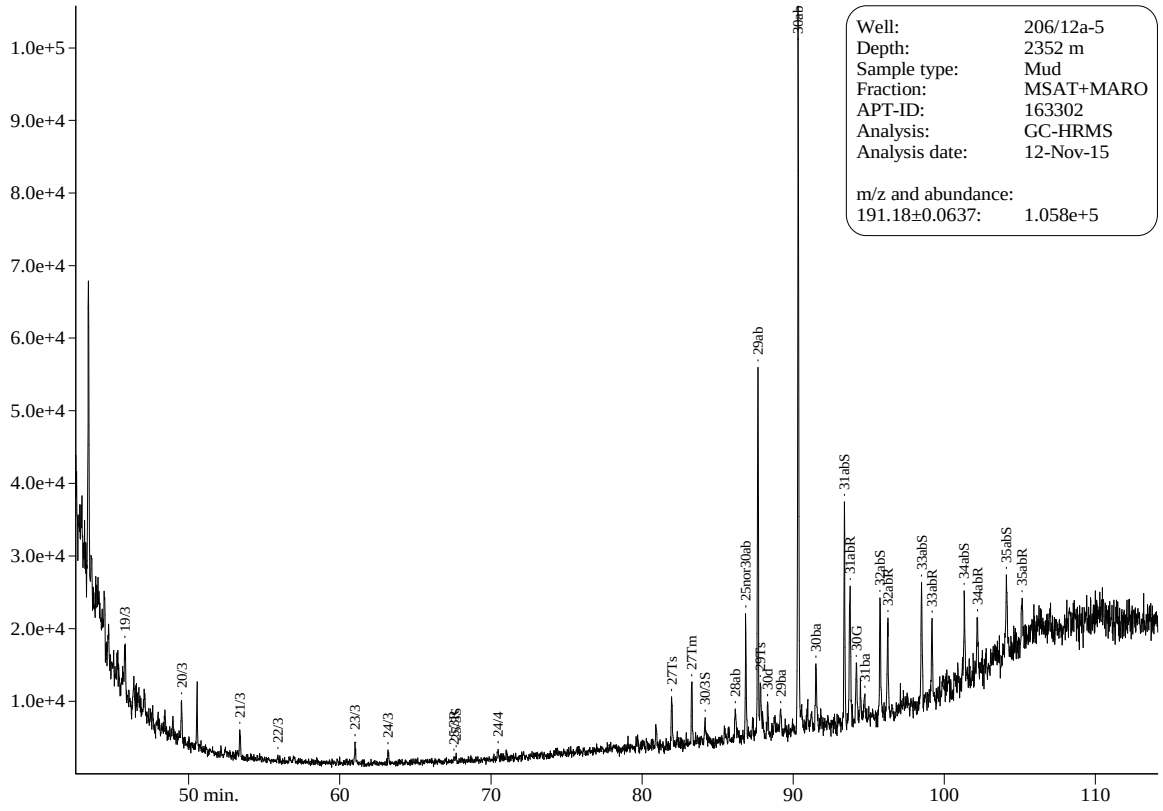
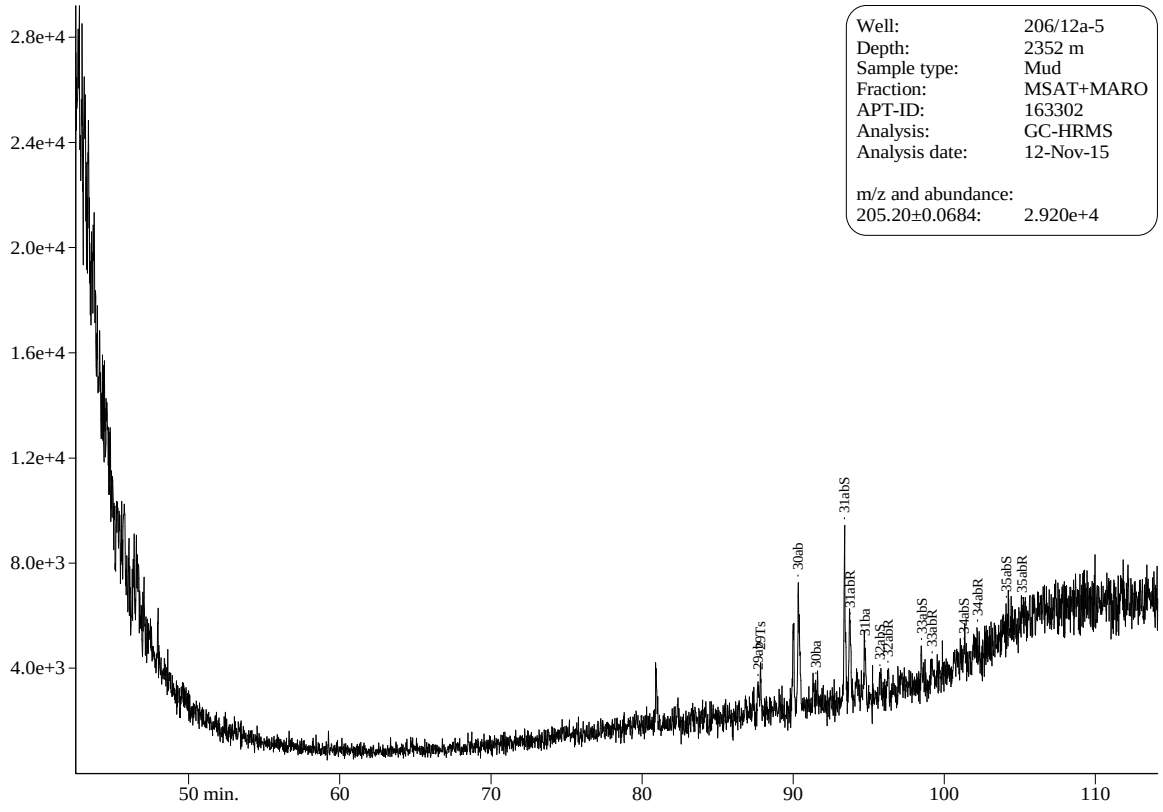


FIGURE Ap7.16 Alkane fraction GC-MS fragmentograms: 206/12a-5, MUD, 2352m (Basement)



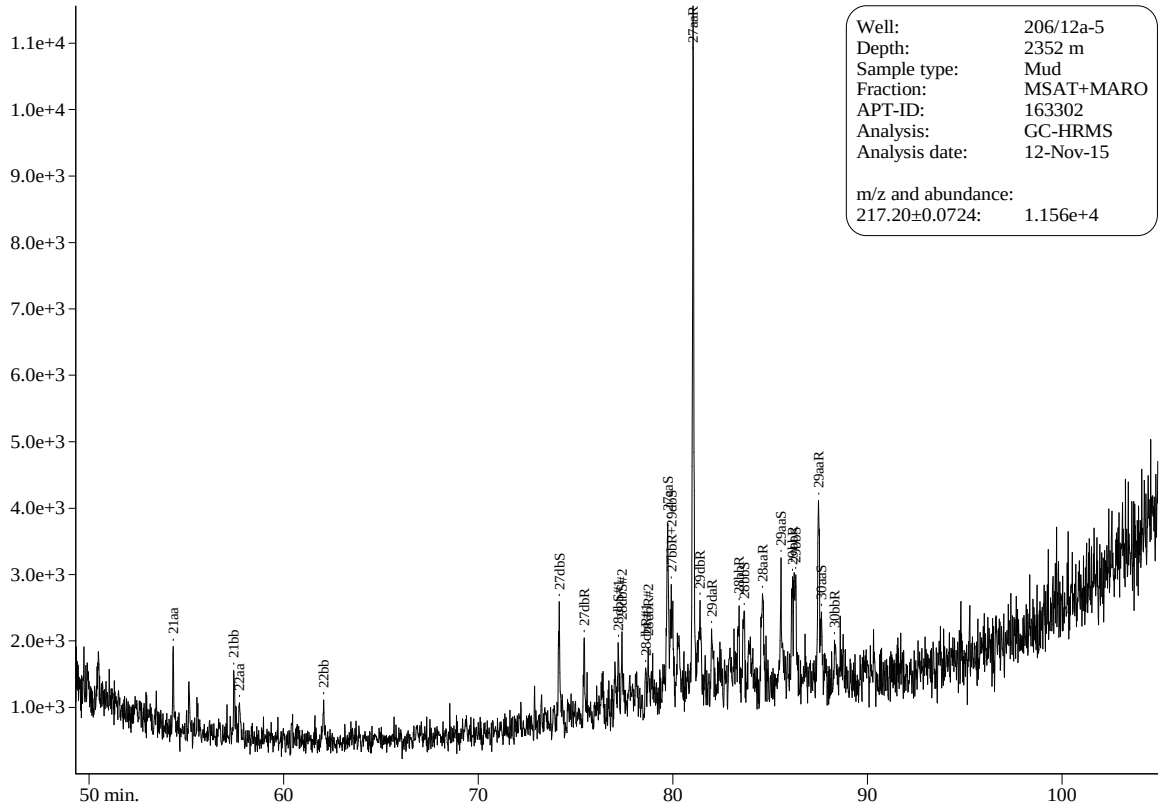
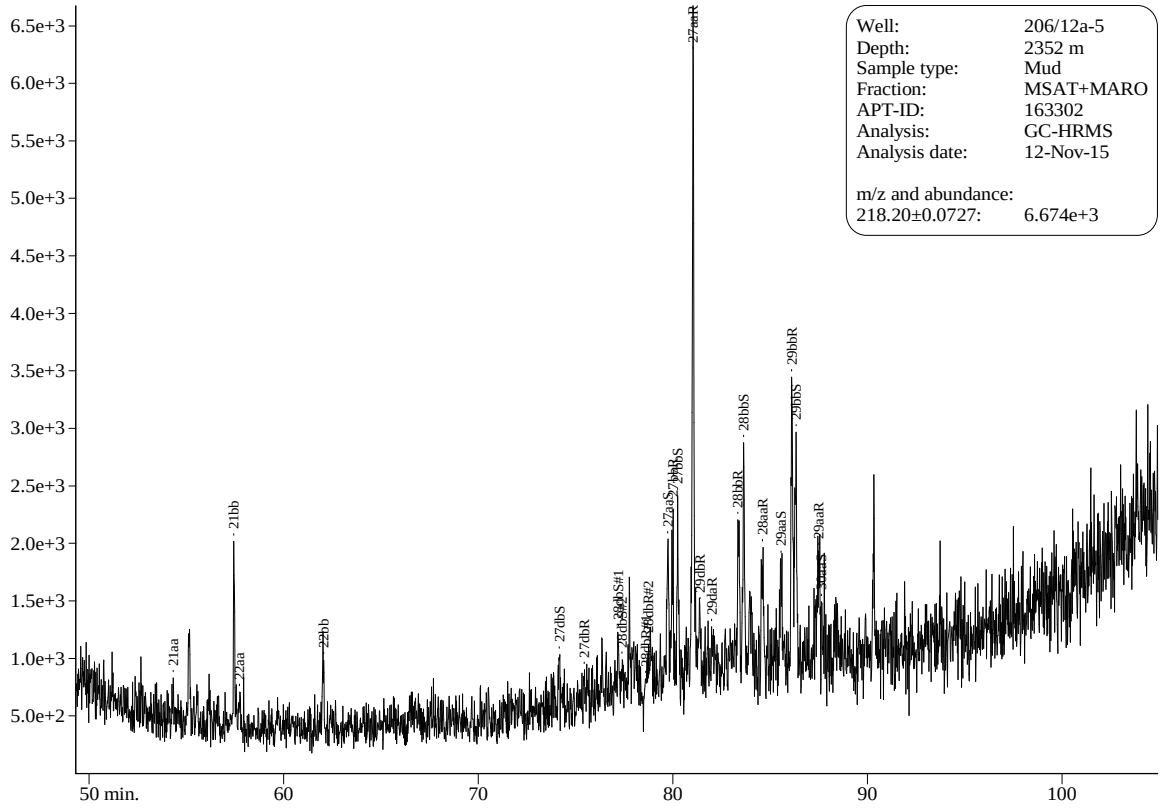


FIGURE Ap7.16 Alkane fraction GC-MS fragmentograms: 206/12a-5, MUD, 2352m (Basement)



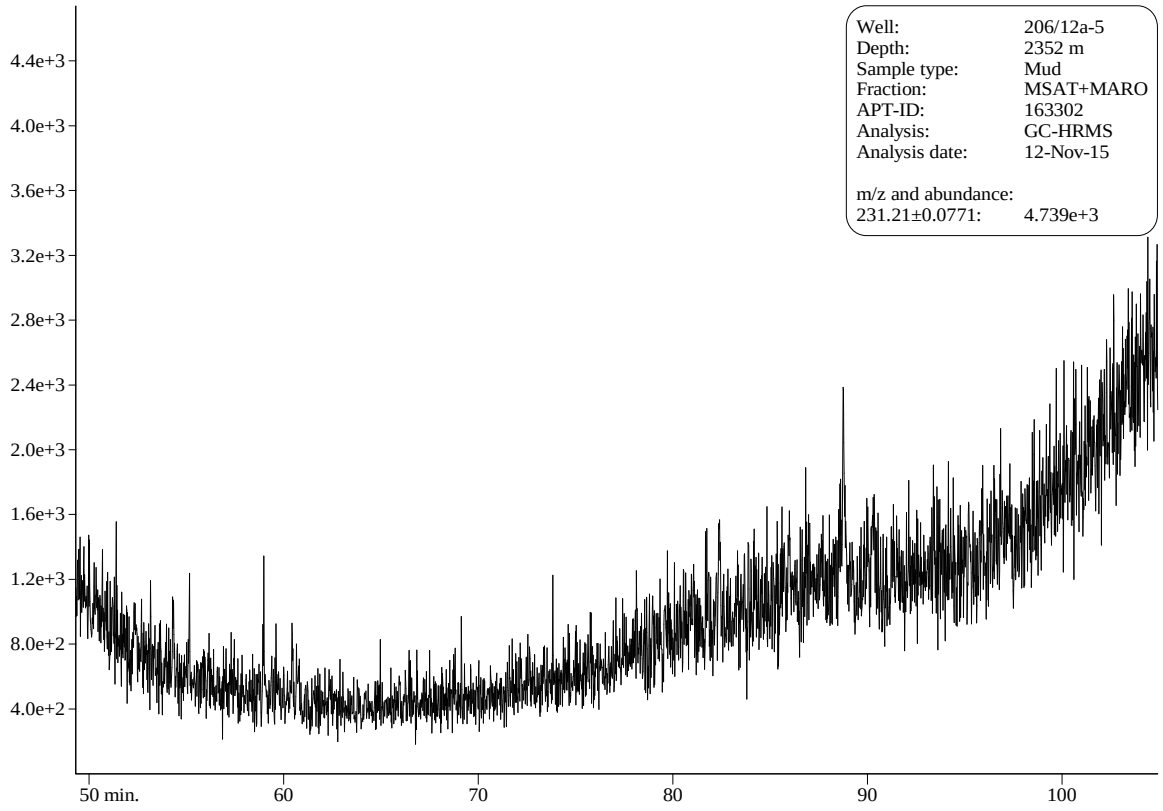
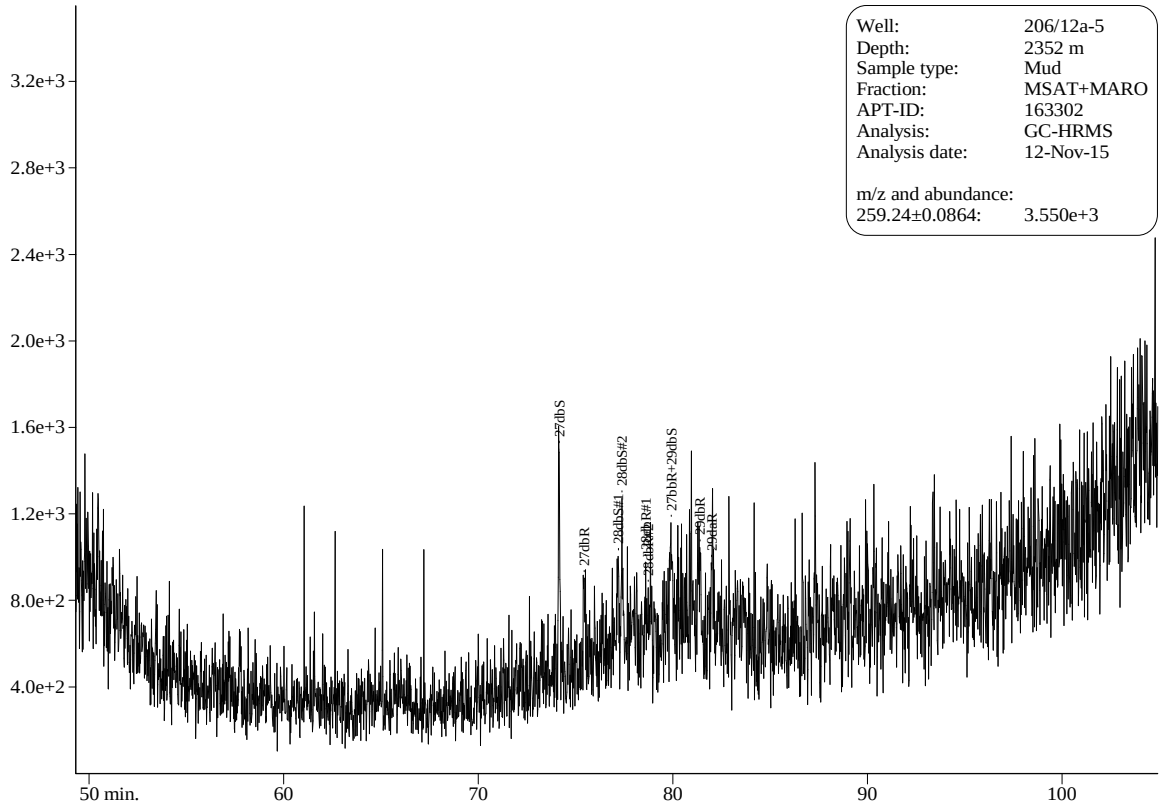


FIGURE Ap7.16 Alkane fraction GC-MS fragmentograms: 206/12a-5, MUD, 2352m (Basement)



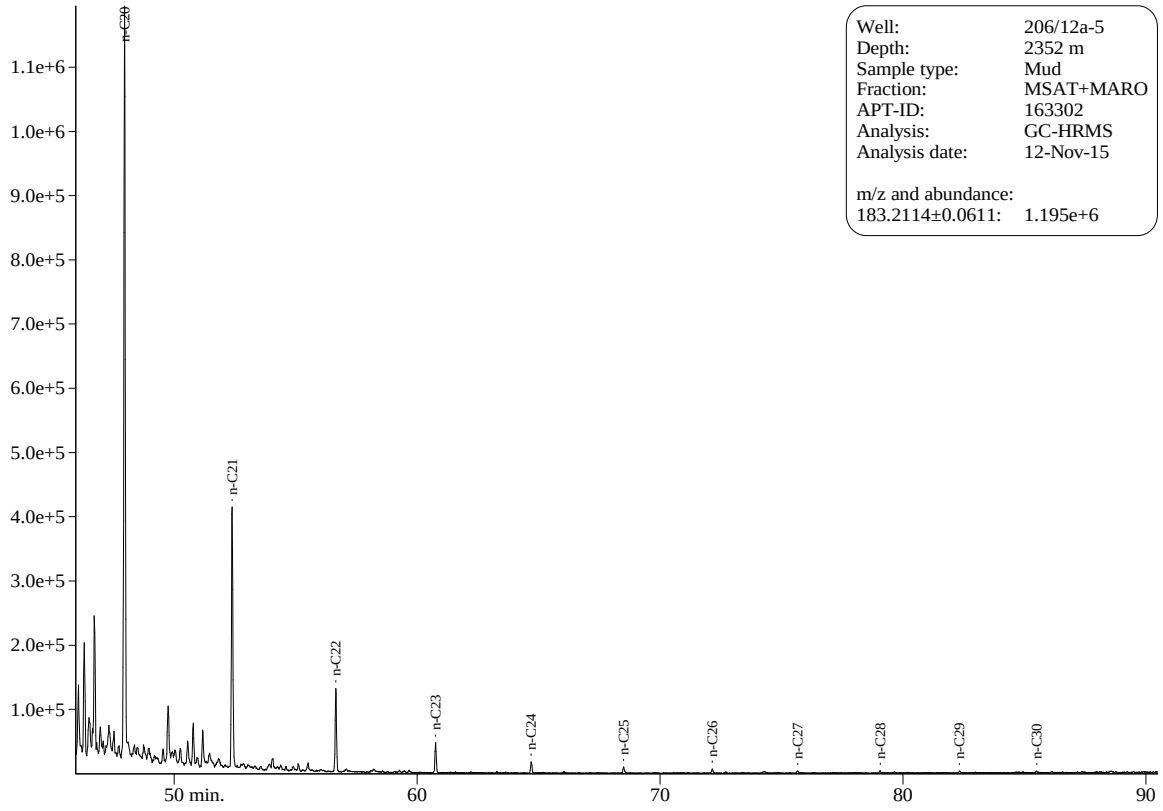


FIGURE Ap7.16 Alkane fraction GC-MS fragmentograms: 206/12a-5, MUD, 2352m (Basement)

APPENDIX 8
Aromatic fraction GC-MS data

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Ap8.3 Aromatic fraction GC-MS amounts in ng/g
Ap8KEY Abbreviations of aromatic fraction biomarkers

Figures

Ap8.1 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72))
Ap8.2 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))
Ap8.3 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret)
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Ap8.10 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1956.67m (Rona)
Ap8.11 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1936-1963m (Rona)
Ap8.12 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE, 1967.35m (Rona)
Ap8.13 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SWC, 1984m (LCG Unit III)
Ap8.14 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1984.5m (LCG Unit III)
Ap8.15 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)
Ap8.16 Aromatic fraction GC-MS fragmentograms: 206/12a-5, MUD, 2352m (Basement)

Well	MD (m)	TVDSS (m)	Sample type	APTID	Naphthalene ratios				Phenanthrene ratios					
					MNR	DNR	TNR1	TNR2	MPI1	MPI2	MPDF1	MPDF2	MSAro 7	MSAro 8
206/12a-5	1897.03	1873.4	Spun out oil	146645	2.47	7.78	1.01	0.43	0.55	0.58	0.48	0.26	0.19	3.17
	1916.67	1893.0	Core plug	146972	2.84	8.84	1.08	0.42	0.61	0.66	0.51	0.27	0.16	3.63
	1922.08	1898.4	Spun out oil	146643	3.13	7.80	1.06	0.38	0.60	0.65	0.49	0.26	0.18	3.73
	1922.40	1898.7	Oil	160302	3.20	9.34	1.09	0.41	0.60	0.63	0.52	0.27	0.17	3.77
	1932.67	1909.0	Core plug	146976	2.98	8.01	1.00	0.43	0.58	0.63	0.50	0.27	0.17	3.35
	1937.32	1913.7	Core	154661	3.10	7.63	1.16	0.47	0.53	0.61	0.49	0.28	0.18	5.01
	1938.00	1914.3	Mud	163301	2.02	9.65	0.76	0.25	0.52	0.61	0.55	0.32	0.05	6.36
	1948.67	1925.0	Spun out oil	146644	3.25	8.42	1.03	0.40	0.61	0.66	0.51	0.27	0.15	3.21
	1956.00	1932.3	Oil	160303	3.17	8.81	1.05	0.40	0.57	0.59	0.51	0.27	0.18	3.57
	1956.67	1933.0	Core plug	146982	3.32	9.83	1.07	0.46	0.60	0.64	0.51	0.27	0.15	3.39

Well	MD (m)	TVDSS (m)	Sample type	APTID	Thiophene ratios		Aromatic sterane ratios							
					MDR1	MDR2	TAS1	TAS2	TASC1	TASC2	TASC3	MSAro 1	MSAro 9	AROM2
206/12a-5	1897.03	1873.4	Spun out oil	146645	2.74	0.49	1.10	0.74	0.50	0.26	0.52	0.24	0.73	0.73
	1916.67	1893.0	Core plug	146972	2.99	0.57	0.95	0.74	0.51	0.26	0.50	0.28	0.74	0.75
	1922.08	1898.4	Spun out oil	146643	2.88	0.55	1.09	0.75	0.51	0.26	0.51	0.25	0.72	0.71
	1922.40	1898.7	Oil	160302	2.80	0.49	1.05	0.76	0.52	0.26	0.50	0.26	0.73	0.74
	1932.67	1909.0	Core plug	146976	2.72	0.51	0.95	0.76	0.47	0.24	0.47	0.24	0.72	0.71
	1937.32	1913.7	Core	154661	2.96	0.45	1.35	0.89	0.72	0.41	0.68	0.42	0.88	0.87
	1938.00	1914.3	Mud	163301	4.39	0.45	0.82	0.86	0.31	0.13	0.30	0.29	0.82	0.78
	1948.67	1925.0	Spun out oil	146644	3.04	0.55	1.19	0.73	0.53	0.27	0.53	0.24	0.72	0.72
	1956.00	1932.3	Oil	160303	2.90	0.48	1.11	0.73	0.51	0.26	0.51	0.27	0.74	0.73
	1956.67	1933.0	Core plug	146982	2.95	0.57	1.02	0.80	0.50	0.25	0.50	0.24	0.72	0.73

see last page of table for key to ratios

TABLE Ap8.1 Aromatic fraction GC-MS parameters

Well	MD (m)	TVDSS (m)	Sample type	APTID	Naphthalene ratios				Phenanthrene ratios					
					MNR	DNR	TNR1	TNR2	MPI1	MPI2	MPDF1	MPDF2	MSAro 7	MSAro 8
206/12a-5	1936-1963	1912.3-1939.3	Oil	160301	3.23	9.90	1.03	0.41	0.61	0.65	0.51	0.28	0.18	3.92
	1967.35	1943.7	Core	154666	3.07	8.48	1.17	0.44	0.54	0.62	0.49	0.28	0.16	4.84
	1984.00	1960.3	SWC	153498	2.43	11.66	1.44	0.70	0.71	0.77	0.49	0.27	0.10	2.74
	1984.50	1960.8	Oil	160304	3.30	9.05	1.01	0.43	0.57	0.62	0.51	0.28	0.18	3.45
	1997.67	1974.0	Core plug	146992	3.28	8.68	1.04	0.45	0.59	0.62	0.51	0.27	0.17	3.08
	2352.00	2336.6	Mud	163302	2.14	10.63	0.73	0.26	0.44	0.49	0.56	0.31	0.04	7.91

Well	MD (m)	TVDSS (m)	Sample type	APTID	Thiophene ratios		Aromatic sterane ratios							
					MDR1	MDR2	TAS1	TAS2	TASC1	TASC2	TASC3	MSAro 1	MSAro 9	AROM2
206/12a-5	1936-1963	1912.3-1939.3	Oil	160301	2.84	0.52	1.03	0.79	0.52	0.26	0.51	0.26	0.73	0.73
	1967.35	1943.7	Core	154666	3.02	0.46	1.27	0.75	0.67	0.38	0.65	0.32	0.81	0.82
	1984.00	1960.3	SWC	153498	3.06	1.08	1.24	0.67	0.51	0.26	0.50	0.29	0.82	0.81
	1984.50	1960.8	Oil	160304	2.92	0.46	1.03	0.74	0.50	0.26	0.50	0.25	0.74	0.73
	1997.67	1974.0	Core plug	146992	2.86	0.51	1.04	0.78	0.53	0.27	0.51	0.26	0.72	0.73
	2352.00	2336.6	Mud	163302	3.59	0.36	0.85	0.48	0.29	0.15	0.27	0.34	0.80	0.79

see last page of table for key to ratios

TABLE Ap8.1 Aromatic fraction GC-MS parameters

KEY: Aromatic GC-MS ratios

MNR : Methylnaphthalene ratio - 2-MN/1-MN {MSAro4}
DNR : Dimethylnaphthalene ratio - (2,6-DMN+2,7-DMN)/1,5-DMN {MSAro5}
TNR1 : Trimethylnaphthalene ratio 1 - 2,3,6-TMN/(1,3,5- + 1,4,6-TMN)
TNR2 : Trimethylnaphthalene ratio 2 - 1,2,5-TMN/1,3,6-TMN
MPI1 : Methylphenanthrene ratio 1 - 1.5*(2-MP + 3-MP)/(P + 1-MP + 9-MP)
MPI2 : Methylphenanthrene ratio 2 - 3*(2-MP)/(P + 1-MP + 9-MP)
MPDF1 : Methylphenanthrene distribution fraction 1 - (2-MP+3-MP)/(1-MP+2-MP+3-MP+9-MP) {MSAro3}
MPDF2 : Methylphenanthrene distribution fraction 2 - 2-MP/(1-MP+2-MP+3-MP+9-MP)
MSAro7 : DBT/P
MSAro8 : 3-MP/Retene
MDR1 : Methyldibenzothiophene ratio 1 - 4-MDBT/1-MDBT {MSAro2}
MDR2 : Methyldibenzothiophene ratio 2 - 4-MDBT/DBT {MSAro6}
TAS1 : Triaromatic sterane ratio 1 - SC26TA/SC28TA
TAS2 : Triaromatic sterane ratio 2 - RC27TA/RC28TA
TASC1 : Triaromatic sidechain cracking 1 - (C20TA)/(C20TA+RC28TA) {Crack1}
TASC2 : Triaromatic sidechain cracking 2 - (C20TA+C21TA)/(C20TA+C21TA+SC26TA+RC26TA+SC27TA+SC28TA+RC27TA+RC28TA) {Crack2}
TASC3 : Triaromatic sidechain cracking 3 - C21TA/(C21TA + RC28TA)
MSAro1 : (C21MA+C22MA)/(C21MA+C22MA+bSC27MA+[bRC27MA+bRC27DMA]+aSC27MA+[bSC28MA+bSC28DMA+aRC27DMA]+aSC27DMA+ ….
… aRC27MA+aSC28MA+aSC29MA+aRC29MA)
MSAro9 : RC28TA/(RC28TA+[aRC28MA+bRC29MA+bRC29DMA])
AROM2 : (C20TA+C21TA+SC26TA+RC26TA+SC27TA+SC28TA+RC27TA+RC28TA/(C20TA+C21TA+SC26TA+RC26TA+SC27TA+SC28TA+RC27TA+RC28TA+C21MA+C22MA+ …
… bSC27MA+[bRC27MA+bRC27DMA]+aSC27MA+[bSC28MA+bSC28DMA+aRC27DMA]+aSC27DMA+aRC27MA+aSC28MA+aSC29MA+aRC29MA)

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 142		m/e 156				
					2-MN	1-MN	2-EN	1-EN	2,6-DMN	2,7-DMN	1,3- + 1,7-DMN
206/12a-5	1897.03	1873.4	Spun out oil	146645	83292690	33730516	6422181	2364901	26926000	26360108	42385381
	1916.67	1893.0	Core plug	146972	72219309	25466554	5453852	1686689	25862245	23891473	37844264
	1922.08	1898.4	Spun out oil	146643	174435626	55677936	11448988	3153947	50065207	35681542	75439826
	1922.40	1898.7	Oil	160302	103463483	32375096	7877987	2145456	39267468	36169862	54043577
	1932.67	1909.0	Core plug	146976	244691517	81985648	23846488	13889988	126470706	130299010	161260836
	1937.32	1913.7	Core	154661	323561527	104293450	21213995	5867474	92387811	86908695	144341036
	1938.00	1914.3	Mud	163301	1174398	582178	45214	11010	241143	196409	459201
	1948.67	1925.0	Spun out oil	146644	273904732	84216860	19561432	5132366	76236924	72973847	122951051
	1956.00	1932.3	Oil	160303	119200439	37576815	8471375	2393495	39396002	37785383	57244382
	1956.67	1933.0	Core plug	146982	99175556	29913772	7443077	2472094	39601688	43025638	49470321
	1936-1963	1912.3-1939.3	Oil	160301	138809122	42926657	8977143	2483844	43728872	43670121	60912281
	1967.35	1943.7	Core	154666	223920519	72909323	17020524	6598573	91939108	79849305	112547306
	1984.00	1960.3	SWC	153498	117299	48265	10149	1726	296165	242719	459086
	1984.50	1960.8	Oil	160304	129154153	39115939	8482172	2446155	41328311	39940044	58058834
	1997.67	1974.0	Core plug	146992	89009535	27171248	6275429	1882920	29452312	29262215	43493093
	2352.00	2336.6	Mud	163302	804127	376508	28142	9268	133394	102803	283733

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 156					m/e 170	
					1,6-DMN	2,3- + 1,4-DMN	1,5-DMN	1,2-DMN	1,8-DMN	1,3,7-TMN	1,3,6-TMN
206/12a-5	1897.03	1873.4	Spun out oil	146645	30441151	15210267	6848487	6664318	185681	12302283	15903125
	1916.67	1893.0	Core plug	146972	25983336	13343228	5629274	6062083	127506	12378163	16023195
	1922.08	1898.4	Spun out oil	146643	51330235	26016782	10997898	10727045	357591	21116183	27544566
	1922.40	1898.7	Oil	160302	36942260	19631813	8080946	8055663	231944	17925579	22634924
	1932.67	1909.0	Core plug	146976	107931428	67691455	32059763	24550674	683553	56590084	72472374
	1937.32	1913.7	Core	154661	107732109	58310976	23485283	24619901	635621	45272418	61660872
	1938.00	1914.3	Mud	163301	302123	110306	45326	42357		92869	136367
	1948.67	1925.0	Spun out oil	146644	84125456	43126014	17730138	17074181	433793	33727354	41678418
	1956.00	1932.3	Oil	160303	40326069	20305952	8757308	8372022	241317	18471700	23014283
	1956.67	1933.0	Core plug	146982	33444108	19474430	8405956	7395246	202006	16801764	20927201
	1936-1963	1912.3-1939.3	Oil	160301	42138774	21827400	8832054	8967841	235461	19423600	24022463
	1967.35	1943.7	Core	154666	84892109	48127453	20253557	18874202	537995	41005889	55493859
	1984.00	1960.3	SWC	153498	223378	175244	46220	27149		464341	516180
	1984.50	1960.8	Oil	160304	39582053	20741301	8980342	8823323	256283	19194072	23460198
	1997.67	1974.0	Core plug	146992	30494630	15282222	6760878	6799144	163796	14253292	17466880
	2352.00	2336.6	Mud	163302	179300	69357	22225	24527	2901	52659	76731

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 170						m/e 178
					1,3,5- + 1,4,6-TMN	2,3,6-TMN	1,2,7-TMN	1,6,7- + 1,2,6-TMN	1,2,4-TMN	1,2,5-TMN	P
206/12a-5	1897.03	1873.4	Spun out oil	146645	11720663	11792668	2988882	11930548	1391456	6843206	57169562
	1916.67	1893.0	Core plug	146972	11040390	11970092	4959535	12448609	1306972	6692604	68969891
	1922.08	1898.4	Spun out oil	146643	18548404	19673029	5613899	18983907	2168892	10455911	79966095
	1922.40	1898.7	Oil	160302	16064504	17518604	7460028	16891413	1928809	9382766	86291512
	1932.67	1909.0	Core plug	146976	55201406	55186119	11496636	61434701	6984862	31072948	270853292
	1937.32	1913.7	Core	154661	43049021	50147864	15804527	46426661	5267341	28790812	410009017
	1938.00	1914.3	Mud	163301	102994	78343	18323	73050	7199	34305	260513
	1948.67	1925.0	Spun out oil	146644	29999920	31003011	7142019	29113772	3386515	16507402	123268176
	1956.00	1932.3	Oil	160303	16083136	16913668	5342750	16591127	1889214	9253724	84029300
	1956.67	1933.0	Core plug	146982	15746774	16829397	10247162	18271140	2153232	9705132	119088560
	1936-1963	1912.3-1939.3	Oil	160301	17168725	17726335	6787058	18176588	2024286	9806209	92313490
	1967.35	1943.7	Core	154666	38572248	44954482	13233969	44233916	4971557	24679548	411183850
	1984.00	1960.3	SWC	153498	536636	771268	47205	753122	25097	361702	17849784
	1984.50	1960.8	Oil	160304	17765188	17984324	6517543	18504586	2086366	10167021	94735490
	1997.67	1974.0	Core plug	146992	12813407	13379370	6504710	13786132	1502149	7811118	76637445
	2352.00	2336.6	Mud	163302	58228	42727	13631	44207	4726	19729	202677

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 192				m/e 206		
					3-MP	2-MP	9-MP	1-MP	2-EP + 9-EP 3,6-DMP	1-EP	2,6- + 2,7- + 3,5-DMP
206/12a-5	1897.03	1873.4	Spun out oil	146645	15895395	18264089	20623089	15896195	3702573	6440146	4258193
	1916.67	1893.0	Core plug	146972	21215733	25285359	25323998	20135170	4750474	9134009	4878319
	1922.08	1898.4	Spun out oil	146643	25526789	29526865	31721673	25238640	5637787	9513702	6350000
	1922.40	1898.7	Oil	160302	25536300	28671569	28331950	21733863	5217042	10055289	5377019
	1932.67	1909.0	Core plug	146976	77823997	91528306	93152547	73275860	16113186	27659822	17299779
	1937.32	1913.7	Core	154661	96081903	131424616	132885352	104241754	18464312	34124203	25928477
	1938.00	1914.3	Mud	163301	53226	73345	65876	36136	5090	10855	6702
	1948.67	1925.0	Spun out oil	146644	38343847	44755997	45789060	35646387	7845510	13514073	8921662
	1956.00	1932.3	Oil	160303	23961510	26121511	27682015	20689714	4702372	8772292	4900572
	1956.67	1933.0	Core plug	146982	36249615	41440463	41535335	33728062	7780497	14366892	7953822
	1936-1963	1912.3-1939.3	Oil	160301	28076710	32678349	32948627	25009897	5565520	10623385	5797269
	1967.35	1943.7	Core	154666	98432402	135825053	138066891	107225798	20029218	35910269	26223000
	1984.00	1960.3	SWC	153498	7589201	8976221	9242787	7994983	1750717	3123838	2152429
	1984.50	1960.8	Oil	160304	26129030	31017544	31667130	23244306	5654995	10641837	5958115
	1997.67	1974.0	Core plug	146992	22589966	25432089	25824335	20383773	5032853	8920205	5077097
	2352.00	2336.6	Mud	163302	35195	43059	37448	24550	3146	6169	4055

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 206						
					1,3- + 2,10- +3,9- + 3,10-DMP	1,6- + 2,5- + 2,9-DMP	1,7-DMP	2,3-DMP	1,9- + 4,9- + 4,10-DMP	1,8-DMP	1,2-DMP
206/12a-5	1897.03	1873.4	Spun out oil	146645	19233109	8934935	8819879	3948094	5413704	2293659	1258937
	1916.67	1893.0	Core plug	146972	25727998	11182852	11599994	5132743	6973335	2797788	1752426
	1922.08	1898.4	Spun out oil	146643	29749222	12864166	13392298	5875559	8245312	3410887	2037845
	1922.40	1898.7	Oil	160302	28240848	12040170	12667320	5563888	8216548	3147676	1983217
	1932.67	1909.0	Core plug	146976	92760176	43467359	40236466	17830221	25846228	8901580	6084010
	1937.32	1913.7	Core	154661	104995274	48189667	53921384	23015049	33343890	14814075	8820459
	1938.00	1914.3	Mud	163301	42251	11067	12815	5008	5075	1143	1291
	1948.67	1925.0	Spun out oil	146644	40640295	19012394	18358170	8015520	11371107	4802868	2779263
	1956.00	1932.3	Oil	160303	25696714	11112313	11690750	5046802	7128219	2938682	1744192
	1956.67	1933.0	Core plug	146982	43071031	19147764	18235540	8117193	11946702	4435751	2945130
	1936-1963	1912.3-1939.3	Oil	160301	30137224	13120059	13179912	6023543	8201960	3274890	2084234
	1967.35	1943.7	Core	154666	114066360	51729899	55826306	23955863	33798161	14269981	8762776
	1984.00	1960.3	SWC	153498	9380974	4240381	4620321	2049107	2720869	1169880	653843
	1984.50	1960.8	Oil	160304	30325512	13114527	13973409	6122318	8709725	3444474	2100898
	1997.67	1974.0	Core plug	146992	27173031	11166316	11797569	5209762	7663579	2930963	1884093
	2352.00	2336.6	Mud	163302	22239	6543	7053	2294	3323	812	729

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 219	m/e 184	m/e 198			m/e 253	
					Retene	DBT	4-MDBT	(3+2)-MDBT	1-MDBT	C21MA	C22MA
206/12a-5	1897.03	1873.4	Spun out oil	146645	5014695	10716370	5244262	2715191	1913007	577184	406841
	1916.67	1893.0	Core plug	146972	5841539	11155524	6389808	3507369	2134109	718657	536787
	1922.08	1898.4	Spun out oil	146643	6842635	14713314	8101714	4492320	2813401	761441	585042
	1922.40	1898.7	Oil	160302	6767892	14972539	7376070	4453530	2632752	796330	576832
	1932.67	1909.0	Core plug	146976	23246885	46842992	23694611	14198537	8696253	2372708	1822648
	1937.32	1913.7	Core	154661	19186341	72009717	32411526	18864540	10950481	790247	502094
	1938.00	1914.3	Mud	163301	8366	11896	5338	2045	1215	336	171
	1948.67	1925.0	Spun out oil	146644	11938713	18092192	10020500	5303736	3292529	1117706	786990
	1956.00	1932.3	Oil	160303	6720097	14843282	7191999	4458032	2476231	754633	520667
	1956.67	1933.0	Core plug	146982	10704257	18190895	10451530	5967281	3542925	1243311	926701
	1936-1963	1912.3-1939.3	Oil	160301	7170879	16348336	8468380	4970574	2983012	844715	616544
	1967.35	1943.7	Core	154666	20318051	64498813	29903577	17999709	9915694	1011444	672582
	1984.00	1960.3	SWC	153498	2766815	1862278	2015623	1049838	659197	208853	103320
	1984.50	1960.8	Oil	160304	7579659	17403088	8039188	4977499	2753805	947972	661344
	1997.67	1974.0	Core plug	146992	7338663	12885139	6583653	3841207	2301227	861584	598624
	2352.00	2336.6	Mud	163302	4447	8174	2960	1393	825	201	92

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 253						
					bSC27MA	bSC27DMA	bRC27MA+ bRC27DMA	aSC27MA	bSC28MA+ bSC28DMA + aRC27DMA	aSC27DMA	aRC27MA
206/12a-5	1897.03	1873.4	Spun out oil	146645	189314	699145	582142	308859	876014	153874	255000
	1916.67	1893.0	Core plug	146972	180150	889070	630948	305589	1013924	148866	262279
	1922.08	1898.4	Spun out oil	146643	272072	859743	796984	427319	1210358	189452	362754
	1922.40	1898.7	Oil	160302	250127	981844	768295	401631	1158299	171474	342247
	1932.67	1909.0	Core plug	146976	894016	2817879	2635152	1226883	3872304	687206	1189829
	1937.32	1913.7	Core	154661	116803	422843	350890	170388	594790	58316	144663
	1938.00	1914.3	Mud	163301	111	277	227	205	448		
	1948.67	1925.0	Spun out oil	146644	411920	1300300	1073290	623381	1740164	276969	538973
	1956.00	1932.3	Oil	160303	221175	900840	691357	328410	1080481	144486	293727
	1956.67	1933.0	Core plug	146982	465322	1542569	1321248	667725	1862101	281054	604978
	1936-1963	1912.3-1939.3	Oil	160301	274751	1041094	808804	422941	1233670	174447	366420
	1967.35	1943.7	Core	154666	220282	885276	719704	343945	1038763	124823	324609
	1984.00	1960.3	SWC	153498	31896	212399	170135	72937	279069	20407	51393
	1984.50	1960.8	Oil	160304	309864	1197275	909143	472044	1364955	203679	410559
	1997.67	1974.0	Core plug	146992	290408	1078693	795476	408657	1208744	171420	377537
	2352.00	2336.6	Mud	163302		109	155	84	167		

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 253						m/e 231
					aSC28MA	bRC28MA+ bRC28DMA	bSC29MA+ bSC29DMA	aSC29MA	aRC28MA+ bRC29MA+ bRC29DMA	aRC29MA	C20TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	359831	571788	603820	192151	503899	153100	1325855
	1916.67	1893.0	Core plug	146972	385346	683241	741727	221654	626756	125851	1829104
	1922.08	1898.4	Spun out oil	146643	454980	742398	804381	232492	652330	202653	1752272
	1922.40	1898.7	Oil	160302	469825	751204	815153	242102	707005	185898	2038133
	1932.67	1909.0	Core plug	146976	1396764	2497257	2832116	813106	2327595	760189	5333165
	1937.32	1913.7	Core	154661	203069	321801	320534	79780	256283	60520	4642718
	1938.00	1914.3	Mud	163301	151	219	258	126	200		421
	1948.67	1925.0	Spun out oil	146644	734836	1085237	1157351	368605	924331	302656	2749027
	1956.00	1932.3	Oil	160303	408394	666068	668163	205210	584582	160159	1745735
	1956.67	1933.0	Core plug	146982	749478	1293077	1372240	441912	1193525	337619	2947733
	1936-1963	1912.3-1939.3	Oil	160301	484837	788735	823070	261992	698912	186794	2055406
	1967.35	1943.7	Core	154666	445683	654800	706667	170388	580255	120906	4773390
	1984.00	1960.3	SWC	153498	86410	169395	166128	32317	127087	21227	582023
	1984.50	1960.8	Oil	160304	546056	865319	949754	294311	796739	222828	2256840
	1997.67	1974.0	Core plug	146992	496095	767187	796915	265491	704473	180584	2103613
	2352.00	2336.6	Mud	163302	162	176	130		145		234

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 231						
					C21TA	SC26TA	RC26TA+ SC27TA	M1	M2	SC28TA	RC27TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	1438675	1374771	2915715	118387	284434	1250535	994483
	1916.67	1893.0	Core plug	146972	1774722	1700395	3653121	128369	329544	1780551	1309690
	1922.08	1898.4	Spun out oil	146643	1772234	1769242	3747153	142175	384308	1628570	1257895
	1922.40	1898.7	Oil	160302	1853421	1929025	3922145	152648	398095	1840493	1419257
	1932.67	1909.0	Core plug	146976	5243793	5305728	11528548	518142	1248899	5592098	4560736
	1937.32	1913.7	Core	154661	3887466	2310244	4899080	153690	376534	1710820	1608693
	1938.00	1914.3	Mud	163301	395	655	2254			797	802
	1948.67	1925.0	Spun out oil	146644	2708211	2641318	5412500	252057	531326	2228838	1784880
	1956.00	1932.3	Oil	160303	1710582	1781573	3563069	126922	376557	1607976	1200688
	1956.67	1933.0	Core plug	146982	2973374	2930781	6270294	231246	628981	2865459	2390398
	1936-1963	1912.3-1939.3	Oil	160301	1983674	1919284	4212470	161661	408607	1867872	1497386
	1967.35	1943.7	Core	154666	4401991	2734623	5593979	233229	492570	2149556	1791784
	1984.00	1960.3	SWC	153498	568350	606242	1304717	24350	89378	487123	380030
	1984.50	1960.8	Oil	160304	2214616	2107183	4576970	176641	477497	2054987	1647648
	1997.67	1974.0	Core plug	146992	1956899	1919951	4161810	152842	419156	1839648	1442052
	2352.00	2336.6	Mud	163302	217	330	930			388	277

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 231						
					M3	M4	RC28TA	3MS-TA	4MS-TA	2,24DMS-TA	3,24DMS-TA +3MR-TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	329625	501596	1348287	230160	374750	64363	447543
	1916.67	1893.0	Core plug	146972	390859	547298	1763598	305244	478609	93550	986770
	1922.08	1898.4	Spun out oil	146643	454166	652280	1669907	290349	505784	86131	569666
	1922.40	1898.7	Oil	160302	451191	619627	1870621	353898	544369	95117	1067353
	1932.67	1909.0	Core plug	146976	1300014	2090925	5998090	1019908	1534624	299303	1905339
	1937.32	1913.7	Core	154661	446227	636158	1812722	317694	589709	70385	638293
	1938.00	1914.3	Mud	163301		369	934		270		244
	1948.67	1925.0	Spun out oil	146644	565210	803264	2429254	466512	789891	144232	906336
	1956.00	1932.3	Oil	160303	404294	554835	1644217	307118	482088	82563	935805
	1956.67	1933.0	Core plug	146982	715376	1032597	2997060	578022	893525	192455	1703314
	1936-1963	1912.3-1939.3	Oil	160301	448354	655845	1899122	365686	561610	96238	1122015
	1967.35	1943.7	Core	154666	563252	810318	2398422	435504	766074	109804	831666
	1984.00	1960.3	SWC	153498	105439	160129	567079	51184	116468	9863	139771
	1984.50	1960.8	Oil	160304	525434	695047	2221762	390866	628559	112327	1191347
	1997.67	1974.0	Core plug	146992	479466	647040	1845125	340119	557806	115736	1045343
	2352.00	2336.6	Mud	163302			581	101	107		113

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 231						
					4,24DMS-TA +4MR-TA	D1-TA	3M24ES-TA	D2-TA	4M24ES-TA	3,24DMR-TA	4,24DMR-TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	729432	389203	302879	902036	354973	254699	400139
	1916.67	1893.0	Core plug	146972	526656	454063	407012	1184930	447647	373758	544867
	1922.08	1898.4	Spun out oil	146643	944775	458356	397370	1192778	438578	325030	493822
	1922.40	1898.7	Oil	160302	628968	538528	432413	1319242	486674	419316	629939
	1932.67	1909.0	Core plug	146976	2918468	1388091	1234536	4024413	1383104	1083107	1635007
	1937.32	1913.7	Core	154661	1032188	449927	328570	1098020	398415	300763	534895
	1938.00	1914.3	Mud	163301	358	293		613	304		
	1948.67	1925.0	Spun out oil	146644	1409490	694918	559385	1842130	627022	469410	780106
	1956.00	1932.3	Oil	160303	535444	462226	375015	1197161	424675	342336	521301
	1956.67	1933.0	Core plug	146982	985630	883800	759413	2197783	868222	672946	1023374
	1936-1963	1912.3-1939.3	Oil	160301	587663	542233	450183	1353019	517885	433989	625707
	1967.35	1943.7	Core	154666	1303920	620862	471458	1596613	544569	434991	671407
	1984.00	1960.3	SWC	153498	260359	106732	74632	361035	71452	59923	123353
	1984.50	1960.8	Oil	160304	656233	609896	504683	1579138	577726	466318	682303
	1997.67	1974.0	Core plug	146992	587585	575684	435743	1373539	461475	396031	594221
	2352.00	2336.6	Mud	163302	144			237			

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 231						
					D3-TA	D4-TA	2M24ER-TA	3M24ER-TA	D5-TA	4M24ER-TA	D6-TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	837443	1034408	64801	277193	682333	296014	1345831
	1916.67	1893.0	Core plug	146972	1009516	1361542	64823	365232	829621	394038	1693074
	1922.08	1898.4	Spun out oil	146643	1035868	1255683	69632	333785	897425	388191	1769784
	1922.40	1898.7	Oil	160302	1154472	1520926	77347	393013	1013608	431070	1980257
	1932.67	1909.0	Core plug	146976	3488034	4426835	274608	1223397	3208022	1291693	6046076
	1937.32	1913.7	Core	154661	948114	1217873	59960	314905	755966	373035	1655639
	1938.00	1914.3	Mud	163301	604	728		226	496	215	1241
	1948.67	1925.0	Spun out oil	146644	1599645	1973539	136865	535017	1310128	573350	2620338
	1956.00	1932.3	Oil	160303	1074795	1351640	65897	359132	905695	371385	1733455
	1956.67	1933.0	Core plug	146982	1927764	2518192	132372	655425	1671384	740296	3286973
	1936-1963	1912.3-1939.3	Oil	160301	1174362	1545120	78895	424450	1025734	439035	2047592
	1967.35	1943.7	Core	154666	1460705	1862406	76363	475838	1222277	488681	2421523
	1984.00	1960.3	SWC	153498	333961	417768	10878	78866	257998	80590	614127
	1984.50	1960.8	Oil	160304	1266156	1686450	87401	488024	1141230	492410	2334405
	1997.67	1974.0	Core plug	146992	1210540	1624782	80233	401767	1030989	430421	2200458
	2352.00	2336.6	Mud	163302	223	339			201	137	403

TABLE Ap8.2 Aromatic fraction GC-MS peak heights

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 142		m/e 156				
					2-MN	1-MN	2-EN	1-EN	2,6-DMN	2,7-DMN	1,3- + 1,7-DMN
206/12a-5	1897.03	1873.4	Spun out oil	146645	386726	156610	29818	10980	125017	122389	196794
	1916.67	1893.0	Core plug	146972	551073	194324	41616	12870	197343	182305	288772
	1922.08	1898.4	Spun out oil	146643	637963	203631	41872	11535	183103	130498	275906
	1922.40	1898.7	Oil	160302	719896	225265	54815	14928	273222	251669	376033
	1932.67	1909.0	Core plug	146976	1077505	361026	105009	61165	556917	573775	710116
	1937.32	1913.7	Core	154661	2121960	683971	139124	38480	605892	569959	946608
	1938.00	1914.3	Mud	163301	5503	2728	212	52	1130	920	2152
	1948.67	1925.0	Spun out oil	146644	867846	266835	61979	16262	241551	231212	389561
	1956.00	1932.3	Oil	160303	740978	233586	52660	14879	244895	234883	355844
	1956.67	1933.0	Core plug	146982	769077	231972	57719	19170	307099	333651	383628
	1936-1963	1912.3-1939.3	Oil	160301	783718	242364	50685	14024	246894	246562	343911
	1967.35	1943.7	Core	154666	1224470	398692	93074	36083	502753	436642	615445
	1984.00	1960.3	SWC	153498	712	293	62	10	1799	1474	2788
	1984.50	1960.8	Oil	160304	743888	225296	48855	14089	238038	230042	334401
	1997.67	1974.0	Core plug	146992	682468	208332	48116	14437	225821	224364	333477
	2352.00	2336.6	Mud	163302	4927	2307	172	57	817	630	1738

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 156					m/e 170	
					1,6-DMN	2,3- + 1,4-DMN	1,5-DMN	1,2-DMN	1,8-DMN	1,3,7-TMN	1,3,6-TMN
206/12a-5	1897.03	1873.4	Spun out oil	146645	141338	70621	31797	30942	862	57119	73838
	1916.67	1893.0	Core plug	146972	198267	101816	42954	46257	973	94452	122266
	1922.08	1898.4	Spun out oil	146643	187730	95151	40223	39232	1308	77228	100739
	1922.40	1898.7	Oil	160302	257043	136598	56227	56051	1614	124726	157493
	1932.67	1909.0	Core plug	146976	475279	298081	141176	108110	3010	249196	319134
	1937.32	1913.7	Core	154661	706522	382411	154020	161461	4168	296903	404380
	1938.00	1914.3	Mud	163301	1416	517	212	198		435	639
	1948.67	1925.0	Spun out oil	146644	266545	136641	56177	54098	1374	106863	132055
	1956.00	1932.3	Oil	160303	250676	126227	54437	52042	1500	114824	143062
	1956.67	1933.0	Core plug	146982	259349	151018	65186	57348	1566	130293	162284
	1936-1963	1912.3-1939.3	Oil	160301	237916	123238	49866	50633	1329	109666	135631
	1967.35	1943.7	Core	154666	464217	263176	110753	103210	2942	224233	303458
	1984.00	1960.3	SWC	153498	1357	1064	281	165		2820	3135
	1984.50	1960.8	Oil	160304	227980	119463	51724	50820	1476	110552	135123
	1997.67	1974.0	Core plug	146992	233813	117174	51838	52131	1256	109285	133925
	2352.00	2336.6	Mud	163302	1099	425	136	150	18	323	470

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 170						m/e 178
					1,3,5- + 1,4,6-TMN	2,3,6-TMN	1,2,7-TMN	1,6,7- + 1,2,6-TMN	1,2,4-TMN	1,2,5-TMN	P
206/12a-5	1897.03	1873.4	Spun out oil	146645	54419	54753	13877	55393	6460	31773	504560
	1916.67	1893.0	Core plug	146972	84244	91338	37844	94990	9973	51068	601923
	1922.08	1898.4	Spun out oil	146643	67837	71950	20532	69430	7932	38240	527299
	1922.40	1898.7	Oil	160302	111776	121894	51907	117530	13421	65285	724485
	1932.67	1909.0	Core plug	146976	243081	243013	50626	270529	30758	136830	1639622
	1937.32	1913.7	Core	154661	282321	328876	103648	304472	34544	188814	2962069
	1938.00	1914.3	Mud	163301	483	367	86	342	34	161	2335
	1948.67	1925.0	Spun out oil	146644	95052	98231	22629	92245	10730	52302	637934
	1956.00	1932.3	Oil	160303	99977	105139	33212	103134	11744	57523	611827
	1956.67	1933.0	Core plug	146982	122112	130507	79464	141687	16698	75260	932688
	1936-1963	1912.3-1939.3	Oil	160301	96935	100083	38320	102625	11429	55366	663723
	1967.35	1943.7	Core	154666	210926	245826	72368	241885	27186	134956	2668054
	1984.00	1960.3	SWC	153498	3259	4684	287	4574	152	2197	124238
	1984.50	1960.8	Oil	160304	102322	103584	37539	106581	12017	58559	611440
	1997.67	1974.0	Core plug	146992	98245	102584	49874	105703	11518	59891	580266
	2352.00	2336.6	Mud	163302	357	262	84	271	29	121	2243

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 192				m/e 206		
					3-MP	2-MP	9-MP	1-MP	2-EP + 9-EP 3,6-DMP	1-EP	2,6- + 2,7- + 3,5-DMP
206/12a-5	1897.03	1873.4	Spun out oil	146645	140288	161193	182013	140295	32678	56839	37581
	1916.67	1893.0	Core plug	146972	185157	220674	221011	175726	41459	79716	42575
	1922.08	1898.4	Spun out oil	146643	168324	194701	209174	166424	37176	62734	41872
	1922.40	1898.7	Oil	160302	214397	240720	237869	182473	43801	84422	45144
	1932.67	1909.0	Core plug	146976	471111	554071	563903	443579	97542	167440	104725
	1937.32	1913.7	Core	154661	694134	949464	960017	753084	133394	246527	187318
	1938.00	1914.3	Mud	163301	477	657	590	324	46	97	60
	1948.67	1925.0	Spun out oil	146644	198436	231620	236966	184476	40602	69938	46171
	1956.00	1932.3	Oil	160303	174466	190194	201556	150644	34238	63872	35682
	1956.67	1933.0	Core plug	146982	283903	324557	325300	264154	60936	112520	62293
	1936-1963	1912.3-1939.3	Oil	160301	201868	234953	236897	179818	40015	76381	41682
	1967.35	1943.7	Core	154666	638700	881330	895876	695757	129964	233011	170154
	1984.00	1960.3	SWC	153498	52822	62476	64331	55647	12185	21742	14981
	1984.50	1960.8	Oil	160304	168641	200193	204385	150023	36498	68684	38455
	1997.67	1974.0	Core plug	146992	171042	192561	195531	154337	38107	67540	38442
	2352.00	2336.6	Mud	163302	390	477	414	272	35	68	45

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 206						
					1,3- + 2,10- +3,9- + 3,10-DMP	1,6- + 2,5- + 2,9-DMP	1,7-DMP	2,3-DMP	1,9- + 4,9- + 4,10-DMP	1,8-DMP	1,2-DMP
206/12a-5	1897.03	1873.4	Spun out oil	146645	169745	78857	77841	34845	47780	20243	11111
	1916.67	1893.0	Core plug	146972	224537	97596	101237	44795	60859	24417	15294
	1922.08	1898.4	Spun out oil	146643	196167	84827	88309	38744	54370	22491	13438
	1922.40	1898.7	Oil	160302	237104	101087	106352	46713	68984	26427	16651
	1932.67	1909.0	Core plug	146976	561528	263132	243573	107936	156461	53886	36830
	1937.32	1913.7	Core	154661	758528	348141	389550	166270	240890	107023	63723
	1938.00	1914.3	Mud	163301	379	99	115	45	45	10	12
	1948.67	1925.0	Spun out oil	146644	210321	98392	95007	41482	58847	24856	14383
	1956.00	1932.3	Oil	160303	187101	80910	85122	36746	51901	21397	12700
	1956.67	1933.0	Core plug	146982	337327	149963	142819	63573	93565	34740	23066
	1936-1963	1912.3-1939.3	Oil	160301	216683	94332	94762	43309	58971	23546	14985
	1967.35	1943.7	Core	154666	740144	335660	362241	155443	219307	92594	56859
	1984.00	1960.3	SWC	153498	65293	29514	32158	14262	18938	8143	4551
	1984.50	1960.8	Oil	160304	195726	84644	90187	39515	56214	22231	13560
	1997.67	1974.0	Core plug	146992	205743	84547	89326	39446	58025	22192	14266
	2352.00	2336.6	Mud	163302	246	72	78	25	37	9	8

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 219	m/e 184	m/e 198			m/e 253	
					Retene	DBT	4-MDBT	(3+2)-MDBT	1-MDBT	C21MA	C22MA
206/12a-5	1897.03	1873.4	Spun out oil	146645	44258	94579	46284	23963	16884	5677	4002
	1916.67	1893.0	Core plug	146972	50981	97358	55766	30610	18625	5278	3942
	1922.08	1898.4	Spun out oil	146643	45121	97020	53423	29622	18552	5530	4249
	1922.40	1898.7	Oil	160302	56822	125706	61928	37391	22104	6039	4374
	1932.67	1909.0	Core plug	146976	140726	283566	143436	85951	52643	16145	12402
	1937.32	1913.7	Core	154661	138610	520227	234154	136285	79111	5709	3627
	1938.00	1914.3	Mud	163301	75	107	48	18	11	7	3
	1948.67	1925.0	Spun out oil	146644	61785	93630	51858	27448	17039	6196	4363
	1956.00	1932.3	Oil	160303	48930	108076	52366	32459	18030	5514	3805
	1956.67	1933.0	Core plug	146982	83835	142469	81855	46735	27748	8328	6207
	1936-1963	1912.3-1939.3	Oil	160301	51558	117543	60887	35738	21447	5661	4132
	1967.35	1943.7	Core	154666	131838	418514	194036	116795	64340	6563	4364
	1984.00	1960.3	SWC	153498	19258	12962	14029	7307	4588	1454	719
	1984.50	1960.8	Oil	160304	48920	112323	51886	32126	17774	6030	4207
	1997.67	1974.0	Core plug	146992	55565	97561	49849	29084	17424	5644	3921
	2352.00	2336.6	Mud	163302	49	90	33	15	9	5	2

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 253						
					bSC27MA	bSC27DMA	bRC27MA+ bRC27DMA	aSC27MA	bSC28MA+ bSC28DMA + aRC27DMA	aSC27DMA	aRC27MA
206/12a-5	1897.03	1873.4	Spun out oil	146645	1862	6877	5726	3038	8616	1513	2508
	1916.67	1893.0	Core plug	146972	1323	6529	4634	2244	7446	1093	1926
	1922.08	1898.4	Spun out oil	146643	1976	6244	5789	3104	8791	1376	2635
	1922.40	1898.7	Oil	160302	1897	7446	5826	3046	8784	1300	2595
	1932.67	1909.0	Core plug	146976	6083	19175	17931	8348	26350	4676	8096
	1937.32	1913.7	Core	154661	844	3055	2535	1231	4297	421	1045
	1938.00	1914.3	Mud	163301	2	6	5	4	9		
	1948.67	1925.0	Spun out oil	146644	2283	7208	5950	3456	9647	1535	2988
	1956.00	1932.3	Oil	160303	1616	6583	5052	2400	7895	1056	2146
	1956.67	1933.0	Core plug	146982	3117	10332	8850	4472	12472	1882	4052
	1936-1963	1912.3-1939.3	Oil	160301	1841	6977	5420	2834	8268	1169	2456
	1967.35	1943.7	Core	154666	1429	5744	4670	2232	6740	810	2106
	1984.00	1960.3	SWC	153498	222	1478	1184	508	1942	142	358
	1984.50	1960.8	Oil	160304	1971	7616	5783	3003	8683	1296	2612
	1997.67	1974.0	Core plug	146992	1902	7066	5211	2677	7918	1123	2473
	2352.00	2336.6	Mud	163302		3	4	2	4		

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 253						m/e 231
					aSC28MA	bRC28MA+ bRC28DMA	bSC29MA+ bSC29DMA	aSC29MA	aRC28MA+ bRC29MA+ bRC29DMA	aRC29MA	C20TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	3539	5624	5939	1890	4956	1506	13041
	1916.67	1893.0	Core plug	146972	2830	5018	5447	1628	4603	924	13433
	1922.08	1898.4	Spun out oil	146643	3305	5392	5842	1689	4738	1472	12727
	1922.40	1898.7	Oil	160302	3563	5697	6181	1836	5361	1410	15456
	1932.67	1909.0	Core plug	146976	9504	16993	19271	5533	15838	5173	36290
	1937.32	1913.7	Core	154661	1467	2325	2316	576	1851	437	33541
	1938.00	1914.3	Mud	163301	3	4	5	3	4		8
	1948.67	1925.0	Spun out oil	146644	4074	6016	6416	2043	5124	1678	15239
	1956.00	1932.3	Oil	160303	2984	4867	4882	1500	4272	1170	12757
	1956.67	1933.0	Core plug	146982	5020	8661	9191	2960	7994	2261	19744
	1936-1963	1912.3-1939.3	Oil	160301	3249	5286	5516	1756	4684	1252	13775
	1967.35	1943.7	Core	154666	2892	4249	4585	1106	3765	785	30973
	1984.00	1960.3	SWC	153498	601	1179	1156	225	885	148	4051
	1984.50	1960.8	Oil	160304	3474	5504	6042	1872	5068	1417	14356
	1997.67	1974.0	Core plug	146992	3250	5025	5220	1739	4615	1183	13780
	2352.00	2336.6	Mud	163302	4	4	3		4		6

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 231						
					C21TA	SC26TA	RC26TA+ SC27TA	M1	M2	SC28TA	RC27TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	14151	13522	28679	1164	2798	12300	9782
	1916.67	1893.0	Core plug	146972	13033	12487	26828	943	2420	13076	9618
	1922.08	1898.4	Spun out oil	146643	12872	12850	27216	1033	2791	11829	9136
	1922.40	1898.7	Oil	160302	14055	14628	29742	1158	3019	13957	10762
	1932.67	1909.0	Core plug	146976	35682	36104	78447	3526	8498	38052	31034
	1937.32	1913.7	Core	154661	28085	16690	35393	1110	2720	12360	11622
	1938.00	1914.3	Mud	163301	8	13	45			16	16
	1948.67	1925.0	Spun out oil	146644	15013	14642	30004	1397	2945	12356	9894
	1956.00	1932.3	Oil	160303	12500	13019	26036	927	2752	11750	8774
	1956.67	1933.0	Core plug	146982	19915	19630	41998	1549	4213	19193	16011
	1936-1963	1912.3-1939.3	Oil	160301	13294	12863	28231	1083	2738	12518	10035
	1967.35	1943.7	Core	154666	28563	17744	36298	1513	3196	13948	11626
	1984.00	1960.3	SWC	153498	3956	4220	9081	169	622	3390	2645
	1984.50	1960.8	Oil	160304	14087	13404	29115	1124	3037	13072	10481
	1997.67	1974.0	Core plug	146992	12819	12577	27262	1001	2746	12051	9446
	2352.00	2336.6	Mud	163302	5	8	23			10	7

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 231						
					M3	M4	RC28TA	3MS-TA	4MS-TA	2,24DMS-TA	3,24DMS-TA +3MR-TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	3242	4934	13262	2264	3686	633	4402
	1916.67	1893.0	Core plug	146972	2870	4019	12951	2242	3515	687	7247
	1922.08	1898.4	Spun out oil	146643	3299	4738	12129	2109	3674	626	4138
	1922.40	1898.7	Oil	160302	3421	4699	14185	2684	4128	721	8094
	1932.67	1909.0	Core plug	146976	8846	14228	40815	6940	10443	2037	12965
	1937.32	1913.7	Core	154661	3224	4596	13096	2295	4260	508	4611
	1938.00	1914.3	Mud	163301		7	19		5		5
	1948.67	1925.0	Spun out oil	146644	3133	4453	13467	2586	4379	800	5024
	1956.00	1932.3	Oil	160303	2954	4054	12015	2244	3523	603	6838
	1956.67	1933.0	Core plug	146982	4792	6916	20074	3872	5985	1289	11409
	1936-1963	1912.3-1939.3	Oil	160301	3005	4395	12728	2451	3764	645	7520
	1967.35	1943.7	Core	154666	3655	5258	15563	2826	4971	712	5396
	1984.00	1960.3	SWC	153498	734	1115	3947	356	811	69	973
	1984.50	1960.8	Oil	160304	3342	4421	14133	2486	3998	715	7578
	1997.67	1974.0	Core plug	146992	3141	4238	12087	2228	3654	758	6848
	2352.00	2336.6	Mud	163302			14	3	3		3

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 231						
					4,24DMS-TA +4MR-TA	D1-TA	3M24ES-TA	D2-TA	4M24ES-TA	3,24DMR-TA	4,24DMR-TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	7175	3828	2979	8872	3491	2505	3936
	1916.67	1893.0	Core plug	146972	3868	3335	2989	8702	3287	2745	4001
	1922.08	1898.4	Spun out oil	146643	6862	3329	2886	8663	3185	2361	3587
	1922.40	1898.7	Oil	160302	4770	4084	3279	10004	3691	3180	4777
	1932.67	1909.0	Core plug	146976	19859	9445	8401	27385	9412	7370	11126
	1937.32	1913.7	Core	154661	7457	3250	2374	7933	2878	2173	3864
	1938.00	1914.3	Mud	163301	7	6		12	6		
	1948.67	1925.0	Spun out oil	146644	7814	3852	3101	10212	3476	2602	4325
	1956.00	1932.3	Oil	160303	3913	3378	2740	8748	3103	2502	3809
	1956.67	1933.0	Core plug	146982	6602	5920	5086	14721	5815	4507	6854
	1936-1963	1912.3-1939.3	Oil	160301	3938	3634	3017	9068	3471	2909	4193
	1967.35	1943.7	Core	154666	8461	4029	3059	10360	3534	2823	4357
	1984.00	1960.3	SWC	153498	1812	743	519	2513	497	417	859
	1984.50	1960.8	Oil	160304	4174	3880	3210	10045	3675	2966	4340
	1997.67	1974.0	Core plug	146992	3849	3771	2854	8997	3023	2594	3892
	2352.00	2336.6	Mud	163302	4			6			

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Well	MD (m)	TVDSS (m)	Sample type	APTID	m/e 231						
					D3-TA	D4-TA	2M24ER-TA	3M24ER-TA	D5-TA	4M24ER-TA	D6-TA
206/12a-5	1897.03	1873.4	Spun out oil	146645	8237	10174	637	2726	6711	2912	13238
	1916.67	1893.0	Core plug	146972	7414	9999	476	2682	6093	2894	12434
	1922.08	1898.4	Spun out oil	146643	7524	9120	506	2424	6518	2819	12854
	1922.40	1898.7	Oil	160302	8755	11533	587	2980	7686	3269	15017
	1932.67	1909.0	Core plug	146976	23735	30123	1869	8325	21829	8789	41141
	1937.32	1913.7	Core	154661	6850	8798	433	2275	5461	2695	11961
	1938.00	1914.3	Mud	163301	12	15		5	10	4	25
	1948.67	1925.0	Spun out oil	146644	8868	10940	759	2966	7263	3178	14526
	1956.00	1932.3	Oil	160303	7854	9877	482	2624	6618	2714	12667
	1956.67	1933.0	Core plug	146982	12912	16867	887	4390	11195	4958	22016
	1936-1963	1912.3-1939.3	Oil	160301	7870	10355	529	2845	6874	2942	13723
	1967.35	1943.7	Core	154666	9478	12085	495	3088	7931	3171	15713
	1984.00	1960.3	SWC	153498	2324	2908	76	549	1796	561	4274
	1984.50	1960.8	Oil	160304	8054	10728	556	3104	7260	3132	14849
	1997.67	1974.0	Core plug	146992	7930	10643	526	2632	6754	2819	14414
	2352.00	2336.6	Mud	163302	6	8			5	3	10

TABLE Ap8.3 Aromatic fraction GC-MS amounts in ng/g

Polycyclic aromatic hydrocarbons and sulphur compounds

Peak	Ion(s)	Compound
MN	m/e 142	methylnaphthalenes
EN	me/ 156	ethylnaphthalenes
DMN	me/ 156	dimethylnaphthalenes
TMN	m/e 170	trimethylnaphthalenes
P	m/e 178	phenanthrene
MP	m/e 192	methylphenanthrenes
EP	m/e 206	ethylphenanthrenes
DMP	m/e 206	dimethylphenanthrenes
Retene	m/e 219	retene
DBT	m/e 184	dibenzothiophene
MDBT	m/e 198	methyldibenzothiophenes

Triaromatic steroids

Peak	Ion(s)	Compound
3MS-TA	245	(20S) 3-methyl triaromatic steroid
4MS-TA	245	(20S) 4-methyl triaromatic steroid
2,24DMS-TA	245	(20S) 2,24-dimethyl triaromatic steroid
3,24DMS+3MR-TA	245	(20S) 3,24-dimethyl triaromatic steroid
4,24DMS+4MR-TA	245	(20S) 4,24-dimethyl triaromatic steroid
3M24ES-TA	245	(20S) 3-methyl-24-ethyl triaromatic
4M24ES-TA	245	(20S) 4-methyl-24-ethyl triaromatic
3,24DMR-TA	245	(20R) 3,24-dimethyl triaromatic steroid
4,24DMR-TA	245	(20R) 4,24-dimethyl triaromatic steroid
2M24ER-TA	245	(20R) 2-methyl-24-ethyl triaromatic
3M24ER-TA	245	(20R) 3-methyl-24-ethyl triaromatic
4M24ER-TA	245	(20R) 4-methyl-24-ethyl triaromatic

Triaromatic dinosteroids

Peak	Ion(s)	Compound
D1-TA	245	triaromatic dinosteroid #1
D2-TA	245	triaromatic dinosteroid #2
D3-TA	245	triaromatic dinosteroid #3
D4-TA	245	triaromatic dinosteroid #4
D5-TA	245	triaromatic dinosteroid #5
D6-TA	245	triaromatic dinosteroid #6
M1	231	unidentified triaromatic steroid
M2	231	unidentified triaromatic steroid
M3	231	unidentified triaromatic steroid
M4	231	unidentified triaromatic steroid

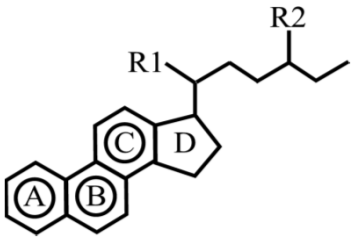
ABC-ring triaromatic steroids

Peak	Ion(s)	Substituent	
		R1	R2
C20TA	231	CH3	H
C21TA	231	CH3	CH3
SC26TA	231	S(CH3)	C6H13
RC26TA+	231	R(CH3)	C6H13
SC27TA		S(CH3)	C7H15
SC28TA	231	S(CH3)	C8H17
RC27TA	231	R(CH3)	C7H15
RC28TA	231	R(CH3)	C8H17

C-ring monoaromatic steroids

Peak	Ion(s)	Substituent			
		R1	R2	R3	R4
C21MA	253				
C22MA	253				
βSC27MA	253	β(H)	CH3	S(CH3)	H
βSC27DMA	253	β(CH3)	H	S(CH3)	H
βRC27DMA+	253	β(CH3)	H	RCH3)	H
βRC27MA		β(H)	CH3	R(CH3)	H
αSC27MA	253	α(H)	CH3	S(CH3)	H
βSC28MA+	253	β(H)	CH3	S(CH3)	CH3
αRC27DMA+		α(CH3)	H	R(CH3)	H
βSC28DMA		β(CH3)	H	S(CH3)	CH3
αSC27DMA	253	α(CH3)	H	S(CH3)	CH3
αRC27MA	253	α(H)	CH3	R(CH3)	H
αSC28MA	253	α(H)	CH3	S(CH3)	CH3
βRC28MA+	253	β(H)	CH3	R(CH3)	CH3
βRC28DMA		β(CH3)	H	R(CH3)	CH3
βSC29MA+	253	β(H)	CH3	S(CH3)	C2H5
βSC29DMA		β(CH3)	H	S(CH3)	C2H5
αSC29MA	253	α(H)	CH3	S(CH3)	C2H5
αRC28MA+	253	α(H)	CH3	R(CH3)	CH3
βRC29MA+		β(H)	CH3	R(CH3)	C2H5
βRC29DMA		β(CH3)	H	R(CH3)	C2H5
αRC29MA	253	α(H)	CH3	R(CH3)	C2H5

ABC-ring triaromatic steroids



C-ring monoaromatic steroids

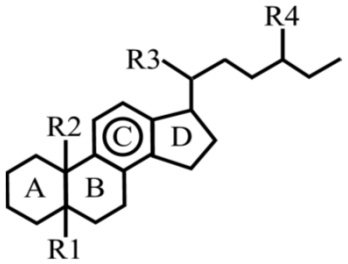
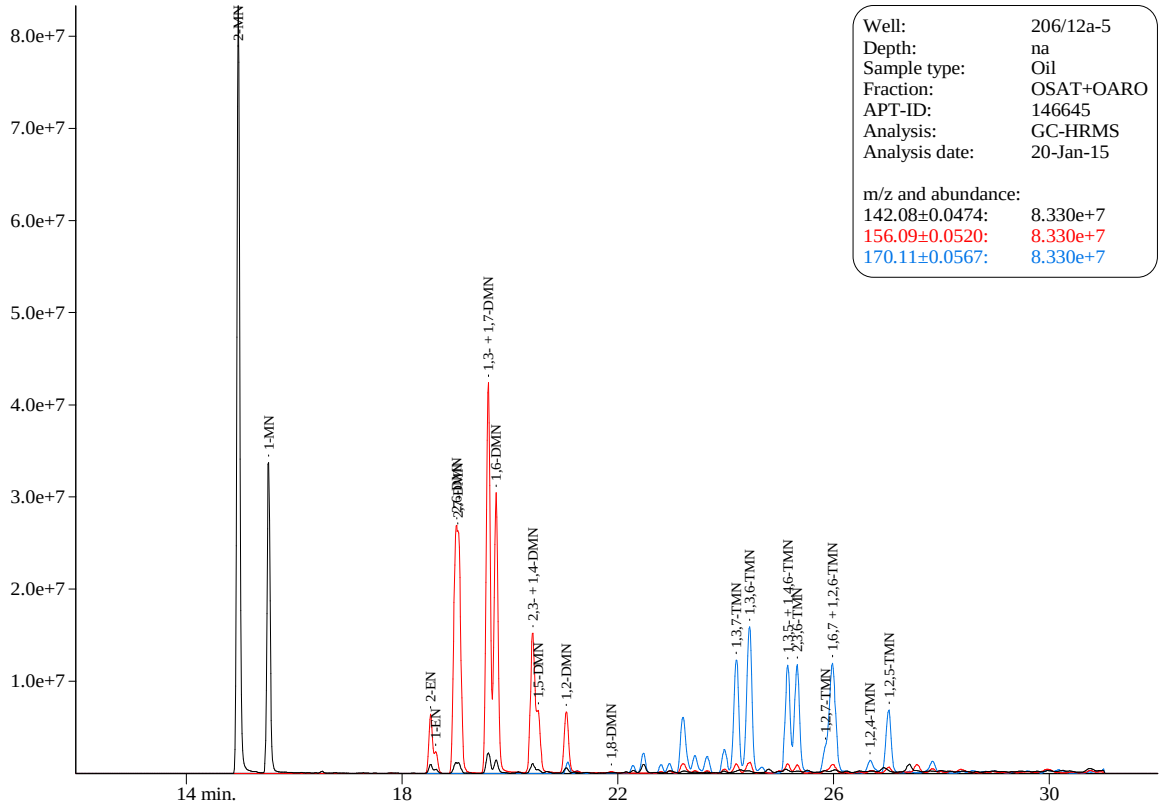


TABLE Ap8KEY Abbreviations of aromatic fraction biomarkers



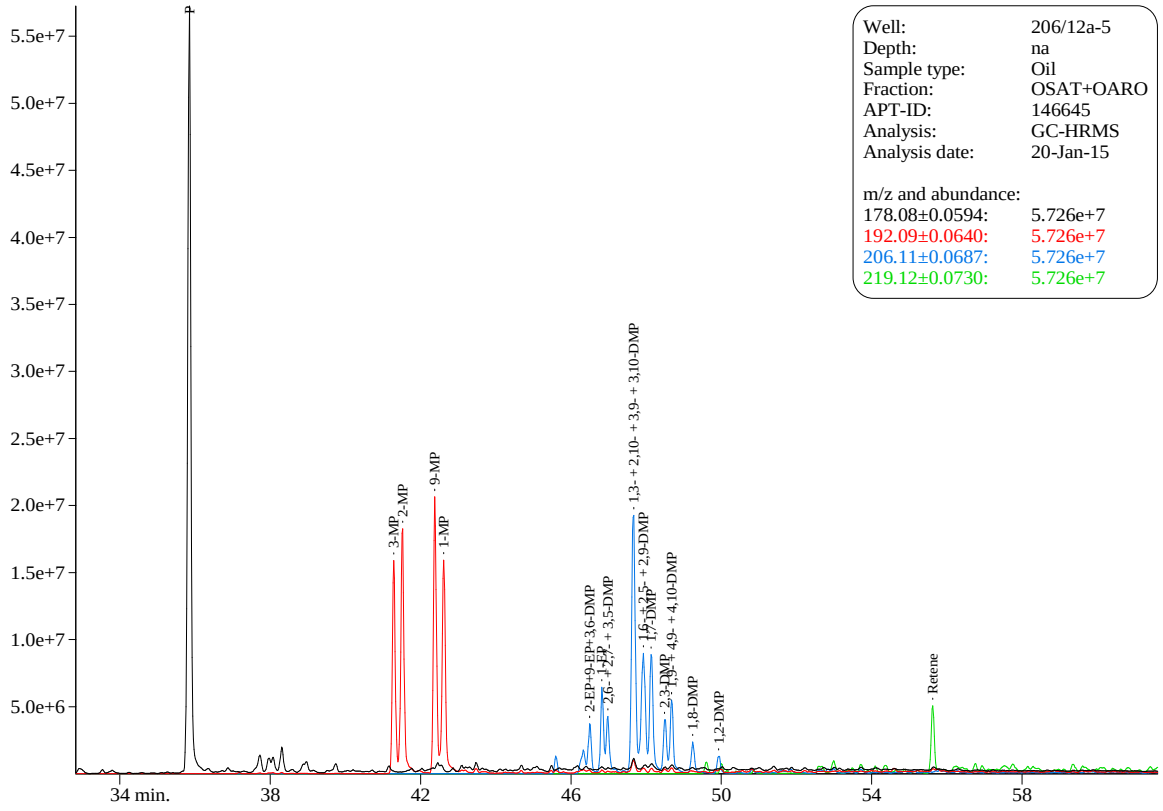
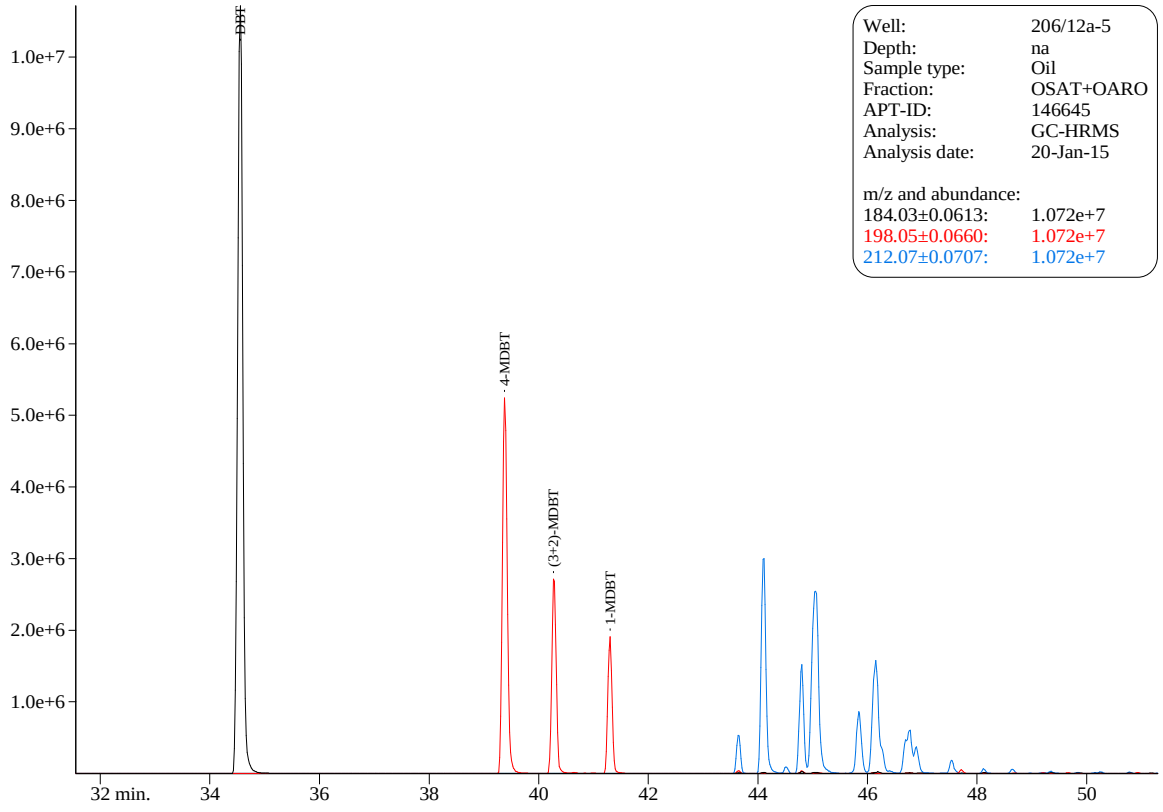


FIGURE Ap8.1 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72))



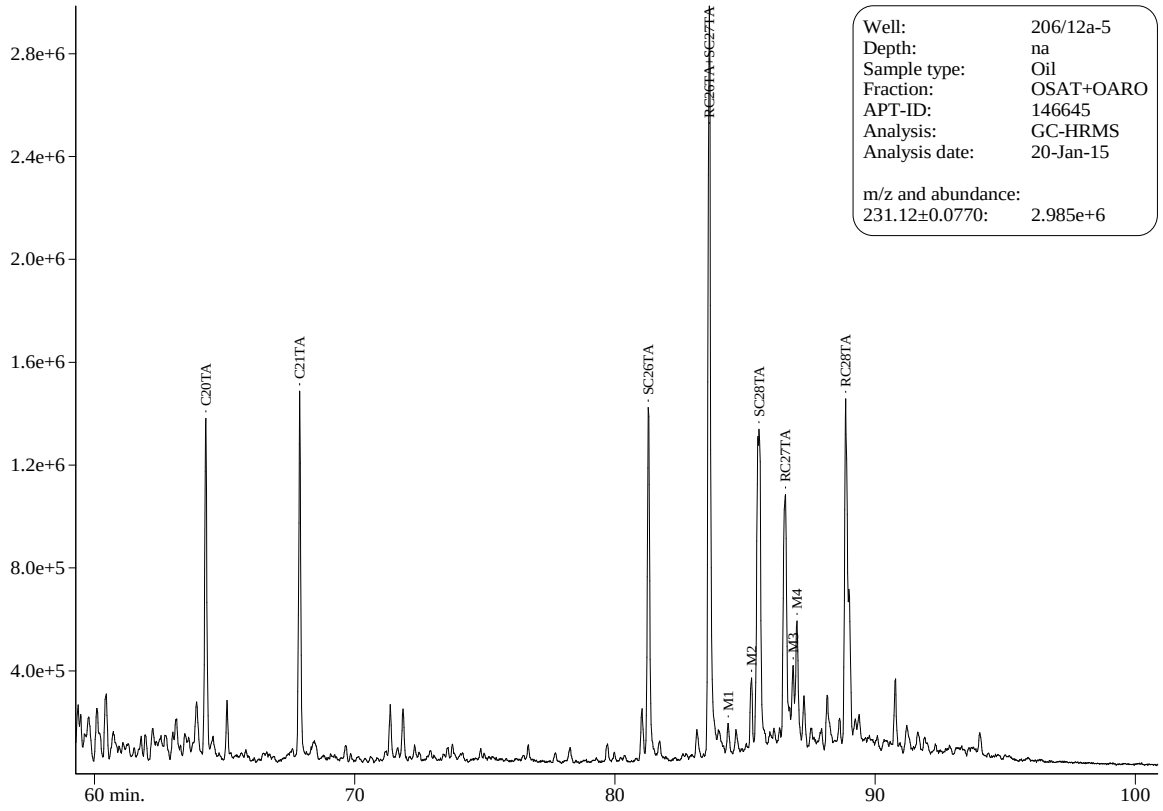


FIGURE Ap8.1 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72))

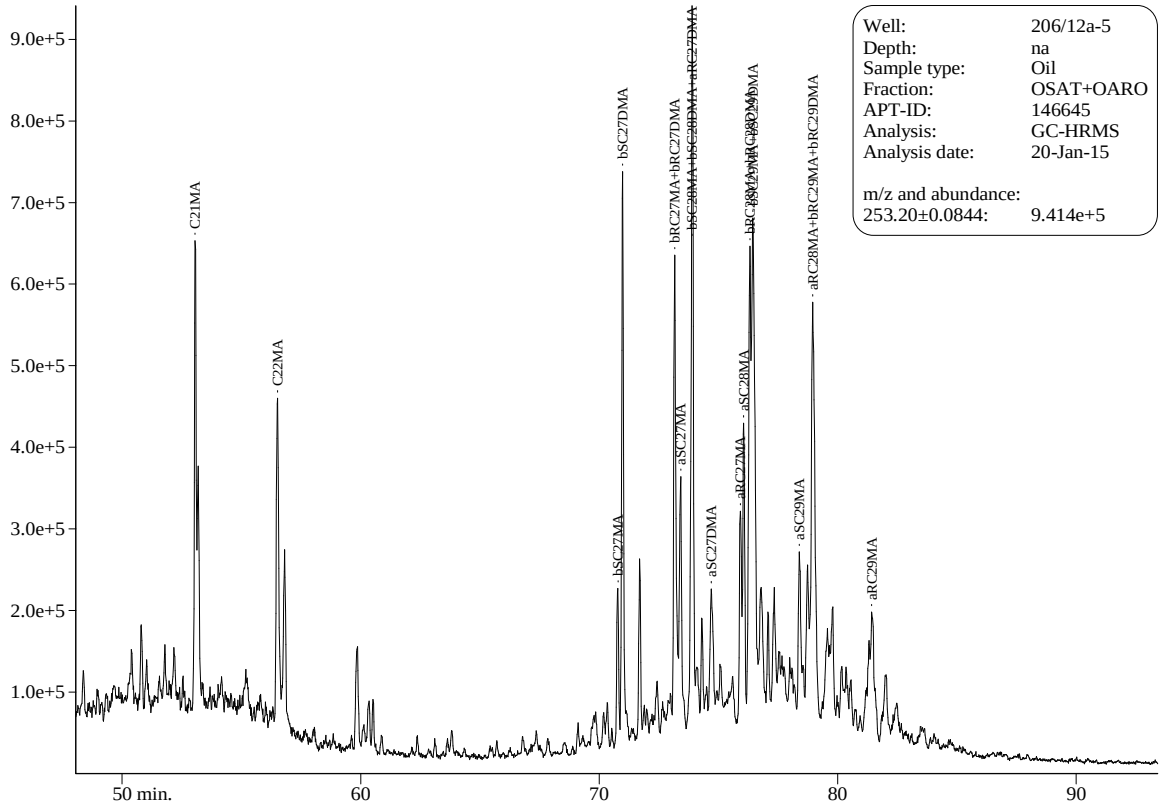
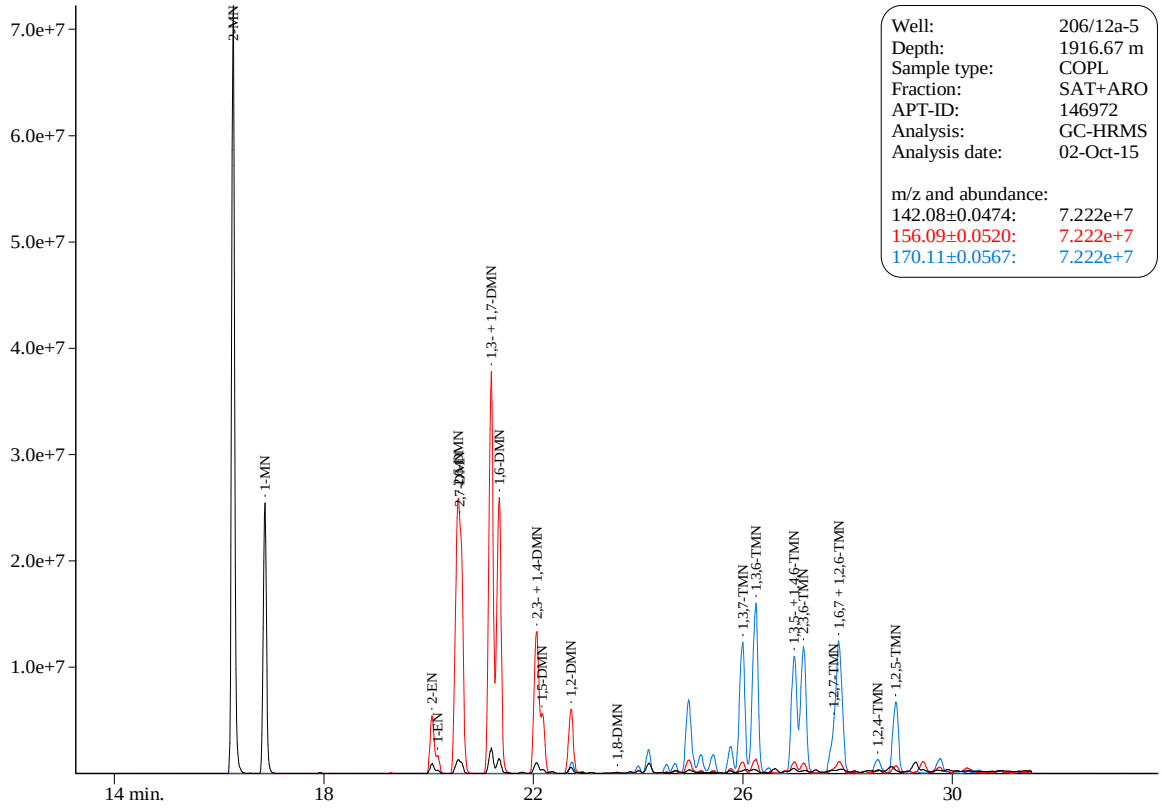


FIGURE Ap8.1 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1897.03m (Turonian (K72))



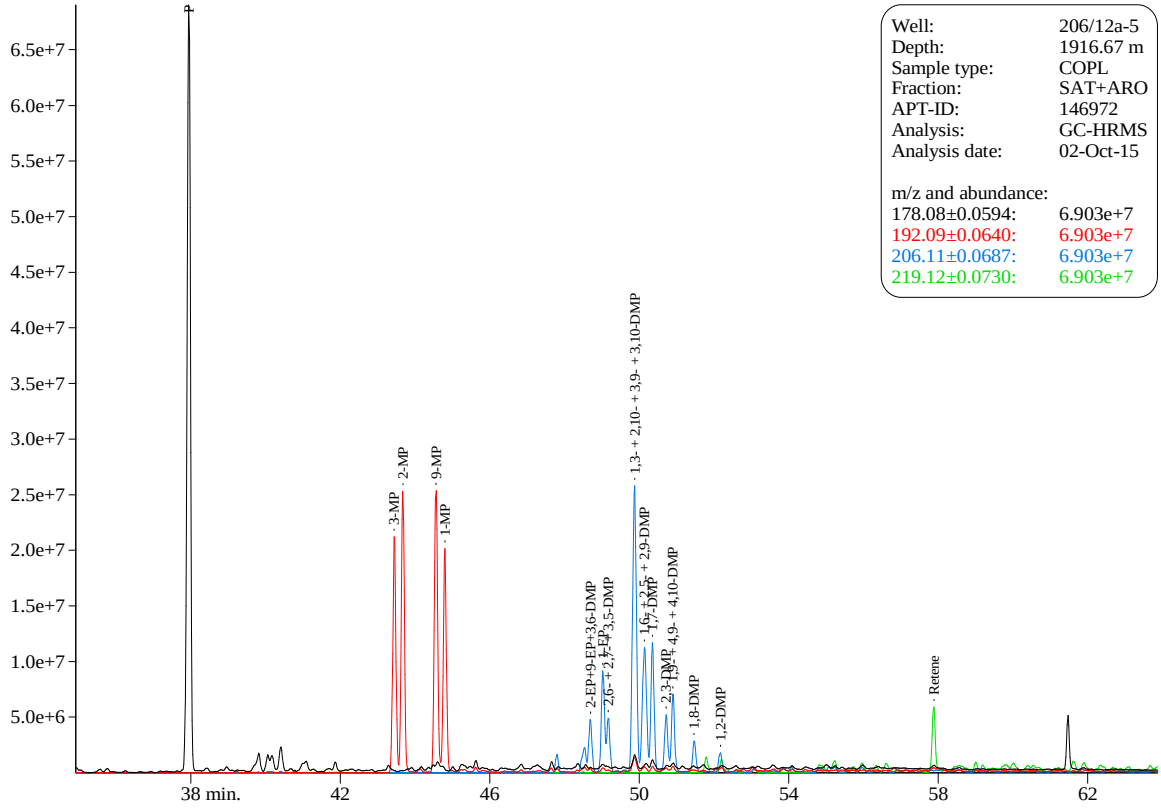
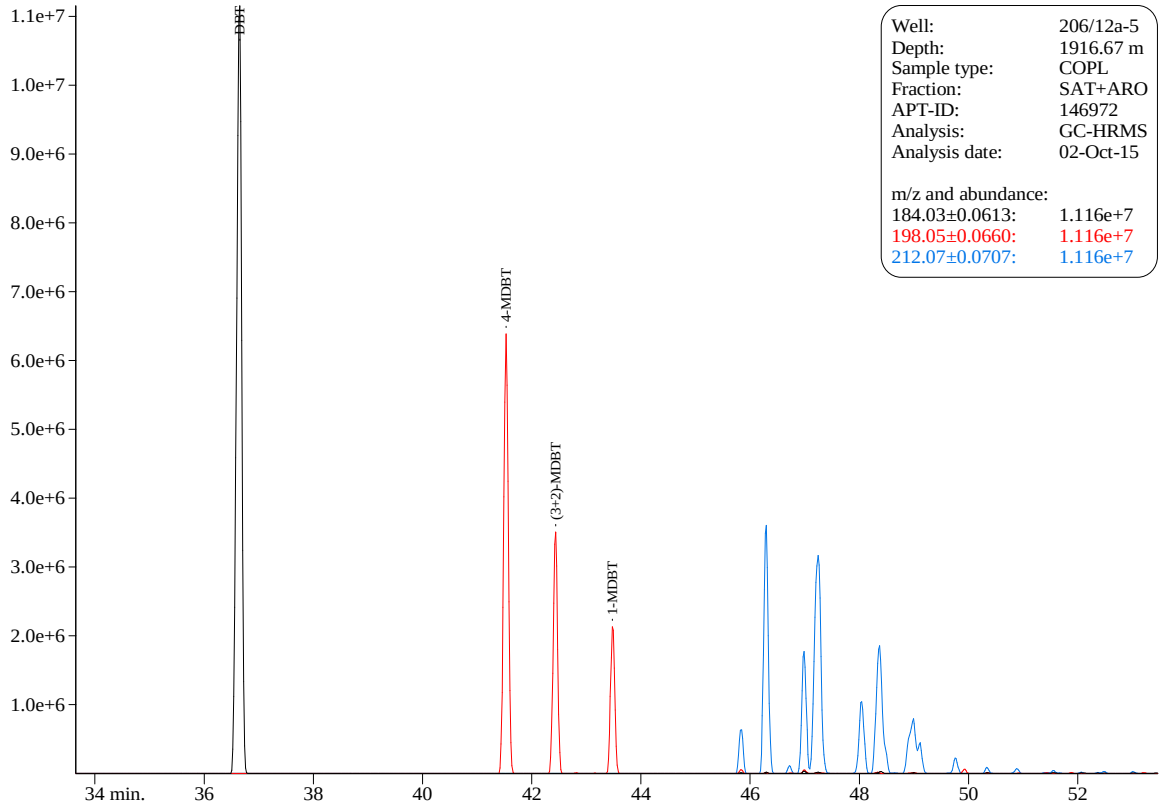


FIGURE Ap8.2 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))



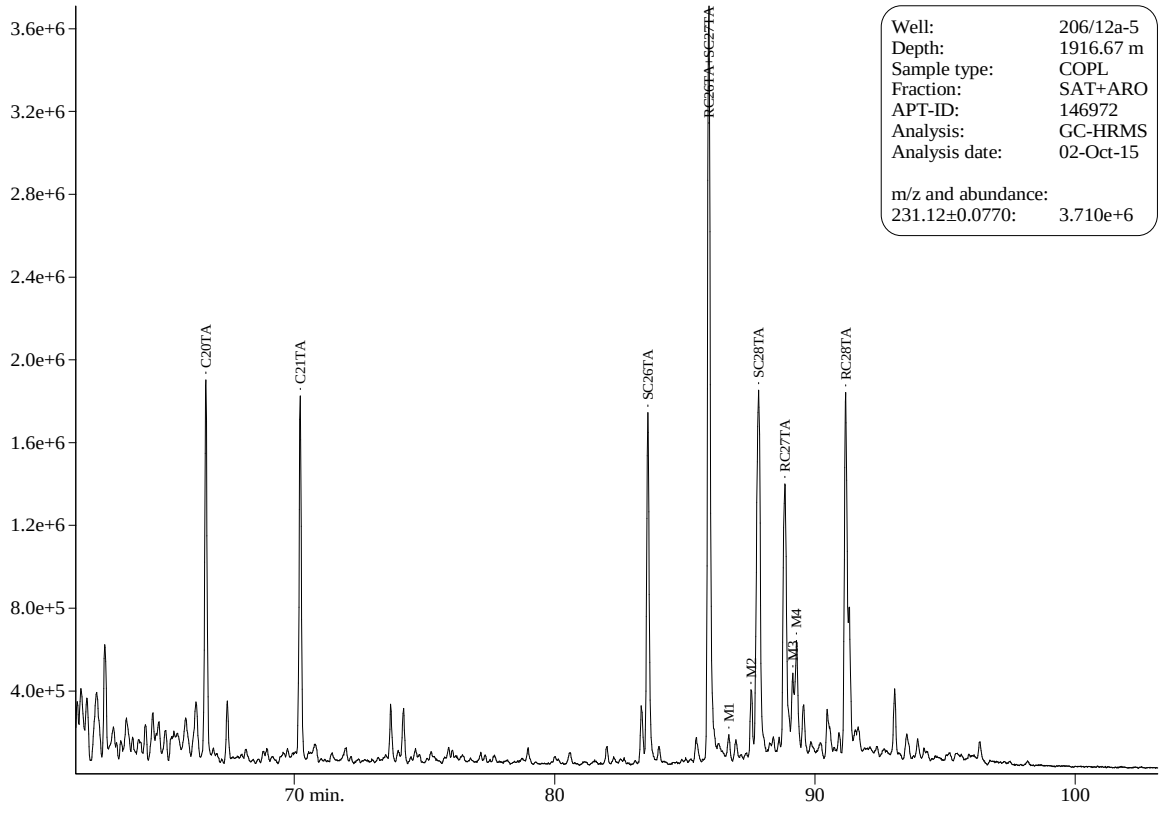


FIGURE Ap8.2 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))

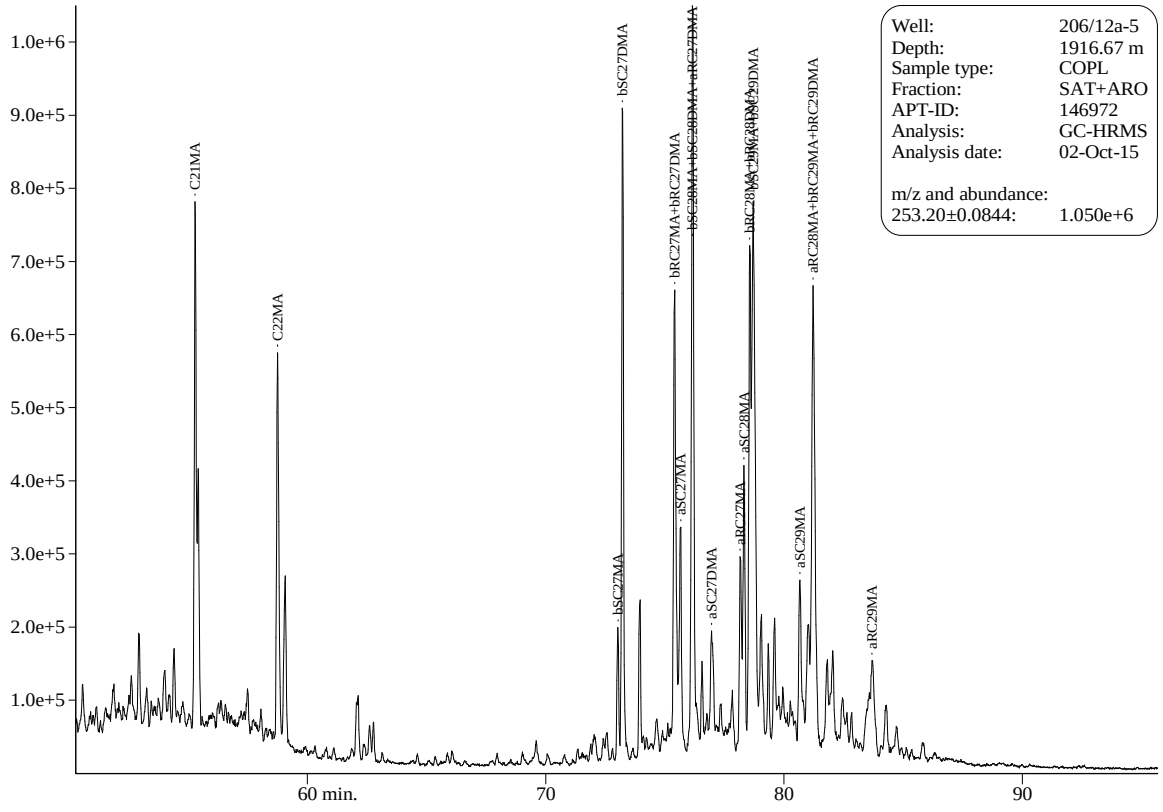
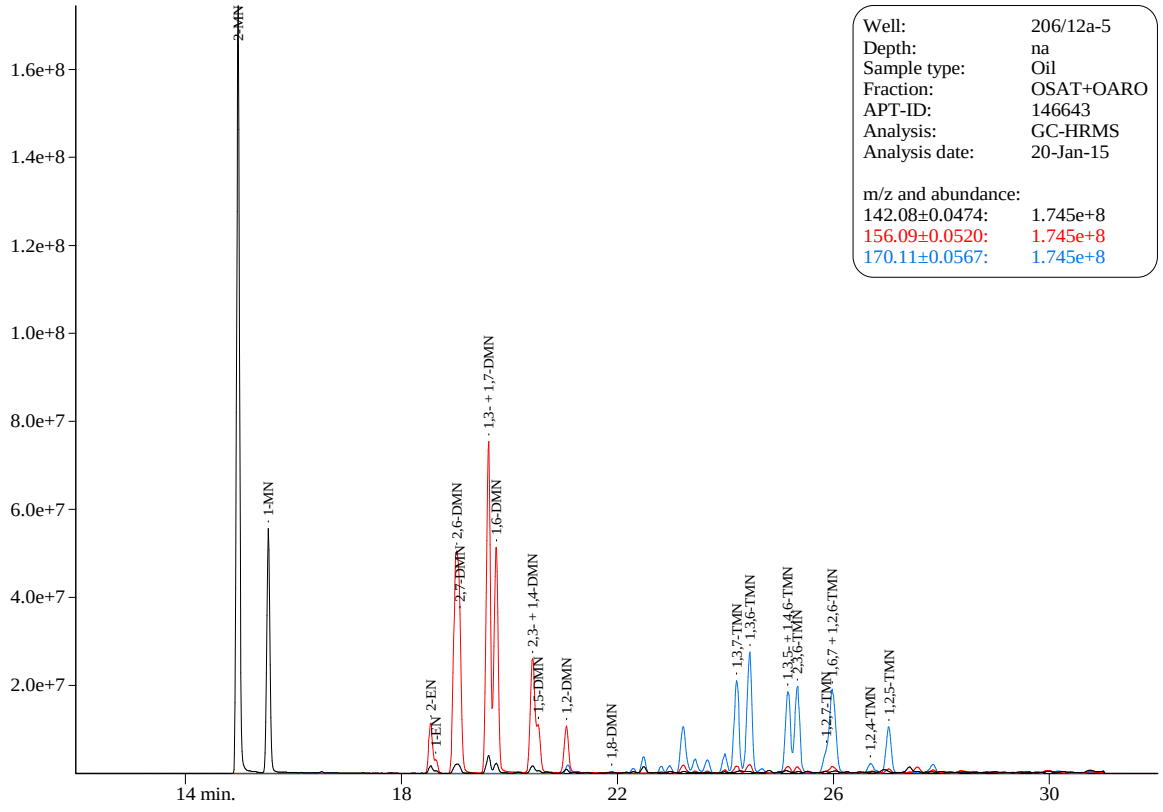


FIGURE Ap8.2 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1916.67m (Turonian (K72))



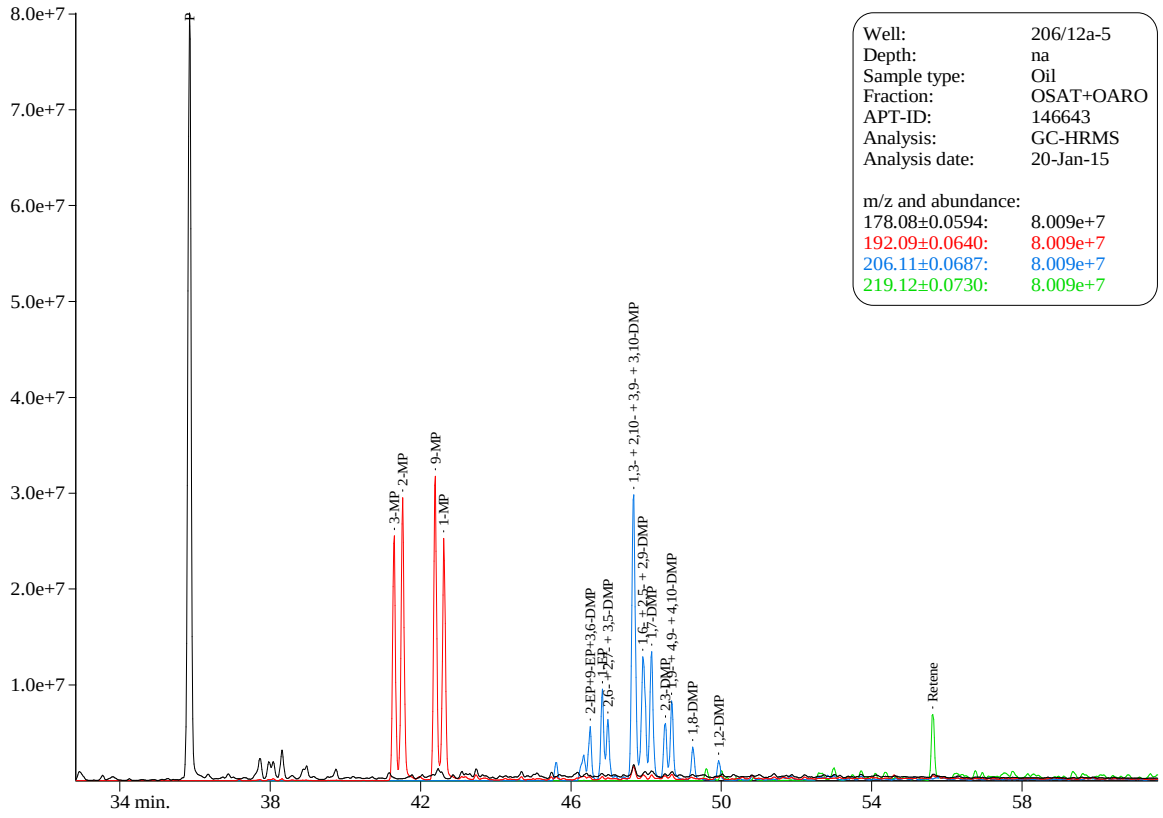
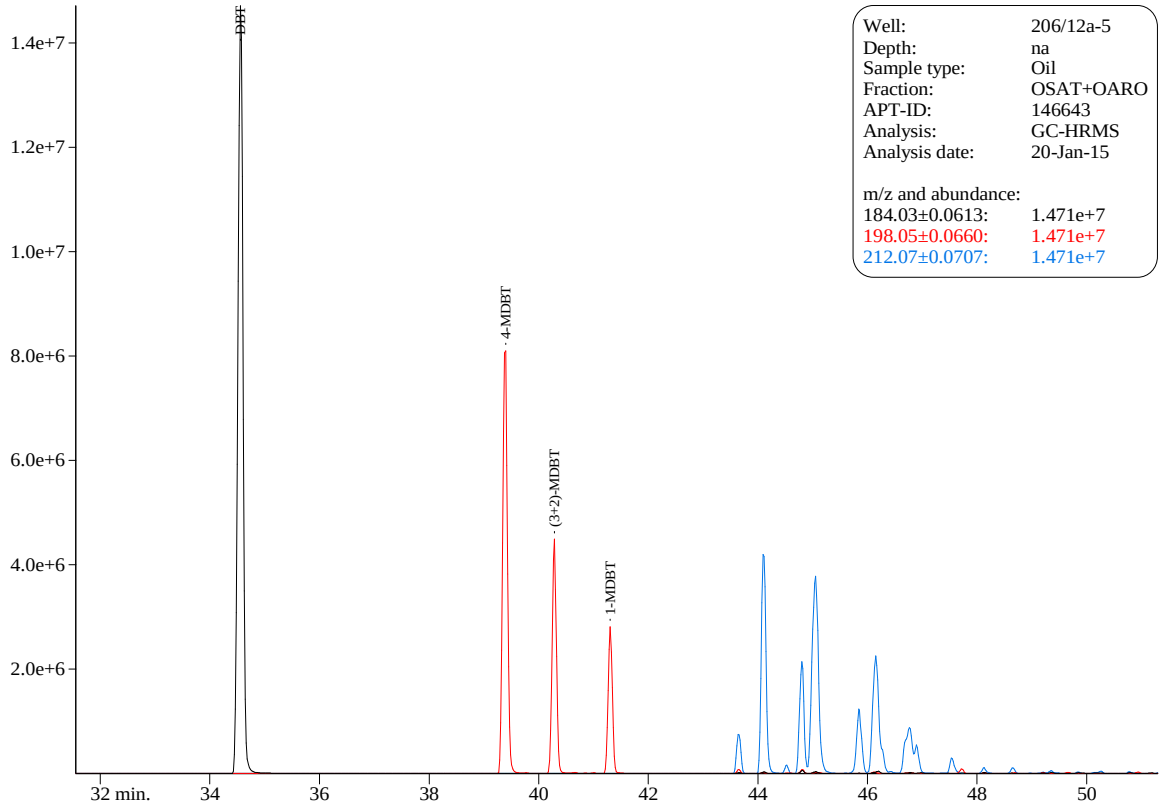


FIGURE Ap8.3 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret)



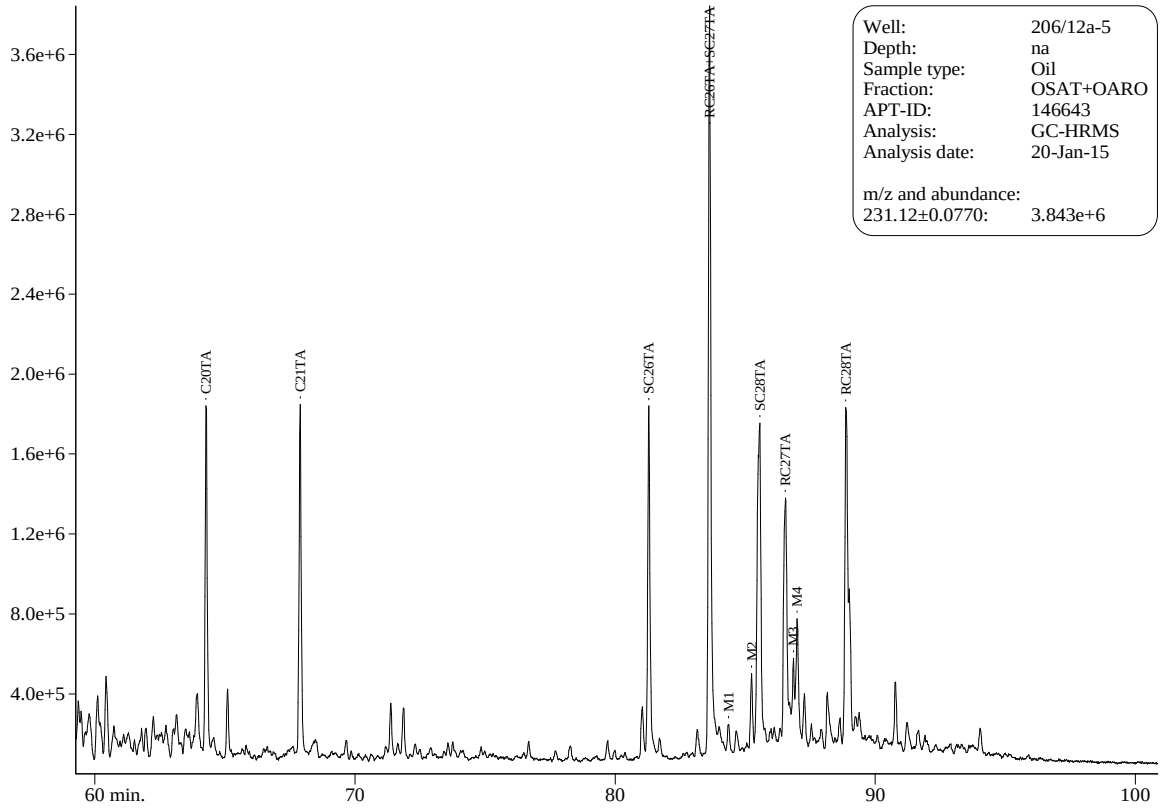


FIGURE Ap8.3 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret)

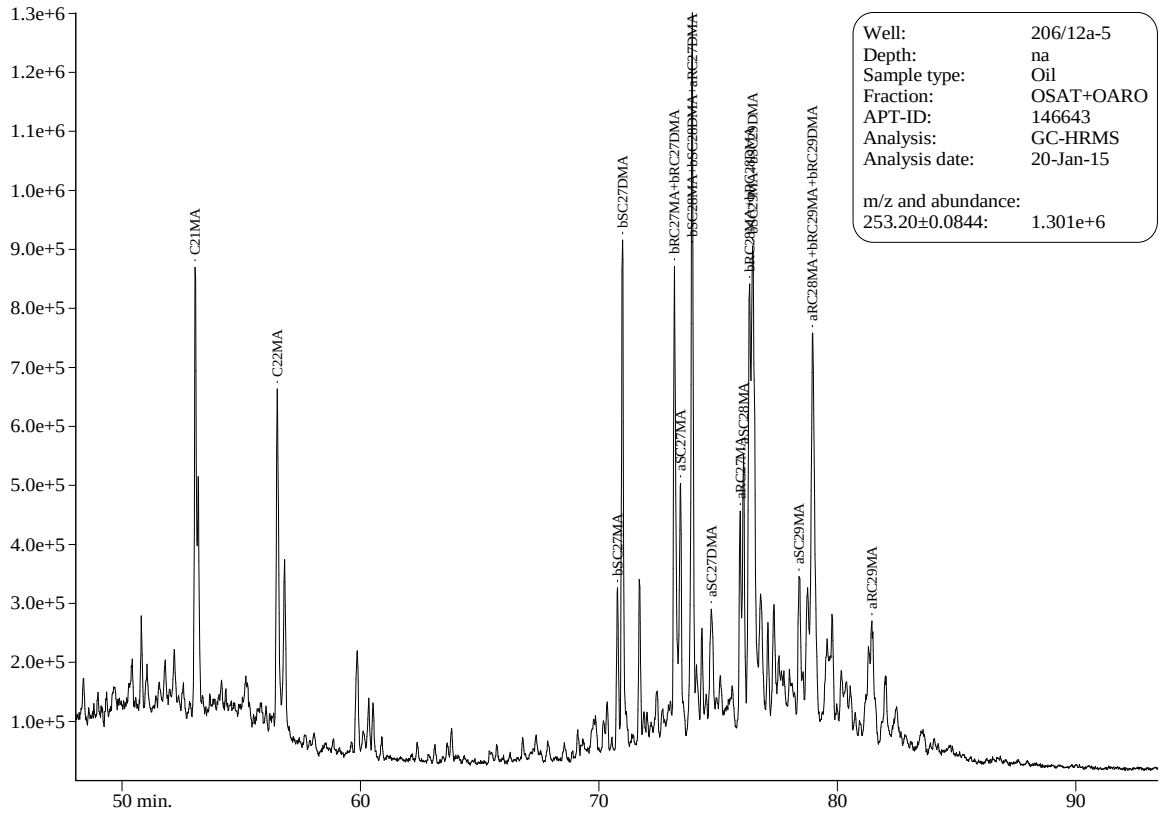
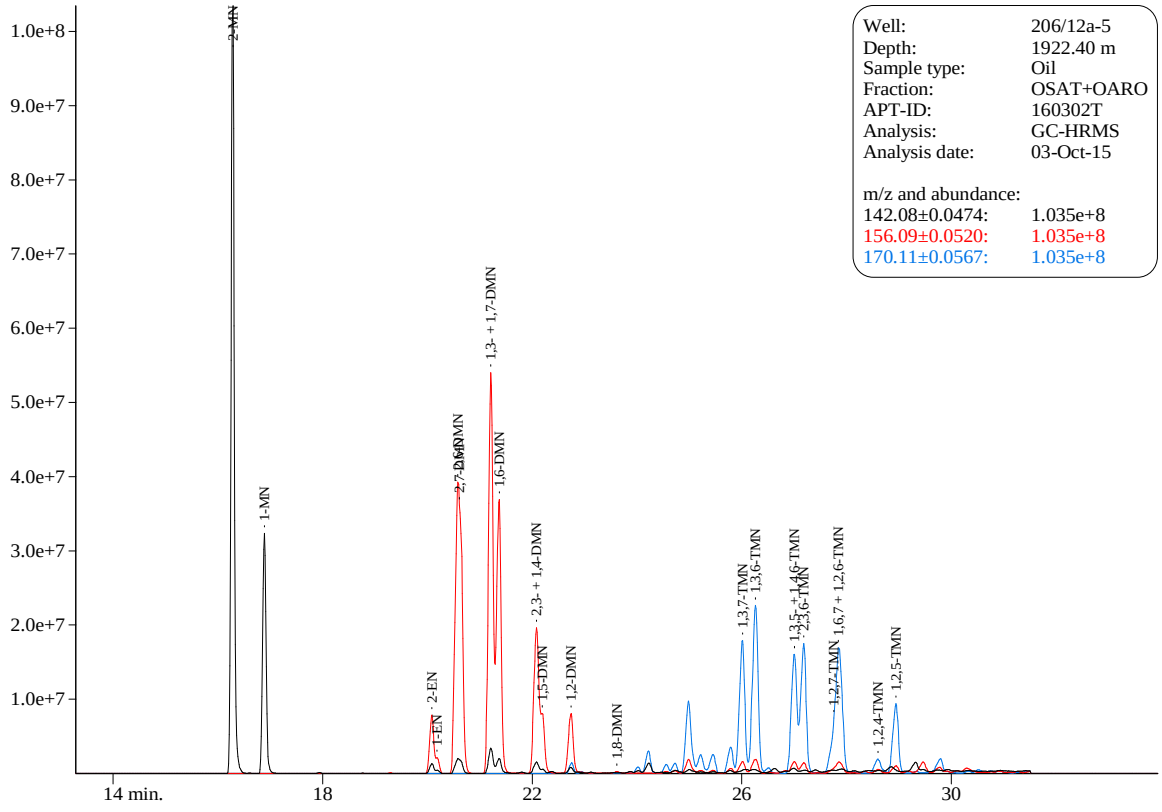


FIGURE Ap8.3 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1922.08m (K30 Sand Cret)



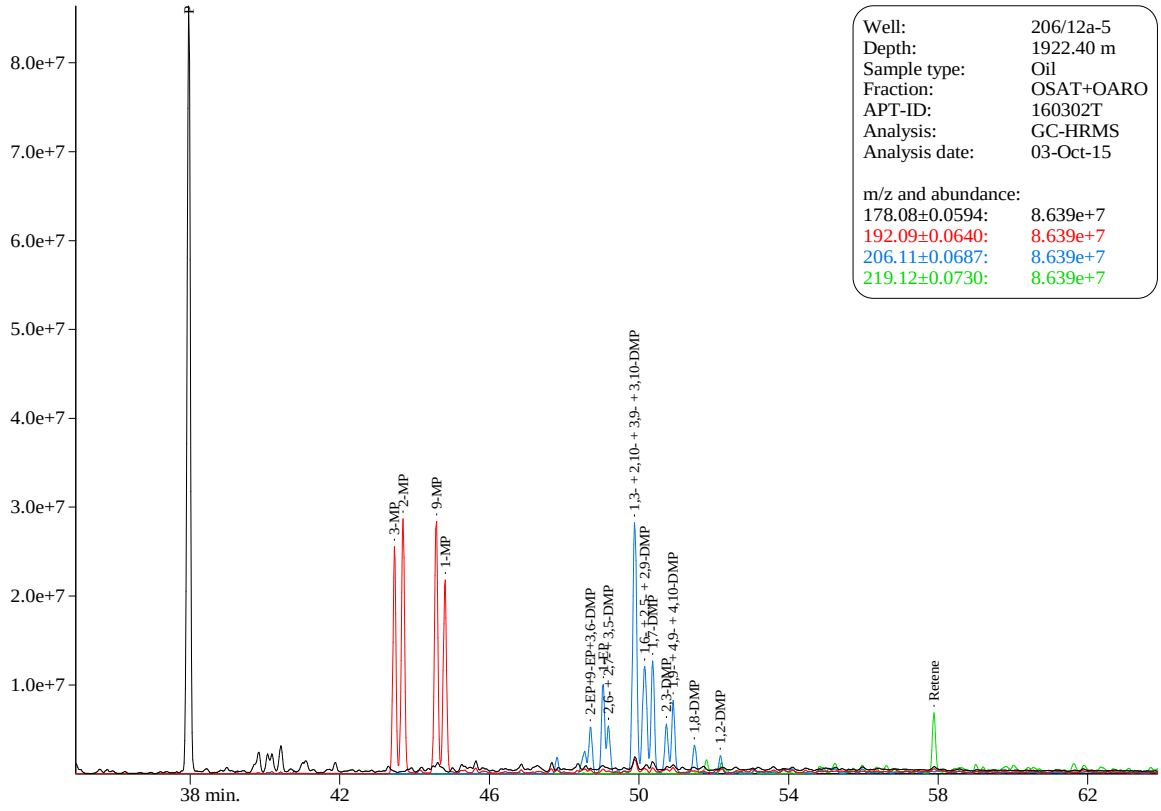
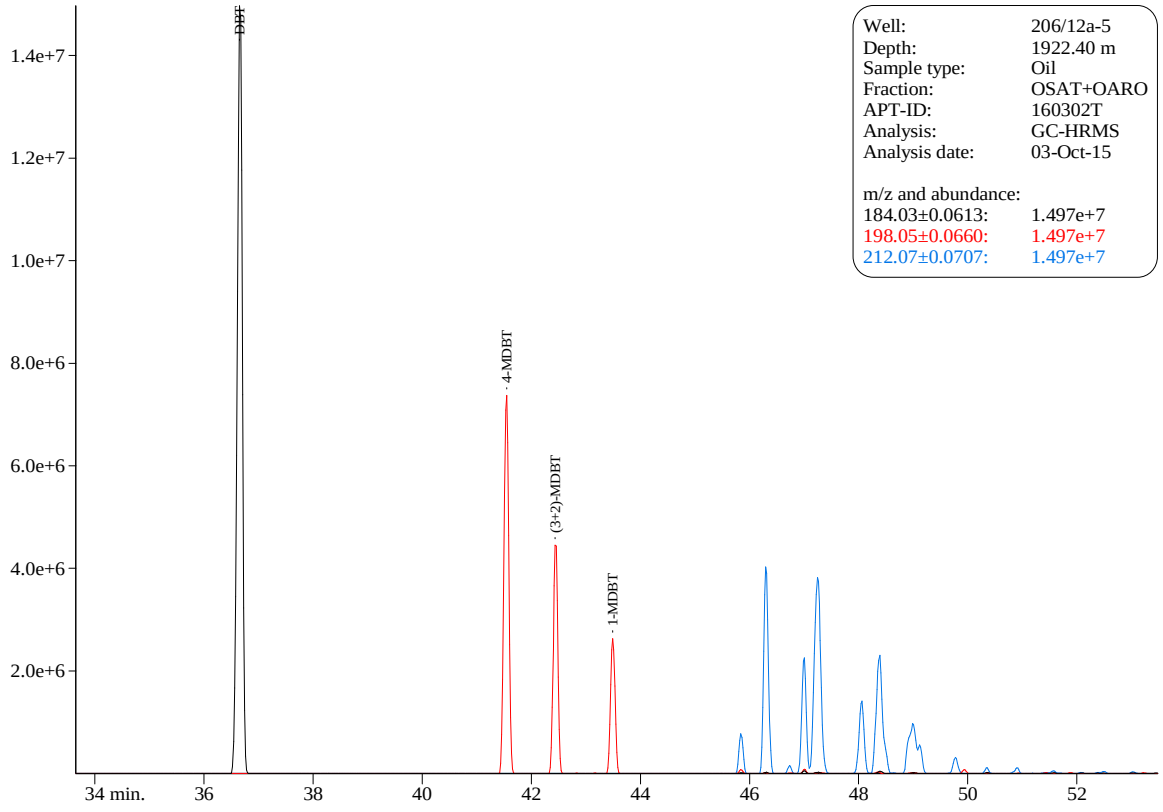


FIGURE Ap8.4 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1922.4m (K30 Sand Cret)



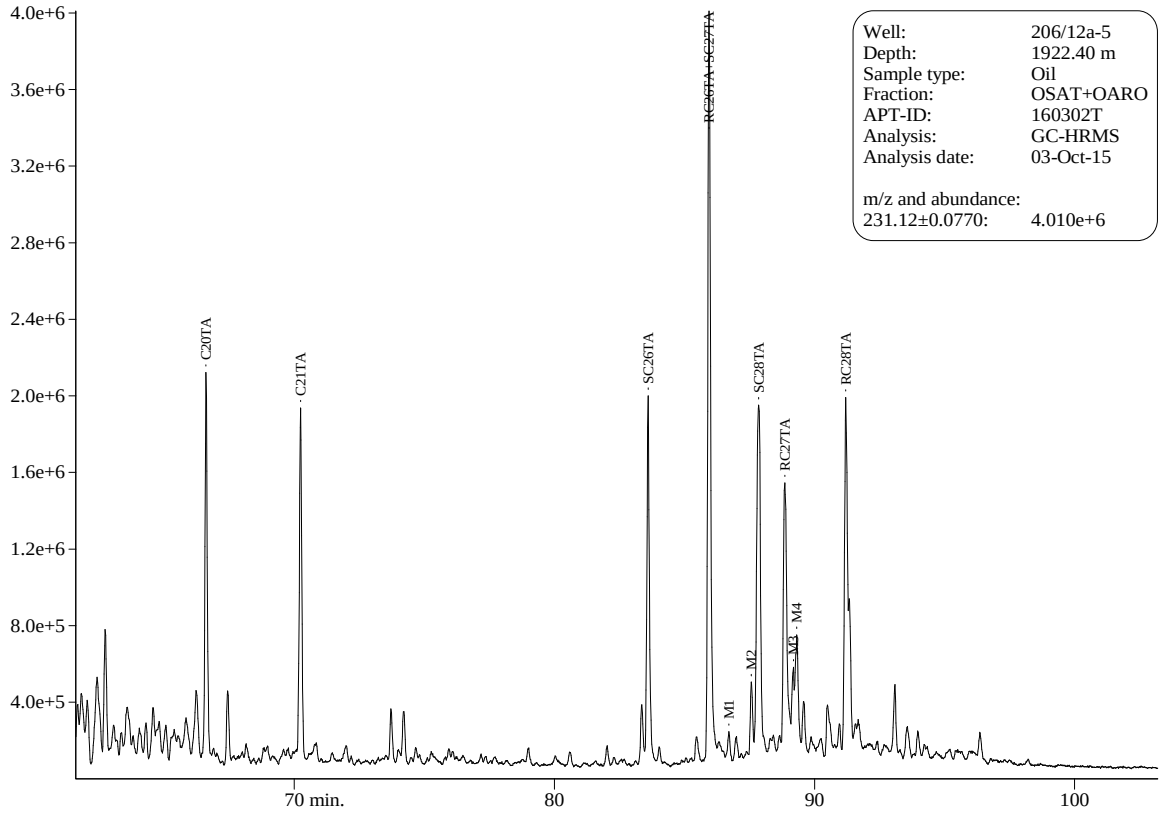


FIGURE Ap8.4 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1922.4m (K30 Sand Cret)

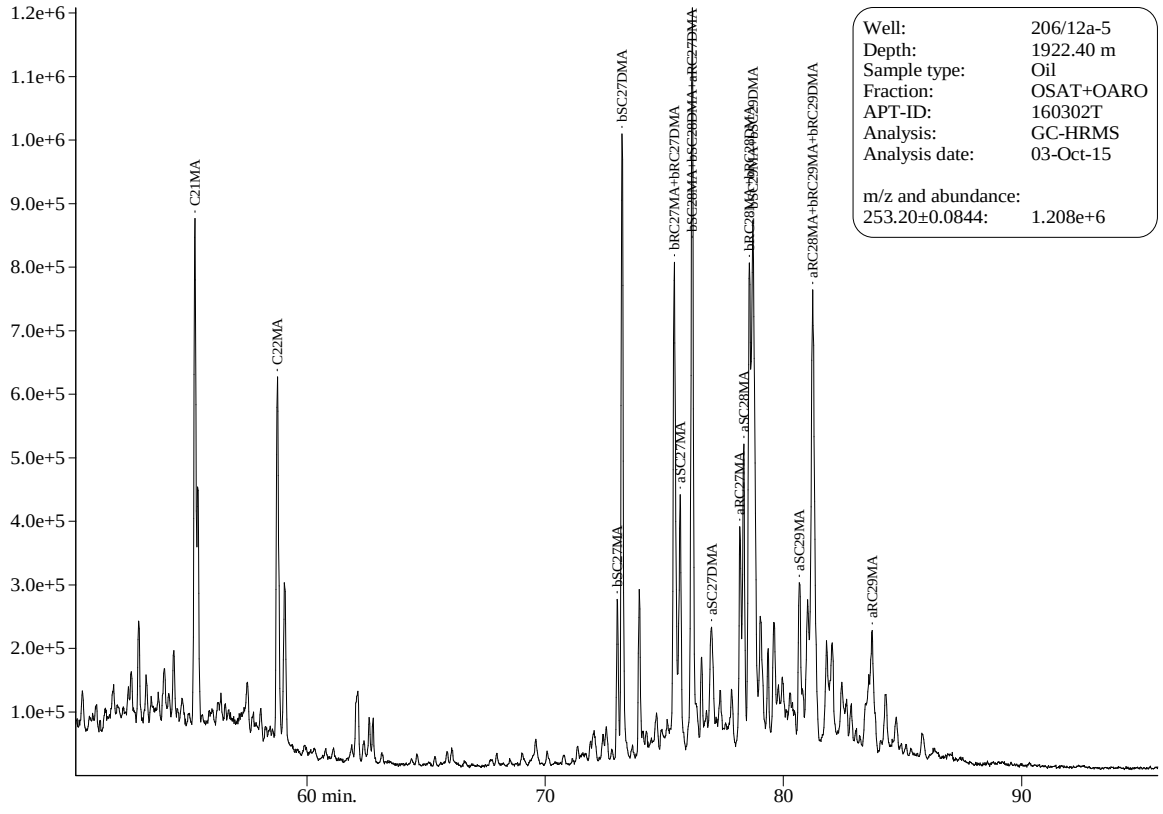
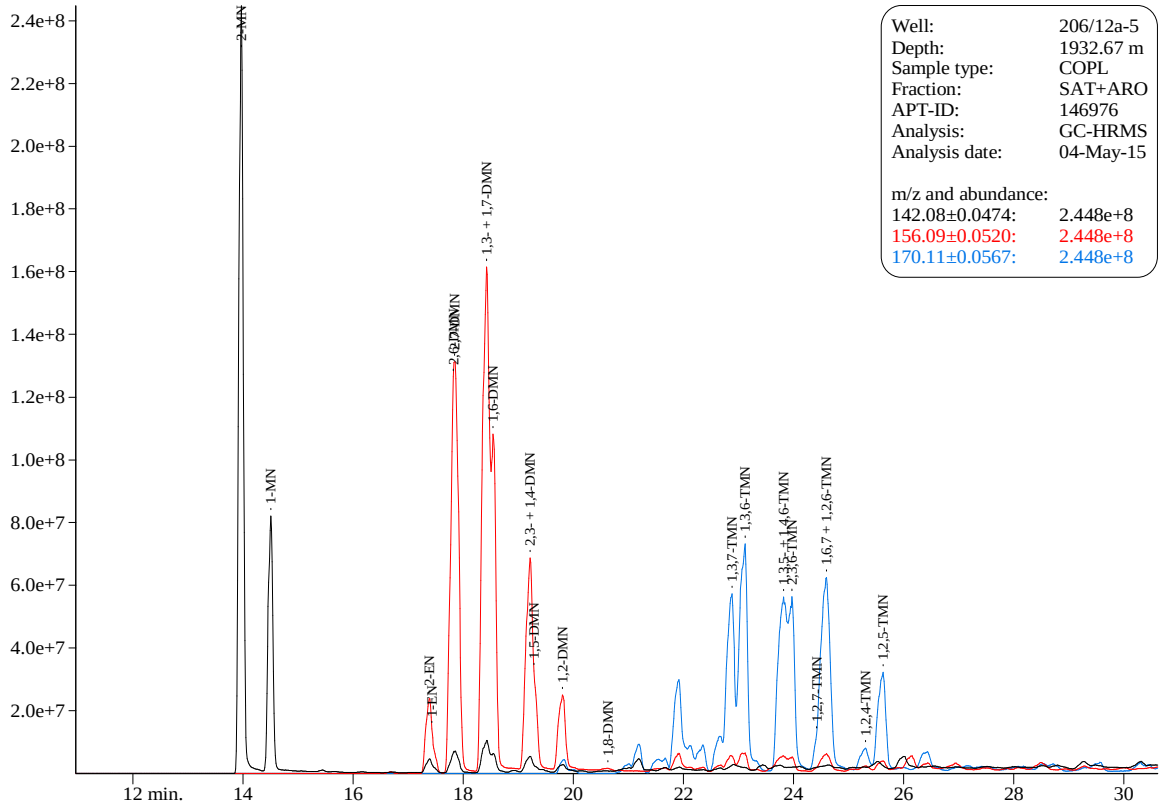


FIGURE Ap8.4 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1922.4m (K30 Sand Cret)



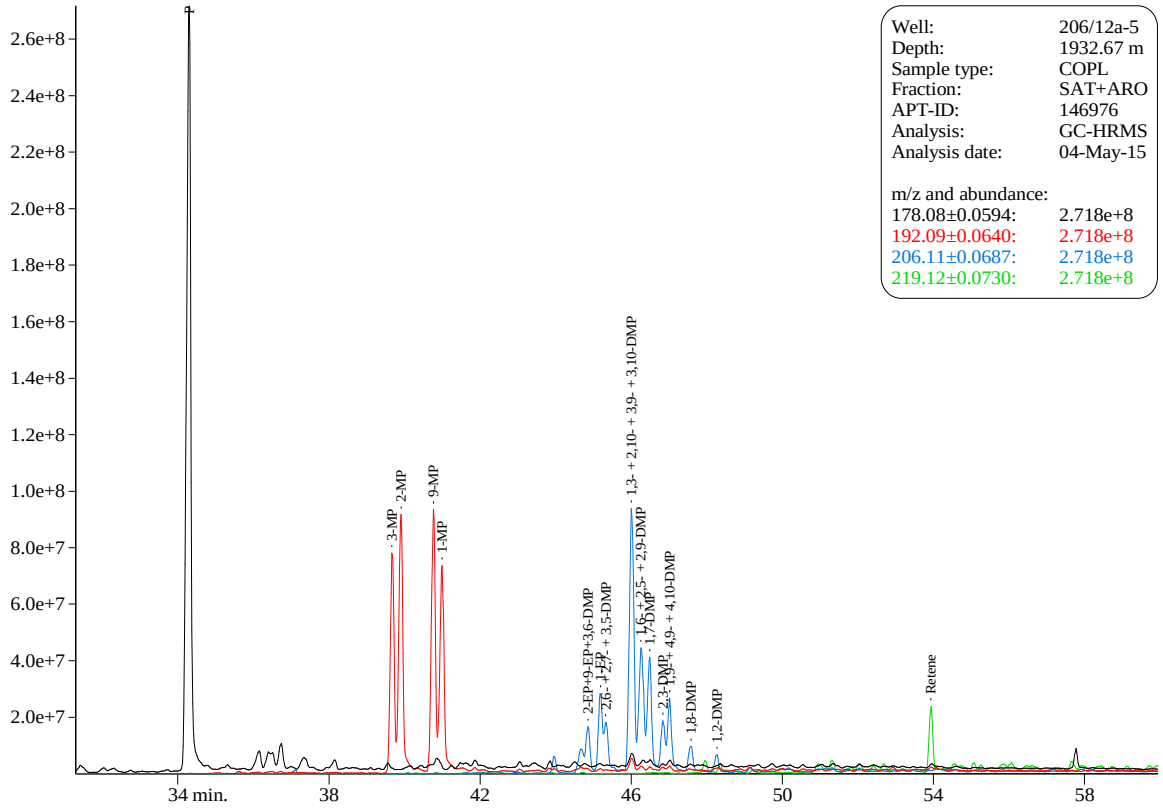
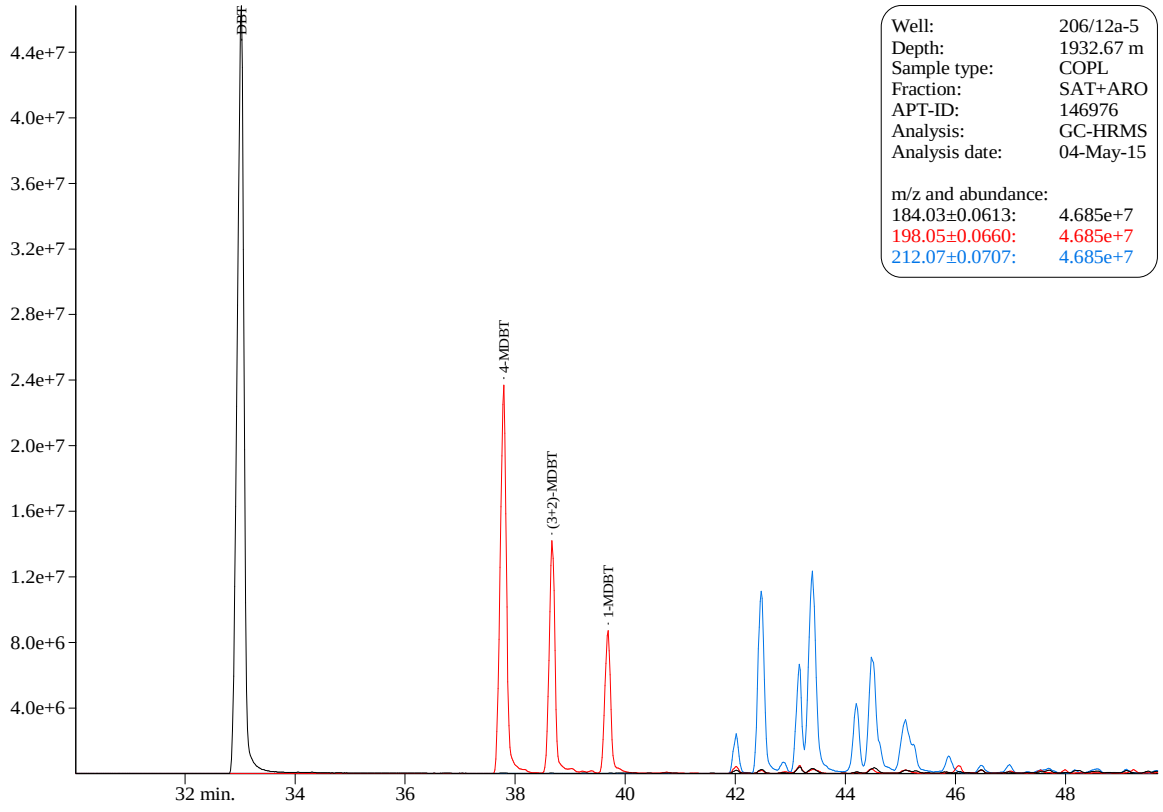


FIGURE Ap8.5 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1932.67m (Kimmeridge Clay Fm.)



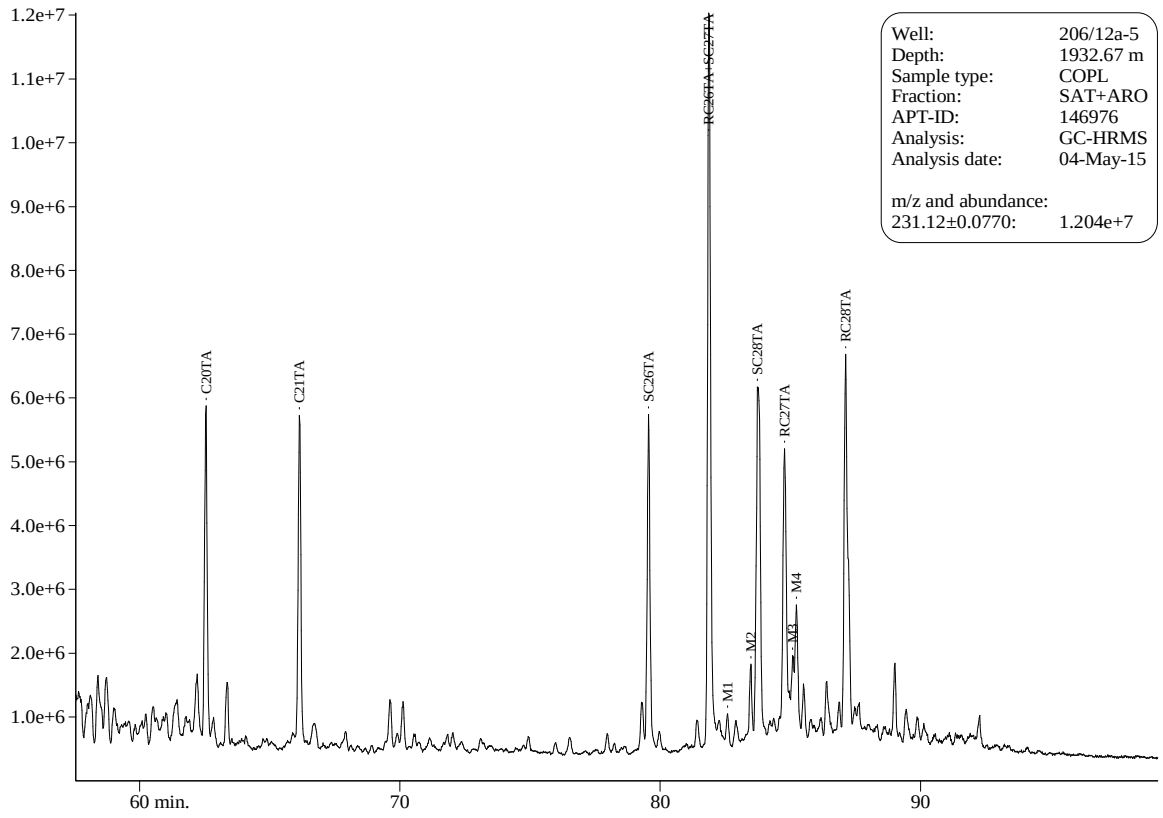


FIGURE Ap8.5 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1932.67m (Kimmeridge Clay Fm.)

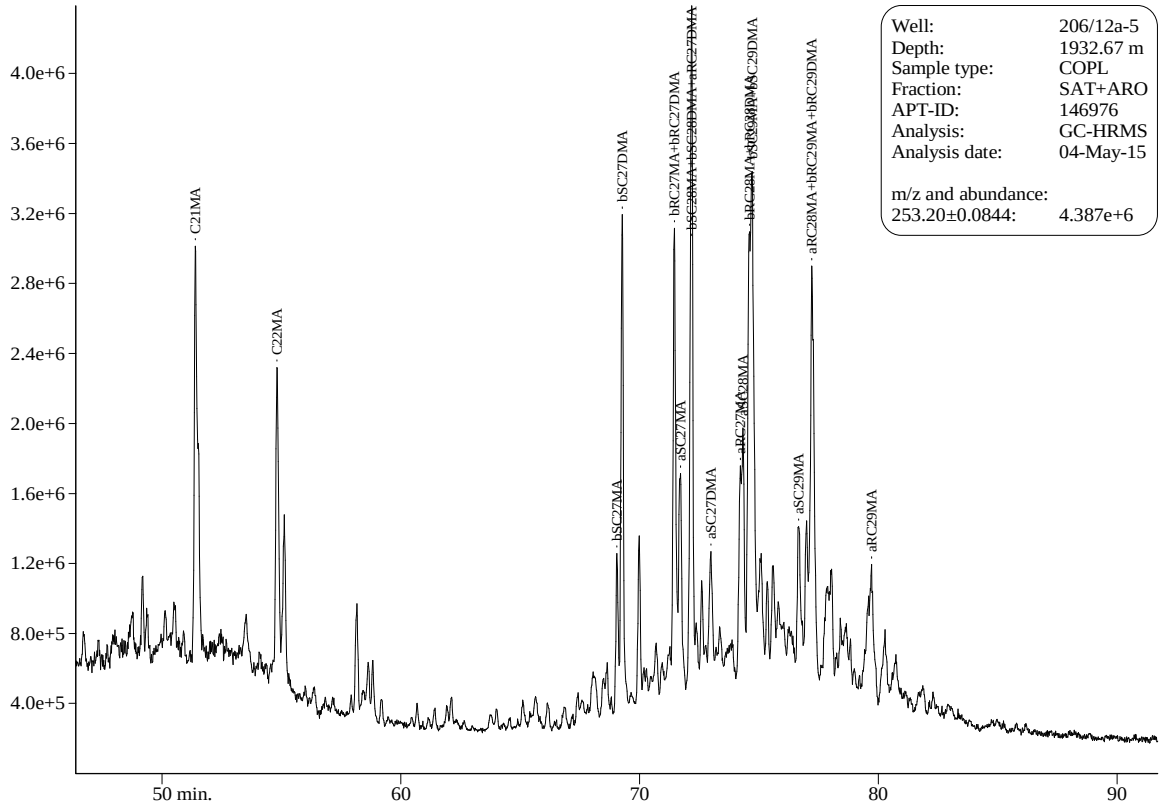
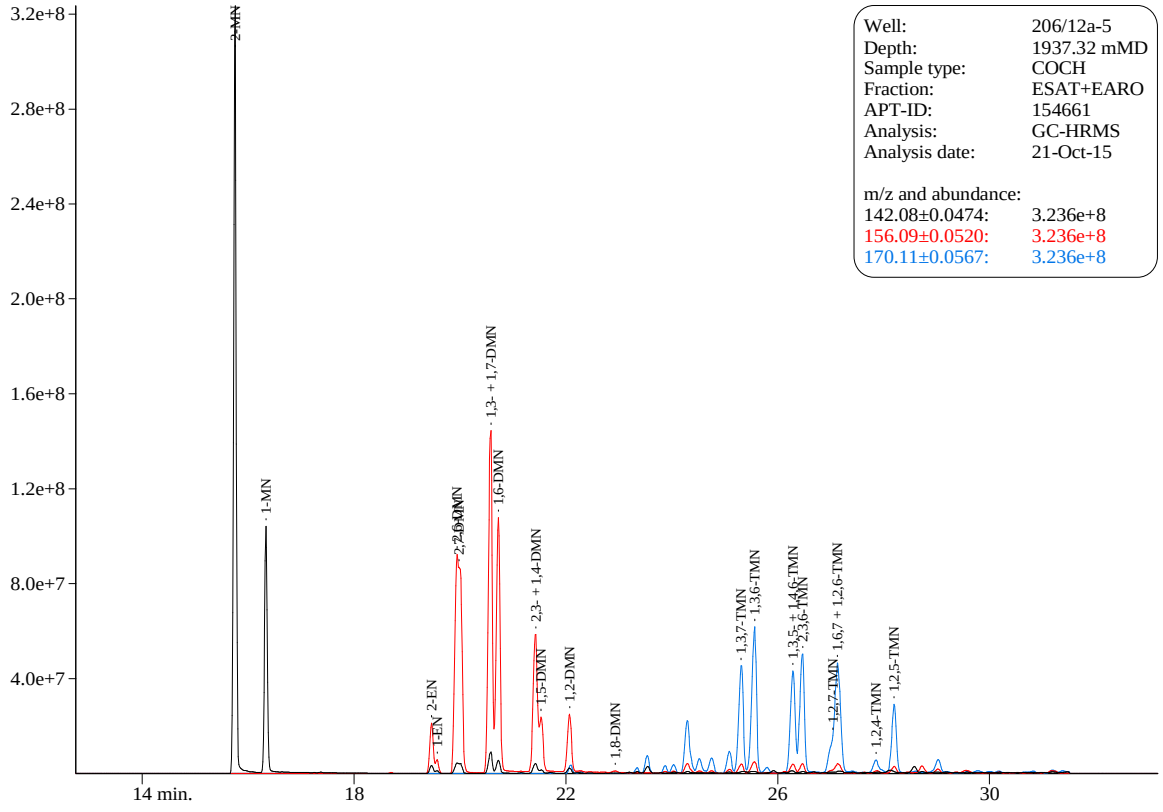


FIGURE Ap8.5 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1932.67m (Kimmeridge Clay Fm.)



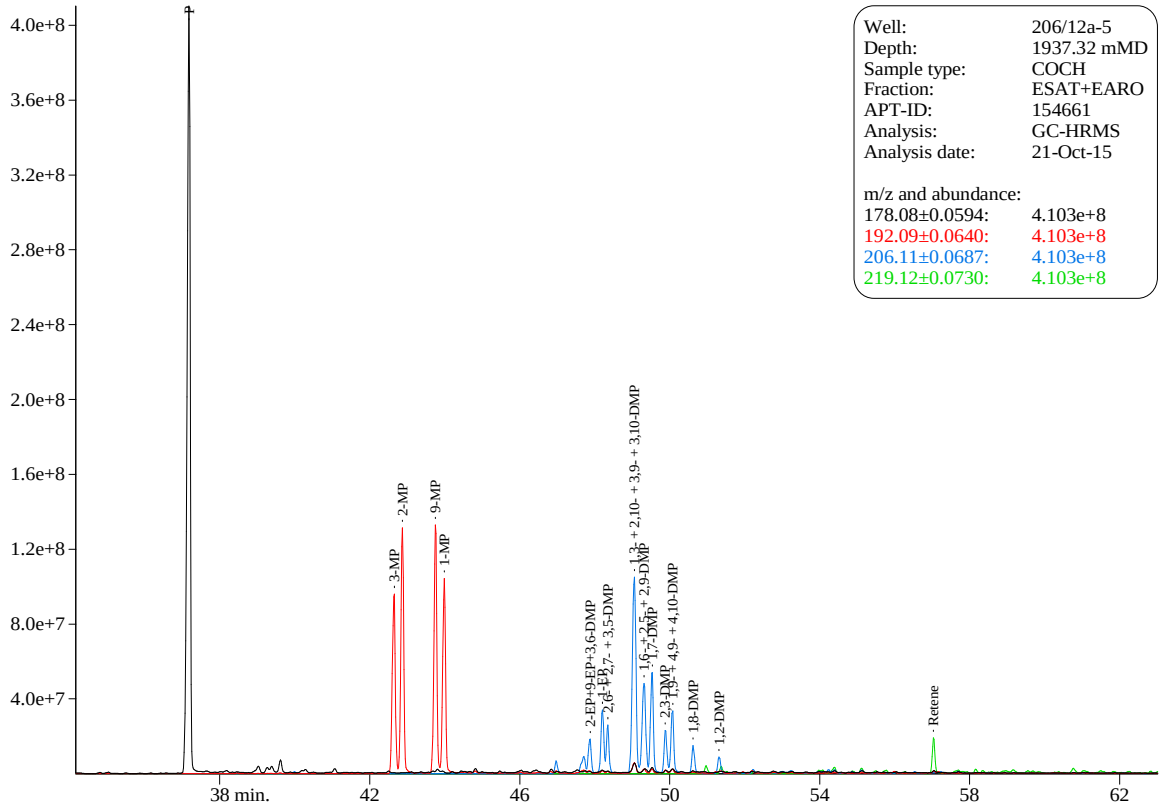
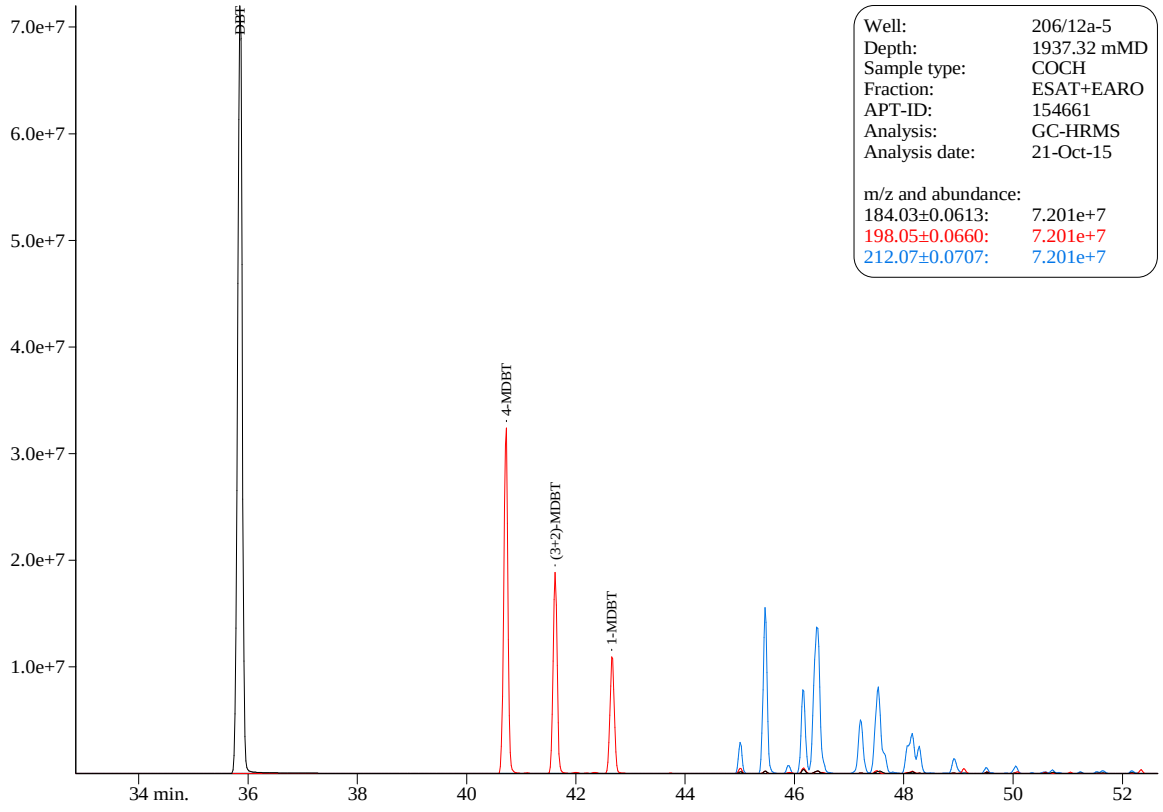


FIGURE Ap8.6 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE, 1937.32m (Rona)



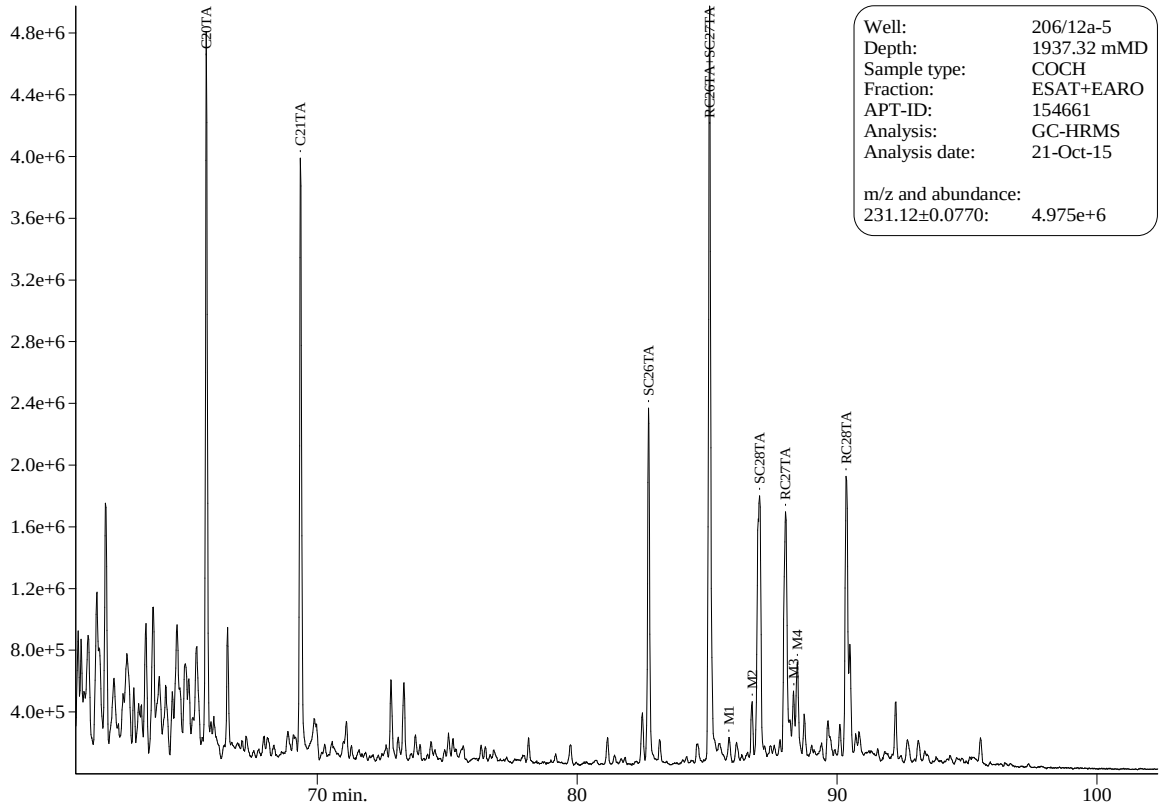


FIGURE Ap8.6 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE, 1937.32m (Rona)

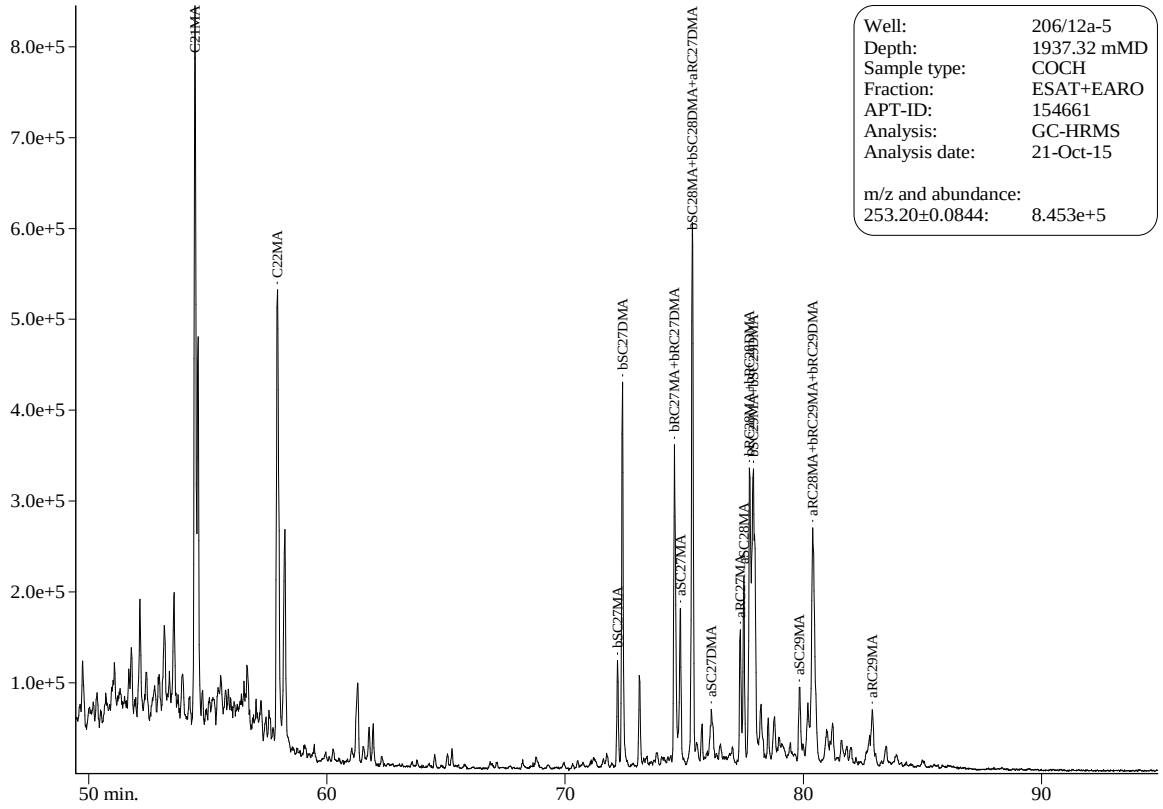
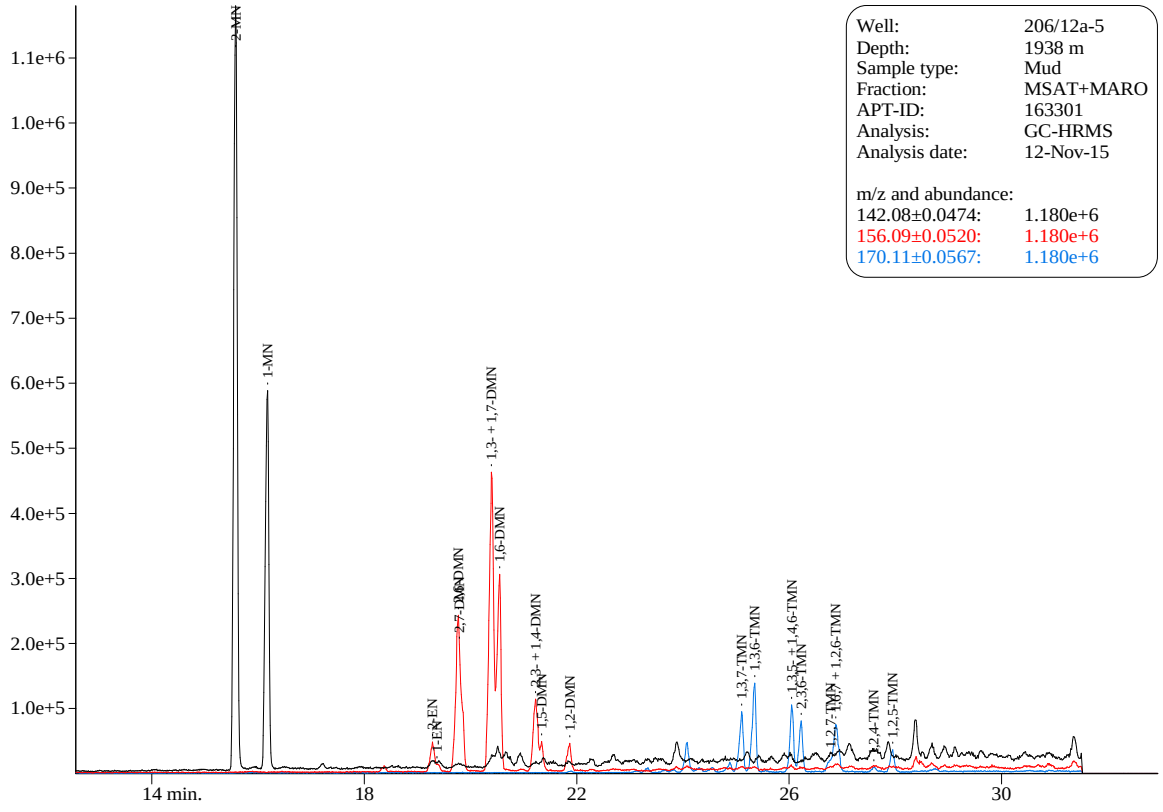


FIGURE Ap8.6 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE, 1937.32m (Rona)



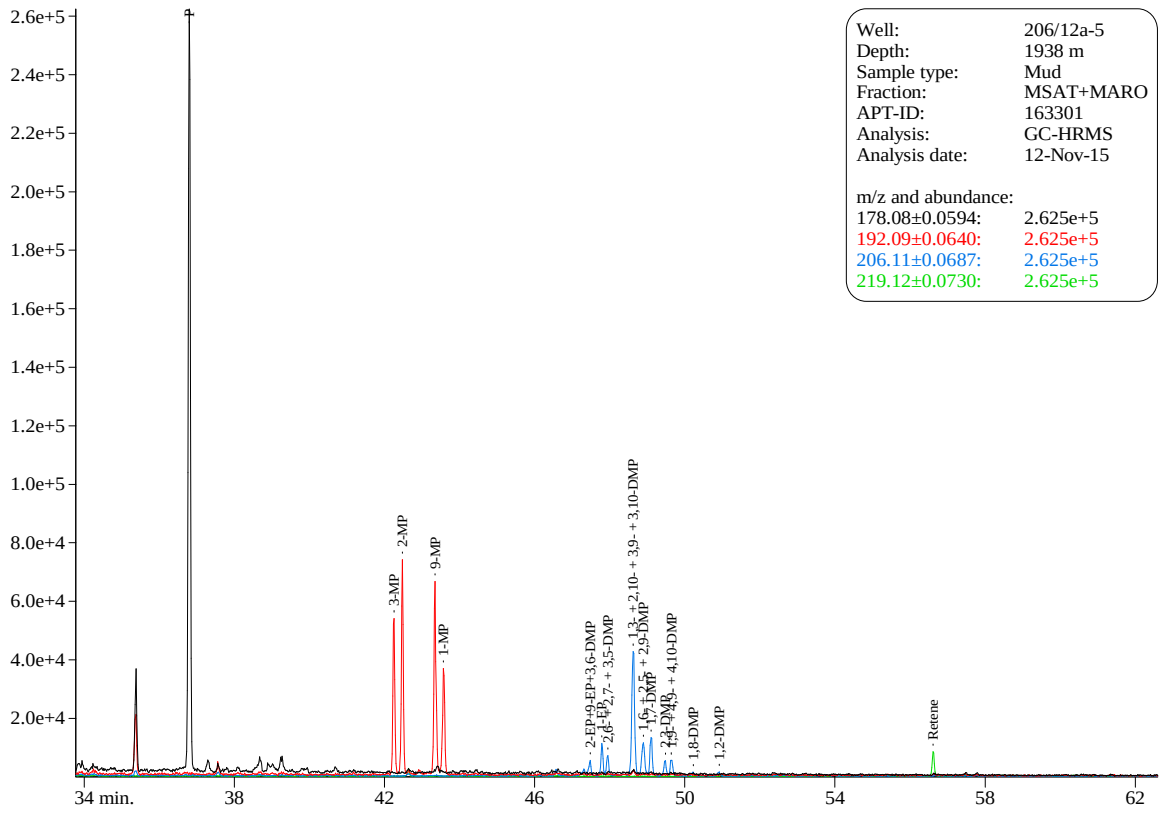
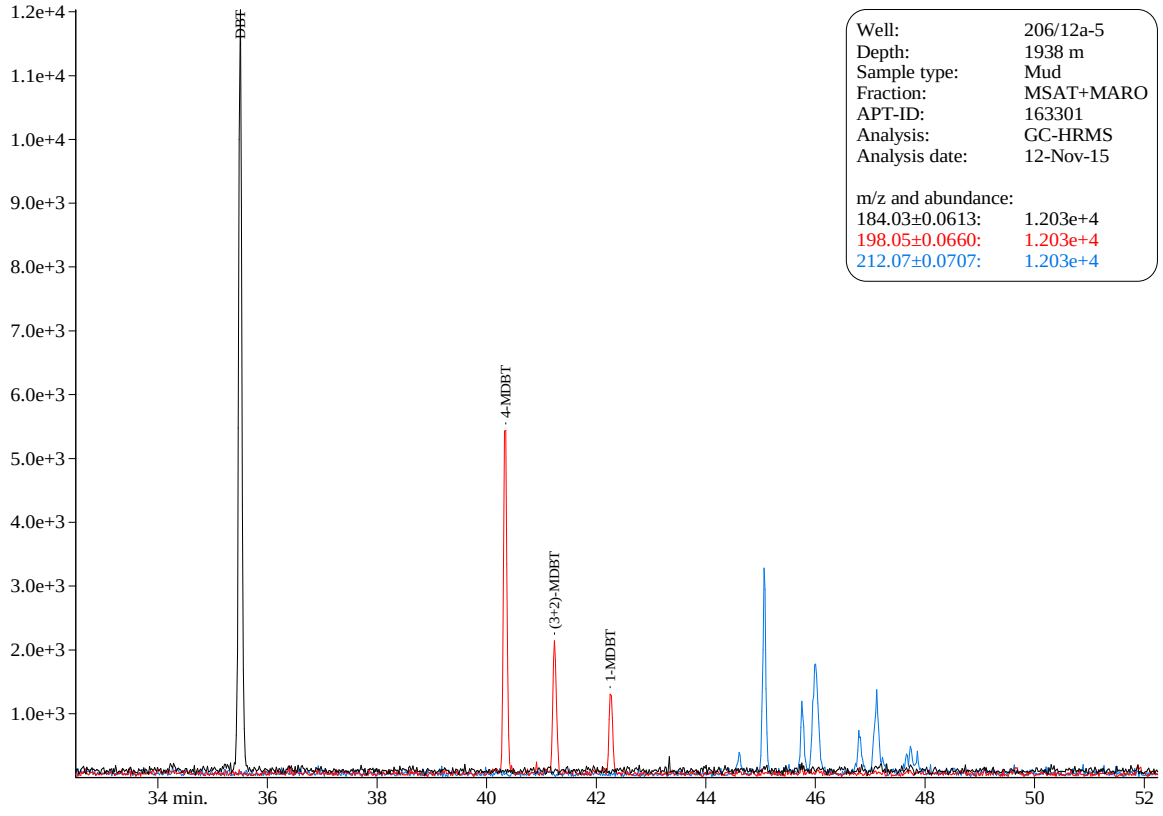


FIGURE Ap8.7 Aromatic fraction GC-MS fragmentograms: 206/12a-5, MUD, 1938m (Rona)



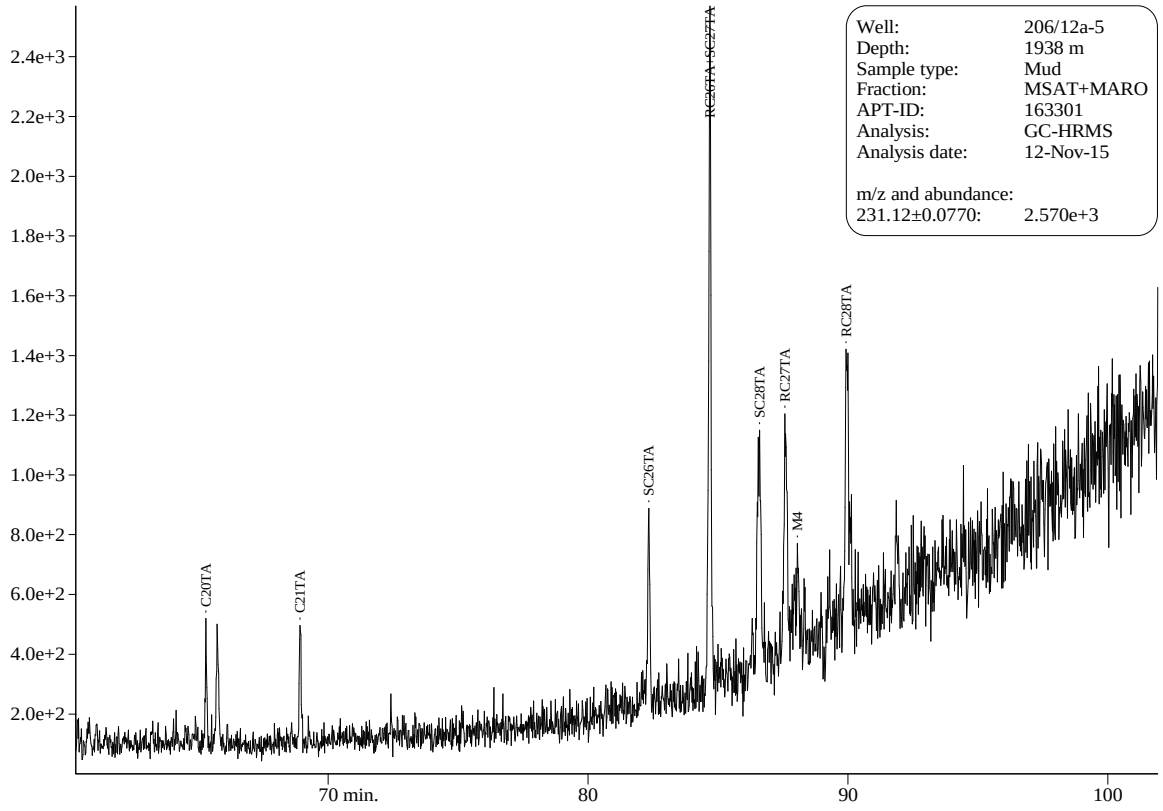


FIGURE Ap8.7 Aromatic fraction GC-MS fragmentograms: 206/12a-5, MUD, 1938m (Rona)

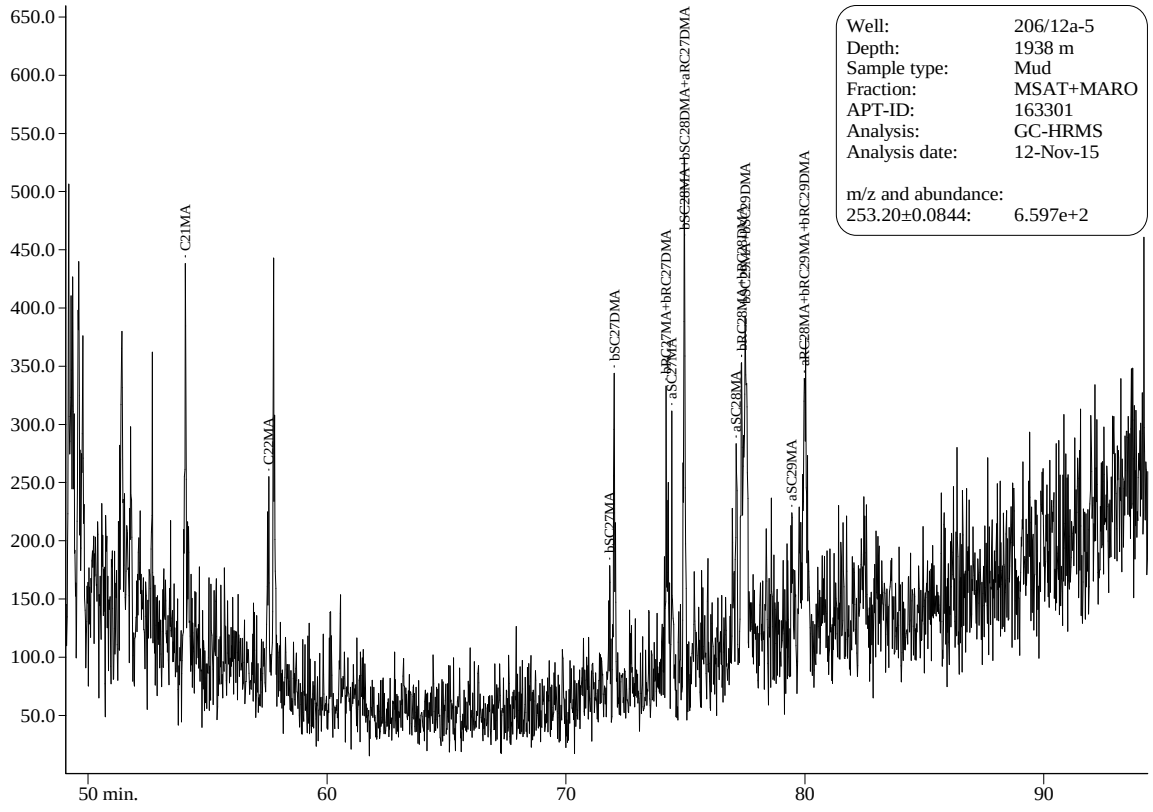
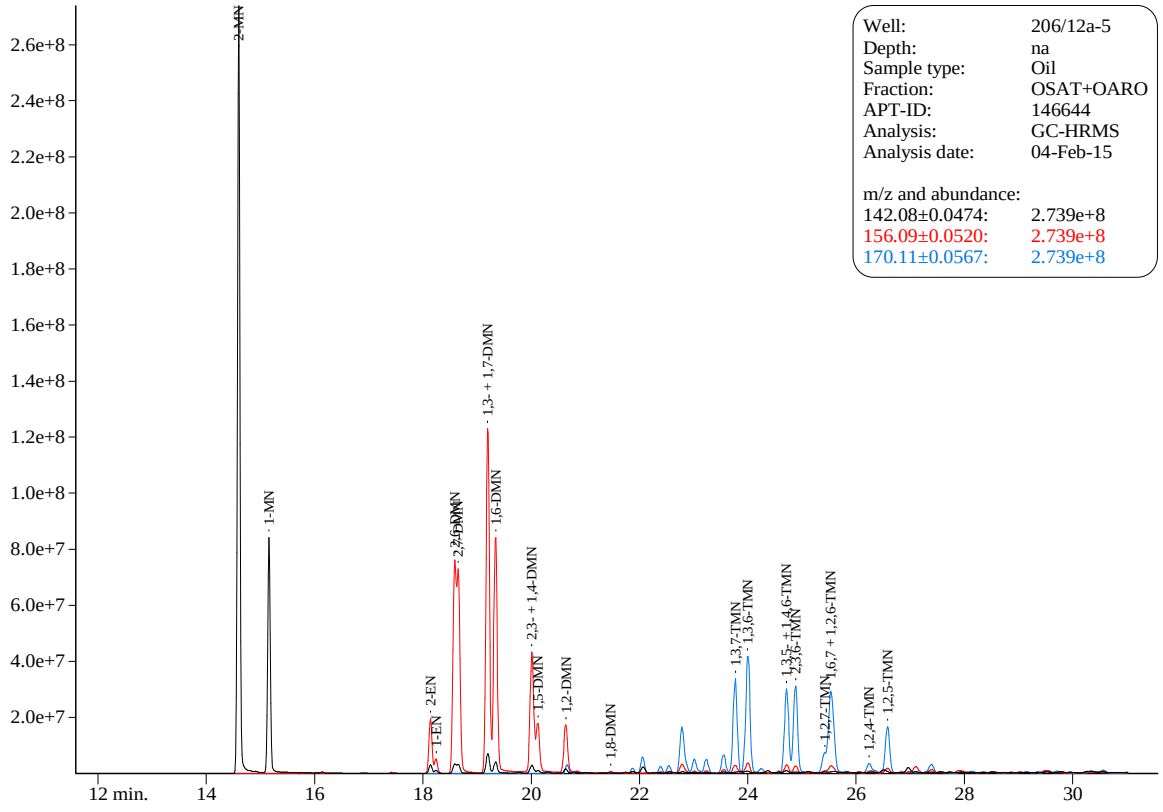


FIGURE Ap8.7 Aromatic fraction GC-MS fragmentograms: 206/12a-5, MUD, 1938m (Rona)



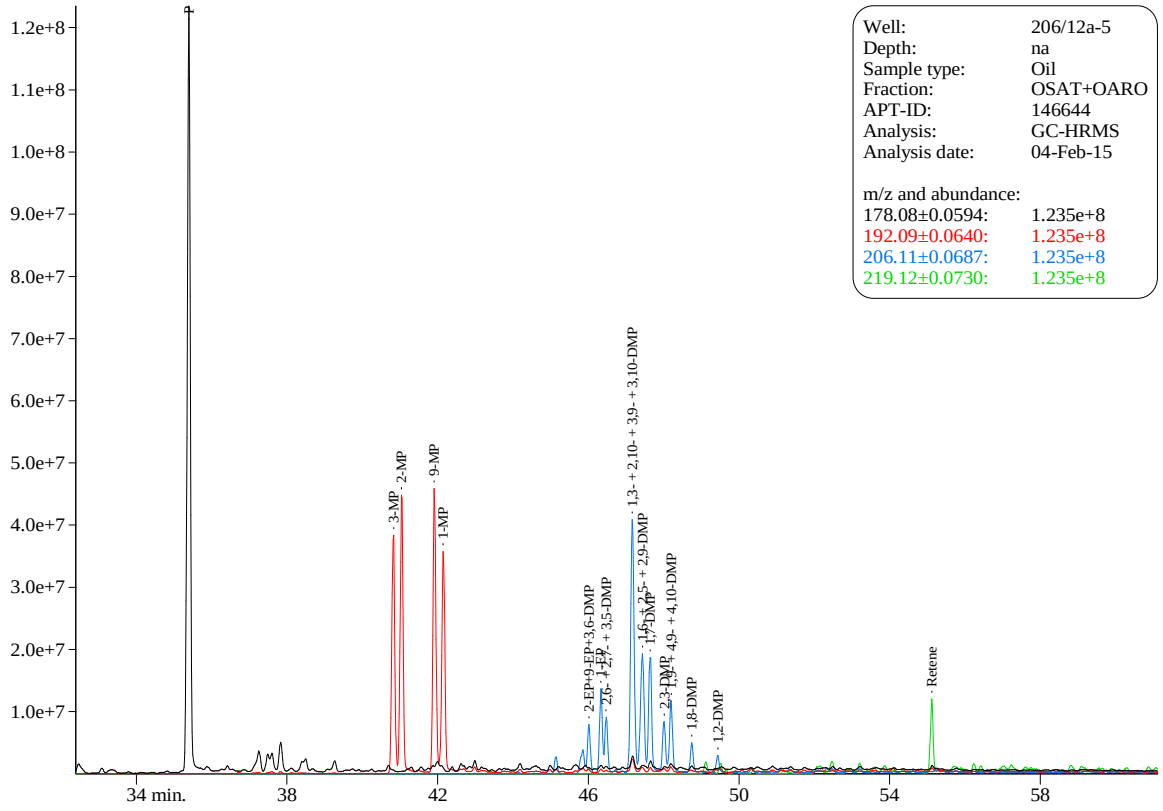
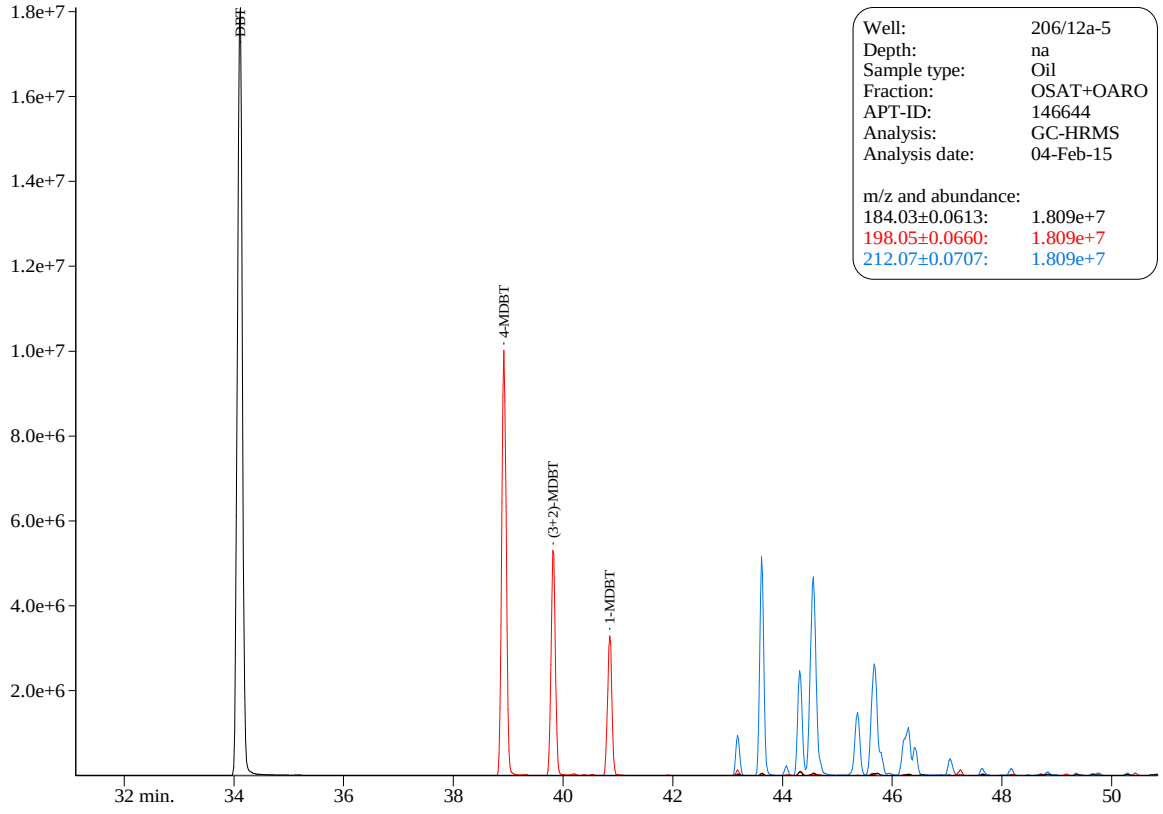


FIGURE Ap8.8 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona)



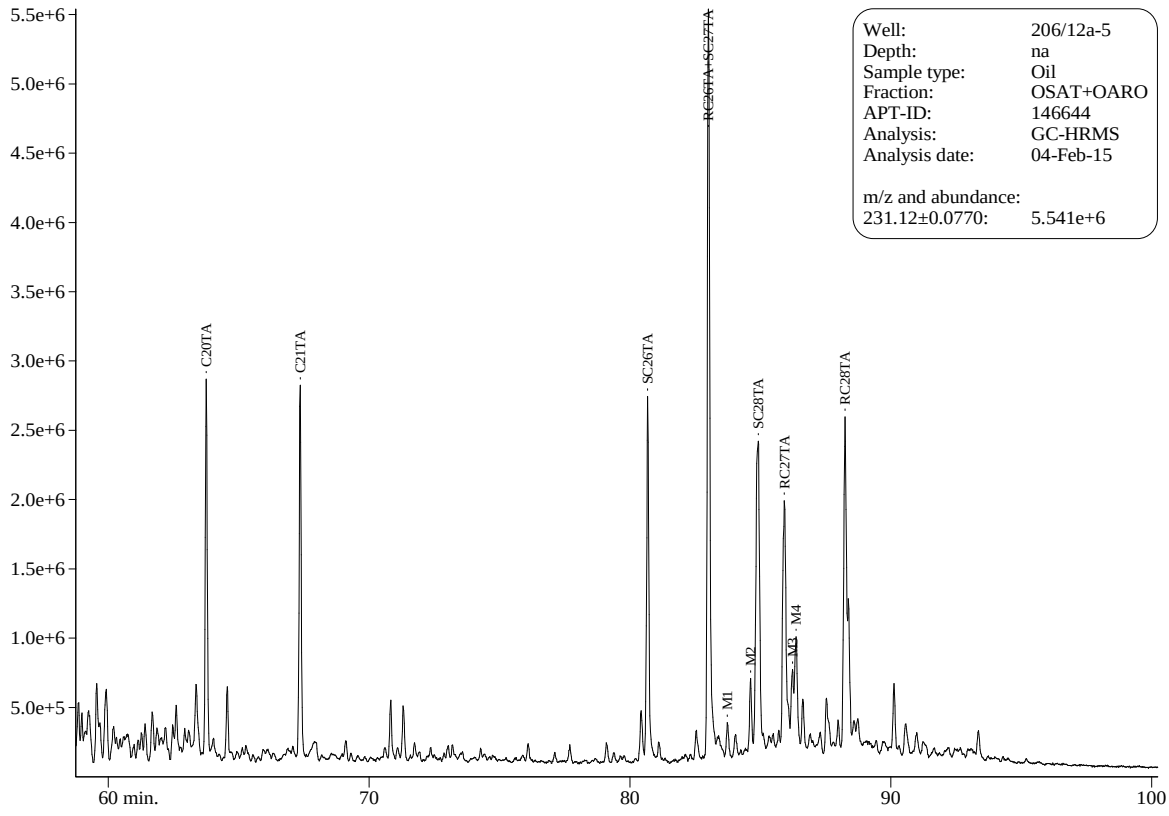


FIGURE Ap8.8 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona)

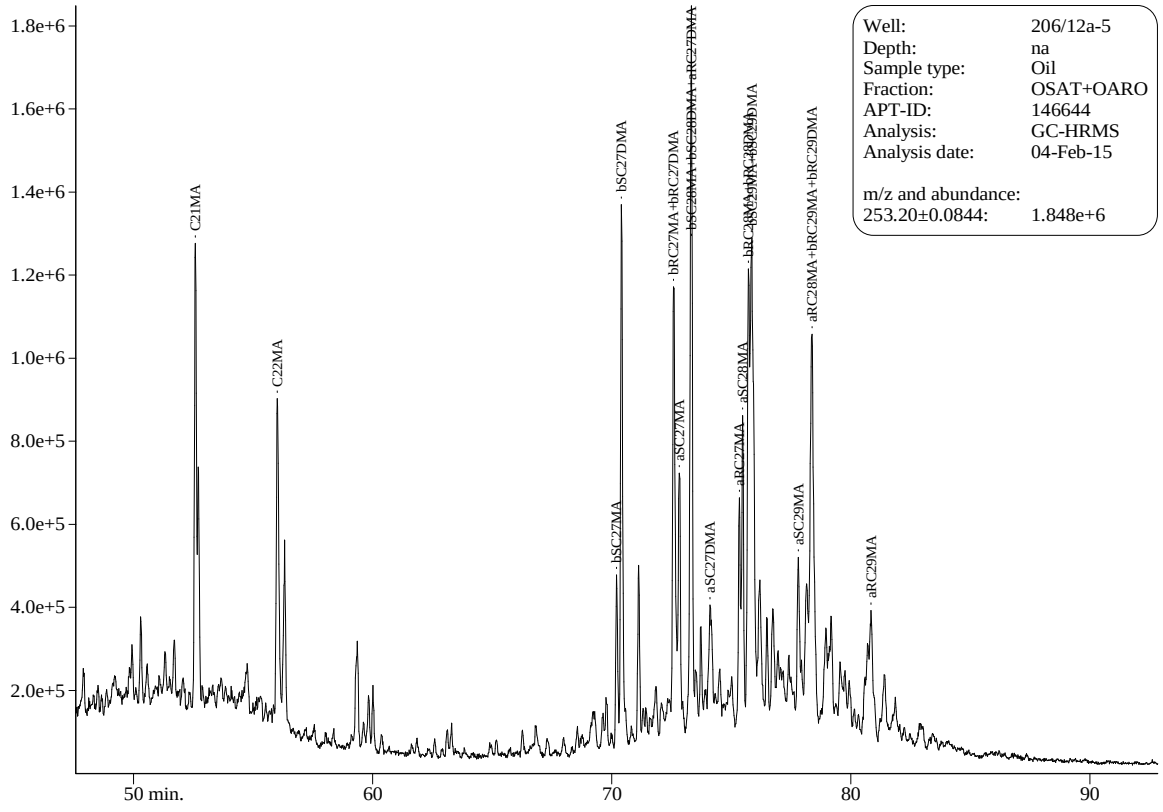
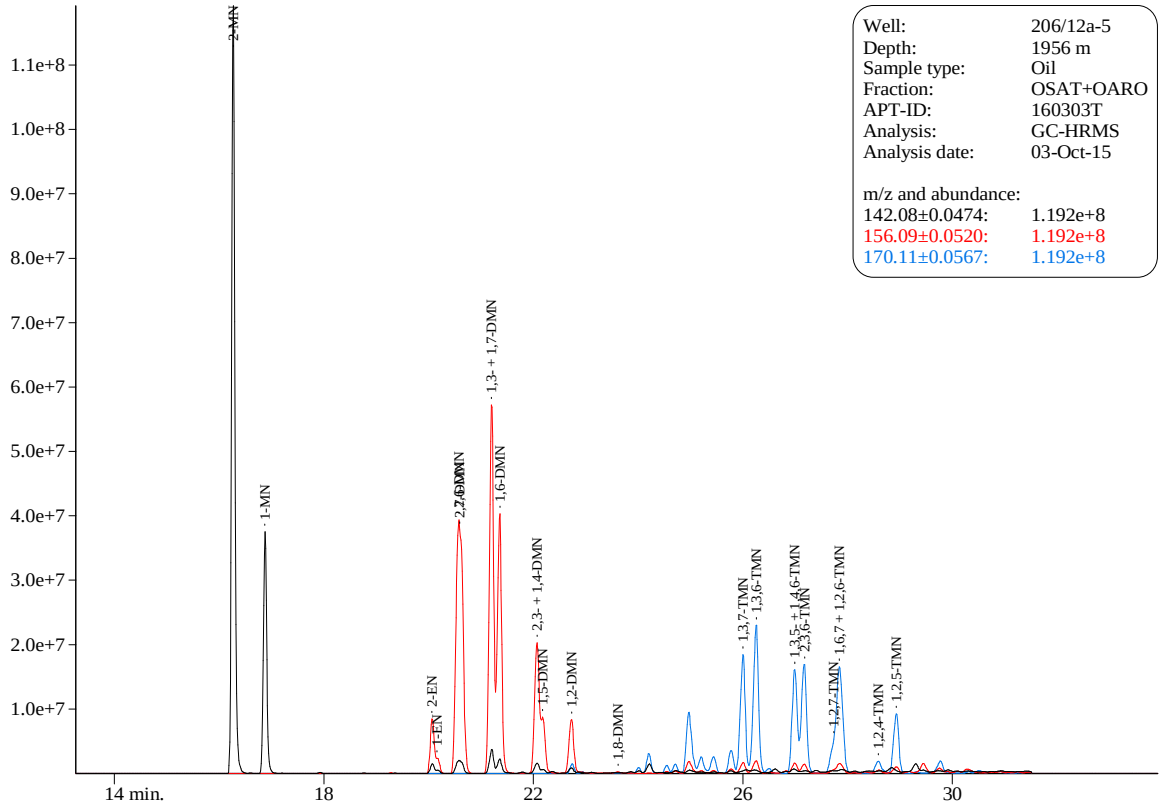


FIGURE Ap8.8 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SPUN OUT OIL, 1948.67m (Rona)



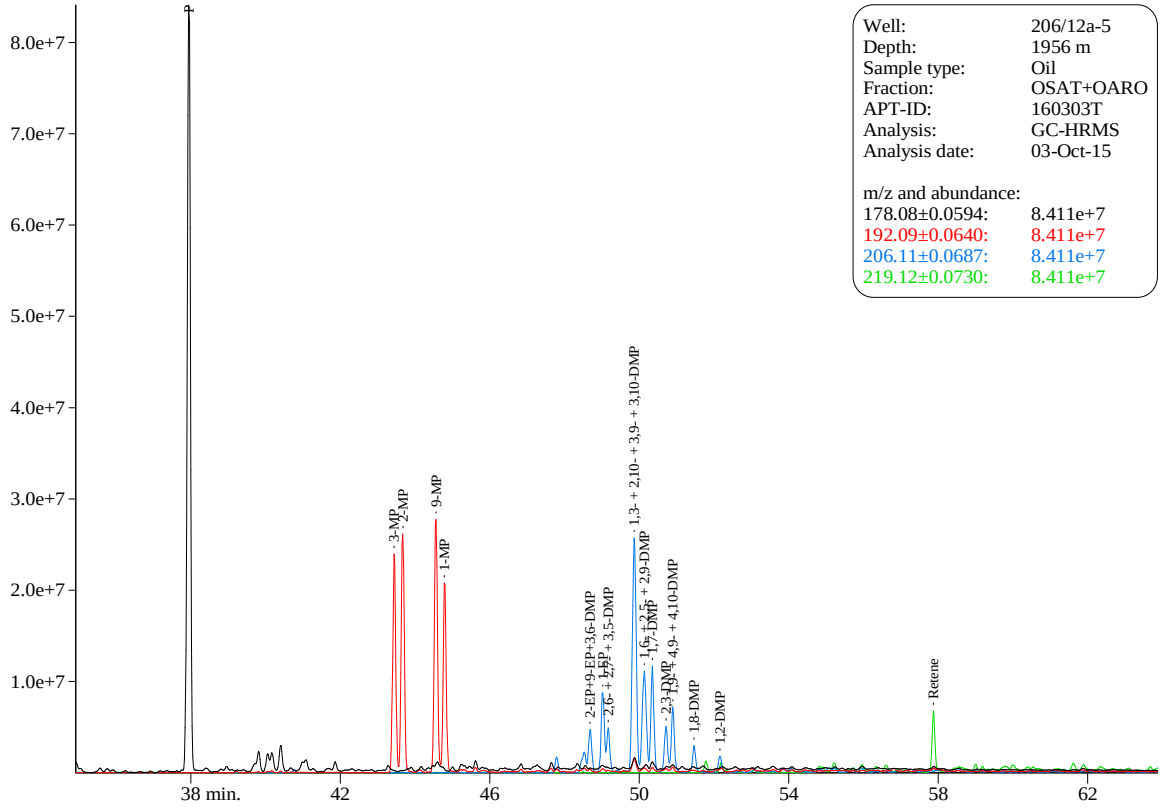
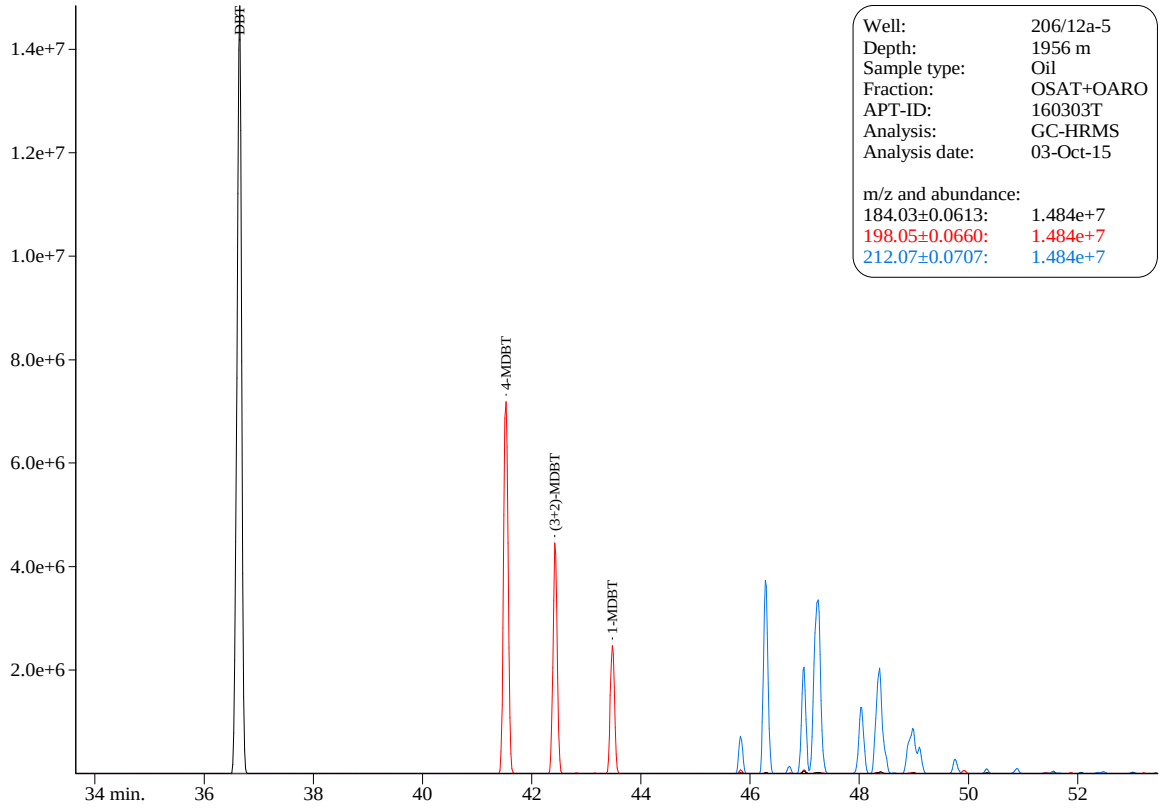


FIGURE Ap8.9 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1956m (Rona)



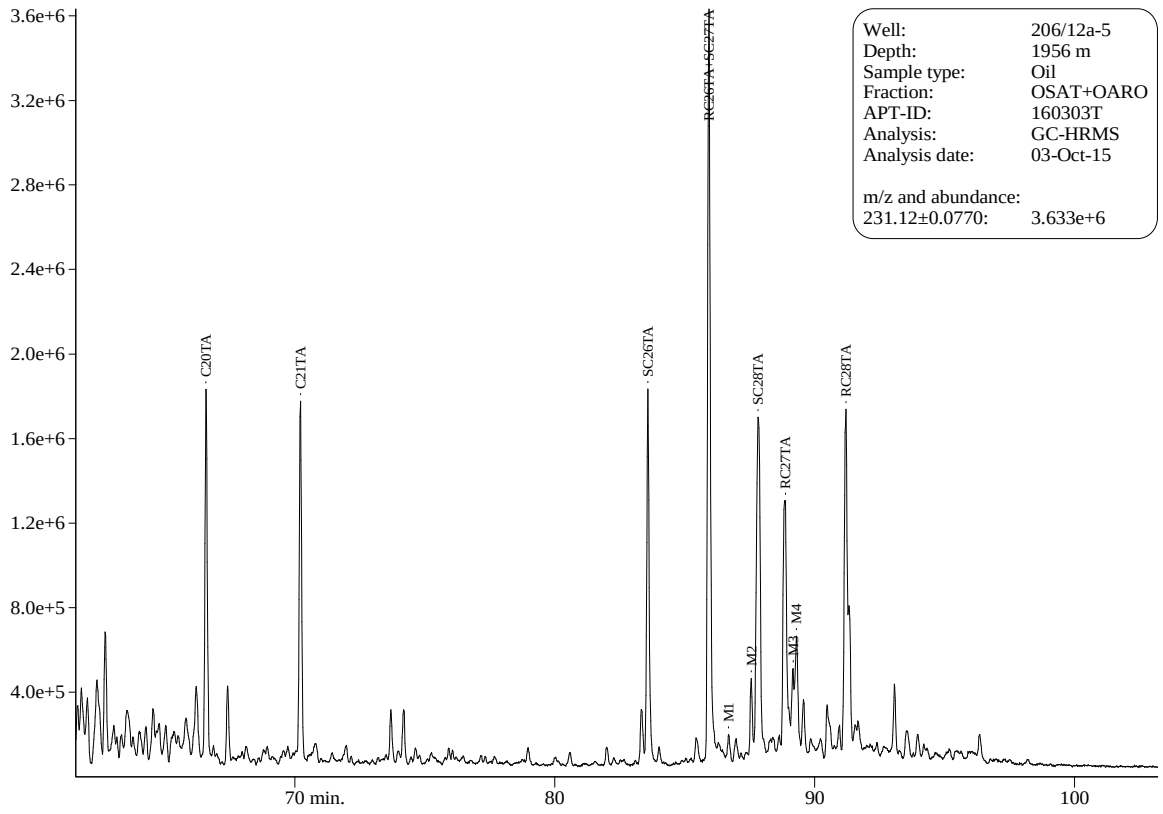


FIGURE Ap8.9 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1956m (Rona)

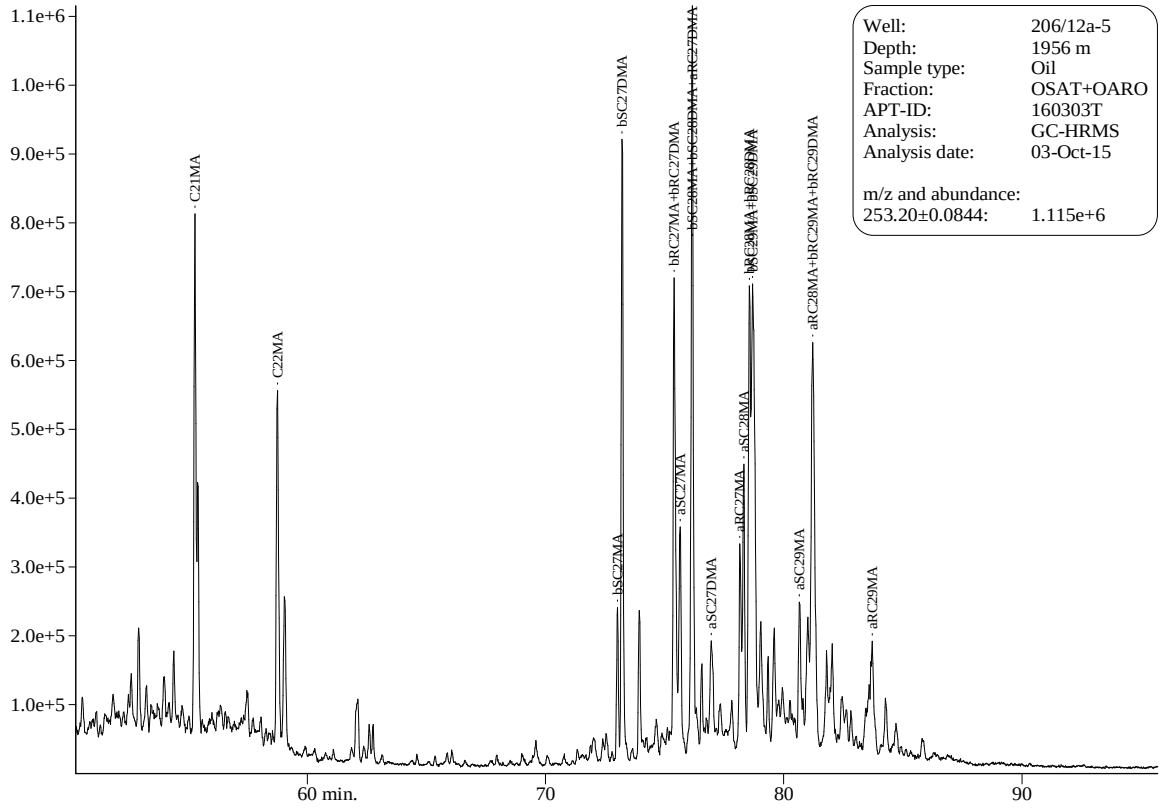
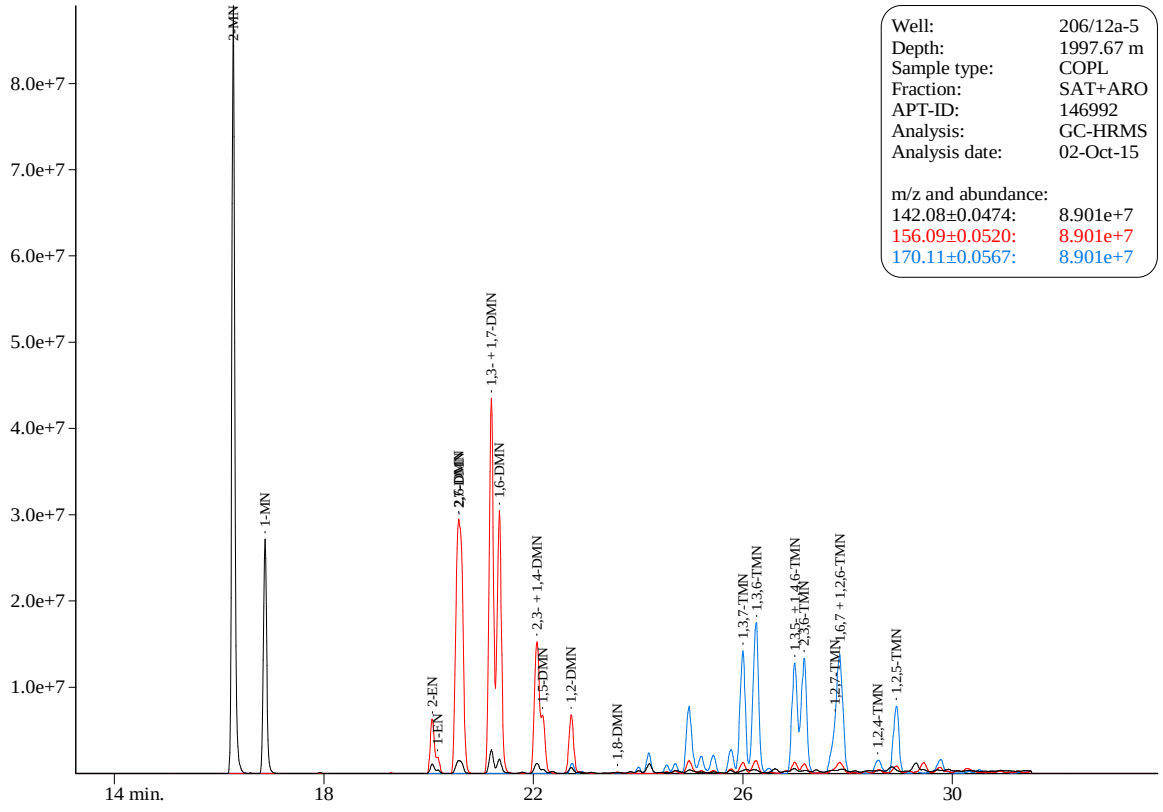


FIGURE Ap8.9 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1956m (Rona)



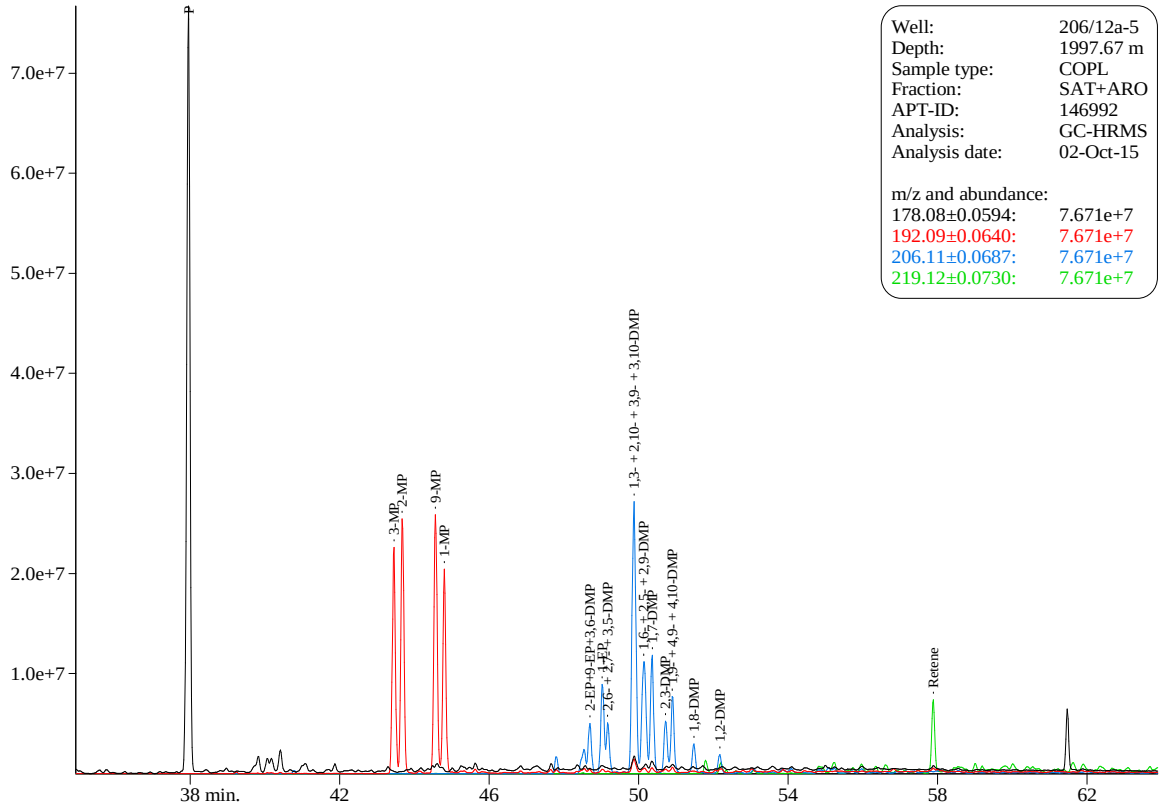
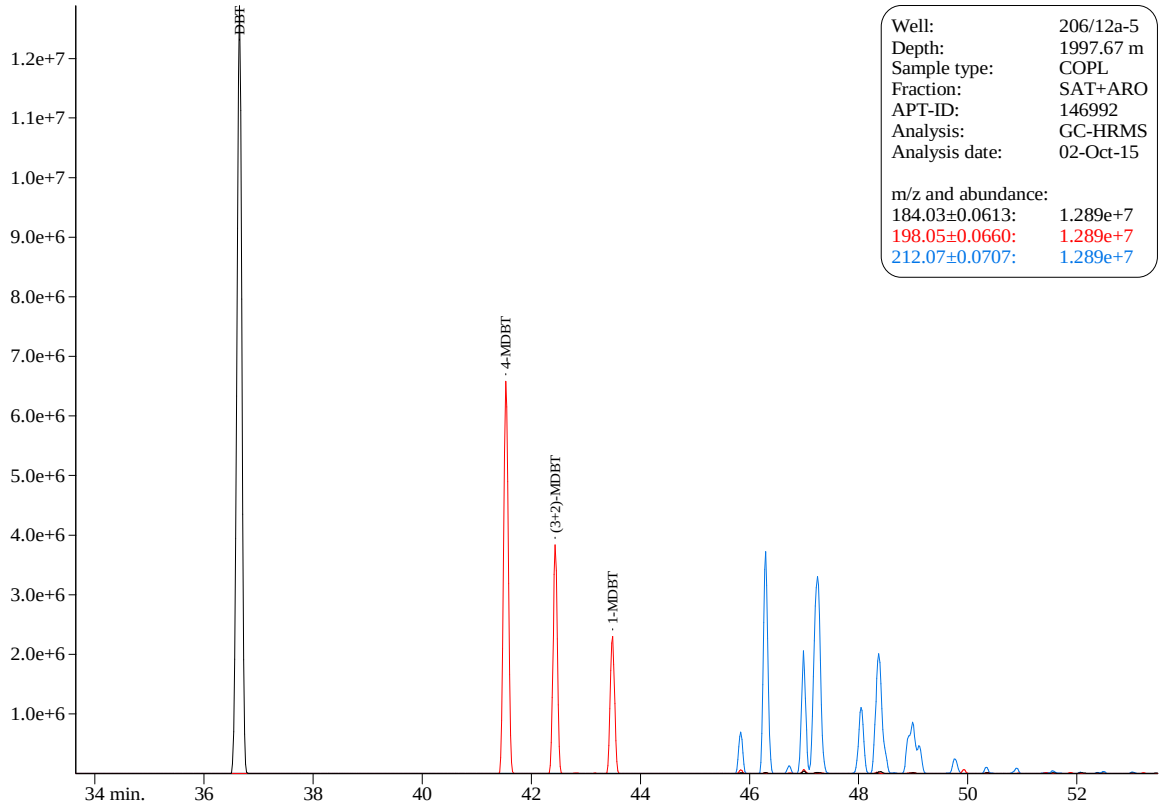


FIGURE Ap8.10 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1956.67m (Rona)



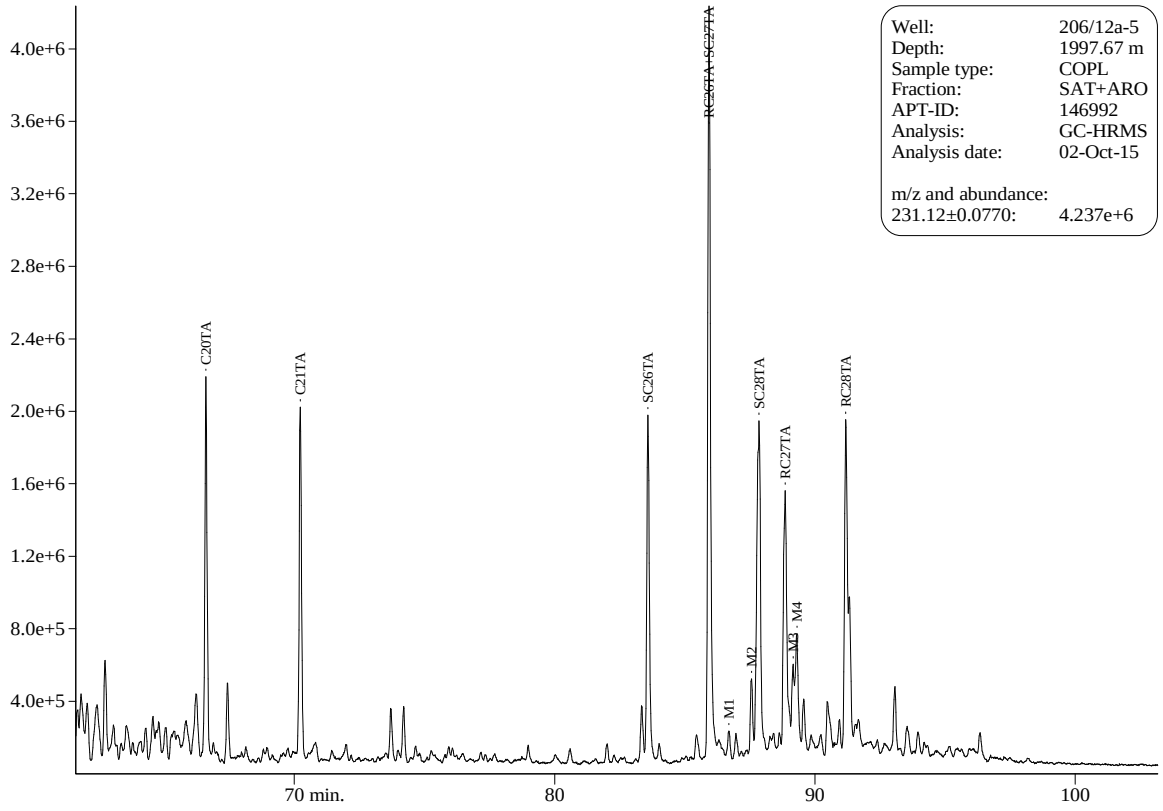


FIGURE Ap8.10 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1956.67m (Rona)

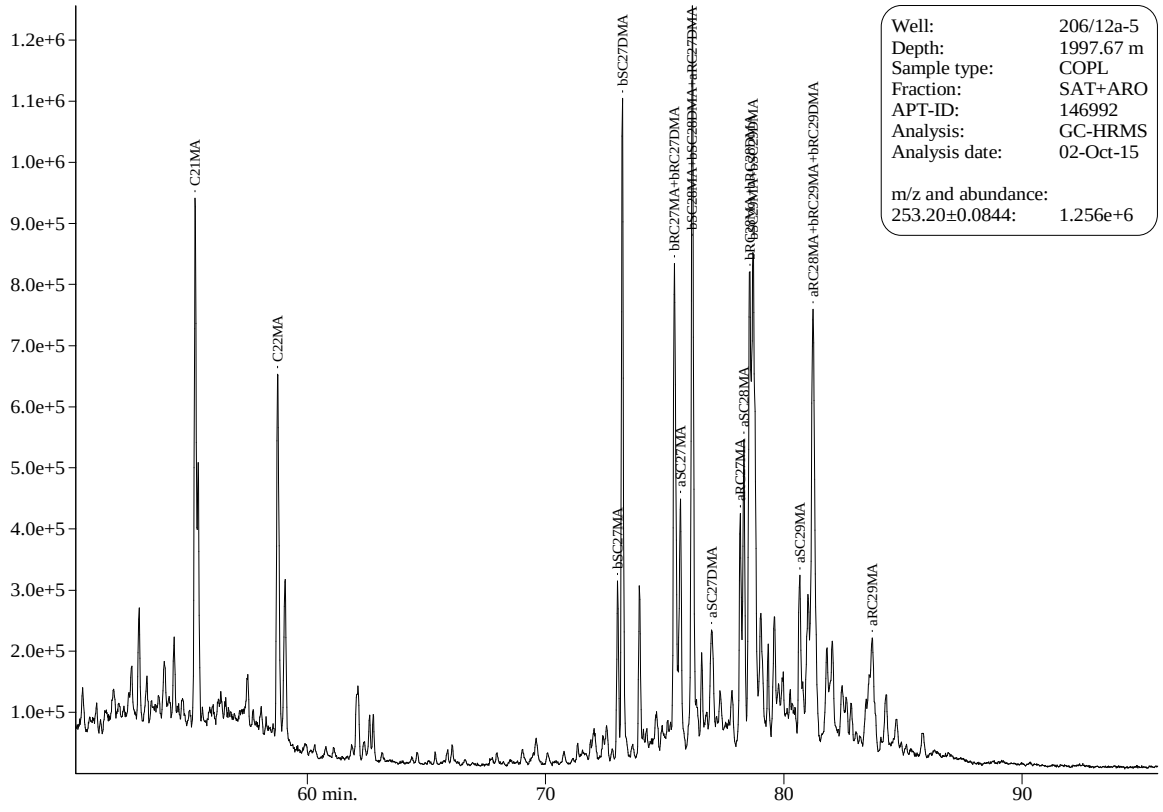
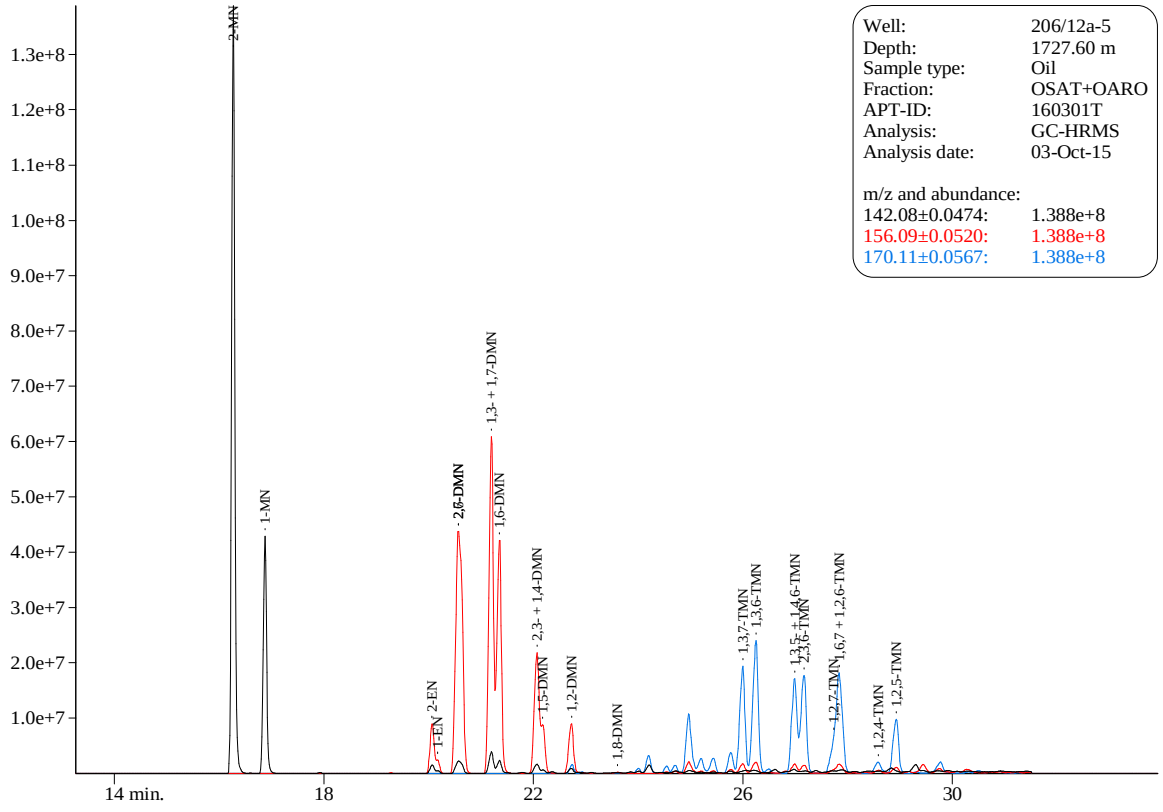


FIGURE Ap8.10 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1956.67m (Rona)



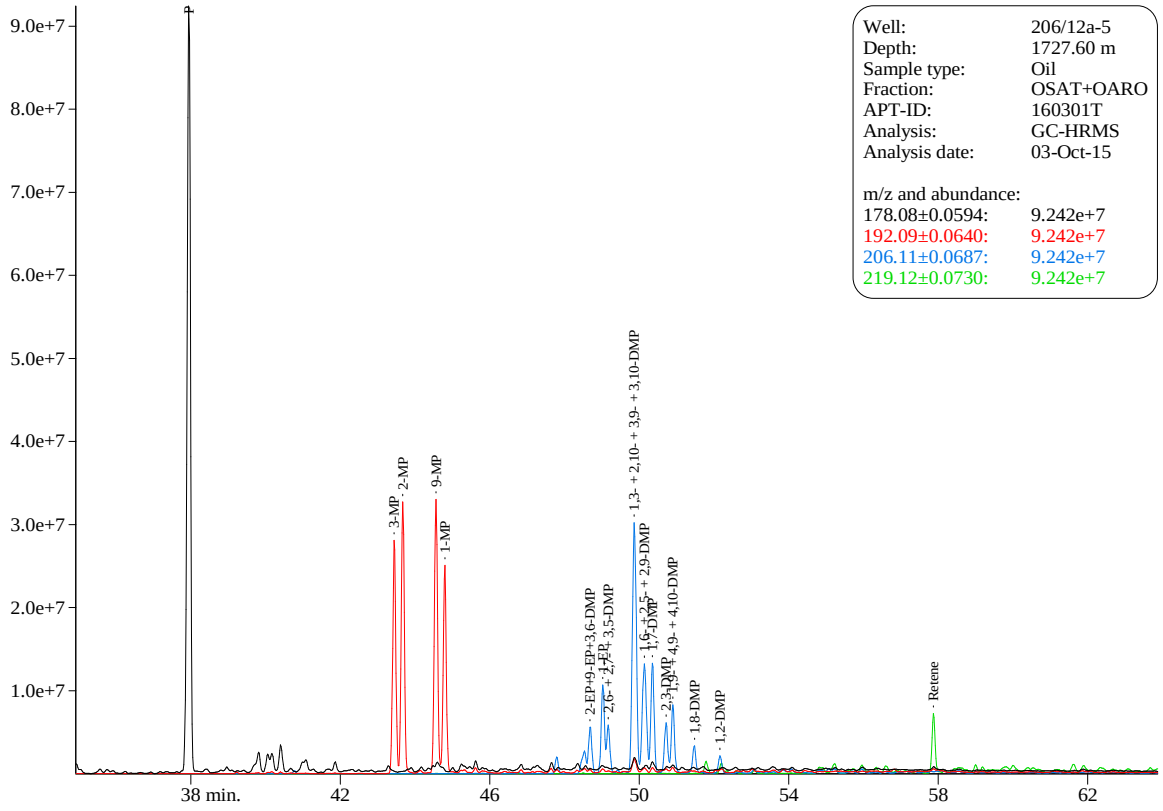
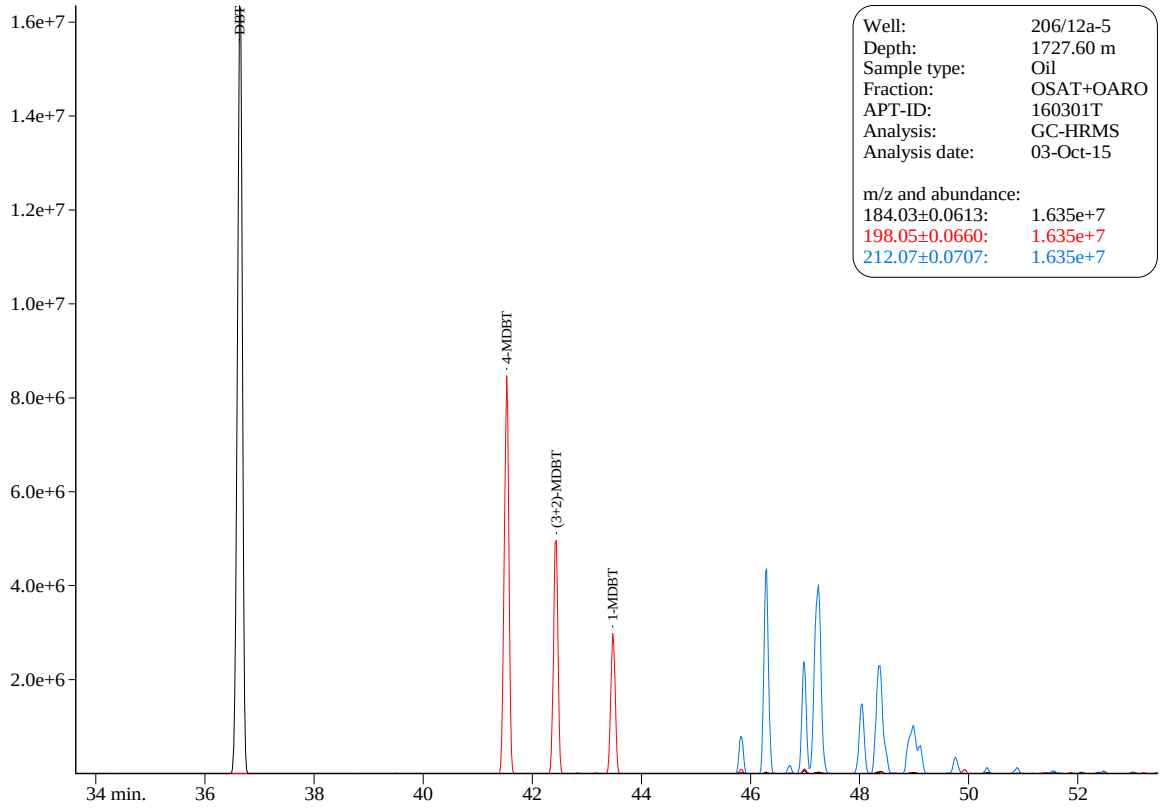


FIGURE Ap8.11 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1936-1963m (Rona)



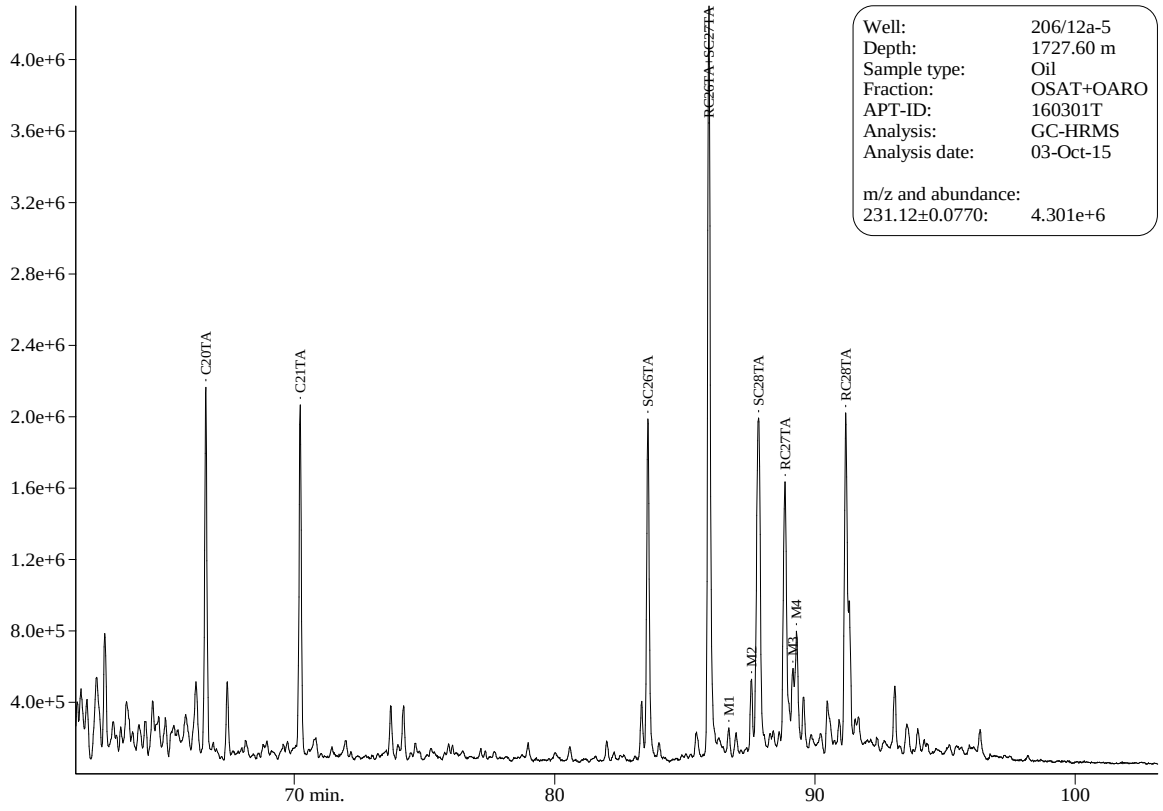


FIGURE Ap8.11 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1936-1963m (Rona)

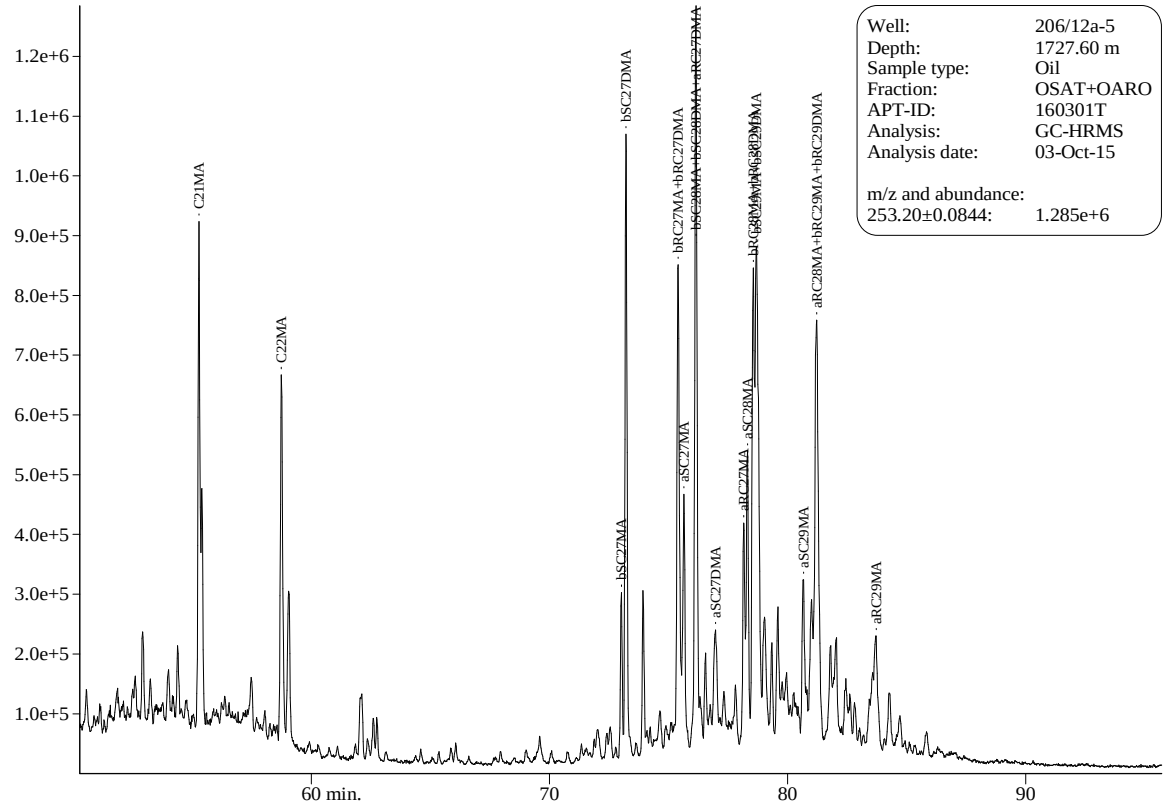
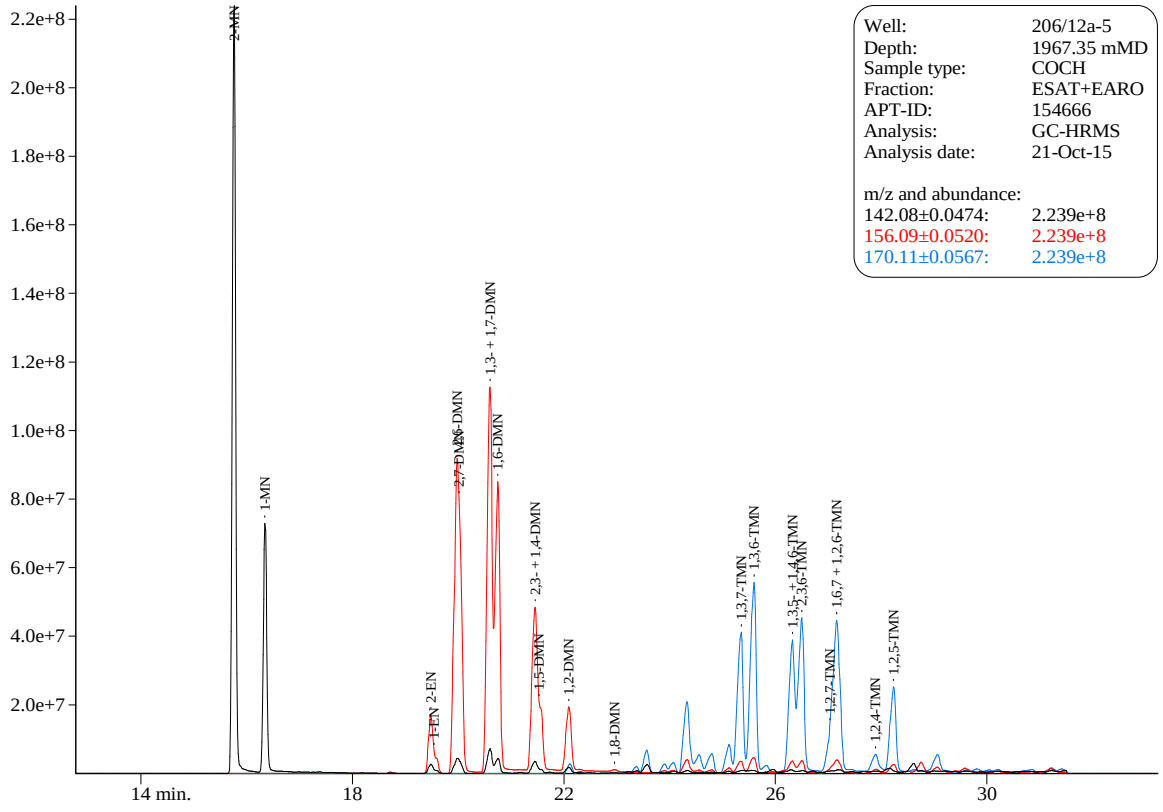


FIGURE Ap8.11 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1936-1963m (Rona)



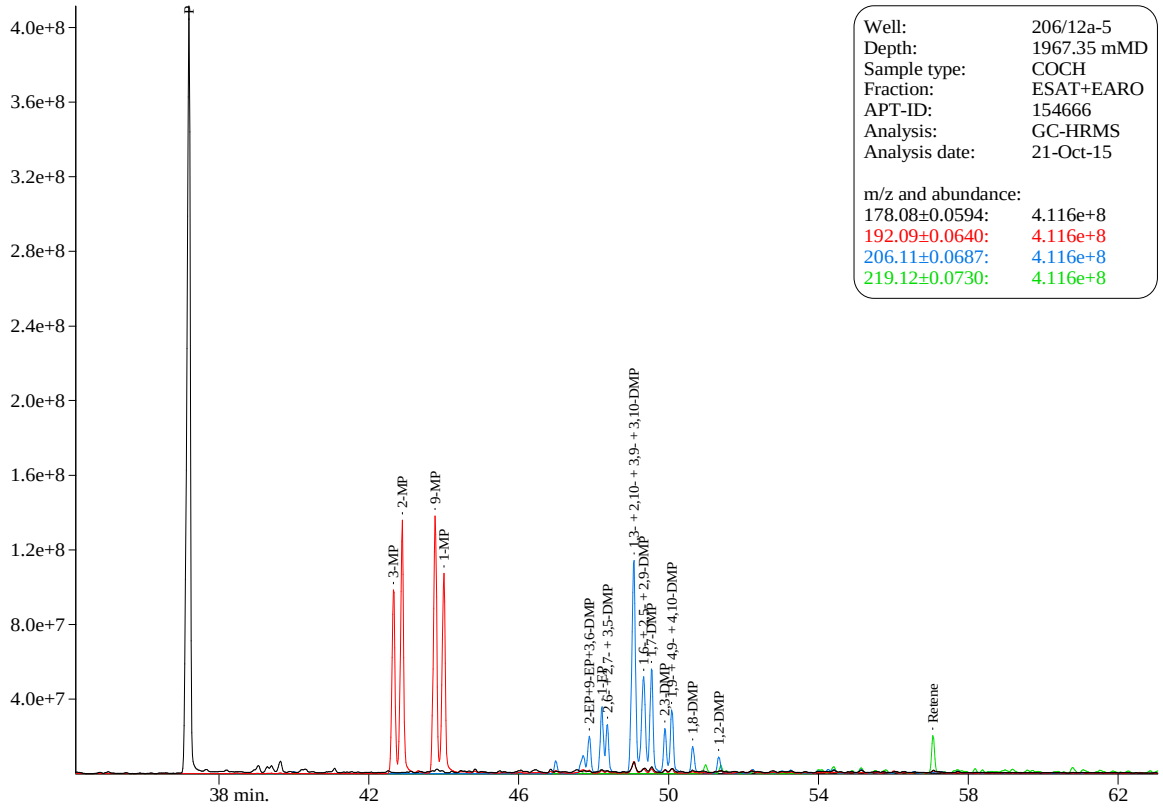
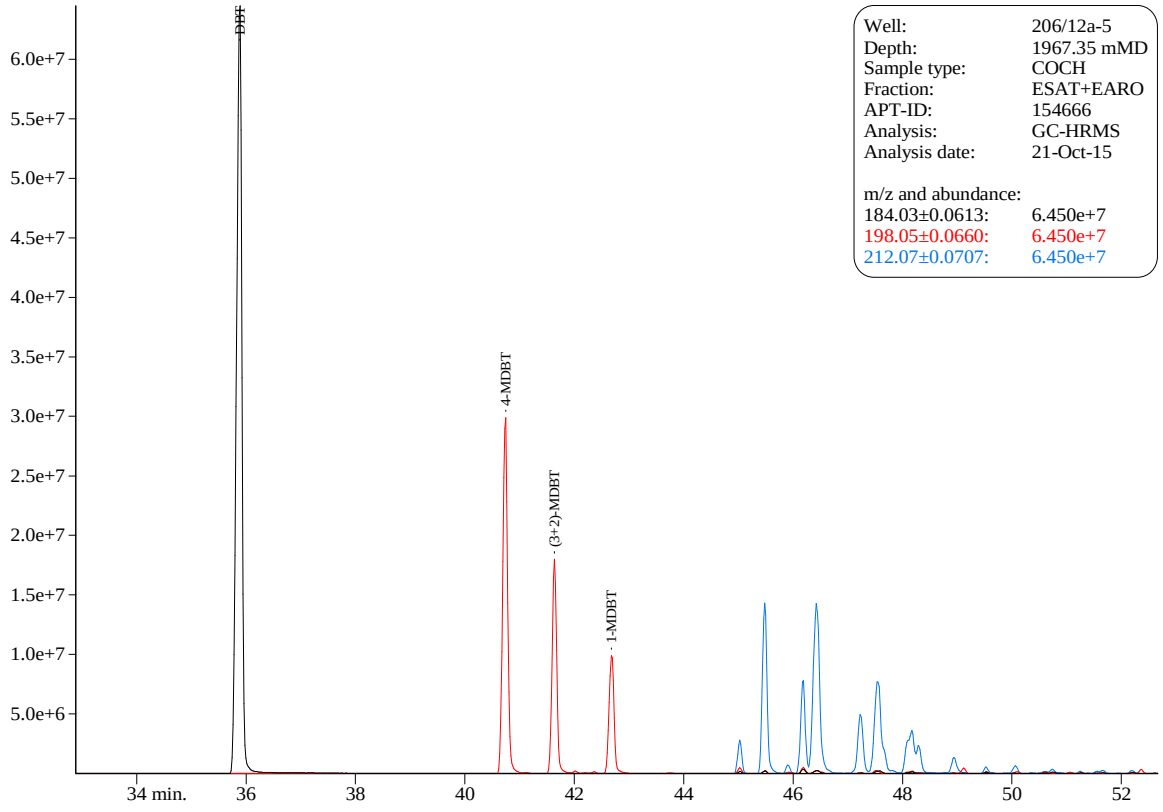


FIGURE Ap8.12 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE, 1967.35m (Rona)



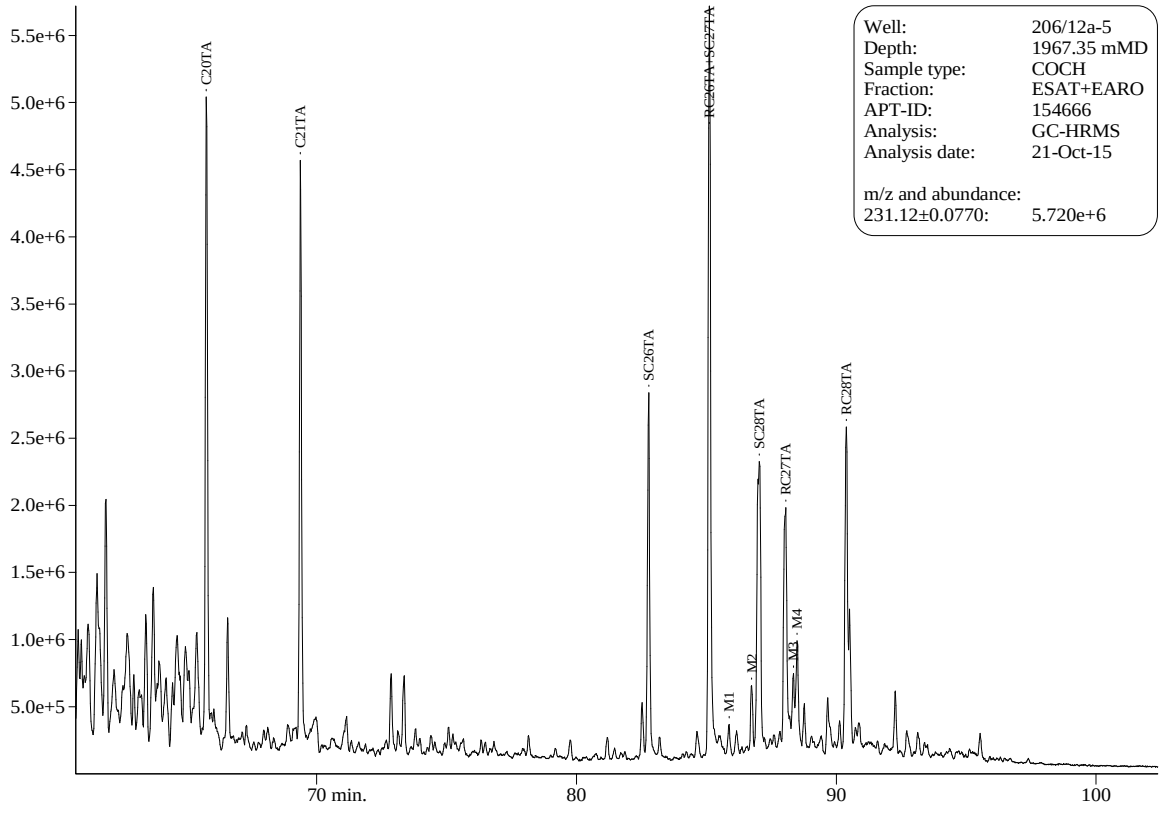


FIGURE Ap8.12 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE, 1967.35m (Rona)

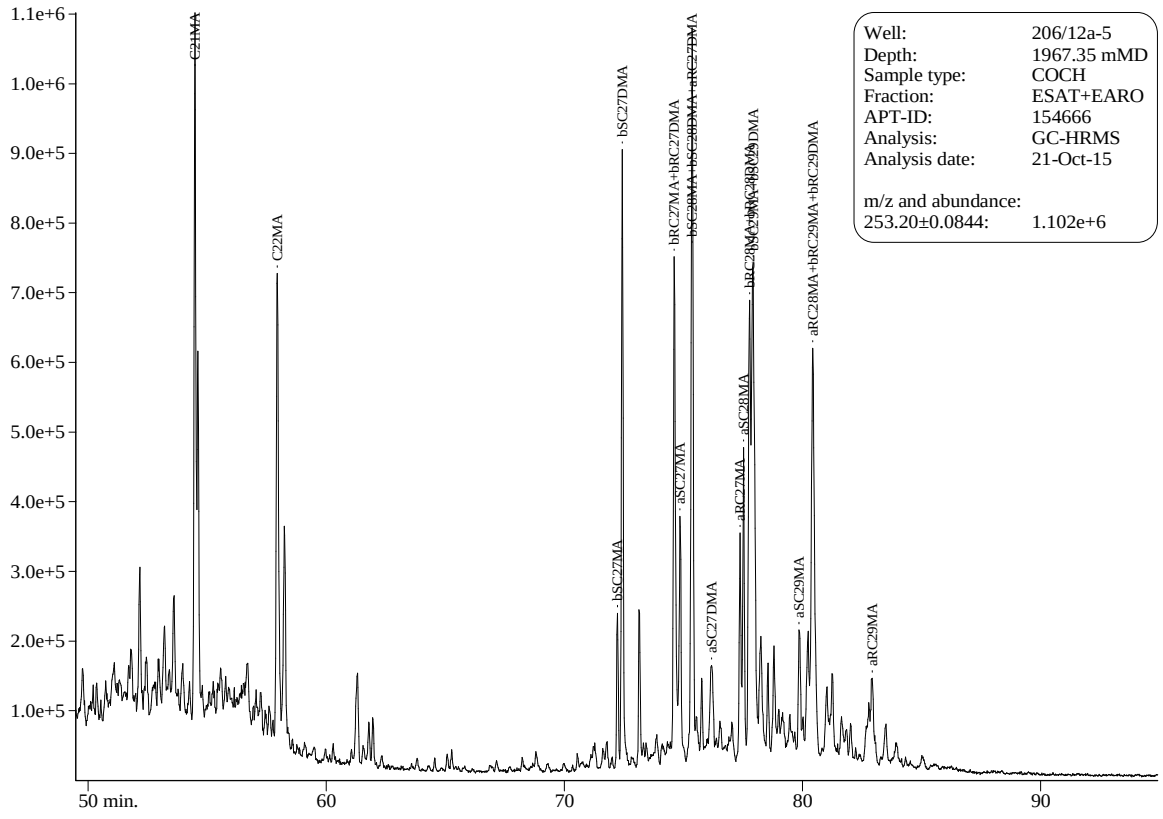
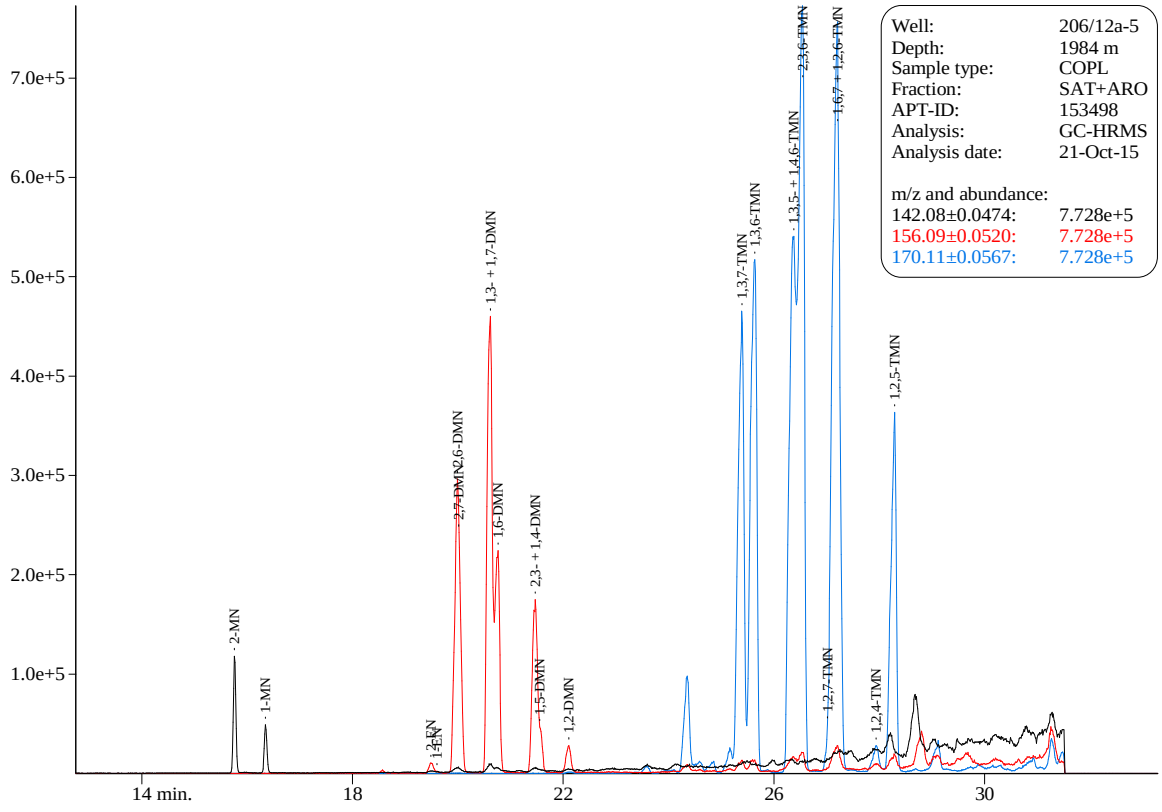


FIGURE Ap8.12 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE, 1967.35m (Rona)



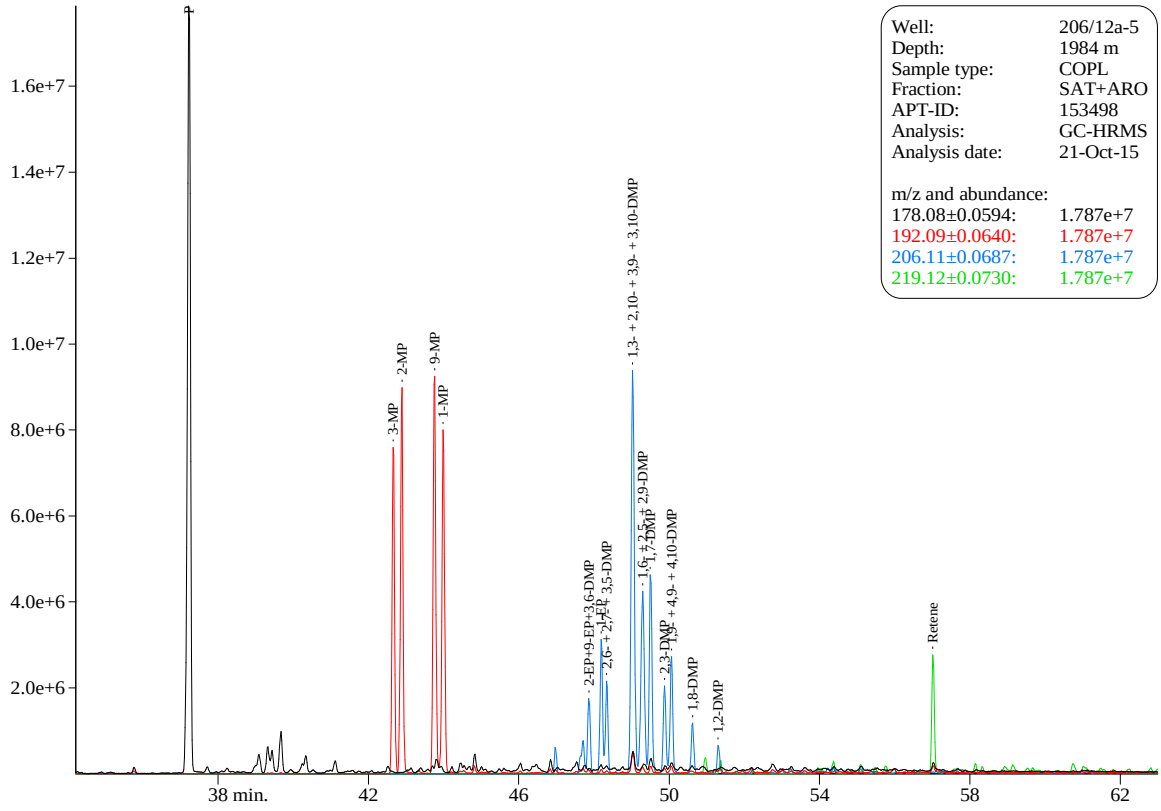
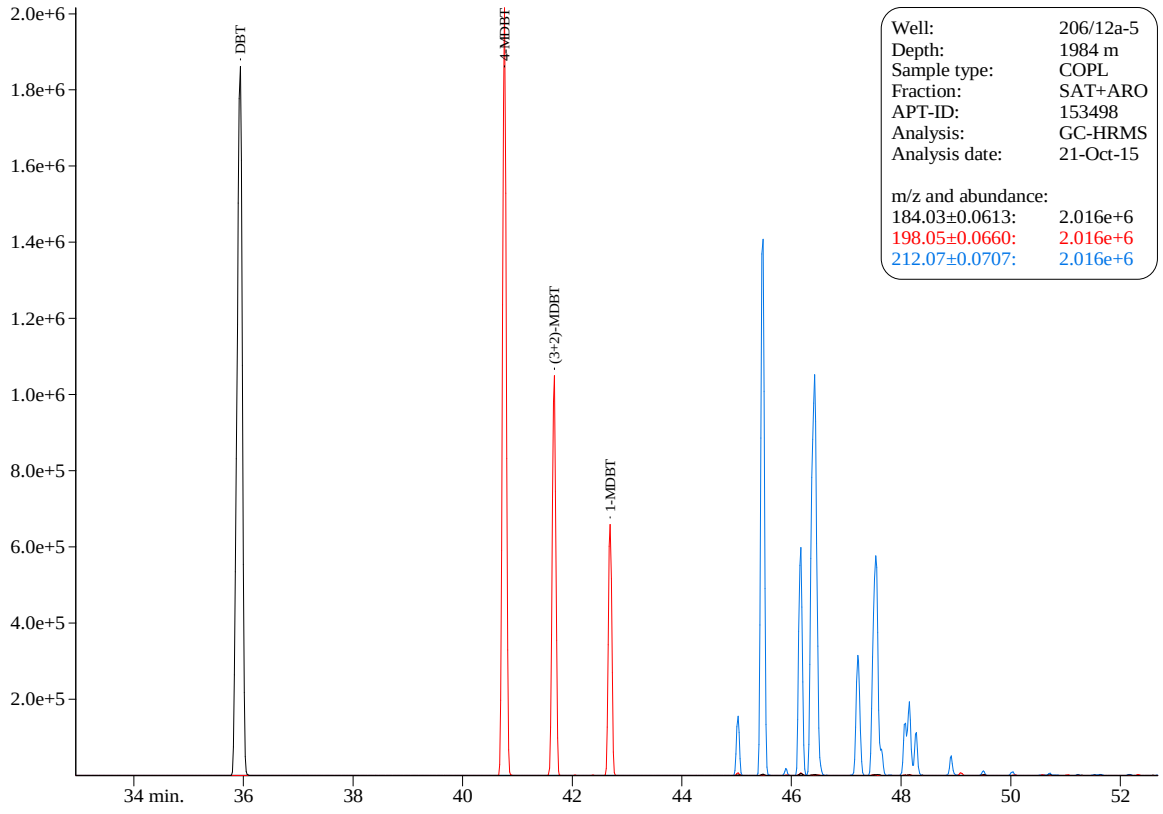


FIGURE Ap8.13 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SWC, 1984m (LCG Unit III)



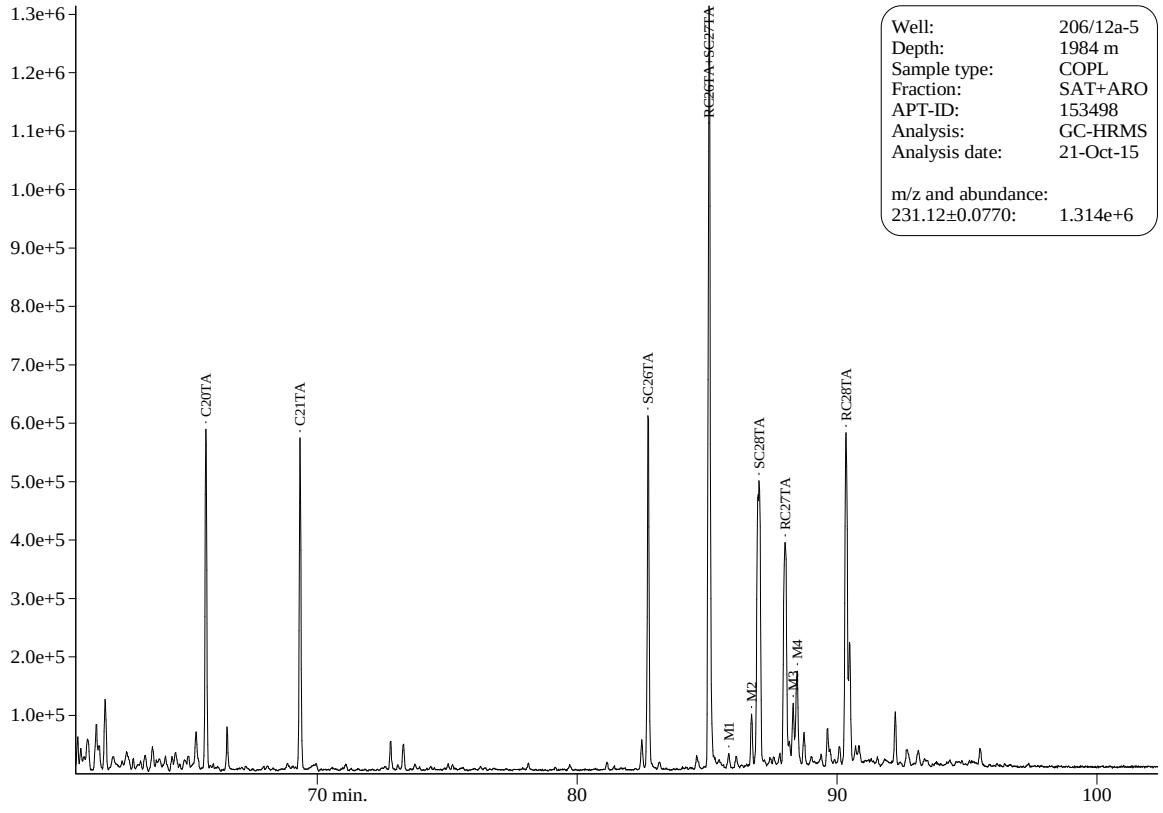


FIGURE Ap8.13 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SWC, 1984m (LCG Unit III)

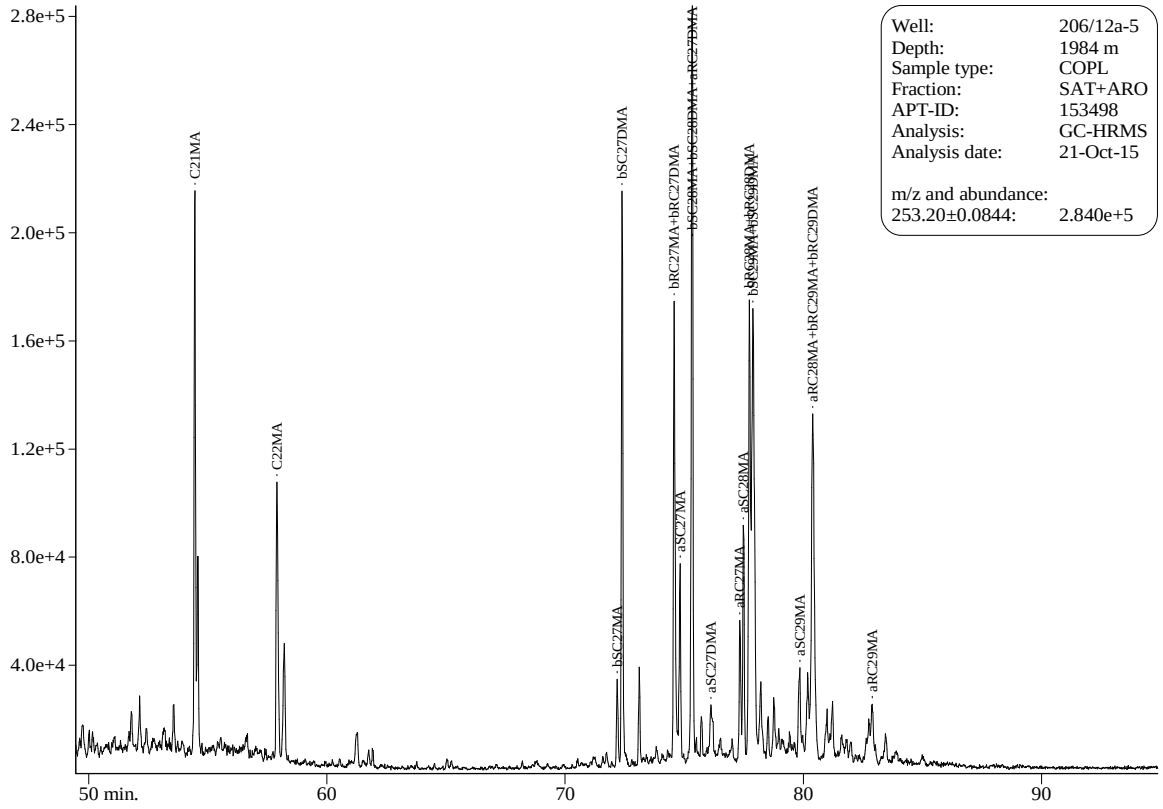
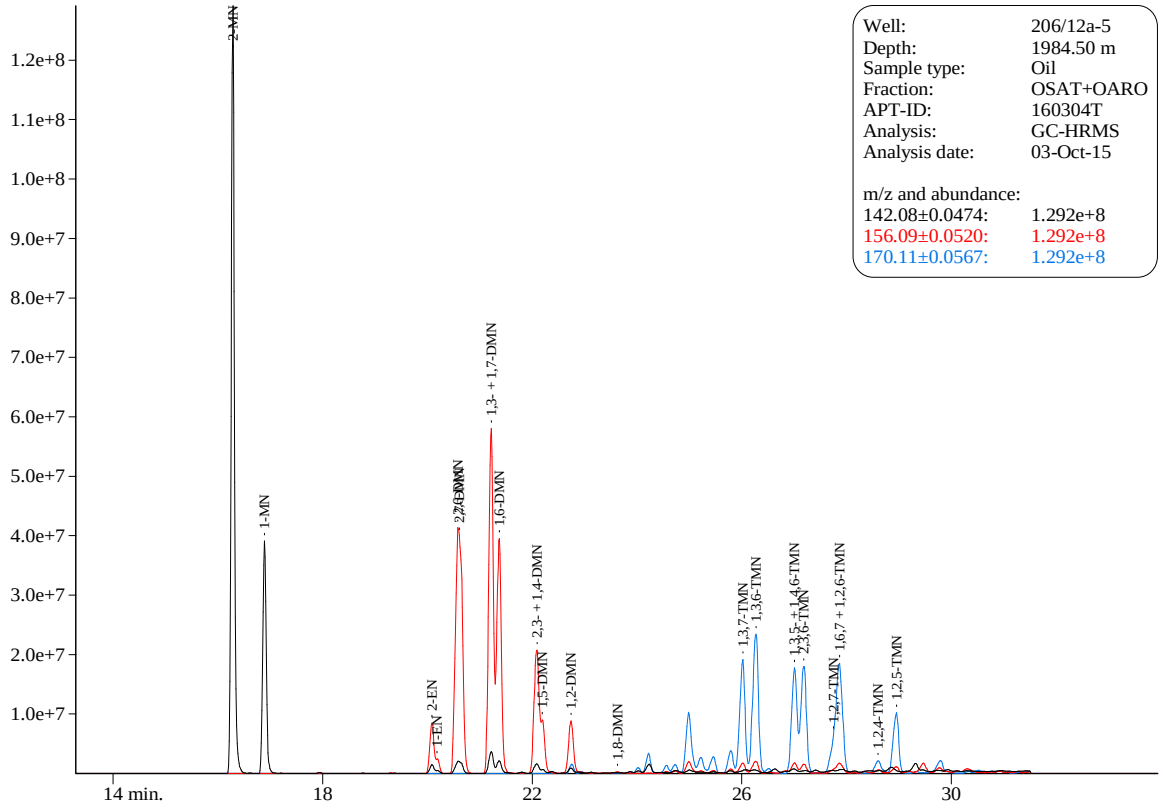


FIGURE Ap8.13 Aromatic fraction GC-MS fragmentograms: 206/12a-5, SWC, 1984m (LCG Unit III)



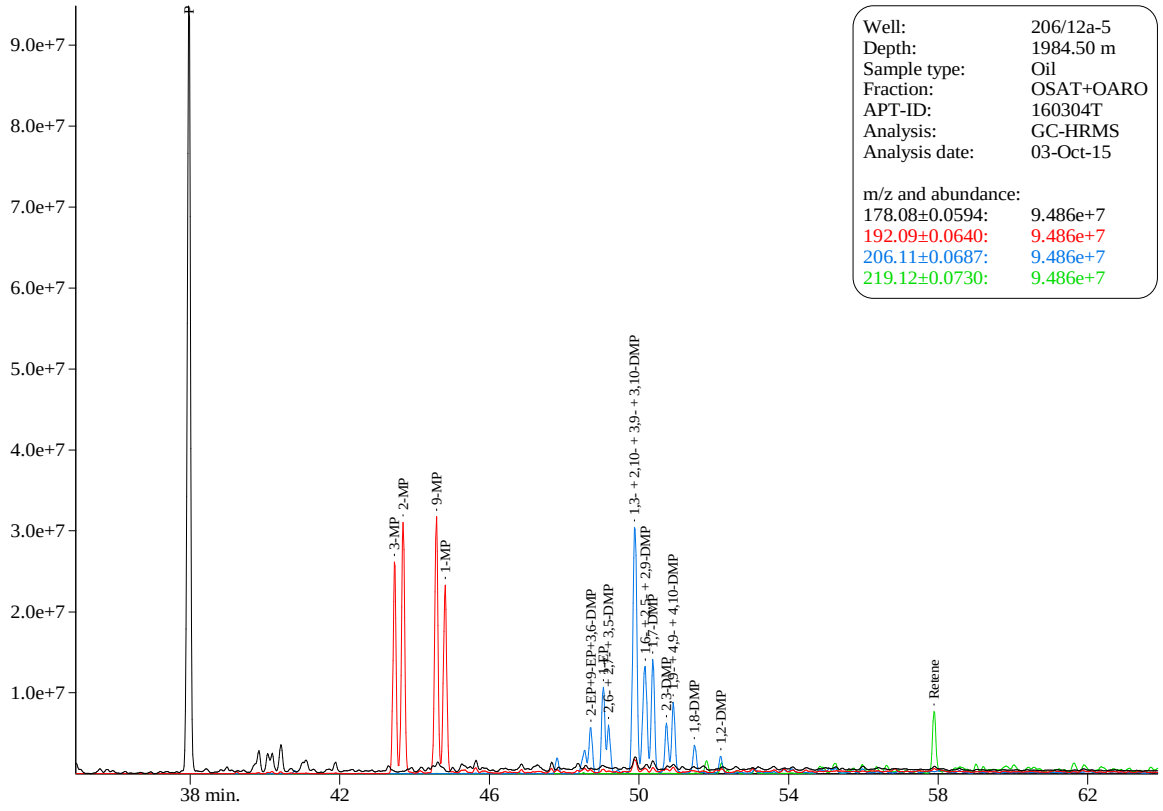
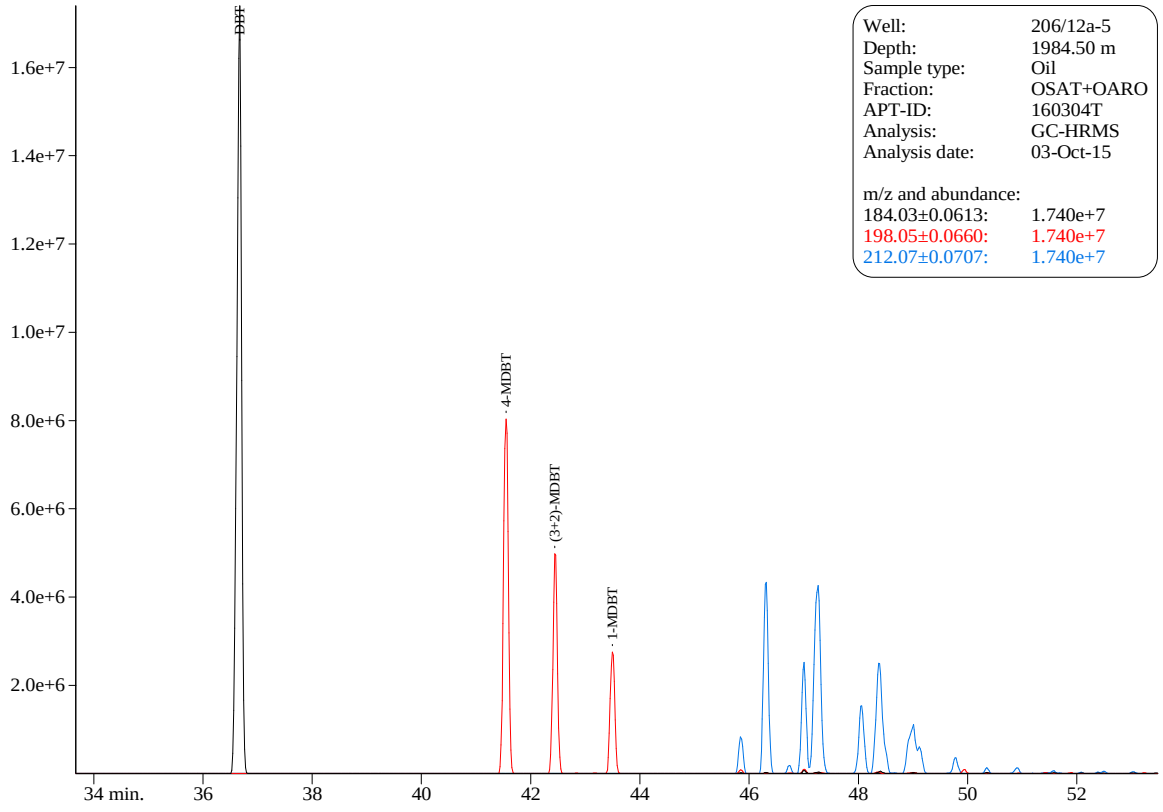


FIGURE Ap8.14 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1984.5m (LCG Unit III)



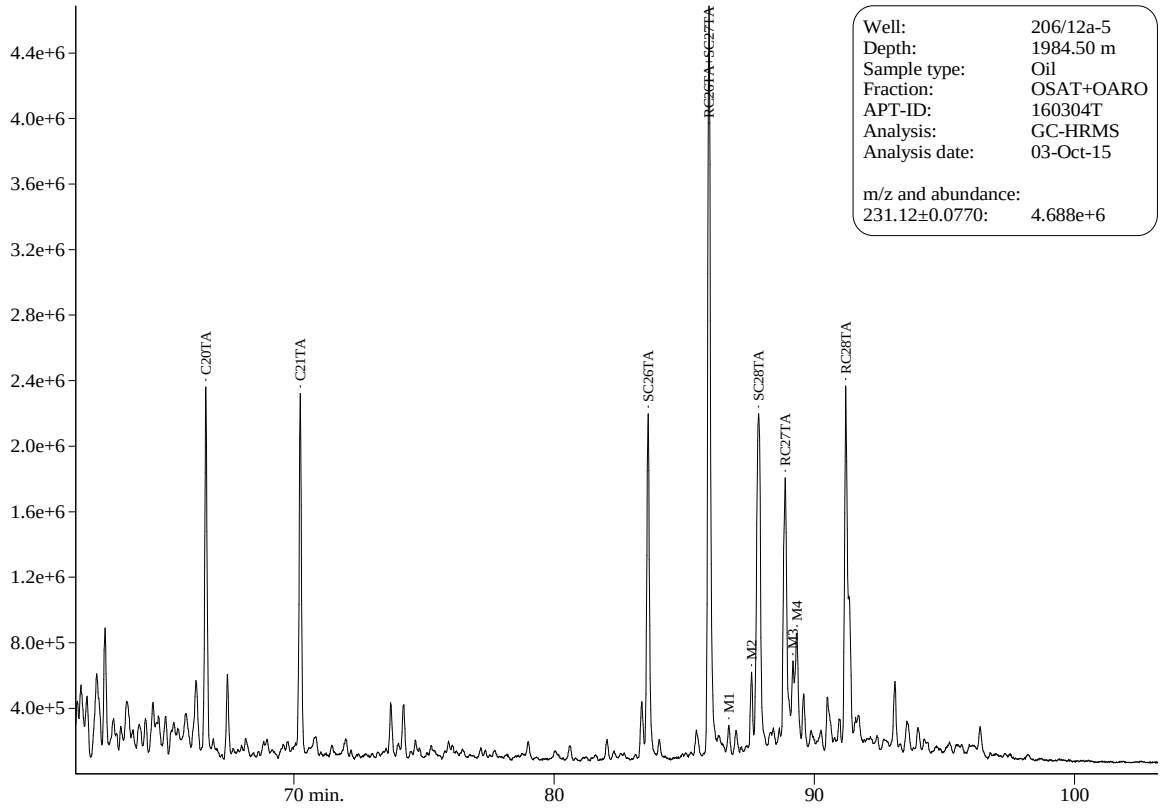


FIGURE Ap8.14 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1984.5m (LCG Unit III)

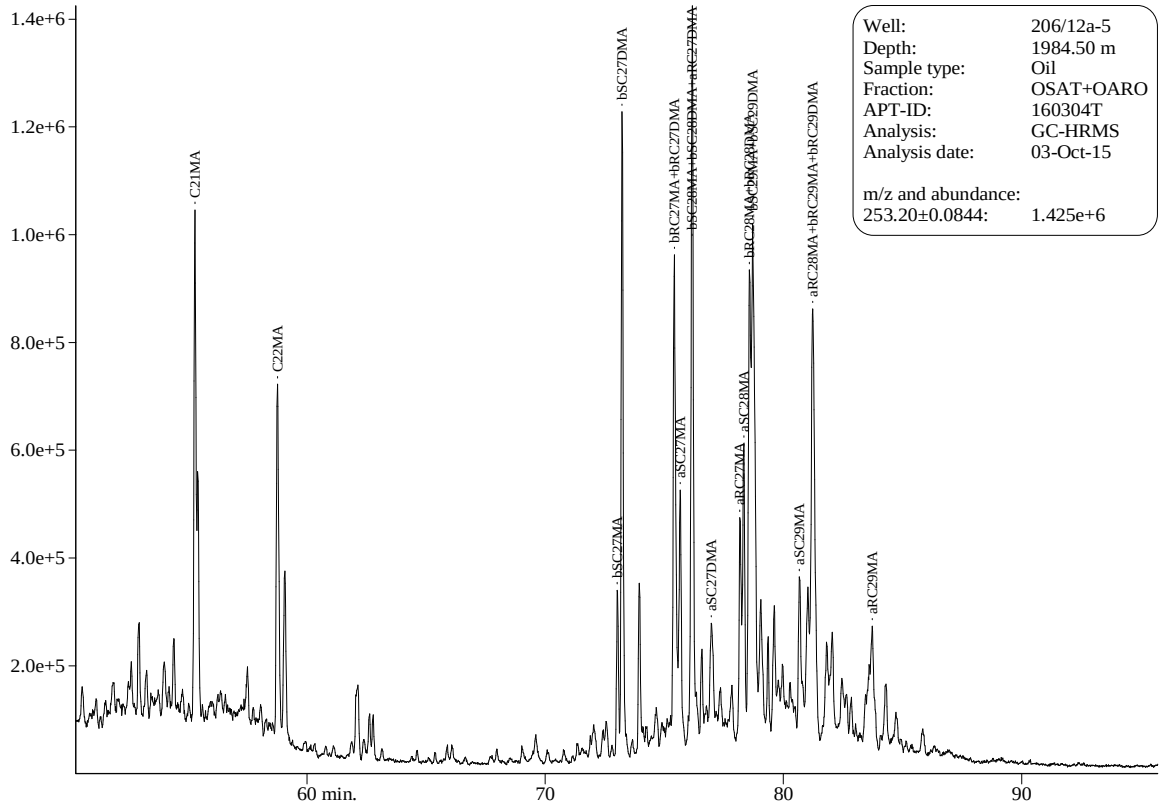
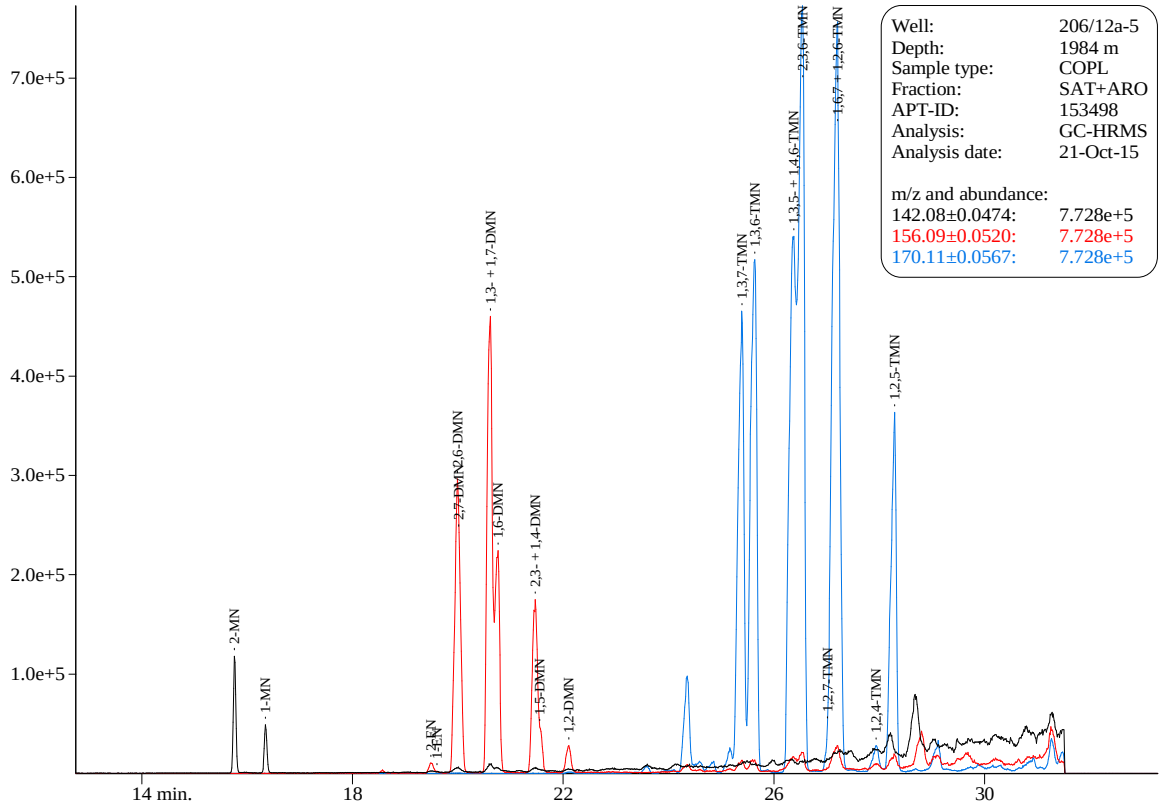


FIGURE Ap8.14 Aromatic fraction GC-MS fragmentograms: 206/12a-5, OIL, 1984.5m (LCG Unit III)



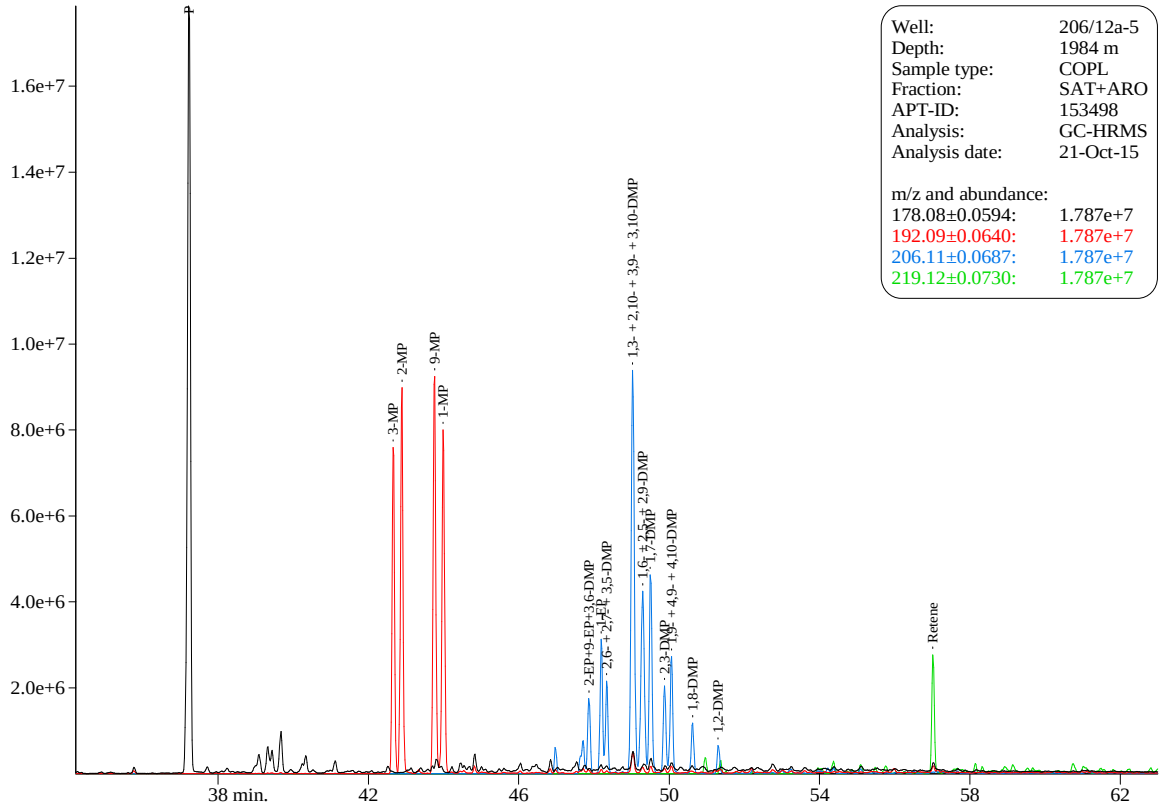
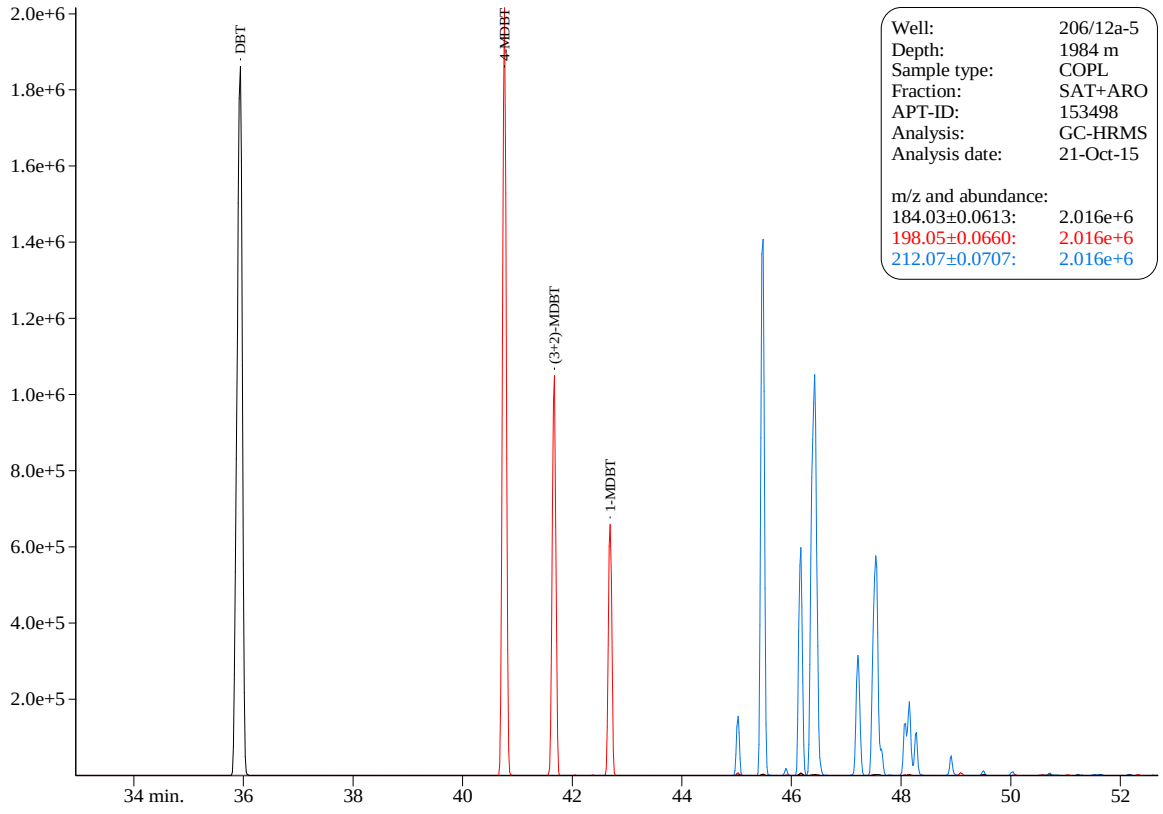


FIGURE Ap8.15 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)



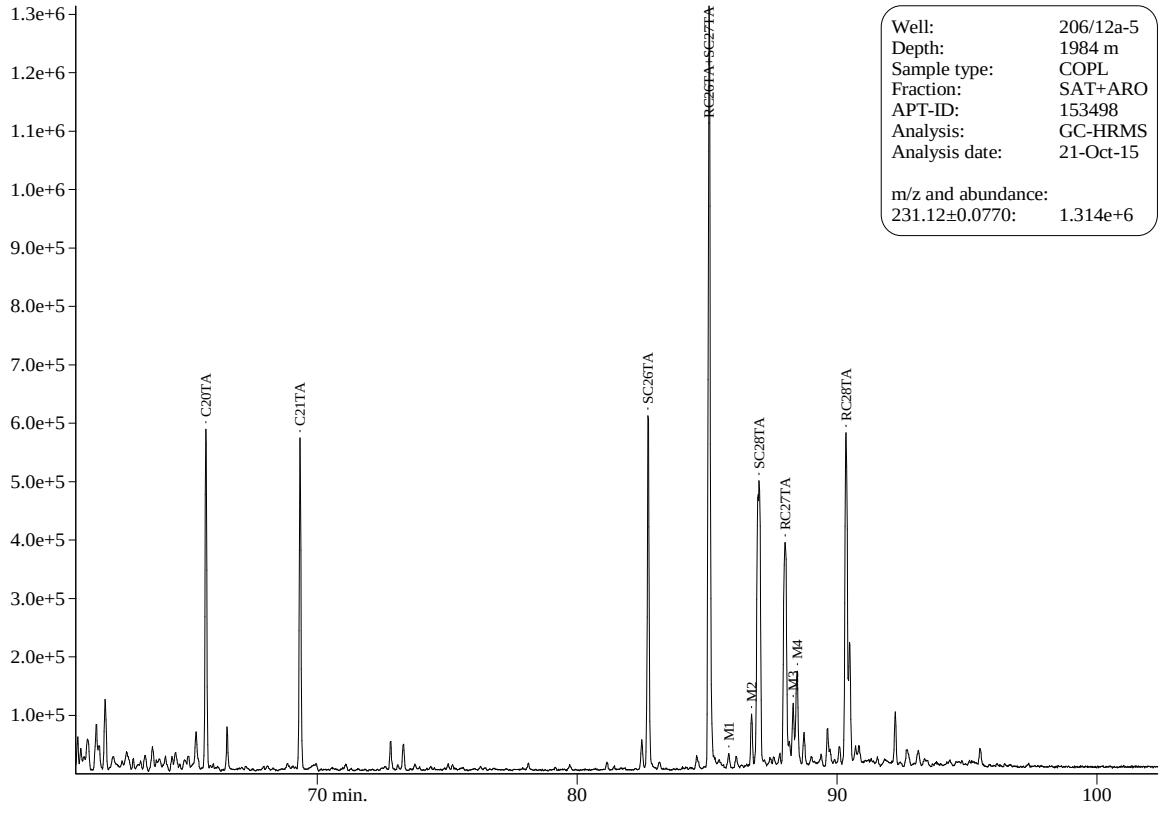


FIGURE Ap8.15 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)

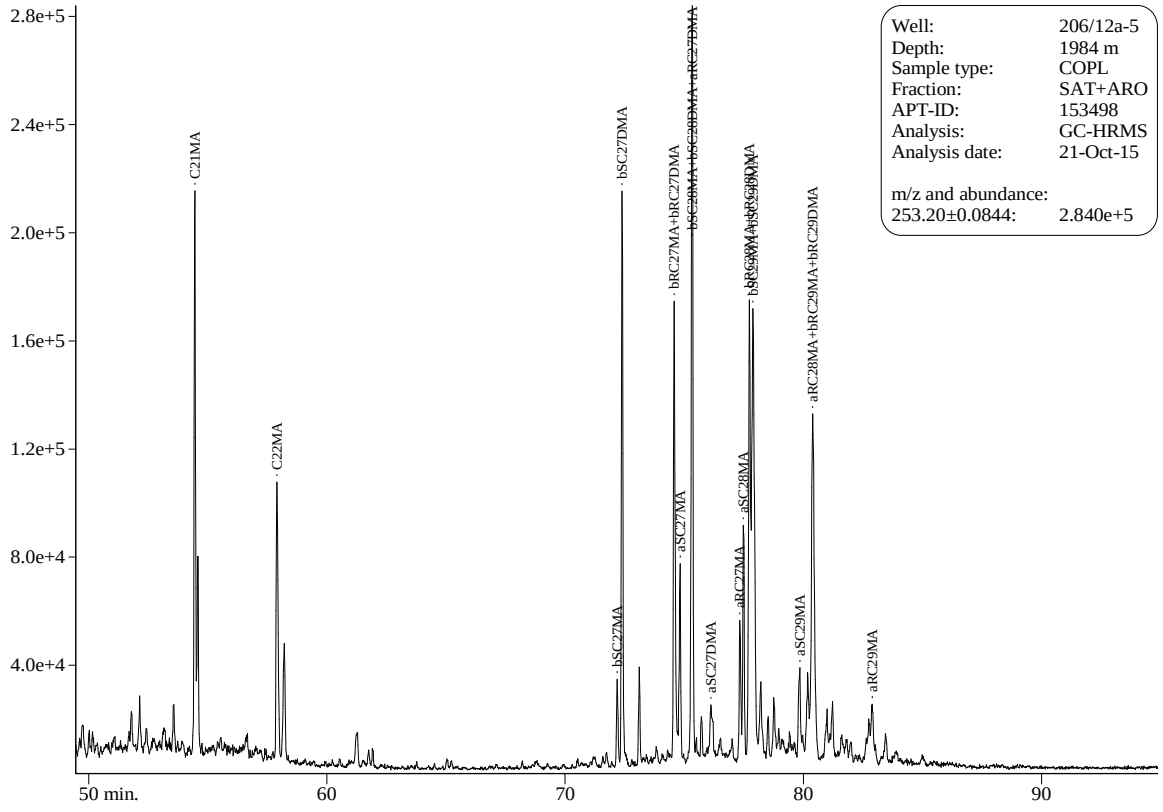
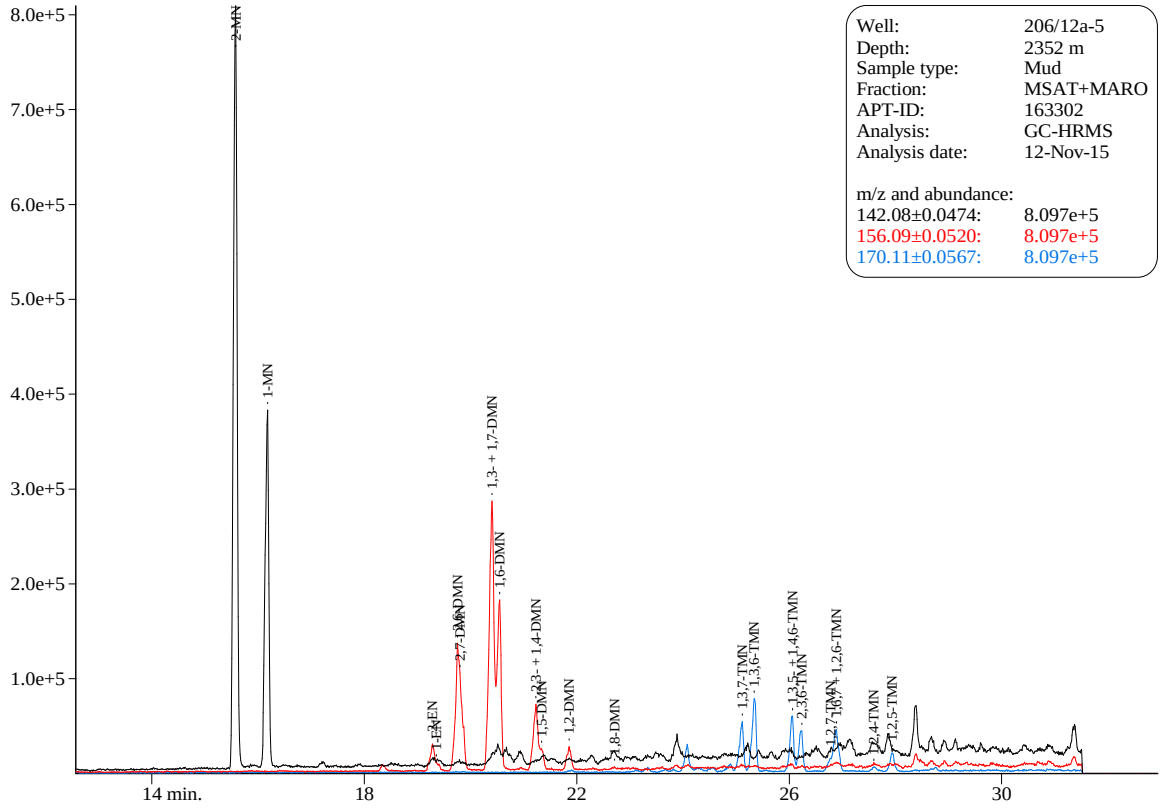


FIGURE Ap8.15 Aromatic fraction GC-MS fragmentograms: 206/12a-5, CORE PLUG, 1997.67m (LCG Unit III)



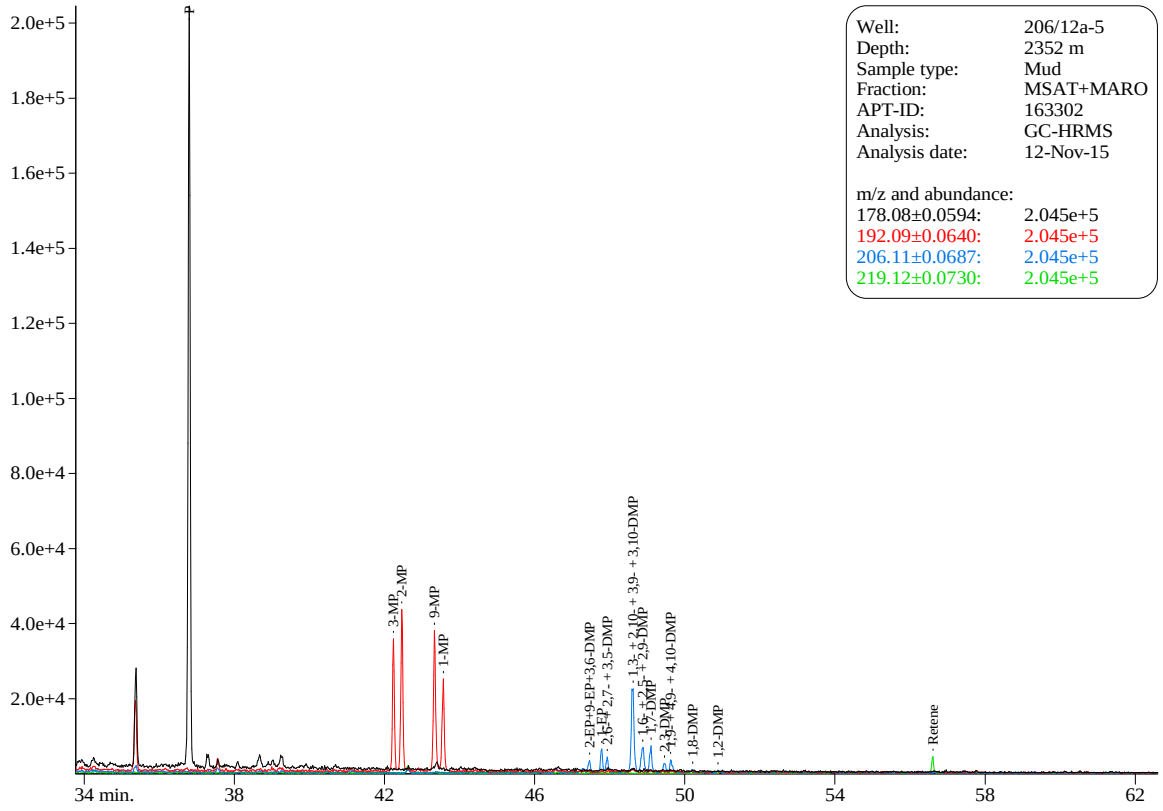
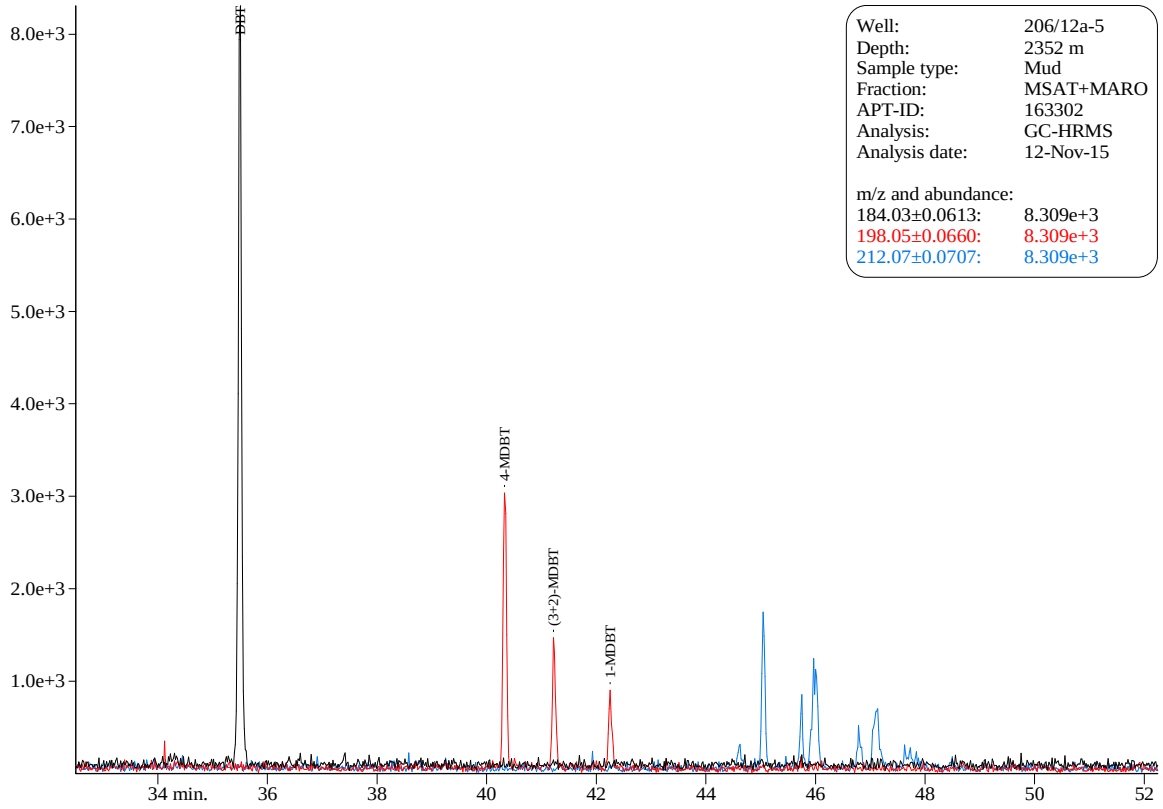


FIGURE Ap8.16 Aromatic fraction GC-MS fragmentograms: 206/12a-5, MUD, 2352m (Basement)



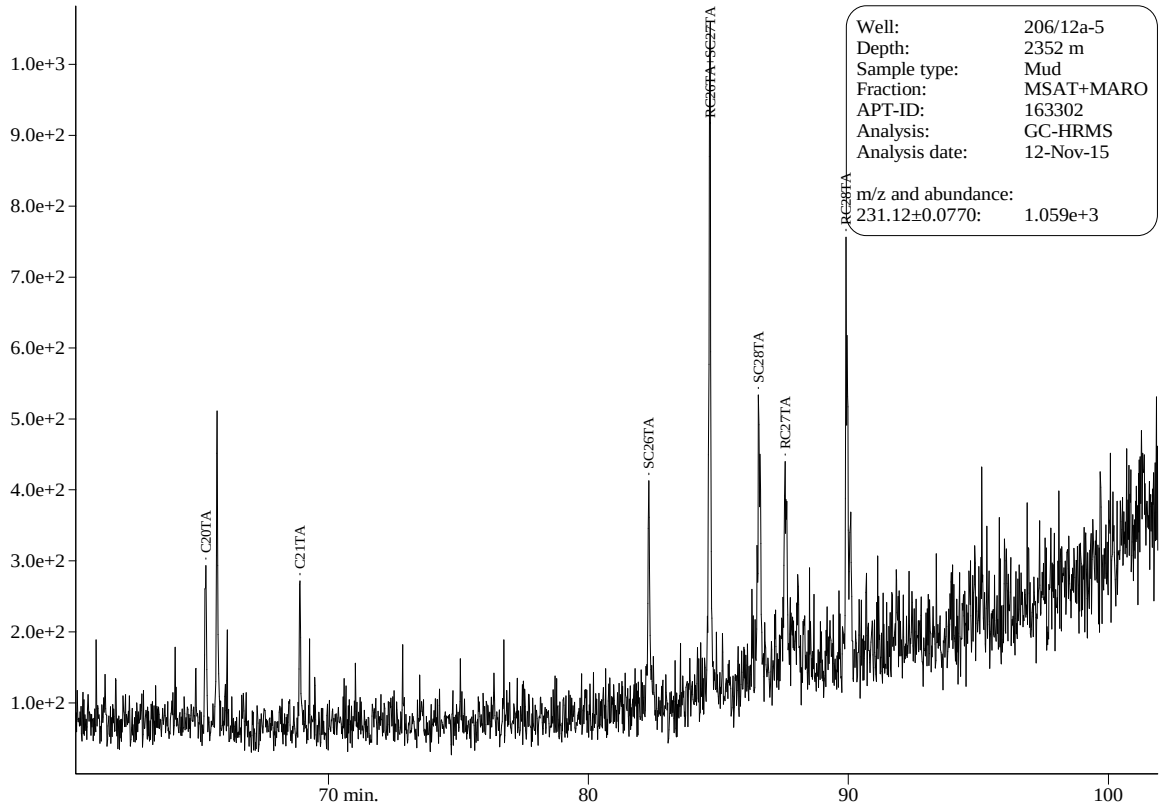


FIGURE Ap8.16 Aromatic fraction GC-MS fragmentograms: 206/12a-5, MUD, 2352m (Basement)

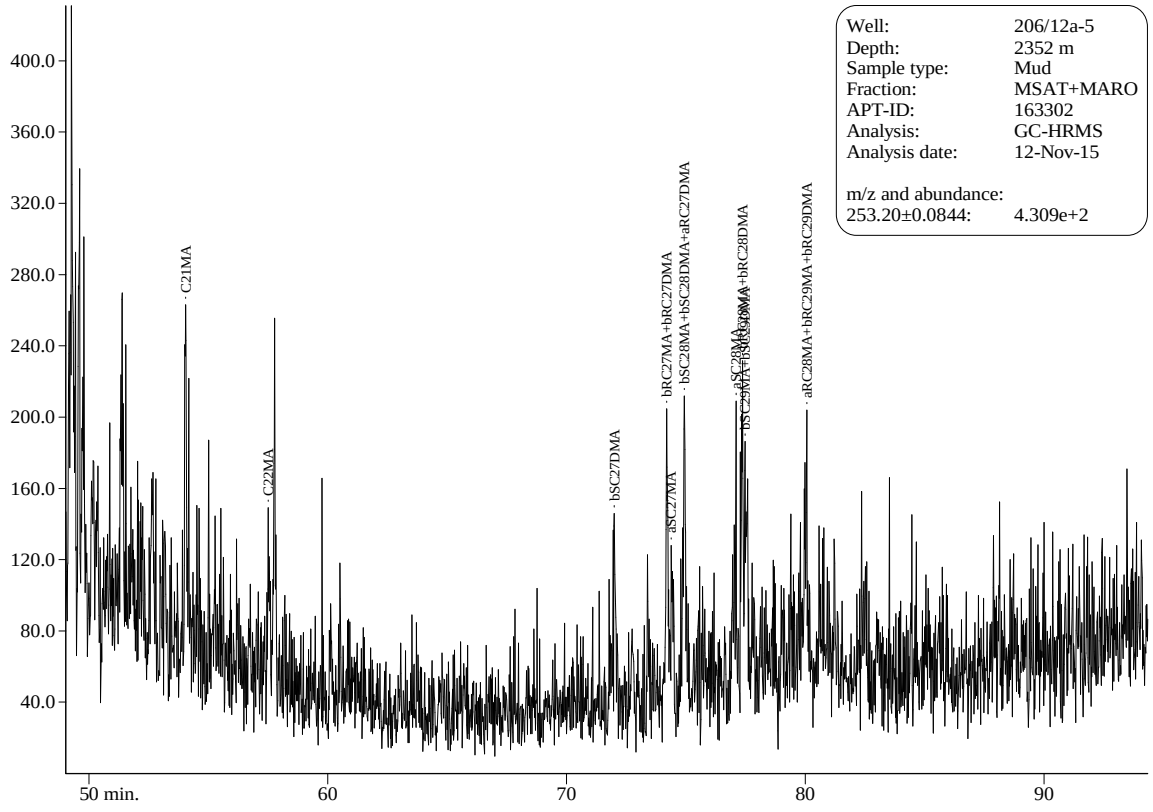


FIGURE Ap8.16 Aromatic fraction GC-MS fragmentograms: 206/12a-5, MUD, 2352m (Basement)

APPENDIX 9
Carbon isotope ratio data

Tables

Ap9.1 Carbon isotope ratios

Well	MD (m)	TVDSS (m)	Sample type	APTID	Carbon isotope ratios (‰ relative to PDB)					
					$\delta^{13}C$ whole oil/EOM	$\delta^{13}C$ alkanes	$\delta^{13}C$ aromatics	$\delta^{13}C$ polars	$\delta^{13}C$ asphaltenes	$\delta^{13}C$ kerogen
206/12a-5	1897.03	1873.4	Spun out oil	146645	-29.90	-30.40	-28.70	-29.4	-29.50	
	1922.08	1898.4	Spun out oil	146643	-29.90	-30.60	-29.10	-29.5	-29.70	
	1922.40	1898.7	Oil	160302	-30.10	-30.50	-29.20	-29.4	-30.00	
	1932.67	1909.0	Core plug	146976	-29.50	-30.70	-29.10	-29.6	-29.70	
	1937.32	1913.7	Core	154661	-29.30	-30.30	-28.20	-29.7	-29.60	
	1948.67	1925.0	Spun out oil	146644	-29.80	-30.50	-29.10	-29.4	-29.70	
	1956.00	1932.3	Oil	160303	-29.90	-30.50	-29.20	-29.5	-30.20	
	1936-1963	1912.3-1939.3	Oil	160301	-30.00	-30.40	-29.10	-29.2	-30.30	
	1984.50	1960.8	Oil	160304	-29.80	-30.40	-29.10	-29.4	-30.10	

TABLE Ap9.1 Carbon isotope ratios: current study samples

APPENDIX 10
Chemical and elemental compositions plus physical properties data

Tables

Ap10.1 Chemical and elemental compositions plus physical properties data

Well	MD (m)	TVDSS (m)	Sample type	APTID	TAN (mg/g KOH)	Sulphur content		Elements content (ppm)									
						ppm	%	Ba	Ca	Mg	P	Zn	B	Na	Si	Li	Al
206/12a-5	1922.4	1898.7	Oil (MDT)	160302	0.7	3248	3.248	7.2	25.0	0.8	0.0	0.0	0.6	1.2	35.0	0.0	0.9
	1936-1963	1912.3-1939.3	Oil (DST)	160301	0.8	3187	3.187	7.9	9.1	1.2	0.8	0.7	0.6	2.6	49.0	0.1	0.6
	1956.0	1932.3	Oil (MDT)	160303	0.8	2922	2.922	2.9	5.6	1.0	1.8	1.7	0.6	1.8	153.0	0.1	0.0
	1984.5	1960.8	Oil (MDT)	160304	0.8	3138	3.138	7.2	10.0	2.1	0.7	2.3	0.6	2.2	163.0	0.1	1.2

Well	MD (m)	TVDSS (m)	Sample type	APTID	Elements content (ppm)											Ni/V Ratio
					Cr	Cu	Fe	Pb	Sn	Mo	Ni	Ti	Ag	Mn	V	
206/12a-5	1922.4	1898.7	Oil (MDT)	160302	0.0	0.6	2.6	0.6	1.1	0.1	8.0	0.2	0.0	0.2	2.1	3.81
	1936-1963	1912.3-1939.3	Oil (DST)	160301	0.0	0.3	1.9	0.0	0.9	0.0	7.1	0.2	0.0	0.6	1.8	3.94
	1956.0	1932.3	Oil (MDT)	160303	0.1	0.4	1.7	0.5	0.6	0.3	8.1	0.3	0.1	0.4	2.0	4.05
	1984.5	1960.8	Oil (MDT)	160304	0.0	0.7	2.5	0.5	0.0	0.0	8.9	0.2	0.1	0.2	2.2	4.05

Well	MD (m)	TVDSS (m)	Sample type	APTID	API	Kinematic viscosity (cSt)			Wax (%)
						20°C	40°C	60°C	
206/12a-5	1922.4	1898.7	Oil (MDT)	160302	25.4	89.7	34.5	16.67	3.23
	1936-1963	1912.3-1939.3	Oil (DST)	160301	25.8	76.7	29.8	14.59	3.09
	1956.0	1932.3	Oil (MDT)	160303	26.7	103.5	33.8	14.86	2.19
	1984.5	1960.8	Oil (MDT)	160304	25.5	117.9	33.9	13.88	3.02

TABLE Ap10.1 Chemical and elemental compositions plus physical properties data

APPENDIX 11
Gas composition and isotope ratio data

Tables

Ap11.1a Composition data (mole%): mud gases - Isotubes
Ap11.1b Composition data (ppm): mud gases - Isotubes
Ap11.1c Isotope ratio data: mud gases - Isotubes
Ap11.2a Composition data (mole%): mud gases - Isojars
Ap11.2b Composition data (ppm): mud gases - Isojars
Ap11.2c Isotope ratio data: mud gases - Isojars
Ap11.3a Composition data (mole%): tested gases
Ap11.3b Composition data (ppm): tested gases
Ap11.3c Isotope ratio data: tested gases

Well	MD (m)	TVDSS (m)	Sample type	%C1 in total gas	%CO_2 in total gas	Hydrocarbon composition (mole%, total hydrocarbons = 100%)									Degree of Wetness (%)
						C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	
206/12a-5	950	926.5	Isotube	0.0002	0.051	100									0.000
	970	946.5	Isotube	0.0002	0.055	100									0.000
	990	966.5	Isotube	0.0002	0.056	100									0.000
	1010	986.5	Isotube	0.0026	0.060	100									0.000
	1050	1026.5	Isotube	0.0058	0.061	100									0.000
	1090	1066.5	Isotube	0.0046	0.057	100									0.000
	1111.5	1088.0	Isotube	0.011	0.073	100									0.000
	1150	1126.5	Isotube	0.076	0.017	99.8	0.12	0.039	0.013				0.018	0.028	0.179
	1190	1166.5	Isotube	0.078	0.019	99.8	0.13		0.037	0.012		0.0092	0.02	0.022	0.218
	1250	1226.4	Isotube	0.11	0.020	99.7	0.17		0.035	0.0088		0.013	0.014	0.016	0.248
	1310	1286.4	Isotube	0.14	0.019	99.7	0.23		0.045	0.0094	0.0089	0.011	0.0099	0.0099	0.315
	1370	1346.4	Isotube	0.1	0.020	99.6	0.31		0.063	0.013	0.012	0.019	0.019	0.015	0.438
	1410	1386.4	Isotube	0.14	0.018	99.5	0.34	0.0072	0.07	0.0046	0.018	0.019	0.015	0.014	0.476
	1450	1426.4	Isotube	0.14	0.019	99.5	0.38		0.082	0.0067	0.018	0.022	0.015	0.012	0.529
	1490	1466.4	Isotube	0.1	0.016	99.3	0.49		0.11	0.0061	0.025	0.035	0.022	0.022	0.704
	1530	1506.4	Isotube	0.097	0.016	99.3	0.47		0.092	0.0099	0.025	0.032	0.026	0.02	0.665
	1570	1546.4	Isotube	0.079	0.016	99.3	0.47		0.081	0.016	0.018	0.028	0.027	0.017	0.641
	1610	1586.4	Isotube	0.096	0.015	99.4	0.42	0.015	0.07	0.01		0.015	0.012	0.016	0.533
	1650	1626.4	Isotube	0.16	0.016	99.2	0.69		0.094	0.0059	0.015	0.021	0.0082	0.0082	0.836
	1750	1726.4	Isotube	0.11	0.020	97.8	1.7		0.31	0.0084	0.063	0.074	0.029	0.027	2.203
	1790	1766.4	Isotube	0.061	0.016	97.6	1.6		0.39	0.026	0.085	0.12	0.068	0.071	2.336
	1810	1786.4	Isotube	0.077	0.021	98.2	1.4		0.24	0.016	0.046	0.062	0.027	0.032	1.807
	1830	1806.4	Isotube	0.1	0.019	97.4	1.6	0.34	0.32	0.081	0.063	0.079	0.037	0.039	2.148
	1850	1826.4	Isotube	0.0004	0.039	100									0.000
	1870	1846.3	Isotube	0.0003	0.039	100									0.000
	1890	1866.3	Isotube	0.13	0.025	95.6	3	0.26	0.74	0.066	0.14	0.15	0.067	0.051	4.158

TABLE Ap11.1a Composition data (mole%): mud gases - Isotubes

Well	MD (m)	TVDSS (m)	Sample type	%C1 in total gas	%CO_2 in total gas	Hydrocarbon composition (mole%, total hydrocarbons = 100%)									Degree of Wetness (%)
						C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	
206/12a-5	1910	1886.3	Isotube	0.099	0.024	95.3	3.1	0.27	0.82	0.077	0.17	0.19	0.091	0.074	4.456
	1930	1906.3	Isotube	0.36	0.027	94.7	3.5	0.21	1	0.062	0.19	0.19	0.079	0.054	5.027
	1950	1926.3	Isotube	0.98	0.029	94.2	3.8	0.22	1.1	0.067	0.22	0.21	0.086	0.055	5.489
	1970	1946.3	Isotube	0.65	0.023	93.5	4.1	0.24	1.3	0.079	0.27	0.27	0.12	0.076	6.158
	1990	1966.3	Isotube	0.39	0.027	92	4.8	0.34	1.7	0.12	0.4	0.4	0.19	0.12	7.640
	1998	1974.3	Isotube	0.37	0.044	93.1	4.1	0.36	1.4	0.11	0.3	0.31	0.14	0.1	6.385
	2016	1992.3	Isotube	0.13	0.022	91.8	4	1.2	1.4	0.23	0.36	0.42	0.28	0.22	6.783
	2025	2001.3	Isotube	0.31	0.032	93.8	3.7	0.45	1.2	0.091	0.25	0.28	0.14	0.11	5.710
	2034	2010.3	Isotube	0.81	0.021	94.7	3.5	0.14	1	0.034	0.21	0.22	0.1	0.074	5.114
	2043	2019.3	Isotube	0.5	0.020	93.8	3.8	0.29	1.2	0.064	0.27	0.29	0.14	0.1	5.823
	2052	2028.3	Isotube	0.44	0.021	93.5	3.9	0.23	1.4	0.056	0.3	0.32	0.15	0.11	6.200
	2061	2037.2	Isotube	0.47	0.024	93.8	3.8	0.19	1.3	0.048	0.28	0.31	0.15	0.11	5.965
	2070	2046.2	Isotube	0.15	0.020	85.6	5.7	3.7	2.3	0.65	0.59	0.71	0.39	0.31	10.460
	2077	2053.2	Isotube	0.095	0.025	89.4	3.8	3.4	1.3	0.56	0.38	0.46	0.34	0.28	6.836
	2079	2055.2	Isotube	0.067	0.038	78.2	6.4	8.7	2.6	1.6	0.7	0.88	0.52	0.42	12.840
	2088	2064.1	Isotube	0.06	0.022	72.7	6.5	13.5	2.3	2.5	0.66	0.86	0.56	0.47	13.504
	2097	2073.1	Isotube	0.019	0.058	66.3	7	17.5	2.3	3.7	0.68	1	0.81	0.74	15.895
	2106	2082.0	Isotube	0.048	0.021	67	6.4	18.9	1.7	3.8	0.47	0.71	0.53	0.46	13.291
	2115	2090.9	Isotube	0.074	0.019	75.1	5	14.3	1.2	2.9	0.31	0.45	0.36	0.32	9.234
	2124	2099.8	Isotube	0.054	0.022	80.5	4.1	9.9	1.1	2.7	0.33	0.51	0.44	0.4	7.874
	2133	2108.7	Isotube	0.058	0.021	85.8	3.2	6.6	0.93	2	0.27	0.43	0.39	0.36	6.106
	2142	2117.6	Isotube	0.044	0.020	87.5	2.9	5.2	0.93	1.8	0.29	0.48	0.47	0.42	5.904
	2160	2135.3	Isotube	0.033	0.022	91.9	2	2.6	0.69	1.2	0.24	0.42	0.5	0.45	4.470
	2187	2161.6	Isotube	0.03	0.026	94.7	1.5	1.3	0.53	0.67	0.18	0.33	0.45	0.39	3.446

TABLE Ap11.1a Composition data (mole%): mud gases - Isotubes

Well	MD (m)	TVDSS (m)	Sample type	%C1 in total gas	%CO_2 in total gas	Hydrocarbon composition (mole%, total hydrocarbons = 100%)									Degree of Wetness (%)
						C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	
206/12a-5	2223	2196.5	Isotube	0.032	0.017	72	5.2	18.2	0.68	3.3	0.07	0.16	0.21	0.2	8.304
	2250	2222.6	Isotube	0.086	0.017	65.5	6.7	22.4	0.83	4.4	0.028	0.11	0.043	0.047	10.590
	2277	2248.8	Isotube	0.11	0.023	60.6	7.7	24.5	1.1	5.8	0.043	0.14	0.027	0.044	12.999
	2304	2275.1	Isotube	0.11	0.019	59.3	8	25	1.2	6.3	0.045	0.17	0.023	0.047	13.789
	2322	2292.8	Isotube	0.22	0.018	62.8	7.1	23.8	0.93	5.1	0.031	0.12	0.018	0.03	11.585
	2331	2301.6	Isotube	0.097	0.019	56.7	9.2	25.2	1.6	7	0.063	0.24	0.033	0.063	16.494
	2340	2310.5	Isotube	0.096	0.030	59.5	10	22.3	1.7	6	0.073	0.27	0.032	0.069	16.950

KEY

C1 : CH_4 methane
C2 : C_2H_6 ethane
C2= : C_2H_4 ethene
C3 : C_3H_8 propane
C3= : C_3H_6 propene
i-C4 : C_3H_7-CH_3 iso-butane
n-C4 : C_4H_{10} normal butane
i-C5 : C_4H_9-CH_3 iso-pentane
n-C5 : C_5H_{12} normal pentane

Degree of Wetness : (sum of C2-C5)/(sum of C1-C5)*100 (excluding ethene and propene)

TABLE Ap11.1a Composition data (mole%): mud gases - Isotubes

Well	MD (m)	TVDSS (m)	Sample type	%C1 in total gas	%CO_2 in total gas	Hydrocarbon composition (ppm)									Degree of Wetness (%)
						C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	
206/12a-5	950	926.5	Isotube	0.0002	0.051	2									0.000
	970	946.5	Isotube	0.0002	0.055	2									0.000
	990	966.5	Isotube	0.0002	0.056	2									0.000
	1010	986.5	Isotube	0.0026	0.060	26									0.000
	1050	1026.5	Isotube	0.0058	0.061	58									0.000
	1090	1066.5	Isotube	0.0046	0.057	46									0.000
	1111.5	1088.0	Isotube	0.011	0.073	110									0.000
	1150	1126.5	Isotube	0.076	0.017	760	0.9	0.3	0.1				0.1	0.2	0.171
	1190	1166.5	Isotube	0.078	0.019	780	1		0.3	0.1		0.1	0.2	0.2	0.230
	1250	1226.4	Isotube	0.11	0.020	1100	2		0.4	0.1		0.1	0.2	0.2	0.263
	1310	1286.4	Isotube	0.14	0.019	1400	3		0.6	0.1	0.1	0.2	0.1	0.1	0.292
	1370	1346.4	Isotube	0.1	0.020	1000	3		0.6	0.1	0.1	0.2	0.2	0.2	0.428
	1410	1386.4	Isotube	0.14	0.018	1400	5	0.1	1	0.1	0.3	0.3	0.2	0.2	0.498
	1450	1426.4	Isotube	0.14	0.019	1400	5		1	0.1	0.3	0.3	0.2	0.2	0.498
	1490	1466.4	Isotube	0.1	0.016	1000	5		1	0.1	0.3	0.4	0.2	0.2	0.705
	1530	1506.4	Isotube	0.097	0.016	970	5		0.9	0.1	0.2	0.3	0.3	0.2	0.706
	1570	1546.4	Isotube	0.079	0.016	790	4		0.6	0.1	0.1	0.2	0.2	0.1	0.654
	1610	1586.4	Isotube	0.096	0.015	960	4	0.1	0.7	0.1		0.1	0.1	0.2	0.528
	1650	1626.4	Isotube	0.16	0.016	1600	11		2	0.1	0.2	0.3	0.1	0.1	0.849
	1750	1726.4	Isotube	0.11	0.020	1100	19		3	0.1	0.7	0.8	0.3	0.3	2.144
	1790	1766.4	Isotube	0.061	0.016	610	10		2	0.2	0.5	0.8	0.4	0.4	2.259
	1810	1786.4	Isotube	0.077	0.021	770	11		2	0.1	0.4	0.5	0.2	0.3	1.836
	1830	1806.4	Isotube	0.1	0.019	1000	16	3	3	0.8	0.6	0.8	0.4	0.4	2.076
	1850	1826.4	Isotube	0.0004	0.039	4									0.000
	1870	1846.3	Isotube	0.0003	0.039	3									0.000
	1890	1866.3	Isotube	0.13	0.025	1300	41	4	10	0.9	2	2	0.9	0.7	4.172

TABLE Ap11.1b Composition data (ppm): mud gases - Isotubes

Well	MD (m)	TVDSS (m)	Sample type	%C1 in total gas	%CO_2 in total gas	Hydrocarbon composition (ppm)									Degree of Wetness (%)
						C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	
206/12a-5	1910	1886.3	Isotube	0.099	0.024	990	32	3	9	0.8	2	2	0.9	0.8	4.505
	1930	1906.3	Isotube	0.36	0.027	3600	133	8	38	2	7	7	3	2	5.013
	1950	1926.3	Isotube	0.98	0.029	9800	395	23	114	7	23	22	9	6	5.488
	1970	1946.3	Isotube	0.65	0.023	6500	285	17	90	5	19	19	8	5	6.151
	1990	1966.3	Isotube	0.39	0.027	3900	203	14	72	5	17	17	8	5	7.627
	1998	1974.3	Isotube	0.37	0.044	3700	163	14	56	4	12	12	6	4	6.400
	2016	1992.3	Isotube	0.13	0.022	1300	57	17	20	3	5	6	4	3	6.810
	2025	2001.3	Isotube	0.31	0.032	3100	122	15	40	3	8	9	5	4	5.718
	2034	2010.3	Isotube	0.81	0.021	8100	299	12	86	3	18	19	9	6	5.119
	2043	2019.3	Isotube	0.5	0.020	5000	203	15	64	3	14	15	7	5	5.803
	2052	2028.3	Isotube	0.44	0.021	4400	184	11	66	3	14	15	7	5	6.203
	2061	2037.2	Isotube	0.47	0.024	4700	190	10	65	2	14	16	8	6	5.981
	2070	2046.2	Isotube	0.15	0.020	1500	100	65	40	11	10	12	7	5	10.394
	2077	2053.2	Isotube	0.095	0.025	950	40	36	14	6	4	5	4	3	6.863
	2079	2055.2	Isotube	0.067	0.038	670	55	75	22	14	6	8	4	4	12.874
	2088	2064.1	Isotube	0.06	0.022	600	54	111	19	21	5	7	5	4	13.545
	2097	2073.1	Isotube	0.019	0.058	190	20	50	7	11	2	3	2	2	15.929
	2106	2082.0	Isotube	0.048	0.021	480	46	135	12	27	3	5	4	3	13.201
	2115	2090.9	Isotube	0.074	0.019	740	49	141	12	29	3	4	4	3	9.202
	2124	2099.8	Isotube	0.054	0.022	540	28	66	7	18	2	3	3	3	7.850
	2133	2108.7	Isotube	0.058	0.021	580	22	45	6	14	2	3	3	2	6.149
	2142	2117.6	Isotube	0.044	0.020	440	15	26	5	9	1	2	2	2	5.782
	2160	2135.3	Isotube	0.033	0.022	330	7	9	2	4	0.9	2	2	2	4.597
	2187	2161.6	Isotube	0.03	0.026	300	5	4	2	2	0.6	1	1	1	3.413
	2223	#REF!	Isotube	0.032	0.017	320	23	81	3	15	0.3	0.7	0.9	0.9	8.257
	2250	#REF!	Isotube	0.086	0.017	860	88	294	11	58	0.4	1	0.6	0.6	10.566

TABLE Ap11.1b Composition data (ppm): mud gases - Isotubes

Well	MD (m)	TVDSS (m)	Sample type	%C1 in total gas	%CO_2 in total gas	Hydrocarbon composition (ppm)									Degree of Wetness (%)
						C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	
206/12a-5	2223	2196.5	Isotube	0.032	0.017	320	23	81	3	15	0.3	0.7	0.9	0.9	8.257
	2250	2222.6	Isotube	0.086	0.017	860	88	294	11	58	0.4	1	0.6	0.6	10.566
	2277	2248.8	Isotube	0.11	0.023	1100	140	445	20	105	0.8	3	0.5	0.8	13.050
	2304	2275.1	Isotube	0.11	0.019	1100	148	464	22	117	0.8	3	0.4	0.9	13.732
	2322	2292.8	Isotube	0.22	0.018	2200	249	834	33	179	1	4	0.6	1	11.597
	2331	2301.6	Isotube	0.097	0.019	970	157	431	27	120	1	4	0.6	1	16.423
	2340	2310.5	Isotube	0.096	0.030	960	161	360	27	97	1	4	0.5	1	16.847

KEY

C1 : CH_4 methane
C2 : C_2H_6 ethane
C2= : C_2H_4 ethene
C3 : C_3H_8 propane
C3= : C_3H_6 propene
i-C4 : C_3H_7-CH_3 iso-butane
n-C4 : C_4H_{10} normal butane
i-C5 : C_4H_9-CH_3 iso-pentane
n-C5 : C_5H_{12} normal pentane

Degree of Wetness : (sum of C2-C5)/(sum of C1-C5)*100 (excluding ethene and propene)

TABLE Ap11.1b Composition data (ppm): mud gases - Isotubes

Well	MD (m)	TVDSS (m)	Sample type	Carbon isotope ratio ($\delta^{13}C$ ‰ relative to PDB)										δD
				C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	CO_2	C1
206/12a-5	950	926.5	Isotube	too small										
	970	946.5	Isotube	too small										
	990	966.5	Isotube	too small										
	1010	986.5	Isotube	-80.1										too small
	1050	1026.5	Isotube	-76.1										
	1090	1066.5	Isotube	-68.8										
	1111.5	1088.0	Isotube	-67.1										too small
	1150	1126.5	Isotube	-62.1										
	1190	1166.5	Isotube	-60.1										
	1250	1226.4	Isotube	-55.6										
	1310	1286.4	Isotube	-53.3										
	1370	1346.4	Isotube	-51.9	-33.8		[-30]							-167
	1410	1386.4	Isotube	-50										
	1450	1426.4	Isotube	-49.4	-33.3		-29		[-29]	[-29]				-163
	1490	1466.4	Isotube	-48.2										
	1530	1506.4	Isotube	-47.5	-32.3		[-29]		[-29]	[-29]				-163
	1570	1546.4	Isotube	-46.5										
	1610	1586.4	Isotube	-46.8										
	1650	1626.4	Isotube	-45.6	-31.5		-29		[-29]	[-30]				-160
	1750	1726.4	Isotube	-43.7	-31.1		-30		[-28]	[-29]	[-28]	[-30]		-161
	1790	1766.4	Isotube	-41.3										
	1810	1786.4	Isotube	-43.4	-30.6		-29		[-28]	[-28]		[-29]		-161
	1830	1806.4	Isotube	-43	-30.7	-34	-29	[-32]	[-29]	[-28]	[-27]	[-29]		-160
	1850	1826.4	Isotube	too small										
	1870	1846.3	Isotube	too small										
	1890	1866.3	Isotube	-43.7	-30.5	-32	-28	[-33]	-28	-28	[-27]	[-28]		-165

TABLE Ap11.1c Isotope ratio data: mud gases - Isotubes

Well	MD (m)	TVDSS (m)	Sample type	Carbon isotope ratio ($\delta^{13}C$ ‰ relative to PDB)										δD
				C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	CO_2	C1
206/12a-5	1910	1886.3	Isotube	-43										
	1930	1906.3	Isotube	-43.2	-30		-28		-28	-28	-27	-27		-163
	1950	1926.3	Isotube	-43.3										
	1970	1946.3	Isotube	-42.8	-30.1		-28	-32	-28	-27	-27	-27		-167
	1990	1966.3	Isotube	-39.4										
	1998	1974.3	Isotube	-43.2	-30		-28	-32	-28	-28	-27	-27		-167
	2016	1992.3	Isotube	-42	-29.9	-32	-28	-32	-28	-27	-27	-27		-164
	2025	2001.3	Isotube	-43	-30.3		-28	-30	-28	-27	-27	-27		-169
	2034	2010.3	Isotube	-42.9	-30.1		-28	-31	-28	-28	-27	-28		-165
	2043	2019.3	Isotube	-42.3	-30		-28	-31	-28	-28	-27	-27		-165
	2052	2028.3	Isotube	-42.2	-29.8		-28		-28	-28	-27	-27		-166
	2061	2037.2	Isotube	-42.2	-29.7		-28	-32	-28	-28	-27	-27		-166
	2070	2046.2	Isotube	-39.3										
	2077	2053.2	Isotube	-42.7	-30.5	-33	-28	-32	-27	-27	-27	-27		-164
	2079	2055.2	Isotube	-38.3										
	2088	2064.1	Isotube	-37.5	-31.1	-32	-28	-33	-28	-28	-28	-27		-145
	2097	2073.1	Isotube	-37.1										
	2106	2082.0	Isotube	-37.4	-31.9	-32	-28	-32	-28	-28	-27	-28		-135
	2115	2090.9	Isotube	-39.1	-32.1	-32	-29	-32	-28	-28	-28	-28		-141
	2124	2099.8	Isotube	-39.8	-31.8	-31	-29	-33	-28	-28	-28	-27		-146
	2133	2108.7	Isotube	-40.1	-31.4	-31	-29	-32	-28	-28	-27	-27		-149
	2142	2117.6	Isotube	-39.8	-31.1	-31	-27	-33	-28	-28	-27	-27		-154
	2160	2135.3	Isotube	-40	-30.8	-31	-28	-33	[-28]	-28	-27	-27		-153
	2187	2161.6	Isotube	-40	-30.6	-31	-27	-33	[-28]	-28	-27	-27		-154

TABLE Ap11.1c Isotope ratio data: mud gases - Isotubes

Well	MD (m)	TVDSS (m)	Sample type	%C1 in total gas	%CO_2 in total gas	Hydrocarbon composition (mole%, total hydrocarbons = 100%)									Degree of Wetness (%)
						C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	
206/12a-5	1120	1096.5	Isojar (Out)	0.0018	<50ppm	85.1	2	3.6	3.2	4.1		0.44	0.35	1.1	7.691
	1832	1808.4	Isojar (Out)	0.15	<50ppm	92.5	5	0.049	1.5	0.032	0.33	0.32	0.12	0.098	7.378
	1833	1809.4	Isojar (Out)	0.033	<50ppm	72.9	14.3	0.17	7.3	0.13	2	1.8	0.84	0.64	26.939
	1873	1849.3	Isojar (Out)	0.0039	<50ppm	18.7	22.8	0.32	27.9	0.28	11.5	9.5	5.6	3.4	81.187
	1883	1859.3	Isojar (Out)	0.023	<50ppm	91.1	4.4	0.61	1.9	0.29	0.49	0.6	0.35	0.3	8.110
	1933	1909.3	Isojar (Out)	0.095	<50ppm	89	5.6	0.3	2.8	0.13	0.82	0.72	0.39	0.24	10.616
	1983	1959.3	Isojar (Out)	0.3	<50ppm	90	5.7	0.39	2.2	0.15	0.55	0.53	0.27	0.18	9.484
	2033	2009.3	Isojar (Out)	0.079	<50ppm	71.1	12.5	0.3	8.4	0.11	3.1	2.3	1.4	0.78	28.600
	2083	2059.2	Isojar (Out)	0.022	<50ppm	73.3	7.4	9.7	3.4	2.3	1	1.3	0.85	0.69	16.648
	2133	2108.7	Isojar (Out)	0.013	<50ppm	83.1	5	5.1	1.7	2.6	0.51	0.74	0.7	0.64	10.055
	2183	2157.7	Isojar (Out)	0.0059	<50ppm	90.2	2.6	1.9	1.1	1.4	0.51	0.62	0.93	0.84	6.818
	2233	2206.1	Isojar (Out)	0.0093	<50ppm	59.1	8.3	23.4	1.5	6.8	0.11	0.27	0.27	0.25	15.330
	2283	2254.6	Isojar (Out)	0.0094	<50ppm	49.4	10.7	25.7	2.5	10.9	0.15	0.45	0.08	0.15	22.119
	2333	2303.6	Isojar (Out)	0.017	<50ppm	59.2	10.3	19.4	2.3	8	0.12	0.45	0.074	0.15	18.451
	2352	2322.2	Isojar (Out)	0.033	0.0093	72.1	8.9	12.7	1.7	4.1	0.068	0.28	0.034	0.11	13.333

KEY

C1 : CH_4 methane
C2 : C_2H_6 ethane
C2= : C_2H_4 ethene
C3 : C_3H_8 propane
C3= : C_3H_6 propene
i-C4 : C_3H_7-CH_3 iso-butane
n-C4 : C_4H_{10} normal butane
i-C5 : C_4H_9-CH_3 iso-pentane
n-C5 : C_5H_{12} normal pentane

Degree of Wetness : (sum of C2-C5)/(sum of C1-C5)*100 (excluding ethene and propene)

TABLE Ap11.2a Composition data (mole%): mud gases - Isojars

Well	MD (m)	TVDSS (m)	Sample type	%C1 in total gas	%CO_2 in total gas	Hydrocarbon composition (ppm)									Degree of Wetness (%)
						C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	
206/12a-5	1120	1096.5	Isojar (Out)	0.0018	<50ppm	18	0.4	0.8	0.7	0.9		0.1	0.1	0.2	7.692
	1832	1808.4	Isojar (Out)	0.15	<50ppm	1500	81	0.8	24	0.5	5	5	2	2	7.350
	1833	1809.4	Isojar (Out)	0.033	<50ppm	330	65	0.8	33	0.6	9	8	4	3	26.991
	1873	1849.3	Isojar (Out)	0.0039	<50ppm	39	48	0.7	58	0.6	24	20	12	7	81.250
	1883	1859.3	Isojar (Out)	0.023	<50ppm	230	11	2	5	0.7	1	2	0.9	0.8	8.257
	1933	1909.3	Isojar (Out)	0.095	<50ppm	950	60	3	30	1	9	8	4	3	10.714
	1983	1959.3	Isojar (Out)	0.3	<50ppm	3000	190	13	73	5	18	18	9	6	9.475
	2033	2009.3	Isojar (Out)	0.079	<50ppm	790	139	3	93	1	34	26	16	9	28.636
	2083	2059.2	Isojar (Out)	0.022	<50ppm	220	22	29	10	7	3	4	3	2	16.667
	2133	2108.7	Isojar (Out)	0.013	<50ppm	130	8	8	3	4	0.8	1	1	1	10.221
	2183	2157.7	Isojar (Out)	0.0059	<50ppm	59	2	1	0.7	0.9	0.3	0.4	0.6	0.5	7.087
	2233	2206.1	Isojar (Out)	0.0093	<50ppm	93	13	37	2	11	0.2	0.4	0.4	0.4	14.991
	2283	2254.6	Isojar (Out)	0.0094	<50ppm	94	20	49	5	21	0.3	0.9	0.2	0.3	22.121
	2333	2303.6	Isojar (Out)	0.017	<50ppm	170	30	56	7	23	0.3	1	0.2	0.4	18.621
	2352	2322.2	Isojar (Out)	0.033	0.0093	330	41	58	8	19	0.3	1	0.2	0.5	13.386

KEY

C1 : CH_4 methane
C2 : C_2H_6 ethane
C2= : C_2H_4 ethene
C3 : C_3H_8 propane
C3= : C_3H_6 propene
i-C4 : C_3H_7-CH_3 iso-butane
n-C4 : C_4H_{10} normal butane
i-C5 : C_4H_9-CH_3 iso-pentane
n-C5 : C_5H_{12} normal pentane

Degree of Wetness : (sum of C2-C5)/(sum of C1-C5)*100 (excluding ethene and propene)

TABLE Ap11.2b Composition data (ppm): mud gases - Isojars

Well	MD (m)	TVDSS (m)	Sample type	Carbon isotope ratio ($\delta^{13}C$ ‰ relative to PDB)										δD
				C1	C2	C2=	C3	C3=	i-C_4	n-C_4	i-C_5	n-C_5	CO_2	C1
206/12a-5	1120	1096.5	Isojar (Out)	-56.2	[-43.4]	[-40]	[-34]	[-32]						Too small
	1832	1808.4	Isojar (Out)	-33.8	-27.2	[-37]	-28	[-34]	-27	-27	-27	-28		
	1833	1809.4	Isojar (Out)	-23.5	-25.2	[-37]	-27	[-32]	-27	-27	-27	-28		-153
	1873	1849.3	Isojar (Out)	-12	-19	[-35]	-24	[-31]	-26	-26	-26	-27		Too small
	1883	1859.3	Isojar (Out)	-39.1	-28.3	-37	-28	[-32]	-28	-28	[-27]	[-29]		-158
	1933	1909.3	Isojar (Out)	-38.1	-27.4	-31	-26	-31	-27	-26	-27	-27		-161
	1983	1959.3	Isojar (Out)	-37.3	-27.8	-29	-27	-31	-27	-27	-27	-28		-163
	2033	2009.3	Isojar (Out)	-29.9	-23.7	-28	-25	-30	-26	-25	-26	-27		-163
	2083	2059.2	Isojar (Out)	-35.6	-29.1	-31	-28	-31	-28	-28	-27	-28		[-149]
	2133	2108.7	Isojar (Out)	-32.7										
	2183	2157.7	Isojar (Out)	-39.4										
	2233	2206.1	Isojar (Out)	-35										
	2283	2254.6	Isojar (Out)	-33.7										
	2333	2303.6	Isojar (Out)	-34.3										
	2352	2322.2	Isojar (Out)	-34.3										

Leaking sample

PDB : Pee Dee Belemnite

δD : Deuterium isotope ratio (dD ‰ relative to SMOW (Standard Mean Ocean Water))

[data] : Isotopes measured from low concentration of gas, near analytical threshold

i/s : Insufficient sample

TABLE Ap11.2c Isotope ratio data: mud gases - Isojars

Well	MD (m)	TVDSS (m)	Sample type	Absolute composition (mole%, corrected for air, total of all measure compounds = 100%)										
				C1	C2	C2=	C3	C3=	i-C_4	n-C_4	neo-C_5	i-C_5	n-C_5	C6+
206/12a-5	1922.4	1898.7	Tested gas	82.1	6.3	0	3.8	0	1.2	1.4	0.017	0.82	0.56	1.5
	1936-1963	1912.3-1939.2	Tested gas (BHS)	81.7	6.3	0	3.8	0	1.2	1.4	0.018	0.86	0.60	1.7
	1936-1963	1912.3-1939.3	Tested gas (SGS)	90.4	4.7	0	1.8	0	0.40	0.38	0.0045	0.14	0.075	0.068
	1956	1932.3	Tested gas	81.1	6.3	0.0003	3.9	0.0003	1.2	1.5	0.019	0.92	0.65	1.8
	1984.5	1960.8	Tested gas	82.1	6.3	0	3.7	0	1.1	1.4	0.019	0.81	0.57	1.6

Well	MD (m)	TVDSS (m)	Sample type	Absolute composition (mole%) cont.				Degree of Wetness (%)	% air found in sample
				N_2	CO_2	H_2	H_2S		
206/12a-5	1922.4	1898.7	Tested gas	2.0	0.32	0	0	14.6392	0.41
	1936-1963	1912.3-1939.2	Tested gas (BHS)	2.0	0.38	0	0	14.7715	0.51
	1936-1963	1912.3-1939.3	Tested gas (SGS)	1.6	0.39	0	0	7.65616	0.88
	1956	1932.3	Tested gas	2.1	0.33	0	0	15.1407	2.6
	1984.5	1960.8	Tested gas	1.9	0.37	0	0	14.4613	7.3

KEY

C1 :	CH_4	methane	**i-C4 :**	C_3H_7-CH_3	iso-butane
C2 :	C_2H_6	ethane	**n-C4 :**	C_4H_{10}	normal butane
C2= :	C_2H_4	ethene	**neo-C5 :**	C_3H_6-$(CH_3)_2$	neo-pentane (dimethylpropane)
C3 :	C_3H_8	propane	**i-C5 :**	C_4H_9-CH_3	iso-pentane
C3= :	C_3H_6	propene	**n-C5 :**	C_5H_{12}	normal pentane

Degree of Wetness : (sum of C2-C5)/(sum of C1-C5)*100 (excluding ethene, propene and neo-pentane)

TABLE Ap11.2b Composition data (ppm): mud gases - Isojars

Well	MD (m)	TVDSS (m)	Sample type	Composition (ppm)										
				Methane	Ethane	Ethene	Propane	Propene	i-C_4	n-C_4	neo-C_5	i-C_5	n-C_5	C6+
206/12a-5	1922.4	1898.7	Tested gas	82100	6300	0	3800	0	1200	1400	17	820	560	1500
	1936-1963	1912.3-1939.2	Tested gas (BHS)	81700	6300	0	3800	0	1200	1400	18	860	600	1700
	1936-1963	1912.3-1939.3	Tested gas (SGS)	90400	4700	0	1800	0	400	380	4.5	140	75	68
	1956	1932.3	Tested gas	81100	6300	0.3	3900	0.3	1200	1500	19	920	650	1800
	1984.5	1960.8	Tested gas	82100	6300	0	3700	0	1100	1400	19	810	570	1600

Well	MD (m)	TVDSS (m)	Sample type	Composition (ppm) cont.				Degree of Wetness (%)
				N_2	CO_2	H_2	H_2S	
206/12a-5	1922.4	1898.7	Tested gas	2000	320	0	0	14.6392
	1936-1963	1912.3-1939.2	Tested gas (BHS)	2000	380	0	0	14.7715
	1936-1963	1912.3-1939.3	Tested gas (SGS)	1600	390	0	0	7.65616
	1956	1932.3	Tested gas	2100	330	0	0	15.1407
	1984.5	1960.8	Tested gas	1900	370	0	0	14.4613

KEY

C1 :	CH_4	methane	**i-C4 :**	C_3H_7-CH_3	iso-butane
C2 :	C_2H_6	ethane	**n-C4 :**	C_4H_{10}	normal butane
C2= :	C_2H_4	ethene	**neo-C5 :**	C_3H_6-$(CH_3)_2$	neo-pentane (dimethylpropane)
C3 :	C_3H_8	propane	**i-C5 :**	C_4H_9-CH_3	iso-pentane
C3= :	C_3H_6	propene	**n-C5 :**	C_5H_{12}	normal pentane

Degree of Wetness : (sum of C2-C5)/(sum of C1-C5)*100 (excluding ethene, propene and neo-pentane)

TABLE Ap11.2b Composition data (ppm): mud gases - Isojars

Well	MD (m)	TVDSS (m)	Sample type	Carbon isotope ratio ($\delta^{13}C$ ‰ relative to PDB)								δD	$\delta^{15}N$	
				C1	C2	C3	i-C_4	n-C_4	i-C_5	n-C_5	CO_2	Methane	un-corrected	corrected for air
206/12a-5	1922.4	1898.7	Tested gas	-42.6	-30.2	-27.9	-27.7	-27.4	-26.9	-27.2	-11.9	-167		
	1936-1963	1912.3-1939.2	Tested gas (BHS)	-42.6	-30.2	-27.9	-27.7	-27.4	-27	-27.3	-12	-168		
	1936-1963	1912.3-1939.3	Tested gas (SGS)	-42.6	-30	-27.8	-27.5	-27.1	-26.5	-26.8	-11.9	-167		
	1956	1932.3	Tested gas	-42.6	-30.1	-27.9	-27.6	-27.4	-27	-27.2	-12.7	-168		
	1984.5	1960.8	Tested gas	-42.6	-30.1	-28	-27.7	-27.5	-27	-27.4	-13	-168		

PDB : Pee Dee Belemnite
δD : Deuterium isotope ratio (dD ‰ relative to SMOW (Standard Mean Ocean Water))
$\delta^{15}N$: Nitrigen isotope ratio ($\delta^{15}N$ ‰ relative to N_2 Air)
[data] : Isotopes measured from low concentration of gas, near analytical threshold
i/s : Insufficient sample

TABLE Ap11.2b Composition data (ppm): mud gases - Isojars

APPENDIX 12
Analytical methods used in the APT AS laboratories (Norway)

AP12.1 INTRODUCTION

Chemical analyses of oil and rock samples are carried out by APT AS in their laboratories in Kjeller, near Oslo, Norway as described below. All experimental procedures follow those published in The Norwegian Industry Guide to Organic Geochemical Analyses 4th Edition (NIGOGA 2000). Below are brief descriptions of procedures/analytical conditions used.

AP12.2 CHEMICAL ANALYSES

Sample preparation
Cuttings samples are washed in water to remove mud. When oil based mud is used, soap (Zalo) is added to the sample and the sample is washed thoroughly in warm water to remove mud and soap.

TOC
A Leco SC-632 instrument is used. Diluted HCl is added to the crushed rock sample to remove carbonate. The sample is then introduced into the Leco combustion oven, and the amount of carbon in the sample is measured as carbon dioxide by an IR-detector.

Rock-Eval
A Rock-Eval 6 instrument is used. Jet-Rock 1 is run as every tenth sample and checked against the acceptable range given in NIGOGA.

Temperature programme
Pyrolysis: 300°C (3 min.) - 25°C/min. - 650°C (0 min.)

Extraction
A Soxtec Tecator instrument is used. Thimbles are pre extracted in dichloromethane with 7% (vol/vol) methanol, 10 min boiling and 20 min rinsing. The crushed sample is weighed accurately in the pre extracted thimbles and boiled for 1 hour and rinsed for 2 hours in approximately 80 cc of dichloromethane with 7% (vol/vol) methanol. Copper blades activated in concentrated hydrochloric acid are added to the extraction cups to cause free sulphur to react with the copper. An aliquot of 10% of the extract is transferred to a pre weighed bottle and evaporated to dryness. The amount of extractable organic matter is calculated from the weight of this 10% aliquot.

Deasphaltening
Extracts are evaporated almost to dryness before a small amount of dichloromethane (3 times the amount of EOM) is added. Pentane is added in excess (40 times the volume of EOM/oil and dichloromethane). The solution is stored for at least 12 hours in a dark place before the solution is filtered or centrifuged and the weight of the asphaltenes measured.

Quantitative MPLC 3 fractions
The MPLC is constructed as described by Radke *et al.* (1980). The system includes two HPLC pumps, sample injector, sample collector and two packed columns. The pre column is filled with Kieselgel 100, which is heated at 600 °C for 2 hours to deactivate it. The main column, a LiChroprep Si60 column, is heated at 120 °C for 2 hours with a helium flow to make it water free.

Approximately 30 mg of deasphaltened oil or EOM diluted in 1 ml hexane is injected into a sample loop. The solvents used are hexane and dichloromethane.

Fraction 1 - Saturates
Hexane through the sample loop, the pre column and the main column is collected until all saturates are collected.

Fraction 2 – Aromatics A
Hexane that back flushes the main column is collected.

Fraction 3 – Polars (NSO-fraction)
Dichloromethane that back flushes the pre column is collected.

Solvents from all fractions are removed until the total volume is 1 ml by using a Turbovap unit. The fractions are transferred to small pre weighed vials and dried carefully. The weights are then measured.

GC of whole oil
An Agilent 7890A instrument is used. The column is a HP PONA, length 50m, i.d. 0.2 mm, film thickness 0.5 µm. 2,2,4-tri-methyl-pentane is used as an internal standard.

Temperature programme
30°C (10 min.) - 2°C/min. - 60°C (10 min.)- 2°C/min - 130°C (0 min.)-4°C/min. - 320°C (25 min.).

GC of EOM fraction
A HP7890A instrument is used. The column is a CP-Sil-5 CB-MS, length 30 m, i.d. 0.25 mm, film thickness 0.25 µm. C20D42 is used as an internal standards.

Temperature programme
50°C (1 min.) - 4°C/min. - 320°C (25 min.).

GC of saturated fraction
A HP7890A instrument is used. The column is a CP-Sil-5 CB-MS, length 30m, i.d. 0.25mm, film thickness 0.25µm. C20D42 is used as an internal standards.

Temperature programme
50°C (1 min.) - 4°C/min. - 320°C (25 min.).

GCMS of saturated and aromatic fractions
A Micromass ProSpec high resolution instrument is used. The instrument is tuned to a resolution of 3000 and data are acquired in Selected Ion Recording (SIR) mode. The column used is a 60m CP-Sil-5 CB-MS with an i.d. of 0.25 mm and a film thickness 0.25 □m.

D4-27ααR is used as internal standard when quantitative results are requested for the saturated compounds. D8-naphthalene and D10-phenanthrene are used as internal standards when quantitative results are required for the aromatic compounds. The aromatic and aliphatic fractions may be analysed together or separately.

Temperature programme
50°C (1 min.) - 20°C/min. - 120°C - 2°C/min - 320°C (20 min.).

Stable carbon and hydrogen isotope analysis of fractions
The samples were dissolved in a known amount of dichloromethane, and 0.2 mg of the sample (or as much as possible) was transferred to a Zn capsule. The solvent was evaporated in an oven at 50 °C. The samples were then combusted in an EuroVector Elemental Analyser EA3028-IRMS at 1030°C. The produced water is trapped on Mg(ClO4)2 and the CO2 is flushed into a Horizon, Isotope Ratio mass spectrometer (IRMS) from NU-Instruments. A standard (NGS NSO-1, topped oil) is analysed for each 10th sample. The δ13C value obtained for this standard is –28.61‰ VPDB. The variation in the isotopic values for the standard by repeated analysis over a period of three years is ± 0.09‰.

$H_2O_{(g)}$ aliquot was reduced with $Zn_{(s)}$ to $H_{2(g)}$ and $ZnO_{(s)}$ in sealed, evacuated quartz vessels at 900ºC. The δD composition was determined by a Micromass Optima Isotope Ratio mass spectrometer (IRMS). Average analysed value for GISP from IAEA is δDVSMOW= -189.71 ± 0.89 ‰ (one standard deviation). Given value from IAEA is -189.73 ± 0.9 ‰.

Ap12.3 MICROSCOPY ANALYSES

Kerogen microscopy is carried out by APT UK in both reflected and transmitted light as described below:

Microscopes
Reflectivity measurements are made using a Zeiss Universal microscope equipped with an Epi 40X Pol oil immersion objective and 10.0X eyepieces, giving a total visual magnification of 400X. The immersion oil used is Cargille Type A (refractive index = 1.515° at 23°C). Measurements are made in light at a wavelength of 546nm (green), which is the ICCP standard. The measuring aperture is circular with a diameter of 1.6 micron or 0.8 microns; the choice of measuring aperture size is dependent on the size and quality of the organic material being measured.

Kerogen colouration and fluorescence analyses are conducted in transmitted white light using a Zeiss Axioplan microscope fitted with Zeiss Epiplan – Neofluar 10X, 40X and 63X objectives and 10X eyepieces, which gives a total visual magnification of 100X, 400X and 630X.

Ultra-violet fluorescence colours are observed in reflected short wavelength blue light using an excitation filter passing light of 365nm wavelength, a barrier filter of 397nm, and a dichromatic beam splitter of 395nm wavelength.

Reflected light studies

The reflectivity studies have been carried in the first instance out using polished samples of isolated kerogen mounted in resin blocks.

Grinding and polishing were carried out on a rotating lap wheel. The surface of the blocks was ground flat on carborundum papers using progressively finer grinding stages of 120, 240, 600 and 1200 grit. Polishing was carried out in three stages; 5/20, 3/50 and gamma alumina (.05µ) on magnetically attached polishing pads using water as lubricant. The finished blocks were mounted on microscope slides using a hand press and a small lump of plasticine, ensuring that the polished surface is normal to the incident light.

Measurements made are of Ro(random), which means they are made in nominally un-polarised light; as against Ro(max), which are made in polarised light. Two of three standards are normally used: Spinel (Ro = 0.588%0, Yttrium-Aluminium-Garnet (Ro = 0.879%) and Gadolinium-Gallium-Garnet (Ro = 1.696%). The choice of standards used is dependent on the expected range of reflectivity of the samples being examined.

Spore colouration studies

The slides used for spore colouration work are prepared using standard palynological techniques (hydrochloric and hydrofluoric acids) but omitting any oxidative stages. The kerogen is floated in zinc bromide after agitation with ultrasonic energy to disaggregate the kerogen components to remove mineral matter. The floated kerogen is then split into unsieved total kerogen and sieved >10µ fractions, each of which are mounted on separate cover slips, which are then affixed to the same microscope slide using Norland Optical Adhesive. Unused kerogen is retained in water for use in preparing polished kerogen blocks for reflectivity studies.

Spore colours are determined using the >10µ fraction while kerogen description is carried out using the total kerogen. Spore colour and kerogen typing analyses are supplemented by observation of exinite fluorescence under reflected blue light excitation.

AP12.4 REFERENCES

NIGOGA, 2000. The Norwegian Industry Guide to Organic Geochemical Analyses. Accessed11 December 2012, <http://www.npd.no/engelsk/nigoga/default.htm>.

APPENDIX 13
Analytical methods used in the ISOLAB BV laboratories (The Netherlands)

AP13.1 INTRODUCTION

Chemical analyses of gas samples are carried out by ISOLAB BV in their laboratories in Neerijnen, The Netherlands, as described below. Below are brief descriptions of procedures/analytical conditions used as stated by Bart van der Haven in May 2014.

AP13.2 CHEMICAL ANALYSES

Composition of mudgas samples
For composition analysis of samples with low gas concentration, like mudgas samples in Isotubes or Isojars, an Agilent 6890N/7890A/7890B GC (Agilent Technologies, Santa Clara, US) equipped with a 30m 0.32mm Gaspro column or 25m 0.32mm Porabond Q column (both from Agilent) is used. Samples are injected into a heated injection valve on the GC using a gastight syringe. The FID detector is coupled to a data system and response is calibrated against a reference gas (Scott Specialty Gases Netherlands BV, Breda, The Netherlands) several times a day.

Minimum concentrations required
ca. 1ppm for methane to 0.1ppm for pentane.

Composition of (pressurised) gas samples
The composition of (pressurized) natural gas samples is analysed on two GC's. The H2, O2+Ar, N2 and Methane are analysed on an Agilent 6890N/7890A/7890B GC (Agilent Technologies, Santa Clara, US) equipped with a 12m, 0.32mm molsieve column (Agilent), a heated injection valve and a TCD detector. Hydrocarbons and CO2 are analysed on an Agilent 6890N/7890A/7890B GC equipped with a 50m, 0.32mm Porabond-Q column (Agilent), a heated injection valve, a heated backflush valve (for C6+), a TCD detector (for CO2) and a FID detector (for hydrocarbons). From the results of all three detectors, one complete composition is calculated. Detector responses are calibrated several times a day using various reference standards.

Minimum concentrations required
ca. 1ppm for hydrocarbons ca. 400ppm for H2 and ca. 100ppm for CO2, O2 and N2.

Analysis of carbon isotopes of methane
Carbon isotopes of methane are analysed with an Agilent 6890N GC (Agilent Technologies, Santa Clara, US) interfaced to a Finigan Delta S IRMS (Finigan is now part of Thermo Scientific, Bremen, Germany) using a Finigan GC-C II interface. The GC is equipped with a 12m, 0.32mm molsieve column (Agilent) and an injection valve. Samples are calibrated regularly against a calibration standard and results are reported in promille vs. vPDB.

Minimum concentration required
25-50ppm.

Analysis of hydrogen isotopes of methane
The dD measurements are done on an Agilent 7890A GC (Agilent Technologies, Santa Clara, US) interfaced to a MAT 253 IRMS (Thermo Scientific, Bremen, Germany) using a GC-Isolink interface from Thermo. The GC is equipped with a 25m, 0.32mm molsieve column (Agilent) and an injection valve. Samples are calibrated regularly against a calibration standard and results are reported in promille vs. vSMOW.

Minimum concentration required
ca. 300-400ppm

Analysis of nitrogen isotopes
Nitrogen isotopes of Nitrogen (N2) are analysed on an Agilent 7890A GC (Agilent Technologies, Santa Clara, US) interfaced to a Finigan MAT 252 IRMS (Finigan is now part of Thermo Scientific, Bremen, Germany) using a GC-C II interface from Finigan. The GC is equipped with a 25m, 0.32mm molsieve column (Agilent) and an injection valve. Samples are calibrated regularly against air and results are reported in promille vs. air N2.

Minimum concentration required
ca. 0.5%

Analysis of ethane-pentane carbon isotopes of mudgas samples (single run screening)
Carbon isotopes of ethane-pentane are analysed on an Agilent 7890A GC (Agilent Technologies, Santa Clara, US) interfaced to a Finigan MAT 252 IRMS (Finigan is now part of Thermo Scientific, Bremen, Germany) using a GC-C II interface from Finigan. The GC is equipped with a 30m 0.32mm Gaspro column or 25m 0.32mm Porabond Q column (both from Agilent). Cold trapping is used to pre-concentrate the sample when necessary. The system is calibrated about once a day using an in house natural gas standard and results are reported in promille vs. vPDB.

Minimum concentration required
ca. 0.5-1ppm

Analysis of CO2 and ethane-pentane carbon isotopes of (pressurized) natural gas samples (triplicate)
Carbon isotopes of ethane-pentane are analysed on an Agilent 7890A GC (Agilent Technologies, Santa Clara, US) interfaced to a MAT 253 IRMS (Thermo Scientific, Bremen, Germany) using a GC-Isolink or a Finigan GC-C III interface from Thermo. The GC is equipped with a 25m, 0.32mm Porabond-Q column (Agilent) and an injection valve. Cold trapping is used to pre-concentrate the sample when necessary. Samples are run at least 3 times after which the average of the results is calculated. The system is calibrated at least once a day using an in house natural gas standard and results are reported in promille vs. vPDB.

Minimum concentrations required
Dependent on sample composition.

Analysis ethane-pentane hydrogen isotopes of (pressurised) natural gas samples (triplicate)
Hydrogen isotopes of ethane-pentane are analysed on an Agilent 7890A GC (Agilent Technologies, Santa Clara, US) interfaced to a MAT 253 IRMS (Thermo Scientific, Bremen,

Germany) using a GC-Isolink or a Finigan GC-C III interface from Thermo. The GC is equipped with a 30m 0.32mm Gaspro column, a 25m 0.32mm Porabond-Q column (both from Agilent) or a combination of both and an injection valve. Cold trapping is used to pre-concentrate the sample when necessary. Samples are run at least 3 times after which the average of the results is calculated. The system is calibrated at least once a day using an in house natural gas standard and results are reported in promille vs. vSMOW.

Minimum concentrations required
depending on sample composition.

Hydrogen isotopes of H2
Hydrogen isotopes of H2 are analysed on an Agilent 7890A GC (Agilent Technologies, Santa Clara, US) interfaced to a MAT 253 IRMS (Thermo Scientific, Bremen, Germany) using a GC-Isolink or a Finigan GC-C III interface from Thermo. The GC is equipped with a 25m 0.32mm molsieve column (Agilent) and an injection valve. Samples are calibrated regularly against a calibration standard and results are reported in promille vs. vSMOW.

Minimum concentration required
500-600ppm.